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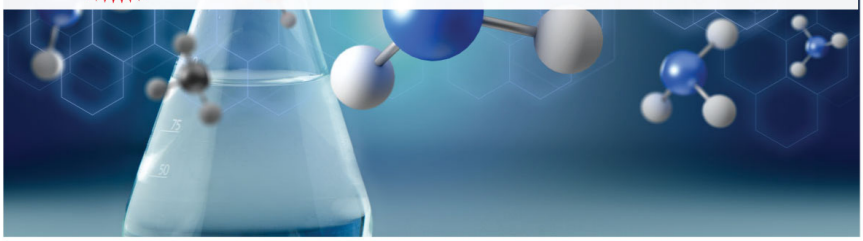
As per the CBSE **Revised Curriculum**
For the Academic Year **2020-21**



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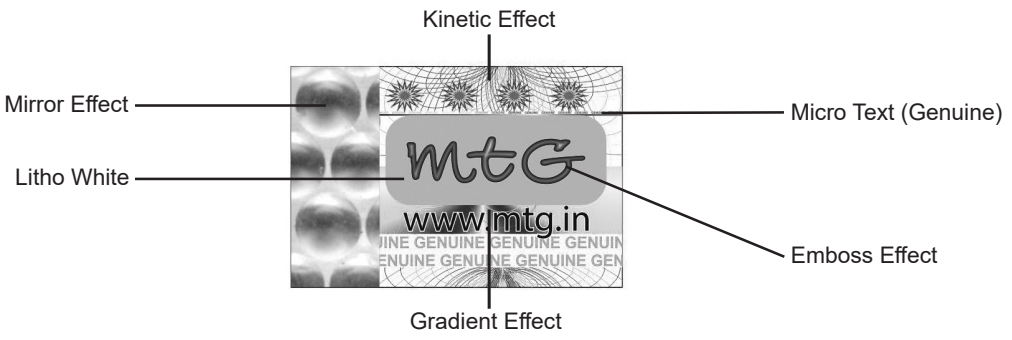

PREFACE

We feel pleased and delighted in presenting the revised edition of the book **"CBSE Champion Chapterwise-Topicwise Chemistry"** as per the CBSE Curriculum for the academic year 2020-21. Special efforts have been put to produce this book in order to equip students with practice material including previous 10 years' CBSE Board Examination questions. It will give them comprehensive knowledge of subject according to the latest syllabus and pattern of CBSE Board Examination. The book will be helpful in imparting students a clear and vivid understanding of the subject.

Salient features

Comprehensive and Lucid Theory : Well explained theory with important formulae, flowcharts and tables for quick recap.

Topicwise Graphical Analysis : Graphical analysis of previous 10 years' CBSE Board papers' questions (1 mark, 2 marks, 3 marks, 5 marks) provided to let students figure out which chapter and which topic is to be revised hard and how much is the weightage of that topic.

Chapterwise-Topicwise Questions and Answers : Theory is followed by chapterwise-topicwise questions pulled from previous 10 years' CBSE-DELHI, ALL INDIA, FOREIGN and COMPARTMENT papers. Answers are given according to the CBSE marking scheme.

Strictly Based on NCERT Pattern : In the previous years' CBSE papers SA, LA I or LA II type questions are generally framed by clubbing together questions from different topics and chapters. These questions are segregated strictly according to NCERT topics. *e.g.*, (*1/3, Delhi 2016*), this question of 1 mark was asked in LA I type category, (*1/5, AI 2015*), this question of 1 mark was asked in LA II type category, (*1/2, Foreign 2016*), this question of 1 mark was asked in SA type category.

Topicwise questions are arranged in descending chronological (2020-2011) order so, that latest years' questions come first in practice and revision.

Key Concepts Highlight : Key concepts have been highlighted for their reinforcement.

Practice Papers with Objective Type Questions : 10 Practice Papers strictly based on the latest design and blue print issued by CBSE Board are also incorporated. Various types of questions like Passage Based, One Word Answer, MCQs, Assertion-Reason, VSA, SA and LA type are included.

Solved CBSE Sample Paper : Solved CBSE sample paper is included with the marking scheme.

We are sure that the value addition done in this book will prove helpful to students in achieving success in board examinations. Every possible effort has been made to make this book error free. Useful suggestions by our readers for the rectification and improvement of the book content would be gracefully acknowledged and incorporated in further editions.

Readers are welcome to send their suggestions at **editor@mtg.in.**

All The Best

MTG Editorial Board

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SYLLABUS*

Unit No.	Title	No. of Periods	Marks
Unit I	Solid State	10	23
Unit II	Solutions	10	
Unit III	Electrochemistry	12	
Unit IV	Chemical Kinetics	10	
Unit V	Surface Chemistry	08	
Unit VI	General Principles and Processes of Isolation of Elements	08	19
Unit VII	*p*- Block Elements	12	
Unit VIII	*d*- and *f*- Block Elements	12	
Unit IX	Coordination Compounds	12	
Unit X	Haloalkanes and Haloarenes	10	28
Unit XI	Alcohols, Phenols and Ethers	10	
Unit XII	Aldehydes, Ketones and Carboxylic Acids	10	
Unit XIII	Amines	10	
Unit XIV	Biomolecules	12	
Unit XV	Polymers	08	
Unit XVI	Chemistry in Everyday Life	06	
	Total	**160**	**70**

Unit I : Solid State **[10 Periods]**

Classification of solids based on different binding forces: molecular, ionic, covalent and metallic solids, amorphous and crystalline solids (elementary idea). Unit cell in two dimensional and three dimensional lattices, calculation of density of unit cell, packing in solids, packing efficiency, voids, number of atoms per unit cell in a cubic unit cell, point defects, electrical and magnetic properties.

Band theory of metals, conductors, semiconductors and insulators and *n* and *p* type semiconductors.

Unit II : Solutions **[10 Periods]**

Types of solutions, expression of concentration of solutions of solids in liquids, solubility of gases in liquids, solid solutions, Raoult's law, colligative properties - relative lowering of vapour pressure, elevation of boiling point, depression of freezing point, osmotic pressure, determination of molecular masses using colligative properties, abnormal molecular mass, Van't Hoff factor.

Unit III : Electrochemistry **[12 Periods]**

Redox reactions, EMF of a cell, standard electrode potential, Nernst equation and its application to chemical cells, Relation between Gibbs energy change and EMF of a cell, conductance in electrolytic solutions, specific and molar conductivity, variations of conductivity with concentration, Kohlrausch's Law, electrolysis and law of electrolysis (elementary idea), dry cell-electrolytic cells and Galvanic cells, lead accumulator, fuel cells, corrosion.

Unit IV : Chemical Kinetics **[10 Periods]**

Rate of a reaction (Average and instantaneous), factors affecting rate of reaction: concentration, temperature, catalyst; order and molecularity of a reaction, rate law and specific rate constant, integrated

rate equations and half-life (only for zero and first order reactions), concept of collision theory (elementary idea, no mathematical treatment), activation energy, Arrhenius equation.

Unit V : Surface Chemistry **[08 Periods]**

Adsorption - physisorption and chemisorption, factors affecting adsorption of gases on solids, catalysis: homogenous and heterogenous, activity and selectivity of solid catalysts; enzyme catalysis, colloidal state: distinction between true solutions, colloids and suspension; lyophilic, lyophobic, multi-molecular and macromolecular colloids; properties of colloids; Tyndall effect, Brownian movement, electrophoresis, coagulation, emulsion - types of emulsions.

Unit VI : General Principles and Processes of Isolation of Elements **[08 Periods]**

Principles and methods of extraction - concentration, oxidation, reduction - electrolytic method and refining; occurrence and principles of extraction of aluminium, copper, zinc and iron.

Unit VII : *p*-Block Elements **[12 Periods]**

Group -15 Elements: General introduction, electronic configuration, occurrence, oxidation states, trends in physical and chemical properties; Nitrogen : preparation, properties and uses; compounds of Nitrogen: preparation and properties of Ammonia and Nitric Acid, Oxides of Nitrogen (Structure only); Phosphorus - allotropic forms, compounds of Phosphorus: Preparation and properties of Phosphine, Halides and Oxoacids (elementary idea only).

Group 16 Elements: General introduction, electronic configuration, oxidation states, occurrence, trends in physical and chemical properties, dioxygen: preparation, properties and uses, classification of Oxides, Ozone, Sulphur -allotropic forms; compounds of Sulphur: preparation, properties and uses of Sulphur dioxide, Sulphuric Acid: industrial process of manufacture, properties and uses; Oxoacids of Sulphur (Structures only).

Group 17 Elements: General introduction, electronic configuration, oxidation states, occurrence, trends in physical and chemical properties; compounds of halogens, Preparation, properties and uses of Chlorine and Hydrochloric acid, interhalogen compounds, Oxoacids of halogens (structures only).

Group 18 Elements: General introduction, electronic configuration, occurrence, trends in physical and chemical properties, uses.

Unit VIII : *d*- and *f*-Block Elements **[12 Periods]**

General introduction, electronic configuration, occurrence and characteristics of transition metals, general trends in properties of the first row transition metals - metallic character, ionization enthalpy, oxidation states, ionic radii, colour, catalytic property, magnetic properties, interstitial compounds, alloy formation, preparation and properties of $K_2Cr_2O_7$ and $KMnO_4$.

Lanthanoids - Electronic configuration, oxidation states, chemical reactivity and lanthanoid contraction and its consequences.

Actinoids - Electronic configuration, oxidation states and comparison with lanthanoids.

Unit IX : Coordination Compounds **[12 Periods]**

Coordination compounds - Introduction, ligands, coordination number, colour, magnetic properties and shapes, IUPAC nomenclature of mononuclear coordination compounds. Bonding, Werner's theory, VBT, and CFT; structure and stereoisomerism, importance of coordination compounds (in qualitative analysis, extraction of metals and biological system).

Unit X : Haloalkanes and Haloarenes **[10 Periods]**

Haloalkanes: Nomenclature, nature of C–*X* bond, physical and chemical properties, optical rotation mechanism of substitution reactions.

Haloarenes: Nature of C–*X* bond, substitution reactions (Directive influence of halogen in monosubstituted compounds only).

Uses and environmental effects of - dichloromethane, trichloromethane, tetrachloromethane, iodoform, freons, DDT.

Unit XI: Alcohols, Phenols and Ethers **[10 Periods]**

Alcohols: Nomenclature, methods of preparation, physical and chemical properties (of primary alcohols only), identification of primary, secondary and tertiary alcohols, mechanism of dehydration, uses with special reference to methanol and ethanol.

Phenols: Nomenclature, methods of preparation, physical and chemical properties, acidic nature of phenol, electrophillic substitution reactions, uses of phenols.

Ethers: Nomenclature, methods of preparation, physical and chemical properties, uses.

Unit XII: Aldehydes, Ketones and Carboxylic Acids **[10 Periods]**

Aldehydes and Ketones: Nomenclature, nature of carbonyl group, methods of preparation, physical and chemical properties, mechanism of nucleophilic addition, reactivity of alpha hydrogen in aldehydes, uses.

Carboxylic Acids: Nomenclature, acidic nature, methods of preparation, physical and chemical properties; uses.

Unit XIII: Amines **[10 Periods]**

Amines: Nomenclature, classification, structure, methods of preparation, physical and chemical properties, uses, identification of primary, secondary and tertiary amines.

Diazonium salts: Preparation, chemical reactions and importance in synthetic organic chemistry.

Unit XIV: Biomolecules **[12 Periods]**

Carbohydrates - Classification (aldoses and ketoses), monosaccahrides (glucose and fructose), *D-L* configuration oligosaccharides (sucrose, lactose, maltose), polysaccharides (starch, cellulose, glycogen); Importance of carbohydrates.

Proteins - Elementary idea of - amino acids, peptide bond, polypeptides, proteins, structure of proteins - primary, secondary, tertiary structure and quaternary structures (qualitative idea only), denaturation of proteins; enzymes. Hormones - Elementary idea excluding structure.

Vitamins - Classification and functions.

Nucleic Acids: DNA and RNA.

Unit XV: Polymers **[08 Period]**

Classification - Natural and synthetic, methods of polymerization (addition and condensation), copolymerization, some important polymers: natural and synthetic like polythene, nylon, polyesters, bakelite, rubber. Biodegradable and non-biodegradable polymers.

Unit XVI: Chemistry in Everyday life **[06 Periods]**

Chemicals in medicines - Analgesics, tranquilizers antiseptics, disinfectants, antimicrobials, antifertility drugs, antibiotics, antacids, antihistamines.

Chemicals in food - Preservatives, artificial sweetening agents, elementary idea of antioxidants.

Cleansing agents- Soaps and detergents, cleansing action.

CHEMISTRY (Code No. 043) QUESTION PAPER DESIGN CLASS-XII (2020-21)

S.No.	Domains	Total Marks	%
1	**Remembering and Understanding:** Exhibit memory of previously learned material by recalling facts, terms, basic concepts and answers. Demonstrate understanding of facts and ideas by organizing, comparing, translating, interpreting, giving descriptions and stating main ideas.	28	40
2	**Applying:** Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	21	30
3	**Analysing, Evaluating and Creating:** Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations. Present and defend opinions by making judgments about information, validity of ideas or quality of work based on a set of criteria. Compile information together in a different way by combining elements in a new pattern or proposing alternative solutions.	21	30

1. No chapterwise weightage. Care to be taken to cover all the chapters.
2. Suitable *internal variations may be made for generating various templates.*

   Choice(s):

   There will be no overall choice in the question paper.

   However, 33% internal choices will be given in all the sections.

CHEMISTRY

LATEST 2020-21 PRACTICE PAPERS

As per the **CBSE Revised Curriculum** for the **Academic Year 2020-21** issued on 7th July 2020

REVISED SYLLABUS*

CLASS XII (THEORY)

Time : 3 Hours **Max. Marks :** 70

Unit	Title	No. of Periods	Marks
I	Solid State	8	23
II	Solutions	8	
III	Electrochemistry	7	
IV	Chemical Kinetics	5	
V	Surface Chemistry	5	
VII	*p*-Block Elements	7	19
VIII	*d*- and *f*-Block Elements	7	
IX	Coordination Compounds	8	
X	Haloalkanes and Haloarenes	9	28
XI	Alcohols, Phenols and Ethers	9	
XII	Aldehydes, Ketones and Carboxylic Acids	10	
XIII	Amines	7	
XIV	Biomolecules	8	
	Total	**98**	**70**

UNIT-I : SOLID STATE (8 Periods)

Classification of solids based on different binding forces: molecular, ionic, covalent and metallic solids, amorphous and crystalline solids (elementary idea). Unit cell in two dimensional and three dimensional lattices, calculation of density of unit cell, packing in solids, packing efficiency, voids, number of atoms per unit cell in a cubic unit cell, point defects.

UNIT-II : SOLUTIONS (8 Periods)

Types of solutions, expression of concentration of solutions of solids in liquids, solubility of gases in liquids, solid solutions, Raoult's law, colligative properties- relative lowering of vapour pressure, elevation of boiling point, depression of freezing point, osmotic pressure, determination of molecular masses using colligative properties.

UNIT-III : ELECTROCHEMISTRY (7 Periods)

Redox reactions, EMF of a cell, standard electrode potential, Nernst equation and its application to chemical cells, Relation between Gibbs energy change and EMF of a cell, conductance in electrolytic solutions, specific and molar conductivity, variations of conductivity with concentration, Kohlrausch's Law, electrolysis.

UNIT-IV : CHEMICAL KINETICS (5 Periods)

Rate of a reaction (Average and instantaneous), factors affecting rate of reaction: concentration, temperature, catalyst; order and molecularity of a reaction, rate law and specific rate constant, integrated rate equations and half-life (only for zero and first order reactions).

UNIT-V : SURFACE CHEMISTRY (5 Periods)

Adsorption- physisorption and chemisorption, factors affecting adsorption of gases on solids, colloidal state: distinction between true solutions, colloids and suspension; lyophilic, lyophobic, multi-molecular and macromolecular colloids; properties of colloids; Tyndall effect, Brownian movement, electrophoresis, coagulation.

UNIT-VII : *p*-BLOCK ELEMENTS (7 Periods)

Group-15 Elements : General introduction, electronic configuration, occurrence, oxidation states, trends in physical and chemical properties; Nitrogen preparation properties and uses; compounds of Nitrogen : preparation and properties of Ammonia and Nitric Acid.

Group 16 Elements : General introduction, electronic configuration, oxidation states, occurrence, trends in physical and chemical properties, dioxygen: preparation, properties and uses, classification of Oxides, Ozone, Sulphur- allotropic forms; compounds of Sulphur: preparation properties and

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uses of Sulphur-dioxide, Sulphuric Acid:properties and uses; Oxoacids of Sulphur (Structures only).

Group 17 Elements : General introduction, electronic configuration, oxidation states, occurrence, trends in physical and chemical properties; compounds of halogens, Preparation, properties and uses of Chlorine and Hydrochloric acid, interhalogen compounds, Oxoacids of halogens (structures only).

Group 18 Elements : General introduction, electronic configuration, occurrence, trends in physical and chemical properties, uses.

UNIT-VIII : *d*- AND *f*-BLOCK ELEMENTS (7 Periods)

General introduction, electronic configuration, occurrence and characteristics of transition metals, general trends in properties of the first row transition metals – metallic character, ionization enthalpy, oxidation states, ionic radii, colour, catalytic property, magnetic properties, interstitial compounds, alloy formation.

Lanthanoids - Electronic configuration, oxidation states and lanthanoid contraction and its consequences.

UNIT-IX : COORDINATION COMPOUNDS (8 Periods)

Coordination compounds- Introduction, ligands, coordination number, colour, magnetic properties and shapes, IUPAC nomenclature of mononuclear coordination compounds. Bonding, Werner's theory, VBT, and CFT.

UNIT-X : HALOALKANES AND HALOARENES (9 Periods)

Haloalkanes : Nomenclature, nature of C–*X* bond, physical and chemical properties, optical rotation mechanism of substitution reactions.

Haloarenes : Nature of C–*X* bond, substitution reactions (Directive influence of halogen in monosubstituted compounds only).

UNIT-XI : ALCOHOLS, PHENOLS AND ETHERS (9 Periods)

Alcohols : Nomenclature, methods of preparation, physical and chemical properties (of primary alcohols only), identification of primary, secondary and tertiary alcohols, mechanism of dehydration.

Phenols : Nomenclature, methods of preparation, physical and chemical properties, acidic nature of phenol, electrophillic substitution reactions, uses of phenols.

Ethers : Nomenclature, methods of preparation, physical and chemical properties, uses.

UNIT-XII : ALDEHYDES, KETONES AND CARBOXYLIC ACIDS (10 Periods)

Aldehydes and Ketones : Nomenclature, nature of carbonyl group, methods of preparation, physical and chemical properties, mechanism of nucleophilic addition, reactivity of alpha hydrogen in aldehydes, uses.

Carboxylic Acids : Nomenclature, acidic nature, methods of preparation, physical and chemical properties, uses.

UNIT-XIII : AMINES (7 Periods)

Amines : Nomenclature, classification, structure, methods of preparation, physical and chemical properties, uses, identification of primary, secondary and tertiary amines.

UNIT-XIV : BIOMOLECULES (8 Periods)

Carbohydrates : Classification (aldoses and ketoses), monosaccahrides (glucose and fructose), *D-L* configuration

Proteins : Elementary idea of - amino acids, peptide bond, polypeptides, proteins, structure of proteins - primary, secondary, tertiary structure and quaternary structures (qualitative idea only), denaturation of proteins.

Nucleic Acids : DNA and RNA.

QUESTION PAPER DESIGN

PART A : Objective type Paper

Type	Marks for each question	No. of Questions	Total Marks	Percentage
Objective	1	19	19	54.29
	2	5	10	28.57
Case Based	3	2	6	17.14
Total		**26**	**35**	**100**

PART B : Descriptive paper

Type	Marks for each question	No. of Questions	Total Marks	Percentage
Short Answer I	2	4	8	22.86
Short Answer II	3	4	12	34.28
Long Answer	5	3	15	42.86
Total		**11**	**35**	**100**

For Practicals & Projects refer to CBSE website

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PRACTICE PAPER 1
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises two parts A and B.

(ii) Part A : Q. no. 1 to 26 are objective type questions.
Q. no. 1 to 19 are objective type questions carrying one mark each.
Q. no. 20 to 24 are objective type questions carrying two marks each.
Q. no. 25 to 26 are case based objective type questions carrying three marks each.

(iii) Part B : Q. no. 27 to 37 are subjective/descriptive type questions.
Q. no. 27 to 30 are short answer type-I questions carrying two marks each.
Q. no. 31 to 34 are short answer type-II questions carrying three marks each.
Q. no. 35 to 37 are long answer type questions carrying five marks each.

(iv) There is no overall choice in the question paper. However, internal choices have been provided in both part A and Part B.

(v) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

PART - A : Objective Type

1. In which of the following, the number of carbon atoms does not remain same when carboxylic acid is obtained by oxidation?
(a) CH_3COCH_3 (b) CCl_3CH_2CHO
(c) $CH_3CH_2CH_2OH$ (d) CH_3CH_2CHO

2. One mole of the complex compound $Co(NH_3)_5Cl_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of $AgNO_3$ solution to yield two moles of $AgCl_{(s)}$. The structure of the complex is
(a) $[Co(NH_3)_5Cl]Cl_2$
(b) $[Co(NH_3)_3Cl_2]\cdot 2NH_3$
(c) $[Co(NH_3)_4Cl_2]Cl\cdot NH_3$
(d) $[Co(NH_3)_4Cl]Cl_2\cdot NH_3$

OR

The oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+$ is
(a) +3 (b) +2 (c) +1 (d) 0

3. Replacement of Cl of chlorobenzene to give phenol requires drastic conditions. But chlorine of 2,4-dinitrochlorobenzene is readily replaced because
(a) NO_2 donates e^- at *meta* position
(b) NO_2 withdraws e^- from *ortho*/*para* positions
(c) NO_2 makes ring electron rich at *ortho* and *para* positions
(d) NO_2 withdraws e^- from meta position.

4. Which of the following is more basic than aniline?
(a) Benzylamine (b) Diphenylamine
(c) Triphenylamine (d) *p*-Nitroaniline

5. Which one of the following statements regarding helium is incorrect?
(a) It is used to fill gas in balloons instead of hydrogen because it is lighter and non-inflammable.
(b) It is used as a cryogenic agent for carrying out experiments at low temperatures.
(c) It is used to produce and sustain powerful superconducting magnets.
(d) It is used in gas-cooled nuclear reactors.

6. The vapour pressure of a solvent decreased by 10 mm of mercury when a non-volatile solute was added to the solvent. The mole fraction of the solute in the solution is 0.2. What should be the mole fraction of the solvent if the decrease in the vapour pressure is to be 20 mm of mercury?
(a) 0.4 (b) 0.6 (c) 0.8 (d) 0.2

OR

A 5% solution of cane sugar (mol. wt. = 342) is isotonic with 1% solution of a substance *X*.

www.examplet.in

The molecular weight of X is
(a) 68.4 (b) 171.2 (c) 34.2 (d) 136.8

7. Which one of the following complexes is an outer orbital complex?
(a) $[Fe(CN)_6]^{4-}$ (b) $[Mn(CN)_6]^{4-}$
(c) $[Co(NH_3)_6]^{3+}$ (d) $[Ni(NH_3)_6]^{2+}$
[Atomic nos.: Mn = 25, Fe = 26, Co = 27, Ni = 28]

8. Increasing order of acid strength among *p*-methoxyphenol, *p*-methylphenol and *p*-nitrophenol is
(a) *p*-nitrophenol, *p*-methoxyphenol, *p*-methylphenol
(b) *p*-methylphenol, *p*-methoxyphenol, *p*-nitrophenol
(c) *p*-nitrophenol, *p*-methylphenol, *p*-methoxyphenol
(d) *p*-methoxyphenol, *p*-methylphenol, *p*-nitrophenol.

9. The ability of anion, to bring about coagulation of a given colloid, depends upon
(a) magnitude of the charge
(b) both magnitude and charge
(c) its charge only
(d) sign of the charge alone.

10. In cube of any crystal A-atom placed at every corners and B-atom placed at every centre of face. The formula of compound is
(a) AB (b) AB_3
(c) A_2B_2 (d) A_2B_3

OR

If a stands for the edge length of the cubic systems: simple cubic, body-centred cubic and face-centred cubic, then the ratio of radii of the spheres in these systems will be respectively
(a) $\frac{1}{2}a : \frac{\sqrt{3}}{2}a : \frac{\sqrt{2}}{2}a$
(b) $1a : \sqrt{3}a : \sqrt{2}a$
(c) $\frac{1}{2}a : \frac{\sqrt{3}}{4}a : \frac{1}{2\sqrt{2}}a$
(d) $\frac{1}{2}a : \sqrt{3}a : \frac{1}{\sqrt{2}}a$

11. The reaction $A \longrightarrow B$ follows first order kinetics. The time taken for 0.8 mole of A to produce 0.6 mole of B is 1 hour. What is the time taken for conversion of 0.9 mole of A to produce 0.675 mole of B ?
(a) 1 hour (b) 0.5 hour
(c) 0.25 hour (d) 2 hours

12. Arrange Ce^{3+}, La^{3+}, Pm^{3+} and Yb^{3+} in increasing order of their ionic radii.
(a) $Yb^{3+} < Pm^{3+} < Ce^{3+} < La^{3+}$
(b) $Ce^{3+} < Yb^{3+} < Pm^{3+} < La^{3+}$
(c) $Yb^{3+} < Pm^{3+} < La^{3+} < Ce^{3+}$
(d) $Pm^{3+} < La^{3+} < Ce^{3+} < Yb^{3+}$

OR

Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them?
(a) $(n-1)d^8ns^2$ (b) $(n-1)d^5ns^1$
(c) $(n-1)d^3ns^2$ (d) $(n-1)d^5ns^2$.

13. A solution contains Fe^{2+}, Fe^{3+} and I^- ions. This solution was treated with iodine at 35 °C. $E°$ for Fe^{3+}/Fe^{2+} is + 0.77 V and $E°$ for $I_2/2I^-$ = 0.536 V. The favourable redox reaction is
(a) I_2 will be reduced to I^-
(b) there will be no redox reaction
(c) I^- will be oxidised to I_2
(d) Fe^{2+} will be oxidised to Fe^{3+}.

14. The atomic numbers of vanadium (V), chromium (Cr), manganese (Mn) and iron (Fe) are respectively 23, 24, 25 and 26. Which one of these may be expected to have the highest second ionisation enthalpy?
(a) V (b) Cr (c) Mn (d) Fe

Questions 15 to 19 :
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

15. Assertion : Equivalent conductance of all electrolytes decreases with increasing concentration.
Reason : Lesser number of ions are available per g equivalent at higher concentration.

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OR

Assertion : A negative value of standard reduction potential means that reduction takes place on this electrode with reference to standard hydrogen electrode.
Reason : The standard electrode potential of a half cell has a fixed value.

16. **Assertion :** The high melting points of transition elements are indicative of high heats of sublimation.
Reason : The ionisation potential of transition metal lies between that for the *s*- and *p*-block elements.

17. **Assertion :** Addition of Grignard reagent to dry ice followed by hydrolysis gives ketones.
Reason : Ketones can be converted into acids by haloform reaction.

OR

Assertion : In formaldehyde, all the four atoms are in the same plane.
Reason : The carbon atom in formaldehyde is sp^2-hybridised.

18. **Assertion :** The value of crystal field splitting depends not only on the ligand but also on the charge on the central metal ion.
Reason : The higher the positive charge on the metal ion, the more closely the ligands are pulled into the metal, the greater the interaction between the ligands and the *d*-electrons of the metal and greater the difference between the t_{2g} and the e_g levels.

19. **Assertion :** In any ionic solid (*MX*) with Schottky defects, the number of positive and negative ions is the same.
Reason : Equal number of cation and anion vacancies are present.

20. Consider the reaction, $2A + B \rightarrow$ products. When concentration of *B* alone was doubled, the half-life did not change. When the concentration of *A* alone was doubled, the rate increased by two times. The unit of rate constant for this reaction is
(a) s^{-1} (b) L mol^{-1} s^{-1}
(c) no unit (d) mol L^{-1} s^{-1}.

OR

For the reaction, $N_2 + 3H_2 \longrightarrow 2NH_3$, if $\frac{d[NH_3]}{dt} = 2\times10^{-4}$ mol $L^{-1}s^{-1}$, the value of $\frac{-d[H_2]}{dt}$ would be
(a) 4×10^{-4} mol L^{-1} s^{-1}
(b) 6×10^{-4} mol L^{-1} s^{-1}
(c) 1×10^{-4} mol L^{-1} s^{-1}
(d) 3×10^{-4} mol L^{-1} s^{-1}

21. Number of isomeric alcohols of molecular formula $C_6H_{14}O$ which give positive iodoform test is
(a) three (b) four
(c) five (d) two.

22. In which of the following arrangements the given sequence is not strictly according to the property indicated against it?
(a) HF < HCl < HBr < HI : increasing acidic strength
(b) $H_2O < H_2S < H_2Se < H_2Te$: increasing pK_a values
(c) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing acidic character
(d) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power

23. The vapour pressure of CCl_4 at 25°C is 143 mm Hg. If 0.5 g of a non-volatile solute (mol. weight = 65) is dissolved in 100 g CCl_4, the vapour pressure of the solution will be
(a) 199.34 mm Hg (b) 143.99 mm Hg
(c) 141.43 mm Hg (d) 94.39 mm Hg.

24. Which one of the following is an inner orbital complex as well as diamagnetic in behaviour?
(a) $[Zn(NH_3)_6]^{2+}$ (b) $[Cr(NH_3)_6]^{3+}$
(c) $[Co(NH_3)_6]^{3+}$ (d) $[Ni(NH_3)_6]^{2+}$
(Atomic number : Zn = 30, Cr = 24, Co = 27, Ni = 28)

OR

$[Cr(H_2O)_6]Cl_3$ (At. no. of Cr = 24) has a magnetic moment of 3.83 B.M. The correct distribution of 3*d* electrons in the chromium of the complex is

(a) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{z^2}$

(b) $3d^1_{(x^2-y^2)}, 3d^1_{z^2}, 3d^1_{xz}$

(c) $3d^1_{xy}, 3d^1_{(x^2-y^2)}, 3d^1_{yz}$

(d) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{xz}$

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Read the following passage and answer the questions 25 and 26.

A 10 g mixture of isobutane and isobutene requires 20 g of bromine (in CCl_4) for complete addition. 10 g of the mixture is catalytically hydrogenated and the entire alkane is monobrominated in the presence of light at 127°C.

25. The final product is
(a) isobutane
(b) tertiary butyl bromide
(c) 1, 2-dibromo-2-methylbutane
(d) None of these.

26. The mass of the final product formed is
(a) 15.85 g (b) 20.12 g
(c) 24.21 g (d) 30 g.

PART - B : Subjective / Descriptive Type

27. Explain the following observations :
(i) Lyophilic colloid is more stable than lyophobic colloid.
(ii) Coagulation takes place when sodium chloride solution is added to a colloidal solution of ferric hydroxide.

28. (i) Arrange the following compounds in the decreasing order of their boiling points :
(a) Pentan-1-ol
(b) 2-Methylbutan-2-ol
(c) 3-Methylbutan-2-ol
(ii) Which type of alcohol does not react with Lucas reagent at room temperature?

OR

Explain the following with an example for each :
(i) Kolbe's reaction,
(ii) Reimer — Tiemann reaction.

29. 0.90 g of a non-electrolyte was dissolved in 87.90 g of benzene. This raised the boiling point of benzene by 0.25°C. If the molecular mass of the non-electrolyte is 103.0 g mol^{-1}, calculate the molar elevation constant for benzene.

30. Write down the structures and names of the products formed when *D*-glucose is treated with
(a) Bromine water
(b) hydrogen iodide (prolonged heating).

31. From the following compounds :
2-Bromopentane, 2-Bromo-2-methylbutane, 1-Bromopentane
(i) Which compound is most reactive towards S_N2 reaction?
(ii) Which compound is optically active?
(iii) Which compound is most reactive towards β-elimination reaction?

32. (a) Which point defect of its crystals decreases the density of a solid?
(b) Calculate the number of unit cells in 8.1 g of aluminium if it crystallizes in a *fcc* structure.
(Atomic mass of Al = 27 g mol^{-1})

OR

Silver crystallises in *fcc* lattice. If the edge length of the unit cell is 4.077×10^{-8} cm and density of the crystal is 10.5 g cm^{-3}, calculate the atomic mass of silver.

33. How will you convert :
(i) Ethanoic acid into methanamine
(ii) Ethanamine into methanamine
(iii) Nitromethane into dimethylamine

34. Explain the following observations :
(a) The electron gain enthalpy of sulphur atom has a greater negative value than that of oxygen atom.
(b) Nitrogen does not form pentahalides.
(c) In aqueous solutions HI is a stronger acid than HCl.

35. (a) (i) What is order of reaction?
(ii) The thermal decomposition of HCOOH is a first order reaction with a rate constant of 2.4×10^{-3} s^{-1} at a certain temperature. Calculate how long will it take for three fourth of initial quantity of HCOOH to decompose.
(Given : log 4 = 0.6021)
(b) For the reaction,
$2N_2O_{5(g)} \longrightarrow 4NO_{2(g)} + O_{2(g)}$, the rate of formation of $NO_{2(g)}$ is 2.8×10^{-3} M s^{-1}. Calculate the rate of disappearance of $N_2O_{5(g)}$.

OR

For a homogeneous gas phase reaction $A_{(g)} \rightarrow B_{(g)} + C_{(g)} + D_{(g)}$, the pressure of the

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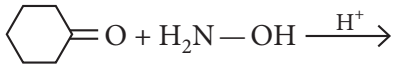
reaction mixture increases by 40% in 20 minute. Calculate rate constant of a reaction.

36. (a) Write the product of the following reactions:

(i)  $+ H_2N-OH \xrightarrow{H^+}$

(ii) $2C_6H_5CHO$ + conc. NaOH $\rightarrow$

(iii) $CH_3COOH \xrightarrow{Cl_2/P}$

(b) Arrange the following compounds in an increasing order of their property as indicated:

(i) Acetaldehyde, acetone, methyl-*tert*-butyl ketone (reactivity towards HCN)

(ii) $CH_3CH_2CH(Br)COOH$, $CH_3CH(Br)CH_2COOH$, $(CH_3)_2CHCOOH$ (acid strength)

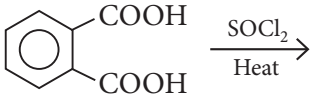
OR

(i) Give a chemical test to distinguish between the following :

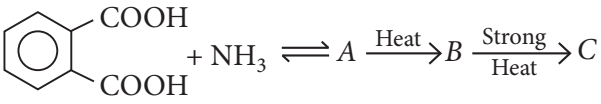
(a) Benzoic acid and ethyl benzoate

(b) Benzophenone and acetophenone.

(ii) Complete the following reaction :



(iii) Identify *A*, *B* and *C* in the following sequence :



37. (a) State reasons for the following :

(i) Transition metals and their many compounds act as good catalyst.

(ii) Unlike Cr^{3+}, Mn^{2+}, Fe^{3+} and the subsequent other M^{2+} ions of the 3*d* series of elements, the 4*d* and the 5*d* series metals generally do not form stable cationic species.

(b) Discuss the general trends in the following properties of the 3*d*-transition elements :

(i) Atomic size

(ii) Oxidation states

(iii) Formation of coloured ions.

OR

On the basis of Lanthanoid contraction, explain the following :

(i) Nature of bonding in La_2O_3 and Lu_2O_3.

(ii) Trends in the stability of oxo salts of lanthanoids from La to Lu.

(iii) Stability of the complexes of lanthanoids.

(iv) Radii of 4*d* and 5*d* block elements.

(v) Trends in acidic character of lanthanoid oxides.

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Detailed Solutions

1. (a) : Ketones on oxidation give carboxylic acids with lesser number of carbon atoms *i.e*,

$CH_3COCH_3 \xrightarrow{[O]} CH_3COOH + CO_2 + H_2O$

2. (a) : $[Co(NH_3)_5Cl]Cl_2 \rightleftharpoons [Co(NH_3)_5Cl]^{2+} + 2Cl^-$

$\Rightarrow$ 3 ions.

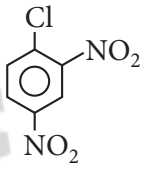
$[Co(NH_3)_5Cl]Cl_2 + 2AgNO_3 \longrightarrow [Co(NH_3)_5Cl](NO_3)_2 + 2AgCl$

OR

(a) : Let the oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+ = x$

$x + 4(0) + 2(-1) = +1$

$\Rightarrow \quad x - 2 = +1 \quad \Rightarrow \quad x = +1 + 2 = +3$

3. (b) : 

Withdrawal of electrons by $-NO_2$ groups from *ortho*/*para* positions cause easier removal of —Cl atom due to the development of positive charge on *o*- and *p*- positions.

4. (a) : Any group (— NO_2, — CN, — SO_3H, – COOH, —Cl, — C_6H_5, etc.) which when present on benzene ring has electron withdrawing effect, decreases basicity of aniline *e.g.*, aniline is more basic than nitroaniline. Lone pair electrons

are more delocalised in diphenylamine and triphenylamine, thus these are less basic than aniline. In benzylamine the electron pair present on nitrogen is not delocalised with the benzene ring hence it is more basic than aniline.

5. (a) : Helium is twice as heavy as hydrogen, its lifting power is 92 percent of that of hydrogen.

6. (b) : x_1 (mole fraction of solute) = 0.2
From Raoult's law,

$$\frac{P^\circ - P_s}{P^\circ} = x_1 \Rightarrow \frac{10}{P^\circ} = 0.2 \Rightarrow P^\circ = 50 \text{ mm}$$

Again, when $P^\circ - P_s = 20$ mm, then

$$\frac{P^\circ - P_s}{P^\circ} = \text{mole fraction of solute} = \frac{20}{50} = 0.4$$

$\Rightarrow$ mole fraction of solvent = 1 – 0.4 = 0.6

OR

(a) :

Cane Sugar	***X***
$W_1 = 5$ g	$W_2 = 1$ g
$V_1 = 100$ mL	$V_2 = 100$ mL
$= 0.1$ L	$= 0.1$ L
$M_1 = 342$	$M_2 = ?$

For isotonic solutions, $C_1 = C_2$

$$\frac{W_1}{M_1 V_1} = \frac{W_2}{M_2 V_2} \Rightarrow \frac{5}{342 \times 0.1} = \frac{1}{M_2 \times 0.1}$$

$$\Rightarrow M_2 = \frac{342}{5} = 68.4$$

7. (d) :

Complex ion	**Hybridization of central ion**
$[Fe(CN)_6]^{4-}$	d^2sp^3 (inner)
$[Mn(CN)_6]^{4-}$	d^2sp^3 (inner)
$[Co(NH_3)_6]^{3+}$	d^2sp^3 (inner)
$[Ni(NH_3)_6]^{2+}$	sp^3d^2 (outer)

8. (d) : $-OCH_3$, $-CH_3$ are electron donating groups and decrease the acidic character of phenols. $-NO_2$, is an electron withdrawing group that tend to increase the acidic character. Thus, the order is

p-methoxyphenol < *p*-methylphenol < *p*-nitrophenol.

9. (b) : Both magnitude and nature of charge effect coagulation of a given colloid. Greater the magnitude of the charge, quicker will be the coagulation.

10. (b) : '*A*' atoms are at '8' corners of the cube.

Thus, no. of '*A*' atoms per unit cell = $8 \times \frac{1}{8} = 1$

'*B*' atoms are at the face centre of six faces. Thus,

no. of '*B*' atoms per unit cell = $6 \times \frac{1}{2} = 3$

The formula is AB_3.

OR

(c) : For simple cubic : $r = a/2$

For body centred : $r = a\sqrt{3}/4$

For face-centred : $r = \frac{a}{2\sqrt{2}}$

where a = edge length, r = radius.

$\therefore$ Ratio of radii of the three will be $\frac{a}{2} : \frac{a\sqrt{3}}{4} : \frac{a}{2\sqrt{2}}$

11. (a) :

In case I		**In case II**	
A	⟶ ***B***	***A***	⟶ ***B***
0.8	0	0.9	0
0.2	0.6	0.225	0.675
1	3	1	3

The time taken for the completion of same fraction of change is independent of initial concentration.

12. (a) : According to their positions in the periods, these values are in the order:

	Yb^{3+}	< Pm^{3+}	< Ce^{3+}	< La^{3+}
At. Nos.	70	61	58	57
Ionic radii (pm)	86	98	103	106

Ionic size decreases from La^{3+} to Lu^{3+} due to lanthanide contraction.

OR

(d) :

	$(n-1)d$					ns
	↑	↑	↑	↑	↑	↑↓

$(n-1)d^5\ ns^2$ can achieve the maximum oxidation state of +7.

13. (c) : Since the reduction potential of Fe^{3+}/Fe^{2+} is greater than that of I_2/I^-, Fe^{3+} will be reduced and I^- will be oxidised.

$$2Fe^{3+} + 2I^- \longrightarrow 2Fe^{2+} + I_2$$

14. (b) : The second ionisation potential value of Cr is highest among the given elements. This is because of the electronic configuration of Cr^+ which is $3d^5$ (half-filled), *i.e.*, for the second ionisation potential, the electron is to be removed from very stable configuration.

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15. (c) : At higher concentration no. of ions per unit volume actually increases. Hence the equivalent conductance decreases on dilution.

OR

(d) : A negative value of standard reduction potential means that oxidation takes place on this electrode with reference to SHE.

16. (b)

17. (d) : Addition of Grignard reagent to dry ice followed by hydrolysis gives carboxylic acid not ketone.

$$R\text{Mg}X + \text{O}=\text{C}=\text{O} \xrightarrow[\text{ether}]{\text{Dry}} R-\overset{\overset{\text{O}}{\|}}{\text{C}}-\text{OMg}X \xrightarrow{\text{H}^+/\text{H}_2\text{O}} R-\overset{\overset{\text{O}}{\|}}{\text{C}}-\text{OH}$$

OR

(a) : Since C — O bonds utilise sp^2-hybrid orbitals, they lie in the same plane and are 120° apart. Hence carbonyl group is planar.

18. (a) **19. (a)**

20. (b) : Rate = $k\,[A]^x\,[B]^y$

When $[B]$ is doubled, keeping $[A]$ constant half-life of the reaction does not change.

Now, for a first order reaction $t_{1/2} = \dfrac{0.693}{k}$

i.e., $t_{1/2}$ is independent of the concentration of the reactant. Hence the reaction is first order with respect to *B*. Now when $[A]$ is doubled, keeping $[B]$ constant, the rate also doubles. Hence the reaction is first order with respect to *A*.

$\therefore$ Rate = $[A]^1[B]^1$

$\therefore$ order = 2

Now for a n^{th} order reaction, unit of rate constant is $(\text{L})^{n-1}\ (\text{mol})^{1-n}\ s^{-1}$; when n = 2, unit of rate constant is L mol^{-1} s^{-1}.

OR

(d) : For reaction, $N_2 + 3H_2 \longrightarrow 2NH_3$

$$\text{Rate} = \frac{1}{2}\frac{d[\text{NH}_3]}{dt} = -\frac{1}{3}\frac{d[\text{H}_2]}{dt} = -\frac{d[\text{N}_2]}{dt}$$

$$\text{Given, } \frac{d[\text{NH}_3]}{dt} = 2\times10^{-4}\ \text{mol L}^{-1}\ \text{s}^{-1}$$

$$\therefore\quad -\frac{d[\text{H}_2]}{dt} = \frac{3}{2}\frac{d[\text{NH}_3]}{dt} = \frac{3}{2}\times2\times10^{-4}$$

$$\Rightarrow\quad -\frac{d[\text{H}_2]}{dt} = 3\times10^{-4}\ \text{mol L}^{-1}\ \text{s}^{-1}$$

21. (b) : The iodoform test is positive for alcohols with formula *R* — CHOH — CH_3. Among $C_6H_{14}O$ isomers, the ones with positive iodoform test are:

I. $CH_3 — CH_2 — CH_2 — CH_2 — CHOH — CH_3$
2-Hexanol

II. $CH_3 — CH_2 — CH(CH_3) — CHOH — CH_3$
3- Methyl-2-pentanol

III. $(CH_3)_2CH — CH_2 — CHOH — CH_3$
4- Methyl-2-pentanol

IV. $(CH_3)_3C — CHOH — CH_3$
3, 3- Dimethyl-2-butanol

22. (b) : Acidic strength of hydrides increase with increase in molecular mass.

Thus, order of acidic strength is

HF < HCl < HBr < HI

$H_2O < H_2S < H_2Se < H_2Te$

$NH_3 < PH_3 < AsH_3 < SbH_3$

and as acidic strength increases, pK_a decreases. Thus order of pK_a

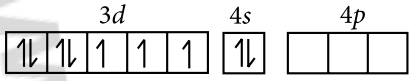
$H_2O > H_2S > H_2Se > H_2Te$

23. (c) : Vapour pressure of pure solvent (P_A°) = 143 mm Hg, weight of solute (w) = 0.5 g, weight of solvent (W) = 100 g, molecular weight of solute (m) = 65 and molecular weight of solvent (M) = 154.

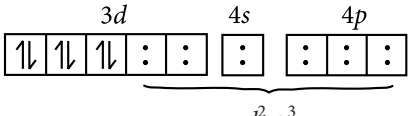
$$\frac{p_A^\circ - p}{p_A^\circ} = \frac{wM}{mW} \quad \text{or} \quad \frac{143-P}{143} = \frac{0.5\times154}{65\times100}$$

or P = 141.31 mm Hg

24. (c) : $[Co(NH_3)_6]^{3+}$: Co(27) $[Ar]^{18}\ 3d^7 4s^2$



Co^{3+} : (24) $[Ar]^{18}\ 3d^6$



electron pair from six ligands(NH_3)

d^2sp^3 → inner octahedral complex and diamagnetic.

OR

(d) : Magnetic moment $= \sqrt{n(n+2)}$

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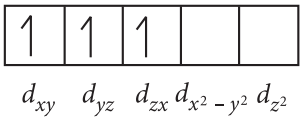
$3.83 = \sqrt{n(n+2)} \Rightarrow (3.83)^2 = n(n+2)$

$\Rightarrow$ $14.6689 = n^2 + 2n$

On solving the equation, $n = 3$

$Cr^{3+} \rightarrow [Ar]3d^3$

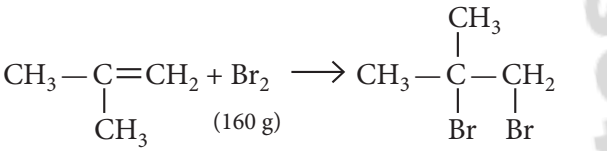


d_{xy} d_{yz} d_{zx} $d_{x^2-y^2}$ d_{z^2}

25. (b)

26. (c) : Let the weight of isobutane is x g. Weight of isobutene is y g. Then $x + y = 10$ g.

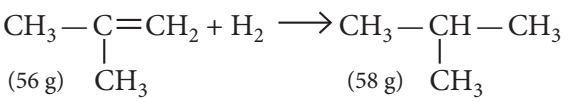
Now, it is isobutene which will add bromine molecule not isobutane.



Isobutylene (56 g) (160 g) 1, 2-dibromo-2-methylbutane

Since 160 g of Br_2 reacts with 56 g of isobutene, hence 20 g of Br_2 reacts with = 7 g of isobutene

i.e., $y = 7$ g, hence $x = 10 - 7 = 3$ g.

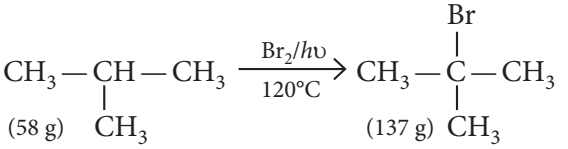


(56 g) (58 g)

Now, 56 g of isobutene gives 58 g of isobutane.

Hence 7 g of isobutene gives $\frac{7 \times 58}{56} = 7.25$ g.

Now total amount of isobutane available for 10 g mixture = 7.25 + 3 = 10.25.



(58 g) (137 g)

Tertiary butylbromide

Now 58 g of isobutane gives 137 g of tertiary butyl bromide.

$\therefore$ 10.25 g of isobutane gives $\frac{10.25 \times 137}{58} = 24.21$ g of tertiary butylbromide.

27. (i) A lyophilic colloid is stable due to the charge as well as solvation of the sol particles. Such a solution can only be coagulated by adding an electrolyte and by adding a suitable solvent which can dehydrate the dispersed phase.

On the other hand, a lyophobic sol is stable due to charge only and hence can be easily coagulated by adding small amount of an electrolyte.

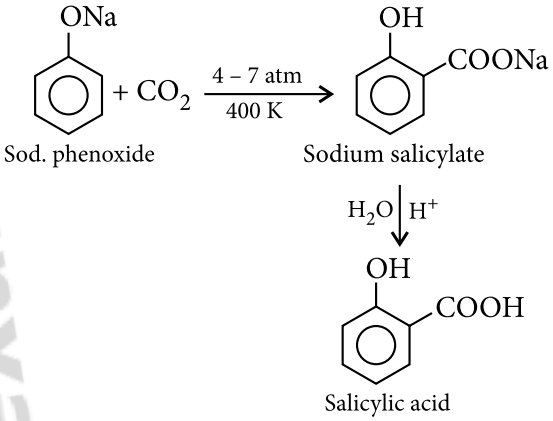
(ii) Ferric hydroxide solution adsorbs Cl^- ions of NaCl and get neutralised. Thus the colloidal particles get precipitated *i.e.*, ferric hydroxide is precipitated when NaCl solution is added.

28. (i) (a) > (c) > (b). This is because with branching the shape becomes spherical and therefore, van der Waals' forces decrease. As a result, boiling point decreases.

(ii) 1°-Alcohol does not react with Lucas reagent at room temperature.

OR

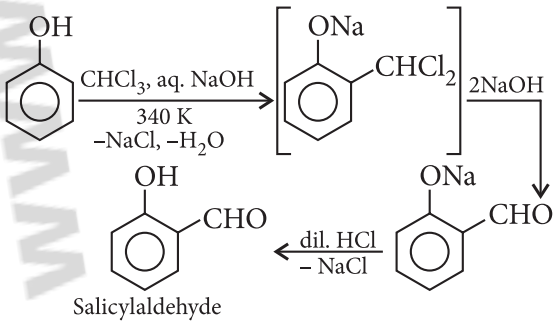
(i) **Kolbe's reaction :** When sodium phenoxide is heated with carbon dioxide under pressure, it gives salicylic acid.



Sod. phenoxide Sodium salicylate

Salicylic acid

(ii) **Reimer–Tiemann reaction :**



Salicylaldehyde

29. $w_1 = 0.90$ g, $w_2 = 87.90$ g, $\Delta T_b = 0.25°C$

$M_1 = 103$ g mol^{-1}, $\Delta T_b = K_b m$

$$\Delta T_b = K_b \times \frac{w_1}{M_1} \times \frac{1000}{w_2}, \; K_b = \frac{\Delta T_b \times M_1 \times w_2}{w_1 \times 1000}$$

$$\Rightarrow \; K_b = \frac{0.25 \times 103 \times 87.90}{0.90 \times 1000} \Rightarrow K_b = \frac{2263.42}{900}$$

$\Rightarrow$ $K_b = 2.5149 = 2.515$ K kg mol^{-1}

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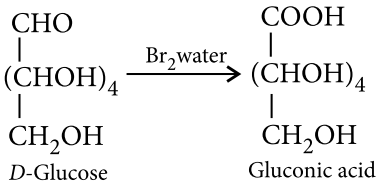
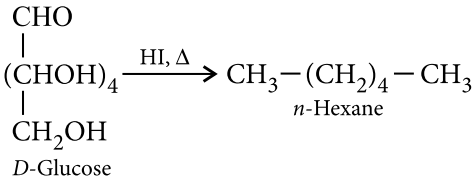
30. (a) *D* - Glucose gets oxidised to six carbon carboxylic acid (gluconic acid) on reaction with bromine water.



D-Glucose Gluconic acid

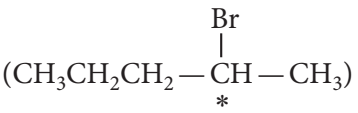
(b) On prolonged heating with HI, *D*-glucose forms *n*-hexane.



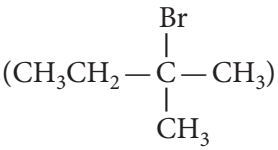
D-Glucose *n*-Hexane

31. (i) 1-Bromopentane
($CH_3CH_2CH_2CH_2CH_2Br$, 1°halide)

(ii) 2-Bromopentane



(iii) 2-Bromo-2-methylbutane



32. (a) Schottky defect

(b) As 27 g of Al will contain 6.023×10^{23} atoms

$\therefore$ 8.1 g Al will contain

$$= \frac{6.023\times10^{23}}{27}\times 8.1 \text{ atoms} = 1.8069 \times 10^{23} \text{ atoms}$$

In *fcc*, 4 atoms are present in one unit cell

$\therefore$ 1.8069×10^{23} atoms will be present in

$$\frac{1.8069\times10^{23}}{4} = 4.517 \times 10^{22} \text{ unit cells}$$

OR

Given, edge length, $a = 4.077 \times 10^{-8}$ cm

density, $d = 10.5$ g cm^{-3}

For *fcc* lattice, $Z = 4$, Atomic mass M = ?

Using formula, $d = \dfrac{Z\times M}{N_A \times a^3}$

or, $M = \dfrac{d\times N_A \times a^3}{Z}$

or $M = \dfrac{10.5\times6.022\times10^{23}\times(4.077\times10^{-8})^3}{4}$

or $M = 107.12$ g mol^{-1}

33. (i) $CH_3COOH \xrightarrow{PCl_5} CH_3-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow{NH_3}$

Ethanoic acid (Acetic acid)

$$CH_3-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{Br_2/\,KOH} CH_3-NH_2$$

Methanamine

(ii) $CH_3-CH_2-NH_2 \xrightarrow[273 - 278\ K]{NaNO_2/\,HCl} CH_3-CH_2-OH$

Ethanamine

$$\xrightarrow{K_2Cr_2O_7\,-/\,H^+} CH_3-COOH \xrightarrow{PCl_5} CH_3-COCl \xrightarrow{NH_3}$$

$$CH_3-CONH_2 \xrightarrow{Br_2/\,KOH} CH_3-NH_2$$

Methanamine

(iii) $CH_3-NO_2 \xrightarrow{Sn\,/\,HCl} CH_3-NH_2 \xrightarrow[\Delta]{CHCl_3\,/\,KOH}$

Nitromethane

$$CH_3-NC \xrightarrow{Na\,/\,C_2H_5OH} CH_3-NH-CH_3$$

Dimethylamine

34. (a) The electron gain enthalpy of oxygen is least negative in this group, due to its small size.

(b) Nitrogen does not form pentahalide, due to the absence of *d*-orbitals in its valence shell.

(c) The strength of an acid depends upon its bond strength. Since bond dissociation energy of HCl is greater than HI therefore, HI is stronger acid than HCl.

35. (a) (i) The sum of powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction. If rate $= kC_A^m \,.\, C_B^n$

Then order of reaction $= m + n$

(ii) Let initial concentration be $[R]_0$.

For a first order reaction, $t = \dfrac{2.303}{k}\log\dfrac{[R]_0}{[R]}$

At time t, $[R] = \dfrac{[R]_0}{4}$, $k = 2.4\times10^{-3}\,s^{-1}$

Thus, $2.4\times10^{-3} = \dfrac{2.303}{t}\log\dfrac{[R]_0}{[R]_0/4}$

$$t = \frac{2.303}{2.4\times10^{-3}}\log 4 = \frac{2.303}{2.4\times10^{-3}}\times 0.6021 = 5.77\times10^2\,s$$

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(b) Given, $\frac{d[NO_2]}{dt} = 2.8\times10^{-3}\ M\ s^{-1}$

According to rate law expression,

$$-\frac{1}{2}\frac{d[N_2O_5]}{dt} = \frac{1}{4}\frac{d[NO_2]}{dt} = \frac{d[O_2]}{dt}$$

$$\therefore \quad -\frac{1}{2}\frac{d[N_2O_5]}{dt} = \frac{1}{4}\times2.8\times10^{-3}$$

$$\frac{-d[N_2O_5]}{dt} = \frac{1}{2}\times2.8\times10^{-3} = 1.4\times10^{-3}\ M\ s^{-1}$$

OR

The given reaction is

	$A_{(g)}$	$\rightarrow$	$B_{(g)}$	+	$C_{(g)}$	+	$D_{(g)}$
Initial pressure	P_o		0		0		0
After 20 min.	$P_o - x$		x		x		x

Total pressure = $P_o - x + x + x + x = P_o + 2x$

$$P_o + 2x = P_o + P_o \times \frac{40}{100}\ ;\ \frac{2x}{1} = \frac{P_o}{2.5} \text{ or } P_o = 5x$$

$$\therefore \quad x = \frac{P_o}{5}$$

Now, $[R]_0 = P_o$

$$[R] = P_o - x = P_o - \frac{P_o}{5} = 0.8\ P_o$$

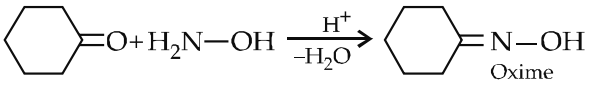
Using formula, $k = \frac{2.303}{t}\log_{10}\frac{[R]_0}{[R]}$

or, $k = \frac{2.303}{20\ min}\log_{10}\frac{P_o}{0.8P_o}$

or, $k = \frac{2.303}{20\ min}\log 1.25$ or, $k = \frac{2.303\times0.0969}{20\ min}$

$k = 1.115\times10^{-2}\ min^{-1}$

36. (a) (i)



(ii) $2C_6H_5CHO$ + conc. NaOH $\rightarrow C_6H_5COONa + C_6H_5CH_2OH$

(iii) $CH_3COOH \xrightarrow{Cl_2/P} CH_2(Cl)COOH$

(b) (i) Greater the number of alkyl groups attached to the carbonyl group, lower is the reactivity towards nucleophilic addition reactions.

$$(CH_3)_3C-\underset{\underset{CH_3}{|}}{C}=O < CH_3-\underset{\underset{CH_3}{|}}{C}=O < CH_3-\underset{\underset{H}{|}}{C}=O$$

Methyl-tert butyl ketone — Acetone — Acetaldehyde

(ii) We know that + *I*-effect decreases while –*I*-effect increases the acid strength of carboxylic acids. The overall acid strength increases in the order :

$(CH_3)_2CHCOOH < CH_3CH(Br)CH_2COOH < CH_3CH_2CH(Br)COOH$

OR

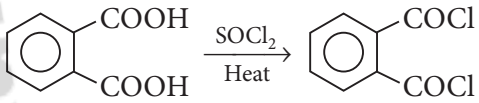
(i) (a) $C_6H_5COOH + NaHCO_3 \longrightarrow C_6H_5COONa + H_2O + CO_2\uparrow$ (effervescence)

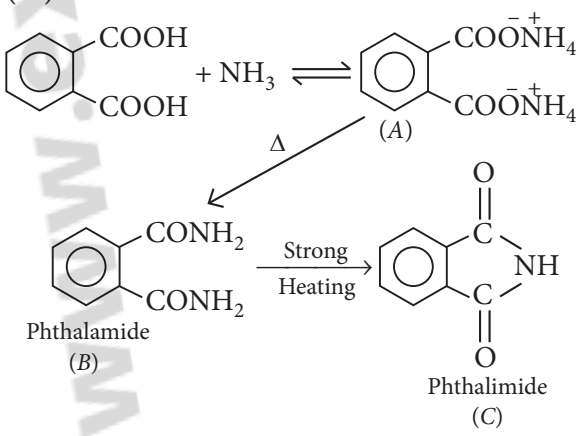
$C_6H_5COOC_2H_5 + NaHCO_3 \longrightarrow$ No reaction

(b) Acetophenone with I_2/NaOH gives yellow precipitate of iodoform but benzophenone does not.

$C_6H_5COCH_3 + 3I_2 + 4NaOH \longrightarrow C_6H_5COONa + CHI_3 + 3NaI + 3H_2O$

$C_6H_5COC_6H_5 + I_2 + NaOH \longrightarrow$ No reaction

(ii) 

(iii) 

37. (a) (i) The transition metals and their compounds, are known for their catalytic activity. This activity is ascribed to their ability to adopt multiple oxidation states, ability to adsorb the reactant on its surface and to form complexes. Vanadium (V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in catalytic hydrogenation) are some of the examples.

(ii) This is because due to lanthanoid contraction the expected increase in size does not occur hence

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they have very high value of ionisation enthalpy.

(b) (i) In 3*d* series atomic radius decreases from $_{21}Sc$ to $_{25}Mn$ and then becomes almost constant for next five elements *i.e.,* $_{26}Fe$ to $_{30}Zn$. This is due to the combined effect of increasing effective nuclear charge and increasing inter electronic repulsions when pairing of electron starts along the period.

(ii) 3*d*-series elements show variable oxidation states due to the participation of 4*s* and 3*d*-electrons in bonding because the energies of 4*s* and 3*d*-subshells are almost equal. Oxidation state increases with the increase in number of unpaired electrons in 3*d*-subshell, reaches a maximum value in the middle and then starts decreasing when pairing of electrons in 3*d*-subshell starts. For example, Mn shows a maximum oxidation state of +7 in 3*d*-series.

(iii) Except Zn^{2+}, 3*d*-series transition metal ions possess incomplete *d*-orbitals which undergo *d-d* transition and impart colour to them.

Some of the ions may impart colour due to charge transfer when metal ion has high +ve charge and anions are polarisable *e.g.,* $KMnO_4$, $K_2Cr_2O_7$.

OR

(i) As the size decreases covalent character increases. Therefore La_2O_3 is more ionic and Lu_2O_3 is more covalent.

(ii) As the size decreases from La to Lu, stability of oxosalts also decreases.

(iii) Stability of complexes increases as the size decreases.

(iv) Radii of 4*d* and 5*d* block elements will be almost same. It is associated with intervention of the 4*f* orbitals which must be filled before 5*d* series of elements begins. The filling of 4*f* before 5*d* results in lanthanoids contraction hence 4*d* and 5*d* series have almost same size.

(v) Acidic character of oxides increases from La to Lu.

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PRACTICE PAPER 2
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises two parts A and B.

(ii) Part A : Q. no. 1 to 26 are objective type questions.
Q. no. 1 to 19 are objective type questions carrying one mark each.
Q. no. 20 to 24 are objective type questions carrying two marks each.
Q. no. 25 to 26 are case based objective type questions carrying three marks each.

(iii) Part B : Q. no. 27 to 37 are subjective/descriptive type questions.
Q. no. 27 to 30 are short answer type-I questions carrying two marks each.
Q. no. 31 to 34 are short answer type-II questions carrying three marks each.
Q. no. 35 to 37 are long answer type questions carrying five marks each.

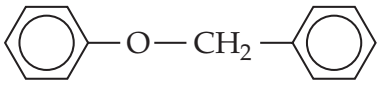
(iv) There is no overall choice in the question paper. However, internal choices have been provided in both part A and Part B.

(v) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

PART - A : Objective Type

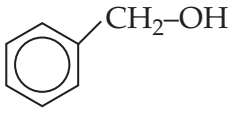
1. The given ether

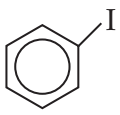


when reacts with cold HI gives

(1)

(2)

(3) 

(4) 

(a) mixture of 3 and 4
(b) mixture of 1 and 2
(c) mixture of 2 and 3
(d) mixture of 1 and 4.

2. The *d*-block elements form coloured ions because these elements
(a) cannot absorb the radiation in the visible region
(b) involve *d-d* transitions which fall in the visible region
(c) allow *d-s* transition
(d) absorb other colours except those required for *d-d* transition.

3. Blood cells retain their normal shape in solution which are
(a) hypotonic to blood
(b) isotonic to blood
(c) hypertonic to blood
(d) equinormal to blood.

OR

The vapour pressure of two liquids *P* and *Q* are 80 and 60 torr, respectively. The total vapour pressure of solution obtained by mixing 3 mole of *P* and 2 mol of *Q* would be
(a) 72 torr (b) 140 torr
(c) 68 torr (d) 20 torr

4. Which of the following is the correct increasing order of basicity of amines in gaseous phase?
(a) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$
(b) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$
(c) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2 > NH_3$
(d) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH > NH_3$

5. The lanthanide contraction is responsible for the fact that
(a) Zr and Hf have about the same radius
(b) Zr and Zn have the same oxidation state
(c) Zr and Y have about the same radius
(d) Zr and Nb have similar oxidation state.

Unsolved Examination Paper

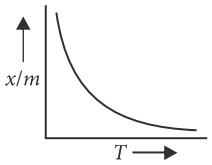
6. Propene, $CH_3CH = CH_2$ can be converted into 1-propanol by oxidation. Indicate which set of reagents amongst the following is ideal for the above conversion?
(a) $KMnO_4$ (alkaline)
(b) Osmium tetroxide (OsO_4/CH_2Cl_2)
(c) B_2H_6 and alk. H_2O_2
(d) O_3/Zn

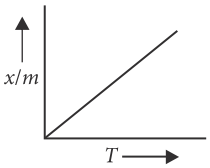
7. Primary and secondary valency of platinum in the complex $[Pt(en)_2Cl_2]$ are
(a) 4, 6 (b) 2, 6
(c) 4, 4 (d) 6, 4

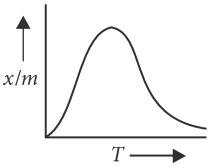
OR

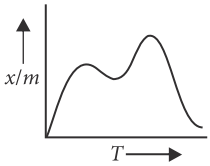
Which among the following will be named as dibromido*bis*(ethylenediamine) chromium(III) bromide?
(a) $[Cr(en)_2Br_2]Br$ (b) $[Cr(en)Br_4]^-$
(c) $[Cr(en)Br_2]Br$ (d) $[Cr(en)_3]Br_3$

8. Which of the following represents physical adsorption?

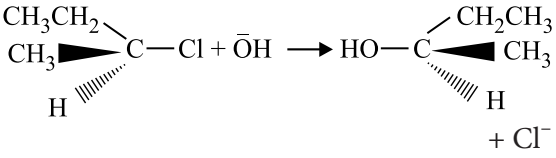
(a) 

(b) 

(c) 

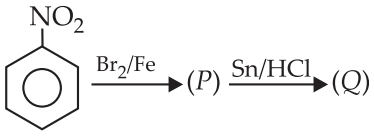
(d) 

9. In the reaction given below :



Which of the following statements is correct?
(a) The reaction proceeds *via* S_N2 mechanism hence inversion of configuration takes place.
(b) The reaction proceeds *via* S_N1 mechanism hence inversion of configuration takes place.
(c) The reaction proceeds *via* S_N2 mechanism hence their is no change in the configuration.
(d) The reaction proceeds *via* S_N1 mechanism hence there is no change in the configuration.

OR

Given : 

The products *P* and *Q* are

	P	***Q***
(a)	*p*-bromonitro-benzene	*p*-bromoaniline
(b)	*o*-bromonitro-benzene	*o*-bromoaniline
(c)	*o*,*p*-dibromonitro-benzene	*o*,*p*-dibromoaniline
(d)	*m*-bromonitro-benzene	*m*-bromoaniline

10. Name of the synthetic radioactive element of group 16 having atomic number 116 is
(a) Livermorium (b) Tennessine
(c) Flerovium (d) Moscovium.

11. Which one of the following alcohols undergoes acid–catalysed dehydration to alkenes most readily?
(a) $(CH_3)_2CHCH_2OH$
(b) $(CH_3)_3COH$
(c) $CH_3CHOHCH_3$
(d) $CH_3CH_2CH_2OH$

12. _____ carbohydrate cannot be hydrolysed into simpler units.
(a) Disaccharide (b) Monosaccharide
(c) Trisaccharide (d) Polysaccharide

13. Which statement is correct about halogens?
(a) They are all diatomic and form univalent ions.
(b) They are all capable of exhibiting several oxidation states.
(c) They are all diatomic and form divalent ions.
(d) They can mutually displace each other from the solution of their compounds with metals.

14. The correct order of reactivity of aldehydes and ketones towards hydrogen cyanide is
(a) $CH_3COCH_3 > CH_3CHO > HCHO$
(b) $CH_3COCH_3 > HCHO > CH_3CHO$
(c) $CH_3CHO > CH_3COCH_3 > HCHO$
(d) $HCHO > CH_3CHO > CH_3COCH_3$

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OR

Which of the following reactions is showing the acidic property of carboxylic acid?

(a) $2R-\overset{\overset{O}{\|}}{C}-OH + 2Na \longrightarrow 2R-\overset{\overset{O}{\|}}{C}-ONa + H_2\uparrow$

(b) $R-\overset{\overset{O}{\|}}{C}-OH + NaOH \longrightarrow R-\overset{\overset{O}{\|}}{C}-ONa + H_2O$

(c) $2R-\overset{\overset{O}{\|}}{C}-OH + 2Na_2CO_3 \longrightarrow 2R-\overset{\overset{O}{\|}}{C}-ONa + H_2O + CO_2\uparrow$

(d) All of the above.

Questions 15 to 19 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is correct, but Reason (R) is incorrect statement.

(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

15. Assertion : The densities, melting and boiling points of the transition elements are generally very high.
Reason : Zn, Hg and Cd have low melting and boiling points as the *d*-block is complete.

16. Assertion : *t*-Butyl methyl ether is not prepared by the reaction of *t*-butyl bromide with sodium methoxide.
Reason : Sodium methoxide is a strong nucleophile.

17. Assertion : Reverse osmosis is used in the desalination of sea water for getting drinking water.
Reason : When pressure more than osmotic pressure is applied, pure water is squeezed out of the sea water through the membrane.

OR

Assertion : In a pressure cooker the water is brought to boil. The cooker is then removed from the stove. Now on removing the lid of the pressure cooker, the water starts boiling again.
Reason : The impurities in water bring down its boiling point.

18. Assertion : $Co(NH_3)_6^{2+}$ is a high spin paramagnetic complex, but $Co(NH_3)_6^{3+}$ is a low spin diamagnetic complex.
Reason : NH_3 is a ligand in the middle of the spectrochemical series and some NH_3 complexes are high spin while others are low spin.

19. Assertion : In comparison to ethyl chloride, it is difficult to carry out nucleophilic substitution on vinyl chloride.
Reason : Vinyl group is electron donating.

OR

Assertion : The presence of nitro group facilitates nucleophilic substitution reactions in aryl halides.
Reason : The intermediate carbanion is stabilised due to the presence of nitro group.

20. Equivalent condudctance of 1M propanoic acid is 10 $ohm^{-1}\ cm^2\ eq^{-1}$ and that at infinite dilution 200 $ohm^{-1}\ cm^2\ eq^{-1}$. pH of the ethanoic acid solution is
(a) 7 (b) 3.3 (c) 1.3 (d) 6.8.

OR

What will be the equilibrium constant for the reaction?

$$2Fe^{+3} + 3I^- \longrightarrow 2Fe^{2+} + I_3^-$$

The standard reduction potentials in acidic conditions are 0.77 and 0.54 V respectively for Fe^{3+}/Fe^{2+} and I_3^-/I^- couple.
(a) 7.07×10^8 (b) 7.07×10^3
(c) 6.07×10^9 (d) 6.07×10^7.

21. Which one of the following order is wrong, with respect to the property indicated?
(a) Benzoic acid > phenol > cyclohexanol (acid strength)
(b) Aniline > cyclohexylamine > benzamide (basic strength)
(c) Formic acid > acetic acid > propanoic acid (acid strength)
(d) Fluoroacetic acid > chloroacetic acid > bromoacetic acid (acid strength)

22. Among the following series of transition metal ions, the one where all metal ions have $3d^2$ electronic configuration is
[At. nos. Ti = 22, V = 23, Cr = 24, Mn = 25]

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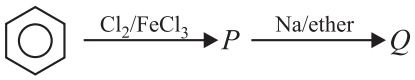
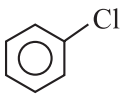
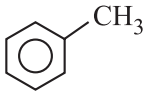
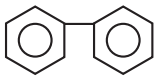
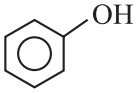
(a) Ti^{3+}, V^{2+}, Cr^{3+}, Mn^{4+}
(b) Ti^{+}, V^{4+}, Cr^{6+}, Mn^{7+}
(c) Ti^{4+}, V^{3+}, Cr^{2+}, Mn^{3+}
(d) Ti^{2+}, V^{3+}, Cr^{4+}, Mn^{5+}

23. The end product (*Q*) in the following sequence of reactions is

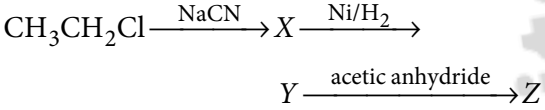


(a) 

(b) 

(c) 
(d) 

OR

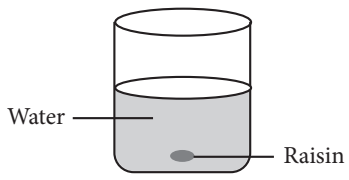


Z in the above reaction sequence is
(a) $CH_3CH_2CH_2NHCOCH_3$
(b) $CH_3CH_2CH_2NH_2$
(c) $CH_3CH_2CH_2CONHCH_3$
(d) $CH_3CH_2CH_2CONHCOCH_3$

24. Among $[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$ species, the hybridisation states at the Ni atom are, respectively
(a) sp^3, dsp^2, dsp^2 (b) sp^3, dsp^2, sp^3
(c) sp^3, sp^3, dsp^2 (d) dsp^2, sp^3, sp^3.
[Atomic number of Ni = 28]

Read the following passage and answer the questions 25 and 26 that follow :

A raisin is added to a beaker containing water as shown in the figure.



25. When kept in water, raisin __(i)__ in size as concentration inside raisin is __(ii)__. This phenomenon is called __(iii)__. Here, (i), (ii) and (iii) are respectively
(a) shrinks, low, reverse osmosis
(b) swells, low, osmosis
(c) shrinks, high, reverse osmosis
(d) swells, high, osmosis

26. A 4% solution of sucrose is isotonic with 3% solution of an unknown organic substance. The molecular mass of the unknown substance is
(a) 350.2 (b) 256.5
(c) 154.8 (d) 175.6

PART - B : Subjective / Descriptive Type

27. Determine the type of cubic lattices to which iron crystal belongs if its cell has an edge length of 286 pm and the density of iron crystal is 7.86 g cm^{-3}.
[Fe = 56 g mol^{-1}, $N_A = 6.023 \times 10^{23}$]

OR

Li crystalises in *bcc* structure. If the edge length of unit cell is 353 pm, calculate (i) radius of Li atom, (ii) packing efficiency of unit cell.

28. Name the purine and pyrimidine bases present in RNA.

29. Show that half life period of a first order reaction is independent of initial concentration of the reactant.

30. How will you obtain an aldehyde by using following process:
(a) Catalytic dehydrogenation,
(b) Catalytic hydrogenation?

31. (a) How will you distinguish between
(i) Propan-2-ol and propan-1-ol
(ii) Phenol and ethanol?
(b) How will you convert phenol to salicylic acid?

32. (i) Mention one important application of Henry's law.
(ii) Calculate the osmotic pressure of a solution obtained by mixing 100 cm^3 of 0.25 m solution of urea and 100 cm^3 of 0.1 m solution of cane-sugar at 293 K.
[R = 0.082 L atm mol^{-1} K^{-1}]

33. Give reasons :
(i) Thermal stability decreases from H_2O to H_2Te.
(ii) Fluoride ion has higher hydration enthalpy than chloride ion.
(iii) Oxygen does not form hexahalide.

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OR

(a) Why all halogens are coloured?

(b) HF is least volatile and HCl is most volatile amongst hydrogen halides. Explain this trend.

34. (a) Haloalkanes react with KCN to give alkyl cyanide while with AgCN form isocyanides as the chief product. Why?

(b) Haloalkanes act as an electrophilic centre. Explain.

35. (a) Write a suitable chemical equation to complete each of the following transformations :

(i) Butan-1-ol to propane

(ii) Toulene to benzene-1,4-dicarboxylic acid

(iii) Benzaldehyde to cinnamaldehyde

(b) Account for the following :

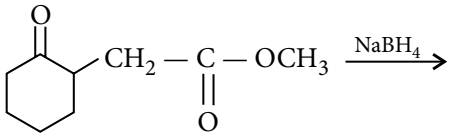
(i) Aromatic carboxylic acids do not undergo Friedel—Crafts reaction.

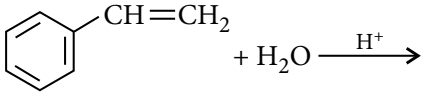
(ii) pK_a value of 4-nitrobenzoic acid is lower than that of benzoic acid.

OR

(a) Of the two hydroxy organic compounds *R*OH and *R*'OH the first one is basic and the other is acidic in behaviour. How is *R* different from *R*'?

(b) Write the structures of the main products in the following reactions :

(i) 

(ii) 

(c) How would you obtain the following:

(i) Benzoquinone from phenol

(ii) 2-Methylpropan-2-ol from methyl magnesium bromide

(iii) Propan-2-ol from propene ?

36. (a) For the reaction :

$2NO_{(g)} + Cl_{2(g)} \longrightarrow 2NOCl_{(g)}$

the following data were collected. All the measurements were taken at 263 K.

Exp. No.	Initial [NO] (M)	Initial [Cl_2] (M)	Initial rate of disapp. of Cl_2 (M/min)
1.	0.15	0.15	0.60
2.	0.15	0.30	1.20
3.	0.30	0.15	2.40
4.	0.25	0.25	?

(i) Write the expression for rate law.

(ii) Calculate the value of rate constant and specify its units.

(iii) What is the initial rate of disappearance of Cl_2 in exp. 4?

(b) A first order reaction is 20% complete in 10 minutes. Calculate the time for 75% completion of the reaction.

OR

(a) A reaction is first order in *A* and second order in *B*.

(i) Write the differential rate equation.

(ii) How is the rate affected on increasing the concentration of *B* three times?

(iii) How is the rate affected when the concentrations of both *A* and *B* are doubled?

(b) With the help of an example explain what is meant by pseudo first order reaction.

37. Using valence bond theory, explain the following in relation to the complexes given below : $[Mn(CN)_6]^{3-}$, $[Co(NH_3)_6]^{3+}$, $[Cr(H_2O)_6]^{3+}$, $[FeCl_6]^{4-}$

(i) Type of hybridisation.

(ii) Inner or outer orbital complex.

(iii) Magnetic behaviour.

(iv) Spin only magnetic moment value.

OR

(i) Explain why $[Fe(H_2O)_6]^{3+}$ has magnetic moment value of 5.92 BM whereas $[Fe(CN)_6]^{3-}$ has a value of only 1.74 BM.

(ii) Arrange following complex ions in increasing order of crystal field splitting energy (Δ_o) :
$[Cr(Cl)_6]^{3-}$, $[Cr(CN)_6]^{3-}$, $[Cr(NH_3)_6]^{3+}$.

(iii) Why do compounds having similar geometry have different magnetic moment?

(iv) $CuSO_4.5H_2O$ is blue in colour while $CuSO_4$ is colourless. Why?

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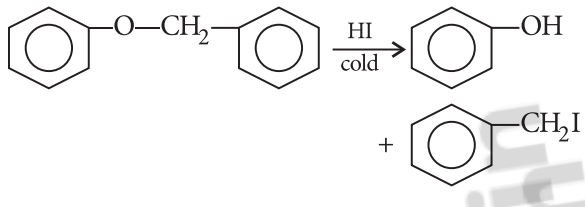
Detailed Solutions

1. (b) :



2. (b)

3. (b) : Blood cells neither swell nor shrink in isotonic solution.

The solutions having same osmotic pressure are called isotonic solutions.

OR

(a) : By Raoult's Law

$P_T = p_P^\circ x_P^\circ + p_Q^\circ x_Q$

where $p_P^\circ = 80$ torr, $p_Q^\circ = 60$ torr, $x_P = \frac{3}{5}$; $x_Q = \frac{2}{5}$

$P_T = 80 \times \frac{3}{5} + 60 \times \frac{2}{5} = 48 + 24 = 72$ torr

4. (b) : Alkyl amines (1°, 2° and 3°) are stronger bases than ammonia. This can be explained in terms of electron releasing inductive effect of alkyl group. With increase in alkyl group, the electron density on the nitrogen atom increases and thus, they can donate the lone pair of electrons more easily than ammonia.

The electron releasing effect is maximum in tertiary amines and minimum in primary amines (in gas phase).

3° amine > 2° amine > 1° amine > NH_3.

5. (a) : Due to lanthanide contraction, the elements of second and third *i.e.*, Zr and Hf transition series resemble more with each other than the elements of first and second transition series.

6. (c) :

$$CH_3-CH=CH_2 \xrightarrow[\text{Hydroboration}]{BH_3,\ THF} \underset{\text{Tripropyl borane}}{(CH_3CH_2CH_2)_3B} \xrightarrow[OH^-]{H_2O_2} 3CH_3CH_2CH_2OH$$

Propan-1-ol

7. (b) : Primary valency corresponds to oxidation number while secondary valency corresponds to coordination number.

OR

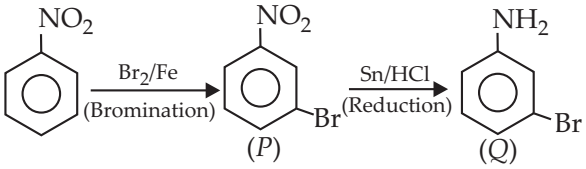
(a)

8. (a)

9. (a) : Inversion of configuration takes place in S_N2 mechanism.

OR

(d) :



$-NO_2$ group is a deactivating and *m*-directing group so it directs the incoming Br^- ion to the *m*-position thereby leading to the formation of *m*-bromonitrobenzene. Upon reduction with Sn/HCl, the $-NO_2$ group reduces to $-NH_2$ *i.e.,* the final product is *m*-bromoaniline.

10. (a)

11. (b) : The order of dehydration of alcohols is 3° > 2° > 1°.

12. (b)

13. (a) : All halogens are diatomic and form univalent ion.

14. (d) : Alkyl group attached to carbonyl carbon increases the electron density on the carbonyl carbon and lowers its reactivity towards nucleophilic addition reactions.

Also, as the number and size of alkyl group increases, the attack of nucleophile on the carbonyl group becomes more and more difficult due to steric hindrance.

Hence, the reactivity order will be :

$HCHO > CH_3CHO > CH_3COCH_3$

OR

All the reactions are showing the acidic property of carboxylic acid. Carboxylic acid forms the sodium salts with all *i.e.*, alkali metals, NaOH and Na_2CO_3 etc. and removes the acidic proton from the carboxylic acid.

15. (b)

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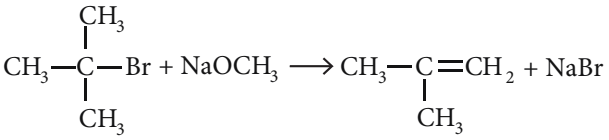
16. (b) : In Williamson's synthesis, if a tertiary alkyl halide is used, an alkene is the only reaction product and no ether is formed.



2-methylpropene

It is because alkoxides are not only nucleophiles but also strong bases as well. They react with alkyl halides leading to elimination reaction.

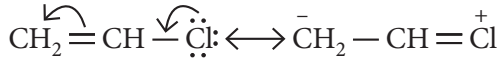
17. (a)

OR

(c) : The pressure cooker works on the principle that boiling point of a liquid increases with increase of vapour pressure above the liquid. Water, along with food to be cooked are heated in a closed vessel, so that the confined water vapour rises the super incumbent pressure. As a result, water boils at a temperature higher than 100°C. When the cooker is removed from the stove and the lid is removed, pressure again decreases due to which its boiling point decreases and water starts boiling again.

18. (a)

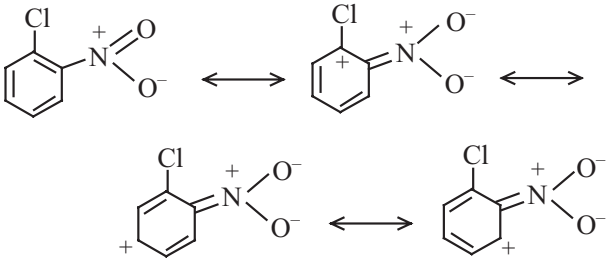
19. (c) : In vinyl chloride, C – Cl bond is stable due to resonance



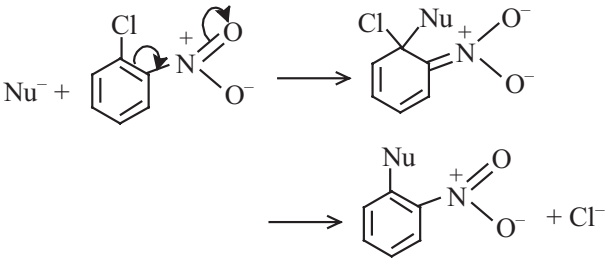
Hence S_N1 reaction in which Cl is replaced by nucleophile is not possible. In addition to this, sp^2 hydridised carbon is more acidic than sp^3 carbon, hence removal of H^+ ion is easier than removal of halide (Cl^-) ion.

OR

(a) : The C-atom of C – Cl in chloronitrobenzene carries a small +ve charge.



Because of the positive charge, attack at this carbon atom by the nucleophile is facilitated.



20. (c) : $\alpha = \dfrac{\Lambda^c_{eq}}{\Lambda^\circ_{eq}} = \dfrac{10}{200} = 0.05$

$[H^+] = c\,\alpha = 1 \times 0.05 = 0.05$

$pH = -\log [H^+] = -\log (5 \times 10^{-2})$

$= 2 \log 10 - \log 5 = 2 - 0.6990 = 1.3.$

OR

(d) : $E^\circ_{cell} = E_c - E_a = 0.77 - 0.54 = 0.23$ V

$\Delta G = -nFE^\circ_{cell} = -RT \ln K$ or $E^\circ_{cell} = \dfrac{0.0591}{n} \log K$

$\log K = \dfrac{0.23 \times 2}{0.0591} = 7.783$

or $K = \text{Antilog } (7.783) = 6.07 \times 10^7$.

21. (b) : Basic strength decreases as,

cyclohexylamine > aniline > benzamide.

Lesser basicity in aniline and benzamide is due to participation of lone pair of electron of $-NH_2$ group in resonance.

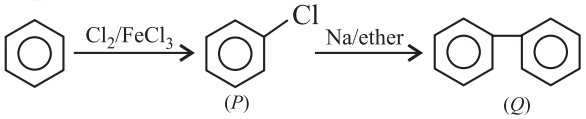
22. (d) : $_{22}Ti = 3d^2\,4s^2$; $Ti^{2+} = 3d^2$

$_{23}V = 3d^3\,4s^2$; $V^{3+} = 3d^2$

$_{24}Cr = 3d^4\,4s^2$; $Cr^{4+} = 3d^2$

$_{25}Mn = 3d^5\,4s^2$; $Mn^{5+} = 3d^2$

23. (c) :



(*P*) (*Q*)

OR

(a) : $CH_3CH_2Cl \xrightarrow{NaCN} \underset{(X)}{CH_3CH_2CN}$

$\xrightarrow{Ni/H_2} \underset{(Y)}{CH_3CH_2CH_2NH_2} \xrightarrow[\text{anhydride}]{\text{acetic}}$

$\underset{(Z)}{CH_3CH_2CH_2NHCOCH_3}$

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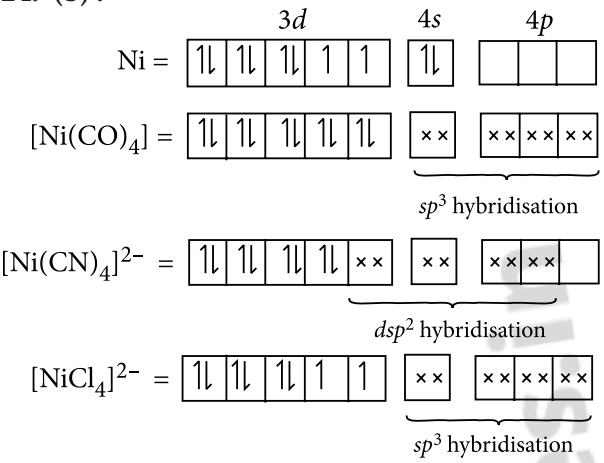
24. (b) :



25. (d) : The walls of raisin (partially dried grapes) consist of semipermeable membrane when kept in water, they swell as concentration inside raisin is high. The phenomenon is called osmosis.

26. (b) : The two solutions are isotonic. They must have same molar concentration,

Concentration of sucrose = 4 g per 100 cm^3 or 40 g L^{-1}

$$\therefore \quad C = \frac{40}{342}\text{M}$$

Concentration of unknown substance = 3 g per 100 cm^3 = 30 g L^{-1} = $\frac{30}{a}$ M

Where, '*a*' is molecular mass of unknown substance.

$$\therefore \quad \frac{40}{342} = \frac{30}{a} \quad \text{or,} \quad a = \frac{30 \times 342}{40} = 256.5$$

27. Given, $a = 286 \text{ pm} = 2.86 \times 10^{-8}$ cm,

$d = 7.86 \text{ g cm}^{-3}$

$M(\text{Fe}) = 56 \text{ g mol}^{-1}$, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$, $Z = ?$

Using formula, $d = \frac{Z \times M}{N_A \times a^3}$

or $Z = \frac{d \times N_A \times a^3}{M}$

$$= \frac{7.86 \text{ g cm}^{-3} \times 6.022 \times 10^{23} \text{ mol}^{-1} \times (2.86 \times 10^{-8} \text{ cm})^3}{56 \text{ g mol}^{-1}}$$

or, $Z \simeq 2 (\text{approx.})$

Hence, structure is *bcc*.

OR

(i) Given : Structure is *bcc*,

$a = 353 \text{ pm} = 353 \times 10^{-12}$ m, $r = ?$

For *bcc* structure,

$$r = \frac{\sqrt{3}\,a}{4} = \frac{1.732 \times 353 \text{ pm}}{4} = 152.85 \text{ pm}$$

(ii) $r = 152.85 \times 10^{-12}$ m

$$\text{Packing efficiency} = \frac{Z \times \frac{4}{3}\pi r^3}{a^3}$$

$$= \frac{2 \times \frac{4}{3} \times 3.14 \times (152.85 \times 10^{-12} \text{ m})^3}{(353 \times 10^{-12} \text{ m})^3} = 0.68$$

28. Adenine and guanine are purine bases present in RNA. Cytosine and Uracil are pyrimidine present in RNA

29. For first order reaction,

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}, \text{ at } t_{1/2}, [R] = \frac{[R]_0}{2}$$

$$\therefore \quad k = \frac{2.303}{t_{1/2}} \log \frac{[R]_0}{[R]_0/2} \text{ or, } t_{1/2} = \frac{2.303}{k} \log 2$$

$$t_{1/2} = \frac{2.303 \times 0.3010}{k} \Rightarrow t_{1/2} = \frac{0.693}{k}$$

In first order reaction, $t_{1/2}$ is independent of initial concentration.

30. (a) Dehydrogenation in the presence of Cu at 573 K of alcohol gives aldehyde.

$$R\text{CH}_2\text{OH} \xrightarrow[573 \text{ K}]{\text{Cu}} R\text{CHO} + \text{H}_2$$

(b) Acid chlorides can be reduced into aldehydes on reacting with hydrogen in presence of Pd or Pt and barium sulphate (catalyst). This reaction is called Rosenmund's reduction.

$$R\text{—}\overset{\overset{\text{O}}{\|}}{\text{C}}\text{—Cl} + \text{H}_2 \xrightarrow{\text{Pd} + \text{BaSO}_4} R\text{—}\overset{\overset{\text{O}}{\|}}{\text{C}}\text{—H}$$

Aldehyde

31. (a) (i)

Propan-2-ol	Propan-1-ol
Propan-2-ol when treated with iodine and alkali gives yellow precipitate of iodoform. $\text{CH}_3\overset{\overset{\text{OH}}{\|}}{\text{CH}}\text{—CH}_3 \xrightarrow{\text{I}_2 + \text{OH}^-} \underset{\substack{\text{Iodoform} \\ \text{(yellow)}}}{\text{CHI}_3} + \text{CH}_3\text{COO}^-$	Propan-1-ol does not react with iodine and alcohol.

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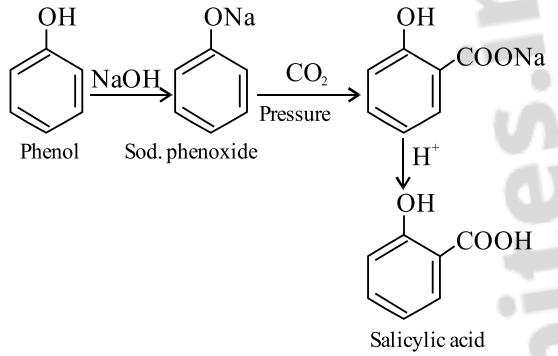
(ii)

Phenol	Ethanol
Phenol reacts with $FeCl_3$ to give brown complex. $FeCl_3 + C_6H_5OH \longrightarrow Fe(C_6H_5O)_3 + 3HCl$ (Brown)	Ethanol does not react with $FeCl_3$. $FeCl_2 + C_2H_5OH \rightarrow$ No reaction

(b)



32. (i) Soft drink and soda water are filled under high pressure.

(ii) Number of moles of urea $= \frac{0.25 \times 100}{1000}$

$= 0.025$ mol

Number of moles of cane sugar

$= \frac{0.1 \times 100}{1000} = 0.01$ mol

Total volume, $V = (100 + 100)\ cm^3$

$= 200\ cm^3 = 0.2\ L$

For osmotic pressure, $\pi = \frac{(n_1 + n_2)}{V} RT$

$= \frac{(0.025 + 0.01)}{0.2} \times 0.082 \times 293$

$= \frac{0.035}{0.2} \times 0.082 \times 293 = 4.2$ atm

33. (i) As the size of central atom increases, M—H bond becomes weaker and longer and hence, thermal stability decreases while going from H_2O to H_2Te.

(ii) As the size of halogen increases down the group *i.e.*, from F to I, the hydration enthalpy of corresponding halide ion decreases. Hence, F^- ion has higher hydration enthalpy (515 kJ mol^{-1}) than Cl^- ion (381 kJ mol^{-1}).

(iii) Oxygen can not expand its octet due to absence of *d*-orbitals.

OR

(a) Halogens absorb a part of the light in the visible region which causes excitation of outer electrons to higher energy levels. The excitation energy depends on the size of the halogen atom, the smaller the atom, the greater is the excitation energy. As different colours are absorbed by halogens, they display different complementary colours.

(b) In covalent compounds, the boiling point increases with increase of molecular mass as van der Waals' forces increase. The volatility thus, decreases in covalent compounds with increase in molecular mass. However, hydrogen bonding is present in HF, due to which, the boiling point of HF is higher and the volatility is less. Thus, the volatility of hydrogen halides can be represented as :

$$\underset{\text{Most volatile}}{HCl} > HBr > HI > \underset{\text{Least volatile}}{HF}$$

34. (a) KCN is predominantly ionic and provides cyanide ion in solution. Although both carbon and nitrogen are in position to donate electron pair but the attack takes place mainly through carbon atom because C—C bond is stronger. Since AgCN is covalent molecule and nitrogen is free to donate electron pair hence isocyanide is main product.

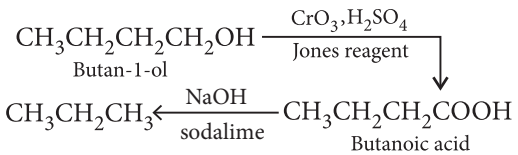
(i) $R\overset{+\delta}{\rightarrow}X\overset{-\delta}{\leftarrow} + CN^- \longrightarrow R—CN + X^-$

(ii) $R\overset{+\delta}{\rightarrow}X\overset{-\delta}{\leftarrow} + AgCN \longrightarrow R—NC + AgX$

(b) Due to greater electronegativity difference between carbon and halogen, electron density shifts towards halogen atom and carbon atom behaves as electron deficient atom or electrophilic centre.

$$—\overset{|}{\underset{|}{C}}\overset{+\delta\ -\delta}{\rightarrow}X$$

35. (a) (i)



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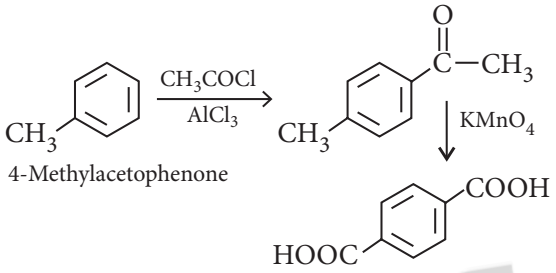
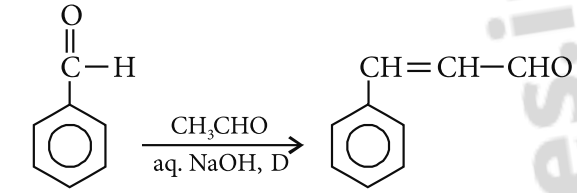
(ii) 

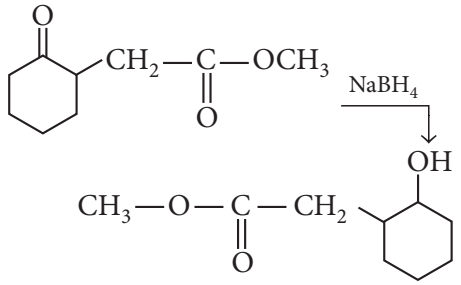
(iii) 

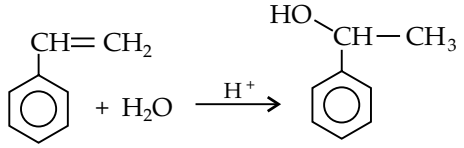
(b) (i) Due to presence of electron withdrawing group (—COOH) in aromatic carboxylic acids, they do not undergo Friedel — Crafts reaction.

(ii) Due to presence of strong electron withdrawing group (—NO_2), 4-nitrobenzoic acid is more acidic than benzoic acid and therefore, pK_a value is lower.

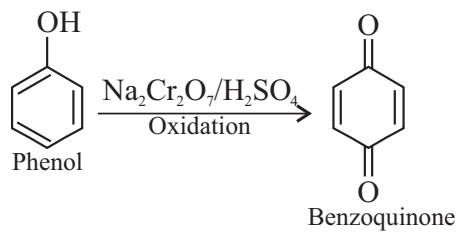
OR

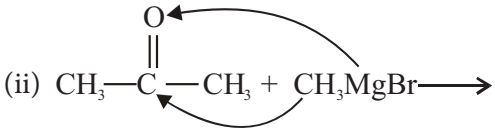
(a) *R* is alkyl group and *R′* is aryl group. *R* must be a group having more electron density than H. *i.e.,* having +*I* effect where as *R′* must be having –*I* effect.

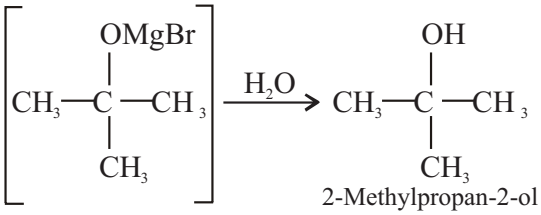
(b) (i) 

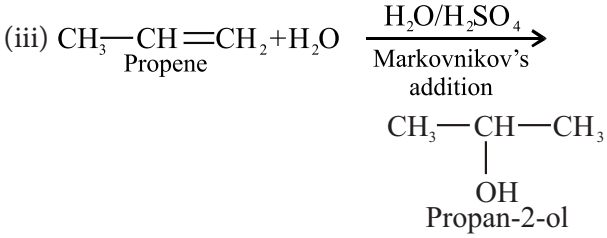
(ii) 

OR

(c) (i) 

(ii) 



(iii) 

36. (a) Given reaction :

$2NO_{(g)} + Cl_{2(g)} \longrightarrow 2NOCl_{(g)}$

Exp. No.	Initial [NO] (M)	Initial [Cl_2] (M)	Initial rate of disapp. of Cl_2 (M/min)
1.	0.15	0.15	0.60
2.	0.15	0.30	1.20
3.	0.30	0.15	2.40
4.	0.25	0.25	?

(i) Let rate of this reaction $r = k[NO]^m[Cl_2]^n$

then $\frac{r_1}{r_2} = \frac{0.60}{1.20} = \frac{k(0.15)^m(0.15)^n}{k(0.15)^m(0.30)^n}$

or $\frac{1}{2} = \left(\frac{1}{2}\right)^n \Rightarrow n = 1$

Again from $\frac{r_2}{r_3} = \frac{1.20}{2.40} = \frac{k(0.15)^m(0.30)^1}{k(0.30)^m(0.15)^1}$

or $\frac{1}{2} = \left(\frac{1}{2}\right)^m \cdot \frac{2}{1}$ or $\frac{1}{4} = \left(\frac{1}{2}\right)^m \Rightarrow m = 2$

Hence, expression for rate law is

$r = k[NO]^2\,[Cl_2]^1$

(ii) Substituting the values in experiment (1),

$0.60\ M\ min^{-1} = k(0.15\ M)^2\ (0.15\ M)^1$

or $k = \frac{0.60\ M\,min^{-1}}{0.0225 \times 0.15\ M^3} = 177.77\ M^{-2}\ min^{-1}$

(iii) $r = 177.7\ M^{-2}\ min^{-1} \times (0.25\ M)^2\ (0.25\ M)$

$= 2.77\ M\ min^{-1}$

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(b) Let initial concentration of the reactant = 100
Thus at t = 10 min, $a - x = 80$
t = ?, when $a - x = 25$
For a first order reaction

$$k = \frac{2.303}{10 \text{ min}} \log \frac{100}{80} = \frac{2.303}{10 \text{ min}} \log \frac{5}{4} \quad ...(i)$$

and $$k = \frac{2.303}{t} \log \frac{100}{25} = \frac{2.303}{t} \log 4 \quad ...(ii)$$

From eqn. (i) and (ii)

$$\frac{2.303}{10 \text{ min}} \log \frac{5}{4} = \frac{2.303}{t} \log 4$$

or $$0.0223 \text{ min}^{-1} = \frac{1.3865}{t}$$

$$t = \frac{1.3865}{0.0223} = 62.2 \text{ min}$$

OR

(a) (i) Reaction is first order in A and second order in B, hence differential rate equation is

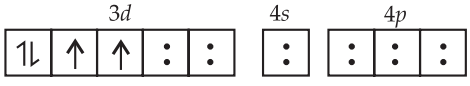
$$\frac{dx}{dt} = k[A][B]^2$$

(ii) Rate = $k[A][B]^2$
If $[B]$ is tripled, Rate = $k[A][3B]^2 = 9[A][B]^2$
Thus, rate increases 9 times.
(iii) If both $[A]$ and $[B]$ are doubled,
Rate = $k\,[2A][2B]^2 = 8[A][B]^2$
Thus, rate of reaction increases 8 times.
(b) The reactions which are of second order but behave like first order reactions are called pseudo first order reactions *e.g.*, during hydrolysis of ethyl acetate (ester) with water, the concentration of water does not alter much during the reaction. So, in the rate equation the concentration of water (the reactant which is in excess) can be taken as constant hence, the reaction behaves as first order reaction.

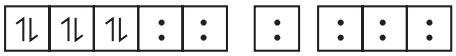
$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

Rate = $k'[CH_3COOC_2H_5][H_2O]$, the term $[H_2O]$ can be taken as constant. Hence, the rate equation becomes
Rate = $k[CH_3COOC_2H_5]$ where $k = k'[H_2O]$

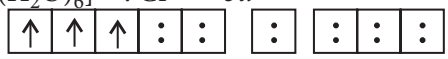
37. $[Mn(CN)_6]^{3-}$: $Mn^{3+} = 3d^4$



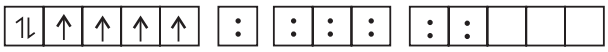
(i) Hybridisation - d^2sp^3
(ii) Inner orbital complex
(iii) Paramagnetic
(iv) Magnetic moment = $\sqrt{2(2+2)} = 2.87$ B.M.
$[Co(NH_3)_6]^{3+}$: $Co^{3+} = 3d^6$



(i) Hybridisation - d^2sp^3
(ii) Inner orbital complex (iii) Diamagnetic
(iv) Magnetic moment = 0
$[Cr(H_2O)_6]^{3+}$: $Cr^{3+} = 3d^3$



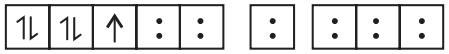
(i) Hybridisation - d^2sp^3
(ii) Inner orbital complex
(iii) Paramagnetic
(iv) Magnetic moment = $\sqrt{3(3+2)} = 3.87$ B.M.
$[Fe(Cl)_6]^{4-}$: $Fe^{2+} = 3d^6$



(i) Hybridisation - sp^3d^2
(ii) Outer orbital complex (iii) Paramagnetic
(iv) Magnetic moment = $\sqrt{4(4+2)} = 4.9$ B.M.

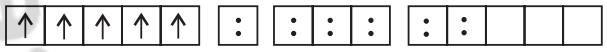
OR

(i) $Fe(CN)_6]^{3-}$:



d^2sp^3, one unpaired electron

$[Fe(H_2O)_6]^{3+}$:



sp^3d^2, 5 unpaired electrons

Due to presence of 5 unpaired electrons $[Fe(H_2O)_6]^{3+}$ has magnetic moment 5.92 B.M whereas $[Fe(CN)_6]^{3-}$ has 1.74 B.M magnetic moment due to presence of one unpaired electron.
(ii) CFSE increases in the order of $[Cr(Cl)_6]^{3-} < Cr(NH_3)_6]^{3-} < [Cr(CN)_6]^{3-}$ depending upon the strength of the field.
(iii) It is due to the presence of weak and strong ligands in complexes. If CFSE is high the complex will show low value of magnetic moment and vice-versa. *e.g.*, $[CoF_6]^{3-}$ is paramagnetic with 3 unpaired electrons while $[Co(NH_3)_6]^{3+}$ is diamagnetic with 0 unpaired electron.
(iv) In $CuSO_4 \cdot 5H_2O$, water acts as ligand and causes crystal field splitting. Hence, *d-d* transition takes place and hydrated $CuSO_4$ shows colour. In anhydrous $CuSO_4$ there is no ligand hence, crystal field splitting is not possible and it is colourless.

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PRACTICE PAPER 3
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises two parts A and B.

(ii) Part A : Q. no. 1 to 26 are objective type questions.
Q. no. 1 to 19 are objective type questions carrying one mark each.
Q. no. 20 to 24 are objective type questions carrying two marks each.
Q. no. 25 to 26 are case based objective type questions carrying three marks each.

(iii) Part B : Q. no. 27 to 37 are subjective/descriptive type questions.
Q. no. 27 to 30 are short answer type-I questions carrying two marks each.
Q. no. 31 to 34 are short answer type-II questions carrying three marks each.
Q. no. 35 to 37 are long answer type questions carrying five marks each.

(iv) There is no overall choice in the question paper. However, internal choices have been provided in both part A and Part B.

(v) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

PART - A : Objective Type

1. Which of the following undergoes nucleophilic substitution exclusively by S_N1 mechanism?
(a) Ethyl chloride (b) Isopropyl chloride
(c) Chlorobenzene (d) Benzyl chloride

OR

Reaction of *t*-butyl bromide with sodium methoxide produces
(a) sodium *t*-butoxide
(b) *t*-butyl methyl ether
(c) isobutane
(d) isobutylene.

2. How many g of dibasic acid (mol. weight 200) should be present in 100 mL of the aqueous solution to give strength of 0.1 N?
(a) 10 g (b) 2 g (c) 1 g (d) 20 g

3. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. The explanation for it is that
(a) concentrated hydrochloric acid emits strongly smelling HCl gas all the time
(b) oxygen in air reacts with the emitted HCl gas to form a cloud of chlorine gas
(c) strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke
(d) due to strong affinity for water, concentrated hydrochloric acid pulls moisture of air towards itself. This moisture forms droplets of water and hence the cloud.

4. The specific conductance of a 0.1 N KCl solution at 23°C is 0.012 ohm^{-1} cm^{-1}. The resistance of cell containing the solution at the same temperature was found to be 55 ohm. The cell constant will be
(a) 0.918 cm^{-1} (b) 0.66 cm^{-1}
(c) 1.142 cm^{-1} (d) 1.12 cm^{-1}

5. Benzoic acid gives benzene on being heated with *X* and phenol gives benzene on being heated with *Y*. Therefore, *X* and *Y* are respectively
(a) soda-lime and copper
(b) Zn dust and NaOH
(c) Zn dust and soda-lime
(d) soda-lime and zinc dust.

OR

The reagent which can be used to distinguish acetophenone from benzophenone is
(a) 2,4-dinitrophenylhydrazine
(b) aqueous solution of $NaHSO_3$
(c) Benedict reagent
(d) I_2 and NaOH.

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6. The IUPAC name for the complex $[Co(NO_2)(NH_3)_5]Cl_2$ is
(a) nitrito-N-pentaamminecobalt(III) chloride
(b) nitrito-N-pentaamminecobalt(II) chloride
(c) pentaammine nitrito-N-cobalt(II) chloride
(d) pentaammine nitrito-N-cobalt(III) chloride.

7. Which is not correct regarding the adsorption of a gas on surface of a solid?
(a) On increasing temperature adsorption increases continuously.
(b) Enthalpy and entropy change is negative.
(c) Adsorption is more for some specific substance.
(d) It is a reversible reaction.

8. Identify the incorrect statement among the following:
(a) 4*f*-and 5*f*-orbitals are equally shielded.
(b) *d*-Block elements show irregular and erratic chemical properties among themselves.
(c) La and Lu have partially filled *d*-orbitals and no other partially filled orbitals.
(d) The chemistry of various lanthanoids is very similar.

9. Which one of the following on reduction with lithium aluminium hydride yields a secondary amine?
(a) Methyl isocyanide (b) Acetamide
(c) Methyl cyanide (d) Nitroethane

10. For a first order reaction $A \longrightarrow B$ the reaction rate at reactant concentration of 0.01 M is found to be 2.0×10^{-5} mol L^{-1} s^{-1}. The half-life period of the reaction is
(a) 30 s (b) 220 s (c) 300 s (d) 347 s.

OR

If 60% of a first order reaction was completed in 60 minutes, 50% of the same reaction would be completed in approximately (log 4 = 0.60, log 5 = 0.69)
(a) 45 minutes (b) 60 minutes
(c) 40 minutes (d) 50 minutes.

11. How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion?
(a) Six (b) Three (c) One (d) Two

12. In context with the transition elements, which of the following statements is incorrect?
(a) In addition to the normal oxidation states, the zero oxidation state is also shown by these elements in complexes.
(b) In the highest oxidation states, the transition metals show basic character and form cationic complexes.
(c) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4*s* and 3*d* electrons are used for bonding.
(d) Once the d^5 configuration is exceeded, the tendency to involve all the 3*d* electrons in bonding decreases.

13. The *fcc* crystal contains how many atoms in each unit cell?
(a) 6 (b) 8 (c) 4 (d) 5

14. The major organic product in the reaction is $CH_3 - O - CH(CH_3)_2 + HI \longrightarrow$ products
(a) $CH_3I + (CH_3)_2CHOH$
(b) $CH_3OH + (CH_3)_2CHI$
(c) $ICH_2OCH(CH_3)_2$
(d) $CH_2OC(CH_3)_2$ (I)

OR

Which one of the following will not form a yellow precipitate on heating with an alkaline solution of iodine?
(a) $CH_3CH(OH)CH_3$
(b) $CH_3CH_2CH(OH)CH_3$
(c) CH_3OH
(d) CH_3CH_2OH

Questions 15 to 19 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

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15. **Assertion :** Oxygen is never more than divalent.
Reason : After forming two covalent bonds, it attains a noble gas configuration and there are no low energy orbitals which can be used to form more bonds.

16. **Assertion :** The freezing point is the temperature at which a solid crystallizes from its solutions.
Reason : The freezing point depression is the difference between the initial temperature and freezing point of a pure solvent.

OR

Assertion : The sum of mole fractions of all the components of a solution is unity.
Reason : Mole fraction is temperature dependent mode of concentration.

17. **Assertion :** When transition metal atoms ionise, the 4*s*-orbital electrons are ionised before the 3*d*-orbital electrons.
Reason : The energy of 3*d*-orbital electrons is same as that of 4*s*-orbitals.

18. **Assertion :** α-Hydrogen atoms in aldehydes and ketones are acidic.
Reason : Anions formed after loss of α-hydrogens are stabilised due to inductive effect.

OR

Assertion : Acetic acid does not undergo haloform reaction.
Reason : Acetic acid has no alpha hydrogen atoms.

19. **Assertion :** Size of the cation is larger in tetrahedral hole than in an octahedral hole.
Reason : In ionic compounds usually anions are present in the packing and cations occupy the voids.

20. A solution has a 1 : 4 mole ratio of pentane to hexane. The vapour pressures of the pure hydrocarbons at 20 °C are 440 mm Hg for pentane and 120 mm Hg for hexane. The mole fraction of pentane in the vapour phase would be

(a) 0.200 (b) 0.549
(c) 0.786 (d) 0.478

OR

Concentrated aqueous sulphuric acid is 98% H_2SO_4 by mass and has a density of 1.80 g mL^{-1}. Volume of acid required to make one litre of 0.1 M H_2SO_4 solution is

(a) 16.65 mL (b) 22.20 mL
(c) 5.55 mL (d) 11.10 mL

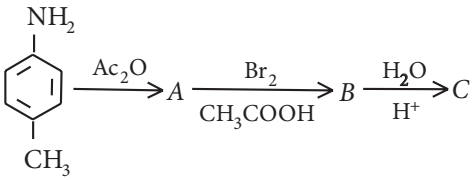
21. Given :
(i) $Cu^{2+} + 2e^- \rightarrow Cu$, $E° = 0.337$ V
(ii) $Cu^{2+} + e^- \rightarrow Cu^+$, $E° = 0.153$ V
Electrode potential, $E°$ for the reaction,
$Cu^+ + e^- \rightarrow Cu$, will be

(a) 0.90 V (b) 0.30 V
(c) 0.38 V (d) 0.52 V

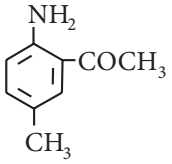
22. Which one of the following does not correctly represent the correct order of the property indicated against it?

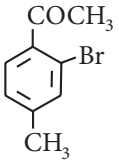
(a) Ti < V < Cr < Mn; increasing number of oxidation states
(b) $Ti^{3+} < V^{3+} < Cr^{3+} < Mn^{3+}$: increasing magnetic moment
(c) Ti < V < Cr < Mn : increasing melting points
(d) Ti < V < Mn < Cr : increasing 2nd ionization enthalpy

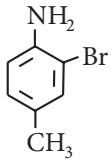
23. The final product *C*, obtained in this reaction

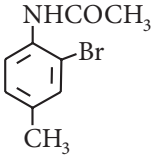


would be

(a) 

(b) 

(c) 

(d) 

OR

An organic compound (C_3H_9N) (*A*), when treated with nitrous acid, gave an alcohol and N_2 gas was evolved. (*A*) on warming with $CHCl_3$ and caustic potash gave (*C*) which on reduction gave isopropylmethylamine. Predict the structure of (*A*).

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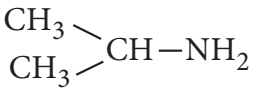
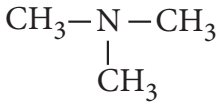
(a) 

(b) $CH_3CH_2 - NH - CH_3$

(c) 

(d) $CH_3CH_2CH_2 - NH_2$

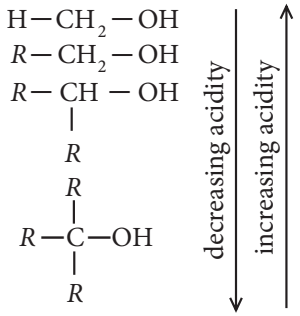
24. Which one of the following is an outer orbital complex and exhibits paramagnetic behaviour?

(a) $[Ni(NH_3)_6]^{2+}$ (b) $[Zn(NH_3)_6]^{2+}$

(c) $[Cr(NH_3)_6]^{3+}$ (d) $[Co(NH_3)_6]^{3+}$

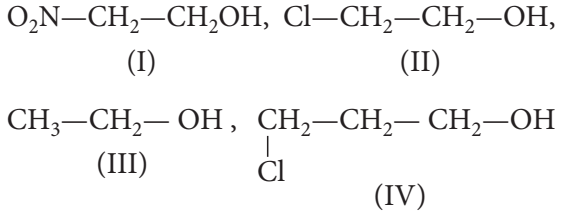
Read the following passage and answer the questions 25 and 26 that follow :

Alcohols react with sodium metal to liberate hydrogen gas. This reaction tells about acidity of alcohols. Alcohols are said to be more acidic than water because alkoxide is less stable than hydroxide ion. This can also be explained on the basis of +I effect. More the +I group attached to adjacent C-atom of –OH group the alcohol will be less acidic.



Conjugate base of weak acid is strong base. –I effect attached at adjacent C-atom of –OH group will increase the acidic strength.

25. Consider the following compounds :

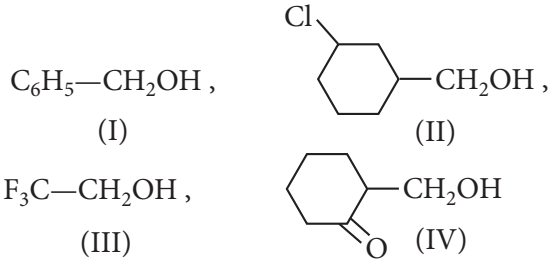


Among these, the most and least acidic compounds are respectively

(a) II, IV (b) I, III

(c) III, I (d) II, III

26. Among the following compounds, the compounds which reacts fastest with Na metal and which will give weakest conjugate base are respectively



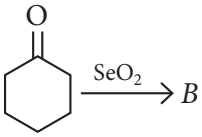
(a) I, III (b) III, IV

(c) III, III (d) II, I

PART - B : Subjective / Descriptive Type

27. Write the IUPAC names of the products (*A*) and (*B*) in the following reactions :

(a) $CH_3COOH \xrightarrow[\Delta]{NH_3} A$

(b) 

OR

How will you convert ethanal into the following compounds?

(i) Butan-1,3-diol (ii) But-2-enal

28. An element crystallises in *fcc* structure. Its density is 7.2 g cm^{-3}. 208 g of this element has 4.283×10^{24} atoms. Calculate the edge length of the unit cell.

29. (a) Draw a schematic graph showing how the rate of a first order reaction changes with change in concentration of the reactant.

(b) Rate of a reaction is given by

Rate = $k[A]^2[B]$

What are the units for the rate and the rate constant for this reaction?

30. When *tert*-butanol and *n*-butanol are separately treated with a few drops of dil. $KMnO_4$ in one case only, the purple colour disappears and a brown precipitate is formed. Which of the two alcohols gives the above reaction and what is the brown precipitate formed?

31. Give reasons :

(a) Racemic mixture is optically inactive.

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(b) The presence of nitro group ($-NO_2$) at *o*/*p* positions increases the reactivity of haloarenes towards nucleophilic substitution reactions.

(c) Alkyl halides, though polar, are immiscible with water.

32. Give reasons for the following observations:

(i) Cu^+ ion is not stable in aqueous solution.

(ii) Mn(II) ion shows maximum paramagnetic character amongst the bivalent ions of first transition series.

(iii) Scandium (at. no. 21) salts are white.

33. Write a suitable chemical equation to complete each of the following transformations :

(i) Butan-1-ol to butanoic acid

(ii) 4-Methylacetophenone to benzene-1,4-dicarboxylic acid

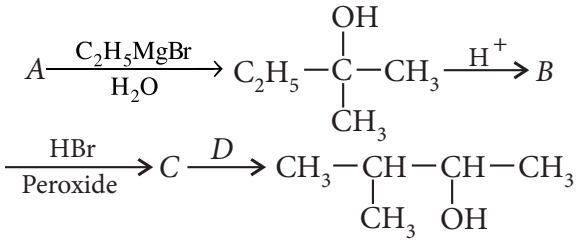
(iii) Benzaldehyde to 3-Phenylpropan-1-ol.

OR

(a) Suggest a suitable oxidising agent for the following conversion :

$(CH_3)_2C=CHCOCH_3 \rightarrow (CH_3)_2C=CHCO_2H$

(b) Write the structural formulae of compounds *A*, *B* and *C* and name the reagent *D* in the following reaction :



34. (a) Using Raoult's law explain how the total vapour pressure over the solution is related to mole fraction of components in the following solutions.

(i) $CHCl_{3\,(l)}$ and $CH_2Cl_{2(l)}$

(ii) $NaCl_{(s)}$ and $H_2O_{(l)}$

(b) Give an example of a solid solution in which the solute is a gas.

35. (a) Explain the terms ideal and non-ideal solutions in the light of forces of interactions operating between molecules in liquid solutions.

(b) What is the effect of temperature on

(i) Molarity

(ii) Molality?

OR

(a) Henry's law constant for CO_2 in water is 1.67×10^8 Pa at 298 K. Calculate the quantity of CO_2 in 500 mL of soda water when packed under 2.5 atm CO_2 pressure at 298 K.

(b) Explain why on addition of 1 mol of NaCl to 1 litre of water, the boiling point of water increases, while addition of 1 mol of methyl alcohol to one litre of water decreases its boiling point.

36. (a) Describe the chemical reaction of chlorine with

(i) dry calcium hydroxide (slaked lime).

(ii) aqueous solution of hot calcium hydroxide (milk of lime).

(b) Give reason :

(i) Xenon does not form fluorides such as XeF_3 and XeF_5.

(ii) Out of nobles gases, xenon is known to form established chemical compounds.

(iii) O_3 acts as a powerful oxidising agent.

OR

(a) Account for the following :

(i) Bi(V) is a stronger oxidizing agent than Sb(V).

(ii) N – N single bond is weaker than P – P single bond.

(iii) Noble gases have very low boiling points.

(b) Draw the structure of the following :

(i) $H_2S_2O_7$ (ii) $HClO_3$

37. (i) Identify *A*, *B* and *C* in the following reactions :

(a) $CH_3Br \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[273\ K]{HNO_2} C$

(b) $CH_3COOH \xrightarrow[\Delta]{NH_3} A \xrightarrow{Br_2 + KOH} B \xrightarrow{CHCl_3 + NaOH} C$

(ii) (a) How are amines prepared from amides?

(b) Acylation of amines with acid chloride is carried out in presence of stronger base. Why?

OR

(a) Name the products with chemical reactions when the following compounds are hydrolysed with dilute HCl.

(i) CH_3CH_2CN (ii) C_2H_5NC

(iii) $C_2H_5NO_2$

(b) Compare the basicities of :

$H_2C=CHCH_2NH_2$, $CH_3CH_2CH_2NH_2$ and $HC\equiv CCH_2NH_2$

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Detailed Solutions

1. (d) : S_N1 reaction is favoured by heavy (bulky) groups on the carbon atom attached to halogens.

OR

(d) : Isobutylene is obtained.

$$H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle Br}{\underset{|}{\overset{|}{C}}}}-CH_3 + CH_3ONa \longrightarrow \underset{\displaystyle CH_3}{\overset{|}{CH_3C}}=CH_2 + CH_3OH + NaBr$$

2. (c) : The strength of the solution is 0.1 N

$$\frac{w}{E}=\frac{V\times N}{1000} \quad \left(\text{Equivalent weight} = \frac{200}{2} = 100\right)$$

$$\Rightarrow \quad w=\frac{100\times 0.1\times 100}{1000}=1\text{ g}$$

3. (c) : HCl gas in presence of moisture in air forms droplets of liquid solution in the form of cloudy smoke.

4. (b) : $\kappa = 0.012\ \text{ohm}^{-1}\text{cm}^{-1}$

$R = 55$ ohm $\Rightarrow C=\frac{1}{R}=\frac{1}{55}\ \text{ohm}^{-1}$

$$\text{Cell Constant } \left(\frac{l}{a}\right)=\frac{\text{Specific Conductance}}{\text{Conductance}}$$

$$=\frac{0.012}{1/55} = 55\times 0.012 = 0.66\ \text{cm}^{-1}$$

5. (d) : $C_6H_5COOH \xrightarrow[(X)]{\text{Soda-lime}} C_6H_6 + Na_2CO_3$

$C_6H_5OH \xrightarrow[(Y)]{\text{Zn dust}} C_6H_6 + ZnO$

X = soda-lime and Y = Zn dust

OR

(d) : Acetophenone reacts with NaOH and I_2 to give yellow ppt. of CHI_3 but benzophenone ($C_6H_5COC_6H_5$) does not. Hence, it can be used to distinguish between them.

$$C_6H_5COCH_3 \xrightarrow{I_2,\ NaOH} \underset{\text{yellow ppt.}}{CHI_3} + C_6H_5COONa$$

6. (d) : $[Co(NO_2)(NH_3)_5]Cl_2$:
Pentaaminenitrito-N-cobalt(III) chloride

7. (a) : Adsorption decreases with increase in temperature.

8. (a) : Shielding of 4*f* is more than that of 5*f*.

9. (a) : Alkyl isocyanides on reduction with lithium aluminium hydride forms secondary amines containing methyl as one of the the alkyl groups.

$$\underset{\text{Methyl isocyanide}}{H_3C-N\rightleftharpoons C} + 4[H] \xrightarrow{LiAlH_4} \underset{\text{Dimethylamine}}{H_3C-NH-CH_3}$$

10. (d) : $A \longrightarrow B$

Rate of reaction $= 2\times 10^{-5}\ \text{mol L}^{-1}\ \text{s}^{-1}$

$\Rightarrow$ order of reaction is n = l, rate = $k\ [A]^n$

$k \longrightarrow$ rate or velocity constant

$[A] = 0.01$ M

$$\Rightarrow \quad k=\frac{2\times 10^{-5}}{0.01}=2\times 10^{-3},\ k=\frac{0.693}{t_{1/2}}$$

$$t_{1/2}=\frac{0.693}{2\times 10^{-3}} = 346.5 \approx 347\text{ s}$$

OR

(a) : For a first order reaction,

$$k=\frac{2.303}{t}\log\frac{a}{a-x}$$

$$k=\frac{2.303}{60}\log\frac{100}{40}=\frac{2.303}{60}\times\log 2.5=0.0153$$

$$\text{Again, } t_{1/2}=\frac{2.303}{k}\log\frac{100}{50}=\frac{2.303}{0.0153}\times\log 2$$

$$= 45.31\text{ min.}$$

11. (c) : EDTA, which has four donor oxygen atoms and two donor nitrogen atoms in each molecule forms complex with Ca^{2+} ion.

12. (b) : When the transition metals are in their highest oxidation state, they no longer have tendency to give away electrons, thus they are not basic but show acidic character and form anionic complexes.

13. (c)

14. (a) : With cold HI, a mixture of alkyl iodide and alcohol is formed. In the case of mixed ethers, the halogen atom attaches to a smaller and less complex alkyl group.

$CH_3OCH(CH_3)_2 + HI \rightarrow CH_3I + (CH_3)_2CHOH$

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OR

(c) : Formation of a yellow precipitate on heating a compound with an alkaline solution of iodine is known as iodoform reaction. Methyl alcohol does not respond to this test. Iodoform test is exhibited by those alcohols which possess $CH_3CH(OH)$ – group.

15. (a)

16. (b)

OR

(c) : $x_1 + x_2 = 1$ *i.e.* mole fraction of solute + mole fraction of solvent = 1. Mole fraction does not depend upon temperature as it does not involve volume.

17. (c)

18. (c) : The anions left after the removal of α-hydrogen is stabilized by resonance effect.

OR

(c) : Haloform reaction is given by only those compounds containing CH_3CO or CH_3CHOH group. The carbonyl group of –COOH does not give the usual reactions of $\diagup\!\!\diagdown C{=}O$ group, mainly due to reduced double bond character of the carbon – oxygen bond (because of resonance). It has 3 α–H atoms.

19. (b)

20. (d) : $\frac{n_{C_5H_{12}}}{n_{C_6H_{14}}} = \frac{1}{4}$

$\Rightarrow x_{C_5H_{12}} = \frac{1}{5}$ and $x_{C_6H_{14}} = \frac{4}{5}$

$p^\circ_{C_5H_{12}} = 440$ mm Hg; $p^\circ_{C_6H_{14}} = 120$ mm Hg

$P_{total} = p^\circ_{C_5H_{12}} x_{C_5H_{12}} + p^\circ_{C_6H_{14}} x_{C_6H_{14}}$

$= 440 \times \frac{1}{5} + 120 \times \frac{4}{5} = 88 + 96 = 184$ mm of Hg

By Raoult's Law, $P_{C_5H_{12}} = p^\circ_{C_5H_{12}} x_{C_5H_{12}}$ ...(1)

$x_{C_5H_{12}} \rightarrow$ mole fraction of pentane in solution

By Dalton's Law, $p_{C_5H_{12}} = x'_{C_5H_{12}} P$ ...(2)

$x'_{C_5H_{12}} \rightarrow$ mole fraction of pentane above the solution.

From (1) and (2),

$p_{C_5H_{12}} = 440 \times \frac{1}{5}$ = 88 mm of Hg

$\Rightarrow 88 = x'_{C_5H_{12}} \times 184$

$x' = \frac{88}{184}$; $x' = 0.478$

OR

(c) : H_2SO_4 is 98% by weight.

Weight of H_2SO_4 = 98 g

Weight of solution = 100 g

$\therefore$ Volume of solution = $\frac{\text{mass}}{\text{density}} = \frac{100}{1.80}$ mL

= 55.55 mL = 0.0555 L

Molarity of solution = $\frac{98}{98 \times 0.0555}$ M = 18.02 M

Let V mL of this H_2SO_4 are used to prepare 1 litre of 0.1 M H_2SO_4.

$\therefore$ mM of concentrated H_2SO_4 = mM of dilute H_2SO_4

or, $V \times 18.02 = 1000 \times 0.1$

$\Rightarrow V = \frac{1000 \times 0.1}{18.02} = 5.55$ mL

21. (d) : Given,

$Cu^{2+} + 2e^- \longrightarrow Cu$; $E_1^\circ = 0.337$ V

$Cu^{2+} + e^- \longrightarrow Cu^+$; $E_2^\circ = 0.153$ V

The required reaction is

$Cu^+ + e^- \longrightarrow Cu$; $E_3^\circ = ?$

Applying, $\Delta G^\circ = -nFE^\circ$, $\Delta G_3^\circ = \Delta G_1^\circ - \Delta G_2^\circ$

$-(n_3 F E_3^\circ) = -(n_1 F E_1^\circ) - (-n_2 F E_2^\circ)$

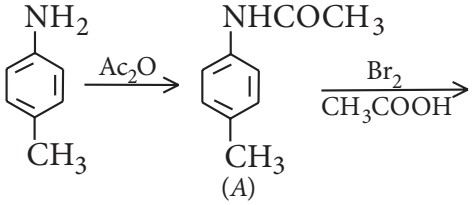
or $E_3^\circ = 2 \times E_1^\circ - E_2^\circ$

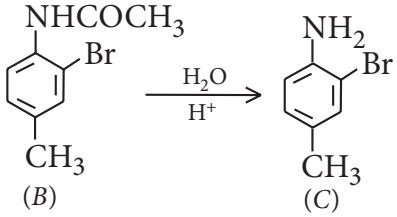
or $E_3^\circ = (2 \times 0.337) - 0.153 = 0.52$ V

22. (c) : Melting point order :

Mn	<	Ti	<	Cr	<	V
1245°C		1668°C		1875°C		1900°C

23. (c) :





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OR

(a) : As *A* gives alcohol on treatment with nitrous acid thus it should be primary amine. C_3H_9N has two possible structure with $-NH_2$ group.

$CH_3 — CH_2 — CH_2 — NH_2$ or $CH_3 — \underset{\displaystyle CH_3}{\underset{|}{CH}} — NH_2$

As it gives isopropylmethylamine thus it should be isopropyl amine not *n*-propyl amine.

$$CH_3 — \underset{\displaystyle CH_3}{\underset{|}{CH}} — NH_2 \xrightarrow{HNO_2} CH_3 — \underset{\displaystyle OH}{\underset{|}{CH}} — CH_3 + N_2\uparrow$$

(*A*)

$$\xrightarrow{CHCl_3/KOH}$$

$$CH_3 — \underset{\displaystyle CH_3}{\underset{|}{CH}} — NC \xrightarrow{\text{Reduction}} CH_3 — \underset{\displaystyle CH_3}{\underset{|}{CH}} — NH — CH_3$$

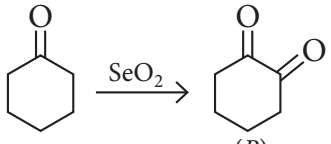
(*C*) Isopropylmethylamine

24. (a) : $[Ni(NH_3)_6]^{2+}$: sp^3d^2 (outer), paramagnetic
$[Zn(NH_3)_6]^{2+}$: sp^3d^2 (outer), diamagnetic
$[Cr(NH_3)_6]^{3+}$: d^2sp^3 (inner), paramagnetic
$[Co(NH_3)_6]^{3+}$: d^2sp^3 (inner), diamagnetic

25. (b) : Nearer the –I effect group to –OH bearing carbon, compound is more acidic.

26. (c) : More acidic alcohol will react faster with Na metal. Stronger the acid, weaker is the conjugate base.

27. (a) $CH_3COOH \xrightarrow[\Delta]{NH_3} \underset{(A)}{CH_3CONH_2}$ Ethanamide

(b) 

(*B*) Cyclohexane-1, 2-dione

OR

(i) $\underset{\text{Ethanal}}{2CH_3CHO} \xrightarrow[\text{Aldol condensation}]{\text{dil. NaOH}}$

$$\xrightarrow[\text{(Reduction)}]{NaBH_4} \overset{4}{CH_3}\overset{3}{CH(OH)}-\overset{2}{CH_2}-\overset{1}{CHO}$$

3-Hydroxybutanal

$$\xrightarrow{} \overset{4}{CH_3}-\overset{3}{CH(OH)}-\overset{2}{CH_2}\overset{1}{CH_2}OH$$

Butane-1, 3-diol

(ii) $\underset{\text{Ethanal}}{2CH_3CHO} \xrightarrow[\text{(ii) } H^+, \Delta]{\text{(i) dil. NaOH}} \underset{\text{But-2-enal}}{CH_3CH=CHCHO}$

28. Given, structure = *fcc*, $d = 7.2\ g\ cm^{-3}$, $M = 208$ g, $N = 4.283 \times 10^{24}$ atoms, a = ?

$$\text{Number of unit cells} = \frac{4.283\times10^{24}}{4} = 1.07\times10^{24}$$

Volume of unit cell = a^3

$\therefore$ Total volume = Number of unit cells × Volume of one unit cell

$= 1.07 \times 10^{24} \times a^3$

Total mass = Total volume × density

$208\ g = 1.07 \times 10^{24} \times a^3 \times 7.2\ g\ cm^{-3}$

or $$a^3 = \frac{208\,g}{1.07\times10^{24}\times7.2\,g\,cm^{-3}}$$

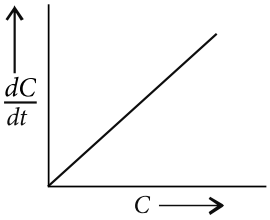
$$= 26.99\times10^{-24}\ cm^3$$

or $a = 3 \times 10^{-8}$ cm = 300 pm

29. (a) : For a first order reaction, rate of reaction is directly proportional to the concentration of the reactant.

Rate = $k \times C$

Hence, graph between rate of reaction and change in concentration will be straight line passing through the origin.



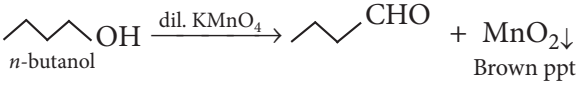
(b) Given rate = $k[A]^2[B]$

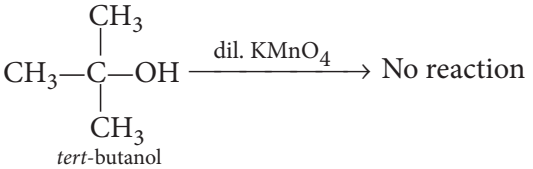
Unit of rate = mol L^{-1} s^{-1}

Unit of rate constant $$k = \frac{\text{Rate}}{[\text{Conc.}]^3}$$

or, $$k = \frac{\text{mol}\,L^{-1}\,s^{-1}}{(\text{mol}\,L^{-1})^3} = \text{mol}^{-2}\,L^2\,s^{-1}$$

30.





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31. (a) : Racemic mixture contains equal amount of *d* and *l* forms, hence rotation due to one enantiomer is cancelled by another.

(b) The presence of nitro group at *o*- and *p*-positions withdraws electrons from the benzene ring and thus, facilitates the attack of the nucleophile on haloarenes. The carbanion thus formed is further stabilised by resonance.

(c) Alkyl halides are polar but are insoluble in water because energy required to break the intermolecular H–bond among water molecules is much higher than energy released by water halide interaction.

32. (i) In aqueous solutions, Cu^+ undergoes disproportionation to form a more stable Cu^{2+} ion.

$2Cu^+_{(aq)} \rightarrow Cu^{2+}_{(aq)} + Cu_{(s)}$

Cu^{2+} in aqueous solutions is more stable than Cu^+ ion because hydration enthalpy of Cu^{2+} is higher than that of Cu^+. It compensates the second ionisation enthalpy of Cu involved in the formation of Cu^{2+} ions.

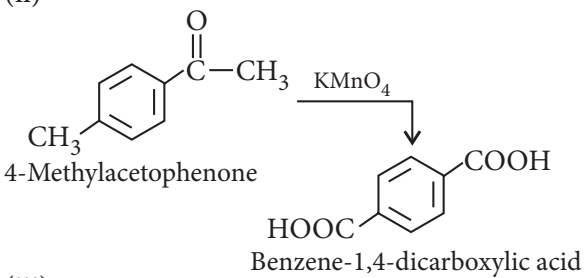
(ii) This is due to the presence of maximum number of unpaired electrons in Mn^{2+} ($3d^5$).

(iii) Only those ions are coloured which have partially filled *d*-orbitals facilitating *d-d* transitions. However, in case of Sc as the *d*-orbital is empty after the formation of compound, there is no *d-d* transition. Thus, the compounds appear white.

33. (i) $CH_3CH_2CH_2CH_2OH \xrightarrow[\text{Jones reagent}]{CrO_3, H_2SO_4} CH_3CH_2CH_2COOH$

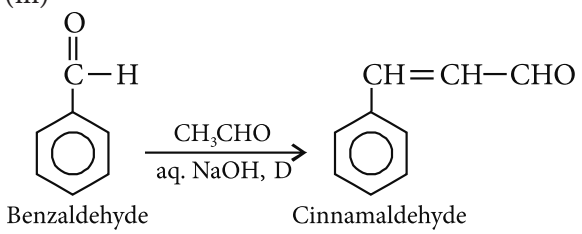
Butan-1-ol — Butanoic acid

(ii)

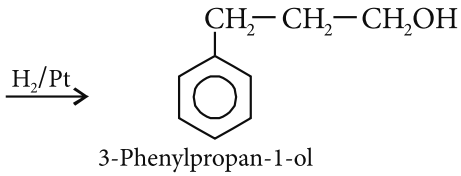


4-Methylacetophenone — Benzene-1,4-dicarboxylic acid

(iii)



Benzaldehyde — Cinnamaldehyde

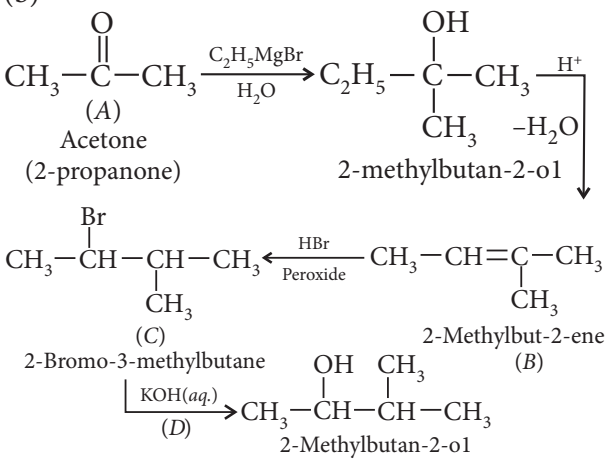


3-Phenylpropan-1-ol

OR

(a) The most suitable reagent for this oxidation is NaOI (I_2/NaOH).

(b)



34. (a) : (i) $CHCl_{3(l)}$ and $CH_2Cl_{2(l)}$

For a binary liquid the total pressure is

$p_1 = x_1 p_1^0$ and $p_2 = x_2 p_2^0$

$P_{total} = p_1 + p_2 = x_1 p_1^0 + x_2 p_2^0$

$= (1 - x_2)\, p_1^0 + x_2 p_2^{\,0}$

$= p_1^0 + (p_2^{\,0} - p_1^0) x_2$

(ii) $NaCl_{(s)}$ and $H_2O_{(l)}$

For non-volatile solute

$P_{total} = p_1 = x_1\, p_1^0$

(b) Solution of hydrogen in palladium and dissolved gases in minerals.

35. (a) : Ideal solutions : The solutions which obey Raoult's law over the entire range of concentration are known as ideal solution. For an ideal solution,

$\Delta_{mix}H = 0$, $\Delta_{mix}V = 0$

$A - B$ interactions $\approx$ $A - A$ interactions and $B - B$ interactions.

Non-ideal solutions : When a solution does not obey Raoult's law over the entire range of concentration, it is called non-ideal solution.

Positive deviations – Vapour pressure of such solutions shows higher value than the predicted value.

$\Delta_{mix}H$ = +ve, $\Delta_{mix}V$ = +ve

$A - B$ interactions < $A - A$ and $B - B$ interactions.

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Negative deviation – Vapour pressure of such solutions shows lower value than expected.

$\Delta_{mix}H$ = –ve, $\Delta_{mix}V$ = –ve

$A - B$ interactions > $A - A$ and $B - B$ interactions.

(b) (i) Molarity : It decreases with the increase in temperature.

(ii) Molality : It does not change with temperature.

OR

(a) Since, 1 atm = 1.01325×10^5 pascal

Hence, 2.5 atm = $2.5 \times 1.01325 \times 10^5$ pascal

$= 2.533125 \times 10^5$ pascal

By Henry's law $p_{CO_2} = K_H x_{CO_2}$

$$x_{CO_2} = \frac{p_{CO_2}}{K_H} = \frac{2.533125 \times 10^5 \text{ Pa}}{1.67 \times 10^8 \text{ Pa}}$$

$= 1.52 \times 10^{-3}$

Also, 500 mL soda water (water containing CO_2) = 500 g (assuming density of water = 1 g mL^{-1})

$$n_{H_2O} = \frac{500}{18} = 27.77 \text{ mol}$$

Let $n_{CO_2} = n$ mol

$\therefore$ Total moles = $27.77 + n$

$\therefore$ $x_{CO_2} = \frac{n}{27.77 + n} = 1.52 \times 10^{-3}$

$\therefore$ $\frac{n}{27.77} = 1.52 \times 10^{-3}$ ($\because$ n <<< 27.77)

$\therefore$ $n_{CO_2} = 1.52 \times 10^{-3} \times 27.77$ mol

$= 0.0422$ mol $= 0.0422$ mol $\times$ 44 g mol^{-1}

$= 1.8568$ g

(b) NaCl is a non-volatile solute hence it lowers the vapour pressure of water and boiling point of water increases. Methyl alcohol is a volatile solute hence it increases the vapour pressure and results in decrease in boiling point of water.

36. (a) : (i)

$$\underset{\text{Slaked lime}}{2Ca(OH)_2} + 2Cl_2 \longrightarrow \underset{\substack{\text{Bleaching}\\ \text{powder}}}{Ca(OCl)_2} + 2H_2O + CaCl_2$$

(ii)

$$\underset{\text{Milk of lime}}{6Ca(OH)_{2(aq)}} + 6Cl_2 \longrightarrow 5CaCl_2 + \underset{\substack{\text{Calcium}\\ \text{chlorate}}}{Ca(ClO_3)_2} + 6H_2O$$

(b) (i) All the filled orbitals of Xe have paired electrons. The promotion of one, two or three electrons from the 5*p*-filled orbitals to the 5*d*-vacant orbitals will give rise to two, four and six half-filled orbitals, respectively. Therefore, Xe can combine with even number of F-atoms but not odd. Thus, it cannot form XeF_3 and XeF_5.

(ii) Except radon which is radioactive, Xe has the least ionization energy among noble gases and hence, it readily forms chemical compounds with strong oxidising agent like O_2 and F_2.

(iii) On heating, ozone readily decomposes to give dioxygen and nascent oxygen.

$$O_3 \xrightarrow{\text{Heat}} O_2 + O(\text{nascent oxygen})$$

Since nascent oxygen is very reactive, therefore, O_3 acts as a powerful oxidising agent.

OR

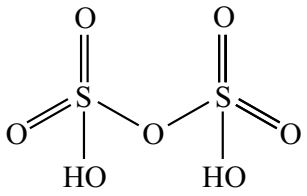
(a) (i) On moving down the group, the stability of + 5 oxidation state decreases while + 3 oxidation state increases due to inert pair effect. Thus, + 5 oxidation state of Bi is less stable than + 5 oxidation state of Sb.

Therefore, Bi(V) is stronger oxidising agent than Sb(V).

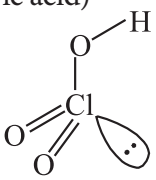
(ii) The single N–N bond is weaker than the single P–P bond because of high interelectronic repulsions of the non-bonding electrons, owing to the small bond length.

(iii) Noble gases being monoatomic have weak dispersion forces and therefore, they are liquefied at very low temperature. Hence, they have low boiling points.

(b) (i) $H_2S_2O_7$



(ii) $HClO_3$ (Chloric acid)



37. (i) (a)

$$CH_3Br \xrightarrow{KCN} \underset{(A)}{CH_3CN} \xrightarrow{LiAlH_4} \underset{(B)}{CH_3CH_2NH_2}$$

$$\xrightarrow[273\text{ K}]{HNO_2} \underset{(C)}{CH_3CH_2OH}$$

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(b)

$$CH_3COOH \xrightarrow[\Delta]{NH_3} \underset{(A)}{CH_3CONH_2} \xrightarrow{Br_2 + KOH} CH_3NH_2 \xrightarrow{CHCl_3 + NaOH} \underset{(C)}{CH_3NC}$$

(ii) (a) Amides on reduction with $LiAlH_4$ or by reaction with Br_2 and alkali give amines.

$$R-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[(ii)\ H_2O]{(i)\ Li\ AlH_4} R-CH_2-NH_2$$

$$R-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{Br_2 + \bar{O}H} R-NH_2$$

(b) During acylation reaction HCl acid is formed. In order to remove HCl and shift the equilibrium to the right hand side a stronger base like pyridine is added.

$$R-NH_2 + RCOCl \xrightarrow{Pyridine} R-NHCOR + HCl$$

OR

(a) (i) $CH_3CH_2CN + 2H_2O \xrightarrow{dil.\ HCl} \underset{\text{Propanoic acid}}{CH_3CH_2COOH} + NH_3$

(ii) $C_2H_5NC + 2H_2O \xrightarrow{dil.\ HCl} \underset{\text{Ethanamine}}{C_2H_5NH_2} + HCOOH$

(iii) $C_2H_5NO_2 + H_2O \xrightarrow{dil.\ HCl} CH_3COOH + NH_2OH$

(b) The significant difference is in the *s* character of hybrid orbital used by the β carbon atom. More the *s* character, more electron withdrawing effect is seen resulting in weakening of base.

$H_2C{=}CHCH_2NH_2$, β carbon uses sp^2 hybrid orbital

$CH_3CH_2CH_2NH_2$, β carbon uses sp^3 hybrid orbtial

$HC{\equiv}C\ CH_2NH_2$, β carbon uses *sp* hybrid orbital

The decreasing order of basicity is :

$CH_3CH_2CH_2NH_2 > H_2C{=}CHCH_2NH_2 > HC{\equiv}CCH_2NH_2$

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PRACTICE PAPER 4
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises two parts A and B.

(ii) Part A : Q. no. 1 to 26 are objective type questions.
Q. no. 1 to 19 are objective type questions carrying one mark each.
Q. no. 20 to 24 are objective type questions carrying two marks each.
Q. no. 25 to 26 are case based objective type questions carrying three marks each.

(iii) Part B : Q. no. 27 to 37 are subjective/descriptive type questions.
Q. no. 27 to 30 are short answer type-I questions carrying two marks each.
Q. no. 31 to 34 are short answer type-II questions carrying three marks each.
Q. no. 35 to 37 are long answer type questions carrying five marks each.

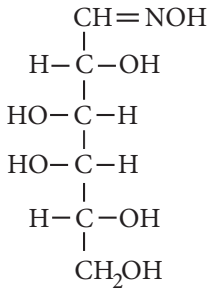
(iv) There is no overall choice in the question paper. However, internal choices have been provided in both part A and Part B.

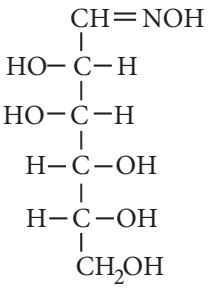
(v) Use of calculators and log tables is not permitted.

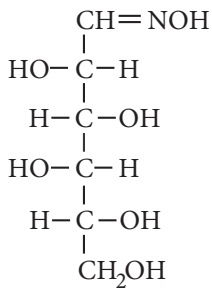
Time allowed : 3 hours **Maximum marks :** 70

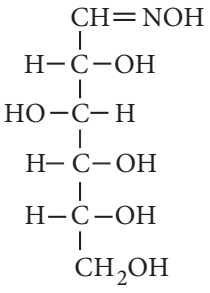
PART - A : Objective Type

1. *D*(+)-glucose reacts with hydroxyl amine and yields an oxime. The structure of the oxime would be

(a) 

(b) 

(c) 

(d) 

2. In this reaction :

$CH_3CHO + HCN \rightarrow CH_3CH(OH)CN \xrightarrow{HOH} CH_3CH(OH)COOH$

an asymmetric centre is generated. The acid obtained would be

(a) *D*-isomer

(b) *L*-isomer

(c) 50% *D* + 50% *L*-isomer

(d) 20% *D* + 80% *L*-isomer.

3. The colour of the coordination compounds depends on the crystal field splitting. What will be the correct order of absorption of wavelength of light in the visible region, for the complexes, $[Co(NH_3)_6]^{3+}$, $[Co(CN)_6]^{3-}$, $[Co(H_2O)_6]^{3+}$?

(a) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

(b) $[Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+} > [Co(CN)_6]^{3-}$

(c) $[Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(CN)_6]^{3-}$

(d) $[Co(NH_3)_6]^{3+} > [Co(CN)_6]^{3-} > [Co(H_2O)_6]^{3+}$

4. For the reaction, $A + B \longrightarrow$ products, it is observed that

(i) on doubling the initial concentration of *A* only, the rate of reaction is also doubled and

(ii) on doubling the initial concentration of both A and *B*, there is a change by a factor of 8 in the rate of the reaction.

The rate of this reaction is given by

(a) rate = $k[A][B]^2$ (b) rate = $k[A]^2[B]^2$

(c) rate = $k[A][B]$ (d) rate = $k[A]^2[B]$

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OR

If the rate of the reaction is equal to the rate constant, the order of the reaction is

(a) 0 (b) 1
(c) 2 (d) 3

5. How many isomers of $C_5H_{11}OH$ will be primary alcohols?

(a) 5 (b) 4
(c) 2 (d) 3

6. Electronic configuration of a transition element *X* in +3 oxidation state is [Ar]$3d^5$. What is its atomic number?

(a) 25 (b) 26
(c) 27 (d) 24

OR

Metallic radii of some transition elements are given below. Which of these elements will have highest density?

Element	Fe	Co	Ni	Cu
Metallic radii/pm	126	125	125	128

(a) Fe (b) Ni (c) Co (d) Cu

7. Which one of the following on treatment with 50% aqueous sodium hydroxide yields the corresponding alcohol and acid?

(a) C_6H_5CHO
(b) $CH_3CH_2CH_2CHO$
(c) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$
(d) $C_6H_5CH_2CHO$

8. For two ionic solids CaO and KI, identify the wrong statement among the following.

(a) CaO has high melting point.
(b) Lattice energy of CaO is much larger than that of KI.
(c) KI has high melting point.
(d) KI is soluble in benzene.

9. On addition of conc. H_2SO_4 to a chloride salt, colourless fumes are evolved but in case of iodide salt, violet fumes come out. This is because

(a) H_2SO_4 reduces HI to I_2
(b) HI is of violet colour
(c) HI gets oxidised to I_2
(d) HI changes to HIO_3.

10. Which of the following is not chiral?

(a) 2-Hydroxypropanoic acid
(b) 2-Butanol
(c) 2,3-Dibromopentane
(d) 3-Bromopentane

OR

In a S_N2 substitution reaction of the type

$$R\text{—Br} + Cl^- \xrightarrow{DMF} R\text{—Cl} + Br^-$$

which one of the following has the highest relative rate?

(a) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2Br$
(b) CH_3CH_2Br
(c) $CH_3CH_2CH_2Br$
(d) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2Br$

11. *A* is a lighter phenol and *B* is an aromatic carboxylic acid. Separation of a mixture of *A* and *B* can be carried out easily by using a solution of

(a) sodium hydroxide
(b) sodium sulphate
(c) calcium chloride
(d) sodium bicarbonate.

12. When chlorine is passed over dry slaked lime at room temperature, the main reaction product is

(a) $Ca(ClO_2)_2$ (b) $CaCl_2$
(c) $CaOCl_2$ (d) $Ca(OCl)_2$

13. The action of nitrous acid on an aliphatic primary amine gives

(a) secondary amine (b) nitro alkane
(c) alcohol (d) alkyl nitrite.

OR

For carbylamine reaction, we need hot alcoholic KOH and

(a) any primary amine and chloroform
(b) chloroform and silver powder
(c) a primary amine and an alkyl halide
(d) a monoalkylamine and trichloromethane.

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14. When 0.1 mol $CoCl_3(NH_3)_5$ is treated with excess of $AgNO_3$, 0.2 mol of AgCl are obtained. The conductivity of solution will correspond to

(a) 1:3 electrolyte (b) 1:2 electrolyte
(c) 1:1 electrolyte (d) 3:1 electrolyte.

Questions 15 to 19 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

15. **Assertion :** Colloidal sols scatter light while true solutions do not.
Reasons : The particles in the colloidal sol are much smaller than that of the true solution.

OR

Assertion : Gases are easily adsorbed on the surface of metals especially transition metals.
Reason : Transition metals have free valencies.

16. **Assertion :** In order to convert *R*–Cl to pure *R*–NH_2, Gabriel phthalimide synthesis can be used.
Reason : With proper choice of alkyl halides, pthalimide synthesis can be used to prepare 1°, 2° or 3° amines.

17. **Assertion :** Transition elements exhibit variable valency.
Reason : In *p*-block elements, the valency either equals the group number or eight minus the group number.

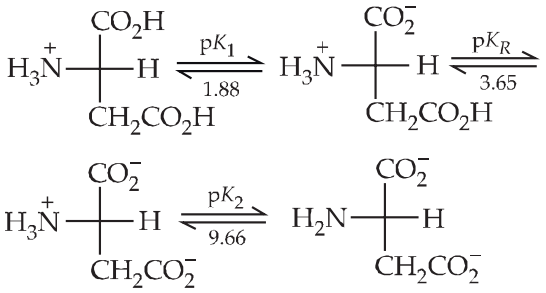
18. **Assertion :** Grignard reagents are prepared in ethers but not in benzene.
Reason : Grignard reagents are soluble in benzene.

OR

Assertion : The boiling points of alcohols are lower than those of hydrocarbons of comparable molecular mass.
Reason : Alcohols show intermolecular hydrogen bonding.

19. **Assertion :** The melting points and boiling points of noble gases are very low.
Reason : The interatomic forces are very weak.

20. Consider the following sequence for aspartic acid :

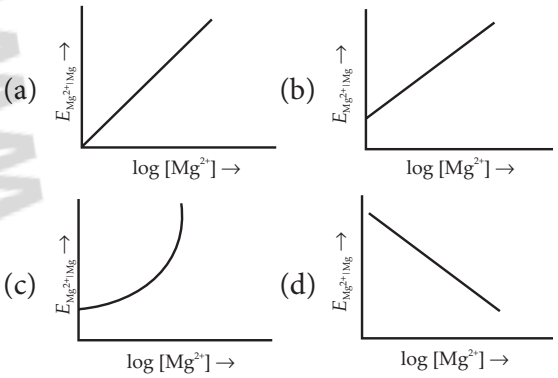


The pI (isoelectric point) of aspartic acid is
(a) 3.65 (b) 2.77 (c) 5.74 (d) 1.88

21. Electrode potential for Mg electrode varies according to the equation

$$E_{Mg^{2+}/Mg} = E^{\circ}_{Mg^{2+}/Mg} - \frac{0.059}{2}\log\frac{1}{[Mg^{2+}]}$$

The graph of $E_{Mg^{2+}|Mg}$ *vs* log $[Mg^{2+}]$ is



OR

Equivalent conductances of Ba^{2+} and Cl^- ions are 127 and 76 ohm^{-1} cm^2 eq^{-1} respectively. Equivalent conductance (in ohm^{-1} cm^2 eq^{-1}) of $BaCl_2$ at infinite dilution is
(a) 139.5 (b) 101.5
(c) 203 (d) 279

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22. Amongst TiF_6^{2-}, CoF_6^{3-}, Cu_2Cl_2 and $NiCl_4^{2-}$, which are the colourless species? (Atomic number of Ti = 22, Co = 27, Cu = 29, Ni = 28)

(a) CoF_6^{3-} and $NiCl_4^{2-}$

(b) TiF_6^{2-} and Cu_2Cl_2

(c) Cu_2Cl_2 and $NiCl_4^{2-}$

(d) TiF_6^{2-} and CoF_6^{3-}

23. The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$. What is the atomic number of the element, which is just below the above element in the periodic table? Identify the element.

(a) 36, Kr (b) 49, In

(c) 33, As (d) 34, Sb

24. In the reaction,

$$CH_3COOH \xrightarrow{LiAlH_4} A \xrightarrow{PCl_5} B \xrightarrow{\text{Alc. KOH}} C,$$

the product *C* is

(a) acetyl chloride (b) acetaldehyde

(c) acetylene (d) ethylene.

OR

The oxidation of toluene with CrO_3 in the presence of $(CH_3CO)_2O$ gives a product *A*, which on treatment with aqueous NaOH produces

(a) C_6H_5COONa

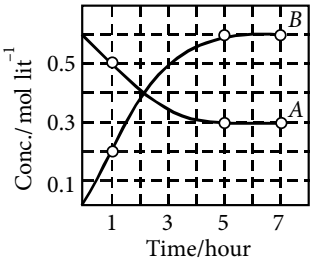
(b) 2, 4-diacetyl toluene

(c) C_6H_5CHO

(d) $(C_6H_5CO)_2O$

Read the given passage and answer the questions 25 and 26 that follow :

The progress of the reaction $A \rightleftharpoons nB$ with time is presented in the figure :



25. The value of *n* is

(a) 1 (b) 2 (c) 3 (d) 4.

26. The unit of rate constant and the initial rate of conversion of *A* are respectively

(a) mol litre^{-1} hr^{-1}, 0.1 mol litre^{-1} hr^{-1}

(b) litre mol^{-1} hr^{-1}, 0.25 mol litre^{-1} hr^{-1}

(c) litre mol^{-1} hr^{-1}, 0.1 mol litre^{-1} hr^{-1}

(d) hr^{-1}, 0.5 mol litre^{-1} hr^{-1}

PART - B : Subjective / Descriptive Type

27. (a) Name the bases that are common in both DNA and RNA.

(b) Despite having an aldehyde group glucose does not give 2,4-DNP test and addition of $NaHSO_3$. Why?

28. (i) When propanone is subjected to Wolff-Kishner reduction, what product will be obtained? Give chemical equation.

(ii) How will you convert acetaldehyde to acetone?

OR

(i) Carboxylic acids are higher boiling liquids as compared to corresponding aldehydes, ketones or alcohols. Explain.

(ii) What happens when primary alcohols are oxidised by potassium permanganate?

29. A first order reaction takes 160 minutes time for 20% completion. Calculate time required for half completion of reaction.

30. Calculate the mass of ascorbic acid (vitamin C, $C_6H_8O_6$) to be dissolved in 75 g acetic acid to lower its melting point by 1.5°C, $K_f = 3.9$ K kg mol^{-1}.

31. Which alkyl halide from the following pairs would you expect to react more rapidly by an S_N2 mechanism? Explain your answer.

(i) $CH_3CH_2CH_2CH_2Br$ or $CH_3CH_2\underset{\displaystyle Br}{\underset{|}{C}}HCH_3$

(ii) $CH_3CH_2\underset{\displaystyle Br}{\underset{|}{C}}HCH_3$ or $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-Br$

(iii) $CH_3\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2CH_2Br$ or $CH_3CH_2\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2Br$

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32. (a) Amongst the following ions which one has the highest magnetic moment value?

(i) $[Cr(H_2O)_6]^{3+}$

(ii) $[Fe(H_2O)_6]^{2+}$

(iii) $[Zn(H_2O)_6]^{2+}$

(b) Amongst the following, the most stable complex is

(i) $[Fe(H_2O)_6]^{3+}$

(ii) $[Fe(NH_3)_6]^{3+}$

(iii) $[Fe(C_2O_4)_3]^{3-}$

(iv) $[FeCl_6]^{3-}$

33. (a) Give a chemical test to distinguish between $CH_3CH_2NH_2$ and $(CH_3)_2NH$.

(b) Complete the following equations :

(i) $CH_3CH_2NH_2 \xrightarrow{HCl}$

(ii) $C_6H_5NH_2 \xrightarrow{Br_2}$

(iii) $C_6H_5NH_2 \xrightarrow[\text{Pyridine}]{(CH_3CO)_2O}$

34. Explain the following terms :

(i) Electrophoresis (ii) Electro-osmosis

(iii) Coagulation.

OR

(a) What is absorption? How is absorption different from adsorption? What is sorption?

(b) On what factors adsorption of a solute from a solution varies?

35. (a) Calculate the equivalent conductivity of acetic acid at infinite dilution, if equivalent conductivity of CH_3COONa, HCl and NaCl at infinite dilution are 91.6, 425.0 and 128.1 S cm^2 eq^{-1} respectively.

(b) On what factors conductivity of electrolytic solution depends?

(c) The resistance of 100 cm^3 aqueous solution of 0.025 M $CuSO_4$ is 520 ohm at 298 K. Calculate the molar conductivity if the cell constant of the conductivity cell is 153.7 m^{-1}.

OR

(a) How can the reduction potential of an electrode be increased?

(b) A Cu rod is dipped in 0.1 M $CuSO_4$ solution. Calculate electrode potential of this half cell.

[$E°_{Cu/Cu^{2+}} = -0.34$ V, Temperature = 25°C]

(c) What are the products of electrolysis of sulphuric acid in dilute solution?

36. (a) Explain Williamson's synthesis.

(b) Give two advantages of Williamson's method.

(c) Why is Williamson's synthesis not applicable to tertiary halides?

OR

(a) Write the reaction and state the conditions for each of the following conversions :

(i) Ethene to ethanol.

(ii) Chlorobenzene to phenol.

(iii) Ethanal to 2-propanol.

(b) Arrange the following compound groups in the increasing order of their property indicated :

(i) *p*-nitrophenol, ethanol, phenol (acidic character)

(ii) propanol, propane, propanal (boiling point)

37. (i) Explain the Ostwald's process for the manufacture of nitric acid.

(ii) Write the balanced reactions when nitric acid reacts with

(a) I_2 (b) C

(c) S_8 (d) P_4

(iii) Give any two uses of HNO_3.

OR

On heating compound (*A*) gives a gas (*B*) which is a constituent of air. This gas when treated with 3 mol of hydrogen (H_2) in the presence of a catalyst gives another gas (*C*) which is basic in nature. Gas *C* on further oxidation in moist condition gives a compound (*D*) which is a part of acid rain. Identify compounds (*A*) to (*D*) and also give necessary equations of all the steps involved.

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Detailed Solutions

1. (d)

2. (c) : Lactic acid ($CH_3CH(OH)COOH$) is an optically active compound due to the presence of asymmetric carbon atom. It exists in *D*-and *L*-form, the ratio of which is found to be (1 : 1), *i.e.*, a racemic mixture is obtained.

3. (c) : As strength of ligand increases crystal field splitting energy increases.

Now, $\Delta E = \frac{hc}{\lambda}$

So, as CFSE increases, ΔE increases and λ decreases.

$[Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(CN)_6]^{3-}$

4. (a) : $R = k[A]^m[B]^n$ ...(i)

$2R = k[2A]^m[B]^n$ ...(ii)

$8R = k[2A]^m[2B]^n$ ...(iii)

from (i), (ii) and (iii), $m = 1$, $n = 2$

So, rate = $k[A][B]^2$

OR

(a) : $A \rightarrow$ products

If $-\frac{dx}{dt} = k$, it means $-\frac{dx}{dt} = k[A]^0 = k$

Hence, order of reaction must be zero.

5. (b)

6. (b) : $X^{3+} = [Ar]3d^5$, $X = [Ar]3d^64s^2$; hence atomic number of element is 26.

OR

(d) : Cu will have highest density.

7. (a) : C_6H_5CHO containing no α-H atom undergoes Cannizzaro reaction to produce benzyl alcohol and sodium benzoate.

8. (d) : KI is an ionic compound while benzene is not.

9. (c) : HI formed during reaction is oxidised to I_2 which is violet in colour.

$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

$2NaI + H_2SO_4 \rightarrow Na_2SO_4 + 2HI$

$2HI \xrightarrow{H_2SO_4} 2H_2O + SO_2 + I_2$

10. (d) : $H_3C-CH_2-\underset{\underset{Br}{|}}{CH}-CH_2-CH_3$

Due to absence of asymmetric carbon atom.

OR

(b) : S_N2 mechanism is followed in case of primary and secondary alkyl halides *i.e.* S_N2 reaction is favoured by small groups on the carbon atoms attached to halogen so,

$CH_3 - X > R - CH_2 - X > R_2CH - X > R_3C - X$.

11. (d) : Carboxylic acids dissolve in $NaHCO_3$ but phenols do not.

12. (c) : $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$

13. (c) :

$$\underset{\text{Primary amine}}{R-NH_2} + \underset{\text{Nitrous acid}}{HNO_2} \longrightarrow \underset{\text{Alcohol}}{ROH} + N_2 + H_2O$$

OR

(a) : $R-NH_2 + CHCl_3 + 3KOH \xrightarrow[\text{(alc.)}]{} RNC + 3KCl + 3H_2O$

14. (b) : 0.1 mol $CoCl_3(NH_3)_5$ when treated with excess of $AgNO_3$ gives 0.2 mol of AgCl. So, the complex is $[Co(NH_3)_5Cl]Cl_2$ which ionises as :

$[Co(NH_3)_5Cl]Cl_2 \longrightarrow [Co(NH_3)_5Cl]^{2+} + 2Cl^-$

Thus the conductivity of the solution will correspond to 1 : 2 electrolyte.

15. (c) : The size of the colloidal particles is large enough to scatter light while particles of a true solution are too small to scatter light.

OR

(a) : As the number of free valencies increases, the surface area for adsorption of gases on solid surface also increases.

16. (c) : Only primary aliphatic amines can be prepared by Gabriel phthalimide reaction.

17. (b)

18. (c) : Grignard reagents are soluble in ether.

OR

(d) : The presence of intermolecular weak hydrogen bonds among alcohols need extra energy to break. That is why they have more b.pt.

19. (a)

20. (b) : $pI = \frac{pK_1 + pK_R}{2} = \frac{1.88 + 3.65}{2} = \frac{5.53}{2}$

$= 2.765 \approx 2.77$

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21. (b) : $E_{Mg^{2+}/Mg} = E^{\circ}_{Mg^{2+}/Mg} + \frac{0.059}{2}\log[Mg^{2+}]$

Compare this equation with the equation of straight line $y = mx + c$.

The graph of $E_{Mg^{2+}/Mg}$ *vs* $\log[Mg^{2+}]$ is a straight line with a positive slope and intercept = $E^{\circ}_{Mg^{2+}/Mg}$.

OR

(a) : $\lambda_{\infty} = \frac{1}{n_+}\lambda^{\infty}_{+} + \frac{1}{n_-}\lambda^{\infty}_{-}$

So, $\lambda_{\infty}(BaCl_2) = \frac{1}{2} \times \lambda^{\infty}_{Ba^{2+}} + \frac{1}{1} \times \lambda^{\infty}_{Cl^-}$

$= \frac{1}{2} \times 127 + 76 = 63 \cdot 5 + 76 = 139 \cdot 5$ ohm^{-1} cm^2 eq^{-1}

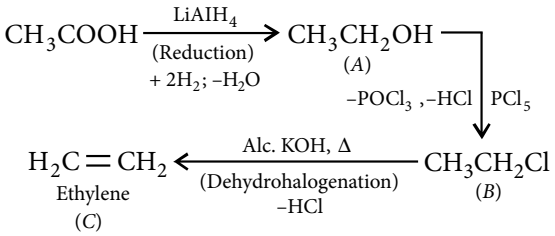
22. (b) : In TiF_6^{2-}, titanium is in +4 state. In Cu_2Cl_2, the copper is in +1 state. Thus, in both cases, transition from one *d*-orbital to other is not possible.

Ti : [Ar] $3d^2 4s^2 \rightarrow Ti^{4+}$: [Ar]$3d^0 4s^0$

Cu : [Ar] $3d^{10} 4s^1 \rightarrow Cu^{+}$: [Ar]$3d^{10} 4s^0$

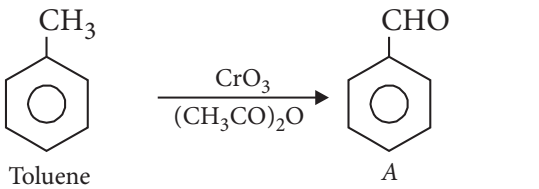
23. (c) : Atomic number of the given element is 15 and it belongs to 5th group. Therefore atomic number of the element below the above element = 15 + 18 = 33. The element is As.

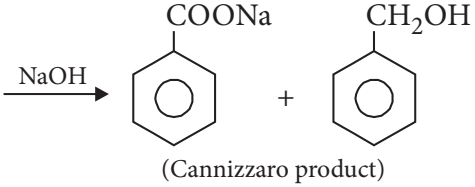
24. (d) :



OR

(a) :





25. (b) : According to figure in the given time of 4 hours (1 to 5) concentration of A falls from 0.5 to 0.3 M, while in the same time concentration of B increases from 0.2 to 0.6 M.

Loss in concentration of A in 4 hours = 0.5 – 0.3 = 0.2 M

Gain in concentration of B in 4 hours = 0.6 – 0.2 = 0.4 M

Thus increase in concentration of B in a given time is twice the decrease in concentration of A. Thus $n = 2$.

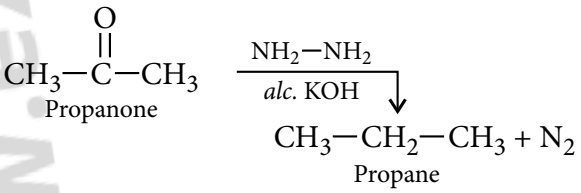
26. (c) : Unit of rate constant for a second order reaction is litre mol^{-1} hr^{-1}

Initial rate of conversion of *A* = Change in conc. of *A* during 1 hour $= \frac{0.6 - 0.5}{1}$ = 0.1 mol litre^{-1} hr^{-1}

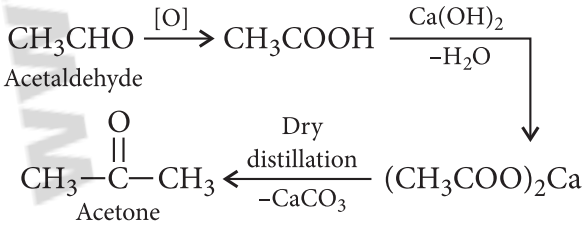
27. (a) Adenine, guanine and cytosine are the bases common in DNA and RNA both.

(b) Glucose does not have open structure and hence it does not have a free –CHO group. Actually –CHO group combines with C_5–OH to form an hemiacetal.

28. (i)



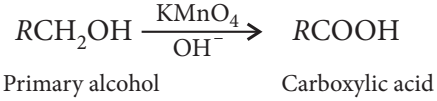
(ii)



OR

(i) Carboxylic acids are higher boiling liquids than aldehydes, ketones and even alcohols of comparable molecular masses due to more extensive association of carboxylic acid molecules by intermolecular hydrogen bonding.

(ii) Primary alcohols on oxidation with potassium permanganate give carboxylic acids.



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29. Given : $[R]_0 = 1$, $[R] = 0.80$, $t = 160$ min
$t_{1/2} = ?$

For first order reaction $k = \frac{2.303}{t} \log_{10} \frac{[R]_0}{[R]}$

or, $k = \frac{2.303}{160 \text{ min}} \log_{10} \frac{1}{0.8}$

or, $k = \frac{2.303}{160 \text{ min}} \times \log 1.25$

or, $k = \frac{2.303 \times 0.0969}{160 \text{ min}} = 1.39 \times 10^{-3} \text{ min}^{-1}$

Again

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{1.39 \times 10^{-3}} \text{ min} = 498 \text{ min}$$

30. Lowering in melting point (ΔT_f) = 1.5°
Mass of solvent (CH_3COOH), $w_1 = 75$ g
Mass of solute, $w_2 = ?$
Molar mass of solvent (CH_3COOH),
$M_1 = 60$ g mol^{-1}
Molar mass of solute, $C_6H_8O_6 = 72 + 8 + 96$
$= 176$ g mol^{-1}

Applying the formula, $\Delta T_f = K_f \times \frac{w_2 \times 1000}{M_2 \times w_1}$

or $w_2 = \frac{M_2 \times w_1 \times \Delta T_f}{1000 \times K_f} = \frac{176 \times 75 \times 1.5}{1000 \times 3.9}$

$= 5.08$ g

31. (i) Reactivity for S_N2 reaction is primary > secondary.

$$CH_3CH_2CH_2CH_2Br \; > \; CH_3CH_2\underset{\displaystyle Br}{\underset{|}{C}}HCH_3$$

(ii) $CH_3CH_2\underset{\displaystyle Br}{\underset{|}{C}}H\,CH_3 > CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-Br$

Reactivity of S_N2 reaction is secondary > tertiary.
(iii)

$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_2-Br > CH_3CH_2\,\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-Br$$

Reactivity of S_N2 reaction decreases with steric hindrance near the leaving group.

32. (a) $Cr^{3+} - 3d^3$, $Fe^{2+} - 3d^6$, $Zn^{2+} - 3d^{10}$
Since Fe^{2+} has 4 unpaired electrons, it has highest magnetic moment.

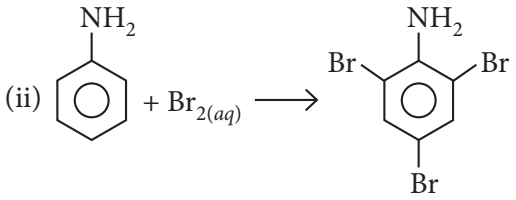
(b) Since $C_2O_4^{2-}$ is a bidentate ligand, it forms the most stable complex.

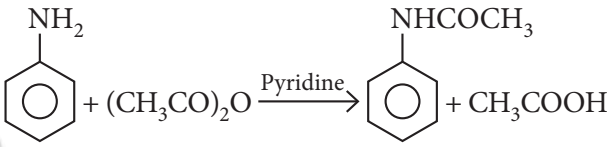
33. (a)

$$\underset{1° \text{ amine}}{CH_3CH_2NH} \xrightarrow{CHCl_3 + KOH} \underset{\text{Carbylamine (pungent smell)}}{CH_3CH_2NC}$$

$$\underset{2° \text{ amine}}{(CH_3)_2NH} \xrightarrow{CHCl_3 + KOH} \text{No reaction}$$

(b) (i) $CH_3CH_2NH_2 \xrightarrow{HCl} CH_3CH_2\overset{+}{N}H_3\overset{-}{Cl}$

(ii) 

(iii) 

34. (i) Electrophoresis : Migration of colloidal sol particles towards oppositely charged electrodes under electric field is called electrophoresis.
(ii) Electroosmosis : Movement of dispersion medium towards an electrode under electric field through semi-permeable membranes is called electro-osmosis.
(iii) Coagulation : The process of setting of colloidal solution particles is known as coagulation.

OR

(a) The process in which a liquid or gas is uniformly distributed into whole of another substance is called absorption. The accumulation of molecular species at the surface rather than in the bulk of a solid or liquid is termed as adsorption.
In adsorption, the adsorbate is concentrated only at the surface and does not penetrate through the surface to the bulk of the adsorbent. While in absorption the substance is uniformly distributed throughout the bulk of the solid.
In a number of cases, both adsorption and absorption take place simultaneously. For such processes the term sorption is used.

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(b) Following factors affect adsorption of solute from solution :
(i) The extent of adsorption decreases with an increase in temperature.
(ii) It increases with an increase of surface area of the adsorbent.
(iii) It also depends on the increase in the concentration of the solute in the solution.
(iv) It also depends on nature of adsorbent and adsorbate.

35. (a) : Given,

$$\Lambda^\circ_{CH_3COONa} = \lambda^\circ_{CH_3COO^-} + \lambda^\circ_{Na^+} = 91.6$$

$$\Lambda^\circ_{HCl} = \lambda^\circ_{H^+} + \lambda^\circ_{Cl^-} = 425.0$$

$$\Lambda^\circ_{NaCl} = \lambda^\circ_{Na^+} + \lambda^\circ_{Cl^-} = 128.1$$

$$\Lambda^\circ_{CH_3COOH} = \lambda^\circ_{CH_3COO^-} + \lambda^\circ_{H^+}$$

$$= \Lambda^\circ_{CH_3COONa} + \Lambda^\circ_{HCl} - \Lambda^\circ_{NaCl}$$

$= 91.6 + 425.0 - 128.1 = 388.5 \text{ S cm}^2 \text{ eq}^{-1}$

(b) Electrolytic conductance depends upon
(i) nature of the electrolyte added
(ii) size of ions produced and their solutions
(iii) the nature of solvent and its viscosity
(iv) concentration of the electrolyte.
(v) temperature (Directly proportional to the temperature).

(c) Given : $V = 100 \text{ cm}^3$, Molarity = 0.025 M, $R = 520$ ohm
$G° = 153.7 \text{ m}^{-1} = 1.537 \text{ cm}^{-1}$, Λ_m = ?

$$\kappa = G° \times \frac{1}{R} = 1.537 \text{ cm}^{-1} \times \frac{1}{520 \text{ ohm}}$$

$$= 2.95 \times 10^{-3} \text{ ohm}^{-1}\text{cm}^{-1}$$

$$\text{Again, } \Lambda_m = \frac{\kappa \times 10^3}{M}$$

$$= \frac{2.95 \times 10^{-3} \text{ ohm}^{-1} \text{ cm}^{-1} \times 10^3}{0.025 \text{ mol cm}^{-3}}$$

$\Lambda_m = 118.0 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$

OR

(a) $M^{n+} + ne^- \longrightarrow M$,

$$E_{M^{n+}/M} = E°_{M^{n+}/M} - \frac{RT}{nF} \ln \frac{1}{[M^{n+}]}$$

$$= E°_{M^{n+}/M} + \frac{RT}{nF} \ln [M^{n+}]$$

Thus, electrode potential can be increased either by increasing metal ion concentration, *i.e.*, $[M^{n+}]$ or temperature T.

(b) Given : $E°_{Cu/Cu^{2+}} = -0.34$ V

$\therefore \quad E°_{Cu^{2+}/Cu} = +0.34$ V

$[Cu^{2+}] = 0.1$ M, $E_{Cu^{2+}/Cu}$ = ?

Using Nernst equation,

$$E_{Cu^{2+}/Cu} = E°_{Cu^{2+}/Cu} - \frac{2.303\ RT}{2F} \log \frac{1}{[Cu^{2+}]}$$

$$\text{or} \quad E = 0.34 - \frac{0.0591}{2} \log \frac{1}{10^{-1}}$$

$= 0.34 - 0.0295 = 0.31$ V

(c) The products of electrolysis of sulphuric acid in dilute H_2SO_4 are hydrogen gas (liberated at cathode) and oxygen gas (liberated at anode).

36. (a) When alkyl halide is allowed to react with sodium alkoxide then ether is formed. This method is useful for preparation of both symmetrical and unsymmetrical ether. This reaction is called Williamson synthesis.

$$R—X + R'—\ddot{\underset{\cdot\cdot}{O}}\text{Na} \longrightarrow R—O—R' + \text{Na}X$$

(b) (i) Both symmetrical and unsymmetrical ethers can be prepared by this method.
(ii) 2° and 3° ethers can also be prepared by this method.
(c) In case of secondary or tertiary alkyl halides elimination competes over substitution and product formed is alkene.
For example,

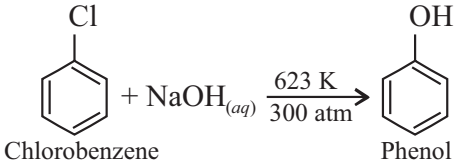
$$CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—Br + Na\ddot{\underset{\cdot\cdot}{O}}—CH_3 \longrightarrow CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}=CH_2 + NaBr + CH_3OH$$

OR

(a) (i)

$$\underset{\text{Ethene}}{CH_2=CH_2} + H_2O \xrightarrow{H_2SO_4} \underset{\text{Ethanol}}{CH_3—CH_2—OH}$$

(ii)



Chlorobenzene Phenol

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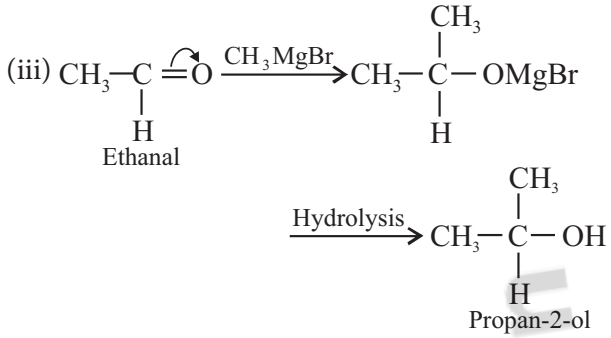
(iii) 

(b) (i) *p*-nitrophenol > phenol > ethanol

(ii) Propanol > propanal > propane

37. (i) This method is based on catalytic oxidation of NH_3 by atmospheric oxygen.

$$4NH_{3(g)} + 5O_{2(g)} \xrightarrow[500\,K,\ 9\,bar]{Pt} 4NO_{(g)} + 6H_2O_{(g)}$$

Nitric oxide thus formed combines with oxygen to give NO_2.

$$2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$$

Nitrogen dioxide so formed dissolves in water to give nitric acid HNO_3.

$$3NO_{2(g)} + H_2O_{(l)} \longrightarrow 2HNO_{3(aq)} + NO_{(g)}$$

(ii) (a) $I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$

(b) $C + 4HNO_3 \longrightarrow CO_2 + 2H_2O + 4NO_2$

(c) $S_8 + 48HNO_{3(conc.)} \longrightarrow 8H_2SO_4 + 48NO_2 + 16H_2O$

(d) $P_4 + 20HNO_{3(conc.)} \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$

(iii) (a) Nitric acid is used to manufacture ammonium nitrate for fertilisers.

(b) It is used for the preparation of trinitrotoluene and other organic nitro compounds.

OR

$A = NH_4NO_2$; $B = N_2$; $C = NH_3$; $D = HNO_3$

(i) $NH_4NO_2 \rightarrow N_2 + 2H_2O$
(A) (B) Constituent of air

(ii) $N_2 + 3H_2 \xrightarrow{Fe} 2NH_3$
(C) Basic in nature

(iii) $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$
$2NO + O_2 \rightarrow 2NO_2$
$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$
(D) Part of acid rain

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PRACTICE PAPER 5
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises two parts A and B.

(ii) Part A : Q. no. 1 to 26 are objective type questions.
Q. no. 1 to 19 are objective type questions carrying one mark each.
Q. no. 20 to 24 are objective type questions carrying two marks each.
Q. no. 25 to 26 are case based objective type questions carrying three marks each.

(iii) Part B : Q. no. 27 to 37 are subjective/descriptive type questions.
Q. no. 27 to 30 are short answer type-I questions carrying two marks each.
Q. no. 31 to 34 are short answer type-II questions carrying three marks each.
Q. no. 35 to 37 are long answer type questions carrying five marks each.

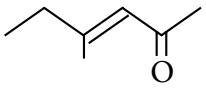
(iv) There is no overall choice in the question paper. However, internal choices have been provided in both part A and Part B.

(v) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

PART - A : Objective Type

1. The IUPAC name of the following compound is



(a) 3-methylhex-3-en-5-one
(b) 4-methylhex-3-en-2-one
(c) 4-methylhex-2-en-2-one
(d) 3-methylhex-2-en-5-one.

OR

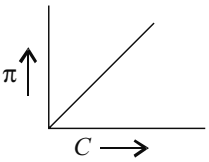
Aldol condensation product of acetone on dehydration gives
(a) but-2-enal
(b) 2-methylpent-3-en-4-one
(c) 4-hydroxy-4-methylpentan-2-one
(d) 4-methylpent-3-en-2-one.

2. If we add dilute aqueous solution of KI dropwise to $AgNO_3$ aqueous solution, the AgI colloid formed will have
(a) negative charge
(b) positive charge
(c) neutral
(d) nothing can be predicted.

3. What are the products obtained when ammonia is reacted with excess chlorine?
(a) N_2 and NCl_3 (b) N_2 and HCl
(c) N_2 and NH_4Cl (d) NCl_3 and HCl

4. RNA and DNA are chiral molecules, their chirality is due to
(a) chiral bases
(b) chiral phosphate ester units
(c) *D*-sugar component
(d) *L*-sugar component.

5. A graph showing variation of osmotic pressure (π) *vs* molar concentration '*C*' of an aqueous solution at temperature *T* is given.



The slope of the line represents
(a) solution constant *R*
(b) absolute temperature *T*
(c) *RT*
(d) degree of ionisation.

OR

For dilute solutions, Raoult's law states that
(a) lowering of vapour pressure is equal to the mole fraction of the solute
(b) relative lowering of vapour pressure is equal to the mole fraction of the solvent
(c) relative lowering of vapour pressure of the solvent is equal to the mole fraction of the solute
(d) vapour pressure of the solution is equal to the vapour pressure of the solvent.

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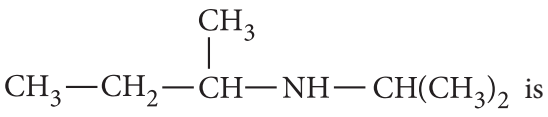
6. The IUPAC name of



is

(a) isobutyldimethanamine
(b) isopropylbutan-2-amine
(c) dimethyl butanamine
(d) *n*-propylbutan-2-amine.

7. In the reaction, $A + 3B \longrightarrow 2C$, the rate of formation of *C* is

(a) the same as rate of consumption of *A*
(b) the same as the rate of consumption of *B*
(c) twice the rate of consumption of *A*
(d) 3/2 times the rate of consumption of *B*.

8. Secondary structure of protein refers to

(a) mainly denatured proteins and structure of prosthetic groups
(b) three-dimensional structure, especially the bond between amino acid residues that are distant from each other in the polypeptide chain
(c) linear sequence of amino acid residues in the polypeptide chain
(d) regular folding patterns of continuous portions of the polypeptide chain.

9. The complex that does not give a precipitate with $AgNO_3$ solution is

(a) $[Co(NH_3)_3Cl_3]$ (b) $[Co(NH_3)_6]Cl_3$
(c) $[Ag(NH_3)_2]Cl$ (d) $[Cr(NH_3)_4Cl_2]Cl$

OR

In an octahedral structure, the pair of *d* orbitals involved in d^2sp^3 hybridisation is

(a) $d_{x^2-y^2}, d_{z^2}$ (b) $d_{xz}, d_{x^2-y^2}$
(c) d_{z^2}, d_{xz} (d) d_{xy}, d_{yz}.

10. Which of the following is the correct increasing order of basicity of amines in aqueous phase?

(a) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$
(b) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$
(c) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2 > NH_3$
(d) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH > NH_3$

11. In Nernst equation, the constant 0.05912 represents

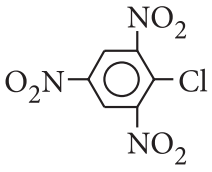
(a) $\frac{2.303}{nF}$ (b) $\frac{2.303\,RT}{F}$
(c) $\frac{RT}{nF}$
(d) $\frac{2.303RT}{nF}\log_{10}\frac{[\text{Reduced state}]}{[\text{Oxidised state}]}$

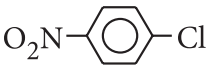
12. Chlorobenzene reacts with Mg in dry ether to give a compound (*A*) which further reacts with ethanol to yield

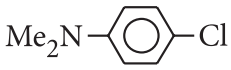
(a) phenol (b) benzene
(c) ethyl benzene (d) phenyl ether.

OR

Which chloro derivative of benzene among the following would undergo hydrolysis most readily with aqueous sodium hydroxide to furnish the corresponding hydroxy derivative?

(a) 

(b) 

(c) 

(d) C_6H_5Cl

13. The common oxidation state of the elements of lanthanoid series is

(a) +2 (b) +3 (c) +4 (d) +1

14. $CH_2{=}CH_2 + B_2H_6 \xrightarrow[H_2O_2]{NaOH}$ Product.

Product in the above reaction is

(a) CH_3CH_2CHO (b) CH_3CH_2OH
(c) CH_3CHO (d) CH_3OH

Questions 15 to 19 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

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15. **Assertion :** Ethoxy benzene undergoes rearrangement (like Fries migration) when heated with Lewis acid.
Reason : *Ortho*, *meta* and *para* ethyl phenol is obtained in rearrangement.

OR

Assertion : The water solubility of the alcohols follow the order : *t*-butyl alcohol> *s*-butyl alcohol> *n*-butyl alcohol.
Reason : Alcohols form H-bonding with water to show soluble nature.

16. **Assertion** : According to Freundlich, $\frac{x}{m} = k.\ P^{1/n}$
Reason : The isotherm shows variation of the amount of gas adsorbed by the adsorbent with temperature.

17. **Assertion :** Tungsten has very high melting point.
Reason : Tungsten is a covalent compound.

18. **Assertion :** Glucose and fructose are reducing sugars.
Reason : Glucose and fructose contain aldehydic and ketonic group adjacent to a group respectively.

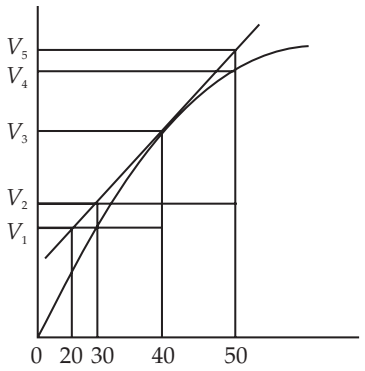
OR

Assertion : Uracil occurs in DNA.
Reason : RNA does not undergo replication.

19. **Assertion :** Ionic and covalent compounds of transition elements are usually coloured.
Reason : Colour is due to incomplete electron shells and the ability to promote an electron from one energy level to another.

20. On heating ammonium dichromate and barium azide separately we get
(a) N_2 in both cases
(b) N_2 with ammonium dichromate and NO with barium azide
(c) N_2O with ammonium dichromate and N_2 with barium azide
(d) N_2O with ammonium dichromate and NO_2 with barium azide.

21. A graph of volume of hydrogen released *vs* time for the reaction between zinc and dil. HCl is given in figure. On the basis of this, mark the correct option.



(a) Average rate upto 40 s is $\frac{V_3 - V_2}{40}$.
(b) Average rate upto 40 seconds is $\frac{V_3 - V_2}{40 - 30}$.
(c) Average rate upto 40 seconds is $\frac{V_3}{40}$.
(d) Average rate upto 40 seconds is $\frac{V_3 - V_1}{40 - 20}$.

OR

Compounds '*A*' and '*B*' react according to the following chemical equation.

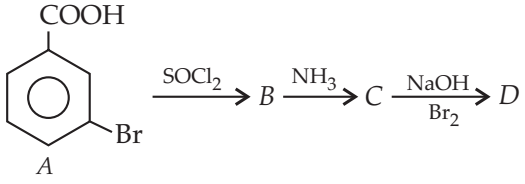
$$A_{(g)} + 2B_{(g)} \longrightarrow 2C_{(g)}$$

Concentration of either '*A*' or '*B*' were changed keeping the concentrations of one of the reactants constant and rates were measured as a function of initial concentration. Following results were obtained. Choose the correct option for the rate equations for this reaction.

Experiment	1	2	3
Initial concentration of $[A]$/mol L^{-1}	0.30	0.30	0.60
Initial concentration of $[B]$/mol L^{-1}	0.30	0.60	0.30
Initial rate of formation of $[C]$/mol L^{-1} s^{-1}	0.10	0.40	0.20

(a) Rate = $k\ [A]^2\ [B]$ (b) Rate = $k\ [A]\ [B]^2$
(c) Rate = $k\ [A]\ [B]$ (d) Rate = $k\ [A]^2\ [B]^0$

22. In a set of reactions *m*-bromobenzoic acid gave a product *D*. Identify the product *D*.



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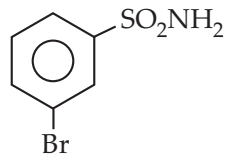
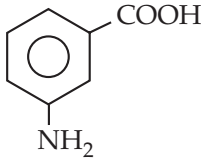
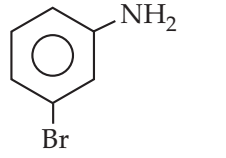
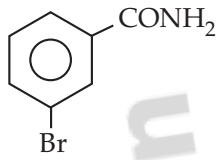
(a) 

(b) 

(c) 

(d) 

23. The name of the compound '*B*' in the following series of reactions, is

Acetone $\xrightarrow{H_2N—OH}$ '*A*' $\xrightarrow{H_2/Pt}$ '*B*'

(a) propan-1-amine (b) propan-2-amine
(c) butan-1-ol (d) butan-2-ol.

OR

Identify '*X*'.

$C_6H_5COCl + (CH_3)_2NH \xrightarrow{Pyridine}$ '*X*'

(a) *N*, *N*-Dimethylbenzamide
(b) *N*, *N*-Dimethylbenzene
(c) *N*-Methyl-N-phenylamine
(d) *N*, *N*-Diphenylmethanamine

24. Which of the following is amphoteric oxides?
Mn_2O_7, CrO_3, Cr_2O_3, CrO, V_2O_5, V_2O_4

(a) V_2O_5, Cr_2O_3 (b) Mn_2O_7, CrO_3
(c) CrO, V_2O_5 (d) V_2O_5, V_2O_4

Read the given passage and answer the questions 25 and 26 that follow :

Standard reduction potentials (SRP) for different systems can be used to decide the spontaneity of a reaction. For example,

$E°2n^{2+}/2n$ = – 0.76 V, hence for the reaction $Zn + 2H^+ \rightarrow Zn^{2+} + H_2$, $\Delta G°$ is negative. It has been found experimentally that if (SRP of an oxidant – SRP of a reductant) is more than 1.7 V, then their combination may lead to explosion (though it may be prevented by kinetic factors).

Now go through the following data and answer the questions.

Data :

$E°_{Ag^+/Ag}$ = 0.80 V ; $E°_{N_2/N^{3-}}$ = –3.09 V
$E°_{ClO_4^-/ClO_3^-}$ = 1.23 V ; $E°_{Na^+/Na}$ = –2.17 V
$E°_{Fe^{3+}/Fe^{2+}}$ = 0.77 V ; $E°_{O_2/H_2O_2}$ = –1.03 V
$E°_{H_2O_2/H_2O}$ = 1.76 V ; $E°_{O_3/O_2}$ = 2.07 V
$E°_{MnO^{4-}/Mn^{2+}}$ = 1.54 V ; $E°_{Cr_2O_7^{2-}/Cr^{3+}}$ = 1.33 V

25. Which of the following ionic combinations may lead to the formation of explosive substance?

(a) Sodium ion and azide ion
(b) Silver ion and perchlorate ion
(c) Silver ion and azide ion
(d) All of these.

26. Which is/are incorrect about the reaction between H_2O_2 and O_3?

I. It is a case of mutual reduction.
II. O_3 will oxidise H_2O_2 into O_2.
III. It is not a redox reaction.
IV. H_2O_2 being a stronger oxidising agent will decompose ozone into oxygen.

(a) Only III and IV
(b) Only I, III and IV
(c) Only II
(d) Only II and III

PART - B : Subjective / Descriptive Type

27. Account for the following :

(i) Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method.
(ii) The boiling points of ethers are lower than those of isomeric alcohols.

28. How does molar conductivity vary with concentration for (i) Weak electrolyte and for (ii) Strong electrolyte? Give reasons for variations.

OR

If $E°$ for copper electrode is 0.34 V, then calculate its emf value when the solution in contact with it is 0.1 M in copper ions? How does emf for copper electrode change when concentration of Cu^{2+} ions in the solution is decreased?

29. An organic compound *A* having molecular formula C_2H_7N on treatment with HNO_2 gave an oily yellow substance. Identify '*A*'. Give equation.

30. Explain why transition metals are

(a) paramagnetic
(b) good catalysts.

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31. (a) Why dried fruits and vegetables slowly swell when placed in water? What will be the effect of temperature on this process?

(b) A solution containing 2.56 g of sulphur in 100 g of CS_2 gave a freezing point lowering of 0.383 K. Calculate the molecular formula of sulphur.
(K_f for CS_2 = 3.83 K kg mol^{-1}, at. wt. of S = 32)

32. (a) Explain why glycine exists as a Zwitter ion ($\overset{+}{N}H_3CH_2COO^-$) but anthranilic acid does not.

(b) Explain the denaturation of proteins.

33. Describe how the following conversions can be brought about :

(i) Cyclohexanol to cyclohexanone.
(ii) Ethylbenzene to benzoic acid.
(iii) Bromobenzene to benzoic acid.

OR

Give examples of the following :

(i) Kolbe's electrolytic reaction
(ii) Kolbe's reaction
(iii) Friedel–Craft's reaction

34. Which halogen :

(i) forms H - bonding
(ii) forms strongest acid
(iii) has largest atom
(iv) has highest ionisation enthalpy
(v) shows some metallic properties
(vi) has highest electron affinity

35. (a) What is crystal field splitting energy? How does the magnitude of Δ_o decide the actual configuration of *d*-orbitals in a coordination entity?

(b) $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ are of different colours in dilute solutions. Why?

OR

(a) What are bidentate ligands? Explain with examples.

(b) Explain the coordination sites of polydentate ligands taking an example of EDTA.

(c) One mole of $Co(NH_3)_5Cl_3$ gives three moles of ions on dissociation in water. One mole of this reacts with two moles of $AgNO_3$ to give two moles of AgCl. Name the complex.

36. (a) Account for the following :

(i) Haloalkanes are more reactive than haloarenes.
(ii) Haloalkanes undergo nucleophilic substitutions whereas haloarenes undergo electrophilic substitution.

(b) How would you convert (Write chemical equations) :

(i) Chlorobenzene to phenol?
(ii) Phenol to bromophenol?

OR

(a) How would you differentiate between S_N1 and S_N2 mechanisms of substitution reaction? Give one example of each.

(b) An optically active compound having molecular formula $C_7H_{15}Br$ reacts with aqueous KOH to give a racemic mixture of products. Write the mechanism involved for this reaction.

37. (a) How are the following properties of crystals affected by Schottky and Frenkel defects?

(i) Density
(ii) Electrical conductivity

(b) The density of copper metal is 8.95 g cm^{-3}. If copper atom has free structure, calculate the radius of copper atom. [Given : Atomic mass of Cu = 63.54 g mol^{-1} and $N_A = 6.02 \times 10^{23}$ mol^{-1}]

OR

(a) Br^- ions form close pack structure. If the radius of Br^- ion is 195 pm, calculate the radius of the cation that just fits in the tetrahedral hole. Can a cation A^+ having a radius of 82 pm be slipped into the octahedral hole of the crystal A^+B^-?

(b) Iron (II) oxide has a cubic structure and edge of the unit cell is 500 pm. If density of oxide is 4 g cm^{-3}, calculate the number of Fe^{2+} and O^{2-} ions present in each unit cell (FeO = 72 g mol^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}).

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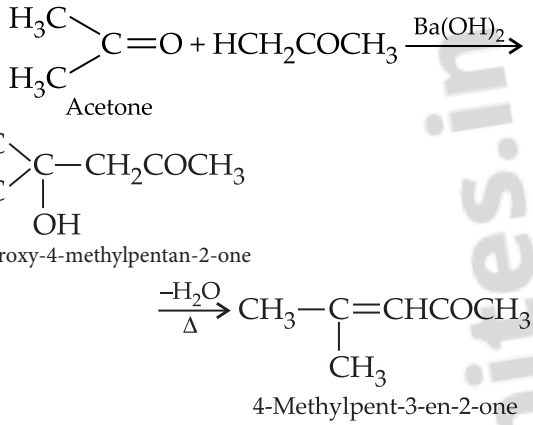
Detailed Solutions

1. (b)

OR

(d) : 

2. (a) : $AgNO_{3(aq)} + KI_{(aq)} \rightarrow \underset{\text{Colloid}}{AgI} + KNO_3$

AgI will absorb I^- from KI electrolyte to form $[AgI]I^-$.

Therefore, it is negatively charged.

3. (d) : $NH_3 + 3Cl_2$ (excess) $\longrightarrow NCl_3 + 3HCl$

4. (c) : The constituents of nucleic acids are nitrogenous bases, sugar and phosphoric acid. The sugar present in DNA is *D*(–)-2-deoxyribose and the sugar present in RNA is *D*(–)-ribose. Due to these *D*(–)-sugar components, DNA and RNA molecules are chiral molecules.

5. (c) : $\pi = CRT$

OR

(c)

6. (b)

7. (c) : $\text{Rate} = -\frac{d[A]}{dt} = -\frac{1}{3}\frac{d[B]}{dt} = \frac{1}{2}\frac{d[C]}{dt}$

8. (d)

9. (a) : $[Co(NH_3)_3Cl_3]$ on ionisation does not give Cl^- ion, hence it does not give a precipitate of AgCl with $AgNO_3$ solution.

OR

(a) : In the formation of d^2sp^3 hybrid orbitals, two $(n-1)d$ orbitals of e_g set [*i.e.* $(n-1)d_z^2$ and $(n-1)d_{x^2-y^2}$ orbitals)], one *ns* and three *np*(np_x, np_y and np_z) orbitals combine together and form six d^2sp^3 hybrid orbitals.

10. (a)

11. (b) : $\frac{2.303RT}{F} = \frac{2.303 \times 8.314 \times 298}{96500}$

= 0.05912

12. (b) : $C_6H_5Cl \xrightarrow{Mg} \underset{(A)}{C_6H_5MgCl} \xrightarrow{CH_3CH_2OH} C_6H_6 + CH_3CH_2OMgCl$

OR

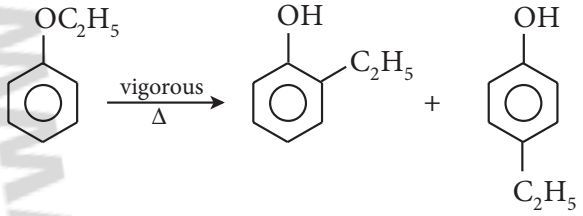
(a) : Cl in 2, 4, 6-trinitrochlorobenzene is activated by three $-NO_2$ groups at *o*-and *p*-positions and hence undergoes hydrolysis most readily.

13. (b)

14. (b) :

$$CH_2=CH_2 + B_2H_6 \longrightarrow B(CH_2CH_3)_3 \xrightarrow{H_2O_2} 3CH_3CH_2OH$$

15. (c) :



OR

(b) : The tendency to show H-bonding decreases with increasing hydrophobic character of carbon chain. The hydrophobic character of carbon chain increases with the length of carbon chain. Also branched chain alcohols show more solubility.

16. (c) : Freundlich adsorption isotherm gives an empirical relationship between the quantity of gas adsorbed by unit mass of solid adsorbent and pressure at a particular temperature.

17. (c) : Tungsten has a large number of unpaired electrons. Hence, metallic bonding in it is quite strong.

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18. (a)

OR

(d) : Uracil occurs in RNA.

19. (a)

20. (a) : $(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + 4H_2O$

$Ba(N_3)_2 \rightarrow 3N_2 + Ba$

21. (c) : Average rate of reaction upto 40 seconds on the basis of the graph is

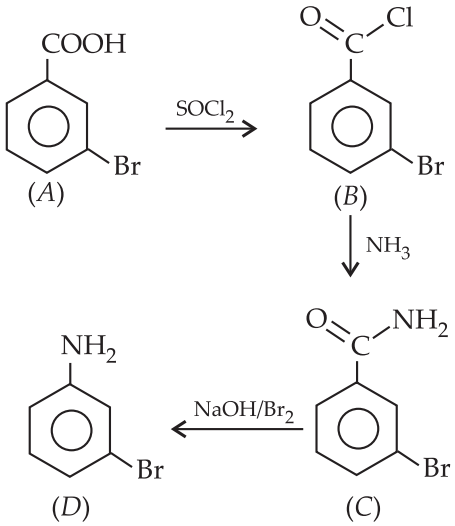
$$\frac{V_3 - 0}{40 - 0} = \frac{V_3}{40}.$$

OR

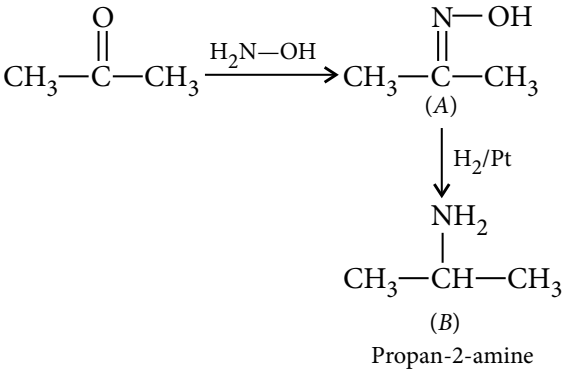
Rate = $k[A]^x[B]^y$

When concentration of *B* is doubled keeping the concentration of *A* constant, then the rate of formation of *C* increases by a factor of four. This indicates that the rate of reaction depends upon the square of concentration of *B*. When concentration of *A* is doubled, the rate of formation of *C* also doubles from the initial value. This shows that the rate depends on first power of concentration of *A*. Hence, Rate = $k[A][B]^2$

22. (c) :

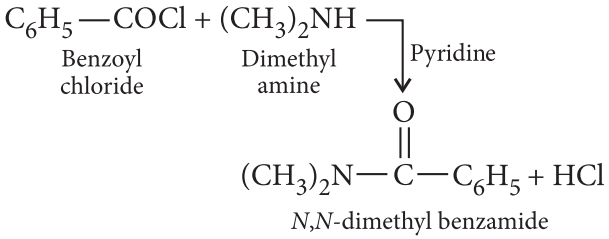


23. (b)



OR

(a) :



24. (a) : V_2O_5 and Cr_2O_3 are amphoteric oxides.

25. (c) : As we know that if SRP of oxidant – SRP of reductant is more than 1.7, then their combination may lead to explosion, therefore in the cases,

$E^o_{N_2/N^{3-}} = -3.09$, and $E^o_{Ag^+/Ag} = 0.80$ V

$E = 0.80 - (-3.09) = +3.89$ V.

As the value of *E* is greater than 1.7,

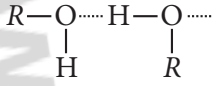
thus this will lead to the explosion.

26. (b) : For the standard reduction potentials given above we may conclude that O_3 is more powerful oxidizing agent than H_2O_2, thus it can oxidise H_2O_2 into O_2.

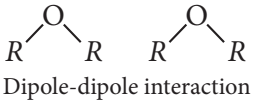
27. (i) : Acid dehydration of 2° and 3° alcohols give alkenes rather than ethers. Due to steric hindrance, the nucleophilic attack by the alcohol molecule on the protonated alcohol molecule does not occur.

The protonated 2° and 3° alcohols lose water molecules to form stable 2° and 3° carbocations.

(ii) The boiling points of ethers are much lower than those of alcohol of comparable molar masses because like alcohols they cannot form intermolecular hydrogen bonds.



Intermolecular hydrogen bond



Dipole-dipole interaction

28. (i) **Weak electrolyte:** When the concentration of weak electrolyte becomes very low, its degree of ionisation rises sharply. There is sharp increase in the number of ions in the solution. Hence, the molar conductivity of a weak electrolyte rises steeply at low concentration.

(ii) **Strong electrolyte :** The molar conductivity of strong electrolyte decreases slightly with the increase in concentration. This decrease is due to increase in attraction as a result of greater number of ions per unit volume. With dilution the ions are

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far apart, interionic attractions become weaker and conductance increases.

OR

(b) : $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$

$$E_{Cu^{2+}/Cu} = E^{\circ}_{Cu^{2+}/Cu} - \frac{0.059}{2}\log\frac{[Cu]}{[Cu^{2+}]}$$

$$= 0.34 - \frac{0.059}{2}\log\frac{1}{0.1}$$

$$= 0.34 - \frac{0.059}{2}\log 10$$

$$= 0.34 - \frac{0.059}{2} \times (1) = 0.34 - 0.0295 = 0.3105 \text{ V}$$

When the concentration of Cu^{2+} ions is decreased, the potential for copper electrode decreases.

29. The compound *A* gives a yellow oily substance on treatment with HNO_2. So, it must be 2°-amine. So, *A* is $(CH_3)_2NH$.

$$(CH_3)_2NH + HNO_2 \longrightarrow \underset{(A)}{(CH_3)_2N{-}N{=}O} + H_2O$$

30. (a) Transition metals are paramagnetic because their atoms or ions contain unpaired electrons.

e.g.,	Sc^{2+}	Ti^{2+}	V^{2+}	Cr^{2+}	Mn^{2+}	Fe^{2+}	Co^{2+}	Ni^{2+}	Cu^{2+}
EC	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	$3d^8$	$3d^9$
n	1	2	3	4	5	4	3	2	1

where, *n* = No. of unpaired electrons.

(b) Transition elements are good catalysts because they possess variable oxidation states and can form unstable intermediate compounds with most of the reactants. Further they also provide large surface area for reaction, *e.g.,* V_2O_5 in the manufacture of sulphuric acid by Contact process.

31. (a) When dried fruits and vegetables are placed in water, they swell due to flow of water into them by osmosis. Increase in temperature speed up the process.

(b) $w_2 = 2.56$ g, $w_1 = 100$ g, $\Delta T_f = 0.383$ K

$K_f = 3.83$ K kg mol^{-1}, $\Delta T_f = K_f \times m$

$$\Rightarrow \Delta T_f = K_f \times \frac{w_2}{M_2} \times \frac{1000}{w_1}$$

$$\Rightarrow M_2 = \frac{w_2 \times 1000}{\Delta T_f \times w_1} \times K_f$$

$$= \frac{2.56 \times 1000}{0.383 \times 100} \times 3.83 = 256 \text{ g mol}^{-1}$$

Molecular mass = Atomic mass × atomicity

$$\Rightarrow \text{Atomicity} = \frac{256}{32} = 8$$

$\therefore$ Sulphur (S) exist as S_8.

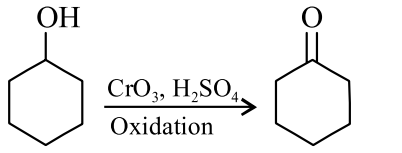
32. (a) : In anthranilic acid (*p*-amino benzoic acid) the electron—withdrawing nature of benzene ring suppresses the transfer of proton from the weak –COOH group to the $-NH_2$ group. Glycine has no such effect hence it exists as a Zwitter ion.

(b) Proteins are very sensitive to the action of heat, change of pH, presence of electrolytes and radiation (particularly short wavelengths).

Whenever proteins are subjected to such changes in the surroundings, they undergo some structural changes leading to the disruption of three dimensional structure. Thus, causes permanent loss of activity of proteins.

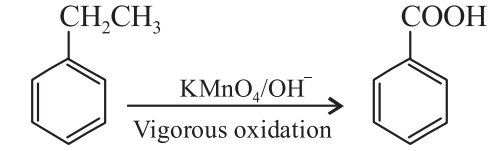
Disruption of native conformation of protein by changing its environment resulting in loss of biological activity is known as denaturation of protein.

33. (b) (i)



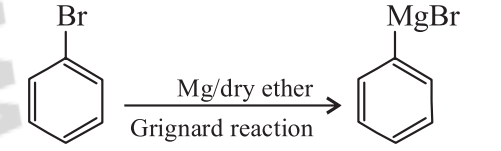
Cyclohexanol — Cyclohexanone

(ii)

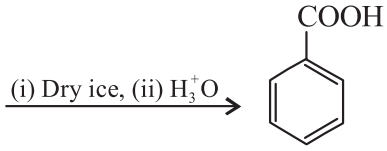


Ethylbenzene — Benzoic acid

(iii)



Bromobenzene — Phenylmagnesium bromide



Benzoic acid

OR

(i) : Kolbe's electrolytic reaction :

$$2CH_3COONa \xrightarrow{\text{Electrolysis}} CH_3{-}CH_3 + 2CO_2 + 2NaOH$$

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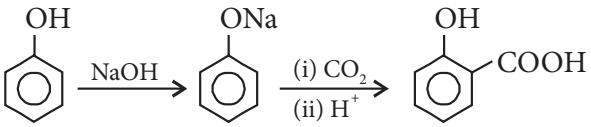
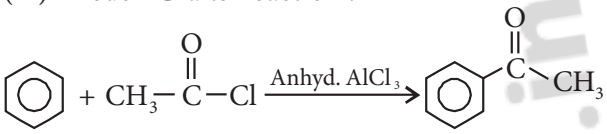
(ii) Kolbe's reaction :



2-Hydroxybenzoic acid
(Salicylic acid)

(iii) Friedel–Crafts' reaction :



Acetophenone

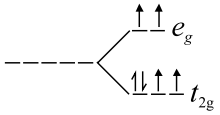
34. (i) Fluorine (ii) Iodine
(iii) Iodine (iv) Fluorine
(v) Iodine (vi) Chlorine

35. (a) : Ligands on approaching the metal ion, split *d*- orbitals into two or more sets of orbitals. The energy difference between these orbitals is called crystal field splitting energy (CFSE). For octahedral complexes, it is represented by Δ_o or 10 Dq.

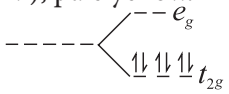
For weak field ligands lying left to the spectrochemical series, magnitude of Δ_o is small and the result is high spin or spin free complexes. Electrons enter into all the five *d*-orbitals, first singly and then pairing starts.

For strong field ligands, Δ_o is large and the complexes formed are low spin or spin paired ones. Electrons first pair up in lower t_{2g} level and then enter into e_g level.

(b) : $[Fe(H_2O)_6]^{2+}$: H_2O is weak field ligand, Δ_o small, light green.

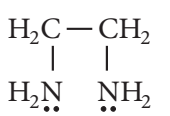


$[Fe(CN)_6]^{4-}$: CN^- is strong field ligand, Δ_o large (absorbs near UV), pale yellow.

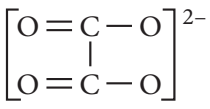


OR

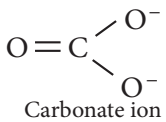
(a) : Ligands which can coordinate with the central metal atom or ion through two donor atoms are known as bidentate ligands. Examples of bidentate ligands are:



Ethylene diamine
(*en*)

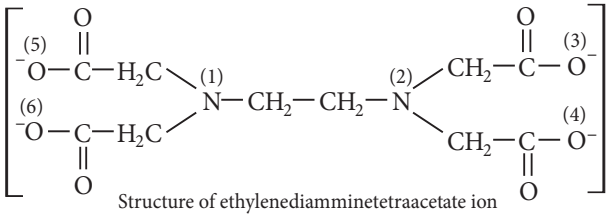


Oxalate anion



Carbonate ion

(b) Ligands which coordinate with the central ion through more than two donor atoms present in the molecule are called polydentate ligands. These are called tridentate (three), tetradentate (four), pentadentate (five) and hexadentate (six) ligands depending upon the number of coordinating donor atoms present in their molecules. A common example of hexadentate ligand is ethylenediamminetetraacetate ion as shown below:



Structure of ethylenediamminetetraacetate ion

(c) The complex is (Pentaaminechlorido-cobalt(III) chloride) $[Co(NH_3)_5Cl]Cl_2$

Only primary valencies are ionised.

$$[Co(NH_3)_5\,Cl]Cl_2 \rightleftharpoons \underbrace{[Co(NH_3)_5\,Cl]^{2+} + 2Cl^-}_{\text{3 ions}}$$

$$\downarrow (2AgNO_3)$$

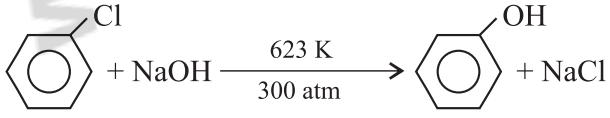
$$2AgCl + [Co(NH_3)_5\,Cl]\,(NO_3)_2$$

36. (a) : (i) Because rupture of "C – *X*" bond in haloarenes is difficult, as it possesses double bond character due to resonance. Haloalkanes have C – *X* single bond.

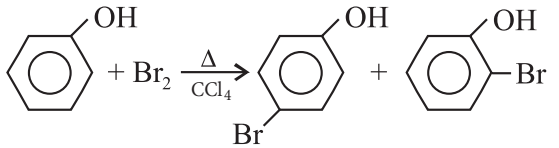
(ii) In haloarenes –ve charge gets localised on arenes using resonance, therefore they undergo electrophilic substitution.

Haloalkanes have electrophilic carbon centre due to polarity of $C \rightarrow X$ bond.

(b) (i)



(ii)



OR

(a) In S_N1 mechanism of substitution reaction, the rate of reaction depends upon the concentration of only reactant. It is two steps process in which carbocation is formed as an intermediate.

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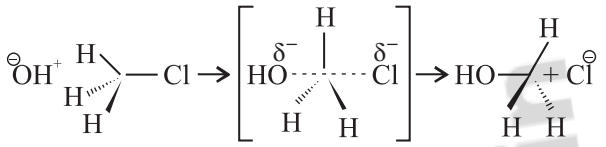
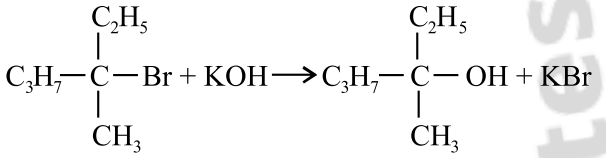
e.g. $(CH_3)_3C\,Br + OH^- \longrightarrow (CH_3)_3COH + Br^-$
2-Bromo-2-methyl propane — 2-Methylpropan-2-ol

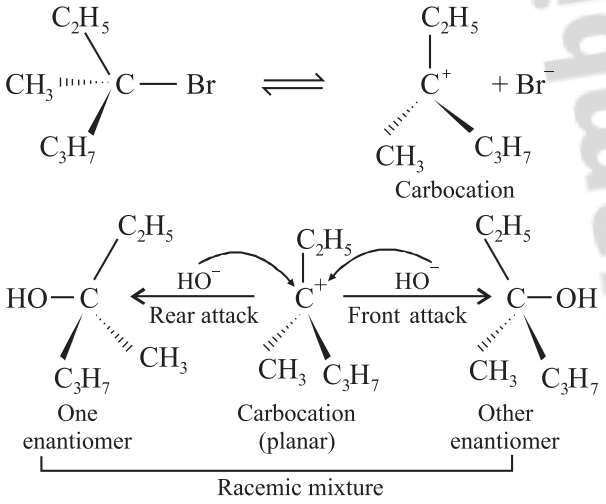
In S_N2 mechanism of substitution, the rate of reaction depends upon the concentration of both the reactant and reagent. It is a single step process.



(b) Clearly $C_7H_{15}Br$ is a tertiary halide.



Mechanism :



37. (a) : (i) Density : It decreases due to Schottky defect but remains same in Frenkel defect.
(ii) Electrical conductivity **:** It increases in both Schottky and Frenkel defects.

(b) $d = \dfrac{Z \times M}{a^3 \times N_A}$

$$8.95 = \frac{4 \times 63.54}{a^3 \times 6.02 \times 10^{23}}$$

$a^3 = 47.145 \times 10^{-24}\ cm^3$

$a = \sqrt[3]{47.145 \times 10^{-24}}$

$a = 3.612 \times 10^{-8}$ cm = 361.2 pm

For *fcc*, $4r = \sqrt{2}a$

$$r = \frac{a}{2\sqrt{2}} = \frac{361.2}{2 \times 1.414} = 127.7\ pm$$

OR

(a) (i) Radius of cation that just fits in the tetrahedral hole

$r_+ = 0.225\ r_-$ or, $r_+ = 0.225 \times 195 = 43.875$ pm

(ii) For octahedral void, $\dfrac{r}{R} = 0.414$ to 0.732

Radius ratio of given ions $\dfrac{r_+}{r_-} = \dfrac{82}{195} = 0.4205$

As the ratio lies in the range of radius of octahedral void, hence, cation A^+ can be slipped into the octahedral void of A^+B^- crystal.

(b) Using formula, $d = \dfrac{Z \times M}{a^3 \times N_A}$

$$4 = \frac{Z \times 72}{(5 \times 10^{-8})^3 \times 6.02 \times 10^{23}}$$

$$Z = \frac{4 \times 125 \times 6.02 \times 10^{-1}}{72} = 4.18 \cong 4$$

Each unit cell has 4 Fe^{2+} ions and 4 O^{2-} ions.

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NOTES

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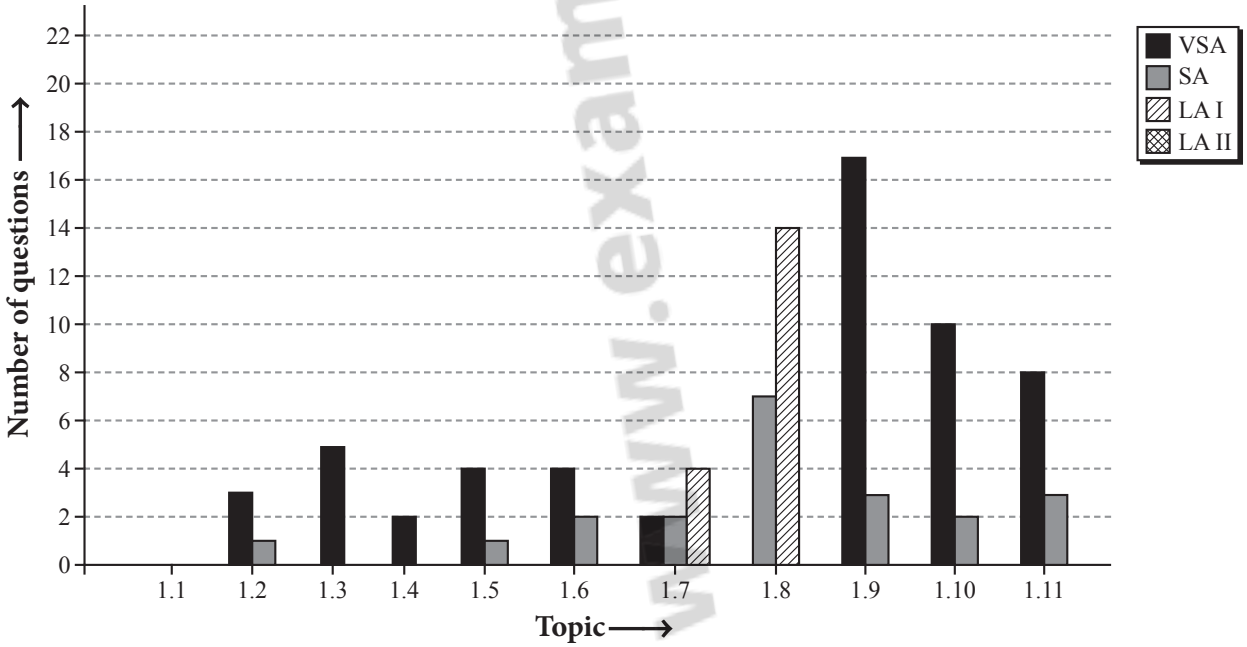
CHAPTER
1

The Solid State

1.1 General Characteristics of Solid State
1.2 Amorphous and Crystalline Solids
1.3 Classification of Crystalline Solids
1.4 Crystal Lattices and Unit Cells
1.5 Number of Atoms in a Unit Cell
1.6 Close Packed Structures
1.7 Packing Efficiency
1.8 Calculations Involving Unit Cell Dimensions
1.9 Imperfections in Solids
1.10 Electrical Properties
1.11 Magnetic Properties

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Calculations Involving Unit Cell Dimensions.*
- Maximum VSA type questions were asked from *Imperfections in Solids.*
- Maximum SA and LA I type questions were asked from *Calculations Involving Unit Cell Dimensions.*

QUICK RECAP

- **Solids :** They have definite mass, volume and shape due to the fixed position of their constituent particles, small intermolecular distances and strong intermolecular forces between them.
- **Solids :** They can be classified as crystalline or amorphous on the basis of the nature of order present in the arrangement of their constituent particles.

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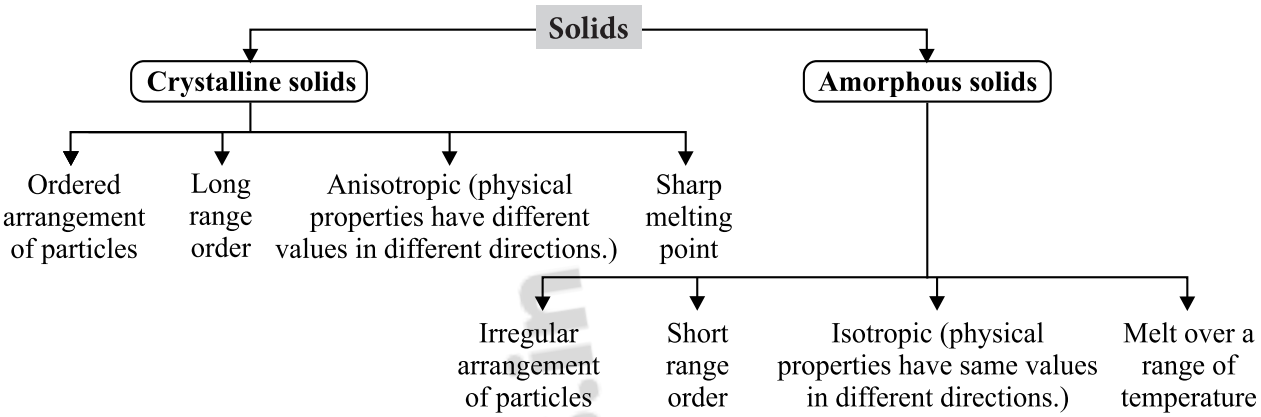


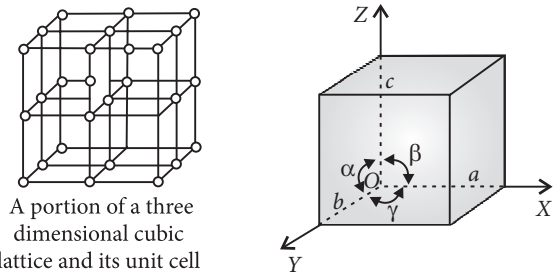
Types of crystalline solids : Classification of crystalline solids on the basis of nature of intermolecular forces :

	Properties	Ionic solid	Metallic solid	Covalent solid	Molecular solid
1.	**Constituent particles are held by**	Ionic bonds	Metallic bonds	Covalent bonds	Weak van der Waals' forces or hydrogen bonds
2.	**Constituent particles**	Ions	Metal ions and sea of electrons	Non-metal atoms	Atoms or molecules
3.	**Physical nature**	Hard and brittle	Hard, malleable and ductile	Hard	Soft
4.	**Electrical conductivity**	Insulator	Conductor	Insulator except graphite	Insulator
5.	**Melting point**	High	Fairly high	Very high	Low
6.	**Examples**	NaCl, ZnS, CaF_2, MgO, etc.	Fe, Cu, Ag, Na, Mg, etc.	Diamond, graphite, quartz, carborundum, AlN, etc.	Ice, I_2, P_4, S_8, naphthalene, camphor, solid CO_2 (dry ice), etc.

Crystal lattice : The regular three dimensional arrangement of constituent particles in a crystal is called *crystal lattice*.

Lattice points : Constituent particles (atoms, molecules or ions) of a solid crystal are represented as points which are known as *lattice points* or *lattice sites*.

Unit cell : The smallest repeating pattern which represents structure of a crystalline solid is called *unit cell*.



A portion of a three dimensional cubic lattice and its unit cell

- It is the smallest portion of a crystal lattice which when repeated in different directions generates the *entire lattice*.
- A unit cell has characteristic dimensions along the three edges *a*, *b* and *c* and interfacial angles α (between *b* and *c*), β (between *a* and *c*) and γ (between *a* and *b*).
- On the basis of primitives or axial distances and interfacial angles there are *seven crystal systems* and *fourteen Bravais lattices*.

Seven primitive unit cells or crystal systems

Crystal system	Axial distances or edge lengths	Axial angles	Examples
Cubic (*most symmetrical*)	$a = b = c$	$\alpha = \beta = \gamma = 90°$	Cu, Zinc blende, KCl, NaCl
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	Sn(White tin), SnO_2, TiO_2, $CaSO_4$
Orthorhombic or Rhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	Rhombic sulphur, KNO_3, $BaSO_4$
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°; \beta \neq 90°$	Monoclinic sulphur, $PbCrO_4$, $Na_2SO_4 \cdot 10H_2O$
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°; \gamma = 120°$	Graphite, ZnO, CdS
Rhombohedral or Trigonal	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	$CaCO_3$ (Calcite), HgS (Cinnabar)
Triclinic (*most unsymmetrical*)	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	$K_2Cr_2O_7$, $CuSO_4 \cdot 5H_2O$, H_3BO_3

Unit cell constant (*Z*) for different types of unit cells

Type of unit cell	Simple cubic	*bcc*	*ccp/fcc*	*hcp*
No. of particles (*Z*)	$8 \times \frac{1}{8} = 1$	$8 \times \frac{1}{8} + 1 = 2$	$8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$	$12 \times \frac{1}{6} + 2 \times \frac{1}{2} + 3 = 6$

Packing in solids

	Arrangement	Space occupied	C. No.	Relation between *d*, *a* and *r*	Examples
Simple cubic packing (*scp*)	*AAA* ... type	52.4%	6	$d = a = 2r$, $r = \frac{a}{2}$	Po
Hexagonal close packing (*hcp*)	*ABABAB* ... type	74%	12	$a = 2r$ $h = 2 \times \sqrt{\frac{2}{3}}\, a$	Mg, Zn, Mo, V, Cd

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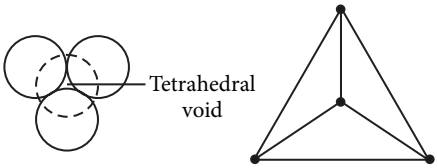
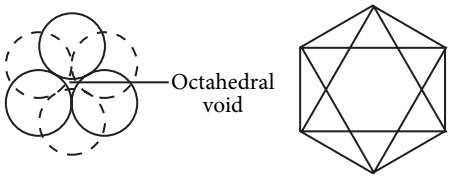
Cubic close packing (*ccp*) or Face centred cubic (*fcc*)	*ABCABC* ... type	74%	12	$d = 2r = \frac{a}{\sqrt{2}}$, $r = \frac{a}{2\sqrt{2}}$	Cu, Ag, Au, Ni, Pt, etc.
Body centred cubic packing (*bcc*)	The spheres in square close packing are slightly opened up, in second layer, the spheres are at the top of hollows and third layer is exactly over the first layer and so on.	68%	8	$d = 2r = \frac{\sqrt{3}a}{2}$, $r = \frac{\sqrt{3}a}{4}$	Li, Na, K, Rb, Cs, Ba, Cd, Fe, Mn, etc.

- In a unit cell some empty space exists between spheres, this is called *void space* or *hole*, also called *interstitial void*, or *interstices*.
- Two types of interstitial voids are present in three dimensional close packing systems :
  - **Tetrahedral voids :** These are holes or interstices surrounded by four spheres present at the corners of a tetrahedron.
  - **Octahedral voids :** These are holes surrounded by six spheres located on a regular octahedron.





- If N is the number of close packed spheres, then the number of octahedral voids generated = N, the number of tetrahedral voids generated = $2N$
- Radius (r) of tetrahedral void = 0.225 R; Radius (r) of octahedral void = 0.414 R where, R is the radius of the spheres in close packing.
- **Radius ratio rules :** For the stability of an ionic compound, each cation should be surrounded by maximum number of anions and *vice versa* for maximum electrostatic forces of attraction.

$$\text{Radius ratio} = \frac{\text{Radius of the cation } (r_+)}{\text{Radius of the anion } (r_-)}$$

Radius ratio	C. No.	Structural arrangement	Example
0.155 – 0.225	3	Planar triangular	Boron trioxide (B_2O_3)
0.225 – 0.414	4	Tetrahedral	Sphalerite (ZnS)
0.414 – 0.732	6	Octahedral	Rock salt (NaCl)
0.732 – 1.0	8	Body centred cubic	Caesium chloride (CsCl)

- **Calculations involving unit cell dimensions :**

$\rho = \frac{Z \times M}{a^3 \times N_A \times 10^{-30}}$ g cm^{-3} ; where Z = no. of atoms per unit cell, a = edge length in pm, ρ = density of

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solid, M = molar mass and N_A = Avogadro's no.

Imperfections in solids : Any deviation from the perfectly ordered arrangement of constituent particles in a crystal is called *imperfection* or *defect*.

Imperfections/Defects in Solids

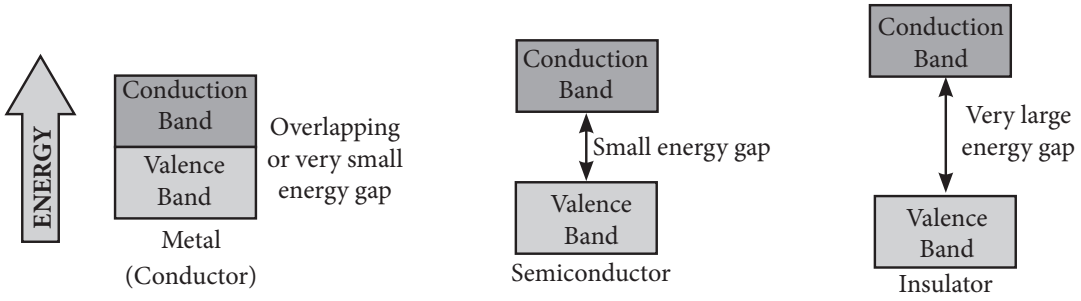
- **Point defects** — Deviations from ideal arrangement around a point or an atom.
  - **Stoichiometric/Intrinsic/Thermodynamic defects** — Do not disturb stoichiometry of solid.
    - **Vacancy defect :** Arises when lattice sites are vacant. Decreases density, shown by non-ionic solids.
    - **Interstitial defect :** Arises when constituent particles occupy interstitial site. Increases density, shown by non-ionic solids.
    - **Frenkel defect :** Smaller ion (usually cation) dislocates from its normal site to an interstitial site. Does not change density, shown by ionic solids, *e.g.*, ZnS, AgCl, AgBr, AgI, etc.
    - **Schottky defect :** Arises when equal number of cations and anions are missing. Decreases density, shown by ionic solids, *e.g.*, NaCl, KCl, CsCl, AgBr, etc.
  - **Impurity defects** — Arise when foreign atoms are present at lattice site *e.g.*, $SrCl_2$, $CdCl_2$, AgCl etc.
  - **Non-stoichiometric defects** — Disturb stoichiometry of solids.
    - **Metal excess defect**
      - **Due to anionic vacancies** — A negative ion is missing from its lattice site, leaving a 'hole' which is occupied by an electron to maintain electrical neutrality, *e.g.*, LiCl, NaCl
      - **Due to the presence of extra cations** — Usually arises when metal oxides are heated, *e.g.*, ZnO.
    - **Metal deficiency defect** — Arises when metal shows variable valency *e.g.*, Fe^{2+}, Fe^{3+}

Electrical properties : The electrical conductivity of solids varies from 10^{-20} to 10^{7} ohm^{-1} m^{-1}. So on the basis of electrical conductivity solids are classified as :

Solids

- **Conductors** — Allow the passage of electric current. **Conductivity :** $10^4 - 10^7$ $ohm^{-1}m^{-1}$
  - **Metallic conductor** — Conducts electricity in solid as well as in molten state. *e.g.*, Cu, Ag.
  - **Electrolytic conductor** — Conducts electricity only in molten state or aqueous solution. *e.g.*, NaCl, KCl.
- **Semiconductors** — Conduct electricity due to presence of impurities. **Conductivity :** 10^{-6} to 10^4 $ohm^{-1}m^{-1}$
- **Insulators** — Do not allow the passage of electric current. **Conductivity :** $10^{-20} - 10^{-10}$ ohm^{-1} m^{-1}

▶ **Distinction among metals, insulators and semiconductors :**



Electrical conductivity of semi-conductors increases with rise in temperature because more

electrons can jump to the conduction band.

Doping : It is a process of adding an appropriate amount of suitable impurity to increase the conductivity of intrinsic semiconductors.

- **By doping with electron rich impurities**, *e.g.*, group 14 elements like Si or Ge with group 15 elements like P, As etc. Free electrons become available which increase conductivity. Such solids are called *n-type semiconductors*.
- **By doping with electron deficit impurities**, *e.g.*, group 14 elements like Si or Ge with group 13 elements like B, Al or Ga. As a result, holes are created. Electrons move to occupy the holes. Thus, holes move towards negatively charged plate as if they carry positive charge. These are called *p-type semiconductors*.

Magnetic properties :

Solids

Paramagnetic
- Possess permanent dipole and attracted by magnetic field.
- They contain unpaired electrons.

e.g., Cu^{2+}, Fe^{3+}, Cr^{3+}, O_2, etc.

Diamagnetic
- Weakly repelled by magnetic field.
- These materials have all paired electrons.

e.g., NaCl, water, benzene, etc.

Ferromagnetic
- Strongly attracted by magnetic field and show permanent magnetism even when magnetic field is removed.
- Magnetic moment of domains in the same direction, ↑↑↑↑

e.g., Fe, Co, Ni, CrO_2, etc.

Antiferromagnetic
- Expected to possess paramagnetism or ferromagnetism but magnetic moment of domains in a compensatory way so as to give a net zero magnetic moment, ↑↓↑↓

e.g., MnO, etc.

Ferrimagnetic
- Expected to have large magnetism but actually have small net magnetic moment as magnetic moment of domains are aligned in parallel and anti-parallel directions in unequal number. ↑↑↓↑↑↓

e.g., Fe_3O_4, $MgFe_2O_4$ and $ZnFe_2O_4$, etc.
- These substances lose ferrimagnetism on heating and become paramagnetic.

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Previous Years' CBSE Board Questions

1.2 Amorphous and Crystalline Solids

VSA (1 mark)

1. Why are crystalline solids anisotropic? (*AI 2014C*)

2. Some of the glass objects recovered from ancient monuments look milky instead of being transparent. Why? *(1/3, AI 2012C)*

3. "Crystalline solids are anisotropic in nature." What does this statement mean? *(Delhi, Foreign 2011)*

SA (2 marks)

4. Write any two differences between amorphous solids and crystalline solids. *(2/5, Delhi 2017)*

1.3 Classification of Crystalline Solids

VSA (1 mark)

5. Based on the nature of intermolecular forces, classify the following solids :
Silicon carbide, Argon *(1/3, AI 2017)*

6. Based on the nature of intermolecular force, classify the following solids :
Benzene, Silver *(1/3, AI 2017)*

7. Based on the nature of intermolecular forces, classify the following solids :
Sodium sulphate, Hydrogen (*1/3, AI 2017*)

8. Give an example each of a molecular solid and an ionic solid. *(AI 2016)*

9. Which one of the following is an example of molecular solid :
CO_2 or SiO_2? *(1/3, Foreign 2014)*

1.4 Crystal Lattices and Unit Cells

VSA (1 mark)

10. Define primitive unit cells. *(AI 2015)*

11. How will you distinguish between the following pair of terms :
Crystal lattice and unit cell? *(AI 2014)*

1.5 Number of Atoms in a Unit Cell

VSA (1 mark)

12. How many atoms per unit cell (*Z*) are present in *bcc* unit cell? *(Delhi 2014C)*

13. How many atoms constitute one unit cell of a face-centred cubic crystal? *(Delhi 2013)*

14. Calculate the number of atoms in a face centred cubic unit cell. *(Delhi 2013C)*

15. How many atoms are there in one unit cell of a body centred cubic crystal? *(Foreign 2011)*

SA (2 marks)

16. Calculate the number of unit cells in 8.1 g of aluminium if it crystallizes in a *fcc* structure. (Atomic mass of Al = 27 g mol^{-1}) *(2/5, Delhi 2017)*

1.6 Close Packed Structures

VSA (1 mark)

17. What is the coordination number of atoms in a (i) *bcc* structure and (ii) *fcc* structure ? *(AI 2019)*

18. What is the formula of a compound in which the element *Y* forms *ccp* lattice and atoms of *X* occupy $1/3^{rd}$ of tetrahedral voids? *(Delhi 2015)*

19. What is the formula of a compound in which the element *Y* forms *ccp* lattice and atoms of *X* occupy $2/3^{rd}$ of octahedral voids? *(Foreign 2015)*

20. How will you distinguish between the following pair of terms :
Tetrahedral and octahedral voids? *(AI 2014)*

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SA (2 marks)

21. A compound forms *hcp* structure. What is the total number of voids in 0.5 mol of it? How many of these are tetrahedral voids? *(AI 2013C)*

22. An alloy of gold and cadmium crystallises with a cubic structure in which gold atoms occupy the corners and cadmium atoms fit into the face centres. Assign formula for this alloy. *(AI 2011C)*

1.7 Packing Efficiency

VSA (1 mark)

23. Express the relationship between atomic radius (r) and the edge length (a) in the *bcc* unit cell. *(Foreign 2014)*

24. Express the relationship between atomic radius (r) and the edge length (a) in the *fcc* unit cell. *(Foreign 2014)*

SA (2 marks)

25. Aluminium crystallises in a *fcc* structure. Atomic radius of the metal is 125 pm. What is the length of the side of the unit cell of the metal? *(AI 2013, Foreign 2011)*

26. Calculate the packing efficiency of a metal for a simple cubic lattice. *(AI 2011)*

LA I (3 marks)

27. Tungsten crystallises in body centred cubic unit cell. If the edge of the unit cell is 316.5 pm, what is the radius of tungsten atom? *(Delhi 2012)*

28. Silver crystallises in face centred cubic unit cell. Each side of this unit cell has a length of 400 pm. Calculate the radius of the silver atom. (Assume the atom just touch each other on the diagonal across the face of the unit cell, that is each face atom is touching the four corner atoms). *(Delhi 2011)*

29. Aluminium crystallises in a cubic close-packed structure. Radius of atom in the metal is 125 pm.

(i) What is the length of the side of the unit cell?

(ii) How many unit cells are there in 1 cm^3 of aluminium? *(Foreign 2011)*

30. Silver crystallises with face-centred cubic unit cell. Each side of this unit cell has a length of 409 pm. What is the radius of silver atom? Assume the atoms just touch each other on the diagonal across the face of the unit cell. *(Foreign 2011)*

1.8 Calculations Involving Unit Cell Dimensions

SA (2 marks)

31. An element with density 11.2 g cm^{-3} forms a *fcc* lattice with edge length of 4×10^{-8} cm. Calculate the atomic mass of the element.
(Given : $N_A = 6.022 \times 10^{23}$ mol^{-1}) *(3/5, 2018C, Delhi 2014)*

32. An element with density 2.8 g cm^{-3} forms a *fcc* unit cell with edge length 4×10^{-8} cm. Calculate the molar mass of the element.
(Given : $N_A = 6.022 \times 10^{23}$ mol^{-1}) *(AI 2014)*

33. An element crystallises in structure having *fcc* unit cell of an edge 200 pm. Calculate the density if 200 g of this element contains 24×10^{23} atoms. *(AI 2013C)*

34. The unit cell of an element of atomic mass 108 u and density 10.5 g cm^{-3} is a cube with edge length, 409 pm. Find the type of unit cell of the crystal. [Given : Avogadro's constant = 6.023×10^{23} mol^{-1}] *(Delhi 2012C)*

35. An element X crystallises in *fcc* structure. 208 g of it has 4.2832×10^{24} atoms. Calculate the edge of the unit cell, if density of X is 7.2 g cm^{-3}. *(Delhi 2012C)*

36. Explain how we can determine the atomic mass of an unknown metal if you know its density and the dimension of unit cell of its crystal. *(AI 2011)*

37. A copper crystal has a face-centred cubic lattice structure. Atomic radius of the copper atom is 128 pm. Calculate the density of copper. (Atomic mass of copper = 63.5) (AI 2011C)

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LA I (3 marks)

38. An element crystallizes in *fcc* lattice with a cell edge of 300 pm. The density of the element is 10.8 g cm^{-3}. Calculate the number of atoms in 108 g of the element.
(Delhi 2019)

39. Chromium crystallises in *bcc* structure. If its edge length is 300 pm, find its density. Atomic mass of chromium is 52 u.
[$N_A = 6.022 \times 10^{23}$ mol^{-1}] *(AI 2019)*

40. An element '*X*' (At. mass = 40 g mol^{-1}) having *fcc* structure, has unit cell edge length of 400 pm. Calculate the density of '*X*' and the number of unit cells in 4 g of '*X*' ($N_A = 6.022 \times 10^{23}$ mol^{-1}). *(2018)*

41. An element has atomic mass 93 g mol^{-1} and density 11.5 g cm^{-3}. If the edge length of its unit cell is 300 pm, identify the type of unit cell. *(3/5, Delhi 2017)*

42. An element crystallises in a *fcc* lattice with cell edge of 250 pm. Calculate the density if 300 g of this element contains 2×10^{24} atoms.
(Delhi 2016)

43. An element crystallises in a *bcc* lattice with cell edge of 500 pm. The density of the element is 7.5 g cm^{-3}. How many atoms are present in 300 g of the element? *(AI 2016)*

44. An element with molar mass 27 g mol^{-1} forms a cubic unit cell with edge length 4.05×10^{-8} cm. If its density is 2.7 g cm^{-3}, what is the nature of the cubic unit cell?
(Delhi 2015)

45. An element *X* (molar mass = 60 g mol^{-1}) has density of 6.23 g cm^{-3}. Identify the type of cubic unit cell, if the edge length of the unit cell is 4×10^{-8} cm. *(Foreign 2015)*

46. The density of copper is 8.95 g cm^{-3}. It has a face centred cubic structure. What is the radius of copper atom? (Atomic mass Cu = 63.5 g mol^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1})
(Delhi 2014C)

47. Iron has a body centred cubic unit cell with a cell dimension of 286.65 pm. The density of iron is 7.87 g cm^{-3}. Use this information to calculate Avogadro's number.
(Gram atomic mass of Fe = 56 g mol^{-1}).
(AI 2014C, 2012, Delhi 2012)

48. An element occurs in *bcc* structure. It has a cell edge length of 250 pm. Calculate the molar mass if its density is 8.55 g cm^{-3}. Also calculate the radius of an atom of this element. *(Delhi 2013C)*

49. Niobium crystallises in body-centred cubic structure. If its density is 8.55 g cm^{-3}, calculate atomic radius of niobium, given its atomic mass 93 u. *(Delhi 2013C)*

50. Copper crystallises with face centred cubic unit cell. If the radius of copper atom is 127.8 pm, calculate the density of copper metal. (Atomic mass of Cu = 63.55 u and Avogadro's number, $N_A = 6.02 \times 10^{23}$ mol^{-1})
(AI 2012)

51. The density of lead is 11.35 g cm^{-3} and crystallise with *fcc* unit cell. Estimate the radius of lead atom.
(At. mass of lead = 207 g mol^{-1} and $N_A = 6.02 \times 10^{23}$ mol^{-1}) *(Delhi 2011)*

1.9 Imperfections in Solids

VSA (1 mark)

52. Analysis shows that FeO has a non-stoichiometric composition with formula $Fe_{0.95}O$. Give reason. *(2018)*

53. Give reason :
In stoichiometric defects, NaCl exhibits Schottky defect and not Frenkel defect.
(1/5, Delhi 2017)

54. ZnO turns yellow on heating. Why?
(1/3, AI 2017)

55. AgCl shows Frenkel defect while NaCl does not. Give reason. *(1/3, AI 2017)*

56. What happens when $CdCl_2$ is doped with AgCl? *(1/3, AI 2017)*

57. Define the following term :
Schottky defect *(1/3, AI 2015)*

58. Which stoichiometric defect does not change the density of the crystal? *(1/2, AI 2014)*

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59. What type of stoichiometric defect is shown by KCl and why? *(1/2, Foreign 2014)*

60. What type of stoichiometric defect is shown by NaCl? *(Delhi 2014C)*

61. What type of stoichiometric defect is shown by AgCl? *(Delhi 2013)*

62. What type of point defect is produced when AgCl is doped with $CdCl_2$? *(Delhi 2013)*

63. Account for the following :
Schottky defects lower the density of related solids. *(1/2, AI 2013)*

64. What change occurs when AgCl is doped with $CdCl_2$? *(1/2, AI 2013)*

65. Which type of ionic substances show Schottky defect in solids? *(Delhi 2013C)*

66. Which stoichiometric defect increases the density of a solid? *(AI 2012, Delhi 2011)*

67. What type of stoichiometric defect is shown by AgBr and AgI? *(AI 2012C)*

68. What type of defect can arise when a solid is heated? *(AI 2012C)*

SA (2 marks)

69. Examine the given defective crystal.

A^+	B^-	A^+	B^-	A^+
B^-	○	B^-	A^+	B^-
A^+	B^-	A^+	○	A^+
B^-	A^+	B^-	A^+	B^-

Answer the following questions :
(i) What type of stoichiometric defect is shown by the crystal?
(ii) How is the density of the crystal affected by this defect?
(iii) What type of ionic substances show such defect? *(Delhi 2014)*

70. (i) What type of non-stoichiometric point defect is responsible for the pink colour of LiCl?
(ii) What type of stoichiometric defect is shown by NaCl? *(AI 2014)*

71. If NaCl is doped with 10^{-3} mole percent $SrCl_2$, what will be the concentration of cation vacancies? ($N_A = 6.02 \times 10^{23}$ mol^{-1}) *(AI 2013C)*

1.10 Electrical Properties

VSA (1 mark)

72. Account for the following :
Conductivity of silicon increases on doping it with phosphorus. *(Delhi 2019, AI 2013)*

73. Give reason :
Silicon on doping with phosphorus forms *n*-type semiconductor. *(1/5, Delhi 2017)*

74. What is meant by groups 12-16 compounds? Give an example. *(1/3, AI 2017)*

75. What type of semiconductor is formed when Ge is doped with Al? *(1/3, AI 2017)*

76. What type of semiconductor is formed when silicon is doped with As? *(1/3, Foreign 2014)*

77. What type of semiconductor is obtained when silicon is doped with boron? *(1/3, Delhi, AI 2013)*

78. What is meant by 'doping' in semiconductor? *(Delhi 2012)*

79. How may the conductivity of an intrinsic semiconductor be increased? *(AI 2012)*

80. What are *n*-type semiconductors? *(AI 2012)*

81. What is meant by an 'intrinsic semiconductor'? *(Foreign 2011)*

SA (2 marks)

82. What is a semiconductor? Describe the two main types of semiconductors. *(Delhi 2013C)*

83. What is a semiconductor? Describe the two main types of semiconductors and contrast their conduction mechanism. *(AI 2013C)*

1.11 Magnetic Properties

VSA (1 mark)

84. Give reason :
Ferrimagnetic substances show better magnetism than antiferromagnetic substances. *(1/5, Delhi 2017)*

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85. What type of magnetism is shown by a substance if magnetic moments of domains are arranged in same direction? *(Delhi 2016)*

86. Define the following term :
Ferromagnetism *(1/2, AI 2015)*

87. Write the type of magnetism observed when the magnetic moments are oppositely aligned and cancel out each other. *(AI 2014)*

88. Write the type of magnetism observed when the magnetic moments are aligned in parallel and anti-parallel directions in unequal numbers. *(AI 2014)*

89. What is meant by 'antiferromagnetism'? *(AI 2014C)*

90. What type of substances would make better magnets, ferromagnetic or ferrimagnetic? *(1/3, Foreign 2014, Delhi 2013)*

91. What type of magnetism is shown in the following alignment of magnetic moments?
↑ ↑ ↑ ↑ ↑ ↑ *(1/3, Delhi 2013)*

SA (2 marks)

92. Explain the following terms with suitable examples : Ferromagnetism and ferrimagnetism *(Delhi 2012C)*

93. Explain the following terms with suitable example of each.
(a) Ferromagnetism
(b) Antiferromagnetism *(Delhi 2011C)*

94. How is ferromagnetism different from paramagnetism? *(Delhi 2011C)*

Detailed Solutions

1. Crystalline solids are anisotropic in nature because of different arrangements of particles in different directions. Some of their physical properties like electrical conductivity, refractive index, etc., shows different values in different directions in the same crystals.

2. Some of the glass objects from ancient monuments look milky instead of being transparent because it undergoes heating during the day and cooling at night *i.e.*, annealing over the years.
As a result, it acquires some crystalline character. They become a bit opaque.

3. *Refer to answer 1.*

4. Differences between crystalline and amorphous solids are as follows :

	Crystalline solid	Amorphous solid
1.	It has definite characteristic geometrical shape.	It has irregular shape.
2.	It has sharp melting point. *e.g.*, Sodium chloride (NaCl).	It melts over a range of temperature. *e.g.*, Plastic.

5. Silicon carbide : Covalent or network solid.
Argon : Non-polar molecular solid.

6. Benzene : Non-polar molecular solid.
Silver : Metallic solid.

7. Sodium sulphate : Crystalline solid
Hydrogen : Non-polar molecular solid.

8. Molecular solids : HCl, SO_2, H_2O (*any one*)
Ionic solids : KCl, $CuSO_4$, ZnS (*any one*)

9. Carbon dioxide (CO_2) is a molecular solid.
Quartz (SiO_2) is a covalent solid.

10. A unit cell in which constituent particles (lattice points) are present only at the corner positions is called primitive or simple unit cell.

11. Arrangement of lattice points in three dimensional space is called crystal lattice.
The smallest repeating unit which represents arrangement of lattice points in a crystal lattice is called unit cell.

12. In a body centred cubic unit cell.

(i) 8 corners $\times \frac{1}{8}$ per corner atom $= 8 \times \frac{1}{8}$
$= 1$ atom

(ii) 1 body centre atom $= 1 \times 1 = 1$ atom

$\therefore$ Total number of atoms per unit cell
$= 1 + 1 = 2$ atoms

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13. In face centred cubic unit cell

(i) No. of atoms contributed from the corners of the unit cell

$= 8 \text{ corner atoms} \times \frac{1}{8}$ atom per unit cell

$= 8 \times \frac{1}{8} = 1$ atom

(ii) No. of atoms contributed from the faces

$= 6 \text{ face centred atoms} \times \frac{1}{2}$ atoms per unit cell

$= 6 \times \frac{1}{2} = 3$ atoms

$\therefore$ Total no. of atoms per unit cell = 4 atoms

14. *Refer to answer 13.*

15. *Refer to answer 12.*

16. As 27 g of Al contains 6.023×10^{23} atoms

$\therefore$ 8.1 g Al will contain $= \frac{6.023 \times 10^{23}}{27} \times 8.1$ atoms

$= 1.8069 \times 10^{23}$ atoms

In *fcc*, 4 atoms are present in one unit cell

$\therefore$ 1.8069×10^{23} atoms will be present in

$\frac{1.8069 \times 10^{23}}{4} = 4.517 \times 10^{22}$ unit cells

17. (i) 8 (ii) 12

18. No. of Y atoms per unit cell $= \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$

No. of tetrahedral voids $= 2 \times 4 = 8$

$\therefore$ No. of X atoms $= \frac{1}{3} \times 8 = \frac{8}{3}$

Formula of the compound $= X_{\frac{8}{3}}Y_4 = X_2Y_3$

19. No. of Y atoms per unit cell in *ccp* lattice

$= \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$

No. of octahedral voids = No. of atoms present in *ccp* lattice

No. of octahedral voids occupied by atoms

$= \frac{2}{3} \times 4 = \frac{8}{3}$

Therefore, formula of the compound $= X_{\frac{8}{3}}Y_4 = X_2Y_3$

20. A void surrounded by four atoms whose centre when joined forms tetrahedron is called tetrahedral void.

Number of tetrahedral voids = 2 × Number of atoms.

A void surrounded by six atoms whose centre when joined forms an octahedron is called octahedral voids.

Number of octahedral voids = Number of atoms.

21. Number of atoms in 0.5 mol *hcp* lattice

$= 0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$

$\therefore$ Number of octahedral voids = Number of atoms

$= 3.011 \times 10^{23}$

Number of tetrahedral voids = 2 × Number of atoms

$= 2 \times 3.011 \times 10^{23} = 6.022 \times 10^{23}$

Total number of voids = Number of octahedral voids + Number of tetrahedral voids

$= 3.011 \times 10^{23} + 6.022 \times 10^{23}$

$= 9.033 \times 10^{23}$ voids

22. When the particles are present not only corners but also at the centre of each face of the unit cell. It is called face centred unit cell.

Number of atoms in a unit cell of the face centred cubic (*fcc*) $\Rightarrow$ 8 atoms on the corners (gold) and 6 atoms of cadmium on the face (one of each face).

Contribution by atoms (gold) on the corners

$= \frac{1}{8} \times 8 = 1$

Contribution by atom (cadmium) on the faces

$= \frac{1}{2} \times 6 = 3$

Number of atoms present in the unit cell = 1 + 3 = 4

Hence, formula for the given alloy = $AuCd_3$

23. In *bcc* unit cell $r = \frac{\sqrt{3}}{4}a$

24. In *fcc* unit cell $r = \frac{a}{2\sqrt{2}}$

25. For *fcc* (or *ccp*), $a = 2\sqrt{2}r$

$= 2 \times 1.414 \times 125$ pm $= 354$ pm

26. Packing efficiency

$= \frac{Z \times \text{Volume of one atom}}{\text{Volume of cubic unit cell}} \times 100$

$= \frac{Z \times \frac{4}{3}\pi r^3}{a^3} \times 100$

For a simple cubic lattice, $a = 2r$ and $Z = 1$

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$\therefore$ Packing efficiency $= \dfrac{1\times\frac{4}{3}\pi r^3}{(2r)^3}\times 100$

$= \dfrac{\pi}{6}\times 100 = 52.4\%$

27. Given : Structure = *bcc*, a = 316.5 pm, r = ?
Using formula,

$$r = \frac{\sqrt{3}a}{4} = \frac{1.732\times 316.5}{4} = 137.04 \text{ pm}$$

28. a = 400 pm

For *fcc*, $r = \dfrac{a}{2\sqrt{2}}$

$$\therefore \quad r = \frac{400}{2\sqrt{2}} = \frac{400}{2\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}} = \frac{400\sqrt{2}}{4} = 100\sqrt{2}$$

$\Rightarrow$ r = 100 × 1.414 = 141.4 pm

29. (i) For *fcc* (or *ccp*), $a = 2\sqrt{2}r = 2\times 1.414\times 125$ pm
= 354 pm

(ii) a = 354 pm = 3.54 × 10^{-8} cm
Volume of one unit cell = a^3 = (3.54 × 10^{-8} cm)^3
= 4.436 × 10^{-23} cm^3

$$\text{Number of unit cells} = \frac{\text{Total volume}}{\text{Volume of one unit cell}}$$

$$= \frac{1 \text{ cm}^3}{4.44\times 10^{-23} \text{ cm}^3} = 2.25\times 10^{22}$$

30. As the atoms just touch each other on the diagonal across the face of unit cell, therefore
$b^2 = a^2 + a^2 = 2a^2$

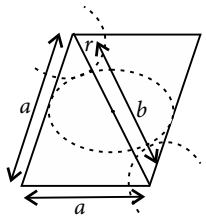
$b = \sqrt{2}a$ ...(i)
Also $b = r + 2r + r = 4r$ ...(ii)

From (i) and (ii), we get

$$4r = \sqrt{2}a \Rightarrow r = \frac{\sqrt{2}}{4}a;$$

$$r = \frac{1.414\times 409 \text{ pm}}{4}$$

= 144.58 pm



31. Given : d = 11.2 g cm^{-3}, Z = 4, a = 4 × 10^{-8} cm, M = ?, N_A = 6.022 × 10^{23} mol^{-1}

$$M = \frac{d\times a^3\times N_A}{Z}$$

$$= \frac{11.2\times(4\times 10^{-8})^3\times 6.022\times 10^{23}}{4} = 108 \text{ g mol}^{-1}$$

32. Given, Density of solid, d = 2.8 g cm^{-3}
For *fcc* unit cell, Z = 4
Edge length, a = 4 × 10^{-8} cm,
Molar mass, M = ?
N_A = 6.022 × 10^{23} mol^{-1}

Using formula, $d = \dfrac{Z\times M}{N_A\times a^3}$

or $M = \dfrac{d\times N_A\times a^3}{Z}$

Substituting these values, we get

$$M = \frac{2.8 \text{ g cm}^{-3}\times 6.022\times 10^{23}\text{ mol}^{-1}\times(4\times 10^{-8}\text{ cm})^3}{4}$$

or $M = \dfrac{2.8\times 6.022\times 6.4}{4} = 26.98 \text{ g mol}^{-1}$

33. Edge length = 200 pm
Volume of the unit cell = (200 × 10^{-10} cm)^3
= 8 × 10^{-24} cm^3
In a *fcc* unit cell there are four atoms per unit cell.

$\therefore$ Mass of unit cell = $\dfrac{200\times 4}{24\times 10^{23}}$ = 33.3 × 10^{-23}g

$$\text{Density} = \frac{\text{Mass of unit cell}}{\text{Volume of unit cell}} = \frac{33.3\times 10^{-23}\text{ g}}{8\times 10^{-24}\text{ cm}^3}$$

= 41.6 g cm^{-3}

34. Atomic mass, M = 108 u, d = 10.5 g/cm^3, a = 409 pm

We know, $d = \dfrac{Z\times M}{a^3\times 10^{-30}\times N_A}$ [$\because$ a is in pm]

$$Z = \frac{d\times a^3\times 10^{-30}\times N_A}{M}$$

$$Z = \frac{10.5\times(409)^3\times 10^{-30}\times 6.022\times 10^{23}}{108} = 4$$

35. For *fcc* structure, Z = 4, d = 7.2 g/cm^3
Let a be the edge length in pm

We know that density, $d = \dfrac{Z\times M}{a^3\times 10^{-30}}$

Mass per atom, $M = \dfrac{208}{4.2832\times 10^{24}}$,

$$a^3 = \frac{Z\times M}{d\times 10^{-30}} \quad (a \text{ is in pm})$$

$$= \frac{4\times 208}{7.2\times 4.2832\times 10^{24}\times 10^{-30}}$$

= 26.97 × 10^6 pm^3 $\Rightarrow$ a = (27 × 10^6)$^{1/3}$ = 300 pm

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36. We can determine the atomic mass of an unknown metal by using the formula of density of its unit cell.

$$d = \frac{Z(\text{No. of atoms per unit cell}) \times M(\text{Atomic mass})}{a^3(\text{Cell edge}) \times N_A\ (\text{Avogadro's number})}$$

By knowing density d, dimension of unit cell we can calculate M, the atomic mass of metal as for a given unit cell, Z is fixed and N_A is a universal constant.

37. For *fcc*, $Z = 4$

$$r = \frac{a}{2\sqrt{2}} \quad \text{or} \quad a = 2\sqrt{2}\, r$$

$a = 2 \times 1.414 \times 128$ pm $= 362$ pm

$a = 362 \times 10^{-10}$ cm

$$d = \frac{Z \times M}{a^3 \times N_A}$$

$$= \frac{4 \times 63.5\ \text{g/mol}}{(362 \times 10^{-10}\ \text{cm})^3 \times (6.02 \times 10^{23}\ \text{mol}^{-1})}$$

$= 8.9\ \text{g cm}^{-3}$

38. $d = \dfrac{Z \times M}{a^3 \times N_A \times 10^{-30}}$ (when a is in pm)

$$\Rightarrow \quad 10.8 = \frac{4 \times M}{(300)^3 \times 6.023 \times 10^{23} \times 10^{-30}}$$

$\Rightarrow \quad M = 43.9$ g

Number of atoms in 108 g of the element

$$= \frac{6.023 \times 10^{23}}{43.9} \times 108 = 14.82 \times 10^{23}$$

39. Density $(d) = \dfrac{Z \times M}{a^3 \times N_0}$

$$= \frac{2 \times 52}{(300 \times 10^{-10})^3 \times (6.022 \times 10^{23})}$$

$$= \frac{2 \times 52}{27 \times 10^{-24} \times 10^{23} \times 6.022} = 6.39\ \text{g cm}^{-3}$$

40. Given, $a = 400$ pm

For *fcc*, $Z = 4$

$$d = \frac{Z \times M}{N_A \times a^3} = \frac{4 \times 40}{6.022 \times 10^{23} \times (400 \times 10^{-10})^3}$$

$$= 4.15\ \text{g cm}^{-3}$$

Mass of one unit cell $= \dfrac{4 \times 40}{6.022 \times 10^{23}}$ g

No. of unit cells $= \dfrac{\text{Weight of cubic crystal}}{\text{Mass of one unit cell}}$

$$= \frac{6.022 \times 10^{23}}{4 \times 40} \times 4$$

$= 1.505 \times 10^{22}$ unit cells

41. Given, atomic mass (M) = 93 g mol^{-1},
density (d) = 11.5 g cm^{-3},
edge length (a) = 300 pm = 300×10^{-10} cm
Z = ?

We know that, $d = \dfrac{Z \times M}{N_A \times a^3}$

$$\text{or,} \quad Z = \frac{d \times N_A \times a^3}{M}$$

$$= \frac{11.5 \times 6.022 \times 10^{23} \times (3 \times 10^{-8})^3}{93} = 2.01 \approx 2$$

So, type of the unit cell is *bcc*.

42. Molar mass of element (M)

$$= \frac{300 \times 6.02 \times 10^{23}}{2 \times 10^{24}} = 90.3\ \text{g mol}^{-1}$$

$$\text{Density } (d) = \frac{Z \times M}{a^3 \times N_A}$$

$$= \frac{4 \times 90.3}{(250 \times 10^{-10})^3 \times 6.02 \times 10^{23}} = 38.4\ \text{g cm}^{-3}$$

43. Given, edge length $a = 500\ \text{pm} = 500 \times 10^{-10}$ cm
Mass of substance = 300 g
Number of atoms = ?
Density, $d = 7.5\ \text{g cm}^{-3}$
Using formula $d = \dfrac{Z \times M}{N_A \times a^3}$

$$\text{or} \quad M = \frac{d \times N_A \times a^3}{Z}$$

$$\text{or} \quad M = \frac{7.5\ \text{g cm}^{-3} \times 6.022 \times 10^{23}\ \text{mol}^{-1} \times (500 \times 10^{-10}\ \text{cm})^3}{2}$$

or $\quad M = 282.28$ g mol^{-1}

$$\text{Number of atoms} = \frac{6.022 \times 10^{23} \times 300}{282.28}$$

$= 6.40 \times 10^{23}$ atoms

44. $M = 27\ \text{g mol}^{-1}$, $a = 4.05 \times 10^{-8}$ cm, $d = 2.7\ \text{g cm}^{-3}$

$$d = \frac{Z \times M}{N_A \times a^3} \quad \therefore \quad Z = \frac{d \times N_A \times a^3}{M}$$

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$$Z = \frac{2.7 \times 6.022 \times 10^{23} \times (4.05 \times 10^{-8})^3}{27} = 4$$

As $Z = 4$, hence cubic unit cell is face centred cubic unit cell.

45. Given, $d = 6.23$ g cm^{-3}, $a = 4 \times 10^{-8}$ cm,
$M = 60$ g mol^{-1}
$N_A = 6.023 \times 10^{23}$, Z = ?

Using formula $d = \frac{Z \times M}{N_A \times a^3}$ or $Z = \frac{d \times N_A \times a^3}{M}$

$$Z = \frac{6.23 \text{ g cm}^{-3} \times 6.023 \times 10^{23} \times (4 \times 10^{-8} \text{ cm})^3}{60 \text{ g mol}^{-1}}$$

or $Z = 4.0$

Hence, structure is *fcc*.

46. Let us assume $Z = 4$ (copper has *fcc*)

$$d = \frac{Z \times M}{a^3 \times N_A}$$

$$8.95 = \frac{4 \times 63.5}{a^3 \times 6.02 \times 10^{23}}$$

$a^3 = 47.145 \times 10^{-24}$ cm^3

$$a = \sqrt[3]{47.145 \times 10^{-24}}$$

$a = 3.612 \times 10^{-8}$ cm = 361.2 pm

For *fcc*, $4r = \sqrt{2}a$

It shows Cu has *fcc* structure,

$$r = \frac{a}{2\sqrt{2}} = \frac{361.2}{2 \times 1.414} = 127.7 \text{ pm}$$

47. Given : For *bcc*, $Z = 2$
$a = 286.65$ pm $= 2.87 \times 10^{-8}$ cm
$d = 7.87$ g cm^{-3}, N_A = ?

Using formula $d = \frac{Z \times M}{N_A \times a^3}$

or, $N_A = \frac{Z \times M}{d \times a^3} = \frac{2 \times 56}{7.87 \times (2.87 \times 10^{-8})^3}$

or, $N_A = 6.022 \times 10^{23}$

48. $a = 250$ pm
$d = 8.55$ g cm^{-3}; $Z = 2$

$$d = \frac{Z \times M}{a^3 \times N_A \times 10^{-30}}$$

$$M = \frac{d \times a^3 \times N_A \times 10^{-30}}{Z}$$

$$= \frac{8.55 \times 250 \times 250 \times 250 \times 6.02 \times 10^{23} \times 10^{-30}}{2}$$

$M = 40.21$ g mol^{-1}

bcc unit cell body diagonal is equal to 4 times the radius of atom.

$\therefore \quad 4r = \sqrt{3}\,a = \sqrt{3} \times 250$

$4r = 1.732 \times 250 \Rightarrow r = 108$ pm

49. Given density (d) = 8.55 g cm^{-3}
Atomic radius (r) = ?
Atomic mass (M) = 93 u = 93 g mol^{-1}
For *bcc* structure, $Z = 2$

Using formula, $d = \frac{Z \times M}{N_A \times a^3}$ or $a^3 = \frac{Z \times M}{N_A \times d}$

or $a^3 = \frac{2 \times 93 \text{ g mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1} \times 8.55 \text{ g cm}^{-3}}$

or $a^3 = 36.1 \times 10^{-24}$ cm^3

or $a = 3.304 \times 10^{-8}$ cm = 330.4 pm

Again for *bcc* $r = \frac{\sqrt{3}a}{4} = \frac{1.732 \times 330.4}{4}$ pm

or $r = 143.1$ pm

50. Given : Structure *fcc*, hence $Z = 4$, $r = 127.8$ pm,
$M = 63.55$ u, $N_A = 6.02 \times 10^{23}$ mol^{-1}, d = ?

Using formula, $d = \frac{Z \times M}{N_A \times a^3}$

we get,

$$d = \frac{4 \times 63.55 \text{ g mol}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1} \times (2\sqrt{2} \times 127.8 \times 10^{-10} \text{ cm})^3}$$

$= 8.940$ g/cm^3

51. $d = \frac{Z \times M}{a^3 \times N_A} \Rightarrow a^3 = \frac{Z \times M}{d \times N_A}$ ...(i)

For *fcc* unit cell, $Z = 4$
Given, $M = 207$ g mol^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}
$d = 11.35$ g cm^{-3}
Substituting these values in equation (i), we get

$$a^3 = \frac{4 \times 207 \text{ g mol}^{-1}}{11.35 \text{ g cm}^{-3} \times 6.02 \times 10^{23} \text{ mol}^{-1}}$$

$$a^3 = \frac{4 \times 207 \times 10}{11.35 \times 6.02 \times 10^{24}} \text{ cm}^3$$

$$\Rightarrow \quad a = \left(\frac{8280}{11.35 \times 6.02}\right)^{1/3} \times 10^{-8} \text{ cm}$$

$\therefore \quad a = 4.949 \times 10^{-8}$ cm $\Rightarrow a = 494.9$ pm

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For *fcc*, $r = \frac{a}{2\sqrt{2}}$

$\therefore \quad r = \frac{494.9}{2\sqrt{2}}\text{ pm} = \frac{494.9\sqrt{2}}{4}\text{ pm} = \frac{494.9 \times 1.414}{4}\text{ pm}$

$\therefore \quad r = 174.95\text{ pm}$

52. In FeO crystal some of the Fe^{2+} ions are replaced by Fe^{3+} ions. Three Fe^{2+} ions are replaced by two Fe^{3+} ions to maintain electrical neutrality. Eventually there will be less amount of metal ($Fe_{0.95}O$) as compared to stoichiometric proportion (FeO).

53. Since Schottky defect is shown by highly ionic compounds having small difference in the size of cations and anions, whereas Frenkel defect is shown by compounds having large difference in the size of cations and anions. Therefore, NaCl exhibits Schottky defect.

54. Initially ZnO is white in colour at room temperature but on heating it loses oxygen which creates anion vacancy defect and it turns yellow.

$$ZnO \xrightarrow{\text{heating}} Zn^{2+} + \frac{1}{2}O_2 + 2e^-$$

55. Frenkel defect is shown by those ionic substances which have large difference in the size of ions. Hence, AgCl shows Frenkel defect but NaCl does not.

56. When $CdCl_2$ is added to AgCl, a cationic vacancy defect is created *i.e.*, an Ag^+ ion from the lattice is absent from its position due to presence of adjacent Cd^{2+} ion.

57. Schottky defect : The defect in which cations and anions are missing in the stoichiometric ratio of compound is called Schottky defect. Schottky defect is actually vacancy defect in ionic solids.

In this defect, electrical neutrality is maintained. In this defect, density of solid decreases. Schottky defect is shown by ionic solids in which the cation and anion are of almost similar sizes.

Examples **:** NaCl, KCl, CsCl, AgBr, etc.

58. Frenkel defect does not change the density of the solid.

59. *Refer to answer 57.*

60. *Refer to answer 57.*

61. Frenkel defect

62. Addition of $CdCl_2$ to AgCl crystal causes impurity defect.

In this defect one Cd^{2+} ion replaces two Ag^+ ions from the crystal. One site is occupied by one Cd^{2+} ion and other site remains vacant.

63. Schottky defects occur when cations and anions are missing from their lattice site. Mass of unit cell decreases which decreases the density of the solid.

64. *Refer to answer 62.*

65. *Refer to answer 57.*

66. Interstitial defect

67. Frenkel defect is shown by ionic solids in which difference in size of cations and anions is large. Examples : ZnS, AgI and AgBr.

68. On heating a solid vacancy defect is produced in the crystal. This is because on heating, some lattice sites become vacant.

69. (i) Schottky defect

(ii) Density of the crystal decreases.

(iii) This defect is shown by ionic substances in which the cation and anion are of almost similar sizes.

70. (i) Metal excess defect is responsible for pink colour of LiCl. It is also known as anion vacancy defect.

(ii) NaCl crystal shows Schottky defect.

71. The number of cation vacancies created in the lattice of NaCl is equal to the number of divalent Sr^{2+} ions added.

Concentration of $Sr^{2+} = 10^{-3}$ mol %

$$= \frac{10^{-3}}{100} = 10^{-5}\text{ mol}$$

1 mole of $Sr^{2+} = 6.023 \times 10^{23} Sr^{2+}$ ions

10^{-5} mol of $Sr^{2+} = 6.023 \times 10^{23} \times 10^{-5}$

$= 6.023 \times 10^{18}\ Sr^{2+}$ ions

Hence, the concentration of cation vacancies is 6.023×10^{18}.

72. Silicon when doped with phosphorus, which contains five valence electrons, they occupy some of the lattice sites in silicon crystal. Four out of five electrons are used in the formation of four covalent bonds with the four neighboring silicon atoms.

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The fifth electron is extra and becomes delocalised. These delocalised electrons increases the conductivity of doped silicon.

73. Since group 15 elements (*e.g.*, phosphorus) have one electron extra then group 14 elements (*e.g.*, silicon) after forming four covalent bonds. Thus, the extra free electron is responsible for the formation of *n*-type semiconductor.

74. Groups 12-16 compounds are formed by the combination of elements of group-12 and group-16. In groups 12-16 compounds, the bonds are not perfectly covalent and the ionic character depends upon the electronegativities of the two elements, *e.g.*, ZnS.

75. When germanium (Ge) is doped with aluminium (Al), *p*-type semiconductor is formed.

76. *n*-type semiconductor is formed when silicon is doped with arsenic.

77. *p*-type semiconductor is obtained when silicon is doped with boron.

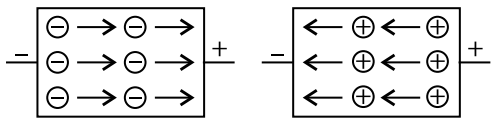
78. Addition of an appropriate amount of suitable impurity in a crystalline solid is called doping.
Doping is done to increase the conductivity of intrinsic semiconductors.

79. The conductivity of intrinsic semiconductor like silicon is too low to be of practical use. This conductivity is increased by adding an appropriate amount of suitable impurity like Al or As which is electron deficient or electron rich.

80. A semiconductor in which doped impurity has more valence electrons than the pure element is called *n*-type semiconductor. *e.g.,* Ge or Si (Group-14) doped with P or As (Group-15).

81. Pure substance which acts as semiconductor is called intrinsic semiconductor. *e.g.,* Si and Ge at high temperature.

82. The solids which have conductivities between 10^{-6} to 10^{4} ohm^{-1} m^{-1} are called semiconductors. *e.g.,* Germanium and silicon. The two main types of semiconductors are as follows :
(i) ***n*-type semiconductor :** When a silicon crystal is doped with atoms of group-15 elements, such as P, As, Sb or Bi then only four of the five valence electrons of each impurity atom participate in forming covalent bonds and fifth electron is almost free to conduct electricity. Group-14 elements doped with a group-15 elements are called *n*-type semiconductors.
(ii) ***p*-type semiconductor :** When a silicon crystal is doped with atoms of group-13 elements, such as B, Al, Ga or In. Each impurity atom form only three covalent bonds with the host atom. The place where the fourth electron is missing is caused a hole which move through the crystal like a positive charge and hence increases its conductivity. Group-14 elements doped with group-13 elements are called *p*-type semiconductors.



n-type semiconductor *p*-type semiconductor

83. *Refer to answer 82.*

84. Ferrimagnetic substances have a net dipole moment due to unequal parallel and antiparallel alignment of magnetic moments whereas antiferromagnetic substances have net magnetic moment zero due to compensatory alignment of magnetic moments. Therefore, ferrimagnetic substances show better magnetism than antiferromagnetic substances.

85. Ferromagnetism

86. Ferromagnetism : Materials which are strongly attracted by magnetic field are called ferromagnetic materials and the property thus exhibited is called ferromagnetism.
e.g., Fe, Co, Ni show ferromagnetism at room temperature.

87. Antiferromagnetism

88. Ferrimagnetism

89. Antiferromagnetism : Paramagnetic substances which show very poor magnetic effect are antiferromagnetic and this property is called antiferromagnetism.
Examples : MnO, Fe_2O_3, Cr_2O_3.

90. Ferromagnetic substances because these substances become a permanent magnet.

91. Ferromagnetism

92. Ferromagnetism : Substances which are very strongly attracted by the magnetic field are called ferromagnetic and this property is called ferromagnetism.

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Examples : Iron, cobalt, nickel, gadolinium and CrO_2.

These substances can be permanently magnetised *i.e.,* they retain magnetic property even in absence of applied magnetic field.

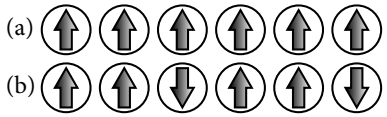
In ferromagnetic substances all the domains (tiny magnet formed by grouping of metal ions in small region) are oriented in the direction of the applied magnetic field and produce a strong magnetic effect. This ordered orientation of domain persists even when the magnetic field is removed.

Ferrimagnetism : Substances which are weakly attracted by the magnetic field as compared to ferromagnetic substances are called ferrimagnetic and this property is called ferrimagnetism.

Examples : Fe_3O_4 (Magnetite), Ferrites ($MgFe_2O_4$, $ZnFe_2O_4$.)

Ferrimagnetism is observed when the magnetic domains are aligned in parallel and antiparallel directions in unequal numbers. Hence, net magnetic moment is never zero.

These substances lose ferrimagnetism and become paramagnetic on heating.



Schematic alignment of magnetic moments in (a) ferromagnetic (b) ferrimagnetic.

93. (a) *Refer to answer 92.*

(b) *Refer to answer 89.*

94. *Refer to answers 92.*

Paramagnetism : Materials which are weakly attracted by magnetic fields are called paramagnetic materials and this property is called paramagnetism.

Paramagnetic substances contain unpaired electrons. *e.g.,* TiO, CuO, O_2 and VO_2, etc.

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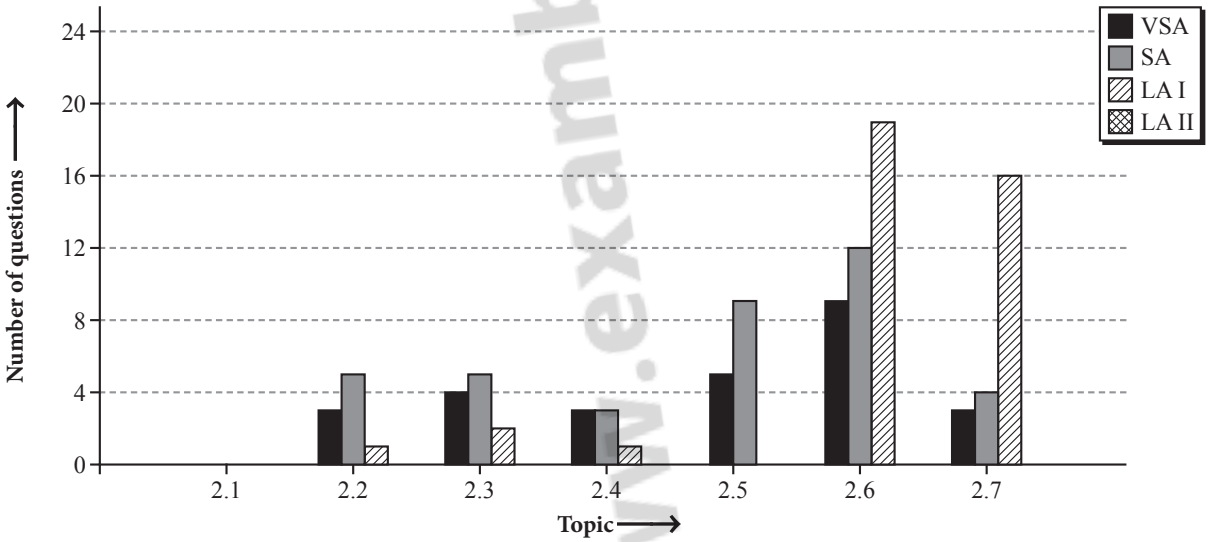
CHAPTER
2

Solutions

2.1 Types of Solutions
2.2 Expressing Concentration of Solutions
2.3 Solubility
2.4 Vapour Pressure of Liquid Solutions
2.5 Ideal and Non-ideal Solutions
2.6 Colligative Properties and Determination of Molar Mass
2.7 Abnormal Molar Masses

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Colligative Properties and Determination of Molar Mass.*
- Maximum VSA type questions were asked from *Colligative Properties and Determination of Molar Mass.*
- Maximum SA type questions were asked from *Ideal and Non-Ideal Solutions* and *Colligative Properties and Determination of Molar Mass.*
- Maximum LA I type questions were asked from *Colligative Properties and Determination of Molar Mass.*

QUICK RECAP

- **Solution :** A perfectly homogeneous mixture of two or more components is called *solution*.
- **Solute :** The component which is present in lesser amount or whose physical state is changed during the formation of solution is called *solute*.

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Solvent : The component which is present in larger amount and determines the physical state of the solution is called *solvent*.

Types of solution : Depending upon the nature of solute and solvent, solutions are classified as follows :

- **Gaseous solutions :** Solutions in which gas acts as solvent.

Solute	Solvent	Examples
Solid	Gas	Iodine vapours in air, dust or smoke particles in air
Liquid	Gas	Water vapours in air (humidity)
Gas	Gas	Air

- **Liquid solutions :** Solutions in which liquids are present in larger amount.

Solute	Solvent	Examples
Solid	Liquid	Salt/sugar in water
Liquid	Liquid	Alcohol in water
Gas	Liquid	Aerated drinks, O_2 in water

- **Solid solutions :** Solutions in which solids are present in larger amount.

Solute	Solvent	Examples
Solid	Solid	Alloys
Liquid	Solid	Hg in Ag, Hg in Zn (Amalgam)
Gas	Solid	Solution of hydrogen in palladium.

Different methods for expressing concentration of solution :

Method of expressing	Formula
Mass percentage	$\left(\frac{w}{W}\right)\% = \frac{w_2}{(w_1 + w_2)} \times 100$
Volume percentage	$\left(\frac{v}{V}\right)\% = \frac{V_2}{(V_1 + V_2)} \times 100$
Mass by volume percentage	$\left(\frac{w}{V}\right)\% = \frac{w_2}{V_{\text{solution}}\,(\text{in mL})} \times 100$
Mass fraction	$x_1 = \frac{w_1}{w_1 + w_2}$ or $x_2 = \frac{w_2}{w_1 + w_2}$
Strength ($g\ L^{-1}$)	$\frac{w_2\ (\text{in g})}{V_{\text{solution}}\,(\text{in mL})} \times 1000$
Parts per million (ppm)	$\frac{w_2}{(w_1 + w_2)} \times 10^6$
Molarity, (M) ($\text{mol}\ L^{-1}$)	$\frac{w_2 \times 1000}{M_2 \times V_{\text{solution}}\,(\text{in mL})}$
Molality, (m) ($\text{mol}\ kg^{-1}$)	$\frac{w_2 \times 1000}{M_2 \times w_1 (\text{in g})}$
Mole fraction, (x)	$x_1 = \frac{n_1}{n_1 + n_2}$ or $x_2 = \frac{n_2}{n_1 + n_2}$ $(x_1 + x_2 = 1)$ In case of gases only, $y_1 = \frac{p_1}{p_1 + p_2}$ or $y_2 = \frac{p_2}{p_1 + p_2}$ $(y_1 + y_2 = 1)$

Solubility : Maximum amount of substance that can be dissolved in a specified amount of solvent at a specified temperature is called its *solubility*.

Factors affecting solubility of a solid in a liquid :

- **Nature of solute and solvent :** Polar solutes dissolve in polar solvents and non-polar solutes dissolve in non-polar solvents. (*i.e.*, like dissolves like).
- **Effect of temperature :**
  - If the dissolution process is endothermic ($\Delta_{sol}H > 0$), the solubility **increases** with **rise** in temperature.
  - If dissolution process is exothermic ($\Delta_{sol}H < 0$), the solubility **decreases** with **rise** in temperature.
- **Effect of pressure :** Pressure does not have any significant effect on solubility of solids in liquids as these are highly incompressible.

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- **Factors affecting solubility of a gas in a liquid :**
  - **Effect of pressure : Henry's law** states that "the partial pressure of the gas in vapour phase (p) is proportional to the mole fraction of the gas (x) in the solution"
    $p = K_H x$ where, K_H is the Henry's law constant and is different for different gases at a particular temperature.
    Higher the value of K_H at a given pressure, the lower is the solubility of the gas in the liquid.
  - **Effect of temperature :** As dissolution is an exothermic process, then according to Le Chatelier's principle, the solubility should decrease with increase of temperature.
- **Raoult's law :** It states that for a solution of volatile liquids, the partial vapour pressure of each component of the solution is directly proportional to its mole fraction present in solution. $p_1 = p_1^\circ x_1$ and $p_2 = p_2^\circ x_2$; where p_1° and p_2° are vapour pressures of pure components 1 and 2 respectively, at the same temperature.
- **Dalton's law of partial pressures :**

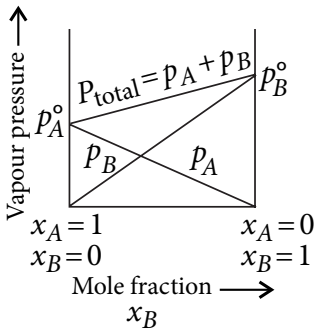
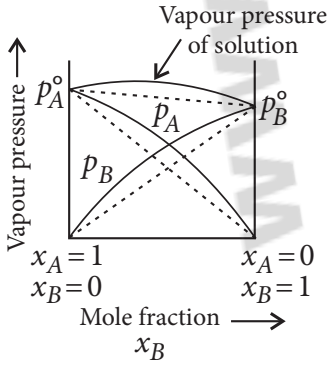
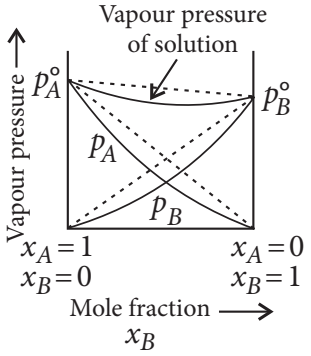
$$P_{total} = p_1 + p_2 = x_1 p_1^\circ + x_2 p_2^\circ$$
$$= (1 - x_2)p_1^\circ + x_2 p_2^\circ = p_1^\circ + (p_2^\circ - p_1^\circ) x_2$$

- If y_1 and y_2 are the mole fractions of the components 1 and 2 respectively in the vapour phase then, $p_1 = y_1 P_{total}$ and $p_2 = y_2 P_{total}$
- **Raoult's law for solid-liquid solutions :** It states that relative lowering in vapour pressure of a solution containing a non-volatile solute is equal to the mole fraction of the solute.

$$\frac{p^\circ - p_s}{p^\circ} = x_2$$

where, p° = vapour pressure of pure solvent
p_s = vapour pressure of solution
x_2 = mole fraction of solute.

- **Ideal and non-ideal solutions :**

Ideal solutions	Non-ideal solutions	
	Positive deviation from Raoult's law	Negative deviation from Raoult's law
		
$A - B$ interactions $\approx A - A$ and $B - B$ interactions	$A - B$ interactions $<< A - A$ and $B - B$ interactions	$A - B$ interactions $>> A - A$ and $B - B$ interactions
$\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$	$\Delta H_{mix} > 0$, $\Delta V_{mix} > 0$	$\Delta H_{mix} < 0$, $\Delta V_{mix} < 0$
e.g., dilute solutions, benzene + toluene, *n*-hexane + *n*-heptane	*e.g.*, acetone + ethanol, acetone + CS_2, water + methanol	*e.g.*, acetone + aniline, acetone + chloroform, HNO_3 + water

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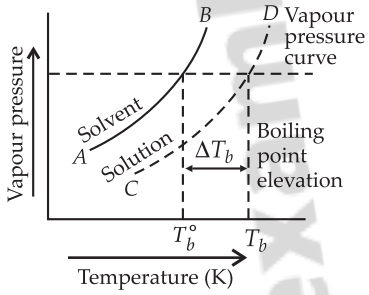
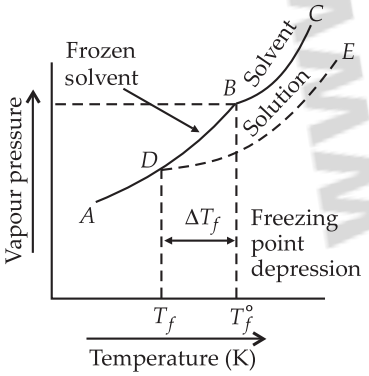
- **Azeotropes :** The mixtures of liquids which boil at constant temperature like a pure liquid and possess same composition of components in liquid as well as vapour phase are called *constant boiling mixtures* or *azeotropic mixtures*.
  - **Minimum boiling azeotropes :** They formed by those liquid pairs which show positive deviations from ideal behaviour *e.g.*, ethanol-water mixture.
  - **Maximum boiling azeotropes :** They are formed by those liquid pairs which show negative deviations from ideal behaviour, *e.g.*, nitric acid-water mixture.
- **Colligative properties :** Properties which depend only on the number of solute particles dissolved in a definite amount of the solvent and do not depend on the nature of the solute are called *colligative properties*.

Colligative properties	Expression
Relative lowering of vapour pressure : When a non-volatile solute is dissolved in a solvent, vapour pressure of the solution is lower than that of the pure solvent which is known as *lowering of vapour pressure*. Relative lowering of vapour pressure is equal to the mole fraction of the solute in the solution.	$\frac{p_1^\circ - p_1}{p_1^\circ} = x_2 = \frac{n_2}{n_1 + n_2}$ $= \frac{n_2}{n_1} = \frac{w_2 \times M_1}{M_2 \times w_1}$ ($\because$ for dilute solutions, $n_2 << n_1$)
Elevation in boiling point : The boiling point of a solution containing a non-volatile solute is always higher than the boiling point of the pure solvent. This increase in boiling point is termed as *elevation in boiling point*. 	$\Delta T_b = T_b - T_b^\circ$ $\Delta T_b \propto m$ or $\Delta T_b = K_b m$ $= K_b \left(\frac{w_2 \times 1000}{M_2 \times w_1 (\text{in g})} \right)$ or $M_2 = \frac{K_b \times w_2 \times 1000}{\Delta T_b \times w_1 (\text{in g})}$ K_b is called *boiling point elevation constant* or *molal elevation constant* or *ebullioscopic constant*, having unit K kg mol^{-1}.
Depression in freezing point : The freezing point of a solution containing a non-volatile solute is always less than the freezing point of the pure solvent. This decrease in freezing point is termed as *depression in freezing point*. 	$\Delta T_f = T_f^\circ - T_f$ $\Delta T_f \propto m$ or $\Delta T_f = K_f m$ $= K_f \left(\frac{w_2 \times 1000}{M_2 \times w_1 (\text{in g})} \right)$ or $M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1 (\text{in g})}$ K_f is known as *freezing point depression constant* or *molal depression constant* or *cryoscopic constant*, having unit K kg mol^{-1}.
Osmotic pressure : The movement of solvent molecules from less concentrated solution to more concentrated solution through semipermeable membrane is termed as *osmosis*. The hydrostatic pressure which develops on account of osmosis is called *osmotic pressure* or the excess pressure that must be applied on the solution to prevent osmosis is called *osmotic pressure*.	$\pi = CRT = \left(\frac{n_2}{V} \right) RT,$ $\pi V = \frac{w_2 RT}{M_2}$ or $M_2 = \frac{w_2 RT}{\pi V}$

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- Two solutions having same osmotic pressure at a given temperature are called *isotonic solutions*.
- If one solution is of lower osmotic pressure, it is called *hypotonic* with respect to the more concentrated solution. The more concentrated solution is said to be *hypertonic* with respect to the dilute solution.
- If a pressure higher than the osmotic pressure is applied on the solution, the solvent will flow from the solution into the pure solvent through the semipermeable membrane and the process is called *reverse osmosis*. It is used in *desalination of sea water*.

Abnormal molecular mass : When the molecular mass of a substance determined by any of the colligative properties comes out to be different than the expected value, the substance is said to show *abnormal molecular mass*.

- Abnormal molecular masses are observed when the solution is non-ideal (not dilute) or the solute undergoes *association* or *dissociation*.

van't Hoff Factor : It is defined as the ratio of the experimental value of the colligative property to the calculated value of the colligative property.

$$i = \frac{\text{Observed value of the colligative property}}{\text{Calculated value of the colligative property}}$$

$$i = \frac{\text{Calculated molecular mass}}{\text{Observed molecular mass}}$$

$$i = \frac{\text{(Total number of moles of particles after association / dissociation)}}{\text{(Total number of moles of particles before association / dissociation)}}$$

- If $i > 1$, solute undergoes dissociation in the solution and if $i < 1$, solute undergoes association in the solution.

$$\alpha_{\text{dissociation}} = \frac{i-1}{n-1}$$

$$\alpha_{\text{association}} = \frac{1-i}{1-\frac{1}{n}}$$

- For substances undergoing association or dissociation in the solution, the various expressions for the colligative properties are modified as follows :

$$\frac{p_1^\circ - p_1}{p_1^\circ} = ix_2 \,;\; \Delta T_b = iK_b m;$$

$$\Delta T_f = iK_f m;$$

$$\pi = iCRT$$

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Previous Years' CBSE Board Questions

2.2 Expressing Concentration of Solutions

VSA (1 mark)

1. Define the following term :
Molality (*m*) *(1/2, Delhi 2017, 1/5 AI 2014C)*

2. Define the following term :
Molarity (*M*) *(1/2, Delhi 2017, 1/5 AI 2014)*

3. Define the following term :
Mole fraction
(1/5 AI 2014 C, 1/2, Delhi 2012, AI 2012)

SA (2 marks)

4. Calculate the molarity of 9.8% (*w*/*W*) solution of H_2SO_4 if the density of the solution is 1.02 g mL^{-1}. (Molar mass of H_2SO_4 = 98 g mol^{-1}) *(2/5, Foreign 2014)*

5. Differentiate between molarity and molality of a solution. How can we change molality value of a solution into molarity value?
(Delhi 2014C)

6. A solution of glucose ($C_6H_{12}O_6$) in water is labelled as 10% by weight. What would be the molality of the solution?
(Molar mass of glucose = 180 g mol^{-1})
(2/5, AI 2013)

7. Differentiate between molarity and molality in a solution. What is the effect of temperature change on molarity and molality in a solution? *(2/5, Delhi 2011, 2/5, AI 2011)*

8. Differentiate between molarity and molality of a solution. Explain how molarity value of a solution can be converted into its molality?
(Foreign 2011)

LA I (3 marks)

9. A solution of glucose (molar mass = 180 g mol^{-1}) in water is labelled as 10% (by mass). What would be the molality and molarity of the solution? (Density of solution = 1.2 g mL^{-1}) *(3/5, AI 2014)*

2.3 Solubility

VSA (1 mark)

10. Why aquatic animals are more comfortable in cold water than in warm water?
(1/3, 2018)

11. Gas (*A*) is more soluble in water than gas (*B*) at the same temperature. Which one of the two gases will have the higher value of K_H (Henry's constant) and why?
(1/2, AI 2016)

12. Explain the following :
Henry's law about dissolution of a gas in a liquid. *(1/5, AI 2012)*

13. State the following :
Henry's law about partial pressure of a gas in a mixture. *(1/5, Delhi, AI 2011)*

SA (2 marks)

14. State Henry's law. Why is air diluted with helium in the tanks used by scuba divers?
(2/5, 2020)

15. State Henry's law. Calculate the solubility of CO_2 in water at 298 K under 760 mm Hg. (K_H for CO_2 in water at 298 K is 1.25×10^6 mm Hg) *(2020)*

16. Give reasons for the following.
(a) Aquatic species are more comfortable in cold water than in warm water.
(b) At higher altitudes people suffer from anoxia resulting in inability to think.
(AI 2019)

17. State Henry's law and mention two of its important applications.
(2/5, AI 2013C, 2012C)

18. Explain why aquatic species are more comfortable in cold water rather than in warm water. *(Delhi 2012C)*

LA I (3 marks)

19. The partial pressure of ethane over a saturated solution containing 6.56×10^{-2} g of ethane is 1 bar. If the solution contains 5.0×10^{-2} g of ethane, then what will be the partial pressure of the gas?
(Delhi 2013C, AI 2012C)

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20. If N_2 gas is bubbled through water at 293 K, how many millimoles of N_2 gas would dissolve in 1 litre of water? Assume that N_2 exerts a partial pressure of 0.987 bar. Given that Henry's law constant for N_2 at 293 K is 76.48 kbar. *(AI 2012C)*

2.4 Vapour Pressure of Liquid Solutions

VSA (1 mark)

21. Identify which liquid will have a higher vapour pressure at 90°C if the boiling points of two liquids *A* and *B* are 140°C and 180°C, respectively. *(One word, 2020)*

22. Define Raoult's law. *(1/5 AI 2014C)*

23. State the following :
Raoult's law in its general form in reference to solutions. *(1/5, Delhi, 1/2, AI 2011)*

SA (2 marks)

24. State Raoult's law for the solution containing volatile components. What is the similarity between Raoult's law and Henry's law?
(Delhi 2014, AI 2013)

25. State Raoult's law for a solution containing volatile components. Name the solution which follows Raoult's law at all concentrations and temperatures. *(2/5, Foreign 2014)*

26. State Raoult's law. How is it formulated for solutions of non-volatile solutes?
(Delhi 2012C)

LA I (3 marks)

27. The vapour pressure of pure liquids *A* and *B* are 450 and 700 mm Hg respectively, at 350 K. Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg. Also find the composition of the vapour phase. *(3/5, AI 2013C)*

2.5 Ideal and Non-ideal Solutions

VSA (1 mark)

28. What happens when acetone is added to pure ethanol? *(1/2, 2020)*

29. Define the following term :
Ideal solution
(1/2, Delhi 2017, 1/5, AI 2017C, 1/5, AI 2013, 2012, 1/2, Delhi 2012)

30. In non-ideal solution, what type of deviation shows the formation of maximum boiling azeotropes? *(1/2, AI 2016)*

31. Some liquids on mixing form 'azeotropes'. What are 'azeotropes'? *(Delhi 2014)*

32. Define the following term :
Azeotrope *(1/5, Foreign 2014)*

SA (2 marks)

33. Write two differences between ideal solutions and non-ideal solutions.
(Delhi 2019, 2/5, AI 2017)

34. What type of azeotropic mixture will be formed by a solution of acetone and chloroform? Justify on the basis of strength of intermolecular interactions that develop in the solution. *(AI 2019)*

35. What is meant by positive deviations from Raoult's law? Give an example. What is the sign of $\Delta_{mix}H$ for positive deviation?
(Delhi 2015)

36. Define azeotropes. What type of azeotrope is formed by positive deviation from Raoult's law? Give an example. *(Delhi 2015)*

37. What is meant by negative deviation from Raoult's law? Give an example. What is the sign of $\Delta_{mix}H$ for negative deviation?
(Foreign 2015)

38. Define azeotropes. What type of azeotrope is formed by negative deviation from Raoult's law? Give an example. *(Foreign 2015)*

39. What type of deviation is shown by a mixture of ethanol and acetone? Give reason.
(2/5, AI 2014)

40. What is meant by positive and negative deviations from Raoult's law and how is the sign of $\Delta_{mix}H$ related to positive and negative deviations from Raoult's law?
(AI 2013 C)

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41. Explain why a solution of chloroform and acetone shows negative deviation from Raoult's law. *(2/5, Delhi 2011C)*

2.6 Colligative Properties and Determination of Molar Mass

VSA (1 mark)

42. **Assertion (A) :** Osmotic pressure is a colligative property.
Reason (R) : Osmotic pressure is directly proportional to molarity.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

43. What happens when a pressure greater than osmotic pressure is applied on the solution side separated from solvent by a semi-permeable membrane? *(1/2, 2020)*

44. Give reason for the following :
Measurement of osmotic pressure method is preferred for the determination of molar masses of macromolecules such as proteins and polymers. *(1/2, 2018)*

45. Define the following term :
Colligative properties *(1/2, Delhi 2017)*

46. Define the following term :
Osmotic pressure *(1/5, AI 2017C, 2013)*

47. What are isotonic solutions?
(Delhi 2014; 1/2, Delhi 2012)

48. Define the following term :
Molal elevation constant (K_b) *(1/5, AI 2014)*

49. How is the vapour pressure of a solvent affected when a non-volatile solute is dissolved in it? *(1/2, Delhi 2014C)*

50. Explain the following :
Boiling point elevation constant for a solvent.
(AI 2012)

SA (2 marks)

51. For a 5% solution of urea (Molar mass = 60 g/mol), calculate the osmotic pressure at 300 K.[R = 0.0821 L atm K^{-1} mol^{-1}] *(2020)*

52. Visha took two aqueous solutions – one containing 7.5 g of urea (Molar mass = 60 g/mol) and the other containing 42.75 g of substance Z in 100 g of water, respectively. It was observed that both the solutions froze at the same temperature. Calculate the molar mass of Z. *(2020)*

53. Give reasons :
(i) 0.1 M KCl has higher boiling point than 0.1 M glucose.
(ii) Meat is preserved for a longer time by salting. *(2020)*

54. Calculate the freezing point of a solution containing 60 g of glucose (molar mass = 180 g mol^{-1}) in 250 g of water.
(K_f of water = 1.86 K kg mol^{-1}) *(2018)*

55. (i) Out of 1 M glucose and 2 M glucose, which one has a higher boiling point and why?
(ii) What happens when the external pressure applied becomes more than the osmotic pressure of solution?
(2/5, Delhi 2016)

56. Blood cells are isotonic with 0.9% sodium chloride solution. What happens if we place blood cells in a solution containing
(i) 1.2% sodium chloride solution?
(ii) 0.4% sodium chloride solution?
(2/5, Delhi 2016)

57. Why does a solution containing non-volatile solute have higher boiling point than the pure solvent? Why is elevation of boiling point a colligative property? *(AI 2015)*

58. Calculate the mass of compound (molar mass = 256 g mol^{-1}) to be dissolved in 75 g of benzene to lower its freezing point by 0.48 K.
(K_f = 5.12 K kg mol^{-1}). *(Delhi 2014)*

59. 18 g of glucose, $C_6H_{12}O_6$ (Molar mass = 180 g mol^{-1}) is dissolved in 1 kg of water in a sauce pan. At what temperature will this solution boil?
(K_b for water = 0.52 K kg mol^{-1}, boiling point of pure water = 373. 15 K) *(Delhi 2013)*

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60. An aqueous solution of sodium chloride freezes below 273 K. Explain the lowering in freezing point of water with the help of a suitable diagram. *(Delhi 2013C)*

61. Define the terms osmosis and osmotic pressure. Is the osmotic pressure of a solution a colligative property? Explain.
(2/5, Delhi 2011)

62. List any four factors on which the colligative properties of a solution depend.
(2/5, AI 2011C)

LA I (3 marks)

63. Calculate the mass of ascorbic acid (Molar mass = 176 g mol^{-1}) to be dissolved in 75 g of acetic acid, to lower its freezing point by 1.5°C .
(K_f = 3.9 K kg mol^{-1}) *(2020)*

64. A 4% solution (*w*/*w*) of sucrose (M = 342 g mol^{-1}) in water has a freezing point of 271.15 K. Calculate the freezing point of 5% glucose (M = 180 g mol^{-1}) in water. (Given : Freezing point of pure water = 273.15 K)
(Delhi 2019)

65. At 300 K, 30 g of glucose present in a litre of its solution has an osmotic pressure of 4.98 bar. If the osmotic pressure of a glucose solution is 1.52 bar at the same temperature, what would be its concentration?*(AI 2019)*

66. A 10% solution (by mass) of sucrose in water has freezing point of 269.15 K. Calculate the freezing point of 10% glucose in water, if freezing point of pure water is 273.15 K. (Given : Molar mass of sucrose = 342 g mol^{-1}, molar mass of glucose = 180 g mol^{-1})
(Delhi 2017)

67. 30 g of urea (M = 60 g mol^{-1}] is dissolved in 846 g of water. Calculate the vapour pressure of water for this solution if vapour pressure of pure water at 298 K is 23.8 mm Hg.
(3/5, AI 2017)

68. Calculate the boiling point elevation for a solution prepared by adding 10 g of $CaCl_2$ to 200 g of water. (K_b for water = 0.52 K kg mol^{-1}, molar mass of $CaCl_2$ = 111 g mol^{-1})
(3/5, AI 2017C, Foreign 2014)

69. Calculate the freezing point of the solution when 31 g of ethylene glycol ($C_2H_6O_2$) is dissolved in 500 g of water.
(K_f for water = 1.86 K kg mol^{-1}) *(AI 2015)*

70. A solution containing 15 g urea (molar mass = 60 g mol^{-1}) per litre of solution in water has the same osmotic pressure (isotonic) as a solution of glucose (molar mass = 180 g mol^{-1}) in water. Calculate the mass of glucose present in one litre of its solution.
(3/5, AI 2014)

71. Define the following terms :
(i) Osmotic pressure
(ii) Colligative properties *(Foreign 2014)*

72. Some ethylene glycol, $HOCH_2CH_2OH$, is added to your car's cooling system along with 5 kg of water. If the freezing point of water-glycol solution is –15.0°C, what is the boiling point of the solution?
(K_b = 0.52 K kg mol^{-1} and K_f = 1.86 K kg mol^{-1} for water)
(Delhi 2014C)

73. 1.00 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant of benzene is 5.12 K kg mol^{-1}. Find the molar mass of the solute.
(AI 2013)

74. A 5% solution (by mass) of cane-sugar in water has freezing point of 271 K. Calculate the freezing point of 5% solution (by mass) of glucose in water if the freezing point of pure water is 273.15 K.
[Molecular masses : Glucose $C_6H_{12}O_6$: 180 amu; Cane-sugar $C_{12}H_{22}O_{11}$: 342 amu]
(3/5, AI 2013C)

75. A solution of glycerol ($C_3H_8O_3$) in water was prepared by dissolving some glycerol in 500 g of water. This solution has a boiling point of 100.42°C while pure water boils at 100°C. What mass of glycerol was dissolved to make the solution?
(K_b for water = 0.512 K kg mol^{-1})
(Delhi 2012, AI 2012)

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76. A solution containing 30 g of non-volatile solute exactly in 90 g of water has a vapour pressure of 2.8 kPa at 298 K. Further 18 g of water is added to this solution. The new vapour pressure becomes 2.9 kPa at 298 K. Calculate
(i) the molecular mass of solute and
(ii) vapour pressure of water at 298 K.
(Delhi 2012C)

77. Calculate the boiling point of a solution prepared by adding 15.00 g of NaCl to 250.00 g of water. (K_b for water = 0.512 K kg mol^{-1}, Molar mass of NaCl = 58.44 g)
(3/5, Delhi 2011)

78. A solution prepared by dissolving 8.95 mg of a gene fragment in 35.0 mL of water has an osmotic pressure of 0.335 torr at 25°C. Assuming the gene fragment is a non-electrolyte, determine its molar mass.
(3/5, Delhi, AI 2011)

79. What would be the molar mass of a compound if 6.21 g of it dissolved in 24.0 g of chloroform to form a solution that has a boiling point of 68.04°C. The boiling point of pure chloroform is 61.7°C and the boiling point elevation constant, K_b for chloroform is 3.63°C/m. *(3/5, Delhi 2011)*

80. The molecular masses of polymers are determined by osmotic pressure method and not by measuring other colligative properties. Give two reasons.
(3/5, AI 2011C)

81. Calculate the boiling point of one molar aqueous solution (density 1.06 g mL^{-1}) of KBr. [Given : K_b for H_2O = 0.52 K kg mol^{-1}, atomic mass : K = 39, Br = 80]
(3/5, AI 2011C)

2.7 Abnormal Molar Masses

VSA (1 mark)

82 Give reason for the following :
Elevation of boiling point of 1 M KCl solution is nearly double than that of 1 M sugar solution. *(1/2, 2018)*

83. Define the following term :
van't Hoff factor
(Delhi 2017, 1/5, Delhi 2012)

84. Define the following term :
Abnormal molar mass *(Delhi 2017)*

SA (2 marks)

85. What is van't Hoff factor? What types of values can it have if in forming the solution the solute molecules undergo
(i) dissociation (ii) association?
(2/5, AI 2014C)

86. Assuming complete dissociation, calculate the expected freezing point of a solution prepared by dissolving 6.00 g of Glauber's salt, $Na_2SO_4 \cdot 10H_2O$ in 0.100 kg of water.
(K_f for water = 1.86 K kg mol^{-1}, atomic masses : Na = 23, S = 32, O = 16, H = 1)
(2/5, AI 2014C)

87. A 1.00 molal aqueous solution of trichloroacetic acid (CCl_3COOH) is heated to its boiling point. The solution has the boiling point of 100.18°C. Determine the van't Hoff factor for trichloroacetic acid.
(K_b for water = 0.512 K kg mol^{-1})
(Delhi 2012)

88. What is van't Hoff factor? What possible value can it have if the solute molecules undergo dissociation? *(2/5, Delhi 2011C)*

LA I (3 marks)

89. A solution contains 5.85 g NaCl (Molar mass = 58.5 g mol^{-1}) per litre of solution. It has an osmotic pressure of 4.75 atm at 27°C. Calculate the degree of dissociation of NaCl in this solution.
(Given : R = 0.082 L atm K^{-1} mol^{-1})
(3/5, 2020)

90. When 19.5 g of $F-CH_2-COOH$ (molar mass = 78 g mol^{-1}) is dissolved in 500 g of water, the depression in freezing point is observed to be 1°C. Calculate the degree of dissociation of $F-CH_2-COOH$.
[Given : K_f for water = 1.86 K kg mol^{-1}]
(3/5, 2020)

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91. The freezing point of a solution containing 5 g of benzoic acid (M = 122 g mol^{-1}) in 35 g of benzene is depressed by 2.94 K. What is the percentage association of benzoic acid if it forms a dimer in solution?
(K_f for benzene = 4.9 K kg mol^{-1}) *(2020)*

92. Calculate the freezing point of solution when 1.9 g of $MgCl_2$ (M = 95 g mol^{-1}) was dissolved in 50 g of water, assuming $MgCl_2$ undergoes complete ionization.
(K_f for water = 1.86 K kg mol^{-1})
(Delhi 2016)

93. When 2.56 g of sulphur was dissolved in 100 g of CS_2, the freezing point lowered by 0.383 K. Calculate the formula of sulphur (S_x).
(K_f the CS_2 = 3.83 K kg mol^{-1}, atomic mass of sulphur = 32 g mol^{-1}) *(3/5 Delhi 2016)*

94. Calculate the boiling point of solution when 4 g of $MgSO_4$ (M = 120 g mol^{-1}) was dissolved in 100 g of water, assuming $MgSO_4$ undergoes complete ionization.
(K_b for water = 0.52 K kg mol^{-1}) *(AI 2016)*

95. 3.9 g of benzoic acid dissolved in 49 g of benzene shows a depression in freezing point of 1.62 K. Calculate the van't Hoff factor and predict the nature of solute (associated or dissociated).
(Given : Molar mass of benzoic acid = 122 g mol^{-1}, K_f for benzene = 4.9 K kg mol^{-1})
(Delhi 2015)

96. Calculate the mass of NaCl (molar mass = 58.5 g mol^{-1}) to be dissolved in 37.2 g of water to lower the freezing point by 2°C, assuming that NaCl undergoes complete dissociation. (K_f for water = 1.86 K kg mol^{-1})
(Foreign 2015)

97. Determine the osmotic pressure of a solution prepared by dissolving 2.5×10^{-2} g of K_2SO_4 in 2 L of water at 25°C, assuming that it is completely dissociated.
(R = 0.0821 L atm K^{-1} mol^{-1}, molar mass of K_2SO_4 = 174 g mol^{-1}) *(Delhi 2013)*

98. Calculate the amount of KCl which must be added to 1 kg of water so that the freezing point is depressed by 2 K (the K_f for water = 1.86 K kg mol^{-1}). *(Delhi 2012)*

99. 15.0 g of an unknown molecular material was dissolved in 450 g of water. The resulting solution was found to freeze at –0.34°C. What is the molar mass of this material?
(K_f for water = 1.86 K kg mol^{-1})
(Delhi 2012, 3/5, AI 2012)

100. Calculate the freezing point of an aqueous solution containing 10.50 g of $MgBr_2$ in 200 g of water. (Molar mass of $MgBr_2$ = 184 g mol^{-1}, K_f for water = 1.86 K kg mol^{-1})
(3/5, Delhi 2011)

101. A 0.561 m solution of an unknown electrolyte depresses the freezing point of water by 2.93°C. What is van't Hoff factor for this electrolyte? The freezing point depression constant (K_f) for water is 1.86°C kg mol^{-1}.
(Foreign 2011)

102. Phenol associates in benzene to a certain extent to form a dimer. A solution containing 20 g of phenol in 1.0 kg of benzene has its freezing point lowered by 0.69 K. Calculate the fraction of phenol that has dimerised
[Given K_f for benzene = 5.1 K m^{-1}]
(3/5, Delhi 2011C)

103. An aqueous solution containing 12.48 g of barium chloride in 1.0 kg of water boils at 373.0832 K. Calculate the degree of dissociation of barium chloride.
[Given K_b for H_2O = 0.52 K m^{-1};
Molar mass of $BaCl_2$ = 208.34 g mol^{-1}]
(3/5, Delhi 2011C)

104. What mass of NaCl must be dissolved in 65.0 g of water to lower the freezing point of water by 7.50°C? The freezing point depression constant (K_f) for water is 1.86°C/m. Assume van't Hoff factor for NaCl is 1.87. (Molar mass of NaCl = 58.5 g mol^{-1})
(AI 2011)

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Detailed Solutions

1. Molality (m) is defined as the number of moles of the solute per kilogram of the solvent and is expressed as :

$$\text{Molality } (m) = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$

2. It is the number of moles of the solute dissolved per litre of the solution. It is denoted by M.

$$\therefore \quad \text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution in litre}}$$

$$= \frac{n}{V(\text{in mL})} \times 1000 = \frac{w_2 \times 1000}{M_2 \times V(\text{in mL})} \quad \left(\therefore n_2 = \frac{w_2}{M_2}\right)$$

3. Mole fraction is the ratio of number of moles of solute or solvent and total number of moles of solution. It is denoted by x.

$$x_{\text{solute}} = \frac{n_2}{n_1 + n_2}, \; x_{\text{solvent}} = \frac{n_1}{n_1 + n_2}$$

4. Mass of solute = 9.8 g
Mass of solution = 100 g
Density of solution = 1.02 g mL^{-1}

$$\therefore \quad \text{Volume of solution} = \frac{\text{Mass of solution}}{\text{Density of solution}}$$

$$= \frac{100 \text{ g}}{1.02 \text{ g mL}^{-1}} = 98.039 \text{ mL} = 0.098 \text{ L}$$

$$\text{Number of moles of solute, } n = \frac{9.8}{98} = 0.1 \text{ mol}$$

$$\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{Volume of solution (in L)}} = \frac{0.1 \text{ mol}}{0.098 \text{ L}} = 1.02 \text{ M}$$

5.

Molarity	Molality
Number of moles of solute dissolved in one litre solution is called molarity.	Number of moles of solute dissolved in one kg solvent is called molality.
$M = \frac{\text{No. of moles of solute}}{\text{Volume of solution in litre}}$	$m = \frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}}$
Molarity depends on temperature as volume depends on temperature. Molarity decreases with rise in temperature.	Molality is independent of temperature as mass does not change with temperature.

If M_B is the molar mass of solute, d is the density of solution then molality (m) of a solution can be converted to molarity (M) by using the formula,

$$M = \frac{md}{\left(1 + \frac{mM_B}{1000}\right)}$$

6. Mass of solution = 100 g
Mass of solute = 10 g
Mass of solvent = 100 – 10 = 90 g = 0.09 kg

$$\text{Number of moles of solute, } n = \frac{10}{180} = 0.055 \text{ mol}$$

$$m = \frac{0.055 \text{ mol}}{0.09 \text{ kg}} = 0.61 \text{ m}$$

7. *Refer to answer 5.*

8. *Refer to answer 5.*

If M_B is the molar mass of solute, d is the density of solution then molarity (M) value of a solution can be converted into its molality (m) by using the following formula,

$$m = \frac{1000 \times M}{(1000 \times d) - (M \times M_B)}$$

9. Given : Mass of solute, W_2 = 10 g
Mass of solvent, W_1 = 90 g
Molar mass of solute, M_2 = 180 g mol^{-1}
Density of solution = 1.2 g mL^{-1}

(i) $$\text{Molality} = \frac{W_2 \times 1000}{M_2 \times W_1} = \frac{10 \times 1000}{180 \times 90} = 0.62 \text{ m}$$

(ii) $$\text{Volume of solution} = \frac{\text{mass}}{\text{density}} = \frac{100 \text{ g}}{1.2 \text{ g mL}^{-1}}$$

$$\text{Molarity} = \frac{W_2 \times 1000}{M_2 \times V}$$

$$M = \frac{10 \times 1000}{180 \times \frac{100}{1.2}} = \frac{10 \times 1000 \times 1.2}{180 \times 100} = 0.67 \text{ M}$$

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10. Increase in temperature decreases the solubility of oxygen in water. As a result, amount of dissolved oxygen decreases. It becomes more difficult to breathe as oxygen is less. Hence, the aquatic species are not comfortable in warm water.

11. According to Henry's law, the solubility of a gas is inversely proportional to the Henry's law constant (K_H) for that gas. Hence, gas (*B*) being less soluble, would have a higher K_H value.

12. Henry's law states that, the partial pressure of the gas in vapour phase (p) is proportional to the mole fraction of the gas (x) in the solution.
$p = K_H \cdot x$ where, K_H = Henry's law constant. Different gases have different K_H values at the same temperature.

13. *Refer to answer 12.*

14. To minimise the painful effects of decompression sickness in deep sea divers, oxygen diluted with less soluble helium gas is used as breathing gas

15. *Refer to answer 12.*

$$\text{Solubility of } CO_2 = \frac{p}{K_H} = \frac{760}{1.25 \times 10^6} = 6.08 \times 10^{-4}$$

16. (a) *Refer to answer 10.*
(b) At high altitudes, the partial pressure of oxygen is less than at the ground level. As a result, there is a low concentration of oxygen in the blood and tissues of the people living at high altitudes. Thus, they feel weak and are unable to think properly.

17. *Refer to answer 12.*
Applications of Henry's law :
(i) To increase the solubility of CO_2 in soft drinks and soda water, the bottle is sealed under high pressure.
(ii) To minimise the painful effects of decompression sickness in deep sea divers, oxygen diluted with less soluble helium gas is used as breathing gas.

18. *Refer to answer 10.*

19. Applying the relationship,
$m = K_H \times p$
In the first case, 6.56×10^{-2} g = $K_H \times 1$ bar
or, $K_H = 6.56 \times 10^{-2}$ g bar^{-1}
In the second case,
5.0×10^{-2} g = $(6.56 \times 10^{-2}$ g bar$^{-1}) \times p$

$$p = \frac{5.0 \times 10^{-2} \text{ g}}{6.56 \times 10^{-2} \text{ g bar}^{-1}} = 0.762 \text{ bar}$$

20. According to Henry's law, $p_{N_2} = K_H \times x_{N_2}$

$$x_{N_2} = \frac{p_{N_2}}{K_H} = \frac{0.987 \text{ bar}}{76480 \text{ bar}} = 1.29 \times 10^{-5}$$

If n moles of N_2 are present in 1 L (*i.e.*, 55.55 moles), of water

$$x_{N_2} = \frac{n}{n + 55.55} \simeq \frac{n}{55.55}$$

$$\therefore \quad \frac{n}{55.55} = 1.29 \times 10^{-5}$$

or $n = 1.29 \times 10^{-5} \times 55.55$ moles
$= 71.659 \times 10^{-5}$ moles = 0.716 millimoles

21. Liquid *A*

22. **Raoult's law :** For a solution of volatile liquids, the partial pressure of each component in the solution is directly proportional to its mole fraction. Thus, for any component, partial vapour pressure, $p \propto x \Rightarrow p = p^\circ . x$
where, p° = vapour pressure of pure component
x = mole fraction of that component

23. *Refer to answer 22.*

24. *Refer to answer 22.*
Henry's law : If gas is the solute and liquid is the solvent, then according to Henry's law,
$p = K_H \cdot x$
i.e., partial pressure of the volatile component (gas) is directly proportional to the mole fraction of that component (gas) in the solution.
Hence, Raoult's law and Henry's law has been identical except that their proportionality constant are different. It is equal to p° for Raoult's law and K_H for Henry's law.
Therefore, Raoult's law becomes a special case of Henry's law in which K_H becomes equal to vapour pressure of pure component p°.

25. *Refer to answer 22.*
Let a solution consists of two volatile liquids *A* and *B* with their mole fractions x_A and x_B respectively. If p_A and p_B are their partial vapour pressures, then,
$p_A \propto x_A \Rightarrow p_A = p_A^\circ x_A$ and $p_B \propto x_B \Rightarrow p_B = p_B^\circ x_B$
where p_A° and p_B° represent the vapour pressures of pure liquid components *A* and *B*.
$P_{total} = p_A + p_B$

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Ideal solutions obey Raoult law at all concentrations and temperature.

26. *Refer to answer 22.*

Raoult's law for solution containing non-volatile solute : It states that partial vapour pressure of a solution of non-volatile solute, $p_{solution}$ is directly proportional to the mole fraction of the solvent in the solution.

Mathematically,

$p_{solution} = p^\circ_{solvent} \times x_{solvent}$

where, $p^\circ_{solvent}$ = vapour pressure of the pure solvent at the given temperature.

or, $\dfrac{p^\circ - p_{solution}}{p^\circ} = x_2$

27. Given : p°_A = 450 mm Hg, p°_B = 700 mm Hg, P_{Total} = 600 mm Hg, x_A = ?

Applying Raoult's law, $p_A = x_A \times p^\circ_A$

$p_B = x_B \times p^\circ_B = (1 - x_A)p^\circ_B$

$P_{Total} = p_A + p_B = x_A \times p^\circ_A + (1 - x_A)p^\circ_B$

$= p^\circ_B + (p^\circ_A - p^\circ_B)x_A$

Substituting the given values, we get

$600 = 700 + (450 - 700)x_A$ or, $250x_A = 100$

or $x_A = \dfrac{100}{250} = 0.40$

Thus, composition of the liquid mixture will be

$x_A = 0.40$; $x_B = 1 - 0.40 = 0.60$

Calculation of composition in the vapour phase,

$p_A = x_A \times p^\circ_A = 0.40 \times 450$ mm Hg = 180 mm Hg

$p_B = x_B \times p^\circ_B = 0.60 \times 700$ mm Hg = 420 mm Hg

Mole fraction of *A* in the vapour phase

$$= \frac{p_A}{p_A + p_B} = \frac{180}{180 + 420} = 0.30$$

Mole fraction of *B* in the vapour phase = 1 – 0.30 = 0.70

28. A mixture of ethanol and acetone shows positive deviation from Raoult's law. Pure ethanol possesses hydrogen bonding. Introduction of acetone between the molecules of ethanol results in breaking of some of these hydrogen bonds. Due to weakening of interactions, the solution shows positive deviation from Raoult's law.

29. A solution which obeys Raoult's law of vapour pressure for all compositions is called ideal solution.

In this solution $\Delta V_{mix} = 0$, $\Delta H_{mix} = 0$

A .... *B* interaction = *A* .... *A* and *B* .... *B* interactions.

30. Non-ideal solutions that show negative deviation from Raoult's law form maximum boiling azeotropes.

31. Azeotropes are the binary mixtures of solutions that have the same composition in liquid and vapour phases and that have constant boiling point.

32. *Refer to answer 31.*

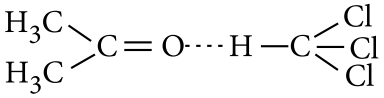
33. The differences between ideal solutions and non ideal solutions are as follows :

(i) In ideal solutions $\Delta V_{mixing} = 0$ and $\Delta H_{mix} = 0$ whereas in non ideal solutions,

$$\Delta V_{mix} \neq 0 \text{ and } \Delta H_{mix} \neq 0.$$

(ii) In ideal solutions, each component obeys Raoult's law at all temperatures and concentrations whereas in non ideal solutions, they do not obey Raoult's law.

34. Mixture of chloroform and acetone shows negative deviation from Raoult's law, thus it forms maximum boiling azeotrope. This is because chloroform molecule is able to form hydrogen bond with acetone molecule as shown :

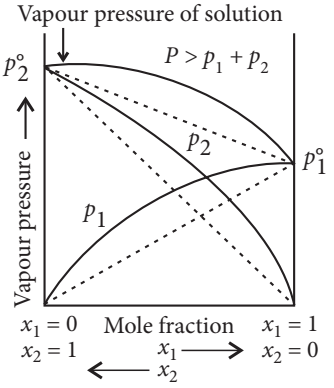


This decreases the escaping tendency of molecules for each component and consequently the vapour pressure decreases resulting in negative deviation from Raoult's law.

35. Positive deviation : For non-ideal solutions, if the vapour pressure is higher, then it is said to exhibit positive deviation.

A—*B* interactions are weaker than *A*—*A* or *B*—*B* interactions. Due to this, vapour pressure increases which results in positive deviation.

In positive deviation, intermolecular force decreases, volume increases, vapour pressures increases. enthalpy increases. Therefore, ΔH_{mix} = +ve, ΔV_{mix} = + ve. *e.g.*, ethanol + acetone and carbon disulphide + acetone show positive deviation.



Plot for non-ideal solution showing positive deviation

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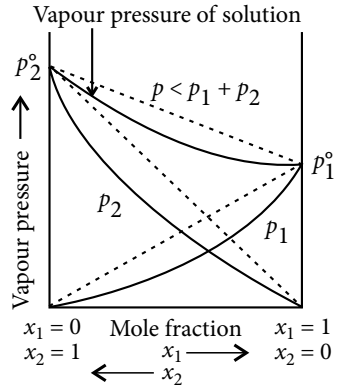
36. *Refer to answer 31.*
A minimum boiling azeotrope is formed by solutions showing a large positive deviation from Raoult's law at a specific composition. For example an ethanol-water mixture containing approximately 95% ethanol by volume.

37. Negative deviation : For non-ideal solution, if the vapour pressure is lower, then it is said to exhibit negative deviation. $A—B$ interactions are stronger than $A—A$ and $B—B$ interactions. Due to this, vapour pressure decreases which results in negative deviation. In negative deviation, intermolecular force increases, volume decreases, vapour pressure decreases and heat is released. Therefore, ΔH_{mix} = –ve, ΔV_{mix} = – ve. For example, phenol + aniline and chloroform + acetone show negative deviation.



Plot for non-ideal solution showing negative deviation

38. *Refer to answer 31.*
A maximum boiling azeotrope is formed by solutions showing a large negative deviation from Raoult's law at a specific composition.
For example, chloroform + acetone mixture.

39. *Refer to answer 28*

40. *Refer to answers 35 and 37.*

41. *Refer to answers 34.*

42. (a) : Colligative properties of ideal solutions depend only on the number of particles of solute dissolved in a definite amount of the solvent and do not depend on the nature of solute.

43. When the external pressure applied becomes more than the osmotic pressure of solution then the solvent molecules from the solution pass through the semi-permeable membrane to the solvent side and the process is called reverse osmosis.

44. In osmotic pressure method, pressure can be measured at room temperature and the molarity of the solution is used instead of molality. That is why this method is used for determination of molar masses of macromolecules as they are generally not stable at higher temperatures.

45. Properties which depend upon the number of solute particles irrespective of their nature relative to the total number of particles present in the solution are called colligative properties.

46. Osmotic pressure is the extra pressure which is applied on the solution to just prevent the flow of solvent into the solution through a semi-permeable membrane.

47. Two solutions having same osmotic pressure at a given temperature are called isotonic solutions.

48. Molal elevation constant may be defined as the elevation in boiling point when the molality of the solution is unity (*i.e.*, 1 mole of the solute is dissolved in 1 kg (1000 g) of the solvent). The units of K_b are therefore, degree/molality *i.e.*, K/m or °C/m or K kg mol^{-1}.

49. When a non-volatile solute is added to a solvent, the vapour pressure of the solvent (above the resulting solution) is lower than the vapour pressure above the pure solvent.

50. *Refer to answer 48.*

51. Molality (m) of urea solution

$$= \frac{w \times 1000}{M \times \text{Mass of solvent}} = \frac{5 \times 1000}{60 \times 95} = 0.877 \text{ m}$$

Osmotic pressure (π) = CRT

$$= 0.877 \times 0.0821 \times 300 = 21.6 \text{ atm}$$

52. Molality (m) of urea = $\dfrac{7.5 \times 1000}{60 \times 100} = 1.25$ m ...(i)

Molality of substance, $Z = \dfrac{42.75 \times 1000}{\text{Molar mass} \times 100}$

$$= \frac{427.5}{\text{Mol. mass}} \quad \text{...(ii)}$$

As both the solutions have same depression in freezing point so they have same molality.
From eqn. (i) and (ii), we get

$$1.25 = \frac{427.5}{\text{Molar mass}}$$

Molar mass of Z = 342 g/mol

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53. (i) As KCl is an electrolyte and one formula unit of KCl dissociates to give two ions (K^+ and Cl^-), therefore molar concentration of particles in the solution = 0.1 × 2 M = 0.2 M
As elevation of boiling point (or any colligative property) is directly proportional to number of particles in solution, hence 0.1 M KCl has higher boiling point that 0.1 M glucose.
(ii) Salting is used because most bacteria, fungi and other potentially pathogenic organism cannot survive in a highly salty environment, due to the hypertonic nature of salt. Any living cell in such an environment will become dehydrated through osmosis and die or become temporarily deactivated.

54. Mass of glucose ($C_6H_{12}O_6$), $W_2 = 60$ g
Mass of water, $W_1 = 250$ g
M_2 (Mol. mass of $C_6H_{12}O_6$) = 180 g mol^{-1}
$K_f = 1.86$ K kg mol^{-1}, $T_f = ?$

Using formula, $K_f \times \dfrac{W_2 \times 1000}{M_2 \times W_1}$

$$= 1.86 \times \frac{31 \times 1000}{62 \times 500} \text{ K}$$

$\Delta T_f = T_f^\circ - T_f$
or $T_f = T_f^\circ - \Delta T_f = 273 - 1.86 = 271.14$ K

55. (i) The elevation in boiling point of a solution is a colligative property which depends on the number of moles of solute added. Higher the concentration of solute added, higher will be the elevation in boiling point. Thus, 2 M glucose has higher boiling point than 1 M glucose solution.
(ii) *Refer to answer 43.*

56. (i) 1.2% sodium chloride solution is hypertonic with respect to 0.9% sodium chloride solution or blood cells thus, on placing blood cells in this solution exosmosis takes place that results in shrinking of cells.
(ii) 0.4% sodium chloride solution is hypotonic with respect to 0.9% sodium chloride solution or blood cells thus, on placing blood cells in this solution endosmosis takes place that results in swelling of cells.

57. The boiling point of the solution is always higher than that of the pure solvent. As the vapour pressure of the solution is lower than that of the pure solvent and vapour pressure increases with increase in temperature, hence, the solution has to be heated more to make the vapour pressure equal to the atmospheric pressure.
Elevation of boiling point is a colligative property because it depends on number of solute particles present in a solution.

58. Given : $W_2 = ?$, $M_2 = 256$ g mol^{-1}, $\Delta T_f = 0.48$ K
$W_1 = 75$ g, $K_f = 5.12$ K kg mol^{-1}

$$\Delta T_f = \frac{K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$W_2 = \frac{\Delta T_f \times M_2 \times W_1}{K_f \times 1000} = \frac{0.48 \times 256 \times 75}{5.12 \times 1000} = 1.8 \text{ g}$$

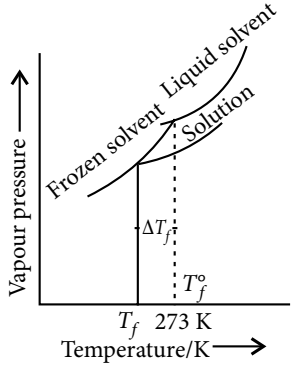
59. Given, $W_1 = 1$ kg = 1000 g, $W_2 = 18$ g,
$M_2 = 180$ g mol^{-1}
$T_b^\circ = 373.15$ K, $K_b = 0.52$ K Kg mol^{-1}, $T_b = ?$
Using formula,

$$\Delta T_b = K_b \times \frac{W_2 \times 1000}{M_2 \times W_1}$$

or $\Delta T_b = \dfrac{0.52 \times 18 \times 1000}{180 \times 1000} = 0.052$ K

$\Delta T_b = T_b - T_b^\circ$
$0.052 = T_b - 373.15$
$T_b = 373.15 + 0.052 = 373.202$ K

60. When a non-volatile solute is added to a solvent, the freezing point of the solution is always lower than that of pure solvent as the vapour pressure of the solvent decreases in the presence of non-volatile solute.
Plot for the lowering in freezing point of water when NaCl is added to it is shown as :



61. Osmosis : The spontaneous movement of the solvent molecules from the pure solvent or from a dilute solution to a concentrated solution through a semi-permeable membrane is called osmosis.

Osmotic Pressure : The minimum excess pressure that has to be applied on the solution to prevent

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the passage of solvent molecules into it through semipermeable membrane is called osmotic pressure.

Osmotic pressure is a colligative property because it depends on the number of solute particles and not on their nature.

62. (i) Number of particles of solute
(ii) Association or dissociation of solute
(iii) Concentration of solute
(iv) Temperature

63. Given, $M_2 = 176$ g mol^{-1}, $\Delta T_f = 1.5°C$
$w_1 = 75$ g, $K_f = 3.9$ K kg mol^{-1}, $w_2 = ?$

$$\Delta T_f = \frac{K_f \times w_2 \times 1000}{M_2 \times w_1}$$

$$w_2 = \frac{\Delta T_f \times M_2 \times w_1}{K_f \times 1000} = \frac{1.5 \times 176 \times 75}{3.9 \times 1000} = 5.077 \text{ g}$$

64. Molality of sucrose solution (m)

$$= \frac{w_2 \times 1000}{M_2 \times w_1} = \frac{4 \times 1000}{342 \times 96} = 0.121 \text{ m}$$

ΔT_f for sucrose solution = 273.15 – 271.15 = 2 K

$$K_f = \frac{\Delta T_f}{m} = \frac{2}{0.121} = 16.52 \text{ K/m}$$

$$\text{Molality of glucose solution} = \frac{5 \times 1000}{180 \times 95} = 0.292 \text{ m}$$

$\Delta T_f = K_f \times m = 16.52 \times 0.292 = 4.82$ K
Freezing point of glucose solution
= 273.15 – 4.82 = 268.33 K

65. $\pi = cRT$, $c = \frac{w_2}{M_2 \times V}$

w_2 = weight of solute
M_2 = molar mass of solute

$\frac{\pi_1}{\pi_2} = \frac{c_1 RT_1}{c_2 RT_2}$. Given, $T_1 = T_2$

$$\frac{\pi_1}{\pi_2} = \frac{c_1}{c_2} \Rightarrow \frac{4.98}{1.52} = \frac{\frac{30}{180 \times 1}}{c_2}$$

$$\Rightarrow c_2 = \frac{30 \times 1.52}{180 \times 4.98} = 0.051 \text{ mol/L}$$

66. Molality (m) of sucrose solution

$$= \frac{w \times 1000}{M \times \text{Mass of solvent}} = \frac{10}{342} \times \frac{1000}{90} = 0.325 \text{ m}$$

ΔT_f for sucrose solution
$= T_f^\circ - T_f = (273.15 - 269.15)$ K = 4 K

$\because \quad \Delta T_f = K_f \times m$

$$\therefore \quad K_f = \frac{\Delta T_f}{m} = \frac{4 \text{ K}}{0.325 \text{ m}} = 12.308 \text{ K/m}$$

$$\text{Molality of glucose solution} = \frac{10}{180} \times \frac{1000}{90} = 0.617 \text{ m}$$

$\Delta T_f = K_f \times m$

$\therefore \quad \Delta T_f = 12.308 \text{ K/m} \times 0.617 \text{ m} = 7.59 = 7.6$ K

$\therefore$ Freezing point of glucose solution,

$T_f^\circ - \Delta T_f = (273.15 - 7.60)$ K = 265.55 K

67. Given, weight of urea (W_2) = 30 g
Weight of water (W_1) = 846 g
Vapour pressure of water p_1° = 23.8 mm Hg

$$n_B = \frac{30}{60} = 0.5, \quad n_A = \frac{846}{18} = 47$$

$$\text{Mole fraction of water } (x_A) = \frac{n_A}{n_A + n_B}$$

$$= \frac{47}{47 + 0.5} = \frac{47}{47.5} = 0.99$$

$P_A = p_A^\circ \times x_A = 23.8 \times 0.99 = 23.5$ mm Hg.

68. Mass of $CaCl_2$ (W_2) = 10 g
Mass of water (W_1) = 200 g
Molar mass of $CaCl_2$ (M_2) = 111 g mol^{-1}
Molal elevation constant (K_b) = 0.512 K kg mol^{-1}

$$m = \frac{W_2 \times 1000}{M_2 \times W_1}$$

$$m = \frac{10}{111} \times \frac{1000}{200} = 0.450 \text{ m}$$

$\Delta T_b = iK_b\, m = 3 \times 0.512 \times 0.450 = 0.6912$ K

69. Mass of ethylene glycol ($C_2H_6O_2$), W_2 = 31 g
Mass of water, W_1 = 500 g
M_2 (Mol. mass of $C_2H_6O_2$) = 62 g mol^{-1},
K_f = 1.86 K kg mol^{-1}, $T_f = ?$

Using formula, $\Delta T_f = K_f \times \frac{W_2 \times 1000}{M_2 \times W_1}$

$$= 1.86 \times \frac{31 \times 1000}{62 \times 500} \text{ K}$$

$\Delta T_f = T_f^\circ - T_f$

or $T_f = T_f^\circ - \Delta T_f = 273 - 1.86 = 271.14$ K

70. Mass of urea = 15 g
Molar mass of urea = 60 g mol^{-1}
Molar mass of glucose = 180 g mol^{-1}
Mass of glucose = ?

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For isotonic solution, osmotic pressure, $\pi_1 = \pi_2$

$n_1 = n_2$ (when volume is same)

or $\frac{W_1}{M_1} = \frac{W_2}{M_2} \Rightarrow \frac{15}{60} = \frac{W_2}{180}$

$\Rightarrow \quad W_2 = \frac{15 \times 180}{60} = 45$ g

71. (i) *Refer to answer 46.*

(ii) *Refer to answer 45.*

72. $T_f = -15°C$, $K_f = 1.86$ K kg mol^{-1}

$\Delta T_f = T_f^\circ - T_f = 0 - (-15°C) = 15°C$

$\Delta T_f = K_f \times m$

$15 = 1.86 \times$ moles of solute $= 40.32$ moles

$\Delta T_b = K_b \times m$

$\Delta T_b = 0.52 \times \frac{\text{moles of solute}}{\text{mass of solvent (kg)}}$

$\Delta T_b = 0.52 \times \frac{40.32}{5\text{ kg}} = 4.19$

$\Delta T_b = T_b - T_b^\circ$

$4.19 = T_b - 373$

$T_b = 373 + 4.19 = 377.19$ K

73. $W_2 = 1.00$ g, $W_1 = 50$ g, $K_f = 5.12$ K kg mol^{-1}, $\Delta T_f = 0.40$ K

$\Delta T_f = \frac{K_f \times W_2 \times 1000}{M_2 \times W_1}$

$M_2 = \frac{K_f \times W_2 \times 1000}{W_1 \times \Delta T_f} = \frac{5.12 \times 1 \times 1000}{50 \times 0.40}$

$= 256$ g mol^{-1}

74. Molality of sugar solution

$= \frac{W_2 \times 1000}{M_2 \times W_1} = \frac{5}{342} \times \frac{1000}{95} = 0.154$ m

$\Delta T_f = T_f^\circ - T_f = 273.15 - 271 = 2.15$ K

$\Delta T_f = K_f \times m \quad \therefore \; K_f = \frac{\Delta T_f}{m} = \frac{2.15}{0.154}$

Molality of glucose solution

$= \frac{W_2 \times 1000}{M_2 \times W_1} = \frac{5}{180} \times \frac{1000}{95} = 0.292$ m

$\therefore \quad \Delta T_f(\text{Glucose}) = K_f \times m = \frac{2.15}{0.154} \times 0.292 = 4.08$

$\therefore$ Freezing point of glucose solution

$= 273.15 - 4.08 = 269.07$ K

75. $W_1 = 500$ g

Boiling point of solution $(T_b) = 100.42°C$

K_b for water $= 0.512$ K kg mol^{-1}

$M_2(C_3H_8O_3)$

$= (3 \times 12) + (8 \times 1) + (3 \times 16)$

$= 92$ g mol^{-1}

$\Delta T_b = T_b - T_b^\circ$

$= 373.42$ K $- 373$ K $= 0.42$ K

$\Delta T_b = \frac{K_b \times W_2 \times 1000}{M_2 \times W_1}$

$W_2 = \frac{\Delta T_b \times M_2 \times W_1}{K_b \times 1000} = \frac{0.42 \times 92 \times 500}{0.512 \times 1000} = 37.73$ g

76. The relative lowering of vapour pressure is given by the following expression,

$(p^\circ_{\text{solvent}} - p_{\text{solution}})/p^\circ_{\text{solvent}} = n_2/(n_1 + n_2)$

for dilute solutions, $n_2 << n_1$, therefore

$(p^\circ_{\text{solvent}} - p_{\text{solution}})/p^\circ_{\text{solvent}} = n_2/n_1$

$= (W_2 \times M_1)/(M_2 \times W_1)$

$(p^\circ_{\text{solvent}} - 2.8)/p^\circ_{\text{solvent}} = (30 \times 18)/(M_2 \times 90)$

$(p^\circ_{\text{solvent}} - 2.8)/p^\circ_{\text{solvent}} = 6/M_2$ ...(1)

Similarly for second case we get,

$(p^\circ_{\text{solvent}} - 2.9)/p^\circ_{\text{solvent}} = (30 \times 18)/(M_2 \times 108)$

$(p^\circ_{\text{solvent}} - 2.9)/p^\circ_{\text{solvent}} = 5/M_2$ ...(2)

Divides eq. (1) by (2), we get

$(p^\circ_{\text{solvent}} - 2.8)/(p^\circ_{\text{solvent}} - 2.9) = 6/5$

$\therefore \quad p^\circ_{\text{solvent}} = 3.4$ kPa

i.e., vapour pressure of water at 298 K is 3.4 kPa

Substituting the value of p°_{solvent} in (1) we get,

$(3.4 - 2.8)/3.4 = 6/M_2$ or $0.6/3.4 = 6/M_2$

$\therefore \quad M_2 = 34$ g

77. $i = 2$, $K_b = 0.512$ K kg mol^{-1}, $W_B = 15$ g

$M_B = 58.44$ g mol^{-1}, $W_A = 250$ g

$\Delta T_b = \frac{i \times K_b \times W_B \times 1000}{M_B \times W_A}$

$\Delta T_b = \frac{2 \times 0.512 \times 15 \times 1000}{58.44 \times 250} = 1.05$ K

Therefore, boiling point of aqueous solution,

$T_b = T_b^\circ + \Delta T_b = 373.15$ K $+ 1.05$ K $= 374.20$ K

78. $W_B = 8.95$ mg $= 8.95 \times 10^{-3}$ g,

$R = 0.0821$ L atm K^{-1} mol^{-1}, $V = 35 \times 10^{-3}$ L

$T = (25 + 273)$ K $= 298$ K, $\pi = \frac{0.335}{760}$ atm

Substituting these values in the equation,

$M_B = \frac{W_B \times R \times T}{\pi \times V}$

$M_B = \frac{8.95 \times 10^{-3}\text{ g} \times 0.0821\text{ L atm K}^{-1}\text{ mol}^{-1} \times 298\text{ K}}{\frac{0.335}{760}\text{ atm} \times 35 \times 10^{-3}\text{ L}}$

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$M_B = 14193.29 \text{ g mol}^{-1}$

79. $W_2 = 6.21$ g, $W_1 = 24.0$ g
$T_b = 68.04$ °C, $T_b^\circ = 61.7$ °C and $K_b = 3.63$ °C/m
$\Delta T_b = T_b - T_b^\circ = 68.04$ °C – 61.7 °C = 6.34 °C

$$\Delta T_b = \frac{K_b \times W_2 \times 1000}{M_2 \times W_1} \text{ or } M_2 = \frac{K_b \times W_2 \times 1000}{\Delta T_b \times W_1}$$

$$M_2 = \frac{3.63^\circ\text{C m}^{-1} \times 6.21\,\text{g} \times 1000}{6.34^\circ\text{C} \times 24.0\,\text{g}} = 148.15\,\text{g mol}^{-1}$$

80. (i) Osmotic pressure can be measured at room temperature and molarity of the solution is used instead of molality.
(ii) Its magnitude is large as compared to other colligative properties even for very dilute solutions.

81. Concentration of the solution = 1 molar
Density of the solution = 1.06 g mL^{-1}
M_2, molar mass of KBr = 39 + 80 = 119 g mol^{-1}
K_b for H_2O = 0.52 K kg mol^{-1}

Molality, $m = \dfrac{M \times 1000}{1000 \times d - M \times M_2}$

$$m = \frac{1 \times 1000}{1000 \times 1.06 - 1 \times 119} = 1.0626 \text{ mol kg}^{-1}$$

$\Delta T_b = K_b m = 0.52$ K kg mol^{-1} × 1.0626 mol kg^{-1}
$\Delta T_b = 0.5525$ K ≈ 0.553 K
$T_b = T_b^\circ + \Delta T_b = 373$ K + 0.553 K = 373.553 K

82. i for 1 M KCl = 2
i for sugar solution = 1
$\therefore \Delta T_b = iK_b m = 2 K_b$ (for KCl)
$\Delta T_b = K_b$ (for sugar)
$\therefore$ ΔT_b of 1 M KCl solution is double than that of 1 M sugar solution.

83. van't Hoff factor : It is defined as the ratio of the experimental value of colligative property to the calculated value of the colligative property and is used to find out the extent of dissociation or association. Mathematically, it is represented as

$$i = \frac{\text{Experimental (or observed) value of colligative property}}{\text{Calculated (or normal) value of colligative property}}$$

84. The molar mass which is either lower or higher than the expected or normal value is known as abnormal molar mass.

85. *Refer to answer 83.*

(i) van't Hoff factor (i) > 1 for solutes undergoing dissociation
(ii) van't Hoff factor (i) < 1 for solutes undergoing association

86. Molecular mass of Glauber's salt ($Na_2SO_4 \cdot 10H_2O$)
= [2 × 23 + 32 + 16 × 4 + 10 × (2 × 1 + 16)]
= (46 + 32 + 64 + 180) g mol^{-1} = 322 g mol^{-1}
$Na_2SO_4 \cdot 10H_2O$ ionises as :

$$Na_2SO_4 \cdot 10H_2O \rightleftharpoons 2Na^+ + SO_4^{2-} + 10H_2O$$

$\Rightarrow$ $i = 3$

$$m = \frac{n_B}{W_A} = \frac{W_B}{M_B \times W_A} = \frac{6.00\,\text{g}}{322\,\text{g mol}^{-1} \times 0.1\,\text{kg}}$$

$= 0.186$ mol kg^{-1} = 0.186 m

Also, $\Delta T_f = i K_f \cdot m$
$= 3 \times 1.86$ K m^{-1} × 0.186 m = 1.04 K
$\Rightarrow$ $T_f = T_f^\circ - \Delta T_f = (273 - 1.04)$ K = 271.96 K

87. Molality of solution, $m = 1.00$ m
Boiling point of solution, $T_b = 100.18°C = 373.18$ K
Boiling point of water (solvent), $T_b^\circ = 100.00°C = 373$ K
$\Delta T_b = T_b - T_b^\circ = 373{\cdot}18$ K – 373 K = 0·18 K
$\Delta T_b = i K_b \cdot m$
0.18 K = $i \times K_b \cdot m$
0.18 K = i × 0.512 K kg mol^{-1} × 1 mol kg^{-1}

$$i = \frac{0.18\text{ K}}{0.512\text{ K kg mol}^{-1} \times 1\text{ mol kg}^{-1}} = 0.35$$

88. *Refer to answer 83 and 85.*

89. $\pi = iCRT = i \cdot \dfrac{n}{V} RT = i \times \dfrac{w}{M} \times \dfrac{1}{V} RT$

or $4.75 = i \times \dfrac{5.85}{58.5} \times \dfrac{1}{1} \times 0.082 \times 300$

or $i = 1.93$

	$NaCl \longrightarrow$	Na^+	+ Cl^-
Initial moles	C	0	0
Moles at eqm.	$C - \alpha$	$C\alpha$	$C\alpha$

Total no. of moles = $C(1 + \alpha)$

$$i = \frac{C(1+\alpha)}{C}$$

or, $i = 1 + \alpha$
or, $1 + \alpha = 1.93$
or, $\alpha = 0.93$ or 93%

90. (a) : Given : $W_2 = 19.5$ g,
$W_1 = 500$ g, $K_f = 1.86$ K kg mol^{-1}
$\Delta T_{f(\text{obs})} = 1.0$

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$$\therefore M_{2(obs)} = \frac{1000 \times K_f \times W_2}{W_1 \times \Delta T_{f(obs)}} = \frac{1000 \times 1.86 \times 19.5}{500 \times 1}$$
$$= 72.54$$

$M_{2(cal)}$ for $F—CH_2—COOH = 78\ g\ mol^{-1}$

$$i = \frac{M_{2(cal)}}{M_{2(obs)}} = \frac{78}{72.54} = 1.0753$$

$$F—CH_2—COOH \rightleftharpoons F—CH_2—COO^- + H^+$$

Initial moles	C	0	0
Moles at eqm.	C – α	Cα	Cα

Total no. of moles = $C(1 + \alpha)$

$$i = \frac{C(1+\alpha)}{C} = 1 + \alpha$$

or $\alpha = i - 1 = 1.0753 - 1 = 0.0753 = 7.53\%$

91. Given, $W_2 = 5$ g, $W_1 = 35$ g
$\Delta T_f = 2.94$ K, $K_f = 4.9$ K kg mol^{-1}

$$\Delta T_f = i\, K_f m = i \times K_f \times \frac{W_2 \times 1000}{M_2 \times W_1}$$

or $$2.94 = i \times 4.9 \times \frac{5 \times 1000}{122 \times 35}$$

or $i = 0.512$

As $i < 1$, solute is associated.

$$2C_6H_5COOH \rightleftharpoons (C_6H_5COOH)_2$$

Initial moles	1	0
Moles at eqm.	1 – α	α/2

Total no. of moles after association : $1 - \alpha + \alpha/2 = 1 - \alpha/2$

$$\therefore \quad i = \frac{1 - \alpha/2}{1} = 1 - \frac{\alpha}{2} \text{ or } 1 - \frac{\alpha}{2} = 0.512$$

or $\alpha = 2 \times (1 - 0.512) = 0.976$ or 97.6%

92. $\Delta T_f = iK_f \cdot m$

i for $MgCl_2 = 3$

$$\text{Molality, } m = \frac{W_2 \times 1000}{M_2 \times W_1} = \frac{1.9 \times 1000}{95 \times 50} = 0.4 \text{ m}$$

$\Delta T_f = 3 \times 1.86 \times 0.4 = 2.232$ K
Freezing point of solution, $T_f = T_f^\circ - \Delta T_f$
$= 273 - 2.232$ K $= 270.77$ K

93. $W_2 = 2.56$ g, $W_1 = 100$ g, $\Delta T_f = 0.383$ K
$K_f = 3.83$ K kg mol^{-1}, $\Delta T_f = K_f \times m$

$$\Rightarrow \quad \Delta T_f = K_f \times \frac{W_2}{M_2} \times \frac{1000}{W_1}$$

$$\Rightarrow \quad M_2 = \frac{W_2 \times 1000}{\Delta T_f \times W_1} \times K_f$$

$$= \frac{2.56 \times 1000}{0.383 \times 100} \times 3.83 = 256 \text{ g mol}^{-1}$$

$$\because \quad i = \frac{\text{Normal atomic mass}}{\text{Observed atomic mass}} = \frac{32}{256} = \frac{1}{8}$$

Sulphur exists as S_8.

94. $W_2 = 4$ g, $M_2 = 120$ g mol^{-1}
$W_1 = 100$ g, $K_b = 0.52$ K kg mol^{-1}
For complete dissociation, $i = 2$
Using formula, $\Delta T_b = iK_b m$

$$\text{or} \quad \Delta T_b = \frac{i \times K_b \times W_2 \times 1000}{M_2 \times W_1}$$

$$= 2 \times 0.52 \times \frac{4 \times 1000}{120 \times 100} = 0.34 \text{ K}$$

$\therefore \quad T_b = T_b^\circ + \Delta T_b = 100 + 0.34 = 100.34°C$

95. $W_2 = 3.9$ g, $W_1 = 49$ g,
$\Delta T_f = 1.62$ K, $M_2 = 122$ g mol^{-1},
$K_f = 4.9$ K kg mol^{-1}

$$\Delta T_f = i\, K_f m = i \times K_f \times \frac{W_2 \times 1000}{M_2 \times W_1}$$

$$\Rightarrow \quad 1.62 = \frac{i \times 4.9 \times 3.9 \times 1000}{122 \times 49}$$

$$\Rightarrow \quad i = \frac{1.62 \times 122 \times 49}{4.9 \times 3.9 \times 1000} = 0.506$$

As $i < 1$, solute is associated.

96. $$\Delta T_f = iK_f m = \frac{i \times K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$2 = \frac{2 \times 1.86 \times W_2 \times 1000}{58.5 \times 37.2}$$

$\Rightarrow \quad W_2 = 1.17$ g

97. Mass of K_2SO_4, $W_2 = 2.5 \times 10^{-2}$ g
Molar mass of K_2SO_4, $M_2 = 174$ g mol^{-1}
$V = 2$ L, $T = 25°C = 298$ K
$R = 0.0821$ L atm K^{-1} mol^{-1}

We know, osmotic pressure, $\pi = \frac{W_2 RT}{M_2 V}$

$$\pi = \frac{2.5 \times 10^{-2} \times 0.0821 \times 298}{174 \times 2}$$

$$= \frac{61.1645 \times 10^{-2}}{348} = 0.1758 \times 10^{-2} \text{ atm}$$

98. $\Delta T_f = 2$ K, $K_f = 1.86$ K kg mol^{-1},
$W_1 = 1$ kg, $\Delta T_f = i\, K_f m$, $M_2(KCl) = 74.5$ g mol^{-1}

$i = 2$ for KCl

$$\Delta T_f = \frac{i \times K_f \times W_2 \times 1000}{M_2 \times W_1}$$

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$$2 = \frac{2 \times 1.86 \times W_2 \times 1000}{74.5 \times 1000}$$

$\Rightarrow \quad W_2 = 40.05$ g

99. $W_1 = 450$ g, $W_2 = 15.0$ g

$\Delta T_f = T_f^\circ - T_f = 273 \text{ K} - 272.66 \text{ K} = 0.34 \text{ K}$

$K_f = 1.86 \text{ K kg mol}^{-1}$

$$\Delta T_f = \frac{K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$M_2 = \frac{K_f \times W_2 \times 1000}{\Delta T_f \times W_1}$$

$$M_2 = \frac{1.86 \times 15 \times 1000}{0.34 \times 450} = 182.35 \text{ g mol}^{-1}$$

100. $W_2 = 10.50$ g, $W_1 = 200$ g

$M_2(MgBr_2) = 184 \text{ g mol}^{-1}$

$K_f = 1.86 \text{ K kg mol}^{-1}$

$MgBr_{2(aq)} \rightarrow Mg^{2+}_{(aq)} + 2Br^-_{(aq)}$, $i = 3$

$$\Delta T_f = iK_f m, \quad \Delta T_f = \frac{i \times K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$\Delta T_f = \frac{3 \times 1.86 \times 10.50 \times 1000}{184 \times 200} = 1.592 \text{ K}$$

Freezing point of solution, $T_f = T_f^\circ - \Delta T_f$

$= 273 - 1.592 = 271.408$ K

101. $m = 0.561$ m, $\Delta T_f = 2.93°C$ and

$K_f = 1.86°C \text{ kg mol}^{-1}$

$\Delta T_f = iK_f m$

$$i = \frac{\Delta T_f}{K_f m} = \frac{2.93°C}{1.86°C \text{ kg mol}^{-1} \times 0.561 \text{ m}} = 2.807$$

102. Here, $n = 2$ because phenol forms dimer on association.

$W_2 = 20$ g, $W_1 = 1$ kg $= 1000$ g, $\Delta T_f = 0.69$ K,

$K_f = 5.1 \text{ K m}^{-1}$

$$\Delta T_f = \frac{K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$M_2 = \frac{K_f \times W_2 \times 1000}{\Delta T_f \times W_1} = \frac{5.1 \times 20 \times 1000}{0.69 \times 1000} = 147.82$$

$M_{2(\text{observed})} = 147.82$

$M_{2(\text{calculated})}$

$C_6H_5OH = 6 \times 12 + 6 \times 1 + 16 = 94 \text{ g mol}^{-1}$

$$i = \frac{M_{2(\text{calculated})}}{M_{2(\text{observed})}} = \frac{94}{147.82} = 0.635$$

$2C_6H_5OH \rightleftharpoons (C_6H_5OH)_2$

$$\alpha = \frac{i-1}{\left(\frac{1}{n}-1\right)} = \frac{0.635-1}{\left(\frac{1}{2}-1\right)} = \frac{0.365}{0.5} = 0.73 = 73\%$$

103. Here, $n = 3$ because 1 molecule of $BaCl_2$ on dissociation gives three ions.

$W_2 = 12.48$ g, $W_1 = 1.0$ kg $= 1000$ g

$T_b = 373.0832$ K, K_b for $H_2O = 0.52 \text{ K m}^{-1}$

and $M_2(BaCl_2) = 208.34 \text{ g mol}^{-1}$

$\Delta T_b = T_b - T_b^\circ = 373.0832 \text{ K} - 373 \text{ K} = 0.0832$ K

$$M_{2(\text{observed})} = \frac{K_b \times W_2 \times 1000}{\Delta T_b \times W_1}$$

$$M_{2(\text{observed})} = \frac{0.52 \times 12.48 \times 1000}{0.0832 \times 1000} = 78$$

$M_{2(\text{observed})} = 78 \text{ g mol}^{-1}$

$$i = \frac{M_{2(\text{calculated})}}{M_{2(\text{observed})}} = \frac{208.34 \text{ g mol}^{-1}}{78 \text{ g mol}^{-1}} = 2.67$$

$$\alpha = \frac{i-1}{n-1} = \frac{2.67-1}{3-1} = \frac{1.67}{2} = 0.835 = 83.5\%$$

104. $W_1 = 65.0$ g, $\Delta T_f = 7.50°C$,

$K_f = 1.86°C/m$, $i = 1.87$ and $M_2 = 58.5 \text{ g mol}^{-1}$

$$\Delta T_f = \frac{i \times K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$W_2 = \frac{\Delta T_f \times M_2 \times W_1}{i \times K_f \times 1000} = \frac{7.50 \times 58.5 \times 65}{1.87 \times 1.86 \times 1000} = 8.199 \text{ g}$$

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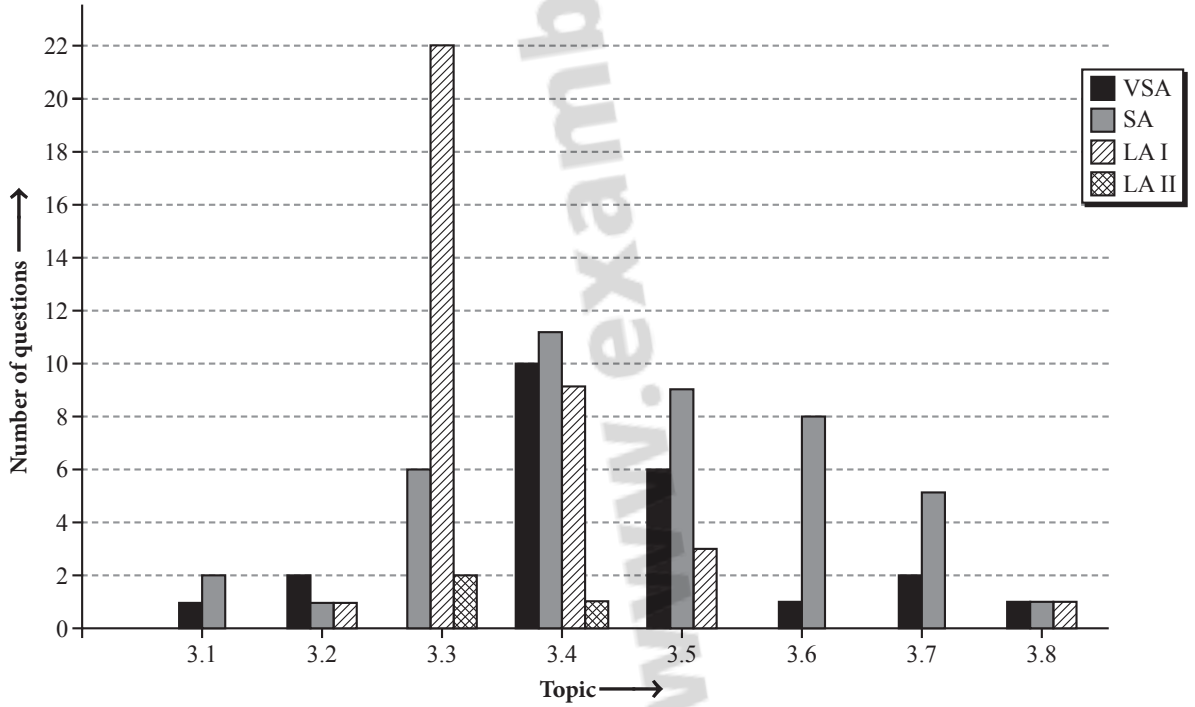
CHAPTER
3

Electrochemistry

3.1 Electrochemical Cells
3.2 Galvanic Cells
3.3 Nernst Equation
3.4 Conductance of Electrolytic Solutions
3.5 Electrolytic Cells and Electrolysis
3.6 Batteries
3.7 Fuel Cells
3.8 Corrosion

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Nernst Equation.*
- Maximum VSA and SA type questions were asked from *Conductance of Electrolytic Solutions*.
- Maximum LA I type questions were asked from *Nernst Equation*.

QUICK RECAP

Electrochemistry : It is the study of production of electricity from energy released during spontaneous chemical reactions and the use of electrical energy to bring about non-spontaneous chemical transformations.

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Differences between electrochemical cell and electrolytic cell

Electrochemical cell (Galvanic or Voltaic cell)	Electrolytic cell
1. It is a device which converts chemical energy into electrical energy.	1. It is a device which converts electrical energy into chemical energy.
2. It is based upon the redox reaction which is spontaneous. *i.e.*, ΔG = –ve	2. The redox reaction is non-spontaneous and takes place only when electrical energy is supplied. *i.e.*, ΔG = +ve
3. Two electrodes are usually set up in two separate beakers.	3. Both the electrodes are suspended in the solution or melt of the electrolyte in the same beaker.
4. The electrolytes taken in the two beakers are different.	4. Only one electrolyte is taken.
5. The electrodes taken are of different materials.	5. The electrodes taken may be of the same or different materials.
6. The electrode on which oxidation takes place is called the *anode* (or –ve pole) and the electrode on which reduction takes place is called the *cathode* (or +ve pole)	6. The electrode which is connected to the –ve terminal of the battery is called the *cathode*; the cations migrate to it which gain electrons and hence, a reduction takes place, the other electrode is called the *anode*.
7. To set up this cell, a salt bridge/porous pot is used.	7. No salt bridge is used in this case.

Nernst equation : For a reduction reaction, $M^{n+}_{(aq)} + ne^- \longrightarrow M_{(s)}$;

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF}\log\frac{1}{[M^{n+}_{(aq)}]}$$

At 298 K,

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{1}{[M^{n+}_{(aq)}]}$$

- For concentration cell, EMF at 298 K is given by

$$E_{cell} = \frac{0.0591}{n}\log\frac{C_2}{C_1}$$

where $C_2 > C_1$

Conductance in electrolytic solutions :

Applications of Nernst equation :

- **To calculate electrode potential of a cell :**

$$aA + bB \xrightarrow{ne^-} xX + yY$$

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[X]^x[Y]^y}{[A]^a[B]^b}\text{ (At 298 K)}$$

- **To calculate equilibrium constant :**

At equilibrium, $E_{cell} = 0$

$$E^\circ_{cell} = \frac{0.0591}{n}\log K_c \text{ at 298 K}$$

Relation between cell potential and Gibbs energy change :

$\Delta G^\circ = -nFE^\circ_{cell}$; $\Delta G^\circ = -2.303\ RT \log K_c$

Property	Formula	Units	Effect of dilution
Conductance (G)	$\frac{1}{R} = \frac{a}{\rho l} = \frac{\kappa a}{l}$	Ohm^{-1} (Ω^{-1})/Siemens (S)	Increases as larger number of ions are produced.
Specific conductance (κ) or conductivity	$\frac{1}{\rho}$ or $G\frac{l}{a}$	Ohm^{-1} cm^{-1}/S m^{-1}	Decreases as number of ions per cm^3 decreases.
Equivalent conductivity (Λ_{eq})	$\kappa \times V$ or $\kappa \times \frac{1000}{N}$	Ω^{-1} cm^2 eq^{-1}/S m^2 eq^{-1}	Increases with dilution due to large increase in V.
Molar conductivity (Λ_m)	$\kappa \times V$ or $\kappa \times \frac{1000}{M}$	Ω^{-1} cm^2 mol^{-1}/S m^2 mol^{-1}	Increases with dilution due to large increase in V.

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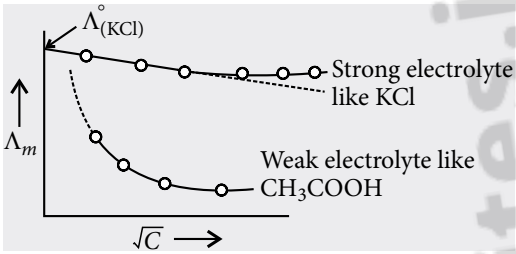
Limiting molar conductivity : When concentration approaches zero *i.e.* ; at infinite dilution, the molar conductivity is known as *limiting molar conductivity* (Λ_m°).

Variation of molar conductivity with concentration : For a strong electrolyte it is shown by *Debye-Huckel Onsager equation* as follows :

$$\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$$



Here, Λ_m° = Molar conductivity at infinite dilution (Limiting molar conductivity)

Λ_m = Molar conductivity at V-dilution

A = Constant which depends upon nature of solvent and temperature

C = Concentration

Plot of Λ_m against $C^{1/2}$ is a straight line with intercept equal to Λ_m° and slope equal to '$-A$'.

Thus, Λ_m decreases linearly with $\sqrt{C}$, when $C = 0$, $\Lambda_m = \Lambda_m^\circ$ and Λ_m° can be determined experimentally.

- **For weak electrolytes :** There is a very large increase in conductance with dilution especially near infinite dilution as no. of ions increases. Λ_m increases as C decreases but does not reach a constant value even at infinite dilution. Hence, their Λ_m° cannot be determined experimentally.
- **For a strong electrolyte :** There is only a small increase in conductance with dilution. This is because a strong electrolyte is completely dissociated in solution and so, the number of ions remain constant and on dilution, interionic attractions decreases as ions move far apart.

Kohlrausch's law of independent migration of ions : It states that limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte.

$\Lambda_m^\circ = \upsilon_+\lambda_+^\circ + \upsilon_-\lambda_-^\circ$; where λ_+° and λ_-° are the limiting molar conductivities of the cation and anion respectively and υ_+ and υ_- are stoichiometric number of cations and anions respectively in one formula unit of the electrolyte.

Applications of Kohlrausch's law :

- **Calculation of molar conductivity of weak electrolytes :**

$$\Lambda_{m\,(CH_3COOH)}^\circ = \lambda_{CH_3COO^-}^\circ + \lambda_{H^+}^\circ$$

$$= \left(\lambda_{H^+}^\circ + \lambda_{Cl^-}^\circ\right) + \left(\lambda_{CH_3COO^-}^\circ + \lambda_{Na^+}^\circ\right) - \left(\lambda_{Na^+}^\circ + \lambda_{Cl^-}^\circ\right)$$

$$= \Lambda_{m\,(HCl)}^\circ + \Lambda_{m\,(CH_3COONa)}^\circ - \Lambda_{m\,(NaCl)}^\circ$$

- **Calculation of degree of dissociation :**

$$\text{Degree of dissociation } (\alpha) = \frac{\Lambda_m}{\Lambda_m^\circ}$$

$$= \frac{\text{Molar conductivity at concentration } c}{\text{Molar conductivity at infinite dilution}}$$

- **Calculation of dissociation constant (K_c) of weak electrolyte :**

	$AB \rightleftharpoons$	A^+	+ B^-
Initial conc.	c	0	0
Conc. at equil.	$(c - c\alpha)$	$c\alpha$	$c\alpha$

$$K_c = \frac{c\alpha \times c\alpha}{(c - c\alpha)} = \frac{c\alpha^2}{(1-\alpha)}$$

Electrolysis : It is the process of decomposition of an electrolyte by passing electricity through its aqueous solution or molten state.

- **Faraday's first law of electrolysis :** The amount of chemical reaction which occurs at any electrode during electrolysis by a current is proportional to the quantity of electricity passed through the electrolyte (solution or melt).

$w \propto Q$ or $w = ZQ = Z \times I \times t$; where, Z is electrochemical equivalent of the substance deposited and $Z = \frac{\text{Eq. wt. of substance}}{96500}$.

- **Faraday's second law of electrolysis :** The amounts of different substances liberated by the same quantity of electricity passing through the electrolytic solution are proportional to their chemical equivalent weights.

$\frac{w_1}{w_2} = \frac{E_1}{E_2}$ where E is the equivalent weight.

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Products of electrolysis :

Electrolyte	Products		Reactions involved	
	At cathode	At anode	At cathode	At anode
Molten NaCl	Na metal	Cl_2 gas	$Na^+_{(l)} + e^- \longrightarrow Na_{(l)}$	$Cl^-_{(l)} \longrightarrow \frac{1}{2} Cl_{2(g)} + e^-$
Aqueous NaCl	H_2 gas	Cl_2 gas	$H_2O_{(l)} + e^- \longrightarrow \frac{1}{2} H_{2(g)} + OH^-_{(aq)}$	$Cl^-_{(aq)} \longrightarrow \frac{1}{2} Cl_{2(g)} + e^-$
Dil. H_2SO_4	H_2 gas	O_2 gas	$H^+_{(aq)} + e^- \longrightarrow \frac{1}{2} H_{2(g)}$	$2H_2O_{(l)} \longrightarrow O_{2(g)} + 4H^+_{(aq)} + 4e^-$
Conc. H_2SO_4	H_2 gas	$S_2O_8^{2-}$	$H^+_{(aq)} + e^- \longrightarrow \frac{1}{2} H_{2(g)}$	$2SO^{2-}_{4(aq)} \longrightarrow S_2O^{2-}_{8(aq)} + 2e^-$

Some commercial cells :

- **Primary cells :** Cells once exhausted cannot be used again *e.g.*, dry cell and mercury cell.
- **Secondary cells :** Rechargeable cell which can be used again and again *e.g.*, nickel-cadmium cell and lead storage battery.
- **Fuel cells :** Cells which can convert the energy of combustion of fuels such as H_2, CO, CH_4, etc. into electrical energy, *e.g.*, $H_2 - O_2$ fuel cell.

Dry cell, lead accumulator and fuel cell :

	Dry cell	Lead storage battery	Fuel cell
Anode	Zinc	Lead	Porous carbon containing catalysts (H_2 passed)
Cathode	Graphite	Lead dioxide	Porous carbon containing catalysts (O_2 passed)
Electrolyte	MnO_2 + C (touching cathode) $NH_4Cl + ZnCl_2$ (touching anode)	H_2SO_4 (38%)	Conc. aqueous KOH
Anode reaction	$Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^-$	$Pb_{(s)} + SO^{2-}_{4(aq)} \longrightarrow PbSO_{4(s)} + 2e^-$	$H_{2(g)} + 2OH^-_{(aq)} \longrightarrow 2H_2O_{(l)} + 2e^-$
Cathode reaction	$MnO_{2(s)} + NH^+_{4(aq)} + e^- \longrightarrow MnO(OH)_{(s)} + NH_{3(g)}$	$PbO_{2(s)} + SO^{2-}_{4(aq)} + 4H^+ + 2e^- \longrightarrow PbSO_{4(s)} + 2H_2O_{(l)}$	$O_{2(g)} + 2H_2O_{(l)} + 4e^- \longrightarrow 4OH^-_{(aq)}$

Corrosion : The slow eating away of metals when exposed to the atmosphere is called *corrosion*.

Corrosion of iron (Rusting) : It is an electrochemical phenomenon which occurs in the presence of moisture and oxygen.

At anode : $2Fe_{(s)} \longrightarrow 2Fe^{2+}_{(aq)} + 4e^-$

At cathode : $O_{2(g)} + 4H^+_{(aq)} + 4e^- \longrightarrow 2H_2O_{(l)}$

Overall reaction : $2Fe_{(s)} + O_{2(g)} + 4H^+_{(aq)} \longrightarrow 2Fe^{2+}_{(aq)} + 2H_2O_{(l)}$.

Methods used for prevention of corrosion : Barrier protection, sacrificial protection, anti-rust solutions.

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Previous Years' CBSE Board Questions

3.1 Electrochemical Cells

VSA (1 mark)

1. An electrochemical cell behave like an electrolytic cell when

(a) $E_{cell} = E_{external}$ (b) $E_{cell} = 0$

(c) $E_{external} > E_{cell}$ (d) $E_{external} < E_{cell}$

(2020)

SA (2 marks)

2. Give two points of differences between electrochemical and electrolytic cells.

(2/5, 2020)

3. Define electrochemical cell. What happens if external potential applied becomes greater than $E°_{cell}$ of electrochemical cell?

(AI 2019, 2/5, AI 2016)

3.2 Galvanic Cells

VSA (1 mark)

4. Represent the galvanic cell in which the reaction

$Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ takes place.

(1/3, Delhi 2013C)

5. What is the necessity to use a salt bridge in a Galvanic cell? *(Delhi 2011C)*

SA (2 marks)

6. Two half-reactions of an electrochemical cell are given below :

$MnO^-_{4(aq)} + 8H^+_{(aq)} + 5e^- \rightarrow Mn^{2+}_{(aq)} + 4H_2O_{(l)}$, $E° = +1.51V$

$Sn^{2+}_{(aq)} \rightarrow Sn^{4+}_{(aq)} + 2e^-$, $E° = +0.15$ V

Construct the redox equation from the standard potential of the cell and predict if the reaction is reactant favoured or product favoured. *(Delhi 2011)*

LA I (3 marks)

7. A cell is prepared by dipping copper rod in 1 M copper sulphate solution and zinc rod in 1 M zinc sulphate solution. The standard reduction potential of copper and zinc are 0.34 V and –0.76 V respectively.

(i) What will be the cell reaction?

(ii) What will be the standard electromotive force of the cell?

(iii) Which electrode will be positive?

(Delhi 2011C)

3.3 Nernst Equation

SA (2 marks)

8. Calculate $\Delta_r G°$ for the reaction :

$Mg_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Mg^{2+}_{(aq)} + Cu_{(s)}$

Given $E°_{cell} = +2.71$ V, 1 F = 96500 C mol^{-1}

(2/3,AI 2014)

9. Equilibrium constant (K_c) for the given cell reaction is 10. Calculate $E°_{cell}$.

$A_{(s)} + B^{2+}_{(aq)} \rightleftharpoons A^{2+}_{(aq)} + B_{(s)}$

(2/3,Foreign 2014)

10. The standard electrode potential ($E°$) for Daniell cell is +1.1 V. Calculate the $\Delta G°$ for the reaction.

$Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

(1 F = 96500 C mol^{-1}) *(AI 2013)*

11. The standard electrode potential for Daniell cell is 1.1 V. Calculate the standard Gibbs energy for the cell reaction.

(F = 96,500 C mol^{-1}) *(Delhi 2013C)*

12. A zinc rod is dipped in 0.1 M solution of $ZnSO_4$. The salt is 95% dissociated at this dilution at 298 K. Calculate the electrode potential.

[$E°_{Zn^{2+}/Zn} = -0.76$ V] *(Delhi 2012C)*

13. Determine the value of equilibrium constant (K_c) and $\Delta G°$ for the following reaction :

$Ni_{(s)} + 2Ag^+_{(aq)} \rightarrow Ni^{2+}_{(aq)} + 2Ag_{(s)}$, $E° = 1.05$ V

(1 F = 96500 C mol^{-1})

(Delhi, Foreign 2011)

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LA I (3 marks)

14. Calculate ΔG° for the reaction,
$Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$
Given : E° for $Zn^{2+}/Zn = -0.76$ V and
E° for $Cu^{2+}/Cu = +0.34$ V
$R = 8.314$ J K^{-1} mol^{-1}, $F = 96500$ C mol^{-1}.
(3/5, 2020)

15. Calculate the maximum work and log K_c for the given reaction at 298 K :
$Ni_{(s)} + 2Ag^+_{(aq)} \rightleftharpoons Ni^{2+}_{(aq)} + 2Ag_{(s)}$
Given :
$E^\circ_{Ni^{2+}/Ni} = -0.25$ V, $E^\circ_{Ag^+/Ag} = +0.80$ V
1F = 96500 C mol^{-1} *(2020)*

16. (a) Calculate e.m.f of the following cell :
$Zn_{(s)}/Zn^{2+}$ (0.1 M) || (0.01 M) $Ag^+/Ag_{(s)}$
Given : $E^\circ_{Zn^{2+}/Zn} = -0.76$ V, $E^\circ_{Ag^+/Ag} = +0.80$ V
[Given : log 10 = 1] *(3/5, 2020)*

17. Calculate $\Delta_r G^\circ$ and log K_c for the following reaction.
$Cd^{2+}_{(aq)} + Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + Cd_{(s)}$
Given : $E^\circ_{Cd^{2+}/Cd} = -0.403$ V ; $E^\circ_{Zn^{2+}/Zn} = -0.763$ V
(AI 2019)

18. Write the cell reaction and calculate the e.m.f. of the following cell at 298 K.
$Sn_{(s)}$ | Sn^{2+} (0.004 M) || H^+ (0.020 M) | $H_{2(g)}$ (1 bar) | $Pt_{(s)}$
(Given : $E^\circ_{Sn^{2+}/Sn} = -0.14$ V) *(3/5, 2018)*

19. For the reaction,
$2AgCl_{(s)} + H_{2(g)}(1\text{ atm}) \rightarrow 2Ag_{(s)} + 2H^+(0.1\text{ M}) + 2Cl^-(0.1\text{ M})$
$\Delta G^\circ = -43600$ J at 25 °C.
Calculate the e.m.f. of the cell.
(log $10^{-n} = -n$) *(3/5, 2018)*

20. Calculate e.m.f. of the following cell at 298 K.
$2Cr_{(s)} + 3Fe^{2+}(0.1M) \rightarrow 2Cr^{3+}(0.01M) + 3Fe_{(s)}$
Given : $E^\circ_{(Cr^{3+}|Cr)} = -0.74$ V, $E^\circ_{(Fe^{2+}|Fe)} = -0.44$ V
(Delhi 2016)

21. Calculate E°_{cell} for the following reaction at 298 K.
$2Al_{(s)} + 3Cu^{2+}(0.01M) \rightarrow 2Al^{3+}(0.01M) + 3Cu_{(s)}$
Given : $E_{cell} = 1.98$ V *(3/5,AI 2016)*

22. Calculate emf of the following cell at 25°C:
$Fe|Fe^{2+}(0.001\text{ M})||H^+(0.01\text{ M})|H_{2(g)}(1\text{ bar})|Pt_{(s)}$
$E^\circ(Fe^{2+}|Fe) = -0.44$ V, $E^\circ(H^+|H_2) = 0.00$ V
(Delhi 2015)

23. Calculate the emf of the following cell at 25°C.
$Zn|Zn^{2+}(0.001\,M)||H^+(0.01M)|H_{2(g)}(1\,bar)|Pt_{(s)}$
$\left(E^\circ_{Zn^{2+}/Zn} = -0.76\text{ V}, E^\circ_{(H^+/H_2)} = 0.00\text{ V}\right)$
(Foreign 2015)

24. For the cell reaction,
$Ni_{(s)}\,|Ni^{2+}_{(aq)}||Ag^+_{(aq)}|Ag_{(s)}$
Calculate the equilibrium constant at 25°C. How much maximum work would be obtained by operation of this cell?
$E^\circ_{(Ni^{2+}/Ni)} = -0.25$ V and $E^\circ_{Ag^+/Ag} = 0.80$V
(3/5, Delhi 2015C)

25. Calculate the standard cell potential of the galvanic cell in which the following reaction takes place :
$Fe^{2+}_{(aq)} + Ag^+_{(aq)} \rightarrow Fe^{3+}_{(aq)} + Ag_{(s)}$
Calculate the $\Delta_r G^\circ$ and equilibrium constant of the reaction.
$\left(E^\circ_{Ag^+/Ag} = 0.80\text{ V} ; E^\circ_{Fe^{3+}/Fe^{2+}} = 0.77\text{ V}\right)$
(3/5, Delhi 2015C)

26. Calculate the emf of following cell at 298 K :
$Mg_{(s)}$ | Mg^{2+}(0.1 M) || Cu^{2+}(0.01 M)|$Cu_{(s)}$
[Given : $E^\circ_{cell} = +2.71$ V, 1 F = 96500 C mol^{-1}]
(3/5, Delhi 2014)

27. Estimate the minimum potential difference needed to reduce Al_2O_3 at 500°C. The Gibbs energy change for the decomposition reaction,
$\frac{2}{3}Al_2O_3 \rightarrow \frac{4}{3}Al + O_2$ is 960 kJ.
($F = 96500$ C mol^{-1}) *(3/5,Delhi 2014C)*

28. Calculate the emf of the following cell at 298 K :
$Fe_{(s)}|Fe^{2+}(0.001\text{ M})||H^+(1\text{ M})|H_{2(g)}$ (1 bar), $Pt_{(s)}$
(Given $E^\circ_{cell} = +0.44$ V) *(Delhi 2013)*

29. Calculate the emf of the following cell at 25°C :
$Ag_{(s)}|Ag^+(10^{-3}\text{ M})||Cu^{2+}(10^{-1}\text{ M})|Cu_{(s)}$
Given : $E^\circ_{cell} = +0.46$ V and log $10^n = n$.
(AI 2013)

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30. In the button cell, widely used in watches, the following reaction takes place.

$Zn_{(s)} + Ag_2O_{(s)} + H_2O_{(l)} \rightarrow Zn^{2+}_{(aq)} + 2Ag_{(s)} + 2OH^-_{(aq)}$

Determine $E°$ and $\Delta G°$ for the reaction.

Given : $E°_{Ag^+/Ag} = +0.80$ V, $E°_{Zn^{2+}/Zn} = -0.76$ V)

(3/5, Delhi 2012)

31. A voltaic cell is set up at 25°C with the following half cells :

Al/Al^{3+} (0.001 M) and Ni/Ni^{2+} (0.50 M)

Write an equation for the reaction that occurs when the cell generates an electric current and determine the cell potential.

$E°_{Ni^{2+}/Ni} = -0.25$ V and $E°_{Al^{3+}/Al} = -1.66$ V.

$(\log 8 \times 10^{-6} = -5.09)$

(3/5, AI 2012, 2011, 3/5, Foreign 2011)

32. The cell in which the following reaction occurs :

$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \rightarrow 2Fe^{2+}_{(aq)} + I_{2(s)}$

has $E°_{cell} = 0.236$ V at 298 K. Calculate the standard Gibbs energy and the equilibrium constant of the cell reaction.

(Antilog of $6.5 = 3.162 \times 10^6$; of $8.0 = 10 \times 10^8$; of $8.5 = 3.162 \times 10^8$)

(Delhi 2012C)

33. Calculate the potential for half-cell containing 0.10 M $K_2Cr_2O_{7(aq)}$, 0.20 M $Cr^{3+}_{(aq)}$ and 1.0×10^{-4} M $H^+_{(aq)}$. The half cell reaction is :

$Cr_2O^{2-}_{7(aq)} + 14H^+_{(aq)} + 6e^- \rightarrow 2Cr^{3+}_{(aq)} + 7H_2O_{(l)}$

and the standard electrode potential is given as $E° = 1.33$ V. *(3/5, AI 2011)*

34. For the cell,

$Zn_{(s)} \mid Zn^{2+}$ (2 M) || Cu^{2+} (0.5 M) $\mid Cu_{(s)}$

(a) Write equation for each half-reaction.

(b) Calculate the cell potential at 25°C

Given :

$E^\circ_{Zn^{2+}/Zn} = -0.76$ V ; $E^\circ_{Cu^{2+}/Cu} = +0.34$V

(Delhi 2011C)

35. Calculate the equilibrium constant, K for the reaction at 298 K,

$Zn_{(s)} + Cu^{2+}_{(aq)} \rightleftharpoons Zn^{2+}_{(aq)} + Cu_{(s)}$

Given : $\Delta G° = -212.300$ kJ mol^{-1}

$E^\circ_{Zn^{2+}/Zn} = -0.76$ V ; $E^\circ_{Cu^{2+}/Cu} = +0.34$ V

(AI 2011C)

LA II (5 marks)

36. $E°_{cell}$ for the given redox reaction is 2.71 V.

$Mg_{(s)} + Cu^{2+}(0.01M) \rightarrow Mg^{2+}(0.001M) + Cu_{(s)}$

Calculate E_{cell} for the reaction. Write the direction of flow of current when an external opposite potential applied is

(i) less than 2.71 V and

(ii) greater than 2.71 V *(Delhi 2019)*

37. Calculate e.m.f and ΔG for the following cell

$Mg_{(s)} \mid Mg^{2+}$ (0.001 M) || Cu^{2+} (0.0001 M) $\mid Cu_{(s)}$

$E^\circ_{(Mg^{2+}/Mg)} = -2 \cdot 37$ V, $E^\circ_{(Cu^{2+}/Cu)} = +0.34$ V

(AI 2015)

3.4 Conductance of Electrolytic Solutions

VSA (1 mark)

38. **Assertion (A) :** Conductivity of an electrolyte decreases with decrease in concentration.

Reason (R) : Number of ions per unit volume increases on dilution.

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is correct, but Reason (R) is incorrect statement.

(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

39. Give reason :

Conductivity of CH_3COOH decreases on dilution. *(1/5, 2018)*

40. Define limiting molar conductivity. Why conductivity of an electrolyte solution decreases with the decrease in concentration?

(1/2, Delhi 2015)

41. State Kohlrausch's law of independent migration of ions. Write its one application.

(1/2, Foreign 2015)

42. Define the following term :

Molar conductivity (Λ_m) *(1/5, Delhi 2015C)*

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43. Define the following term :
Kohlrausch's law of independent migration of ions. *(1/5, Delhi 2015C)*

44. Define the following term :
Limiting molar conductivity *(1/5, Delhi 2014)*

45. State and explain Kohlrausch's law.
(1/3, Delhi 2013C)

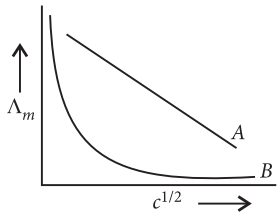
46. Express the relation between conductivity and molar conductivity of a solution held in a cell? *(Delhi 2011)*

47. Express the relation among the conductivity of solution in the cell, the cell constant and the resistance of solution in the cell.
(Delhi 2011)

SA (2 marks)

48. Predict the products of electrolysis of an aqueous solution of $CuCl_2$ with platinum electrodes.
(Given : $E^{\circ}_{Cu^{2+}/Cu} = + 0.34$ V,
$E^{\circ}_{(1/2Cl_2/Cl^-)} = + 1.36$ V
$E^{\circ}_{H^+/H_2(g),\,Pt} = 0.00$ V, $E^{\circ}_{(1/2\,O_2/H_2O)} = + 1.23$ V)
(2/5, 2020)

49. In the plot of molar conductivity (Λ_m) *vs* square root of concentration ($c^{1/2}$),following curves are obtained for two electrolytes *A* and *B*



Answer the following :
(i) Predict the nature of electrolytes *A* and *B*.
(ii) What happens on extrapolation of Λ_m to concentration approaching zero for electrolytes *A* and *B*? *(2/5, Delhi 2019)*

50. Calculate the degree of dissociation (α) of acetic acid if its molar conductivity (Λ_m) is 39.05 S cm^2 mol^{-1}.
(Given : $\lambda^{\circ}(H^+) = 349.68$ S cm^{2}mol^{-1} and $\lambda^{\circ}(CH_3COO^-) = 40.9$ S cm^2 mol^{-1})
(Delhi 2017)

51. Define the term degree of dissociation. Write an expression that relates the molar conductivity of a weak electrolyte to its degree of dissociation.
(2/5, Delhi 2015C)

52. Define conductivity and molar conductivity for the solution of an electrolyte. Discuss their variation with concentration.
(2/5, AI 2015C)

53. State Kohlrausch law of independent migration of ions. Why does the conductivity of a solution decrease with dilution?
(AI 2014)

54. Define the terms conductivity and molar conductivity for the solution of an electrolyte. Comment on their variation with temperature. *(Delhi, AI 2014C)*

55. The resistance of 0.01 M NaCl solution at 25°C is 200 Ω. The cell constant of the conductivity cell used is unity. Calculate the molar conductivity of the solution.
(2/3, AI 2014C)

56. The conductivity of 0.20 M solution of KCl at 298 K is 0.025 S cm^{-1}. Calculate its molar conductivity. *(Delhi 2013)*

57. The conductivity of 0.001 M acetic acid is 4×10^{-5} S/cm. Calculate the dissociation constant of acetic acid, if molar conductivity at infinite dilution for acetic acid is 390 S cm^2/mol. *(2/3, Delhi 2013C, 2012C)*

58. Express the relation among cell constant, resistance of the solution in the cell and conductivity of the solution. How is molar conductivity of a solution related to its conductivity? *(AI 2012)*

LA I (3 marks)

59. Out of the following pairs, predict with reason which pair will allow greater conduction of electricity.
(i) Silver wire at 30°C or silver wire at 60°C.
(ii) 0.1 M CH_3COOH solution or 1 M CH_3COOH solution.
(iii) KCl solution at 20°C or KCl solution at 50°C. *(3/5, 2020)*

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60. The electrical resistance of a column of 0.05 M KOH solution of length 50 cm and area of cross-section 0.625 cm^2 is 5×10^3 ohm. Calculate its resistivity, conductivity and molar conductivity. *(3/5, 2020)*

61. The conductivity of 0.001 mol L^{-1} solution of CH_3COOH is 3.905×10^{-5} S cm^{-1}. Calculate its molar conductivity and degree of dissociation (α).
[Given : $\lambda°$ (H^+) = 349.6 S cm^2 mol^{-1} and $\lambda°(CH_3COO^-)$ = 40.9 S cm^2 mol^{-1}]
(3/5, AI 2016)

62. The conductivity of 0·20 mol L^{-1} solution of KCl is 2.48×10^{-2} S cm^{-1}. Calculate its molar conductivity and degree of dissociation (α).
Given : λ° (K^+) = 73.5 S cm^2 mol^{-1} and λ° (Cl^-) = 76.5 S cm^2 mol^{-1}. *(AI 2015)*

63. Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100 Ω. If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520 Ω, calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} KCl solution is 1.29×10^{-2} Ω^{-1} cm^{-1}. *(3/5, AI 2014)*

64. The value of $\Lambda°_m$ of $Al_2(SO_4)_3$ is 858 S cm^2 mol^{-1}, while λ° SO_4^{2-} is 160 S cm^2 mol^{-1}, calculate the limiting ionic conductivity of Al^{3+}.
(AI 2013C)

65. The electrical resistance of a column of 0.05 M NaOH solution of diameter 1 cm and length 50 cm is 5.55 × 10^3 ohm. Calculate its resistivity, conductivity and molar conductivity. *(AI 2012)*

66. When a certain conductance cell was filled with 0.1 M KCl, it has a resistance of 85 ohms at 25°C. When the same cell was filled with an aqueous solution of 0.052 M unknown electrolyte, the resistance was 96 ohms. Calculate the molar conductance of the electrolyte at this concentration.
[Specific conductance of 0.1 M KCl
= 1.29×10^{-2} ohm^{-1} cm^{-1}]
(AI 2012C)

67. Calculate the degree of dissociation of acetic acid at 298 K, given that :
Λ_m (CH_3COOH) = 11.7 S cm^2 mol^{-1}
$\Lambda°_m(CH_3COO^-)$ = 49.9 S cm^2 mol^{-1}
$\Lambda°_m(H^+)$ = 349.1 S cm^2 mol^{-1} *(Delhi 2011C)*

LA II (5 marks)

68. (a) Define molar conductivity of a solution and explain how molar conductivity changes with change in concentration of solution for a weak and a strong electrolyte.
(b) The resistance of a conductivity cell containing 0.001 M KCl solution at 298 K is 1500 Ω. What is the cell constant if the conductivity of 0.001 M KCl solution at 298 K is 0.146×10^{-3} S cm^{-1}? *(AI 2012)*

3.5 Electrolytic Cells and Electrolysis

VSA (1 mark)

69. How many coulombs are required for the oxidation of 1 mol of H_2O to O_2?
(One word, 2020)

70. Give reason :
On the basis of $E°$ values, O_2 gas should be liberated at anode but it is Cl_2 gas which is liberated in the electrolysis of aqueous NaCl.
(1/5, 2018)

71. Following reactions occur at cathode during the electrolysis of aqueous silver chloride solution : $Ag^+_{(aq)} + e^- \longrightarrow Ag_{(s)}$, $E° = +0.80$ V

$H^+_{(aq)} + e^- \longrightarrow \frac{1}{2}H_{2(g)}$, $E° = 0.00$ V

On the basis of their standard reduction electrode potential ($E°$) values, which reaction is feasible at the cathode and why?
(AI 2017C, 1/2, Delhi 2015)

72. How much charge is required for the reduction of 1 mol of Zn^{2+} to Zn?
(Delhi 2015)

73. Following reactions occur at cathode during the electrolysis of aqueous copper (II)

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chloride solution :

$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$; $E^\circ = +0.34$ V

$H^+_{(aq)} + e^- \longrightarrow \frac{1}{2}H_{2(g)}$; $E^\circ = 0.00$ V

On the basis of their standard reduction electrode potential (E°) values, which reaction is feasible at the cathode and why? *(1/2, Foreign 2015)*

74. State the Faraday's first law of electrolysis. *(Delhi 2015C)*

SA (2 marks)

75. *X* and *Y* are two electrolytes. On dilution molar conductivity of '*X*' increases 2.5 times while that *Y* increases 25 times. Which of the two is a weak electrolyte and why? *(2/5, 2020)*

76. Calculate the mass of Ag deposited at cathode when a current of 2 ampere was passed through a solution of $AgNO_3$ for 15 minutes.
(Given : Molar mass of Ag = 108 g mol^{-1}, 1 F = 96500 C mol^{-1}) *(2/3, Delhi 2017)*

77. State Faraday's first law of electrolysis. How much charge in terms of Faraday is required for the reduction of 1 mol of Cu^{2+} to Cu? *(2/5, Delhi 2014)*

78. A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using a current of 5.0 ampere for 20 minutes. What mass of nickel will be deposited at the cathode?
(Given : At. mass of Ni = 58.7 g mol^{-1}, 1F = 96500 C mol^{-1}) *(Foreign 2014)*

79. Predict the products of electrolysis in each of the following :
(i) An aqueous solution of $AgNO_3$ with platinum electrodes.
(ii) An aqueous solution of H_2SO_4 with platinum electrodes. *(2/5, Delhi 2014C)*

80. How much electricity in terms of Faradays is required to produce 20 g of calcium from molten $CaCl_2$? *(2/3, Delhi 2013C)*

81. Silver is uniformly electrodeposited on a metallic vessel of surface area of 900 cm^2 by passing a current of 0.5 ampere for 2 hours. Calculate the thickness of silver deposited. Given : The density of silver is 10.5 g cm^{-3} and atomic mass of Ag = 108 amu. *(2/3, AI 2013C)*

82. How many coulombs are required to reduce 1 mole $Cr_2O_7^{2-}$ to Cr^{3+}? *(2/3, Delhi 2012C)*

83. How many moles of mercury will be produced by electrolysing 1.0 M $Hg(NO_3)_2$ solution with a current of 2.00 A for 3 hours? *(2/5, AI 2011)*

LA I (3 marks)

84. A steady current of 2 amperes was passed through two electrolytic cells *X* and *Y* connected in series containing electrolytes $FeSO_4$ and $ZnSO_4$ until 2.8 g of Fe deposited at the cathode of cell *X*. How long did the current flow? Calculate the mass of Zn deposited at the cathode of cell *Y*. (Molar mass : Fe = 56 g mol^{-1}, Zn = 65.3 g mol^{-1}, 1 F = 96500 C mol^{-1}) *(3/5, Delhi 2019)*

85. Chromium metal is electroplated using an acidic solution containing CrO_3 according to the following equation :
$CrO_{3(aq)} + 6H^+ + 6e^- \longrightarrow Cr_{(s)} + 3H_2O$
Calculate how many grams of chromium will be electroplated by 24,000 coulombs. How long will it take to electroplate 1.5 g chromium using 12.5 A current?
[Atomic mass of Cr = 52 g mol^{-1}, 1 F = 96500 C mol^{-1}] *(AI 2019)*

86. (a) The cell in which the following reaction occurs :

$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \longrightarrow 2Fe^{2+}_{(aq)} + I_{2(s)}$

has $E^\circ_{cell} = 0.236$ V at 298 K. Calculate the standard Gibbs energy of the cell reaction. (Given : 1 F = 96,500 C mol^{-1})

(b) How many electrons flow through a metallic wire if a current of 0.5 A is passed for 2 hours?
(Given : 1 F = 96,500 C mol^{-1}) *(AI 2017)*

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3.6 Batteries

VSA (1 mark)

87. Define : Secondary batteries
(1/5, Delhi 2015C)

SA (2 marks)

88. Write the name of the cell which is generally used in hearing aids. Write the reactions taking place at the anode and the cathode of this cell. *(AI 2017)*

89. Write the name of the cell which is generally used in inverters. Write the reactions taking place at the anode and the cathode of this cell. *(AI 2017)*

90. Write the name of the cell which is generally used in transistors. Write the reactions taking place at the anode and the cathode of this cell. *(AI 2017)*

91. From the given cells :
Lead storage cell, Mercury cell, Fuel cell and Dry cell
Answer the following :
(i) Which cell is used in hearing aids?
(ii) Which cell was used in Apollo Space Programme?
(iii) Which cell is used in automobiles and inverters?
(iv) Which cell does not have long life?
(Delhi 2016)

92. What type of battery is mercury cell? Why is it more advantageous than dry cell?
(2/5, AI 2015)

93. What type of a battery is the lead storage battery? Write the anode and the cathode reactions and the overall reaction occurring in a lead storage battery when current is drawn from it. *(2/5, Delhi 2012, AI 2011)*

94. Write the reactions taking place at cathode and anode in lead storage battery when the battery is in use. What happens on charging the battery? *(AI 2012C)*

95. Mention the reactions occurring at (i) anode, (ii) cathode, during working of a mercury cell. Why does the voltage of a mercury cell remain constant during its operation?
(Delhi 2011C)

3.7 Fuel Cells

VSA (1 mark)

96. Name the type of cell which was used in Apollo space programme.
(2020, 1/3, AI 2014)

97. Define : Fuel cell
(Delhi 2017, 1/5, Delhi 2015C, 2014)

SA (2 marks)

98. Give two advantages of fuel cells. *(2/5, 2020)*

99. Define fuel cell and write its two advantages.
(2/5, 2018)

100. Write two advantages of H_2—O_2 fuel cell over ordinary cell. *(2/3, Foreign 2014)*

101. What are fuel cells? Explain the electrode reactions involved in the working of H_2—O_2 fuel cell. *(2/3, Delhi 2013C)*

102. Give an example of a fuel cell and write the cathode and anode reactions for it.
(AI 2011C)

3.8 Corrosion

VSA (1 mark)

103. Using the E° values of A and B predict which is better for coating the surface of iron

$\left[E^\circ_{(Fe^{2+}/Fe)} = -0.44\ V\right]$

to prevent corrosion and why?

$E^\circ_{(A^{2+}/A)} = -2.37\ V; E^\circ_{(B^{2+}/B)} = 0.14\ V$

(AI 2016)

SA (2 marks)

104. The chemistry of corrosion of iron is essentially an electrochemical phenomenon. Explain the reactions occurring during the corrosion of iron in the atmosphere.
(Delhi 2011)

LA I (3 marks)

105. What is corrosion? Explain the electrochemical theory of rusting of iron and write the reactions involved in the rusting of iron.
(Delhi 2012C)

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Detailed Solutions

1. (c) : The device which converts the chemical energy liberated during the chemical reaction to electrical energy is called electrochemical cell.
If external potential applied becomes greater than $E°_{cell}$ of electrochemical cell then the cell behaves as an electrolytic cell and the direction of flow of current is reversed.

2.

S.No.	Electrochemical cell	Electrolytic cell
1.	It is a device to convert chemical energy into electrical energy, *i.e.*, electrical energy is produced as a result of the redox reaction.	It is a device to convert electrical energy into chemical energy, *i.e.*, electrical energy is supplied to the electrolytic solution to bring about the redox reaction.
2.	It is based upon the redox reaction which is spontaneous *i.e.*, ΔG = –ve	The redox reaction is non-spontaneous and takes place only when electrical energy is supplied *i.e.*, ΔG = +ve.

3. *Refer to answer 1.*

4. Representation of the galvanic cell for the given reaction is :

$$\underset{\text{Anode}}{Zn} \mid Zn^{2+}_{(aq)} \;\underset{\text{bridge}}{\|}\; \underset{\text{Cathode}}{Cu^{2+}_{(aq)} \mid Cu}$$

5. The salt bridge allows the movement of ions from one solution to the other without mixing of the two solutions. Moreover, it helps to maintain the electrical neutrality of the solutions in the two half cells.

6. At anode : $Sn^{2+}_{(aq)} \longrightarrow Sn^{4+}_{(aq)} + 2e^-] \times 5$

At cathode : $MnO^-_{4(aq)} + 8H^+_{(aq)} + 5e^- \longrightarrow Mn^{2+}_{(aq)} + 4H_2O_{(l)}] \times 2$

Net cell reaction :

$$2MnO^-_{4(aq)} + 5Sn^{2+}_{(aq)} + 16H^+_{(aq)} \longrightarrow 2Mn^{2+}_{(aq)} + 5Sn^{4+}_{(aq)} + 8H_2O_{(l)}$$

$E°_{cell} = E°_{cathode} - E°_{anode}$ = 1.51 V – 0.15 V = 1.36 V
Since, cell potential is positive therefore the reaction is product favoured.

7. (i) The cell reactions are :

$Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^-$ (Anode)

$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$ (Cathode)

Net reaction :

$Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

(ii) $E°_{cell} = E°_{right} - E°_{left}$ = 0.34 V – (– 0.76 V)
= 1.10 V

(iii) Copper electrode will be positive on which reduction takes place.

8. Given : $E°_{cell}$ = 2.71 V

For the reaction,

$Mg_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Mg^{2+}_{(aq)} + Cu_{(s)}$

n = 2, $\Delta_r G°$ = ?

Using formula, $\Delta_r G° = -nFE°_{cell}$

$\Delta_r G° = -2 \times 96500\ C\ mol^{-1} \times 2.71\ V$

or $\Delta_r G° = 523.03\ kJ\ mol^{-1}$

9. $A_{(s)} + B^{2+}_{(aq)} \rightleftharpoons A^{2+}_{(aq)} + B_{(s)}$

Here, n = 2

Using formula,

$$E°_{cell} = \frac{0.059}{n} \log K_c$$

$$E°_{cell} = \frac{0.059}{2} \log 10$$

$E°_{cell}$ = 0.0295 V

10. Here n = 2, $E°_{cell}$ = 1.1 V, F = 96500 C mol^{-1}

$\Delta_r G° = -nFE°_{cell}$

$\Delta_r G° = -2 \times 1.1 \times 96500 = -212300\ J\ mol^{-1}$
$= -212.3\ kJ\ mol^{-1}$

11. *Refer to answer 10.*

12. The electrode reaction is

$Zn^{2+} + 2e^- \longrightarrow Zn$ (n = 2)

Applying Nernst equation, we get,

$$\left[E_{Zn^{2+}/Zn} = E°_{Zn^{2+}/Zn} - \frac{0.0591}{2} \log \frac{1}{Zn^{2+}}\right]$$

As 0.1 M $ZnSO_4$ solution is 95% dissociated, this means that in the solution,

$$[Zn^{2+}] = \frac{95}{100} \times 0.1 = 0.095\ M$$

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$\therefore \quad E_{Zn^{2+}/Zn} = -0.76 - \frac{0.0591}{2}\log\frac{1}{0.095}$

$= -0.76 - 0.02955 (\log 1000 - \log 95)$

$= -0.76 - 0.0295 (3 - 1.9777) = -0.79015$ V

13. $Ni_{(s)} + 2Ag^{+}_{(aq)} \rightarrow Ni^{2+}_{(aq)} + 2Ag_{(s)}$, $E^\circ = 1.05$ V

Here, $n = 2$

Using formula, $\log K_c = \frac{nE^\circ_{cell}}{0.059}$

or $\log K_c = \frac{2 \times 1.05}{0.059} = 35.5932$

K_c = antilog 35.5932 or $K_c = 3.92 \times 10^{35}$

Again, $\Delta G^\circ = -nFE^\circ_{cell}$

$\Delta G^\circ = -2 \times 96500 \times 1.05 = -202650$ J mol^{-1}

$\Delta G^\circ = -202.65$ kJ mol^{-1}

14. $Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

Here, $n = 2$

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$= 0.34 - (-0.76) = 1.1$ V

$F = 96500$ C mol^{-1}

$\Delta_r G^\circ = -nFE^\circ_{cell} = -2 \times 1.1 \times 96500 = -212300$ J mol^{-1}

$= -212.3$ kJ mol^{-1}

15. At anode : $Ni \longrightarrow Ni^{2+} + 2e^-$

At cathode : $[Ag^+ + e^- \longrightarrow Ag] \times 2$

Cell reaction : $Ni + 2Ag^+ \longrightarrow Ni^{2+} + 2Ag$

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$= E^\circ_{Ag^+/Ag} - E^\circ_{Ni^{2+}/Ni} = 0.80$ V $- (-0.25)$

$E^\circ_{cell} = 1.05$ V

$E^\circ_{cell} = \frac{0.0591}{n}\log K_c$

$\log K_c = \frac{E^\circ_{cell} \times n}{0.0591} = \frac{1.05 \times 2}{0.0591}$

$\log K_c = 35.53$

K_c = antilog 35.53 $= 3.38 \times 10^{35}$

$\Delta G^\circ = -nFE^\circ = -2 \times 96500 \times 1.05$

$= -202650$ J mol^{-1} $= -202.65$ kJ mol^{-1}

The maximum work that can be obtained from the cell is 202.65 kJ

16. At anode : $Zn \rightarrow Zn^{2+} + 2e^-$

At cathode : $2Ag^+ + 2e^- \rightarrow 2Ag$

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{Ag^+/Ag} - E^\circ_{Zn^{2+}/Zn}$

$= 0.80 - (-0.76) = 1.56$ V

$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Zn^{2+}]}{[Ag^+]^2}$

$= 1.56 - \frac{0.059}{2}\log\frac{0.1}{(0.01)^2} = 1.4715$ V

17. $E^\circ_{cell} = E^\circ_{right} - E^\circ_{left} = -0.403 - (-0.763) = 0.36$ V

$\Delta_r G^\circ = -nFE^\circ_{cell} = -2 \times 96500 \times 0.36$

$= -69480$ J mol^{-1} $= -69.48$ kJ mol^{-1}

Using formula, $\log K_c = \frac{nE^\circ_{cell}}{0.059}$

$= \frac{2 \times 0.36}{0.059} = 12.20$

K_c = antilog 12.20 $= 1.58 \times 10^{12}$

18. The electrode reactions are

At anode : $Sn_{(s)} \rightarrow Sn^{2+}$ (0.004 M) $+ 2e^-$

At cathode : $2H^+$ (0.02 M) $+ 2e^- \rightarrow H_2$ (1 bar)

Net reaction :

$Sn_{(s)} + 2H^+$ (0.02 M) $\rightarrow Sn^{2+}$ (0.004 M) $+ H_2$ (1 bar)

The Nernst equation of this cell at 25 °C

$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Sn^{2+}](p_{H_2})}{[H^+]^2}$

$E^\circ_{cell} = E^\circ_{H^+|H_2} - E^\circ_{Sn^{2+}|Sn}$

$= 0.000$ V $- (-0.14$ V$) = +0.14$ V

or, $E_{cell} = E^\circ_{cell} - 0.0296\log\frac{0.004 \times 1}{(0.02)^2}$

$= E^\circ_{cell} - 0.0296\log\left(\frac{0.004}{0.0004}\right)$

$= E^\circ_{cell} - 0.0296 (\log 10)$

$= E^\circ_{cell} - 0.0296 \times 1 = E^\circ_{cell} - 0.0296$

$E_{cell} = 0.14 - 0.0296 = 0.1104$ V

19. $\Delta G^\circ = -nFE^\circ_{cell}$

$E^\circ_{cell} = \frac{\Delta G^\circ}{-nF} = \frac{-43600}{-2 \times 96500} = 0.226$ V

$2AgCl_{(s)} + H_{2(g)}$ (1 atm) $\longrightarrow 2Ag_{(s)} + 2H^+$(0.1 M) $+ 2Cl^-$(0.1 M)

$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[Product]}{[Reactant]}$

$= 0.226 - \frac{0.0591}{2}\log\frac{(0.1)^2}{(1)}$

$= 0.226 - \frac{0.0591}{2}\log(10^{-2})$

$= 0.226 - \frac{0.0591}{2}(-2)$

$= 0.226 + 0.0591 = 0.2851$ V

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20. $E^\circ_{cell} = E^\circ_{right} - E^\circ_{left} = -0.44 - (-0.74) = 0.30\ V$

$$E_{cell} = 0.30 - \frac{0.0591}{6}\log\frac{(0.1)^2}{(0.01)^3}$$

$$= 0.30 - 0.0394 = 0.26\ V$$

21. Given cell,

$2Al_{(s)} + 3Cu^{2+}(0.01\ M) \longrightarrow 2Al^{3+}(0.01\ M) + 3Cu_{(s)}$

$E_{cell} = 1.98\ V,\ E^\circ_{cell} = ?$

Using Nernst equation at 298 K

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{6}\log\frac{[Al^{3+}]^2}{[Cu^{2+}]^3}$$

$$1.98\ V = E^\circ_{cell} - \frac{0.0591}{6}\log\frac{[10^{-2}]^2}{[10^{-2}]^3}$$

$$1.98\ V = E^\circ_{cell} - \frac{0.0591}{6}\log 10^2$$

$$1.98\ V = E^\circ_{cell} - \frac{0.0591}{6}\times 2$$

$$\therefore \quad E^\circ_{cell} = 1.98 + \frac{0.0591}{6} = 1.99\ V$$

22. The cell reaction is

$Fe_{(s)} + 2H^+_{(aq)} \longrightarrow Fe^{2+}_{(aq)} + H_{2(g)}$

$E^\circ_{cell} = 0.00 - (-0.44) = 0.44\ V$

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Fe^{2+}]}{[H^+]^2}$$

$$= 0.44 - \frac{0.0591}{2}\log\frac{0.001}{(0.01)^2}$$

$$= 0.44 - 0.02955 = 0.41045\ V$$

23. The cell reaction is

$Zn_{(s)} + 2H^+_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + H_{2(g)}$

$E^\circ_{cell} = 0.00 - (-0.76) = 0.76\ V$

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Zn^{2+}]}{[H^+]^2}$$

$$= 0.76 - \frac{0.0591}{2}\log\frac{(0.001)}{(0.01)^2}$$

$$= 0.76 - 0.02955 = 0.730\ V$$

24. At anode : $Ni \longrightarrow Ni^{2+} + 2e^-$

At cathode : $[Ag^+ + e^- \longrightarrow Ag] \times 2$

Cell reaction : $Ni + 2Ag^+ \longrightarrow Ni^{2+} + 2Ag$

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$= E^\circ_{Ag^+/Ag} - E^\circ_{Ni^{2+}/Ni} = 0.80\ V - (-0.25)$

$E^\circ_{cell} = 1.05\ V$

$$E^\circ_{cell} = \frac{0.0591}{n}\log K_c$$

$$\log K_c = \frac{E^\circ_{cell}\times n}{0.0591} = \frac{1.05\times 2}{0.0591}$$

$\log K_c = 35.53$

$K_c = \text{antilog } 35.53 = 3.38 \times 10^{35}$

$\Delta G^\circ = -nFE^\circ = -2 \times 96500 \times 1.05$

$= -202650\ J\ mol^{-1}$

$= -202.65\ kJ\ mol^{-1}$

The maximum work that can be obtained = 202.65 kJ.

25. $E^\circ_{cell} = +0.80\ V - 0.77\ V = +0.03\ V$

$\Delta_r G^\circ = -nFE^\circ_{cell} = -1 \times 96500 \times 0.03$

$= -2895\ J\ mol^{-1} = -2.895\ kJ\ mol^{-1}$

$\Delta G^\circ = -2.303\ RT \log K_c$

$-2895 = -2.303 \times 8.314 \times 298 \times \log K_c$

or $\log K_c = 0.5074$

$K_c = \text{Antilog } (0.5074) = 3.22$

26. The cell reaction can be represented as :

$Mg_{(s)} + Cu^{2+}_{(aq)} \rightarrow Mg^{2+}_{(aq)} + Cu_{(s)}$

Given: $E^\circ_{cell} = +2.71\ V,\ T = 298\ K$

According to the Nernst equation :

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[Mg^{2+}_{(aq)}]}{[Cu^{2+}_{(aq)}]}$$

$$= 2.71 - \frac{0.0591}{2}\log\frac{0.1}{0.01} = 2.6805\ V$$

27. $Al_2O_3\ (2Al^{3+} + 3O^{2-}) \longrightarrow 2Al + \frac{3}{2}O_2,\ n = 6e^-$

$$\therefore \ \frac{2}{3}Al_2O_3 \longrightarrow \frac{4}{3}Al + O_2,\ n = \frac{2}{3}\times 6e^- = 4e^-$$

$\Delta G = 960 \times 1000 = 960000\ J$

Now, $\Delta G = -nFE^\circ_{cell}$

$$E^\circ_{cell} = -\frac{\Delta G}{nF} = \frac{-960000}{4\times 96500} = -2.487\ V$$

Minimum potential difference needed to reduce Al_2O_3 is –2.487 V.

28. $Fe_{(s)}|Fe^{2+}(0.001M)||H^+(1M)|H_{2(g)}(1bar)|Pt_{(s)}$

Reactions :

Anode : $Fe_{(s)} \longrightarrow Fe^{2+}_{(aq)} + 2e^-$

Cathode : $2H^+_{(aq)} + 2e^- \longrightarrow H_{2(g)}$

Cell reaction : $Fe_{(s)} + 2H^+_{(aq)} \longrightarrow Fe^{2+}_{(aq)} + H_{2(g)}$

$n = 2.$

Using Nernst equation at 298 K

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Fe^{2+}]\times p_{H_2}}{[H^+]^2}$$

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For the given cell,

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{H^+/H_2} - E^\circ_{Fe^{2+}/Fe}$

$= 0 - (-0.44) = + 0.44$ V

Given : $[Fe^{2+}] = 0.001$ M; $[H^+] = 1$ M; $P_{H_2} = 1$ bar

Putting in Nernst equation

$$E_{cell} = 0.44 - 0.0295 \log \frac{0.001 \times 1}{1^2}$$

$= 0.44 - 0.0295 \log 10^{-3}$

$= 0.44 - [(0.0295) \times (-3)]$

$= 0.44 + 0.0885 = 0.53$ V

29. The cell may be represented as

$Ag_{(s)} |Ag^+ (10^{-3} M)||Cu^{2+} (10^{-1} M)| Cu_{(s)}$

Using formula $E_{cell} = E^\circ_{cell} - \frac{0.0591}{2} \log \frac{[Ag^+]^2}{[Cu^{2+}]}$

$$E_{cell} = 0.46 \text{ V} - 0.0295 \log \frac{(10^{-3})^2}{10^{-1}}$$

$= 0.46 - 0.0295 \log 10^{-5} = 0.46 - 0.0295 (-5)$

$= 0.46 + 0.0295 \times 5 = 0.6075$ V

30. The cell reaction in button cell :

$Zn_{(s)} + Ag_2O_{(s)} + H_2O_{(l)} \longrightarrow Zn^{2+}_{(aq)} + 2Ag_{(s)} + 2OH^-_{(aq)}$

(i) Calculation of E°_{cell}

Reactions :

Anode : $Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^-$

Cathode :

$Ag_2O_{(s)} + H_2O_{(l)} + 2e^- \longrightarrow 2Ag_{(s)} + 2OH^-_{(aq)} + 2e^-$

$n = 2$

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{Ag_2O/Ag} - E^\circ_{Zn^{2+}/Zn}$

$= + 0.80 - (- 0.76)$ V $= 1.56$ V

(ii) Calculation of $\Delta_r G^\circ$

$\Delta_r G^\circ = -nFE^\circ_{cell}$

$= - 2 \times 96500$ C mol$^{-1} \times 1.56$ V

$= - 301080$ C V mol^{-1}

$= - 301080$ J mol$^{-1} = - 301$ kJ mol^{-1}

31. At anode : $Al_{(s)} \rightarrow Al^{3+}_{(aq)} + 3e^-] \times 2$

At cathode : $Ni^{2+} + 2e^- \rightarrow Ni_{(s)}] \times 3$

Cell reaction : $2Al_{(s)} + 3Ni^{2+}_{(aq)} \rightarrow 2Al^{3+}_{(aq)} + 3Ni_{(s)}$

Applying Nernst equation to the above cell reaction,

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2 \times 3} \log \frac{[Al^{3+}]^2}{[Ni^{2+}]^3}$$

Now, $E^\circ_{cell} = E^\circ_{Ni^{2+}/Ni} - E^\circ_{Al^{3+}/Al}$

$= - 0.25$ V $- (-1.66) = 1.41$ V

$$\therefore \quad E_{cell} = 1.41 \text{ V} - \frac{0.0591}{6} \log \frac{(10^{-3})^2}{(0.5)^3}$$

$$= 1.41 \text{ V} - \frac{0.0591}{6} \log (8 \times 10^{-6})$$

$$= 1.41 \text{ V} - \frac{0.0591}{6} (-5.09)$$

$= 1.41$V $+ 0.050$ V $= 1.46$ V

32. $2Fe^{3+} + 2e^- \longrightarrow 2Fe^{2+}$ and $2I^- \longrightarrow I_2 + 2e^-$

Hence, for the given cell reaction, $n = 2$

$\Delta_r G^\circ = - nFE^\circ_{cell} = - 2 \times 96500 \times 0.236$

$= - 45.55$ kJ mol^{-1}

$\Delta_r G^\circ = - 2.303\, RT \log K_c$

or $\log K_c = -\frac{\Delta_r G^\circ}{2.303RT}$

$$= -\frac{-45.55 \text{ kJ mol}^{-1}}{2.303 \times 8.314 \times 10^{-3} \text{ kJ K}^{-1}\text{mol}^{-1} \times 298 \text{ K}} = 7.983$$

$\therefore \quad K_c = \text{Antilog } (7.983) = 9.616 \times 10^7$

33. For half cell reaction,

$Cr_2O^{2-}_{7(aq)} + 14H^+_{(aq)} + 6e^- \rightarrow 2Cr^{3+}_{(aq)} + 7H_2O_{(l)}$

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[Cr^{3+}]^2}{[Cr_2O_7^{2-}][H^+]^{14}}$$

Given, $E^\circ_{cell} = 1.33$ V, $n = 6$, $[Cr^{3+}] = 0.2$ M

$[Cr_2O_7^{2-}] = 0.1$ M, $[H^+] = 1 \times 10^{-4}$ M

$$\Rightarrow \quad E_{cell} = 1.33 \text{V} - \frac{0.0591}{6} \log \frac{(0.20)^2}{(0.1)\,(10^{-4})^{14}}$$

$$= 1.33 \text{V} - \frac{0.0591}{6} \log (4 \times 10^{55})$$

$$= 1.33 \text{ V} - \frac{0.0591}{6} [\log 4 + \log 10^{55}]$$

$$= 1.33 \text{ V} - \frac{0.0591}{6} [\log 4 + 55 \log 10]$$

$$= 1.33 \text{ V} - \frac{0.0591}{6} [0.602 + 55]$$

$= 1.33$ V $- 0.548$ V $= 0.782$ V

34. (a) Oxidation half reaction :

$Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^-$

Reduction half reaction :

$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$

(b) $E^\circ_{cell} = 0.34 - (-0.76) = 1.10$ V

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$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$\Rightarrow \quad E_{cell} = 1.10 - \frac{0.0591}{2}\log\frac{2}{0.5}$$

$$= 1.10 - \frac{0.0591}{2}\log\frac{2}{0.5}$$

$$= 1.10 - \frac{0.059}{2} \times 0.6021$$

$$= 1.10 - 0.0177 = 1.0823 \text{ V}$$

35. $\Delta G^\circ = -RT \ln K_c = -2.303\, RT \log K_c$

$\Rightarrow -212300 = -2.303 \times 8.314 \times 298 \times \log K_c$

or $\log K_c = \dfrac{212300}{2.303 \times 8.314 \times 298} = 37.2074$

K_c = Antilog 37.2074 = 1.6×10^{37}.

36. $E_{cell} = E^\circ_{cell} - \dfrac{0.0591}{n}\log\dfrac{[Mg^{2+}]}{[Cu^{2+}]}$

$$= 2.71 - \frac{0.0591}{2}\log\frac{0.001}{0.01} = 2.73955 \text{ V}$$

(i) If external opposing potential is less than 2.71 V then current will flow from Cu to Mg.

(ii) If external opposing potential is greater than 2.71 V then current will flow in opposite direction *i.e.* from Mg to Cu.

37. $Mg_{(s)} \,|\, Mg^{2+}(0.001\text{ M}) \,||\, Cu^{2+}(0.0001\text{ M}) \,|\, Cu_{(s)}$

Reactions :

Anode : $Mg_{(s)} \longrightarrow Mg^{2+}_{(aq)} + 2e^-$

Cathode : $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$

Net cell reaction : $Mg_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Mg^{2+}_{(aq)} + Cu_{(s)}$

$\therefore \quad n = 2$

Using Nernst equation :

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF}\log\frac{[Mg^{2+}]}{[Cu^{2+}]}$$

For the given cell

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{Cu^{2+}/Cu} - E^\circ_{Mg^{2+}/Mg}$

$= 0.34 \text{ V} - (-2.37 \text{ V}) = 2.71 \text{ V}$

Given $[Mg^{2+}] = 0.001$ M, $[Cu^{2+}] = 0.0001$ M

Putting in Nernst equation at 298 K

$$E_{cell} = 2.71 \text{ V} - \frac{0.059}{2}\log\frac{0.001}{0.0001}$$

$E_{cell} = 2.71 - 0.0295 \log 10 = 2.68$ V

$\Delta_r G = -nFE_{cell} = -2 \times 96500 \text{ C mol}^{-1} \times 2.68$

$= -517.240 \text{ J mol}^{-1} = -517.24 \text{ kJ mol}^{-1}$

38. (c) : Conductivity decreases with decrease in concentration for both weak and strong electrolytes, because the number of ions per unit volume that carry the current in a solution decreases on dilution.

39. Conductivity of CH_3COOH (weak electrolyte) decreases with dilution because the number of current carrying particles *i.e.,* ions present per cm^3 of the solution becomes less and less on dilution.

40. The limiting molar conductivity of an electrolyte is defined as its molar conductivity when the concentration of the electrolyte in the solution approaches zero.

Conductivity of an electrolyte decreases with dilution because the number of current carrying particles *i.e.,* ions present per cm^3 of the solution becomes less and less on dilution.

41. Kohlrausch's law of independent migration of ions : It states that limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte.

If $\lambda^\circ_{Na^+}$ and $\lambda^\circ_{Cl^-}$ are limiting molar conductivities of the sodium and chloride ions respectively then the limiting molar conductivity for sodium chloride is given by

$\Lambda^\circ_{m\,(NaCl)} = \lambda^\circ_{Na^+} \lambda^\circ_{Cl^-}$

Kohlrausch's law helps in the calculation of degree of dissociation of weak electrolyte like acetic acid. The degree of dissociation α is given by

$$\alpha = \frac{\Lambda_m}{\Lambda^\circ_m}$$

where, Λ_m is the molar conductivity and Λ°_m is the limiting molar conductivity.

42. Molar Conductivity : Molar conductivity of a solution at a dilution V is the conductance of all the ions produced from one mole of the electrolyte dissolved in V cm^3 of the solution when the electrodes are one cm apart and the area of the electrodes is so large that the whole solution is contained between them.

$\Lambda_m = \kappa V$

It units is $S\ cm^2 mol^{-1}$

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43. *Refer to answer 41.*

44. *Refer to answer 40.*

45. *Refer to answer 41.*

46. $\Lambda_m = \dfrac{\kappa \times 1000}{M}$ in CGS units

$\Lambda_m = \dfrac{\kappa \times 10^{-3}}{M}$ in SI units

where, κ is the conductivity, M is the molar concentration and Λ_m is molar conductivity.

47. $\kappa = \dfrac{1}{R} \times \dfrac{l}{a}$

where, κ is the conductivity R is resistance and l/a is the cell constant.

48. In aqueous solution, CuCl2 ionises into

$CuCl_2 \longrightarrow Cu^{2+} + 2Cl^-$

At cathode : $Cu^{2+} + 2e^- \longrightarrow Cu$

At anode : $2Cl^- \longrightarrow Cl_2 + 2e^-$

At the cathode, the copper ion will be deposited because it has a higher reduction potential than the water molecule.

At the anode, the lower electrode potential value will be preferred but due to over potential of oxygen, chloride ion gets oxidized at the anode.

49. (i) Electrolyte *A* is a strong electrolyte while electrolyte *B* is a weak electrolyte.

(ii) For electrolyte *A*, the plot becomes linear near high dilution and thus can be extrapolated to zero concentration to get the molar conductivity at infinite dilution.

For weak electrolyte *B*, Λ_m increases steeply on dilution and extrapolation to zero concentration is not possible. Hence, molar conductivity at infinite dilution cannot be determined.

50. Degree of dissociation (α) = $\dfrac{\Lambda_m}{\Lambda^\circ_m}$

$$\alpha = \frac{39.05 \text{ S cm}^2 \text{ mol}^{-1}}{(349.68 + 40.9) \text{ S cm}^2 \text{ mol}^{-1}} = 0.1$$

51. The fraction of the total number of molecules present in solution as ions is known as degree of dissociation.

Molar conductivity (λ_m) = $\alpha\lambda^\circ_m$

where λ°_m is the molar conductivity at infinite dilution.

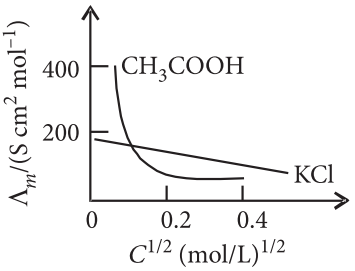
52. The reciprocal of resistivity is known as specific conductance or simply conductivity. It is denoted by κ (kappa).

$\kappa = \dfrac{1}{\rho}$ or $\kappa = G \times \dfrac{l}{a}$

Hence, conductivity of a solution is defined as the conductance of a conductor of 1 cm length and having 1 sq. cm as the area of cross section. Alternatively, it may be defined as conductance of one centimetre cube of the solution of the electrolyte. Molar conductivity of a solution at a dilution V is the conductance of all the ions produced from 1 mole of the electrolyte dissolved in V cm^3 of the solution. It is represented by Λ_m.

$\Lambda_m = \kappa V$

Variation of conductivity and molar conductivity with concentration : Conductivity always decreases with decrease in concentration, for both weak and strong electrolytes. Because the number of ions per unit volume that carry the current in a solution decreases on dilution.



Molar conductivity *vs* $C^{1/2}$ for acetic acid (weak electrolyte) and potassium chloride (strong electrolyte) in aqueous solutions

Molar conductivity increases with decrease in concentration.Because the total volume, V of solution containing one mole of electrolyte also increases. It has been found that decrease in κ on dilution of a solution is more than compensated by increase in its volume.

53. *Refer to answers 41 and 40.*

54. *Refer to answer 52.*

Variation of conductivity and molar conductivity with temperature : Both increase with increase in temperature as degree of ionisation increases.

55. Conductivity (κ) = $\dfrac{1}{R} \times G = \dfrac{1}{200} \times 1$

$= 5 \times 10^{-3} \ \Omega^{-1} \text{ cm}^{-1}$

Molar conductivity (Λ_m)

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$$= \frac{\kappa \times 1000}{M} = \frac{5 \times 10^{-3} \times 1000}{0.01}$$

$= 500\ \Omega^{-1}\ cm^2\ mol^{-1}$

56. Here, $\kappa = 0.025\ S\ cm^{-1}$, Molarity = 0.20 M

Molar conductivity $\Lambda_m^c = \frac{\kappa \times 1000}{\text{Molarity}}$

$= \frac{0.025 \times 1000}{0.20} = 125\ S\ cm^2\ mol^{-1}$

57. $C = 0.001$ M, $\kappa = 4 \times 10^{-5}\ S\ cm^{-1}$, $\Lambda_m^\infty = 390\ S\ cm^2/mol$

$\Lambda_m^c = \frac{\kappa \times 1000}{C}$

Substituting the values,

$\Lambda_m^c = \frac{4 \times 10^{-5} \times 1000}{0.001} = 40\ S\ cm^2/mol$

$\alpha = \frac{\Lambda_m^c}{\Lambda_m^\infty} = \frac{40}{390} = 0.10256 \simeq 0.103$

$$\underset{c}{CH_3COOH} \rightleftharpoons \underset{0}{CH_3COO^-} + \underset{0}{H^+}$$
$$\underset{c(1-\alpha)}{} \qquad \underset{c\alpha}{} \qquad \underset{c\alpha}{}$$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{c\alpha \cdot c\alpha}{c(1-\alpha)} = \frac{c\alpha^2}{1-\alpha}$$

$$K_a = \frac{0.001(0.103)^2}{(1-0.103)} = \frac{1.061 \times 10^{-5}}{0.897} = 1.18 \times 10^{-5}$$

58. *Refer to answers 47 and 46.*

59. (i) Silver wire at 30°C allows greater conduction of electricity than 60°C because with increase in temperature metallic conduction decreases due to vibration of kernels.

(ii) 0.1 M acetic acid solution allows greater conduction of electricity because with dilution degree of dissociation increases and hence, number of ions increases.

(iii) KCl solution at 50°C will have greater conductance. This is because ionic mobilities increases with increase in temperature.

60. Given : $a = 0.625\ cm^2$, $l = 50$ cm

$R = 5 \times 10^3$ ohm, M = 0.05 M

Resistivity, $\rho = R \times \frac{a}{l} = 5 \times 10^3 \times \frac{0.625}{50}$ ohm cm

$= 62.5$ ohm cm

Conductivity, $\kappa = \frac{1}{\rho} = \frac{1}{62.5} = 0.016\ ohm^{-1}\ cm^{-1}$

Molar conductivity, $\Lambda_m = \frac{\kappa \times 1000}{M}$

$= \frac{0.016 \times 1000}{0.05} = 320\ ohm^{-1}\ cm^2\ mol^{-1}$

61. Using formula, $\Lambda_m^c = \frac{\kappa \times 1000}{C}$

Given $\kappa = 3.905 \times 10^{-5}\ S\ cm^{-1}$

$C = 0.001\ mol\ L^{-1}$

$\therefore\ \Lambda_m^c = \frac{3.905 \times 10^{-5} \times 1000}{0.001} = 39.05\ S\ cm^2\ mol^{-1}$

The degree of dissociation,

$\alpha = \frac{\Lambda_m^c}{\Lambda_m^\circ} = \frac{39.05}{390.5} = 0.1$

$[\because \Lambda_m^\circ = 349.6 + 40.9 = 390.5\ S\ cm^2\ mol^{-1}]$

62. Given : Conductivity, $\kappa = 0.0248\ S\ cm^{-1}$

Molarity, $C = 0.20$ M $= 0.20\ mol\ L^{-1}$

Using formula, $\Lambda_m^c = \frac{1000 \times \kappa}{C}$

$\Lambda_m^c = \frac{1000 \times 0.0248}{0.20}$

$= 124\ S\ cm^2\ mol^{-1}$

$\alpha = \frac{\Lambda_m^c}{\Lambda_m^\circ} = \frac{124}{73.5 + 76.5} = 0.82$

63. Resistance of 0.1 M KCl solution $R = 100\ \Omega$

Conductivity $\kappa = 1.29\ S\ m^{-1}$

Cell constant $G^* = \kappa \times R = 1.29 \times 100 = 129\ m^{-1}$

Resistance of 0.02 M KCl solution, $R = 520\ \Omega$

Conductivity, $\kappa = \frac{\text{cell constant}}{R} = \frac{129\ m^{-1}}{520\ \Omega}$

$= 0.248\ S\ m^{-1}$

Concentration, $C = 0.02\ mol\ L^{-1}$

$= 1000 \times 0.02\ mol\ m^{-3} = 20\ mol\ m^{-3}$

Molar conductivity, $\Lambda_m = \frac{\kappa}{C} = \frac{0.248\ S\ m^{-1}}{20\ mol\ m^{-3}}$

$= 0.0124\ S\ m^2\ mol^{-1}$

64. $\Lambda^\circ_m\ Al_2(SO_4)_3 = 2\lambda^\circ_m Al^{3+} + 3\lambda^\circ_m SO_4^{2-}$

$\Rightarrow \quad 858 = 2\lambda^\circ_m Al^{3+} + 3 \times 160$

$\Rightarrow \quad \lambda^\circ_m Al^{3+} = \frac{858 - 480}{2} = 189\ S\ cm^2\ mol^{-1}$

65. Given : Diameter = 1 cm, length = 50 cm

$R = 5.5 \times 10^3$ ohm, $M = 0.05$ M

$\rho = ?\quad \kappa = ?\quad \Lambda_m = ?$

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Area of the column,

$$a = \pi r^2 = 3.14 \times \left(\frac{1}{2}\text{ cm}\right)^2 = \frac{3.14}{4}\text{ cm}^2$$

Resistivity,

$$\rho = R \cdot \frac{a}{l} = 5.5 \times 10^3 \text{ ohm} \times \frac{3.14\text{ cm}^2}{4 \times 50\text{ cm}}$$

$$= 86.35 \text{ ohm cm}$$

Again, conductivity, $\kappa = \frac{1}{\rho}$

$$= \frac{1}{86.35} = 1.158 \times 10^{-2} \text{ ohm}^{-1}\text{ cm}^{-1}$$

and molar conductivity, $\Lambda_m = \kappa \cdot \frac{10^3}{M}$

$$= 1.158 \times 10^{-2} \times \frac{10^3}{5 \times 10^{-2}}$$

$= 231.6 \text{ ohm}^{-1}\text{ cm}^2\text{ mol}^{-1}$

66. $\kappa = 1.29 \times 10^{-2} \text{ ohm}^{-1}\text{ cm}^{-1}$

$\kappa = \frac{1}{R} \times$ Cell constant

$\Rightarrow$ Cell constant $= \kappa \times R$

$= 1.29\ \Omega^{-1}\text{ m}^{-1} \times 85\ \Omega = 109.65\text{ m}^{-1}$

For second solution,

$$\kappa = \frac{1}{R} \times \text{Cell constant} = \frac{1}{96\ \Omega} \times 109.65\text{ m}^{-1}$$

$$= 1.142\ \Omega^{-1}\text{m}^{-1}$$

$$\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{1.142\ \Omega^{-1}\text{m}^{-1} \times 1000\text{ cm}^3}{0.052\text{ mol}}$$

$$\Lambda_m = \frac{1.142\ \Omega^{-1}\text{cm}^{-1} \times 10^{-2} \times 1000\text{ cm}^3}{0.052\text{ mol}}$$

$= 219.61\text{ S cm}^2\text{ mol}^{-1}$

67. According to Kohlrausch's law,

$\Lambda°CH_3COOH = \lambda°CH_3COO^- + \lambda°H^+$

Degree of dissociation, $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$

$$= \frac{11.7\text{ S cm}^2\text{mol}^{-1}}{(49.9 + 349.1)\text{ S cm}^2\text{mol}^{-1}} = \frac{11.7}{399} = 3 \times 10^{-2}$$

68. (a) *Refer to answers 52.*

(b) Here, conductivity (κ) $= 0.146 \times 10^{-3}$ S cm^{-1}, resistance (R) $= 1500\ \Omega$

Cell constant $= \frac{\text{Conductivity}}{\text{Conductance}}$

= Conductivity × Resistance

$= \kappa \times R \qquad \left[\because \text{conductance} = \frac{1}{\text{resistance}}\right]$

$= 0.146 \times 10^{-3} \times 1500 = 0.219\text{ cm}^{-1}$

69. 193000 Coulomb

70. The reaction at anode with lower value of $E°$ is preferred *i.e.*, O_2 gas should be liberated but on account of overpotential of oxygen reaction at anode, preferred reaction is

$$Cl^-_{(aq)} \rightarrow \frac{1}{2}Cl_{2(g)} + e^-$$

i.e., Cl_2 gas is liberated at anode in the electrolysis of aq. NaCl.

71. The species that get reduced at cathode is the one having higher value of standard reduction potential. Hence, the reaction that will occur at cathode is $Ag^+_{(aq)} + e^- \longrightarrow Ag_{(s)}$.

72. $\underset{1\text{ mol}}{Zn^{2+}_{(aq)}} + \underset{2\text{ mol}}{2e^-} \longrightarrow Zn_{(s)}$

One mole of Zn^{2+} requires 2 moles of electrons for reduction *i.e.*

$Q = 2 \times F = 2 \times 96500 = 193000$ C

73. The species that get reduced at cathode is the one which have higher value of standard reduction potential. Hence, the reaction that will occur at cathode is

$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$

74. Faraday's first law of electrolysis : During electrolysis the amount of any substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed through the electrolyte *i.e.*,

$w \propto Q$ or $w \propto I \times t \qquad [\because Q = I \times t]$

$w = Z \times I \times t$

where, Z is a constant of proportionality known as electrochemical equivalent of the substance deposited.

75. *Y* is a weak electrolyte. As the concentration of the weak electrolyte is reduced, more of it ionises. Hence a weak electrolyte shows a very large increase in concentration with dilution. A strong electrolyte shows only small increases in conductance with dilution, because a strong electrolyte is completely dissociated in solution so the number of ions remain constant.

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76. Given : $I = 2$ A, $t = 15$ min $= 15 \times 60$ s $= 900$ s
$w = ?$
$Q = I \times t = 2 \times 900 = 1800$ C
Reaction for deposition of Ag is as follows :

$$Ag^+ + 1e^- \longrightarrow Ag$$
$$1\text{ F} \qquad 1\text{ mol}$$
$$1 \times 96500\text{ C} \qquad 108\text{ g}$$

Thus, 1×96500 C of electricity is required to deposit 108 g of Ag.
$\therefore$ 1800 C of electricity would deposit

$$= \frac{108 \times 1800}{1 \times 96500} = 2.014\text{ g of Ag}$$

77. *Refer to answers 74.*
The electrode reaction is $Cu^{2+} + 2e^- \rightarrow Cu$
$\therefore$ Quantity of charge required for reduction of 1 mol of $Cu^{2+} = 2F$

$$= 2 \times 96500 = 193000\text{ C}$$

78. Given : Current $I = 5$ A; $t = 20 \times 60$ s, $w = ?$
$Q = I \times t = 5 \times 20 \times 60 = 6000$ C
Reaction for deposition of Ni,

$$Ni^{2+} + 2e^- \longrightarrow Ni$$
$$2\text{ mol} \qquad 1\text{ mol}$$
$$2 \times 96500\text{ C} \qquad 58.7\text{ g}$$

Thus, 2×96500 C of electricity produces 58.7 g Ni
$\therefore$ 6000 C of electricity would produce

$$= \frac{58.7 \times 6000}{2 \times 96500} = 1.825\text{ g}$$

79. (i) At the cathode the following reduction reactions compete to take place at the cathode.

$$Ag^+_{(aq)} + e^- \longrightarrow Ag_{(s)}\ ;\ E^\circ = 0.80\text{ V}$$

$$H^+_{(aq)} + e^- \longrightarrow \frac{1}{2}H_{2(g)}\ ;\ E^\circ = 0.00\text{ V}$$

The reaction with a higher value of E° takes place at the cathode. Therefore, the deposition of silver will take place at the cathode.
Since, Pt electrodes are inert, the anode is not attacked by NO_3^- ions. Therefore, OH^- or NO_3^- ions can be oxidized at the anode. But OH^- ions having a lower discharge potential get preference and decompose to liberate O_2.

$$OH^- \longrightarrow OH + e^-$$
$$4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$$

(ii) At the cathode, the following reduction reaction occurs to produce H_2 gas.

$$H^+_{(aq)} + e^- \longrightarrow \frac{1}{2}H_{2(g)}$$

At the anode, the following processes are possible :

$$2H_2O_{(l)} \longrightarrow O_{2(g)} + 4H^+_{(aq)} + 4e^-\ ;\ E^\circ = +1.23\text{ V} \quad ...(1)$$

$$2SO^{2-}_{4(aq)} \longrightarrow S_2O^{2-}_{8(aq)} + 2e^-\ ;\ E^\circ = +1.96\text{ V} \quad ...(2)$$

For dilute sulphuric acid, reaction (1) is preferred to produce O_2 gas but for concentrated sulphuric acid, (2) occurs.

80. Reaction for production of Ca from molten $CaCl_2$:

$$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$$
$$Ca^{2+} + 2e^- \longrightarrow Ca$$
$$2F \qquad 40\text{ g}$$

Electricity required to produce 40 g = 2 F
$\therefore$ Electricity required to produce 20 g

$$= 0.5 \times 2\text{ F} = 1\text{ F}$$

81. Calculation of mass of Ag deposited
The electrode reaction is $Ag^+ + e^- \longrightarrow Ag$
The quantity of electricity passed

$$= \text{Current} \times \text{Time}$$
$$= 0.5\text{ (amp)} \times 2 \times 60 \times 60\text{ (sec)} = 3600\text{ C}$$

From the electrode reaction, it is clear that
96500 C of electricity deposit Ag = 108 g
3600 C of electricity will deposit Ag

$$= \frac{108}{96500} \times 3600 = 4.03\text{ g}$$

Calculation of thickness :
Let the thickness of deposit be x cm
Mass = volume × density = Area × thickness × density
[$\because$ volume = area × thickness]
$\therefore$ $4.03\text{ g} = 900\text{ (cm}^2\text{)} \times x\text{ (cm)} \times 10.5\text{ (g cm}^{-3}\text{)}$

$$\therefore \quad x = \frac{4.03}{900 \times 10.5}\text{ cm} = 4.26 \times 10^{-4}\text{ cm}$$

82. The given reaction is

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$$

one mole $Cr_2O_7^{2-}$ requires 6 mol of electrons for reduction. Hence, quantity of electricity required $= 6\text{ mol} \times 96500\text{ C mol}^{-1} = 5.79 \times 10^5$ coulomb

83. Quantity of electricity passed
$Q = I \times t = 2.0\text{ A} \times 3 \times 60 \times 60\text{ s} = 21600$ C

$$Hg^{2+} + 2e^- \longrightarrow Hg$$
$$2\text{ F} \qquad 1\text{ mol}$$

2×96500 C electricity produces 1 mole Hg
$\therefore$ 21600 C will produce

$$= \frac{21600}{2 \times 96500}$$
$$= 0.112\text{ mole of Hg}$$

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84. According to Faraday's first law of electrolysis,
mass of substance deposited $\propto$ quantity of electricity passed

$$\Rightarrow \quad W = ZQ = \frac{E}{F} \times Q \Rightarrow \quad 2.8 = \frac{28}{96500} \times Q$$

$$\Rightarrow \quad Q = \frac{2.8 \times 96500}{28} = 9650\ C$$

$$\text{Now } Q = It \Rightarrow \quad t = \frac{Q}{I} = \frac{9650}{2} = 4825\ \text{s}$$

According to Faraday's second law of electrolysis,

$$\frac{\text{Mass of Fe deposited}}{\text{Mass of Zn deposited}} = \frac{\text{Eq.wt.of Fe}}{\text{Eq.wt.of Zn}}$$

$$\Rightarrow \quad \frac{2.8}{\text{Mass of Zn deposited}} = \frac{28}{32.65}$$

$$\Rightarrow \quad \text{Mass of Zn deposited} = \frac{2.8}{28} \times 32.65$$

$$= 3.265\ \text{g}$$

85. Charge = 24,000 coulombs
According to reaction,
$Cr^{6+} + 6e^- \rightarrow Cr_{(s)}$
We require 6F or 6 × 96500 C to deposit 1 mole or 52 g of Cr
For 24,000 C, the mass of Cr deposited

$$= \frac{52 \times 24{,}000}{6 \times 96500} = 2.15\ \text{g}$$

According to Faraday's second law of electrolysis,

$$\frac{w_1}{w_2} = \frac{Q_1}{Q_2} \quad \text{or} \quad \frac{2.15}{1.5} = \frac{24{,}000}{12.5 \times t}$$

$$t = \frac{24{,}000 \times 1.5}{12.5 \times 2.15} = 1340\ \text{s}$$

86. *Refer to answer 32.*
(b) Given, I = 0.5 A, t = 2 hrs.
Number of electrons = ?
Total charge (Q) = $I \times t$

$$= 0.5 \times 2 \times 60 \times 60 = 3600\ \text{C}$$

$$\therefore \quad \text{Number of electrons} = \frac{\text{Total charge}}{\text{Charge of one electron}}$$

$$= \frac{3600}{1.6 \times 10^{-19}}$$

$$= 2.25 \times 10^{22} \text{ electrons.}$$

87. Secondary batteries : The batteries which can be recharged again and again are called as secondary batteries. *e.g.*, lead storage battery.

88. The mercury cell is generally used in hearing aids.
Reaction at cathode :

$$HgO + H_2O + 2e^- \longrightarrow Hg + 2OH^-$$

Reaction at anode :

$$Zn(Hg) + 2OH^- \longrightarrow ZnO + H_2O + 2e^-$$

89. The cell which is generally used in inverters is secondary cell *i.e.*, lead storage battery.
At cathode : $PbO_{2(s)} + SO^{2-}_{4(aq)} + 4H^+_{(aq)} + 2e^- \longrightarrow PbSO_{4(s)} + 2H_2O_{(l)}$
At anode : $Pb_{(s)} + SO^{2-}_{4(aq)} \longrightarrow PbSO_{4(s)} + 2e^-$

90. Mercury cell is generally used in transistors.
At cathode : $HgO + H_2O + 2e^- \longrightarrow Hg + 2OH^-$
At anode : $Zn(Hg) + 2OH^- \longrightarrow ZnO + H_2O + 2e^-$

91. (i) Mercury cell is used for low current devices like watches and hearing aids.
(ii) The hydrogen oxygen fuel cell was used in Apollo space programme.
(iii) Lead storage battery is used in automobiles and inverters.
(iv) Dry cell

92. Mercury cell is a primary battery. Hence, it can be used only once and cannot be recharged.
Advantage : The cell potential remains constant during its life time. Hence, it is useful for devices requiring constant current *e.g.*, hearing aids and watches.

93. Lead storage battery is a secondary cell.
Cell reactions during operation
At anode : $Pb_{(s)} + SO^{2-}_{4(aq)} \rightarrow PbSO_{4(s)} + 2e^-$
At cathode :
$PbO_{2(s)} + SO^{2-}_{4(aq)} + 4H^+_{(aq)} + 2e^- \rightarrow PbSO_{4(s)} + 2H_2O_{(l)}$
Overall reaction :
$Pb_{(s)} + PbO_{2(s)} + 2H_2SO_{4(aq)} \rightarrow 2PbSO_{4(s)} + 2H_2O_{(l)}$

94. The lead storage battery is most important secondary cell. The cell reactions when the battery is in use.
At anode:
$Pb_{(s)} + SO^{2-}_{4(aq)} \rightarrow PbSO_{4(s)} + 2e^-$
At cathode :
$PbO_{2(s)} + SO^{2-}_{4(aq)} + 4H^+_{(aq)} + 2e^- \rightarrow PbSO_{4(s)} + 2H_2O_{(l)}$
The overall cell reaction is
$Pb_{(s)} + PbO_{2(s)} + 2H_2SO_{4(aq)} \rightarrow 2PbSO_{4(s)} + 2H_2O_{(l)}$
Following reaction takes place during recharging
At cathode : $PbSO_{4(s)} + 2e^- \rightarrow Pb_{(s)} + SO^{2-}_{4(aq)}$

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At anode : $PbSO_{4(s)} + 2H_2O_{(l)} \rightarrow PbO_{2(s)} + SO_{4\,(aq)}^{2-} + 4H_{(aq)}^{+} + 2e^-$

Net reaction : $2PbSO_{4(s)} + 2H_2O_{(l)} \rightarrow Pb_{(s)} + PbO_{2(s)} + 2H_2SO_{4(aq)}$

95. Mercury cell : It is a miniature cell which finds a frequent use these days to supply energy for watches, video cameras, hearing aids and other compact devices. In mercury cell the anode is zinc-mercury amalgam, and the cathode is a paste of mercury (II) oxide and carbon, electrolyte is a moist paste of KOH - ZnO.

The cell reactions are as follows :

Anode :

$Zn(Hg) + 2OH^- \longrightarrow ZnO_{(s)} + H_2O_{(l)} + 2e^-$

(Amalgam)

Cathode : $HgO_{(s)} + H_2O_{(l)} + 2e^- \longrightarrow Hg_{(l)} + 2OH^-$

Net reaction :

$Zn(Hg) + HgO_{(s)} \longrightarrow ZnO_{(s)} + Hg_{(l)}$

The cell potential remains constant during its life as the overall reaction does not involve any ion in solution whose concentration can change during its use.

96. *Refer to answer 91(ii).*

97. Those galvanic cells which give us direct electrical energy by the combustion of fuels like hydrogen, methane, methanol etc. are called fuel cells.

98. The two advantages of fuel cells are :

(i) They produce electricity with an efficiency of about 70% compared to thermal plants whose efficiency is about 40%.

(ii) They are pollution free.

99. *Refer to answer 97 and 98.*

100. (i) It is pollution free.

(ii) It has high efficiency of 70 - 75% and its rate can be controlled.

101. Fuel cells : *Refer to answer 97.*

The reactions taking place in hydrogen - oxygen fuel cell :

At cathode : $O_{2(g)} + 2H_2O_{(l)} + 4e^- \rightarrow 4OH^-_{(aq)}$

At anode : $2H_{2(g)} + 4OH^-_{(aq)} \rightarrow 4H_2O_{(l)} + 4e^-$

Net reaction : $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$

102. *Refer to answer 101.*

103. Metals of lower electrode potential value when connected with iron protect it from oxidation and prevent corrosion. Hence, coating of metal *A* having lower electrode potential will be better than *B* which has higher $E^\circ_{(B^{2+}/B)} = -0.14$ V.

104. According to electrochemical theory of rusting the impure iron surface behaves like small electrochemical cell. In this any point of iron acts as anode and other iron surface acts as cathode. Moisture having dissolved CO_2 or O_2 acts as an electrolyte. The reactions are given below.

At anode :

$$Fe \rightarrow Fe^{2+} + 2e^-;\ E^\circ_{Fe^{2+}/Fe} = -0.44\ V$$

At cathode :

$$2H^+ + \frac{1}{2}O_2 + 2e^- \rightarrow H_2O;\ E^\circ_{H^+/O_2/H_2O} = 1.23\ V$$

Overall reaction :

$$Fe + 2H^+ + \frac{1}{2}O_2 \rightarrow Fe^{2+} + H_2O;\ E^\circ_{cell} = 1.67\ V$$

The Fe^{2+} ions are further oxidised by atmospheric oxygen to Fe^{3+} ions, which form hydrated ferric oxide (rust).

$$2Fe^{2+} + \frac{1}{2}O_2 + 2H_2O \rightarrow Fe_2O_3 + 4H^+$$

$Fe_2O_3 + xH_2O \rightarrow Fe_2O_3 \cdot xH_2O$ (Rust)

105. Corrosion is the process of slowly eating away of the metal due to attack of the atmospheric gases on the surface of the metal resulting into the formation of compounds such as oxides, sulphides, carbonates, etc. The corrosion of iron is called rusting.

Electrochemical theory of rusting of iron : *Refer to answer 104.*

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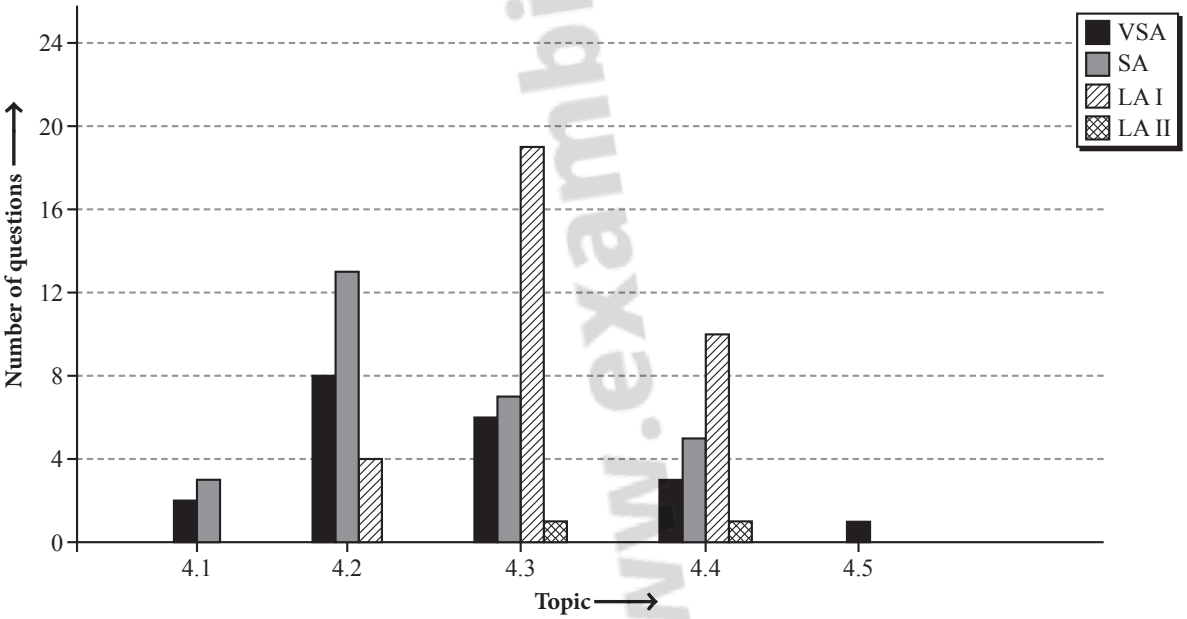
CHAPTER
4

Chemical Kinetics

4.1 Rate of a Chemical Reaction
4.2 Factors Influencing Rate of a Reaction
4.3 Integrated Rate Equations
4.4 Temperature Dependence of the Rate of a Reaction
4.5 Collision Theory of Chemical Reactions

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Integrated Rate Equations.*
- Maximum VSA and SA type questions were asked from *Factors Influencing Rate of a Reaction.*
- Maximum LA I type questions were asked from *Integrated Rate Equations.*

QUICK RECAP

- **Chemical kinetics :** It is the branch of chemistry which deals with the study of reaction rates and their mechanisms.
- **Rate of a reaction :** The rate of a reaction can be defined as the change in concentration of a reactant or a product in unit time. For the reaction,

$$R \longrightarrow P, \text{ Rate} = -\frac{\Delta[R]}{\Delta t} \text{ or } +\frac{\Delta[P]}{\Delta t}$$

- **Units of rate :** Concentration time^{-1} *i.e.*, mol L^{-1} s^{-1} or atm s^{-1} for gaseous reactions.

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Average rate of reaction : It is the average value during a large time interval.

$$r_{av} = \frac{-\Delta[R]}{\Delta t} = \frac{+\Delta[P]}{\Delta t}$$

Instantaneous rate of reaction : It is the rate of a reaction at a particular instant of time *i.e.*, when Δt approaches zero.

$$r_{inst} = \frac{-d[R]}{dt} = \frac{+d[P]}{dt}$$

Factors influencing rate of a reaction :

- **Concentration :** Greater the concentrations of the reactants, faster is the rate of reaction.
- **Physical state of reactants :** Reactions involving gaseous reactants are faster than reactions containing solid and liquid reactants.
- **Temperature :** The rate of reaction increases with increase of temperature. For most of the reactions, rate of reaction becomes almost double with 10° C rise of temperature.
- **Presence of catalyst :** A catalyst generally increases the speed of a reaction.
- **Surface area of reactants :** For a reaction involving a solid reactant or catalyst, the greater is the surface area, the faster is the reaction.
- **Presence of light :** Photochemical reactions take place in the presence of light only.
- **Activation energy :** Lower the activation energy faster is the reaction.

Rate law and rate constant : The equation that correlates the rate of reaction with concentration of reactants is known as *rate law*.

- For a simple reaction, $A + B \rightarrow C + D$
  Rate = $k[A][B]$ where k is the rate constant which is equal to the rate of reaction when concentration of each of the reactant is unity.

Order and molecularity :

Molecularity of reaction	Order of reaction
It is the total number of species taking part in a chemical reaction.	It is the sum of the powers of the concentration terms of reacting species in the rate law equation.
It is a theoretical concept.	It is an experimental quantity.
It is derived from the mechanism of reaction.	It is derived from the rate expression.
It can neither be zero nor fractional. It is always a whole number.	It may be zero, fractional or an integer (may range from 0 to 3).
It is applicable only to elementary reactions. The overall molecularity of a complex reaction has no significance.	It is applicable to elementary as well as complex reactions.

Half life of reaction : The time in which the concentration of a reactant is reduced to one half of its initial concentration is called *half life of the reaction*.

$t_{1/2} \propto \frac{1}{a^{n-1}}$; where n is the order of the reaction.

Rate law, integrated rate law, half-life, units of rate constant and graph for the reactions of different orders :

Order	Rate law	Integrated rate law	Half-life	Units of rate constant	Graph
0	Rate = $k[A]^0$	$[A]_t = -kt + [A]_0$	$t_{1/2} = [A]_0/2k$	mol L^{-1} s^{-1}	$[A]$ *vs* t; slope = $-k$
1	Rate = $k[A]^1$	$\ln[A]_t = -kt + \ln[A]_0$	$t_{1/2} = 0.693/k$	s^{-1}	$\ln[A]$ *vs* t; slope = $-k$
2	Rate = $k[A]^2$	$1/[A]_t = kt + 1/[A]_0$	$t_{1/2} = 1/k\,[A]_0$	L mol^{-1} s^{-1}	$1/[A]$ *vs* t; slope = k
2	Rate = $k[A]\,[B]$	$kt = \frac{1}{[A]_0 - [B]_0} \ln \frac{[B]_0[A]}{[A]_0[B]}$	–	L mol^{-1} s^{-1}	–
n	Rate = $k[A]^n$	$(n-1)kt = \frac{1}{[A]^{n-1}} - \frac{1}{[A_0]^{n-1}}$	$t_{1/2} = \frac{2^{n-1}-1}{k(n-1)[A]_0^{n-1}}$	$(\text{mol L}^{-1})^{1-n}$ s^{-1}	$\frac{1}{[A]^{n-1}}$ *vs* t; slope = k

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Pseudo first order reactions : Those reactions which are not truly of the first order but under certain conditions become reactions of the first order are called *pseudo first order reactions. e.g.,*

- **Acid hydrolysis of ethyl acetate :**

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

Rate $= k'[CH_3COOC_2H_5][H_2O]$
$= k[CH_3COOC_2H_5]$
where, $k = k'[H_2O]$

- **Acid catalysed inversion of cane sugar :**

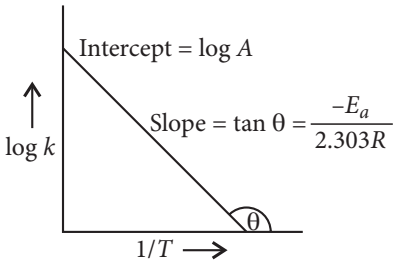
$$\underset{\text{Cane sugar}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

Rate $= k[C_{12}H_{22}O_{11}]$

Effect of temperature on rate of reaction :

- For a chemical reaction with rise in temperature by 10° C, the rate constant is nearly doubled.
- **Arrhenius equation :** $k = Ae^{-E_a/RT}$

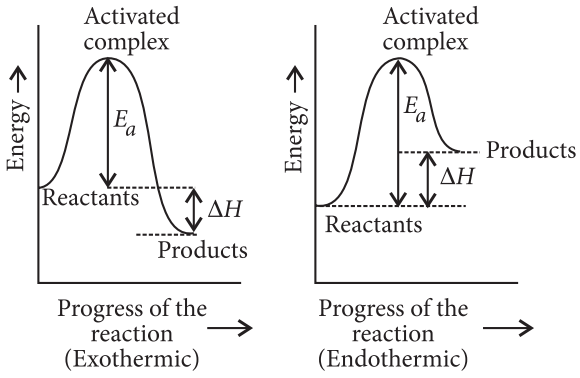
$$\log k = \log A - \frac{E_a}{2.303\,RT}$$



where, k = Rate constant,
A = Pre-exponential factor (frequency factor),
E_a = Activation energy,
T = Temperature

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

Activation energy : The minimum amount of energy required by reactant molecules to participate in a reaction is called *activation energy* (E_a).



- Activation energy = Threshold energy – Average kinetic energy of reacting molecules
- Activation energy (E_a) $= E_{(\text{activated complex})} - E_{(\text{ground state})}$
- ΔH = Activation energy of forward reaction – Activation energy of backward reaction

Collision theory of chemical reactions :

- Reaction occurs due to collision of molecules.
- All collisions are not effective.
- *Effective collisions* are those collisions in which molecules collide with sufficient kinetic energy (called *threshold energy* which is equal to activation energy + energy possessed by reacting species) and proper orientation.
- Rate $= PZ_{AB}e^{-E_a/RT}$ where, P = probability or steric factor which takes into account the fact that in a collision, molecules must be properly oriented and Z_{AB} = collision frequency (*i.e.*, the no. of collisions per second per unit volume of the reaction mixture) of reactants, A and B.

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Previous Years' CBSE Board Questions

4.1 Rate of a Chemical Reaction

VSA (1 mark)

1. Express the rate of the following reaction in terms of the formation of ammonia.
$N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$ *(Delhi 2013C)*

2. Define elementary step in a reaction. *(1/5, Foreign 2011)*

SA (2 marks)

3. In the given reaction
$A + 3B \rightarrow 2C$, the rate of formation of C is 2.5×10^{-4} mol L^{-1} s^{-1}.
Calculate the
(i) rate of reaction, and
(ii) rate of disappearance of B. *(2020)*

4. For the reaction, $2N_2O_{5(g)} \longrightarrow 4NO_{2(g)} + O_{2(g)}$, the rate of formation of $NO_{2(g)}$ is 2.8×10^{-3} M s^{-1}. Calculate the rate of disappearance of $N_2O_{5(g)}$. *(2018)*

5. What is meant by rate of reaction? Differentiate between average rate and instantaneous rate of reaction. (*AI 2012C*)

4.2 Factors Influencing Rate of a Reaction

VSA (1 mark)

6. **Assertion (A) :** The molecularity of the reaction $H_2 + Br_2 \longrightarrow 2HBr$ appears to be 2.
Reason (R) : Two molecules of the reactants are involved in the given elementary reaction.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

7. For a reaction $R \longrightarrow P$, half-life ($t_{1/2}$) is observed to be independent of the initial concentration of reactants. What is the order of reaction? *(Delhi 2017)*

8. Define the rate constant. *(1/5, Foreign 2015, Delhi 2015C)*

9. Define the specific rate of reaction. (*AI 2014C*)

10. For a reaction $A + B \rightarrow P$, the rate law is given by, $r = k[A]^{1/2}[B]^2$
What is the order of this reaction? *(AI 2013)*

11. If the rate constant of reaction is $k = 3 \times 10^{-4} s^{-1}$, then identify the order of the reaction. (*Delhi 2013C, 2011C*)

12. Define 'order of a reaction'. *(AI 2011)*

13. Identify the reaction order from the following rate constant :
$k = 2.3 \times 10^{-5}$ L $mol^{-1}s^{-1}$ *(Delhi, AI 2011C)*

SA (2 marks)

14. For a reaction

$$2H_2O_2 \xrightarrow[\text{alkaline medium}]{I^-} 2H_2O + O_2$$

the proposed mechanism is as given below :

(1) $H_2O_2 + I^- \longrightarrow H_2O + IO^-$ (slow)

(2) $H_2O_2 + IO^- \longrightarrow H_2O + I^- + O_2$ (fast)

(i) Write rate law for the reaction.
(ii) Write the overall order of reaction.
(iii) Out of steps (1) and (2), which one is rate determining step? *(Delhi 2019)*

15. For a reaction : $2NH_{3(g)} \xrightarrow{Pt} N_{2(g)} + 3H_{2(g)}$; Rate = k
(i) Write the order and molecularity of this reaction.
(ii) Write the unit of k. *(Delhi 2016)*

16. For a reaction : $H_2 + Cl_2 \xrightarrow{h\upsilon} 2HCl$; Rate = k
(i) Write the order and molecularity of this reaction
(ii) Write the unit of k. *(AI 2016)*

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17. For a reaction $A + B \longrightarrow P$, the rate is given by Rate = $k[A][B]^2$
(i) How is the rate of reaction affected if the concentration of B is doubled?
(ii) What is the overall order of reaction if A is present in large excess?
(2/5, Delhi 2015)

18. Define rate of reaction. Write two factors that affect the rate of reaction. *(AI 2015)*

19. Write units of rate constants for zero order and for the second order reactions if the concentration is expressed in mol L^{-1} and time in second. *(AI 2015C)*

20. Write two differences between 'order of reaction' and 'molecularity of reaction'.
(Delhi 2014, 2011C)

21. (i) For a reaction, $A + B \rightarrow$ Product, the rate law is given by, Rate = $k[A]^1[B]^2$. What is the order of the reaction?
(ii) Write the unit of rate constant 'k' for the first order reaction. *(Delhi 2014C)*

22. A reaction is of second order with respect to its reactant. How will its reaction rate be affected if the concentration of the reactant is (i) doubled (ii) reduced to half ?
(AI 2014C, 2012)

23. What do you understand by the 'order of a reaction'? Identify the reaction order from each of the following units of reaction rate constant :
(i) L^{-1} mol s^{-1} (ii) L mol^{-1} s^{-1}
(Delhi 2012)

24. Distinguish between 'rate expression' and 'rate constant' of a reaction. *(Delhi 2011)*

25. What do you understand by the rate law and rate constant of a reaction? Identify the order of a reaction if the units of its rate constant are :
(i) L^{-1} mol s^{-1} (ii) L mol^{-1} s^{-1} *(AI 2011)*

26. Express clearly what do you understand by 'rate expression' and 'rate constant' of a reaction. *(2/5, Foreign 2011)*

LA I (3 marks)

27. How will the rate of the reaction be affected when
(a) surface area of the reactant is reduced,
(b) catalyst is added in a reversible reaction, and
(c) temperature of the reaction is increased?
(2020)

28. A reaction is second order in A and first order in B.
(i) Write the differential rate equation.
(ii) How is the rate affected on increasing the concentration of A three times?
(iii) How is the rate affected when the concentration of both A and B are doubled? *(2018C, 3/5, Delhi 2013)*

29. The reaction, $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ contributes to air pollution whenever a fuel is burnt in air at a high temperature. At 1500 K, equilibrium constant K for it is 1.0×10^{-5}. Suppose in a case $[N_2] = 0.80$ mol L^{-1} and $[O_2] = 0.20$ mol L^{-1} before any reaction occurs. Calculate the equilibrium concentrations of the reactants and the product after the mixture has been heated to 1500 K. *(AI 2012)*

30. (i) A reaction is first order in A and second order in B.
(a) Write differential rate equation.
(b) How is rate affected when concentration of B is tripled?
(c) How is rate affected when concentration of both A and B is doubled?
(ii) What is molecularity of a reaction?
(AI 2012C)

4.3 Integrated Rate Equations

VSA (1 mark)

31. Write the slope value obtained in the plot of $\ln[R]$ *vs.* time for a first order reaction.
(One word, 2020)

32. The half-life period for a zero order reaction is equal to

(a) $\frac{0.693}{k}$ (b) $\frac{2k}{[R]_0}$

(c) $\frac{2.303}{k}$ (d) $\frac{[R]_0}{2k}$

(where $[R]_0$ is initial concentration of reactant and k is rate constant.) *(2020)*

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33. **Assertion (A) :** Hydrolysis of an ester follows first order kinetics.
Reason (R) : Concentration of water remains nearly constant during the course of the reaction.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

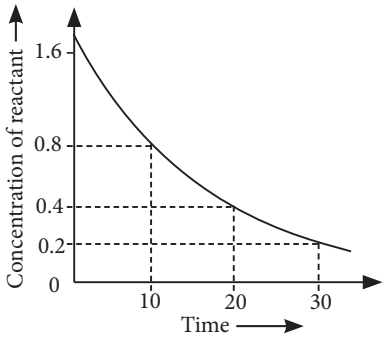
34. Define the half-life period of reaction ($t_{½}$). *(Delhi 2014)*

35. Define the following term :
Pseudo first order reaction *(1/2, Delhi 2014)*

36. If half-life period of a first order reaction is x and $3/4^{th}$ life period of the same reaction is y, how are x and y related to each other? *(1/5, Delhi 2013C)*

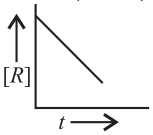
SA (2 marks)

37. Analyse the given graph, drawn between concentration of reactant *vs.* time.



(a) Predict the order of reaction.
(b) Theoretically, can the concentration of the reactant reduce to zero after infinite time. Explain. *(2020)*

38. For a chemical reaction $R \rightarrow P$, the variation in the concentration $[R]$ *vs.* time (t) plot is given as



(i) Predict the order of the reaction.
(ii) What is the slope of the curve? *(2/5, AI 2019, AI 2014)*

39. For a first order reaction, show that time required for 99% completion is twice the time required for the completion of 90% of reaction. *(AI 2017C, 2/5, Delhi 2013)*

40. Define half-life of a reaction. Write the expression of half-life for
(i) zero order reaction and
(ii) first order reaction. *(Foreign 2014)*

41. Rate constant k for a first order reaction has been found to be 2.54×10^{-3} sec^{-1}. Calculate its $3/4^{th}$ life. (log 4 = 0.6020) *(Delhi 2013C)*

42. The thermal decomposition of HCO_2H is a first order reaction with a rate constant of 2.4×10^{-3} s^{-1} at a certain temperature. Calculate how long will it take for three-fourth of initial quantity of HCO_2H to decompose.
(log 0.25 = – 0.6021) *(AI 2011)*

43. In a first order reaction, the concentration of the reactant is reduced from 0.6 mol L^{-1} to 0.2 mol L^{-1} in 5 minutes. Calculate the rate constant of the reaction. *(AI 2011C)*

LA I (3 marks)

44. A first order reaction is 40% complete in 80 minutes. Calculate the value of rate constant (k). In what time will the reaction be 90% completed?
[Given : log 2 = 0.3010, log 3 = 0.4771, log 4 = 0.6021, log 5 = 0.6771, log 6 = 0.7782] *(2020)*

45. The decomposition of NH_3 on platinum surface is zero order reaction. If rate constant (k) is 4×10^{-3} M s^{-1}, how long will it take to reduce the initial concentration of NH_3 from 0.1 M to 0.064 M ? *(Delhi 2019)*

46. A first order reaction takes 30 minutes for 20% decomposition. Calculate $t_{1/2}$.
[log 2 = 0.3010] *(3/5, AI 2019)*

47. Following data are obtained for the reaction :

$$N_2O_5 \longrightarrow 2NO_2 + \frac{1}{2}O_2$$

***t*/s**	0	300	600
$[N_2O_5]$/ mol L^{-1}	1.6×10^{-2}	0.8×10^{-2}	0.4×10^{-2}

(a) Show that it follows first order reaction.
(b) Calculate the half-life.
(Given : log 2 = 0.3010, log 4 = 0.6021) *(Delhi 2017)*

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48. A first order reaction takes 20 minutes for 25% decomposition. Calculate the time when 75% of the reaction will be completed.
(Given : log 2 = 0.3010, log 3 = 0.4771, log 4 = 0.6021) *(AI 2017)*

49. For the first order thermal decomposition reaction, the following data were obtained :
$C_2H_5Cl_{(g)} \longrightarrow C_2H_{4(g)} + HCl_{(g)}$

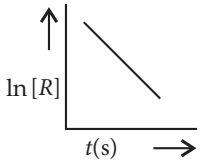
Time/sec	Total pressure/atm
0	0.30
300	0.50

Calculate the rate constant.
(Given : log 2 = 0.301, log 3 = 0.4771, log 4 = 0.6021) *(AI 2016)*

50. A first order reaction takes 30 minutes for 50% completion. Calculate the time required for 90% completion of this reaction. (log 2 = 0.3010) *(3/5, Delhi 2015)*

51. A first order reaction takes 10 minutes for 25% decomposition. Calculate $t_{1/2}$ for the reaction.
(Given : log 2 = 0.3010, log 3 = 0.4771, log 4 = 0.6021) *(3/5, Foreign 2015)*

52. For a chemical reaction $R \longrightarrow P$, the variation in the concentration, ln [R] *vs.* time (s) plot is given as



(i) Predict the order of the reaction.
(ii) What is the slope of the curve?
(iii) Write the unit of the rate constant for this reaction. *(3/5, Foreign 2015)*

53. The rate constant for a first order reaction is 60 s^{-1}. How much time will it take to reduce the initial concentration of the reactant to its $1/10^{th}$ value? *(AI 2015C)*

54. The following data were obtained during the first order thermal decomposition of SO_2Cl_2 at a constant volume :
$SO_2Cl_{2(g)} \longrightarrow SO_{2(g)} + Cl_{2(g)}$

Experiment	Time/s	Total pressure/atm
1	0	0.4
2	100	0.7

Calculate the rate constant.
(Given : log 4 = 0.6021, log 2 = 0.3010) *(Delhi, AI, Foreign 2014)*

55. Hydrogen peroxide, $H_2O_{2(aq)}$ decomposes to $H_2O_{(l)}$ and $O_{2(g)}$ in a reaction that is first order in H_2O_2 and has a rate constant $k = 1.06 \times 10^{-3}$ min^{-1}.
(i) How long will it take for 15% of a sample of H_2O_2 to decompose?
(ii) How long will it take for 85% of the sample to decompose? *(Delhi 2014C)*

56. The first order reaction take 100 minutes for completion of 60% of the reaction. Find the time when 90% of the reaction will be completed. *(3/5, Delhi 2013C)*

57. For the reaction :
$2NO_{(g)} + Cl_{2(g)} \longrightarrow 2NOCl_{(g)}$
The following data were collected. All the measurements were taken at 263 K.

Exp. No.	Initial [NO] (M)	Initial [Cl_2] (M)	Initial rate of disapp. of Cl_2 (M/min)
1.	0.15	0.15	0.60
2.	0.15	0.30	1.20
3.	0.30	0.15	2.40
4.	0.25	0.25	?

(i) Write the expression for rate law.
(ii) Calculate the value of rate constant and specify its units.
(iii) What is the initial rate of disappearance of Cl_2 in exp. 4? *(Delhi 2012)*

58. A first order reaction is 15% completed in 20 minutes. How long will it take to complete 60% of the reaction? *(3/5, Delhi 2012C)*

59. Nitrogen pentoxide decomposes according to equation :
$2N_2O_{5(g)} \longrightarrow 4NO_{2(g)} + O_{2(g)}$
This first order reaction was allowed to proceed at 40° C and the data below were collected :

$[N_2O_5]$ (M)	Time (min)
0.400	0.00
0.289	20.0
0.209	40.0
0.151	60.0
0.109	80.0

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(i) Calculate the rate constant. Include units with answer.
(ii) What will be the concentration of N_2O_5 after 100 minutes?
(iii) Calculate the initial rate of reaction.
(Delhi 2011)

60. Nitrogen pentoxide decomposes according to the equation :

$$2N_2O_{5(g)} \longrightarrow 4NO_{2(g)} + O_{2(g)}$$

This first order reaction was allowed to proceed at 40°C and the data given below were collected :

$[N_2O_5]$ (M)	Time (min)
0.400	0.00
0.289	20.00
0.209	40.00
0.151	60.00
0.109	80.00

(i) Calculate the rate constant for the reaction. Include units with your answer.
(ii) Calculate the initial rate of reaction.
(iii) After how many minutes will $[N_2O_5]$ be equal to 0.350 M? *(3/5, Delhi 2011)*

61. A first order reaction has a rate constant value of 0.00510 min^{-1}. If we begin with 0.10 M concentration of the reactant, how much of the reactant will remain after 3.0 hours? *(3/5, Foreign 2011)*

62. A reactant has a half-life of 10 minutes.
(i) Calculate the rate constant for the first order reaction.
(ii) What fraction of the reactant will be left after an hour of the reaction has occurred? *(Delhi 2011C)*

LA II (5 marks)

63. For the hydrolysis of methyl acetate in aqueous solution, the following results were obtained :

t/s	0	30	60
$[CH_3COOCH_3]$/mol L^{-1}	0.60	0.30	0.15

(i) Show that it follows pseudo first order reaction, as the concentration of water remains constant.
(ii) Calculate the average rate of reaction between the time interval 30 to 60 seconds.
(Given : log 2 = 0.3010, log 4 = 0.6021)
(Delhi 2015)

4.4 Temperature Dependence of the Rate of a Reaction

VSA (1 mark)

64. Will the rate constant of the reaction depend upon T if the E_{act} (activation energy) of the reaction is zero? *(One word, 2020)*

65. What is the effect of adding a catalyst on
(a) activation energy (E_a), and
(b) Gibbs energy (ΔG) of a reaction?
(AI 2017)

66. Define the following :
Energy of activation of reaction.
(1/5, Foreign 2015, AI 2014C)

SA (2 marks)

67. Draw the plot of ln k *vs* $1/T$ for a chemical reaction. What does the intercept represent? What is the relation between slope and E_a?
(2/5, AI 2019)

68. How does a change in temperature affect the rate of a reaction? How can this effect on the rate constant of reaction be represented quantitatively? *(AI 2014C)*

69. With the help of diagram, explain the physical significance of energy of activation (E_a) in chemical reactions. *(AI 2013C)*

70. With the help of a labelled diagram explain the role of activated complex in a reaction.
(2/5, Delhi 2012C)

71. The rate of most of the reactions becomes double when their temperature is raised from 298 K to 308 K. Calculate their activation energy.
[Given R = 8.314 J mol^{-1} K^{-1}] *(Delhi 2011C)*

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LA I (3 marks)

72. The rate constant for the first order decomposition of N_2O_5 is given by the following equation :
$k = (2.5 \times 10^{14}\ s^{-1})\ e^{(-25000\ K)/T}$
Calculate E_a for this reaction and rate constant if its half-life period be 300 minutes. *(2020)*

73. The rate of a reaction becomes four times when the temperature changes from 293 K to 313 K. Calculate the energy of activation (E_a) of the reaction assuming that it does not change with temperature.
[$R = 8.314\ J\ K^{-1}\ mol^{-1}$, log 4 = 0.6021] *(3/5, AI 2019, AI 2013)*

74. A first order reaction is 50% completed in 40 minutes at 300 K and in 20 minutes at 320 K. Calculate the activation energy of the reaction. (Given : log 2 = 0.3010, log 4 = 0.6021, $R = 8.314\ J\ K^{-1}\ mol^{-1}$) *(2018)*

75. The rate constant for the first order decomposition of H_2O_2 is given by the following equation :

$$\log k = 14.2 - \frac{1.0 \times 10^4}{T} K$$

Calculate E_a for this reaction and rate constant k if its half-life period be 200 minutes.
(Given : $R = 8.314\ J\ K^{-1}\ mol^{-1}$) *(Delhi 2016)*

76. The rate constant of a first order reaction increases from 2×10^{-2} to 4×10^{-2} when the temperature changes from 300 K to 310 K. Calculate the energy of activation (E_a).
(log 2 = 0.301, log 3 = 0.4771, log 4 = 0.6021) *(Delhi 2015)*

77. The rate constants of reaction at 500 K and 700 K are 0.02 s^{-1} and 0.07 s^{-1} respectively. Calculate the value of activation energy, E_a ($R = 8.314\ J\ K^{-1}\ mol^{-1}$). *(Delhi 2015C)*

78. For a decomposition reaction, the values of k at two different temperatures are given below:
$k_1 = 2.15 \times 10^{-8}$ L/(mol.s) at 650 K
$k_2 = 2.39 \times 10^{-8}$ L/(mol.s) at 700 K
Calculate the value of E_a for the reaction.
($R = 8.314\ J\ K^{-1}\ mol^{-1}$) *(AI 2014C)*

79. Rate constant 'k' of a reaction varies with temperature T according to the equation :

$$\log k = \log A - \frac{E_a}{2.303R}\left(\frac{1}{T}\right)$$

where E_a is the activation energy. When a graph is plotted for log k *vs* $\frac{1}{T}$, a straight line with a slope of – 4250 K is obtained. Calculate 'E_a' for the reaction.
($R = 8.314\ J\ K^{-1}\ mol^{-1}$). *(3/5, Delhi 2013)*

80. The decomposition of A into products has a value of k as $4.5 \times 10^3\ s^{-1}$ at 10°C and energy of activation is 60 kJ mol^{-1}. At what temperature would k be $1.5 \times 10^4\ s^{-1}$?
(3/5, Delhi 2013C)

81. In general it is observed that the rate of a chemical reaction doubles with every 10 degree rise in temperature. If the generalisation holds good for a reaction in the temperature range 295 K to 305 K, what would be the value of activation energy for this reaction?
[$R = 8.314\ J\ K^{-1}\ mol^{-1}$] *(AI 2012C)*

LA II (5 marks)

82. (i) What is the physical significance of energy of activation? Explain with diagram.
(ii) In general, it is observed that the rate of a chemical reaction doubles with every 10 degree rise in temperature. If the generalisation holds good for the reaction in the temperature range of 295 K to 305 K, what would be the value of activation energy for this reaction?
[$R = 8.314\ J/K^{-1}\ mol^{-1}$] *(Delhi 2012C)*

4.5 Collision Theory of Chemical Reactions

VSA (1 mark)

83. In some cases it is found that a large number of colliding molecules have energy more than threshold energy, yet the reaction is slow, Why? *(Delhi 2013C)*

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Detailed Solutions

1. $N_{2\,(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$

$$\frac{-d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = +\frac{1}{2}\frac{d[NH_3]}{dt}$$

2. Elementary step : Each step of a complex reaction is called an elementary step.

3. Given, $\frac{d[C]}{dt} = 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}

According to rate law expression,

$$-\frac{d[A]}{dt} = -\frac{1}{3}\frac{d[B]}{dt} = \frac{1}{2}\frac{d[C]}{dt}$$

(i) Rate of reaction $= \frac{1}{2}\frac{d[C]}{dt}$

$= \frac{1}{2} \times 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}

$= 1.25 \times 10^{-4}$ mol L^{-1} s^{-1}

(ii) Rate of disappearance of $B = -\frac{d[B]}{dt} = \frac{3}{2}\frac{d[C]}{dt}$

$= \frac{3}{2} \times 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}

$= 3.75 \times 10^{-4}$ mol L^{-1} s^{-1}

4. Given, $\frac{d[NO_2]}{dt} = 2.8 \times 10^{-3}$ M s^{-1}

According to rate law expression,

$$-\frac{1}{2}\frac{d[N_2O_5]}{dt} = \frac{1}{4}\frac{d[NO_2]}{dt} = \frac{d[O_2]}{dt}$$

$$\therefore \quad -\frac{d[N_2O_5]}{dt} = \frac{2}{4} \times 2.8 \times 10^{-3}$$

$$\frac{-d[N_2O_5]}{dt} = \frac{1}{2} \times 2.8 \times 10^{-3} = 1.4 \times 10^{-3} \text{ M s}^{-1}$$

5. Change in concentration (*i.e.*, either decrease in concentration of reactant or increase in concentration of product) per unit time is called rate of reaction.

$$\text{Rate of reaction} = \frac{C_2 - C_1}{t_2 - t_1} = \frac{\Delta C}{\Delta t}$$

The ratio of change of concentration of reactants to the time consumed in that change is called average rate of reaction.

$$r_{av} = \frac{\Delta x}{\Delta t} = -\frac{C_2 - C_1}{t_2 - t_1}$$

The rate of reaction at a particular instant (time) is called instantaneous rate of reaction.

$$r_{ins} = \frac{dx}{dt}$$

dx = small change in concentration

dt = small time interval

6. (a) : Molecularity is two because two molecules of the reactants are involved in the given elementary reaction.

7. Half-life of first order reaction is independent of the initial concentration of reactants.

$$t_{1/2} = \frac{0.693}{k}$$

8. Rate constant is the proportionality factor in the rate law expression for a chemical reaction. It is defined as the rate of a chemical reaction for which the concentration of each of the reacting species is unity.

9. At a given temperature, rate is equal to the rate constant of reaction when concentration of the reactant in unity. Thus rate constant is also known as specific reaction rate.

In the case of two reactants, the reaction may be written as :

$$A + B \rightarrow \text{Products}$$

$$r = \frac{dx}{dt} = k\,C_A C_B$$

where all the terms have their usual meaning.

if $C_A = C_B = 1$ then $r = k$.

10. Rate law, $r = k[A]^{1/2}\,[B]^2$

Order of reaction is sum of the powers of concentration terms.

$$\therefore \quad \text{Order of reaction} = \frac{1}{2} + 2 = \frac{5}{2} = 2.5$$

11. First order reaction.

12. It is defined as "the sum of the powers or exponents to which the concentration terms are raised in the rate law expression."

If rate $= k[A]^m\,[B]^n$, then order $= m + n$.

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13. Second order reaction :

$$\text{Unit of rate constant} = \frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^2}$$

$$= \text{L mol}^{-1}\text{s}^{-1}$$

14. (i) Rate = $k[H_2O_2][I^-]$
(ii) Overall order of reaction is 2.
(iii) Step (1) being the slow step is the rate determining step of the reaction.

15. (i) The decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction at high pressure.
In this reaction, platinum metal acts as a catalyst. At high pressure, the metal surface gets saturated with gas molecules. So, a further change in reaction conditions is unable to alter the amount of ammonia on the surface of the catalyst making rate of the reaction independent of its concentration.
However, two molecules of ammonia react to give products thus, the molecularity is two.
(ii) For a zero order reaction, unit of rate constant is mol L^{-1} s^{-1}.

16. (i) Experimentally, it is found that the rate of reaction is independent of the concentration of H_2 and Cl_2. Hence, it is a zero order reaction.
Molecularity = 2
(ii) Unit of k = mole litre^{-1} sec^{-1}

17. (i) From the rate law equation, order of reaction w.r.t. B is 2. Hence, if concentration of B is doubled, rate will become four times.
(ii) If A is present in large excess, rate of reaction will be independent of concentration of A and hence, order of reaction will be 2.

18. *Refer to answer 5.*

Following factors affect the rate of a chemical reaction :
(i) Concentration of reactants : Higher the concentration of reactants, faster would be the rate of reaction.
Rate = $k\,C^n$, where C = concentration of reactant.
(ii) Temperature : The rate of reaction increases with the temperature. For every 10°C rise in temperature rate of reaction increases 2 to 3 times.

19. Unit of rate constant (k), for zero order reaction.
Rate = $k[A]^0 \Rightarrow k$ = mol L^{-1} s^{-1}
Unit of rate constant (k), for second order reaction
Rate = $k[A]^2 \Rightarrow k$ = L mol^{-1} s^{-1}

20. Distinction between order and molecularity of a reaction :

	Order of a reaction	Molecularity of a reaction
1.	It is the sum of powers of the concentration of the reactants in the rate law expression.	It is the number of reacting species (atoms, ions or molecules) taking part in an elementary reaction which must collide simultaneously in order to bring about a chemical reaction.
2.	It can be zero or even a fraction.	It is always a whole number.

Generally, in a complex reaction the order of reaction is equal to the molecularity of the slowest step.

21. (i) Order of reaction is sum of powers of concentration terms,
$\therefore$ Order of reaction = 1 + 2 = 3
(ii) Unit of rate constant for first order reaction is s^{-1}.

22. Let the concentration of the reactant $[A] = a$
Order of reaction = 2
so rate of reaction = $k\,[A]^2 = ka^2$ ...(1)
(i) Given that concentration of the reactant is doubled
So, $[A] = 2a$,
Putting the value in equation (1) we get
New rate of reaction, $R_1 = k(2a)^2 = 4ka^2$
Hence, rate of reaction will increased to 4 times.
(ii) Given that concentration of the reactant is reduced to half
So, $[A] = (1/2)a$
Putting the value in equation (1), we get
New rate of reaction $R_2 = k((1/2)a)^2 = (1/4)ka^2$
Hence, rate of reaction will reduce to 1/4.

23. Order of a reaction : *Refer to answer 12.*
(i) L^{-1}mol s^{-1} – Zero order reaction
(ii) L mol^{-1} s^{-1} – Second order reaction

24. Rate expression is a way of expressing rate of reaction in terms of concentration of reactants, *e.g.*, for a general reaction, $aA + bB \longrightarrow cC + dD$

$$\text{Rate} = k[A]^x\,[B]^y$$

Rate constant (k) is equal to the rate of reaction when molar concentration of reactant is unity. Its units depends upon the order of reaction.

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25. *Refer to answers 24 and 23.*

26. *Refer to answer 24.*

27. (a) For a reaction involving a solid reactant or catalyst, the smaller is the particle size *i.e.*, greater is the surface area, the faster is the reaction. Therefore, on reducing the surface area, rate of reaction is reduced.

(b) A catalyst generally increases the speed of a reaction without itself being consumed in the reaction. In case of reversible reactions, a catalyst helps to attain the equilibrium quickly without disturbing the state of equilibrium.

(c) The rate of reaction increases with increase of temperature. In most of the cases, the rate of reaction becomes nearly double for 10° rise of temperature. In some cases, reactions do not take place at room temperature but takes place at higher temperature.

28. Since the reaction is second order w.r.t. A and first order w.r.t. B

(i) Differential rate equation is

Rate = $k[A]^2[B]$

(ii) The new concentration of A = $[3A]$

$\therefore$ New rate = $k[3A]^2[B] = 9k[A]^2[B]$

$\therefore$ New rate = 9 times of the original rate

(iii) New concentration of A = $[2A]$

New concentration of B = $[2B]$

$\therefore$ New rate = $k[2A]^2[2B] = 8k[A]^2[B]$

$\therefore$ New rate = 8 times of the original rate.

29. Given reaction is

$$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}\ ; K = 1.0 \times 10^{-5}$$

	N_2	O_2	$2NO$
Initially :	0.80	0.20	0
At equil.:	$0.80 - x$	$0.20 - x$	$2x$

According to law of mass action,

$$K = \frac{[NO]^2}{[N_2][O_2]}$$

$$\text{or} \quad 1.0\times10^{-5} = \frac{(2x)^2}{(0.80-x)(0.20-x)}$$

x is very small hence can be neglected in the denominator. Thus,

$$1.0\times10^{-5} = \frac{4x^2}{0.8\times0.2} \text{ or } 1.0\times10^{-5} = \frac{4x^2}{0.16}$$

or, $4x^2 = 1.0 \times 10^{-5} \times 0.16$ or $4x^2 = 16 \times 10^{-7}$

or, $x^2 = 4 \times 10^{-7}$ or $x = 6.32 \times 10^{-4}$

$\therefore$ $[N_2] = 0.80 - 0.000632 = 0.7994$ mol L^{-1}

$[O_2] = 0.20 - 0.000632 = 0.199$ mol L^{-1}

$[NO] = 2x = 2 \times 0.000632 = 0.001264$ mol L^{-1}

30. (i) (a) Differential rate equation of reaction is

$$\frac{dx}{dt} = k[A]^1[B]^2 = k[A][B]^2$$

(b) When conc. of B is tripled, it means conc. of B becomes $[3B]$

$\therefore$ New rate of reaction, $\frac{dx'}{dt} = k[A][3B]^2$

$= 9k[A][B]^2$

i.e., the rate of reaction will become 9 times.

(c) When conc. of A is doubled and that of B is also doubled, then conc. of A becomes $[2A]$ and that of B becomes $[2B]$.

$\therefore$ Now rate of reaction, $\frac{dx'}{dt} = k[2A][2B]^2$

$= 8k\ [A][B]^2$

i.e., the rate of reaction will become 8 times.

(ii) Molecularity of a reaction is the number of reacting particles which collide simultaneously to bring about the chemical change. It is a theoretical concept.

31. Slope = $-k$

32. (d) : For zero order reaction rate constant is given by :

$$k = \frac{[R]_0 - [R]}{t}, \quad \text{At } t = t_{1/2}, [R] = \frac{[R]_0}{2}$$

$$k = \frac{[R]_0 - \frac{[R]_0}{2}}{t_{1/2}} \Rightarrow k = \frac{[R]_0}{2t_{1/2}} \Rightarrow t_{1/2} = \frac{[R]_0}{2k}$$

In zero order reaction, $t_{1/2}$ is directly proportional to $[R]_0$.

33. (a) : Hydrolysis of an ester is a pseudo first order reaction.

34. The time taken for half of the reaction to complete, *i.e.,* the time in which the concentration of a reactant is reduced to half of its original value is called half-life period of the reaction.

$$t = t_{1/2} \text{ when } [R] = \frac{[R_0]}{2}$$

35. Those reactions which are not truly of the first order but under certain conditions become reactions of the first order are called *pseudo first order reactions.*

e.g., Inversion of cane sugar :

$$\underset{\text{Cane sugar}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

Rate = $k[C_{12}H_{22}O_{11}]^1[H_2O]^0$

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36. For a first order reaction, $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{x}$

For $\frac{3}{4}$ th life period,

$a - x = a - \frac{3}{4}a = \frac{1}{4}a$

$\therefore \quad k = \frac{2.303}{y}\log\frac{a}{(a-x)}$

$\frac{0.693}{x} = \frac{2.303}{y}\log\frac{a}{a/4}$

$\frac{0.693}{x} = \frac{2.303}{y}2\log 2$

$\frac{0.693}{x} = \frac{0.693\times 2}{y}$

$y = 2x$

37. (a) The reaction is of first order.

(b) The rate of first order reactions and the concentration of the reactant vary exponentially with time. Hence, theoretically it will take infinite time for the reaction to go to completion (though after a finite period, the concentration becomes nearly zero).

38. (i) The reaction is of zero order.

(ii) Slope of the curve $= -k = \frac{d[R]}{dt}$

39. For first order reaction, $t = \frac{2.303}{k}\log\frac{[R_0]}{[R_t]}$

For 99% completion of reaction

$t = t_{0.99}$, $[R]_0 = 1$, $[R]_t = (1 - 0.99) = 0.01 = 10^{-2}$

$t_{0.99} = \frac{2.303}{k}\log\frac{1}{10^{-2}} = \frac{2.303}{k}\log 10^2 = \frac{2.303}{k}\times 2$ ...(i)

For 90% completion of reaction

$t = t_{0.90}$, $[R]_0 = 1$, $[R]_t = (1-0.9) = 0.1 = 10^{-1}$

$t_{0.90} = \frac{2.303}{k}\log\frac{1}{10^{-1}} = \frac{2.303}{k}\log 10 = \frac{2.303}{k}$ ...(ii)

Comparing equations (i) and (ii),

$t_{0.99} = 2 \times t_{0.90}$

40. *Refer to answer 34.*

(i) For zero order reaction rate constant is given by :

$k = \frac{[R]_0 - [R]}{t}$, At $t = t_{1/2}, [R] = \frac{[R]_0}{2}$

$k = \frac{[R]_0 - \frac{[R]_0}{2}}{t_{1/2}} \Rightarrow k = \frac{[R]_0}{2t_{1/2}} \Rightarrow t_{1/2} = \frac{[R]_0}{2k}$

In zero order reaction, $t_{1/2}$ is directly proportional to $[R]_0$.

(ii) For first order reaction,

$k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$, At $t = t_{1/2}, [R] = \frac{[R]_0}{2}$

$\therefore\ k = \frac{2.303}{t_{1/2}}\log\frac{[R]_0}{[R]_0/2}$ or, $t_{1/2} = \frac{2.303}{k}\log 2$

$t_{1/2} = \frac{2.303\times 0.3010}{k} \Rightarrow t_{1/2} = \frac{0.693}{k}$

In first order reaction, $t_{1/2}$ is independent of initial concentration.

41. The integrated rate equation for first order reaction is

$k = \frac{2.303}{t}\log\frac{a}{a-x}$

$k = \frac{2.303}{t}\log\frac{a}{a-\frac{3}{4}a}$

$k = \frac{2.303}{t}\log\frac{a}{0.25a}$

$t_{3/4} = \frac{2.303}{2.54\times 10^{-3}\ \text{sec}^{-1}}\times\log\frac{1}{0.25}$

$t_{3/4} = 546$ s

Therefore, the 3/4th life of the reaction is 546 seconds.

42. For a first order reaction,

$t = \frac{2.303}{k}\log\frac{[R]_0}{[R]}$

Given $k = 2.4 \times 10^{-3}\ \text{s}^{-1}$

$[R] = \frac{[R]_0}{4}$, $t = ?$

Substituting these values in the equation, we get

$t = \frac{2.303}{2.4\times 10^{-3}\ \text{s}^{-1}}\log\frac{[R]_0}{[R]}$

$t = \frac{2.303}{2.4\times 10^{-3}\ \text{s}^{-1}}\log 4 = \frac{2.303}{2.4\times 10^{-3}}\times 0.6021$ s

$t = 577.7$ s $= 578$ s

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43. $a = 0.6$ mol/L, $(a - x) = 0.4$ mol/L
$t = 5$ min

$$k = \frac{2.303}{5\text{ min}} \log \frac{0.6}{0.4} = 0.0811 \text{ min}^{-1}$$

44. According to the 1st order kinetics,

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

or, $$k = \frac{2.303}{80} \log \frac{100}{60} = 6.386 \times 10^{-3} \text{ min}^{-1}$$

Time required for 90% completion of the reaction,

$$t = \frac{2.303}{k} \log \frac{100}{10} = \frac{2.303}{6.386 \times 10^{-3}} \log 10$$

= 360.63 minutes.

45. For a zero order reaction, $k = \frac{1}{t}([A]_0 - [A])$

$$\Rightarrow \quad t = \frac{1}{4 \times 10^{-3}} \times (0.1 - 0.064) = 9 \text{ s}$$

46. For a first order reaction, $k = \frac{2.303}{t} \log \frac{[R]_0}{R}$

$$k = \frac{2.303}{30} \log \frac{100}{100 - 20} = \frac{2.303}{30} \times 0.097 = 7.4 \times 10^{-3}$$

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{7.4 \times 10^{-3}} = 93.65 \text{ min}$$

47. (a) The formula of rate constant for first order reaction is

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

$$k_1 = \frac{2.303}{300 \text{ s}} \log \frac{(1.6 \times 10^{-2}) \text{ mol L}^{-1}}{(0.8 \times 10^{-2}) \text{ mol L}^{-1}} = 2.3 \times 10^{-3} \text{ s}^{-1}$$

Similarly, $$k_2 = \frac{2.303}{600 \text{ s}} \log \frac{(1.6 \times 10^{-2}) \text{ mol L}^{-1}}{(0.4 \times 10^{-2}) \text{ mol L}^{-1}}$$

$$= 2.3 \times 10^{-3} \text{ s}^{-1}$$

Unit and magnitude of rate constant shows the given reaction is of first order.

(b) The formula for half-life for first order reaction is

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{2.3 \times 10^{-3} \text{ s}^{-1}} = 301.30 \text{ s}$$

48. According to the 1st order kinetics

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A_t]}$$

$$\Rightarrow \quad k = \frac{2.303}{20 \text{ min}} \log \frac{[100]}{[75]} \Rightarrow k = 0.014 \text{ min}^{-1}$$

Similarly, for 2nd process, $0.014 = \frac{2.303}{t} \log\left(\frac{100}{25}\right)$

$t = 99$ min

49. The given reaction is

	$C_2H_5Cl_{(g)}$	$\longrightarrow$ $C_2H_{4(g)}$	+ $HCl_{(g)}$
At time $t = 0$	0.30 atm	0	0
At time $t = 300$ sec	$0.30 - x$	x	x

Total pressure $= 0.30 - x + x + x = 0.50$

or $0.30 + x = 0.50$

$\therefore \quad x = 0.50 - 0.30 = 0.20$

$\therefore$ Initial pressure, $P_0 = 0.30$ atm

Pressure of C_2H_5Cl after 300 sec,

$P_t = 0.30 - 0.20 = 0.10$ atm

Using formula for first order reaction,

$$k = \frac{2.303}{t} \log \frac{P_0}{P_t}$$

$$k = \frac{2.303}{300} \log \frac{0.30}{0.10}$$

$$k = \frac{2.303}{300} \log 3 = \frac{2.303 \times 0.4771}{300}$$

$= 3.66 \times 10^{-3} \text{ sec}^{-1}$

50. $t_{50\%}$ or $t_{1/2} = 30$ minutes

$$\Rightarrow \quad k = \frac{0.693}{t_{1/2}} = \frac{0.693}{30} = 0.0231 \text{ min}^{-1}$$

$$t_{90\%} = \frac{2.303}{k} \log \frac{100}{100 - 90} = \frac{2.303}{0.0231} \log 10$$

= 99.69 minutes

51. For a first order reaction,

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

$$k = \frac{2.303}{10} \log \frac{100}{100 - 25}$$

$$k = \frac{2.303}{10} \times 0.124 = 0.03$$

$$\therefore \quad t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.03} = 23.1 \text{ min}$$

52. (i) The reaction is of 1st order.

(ii) For first order reaction

$\ln[R] = -kt + \ln [R]_0$

comparing eqn. $y = m \times x + c$

we get a straight line with slope $= -k$ and intercept equal to $\ln[R]_0$.

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(iii) Unit of rate constant for first order reaction

$$\frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^1} = \text{s}^{-1}$$

53. $t = \dfrac{2.303}{k} \log \dfrac{[R]_0}{[R]}$

$\Rightarrow \; k = 60\,\text{s}^{-1}, [R] = \dfrac{1}{10}[R]_0$

or, $[R]_0 = 10\,[R]$

$$t = \frac{2.303}{60} \log \frac{10[R]}{[R]} \Rightarrow \; t = \frac{2.303}{60} \log 10$$

$$\Rightarrow \; t = \frac{2.303 \times 1}{60}$$

$t = 0.0384$ second

54. The given reaction is

	$SO_2Cl_{2(g)} \longrightarrow$	$SO_{2(g)}$ +	$Cl_{2(g)}$
At $t = 0$	0.4 atm	0	0
At time t	$(0.4 - x)$ atm	x atm	x atm

Total pressure at time t will be

$P_T = (0.4 - x) + x + x = 0.4 + x$

$x = (P_T - 0.4)$

Pressure of SO_2Cl_2 at time t will be

$p_{SO_2Cl_2} = 0.4 - x = 0.4 - (P_T - 0.4) = 0.8 - P_T$

At time $t = 100$ s, $P_T = 0.7$ atm

$\therefore \; p_{SO_2Cl_2} = 0.8 - 0.7 = 0.1$ atm

According to first order kinetic equation

$$k = \frac{2.303}{t} \log \frac{p_{SO_2Cl_2(\text{initial})}}{p_{SO_2Cl_2(\text{after reaction})}}$$

$$= \frac{2.303}{100} \log\left(\frac{0.4}{0.1}\right) = 1.3 \times 10^{-2}\ \text{s}^{-1}$$

55. (i) $t = \dfrac{2.303}{k} \log \dfrac{[A]_0}{[A]}$

Given $k = 1.06 \times 10^{-3}\,\text{min}^{-1}$, $\dfrac{[A]_0}{[A]} = \dfrac{100}{85}$

$$t = \frac{2.303}{1.06 \times 10^{-3}\,\text{min}^{-1}} \log \frac{100}{85}$$

$$t = \frac{2303}{1.06}[2 \log 10 - \log 85]\,\text{min}$$

$$t = \frac{2303}{1.06}[2 \times 1 - 1.9294] = \frac{2303 \times 0.0706}{1.06}$$

$t = 153.39$ min $= 153.4$ min.

(ii) Given $k = 1.06 \times 10^{-3}\,\text{min}^{-1}$, $\dfrac{[A]_0}{[A]} = \dfrac{100}{15}$

$$t = \frac{2.303}{1.06 \times 10^{-3}\,\text{min}^{-1}} \log \frac{100}{15}$$

$$= \frac{2303}{1.06}[2 \log 10 - \log 15]\,\text{min}$$

$$= \frac{2303}{1.06}[2 \times 1 - 1.1761] = \frac{2303 \times 0.8239}{1.06}\,\text{min}$$

$= 1790$ min

56. For the first order reaction,

Rate constant, $k = \dfrac{2.303}{100} \log \dfrac{a}{a - x}$

For 60% completion of the reaction, if $a = 100\%$

$a - x = 100 - 60 = 40\%$

Then, $k = \dfrac{2.303}{100} \log \dfrac{100}{40}$ ...(i)

For 90% completion of the reaction,

$a = 100\%$

and $a - x = 100 - 90 = 10\%$

Then, $k = \dfrac{2.303}{t} \log \dfrac{100}{10}$ ...(ii)

Substituting the value of k in eq. (ii),

we have, $\dfrac{2.303}{100} \log \dfrac{100}{40} = \dfrac{2.303}{t}$ $[\because \log 10 = 1]$

or $\dfrac{1}{t} = \dfrac{1}{100} \log \dfrac{100}{40}$

$$\frac{1}{t} = \frac{0.3979}{100}$$

$$t = \frac{100}{0.3979}$$

$t = 251$ min

57. Given reaction :

$2NO_{(g)} + Cl_{2(g)} \longrightarrow 2NOCl_{(g)}$

Exp. No.	Initial [NO] (M)	Initial $[Cl_2]$ (M)	Initial rate of disapp. of Cl_2 (M/min)
1.	0.15	0.15	0.60
2.	0.15	0.30	1.20
3.	0.30	0.15	2.40
4.	0.25	0.25	?

(i) Let rate of this reaction $r = k[NO]^m[Cl_2]^n$

then $\dfrac{r_1}{r_2} = \dfrac{0.60}{1.20} = \dfrac{k(0.15)^m(0.15)^n}{k(0.15)^m(0.30)^n}$

or $\dfrac{1}{2} = \left(\dfrac{1}{2}\right)^n \Rightarrow \; n = 1$

Again from $\dfrac{r_2}{r_3} = \dfrac{1.20}{2.40} = \dfrac{k(0.15)^m(0.30)^1}{k(0.30)^m(0.15)^1}$

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or $\frac{1}{2}=\left(\frac{1}{2}\right)^m\cdot\frac{2}{1}$ or $\frac{1}{4}=\left(\frac{1}{2}\right)^m \Rightarrow m=2$

Hence, expression for rate law is

$r = k[NO]^2\,[Cl_2]^1$

(ii) Substituting the values in experiment (1),

$0.60\ M\ min^{-1} = k(0.15\ M)^2\,(0.15\ M)^1$

$$\text{or } k=\frac{0.60\ M\,min^{-1}}{0.0225\times 0.15\ M^3}=177.77\ M^{-2}\ min^{-1}$$

(iii) $r = 177.7\ M^{-2}\ min^{-1} \times (0.25\ M)^2\,(0.25\ M)$

$= 2.77\ M\ min^{-1}$

58. $$k=\frac{2.303}{20\,min}\log\frac{100}{100-15}=\frac{2.303\times 0.0706}{20}\,min^{-1}$$

$$t=\frac{2.303}{k}\log\frac{100}{100-60}=\frac{2.303\times 20}{2.303\times 0.0706}\log\frac{100}{40}$$

$$=\frac{20}{0.0706}\times 0.3979 = 112.7\ min$$

59. (i) When $t = 20$ min, $[R] = 0.289$ mol L^{-1}

$[R]_0 = 0.400$ mol L^{-1}

For a first order reaction,

$$k=\frac{2.303}{t}\log\frac{[R]_0}{[R]}$$

$$\therefore\quad k=\frac{2.303}{20}\log\frac{0.400}{0.289}$$

$$\Rightarrow\quad k=\frac{2.303}{20}\log\frac{4.00}{2.89}$$

$$\Rightarrow\quad k=\frac{2.303}{20}[\log 4.00-\log 2.89]$$

$$\Rightarrow\quad k=\frac{2.303}{20}[0.6021-0.4609]$$

$$\Rightarrow\quad k=\frac{2.303}{20}\times 0.1412$$

$\Rightarrow\ k = 2.303 \times 0.00706$

$\Rightarrow\ k = 1.626 \times 10^{-2}\ min^{-1}$

(ii) $$t=\frac{2.303}{k}\log\frac{[R]_0}{[R]}$$

Here, $[R]_0 = 0.400$ mol L^{-1}, $t = 100$ min,

$k = 1.626 \times 10^{-2}\ min^{-1}$, $[R] = ?$

$$\therefore\quad 100=\frac{2.303}{1.626\times 10^{-2}}\log\frac{0.400}{[R]}$$

$$\frac{100\times 1.626\times 10^{-2}}{2.303}=\log\frac{0.4}{[R]}$$

$$0.7060=\log\frac{0.4}{[R]}$$

or $\text{Antilog}\ (0.7060)=\frac{0.4}{[R]}$

$$5.082=\frac{0.4}{[R]}\Rightarrow [R]=\frac{0.4}{5.082}=0.079\ M$$

(iii) Initial rate, *i.e.*, rate of reaction when $t = 0$

When $t = 0.00$ min, $[R] = 0.400$ mol L^{-1}

Also, $k = 1.626 \times 10^{-2}\ min^{-1}$

$\therefore$ Initial rate $= k[R]$

$= 1.626 \times 10^{-2}\ min^{-1} \times 0.400$ mol L^{-1}

$= 6.504 \times 10^{-3}$ mol $L^{-1}\ min^{-1}$

60. (i) *Refer to answer 59(i).*

(ii) *Refer to answer 59(iii).*

(iii) $$t=\frac{2.303}{k}\log\frac{[R]_0}{[R]}$$

Here, $k = 1.626 \times 10^{-2}\ min^{-1}$, $[R]_0 = 0.400$ M,

$[R] = 0.350$ M

Substituting these values in the equation, we get

$$t=\frac{2.303}{1.626\times 10^{-2}}\log\frac{0.400}{0.350}$$

$$=\frac{2.303}{1.626\times 10^{-2}}[\log 40-\log 35]$$

$= 141.635\,[1.6021 - 1.5441] = 141.635 \times 0.0580$

$= 8.21$ min

61. Given : $k = 5.10 \times 10^{-3}\ min^{-1}$,

$t = 3 \times 60$ min $= 180$ min

$[R]_0 = 0.1$ M

Substituting these values in the equation

$$t=\frac{2.303}{k}\log\frac{[R]_0}{[R]}$$

We get, $$180\,min=\frac{2.303}{5.1\times 10^{-3}\ min^{-1}}\log\frac{0.1}{[R]}$$

$$\log\frac{0.1}{[R]}=\frac{180\,min\times 5.1\times 10^{-3}\,min^{-1}}{2.303}=0.398$$

$$\frac{0.1}{[R]}=\text{Antilog}\,(0.398)=2.50$$

$[R] = 0.1/2.5 = 0.04$ M

62. (i) $$k=\frac{0.693}{t_{1/2}}=\frac{0.693}{10}=0.0693\,min^{-1}$$

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(ii) $N = N_0\left(\frac{1}{2}\right)^n$

where, $n = \frac{\text{Total time}}{t_{1/2}} = \frac{60}{10} = 6$

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^6 = \frac{1}{64} \quad ...(1)$$

[where N_0 = initial amount of reactant and N = amount of reactant left after time, t].

63. (i) For a first order reaction,

$$k = \frac{2.303}{t}\log\frac{[A]_0}{[A]}$$

When t = 30 s

$$k = \frac{2.303}{30}\log\frac{0.60}{0.30} = \frac{2.303}{30}\times 0.3010 = 0.0231\ s^{-1}$$

When t = 60 s

$$k = \frac{2.303}{60}\log\frac{0.60}{0.15} = \frac{2.303}{60}\times 0.602 = 0.0231\ s^{-1}$$

As the value of k is constant at different time intervals, the reaction is first order w.r.t. ester when $[H_2O]$ is constant. Hence, it is pseudo first order reaction.

(ii) Average rate $= -\frac{C_2 - C_1}{t_2 - t_1} = \frac{-(0.15 - 0.30)}{60 - 30}$

$$= \frac{0.15}{30} = 5\times 10^{-3}\ \text{mol}\ L^{-1}s^{-1}$$

64. No

65. (a) A catalyst lowers the activation energy (E_a) by providing an alternate pathway or reaction mechanism.

(b) Catalyst does not affect the Gibbs energy (ΔG) of a reaction.

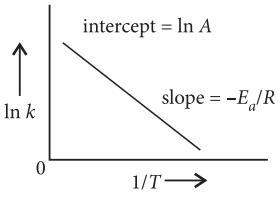
66. The minimum extra energy which must be supplied to the reactants to enable them to cross over the potential energy barrier between reactants and products is called activation energy. It is denoted by E_a.

$E_a = E_T - E_R$

67. $\ln k = -\frac{E_a}{RT} + \ln A$

Intercept = $\ln A$

Slope = $-\frac{E_a}{R}$



68. The rate of reaction is nearly doubled with a rise in temperature by 10° for a chemical reaction. The temperature effect on the rate constant can be represented quantitatively by Arrherius equation :

$k = Ae^{-E_a/RT}$

where $k \rightarrow$ Rate constant

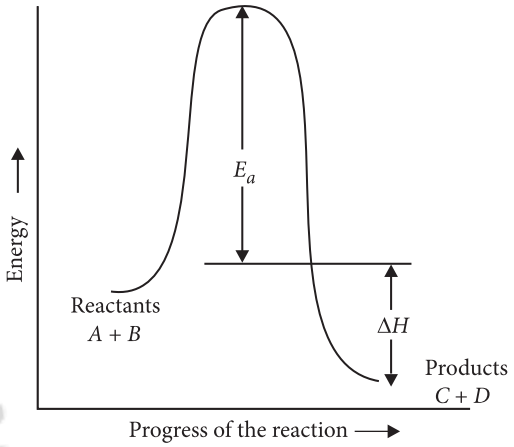
$A \rightarrow$ Arrhenius factor

$R \rightarrow$ Gas constant

$T \rightarrow$ Temperature

$E_a \rightarrow$ Energy of activation for the reaction.

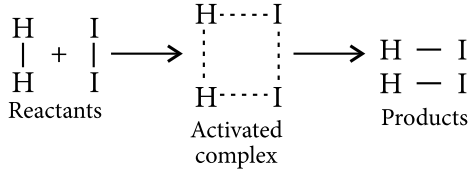
69. The energy required to form activated complex is called activation energy. It is the difference between the threshold energy and the average energy possessed by the reacting molecules.



Activation energy (E_a) = Threshold energy – Average energy possessed by reacting molecules.

70. When the colliding molecules possess the kinetic energy equal to E_a, the atomic configuration of species formed at this stage is different from the reactant as well as the products. This stage is called the activated state and the specific configuration bearing state is called activated complex.

For example, in the reaction between $H_{2(g)}$ and $I_{2(g)}$, activated complex has configuration in which H—H and I—I bonds are breaking and H—I bonds, are forming as shown :



Potential energy diagram of this reaction is shown as :

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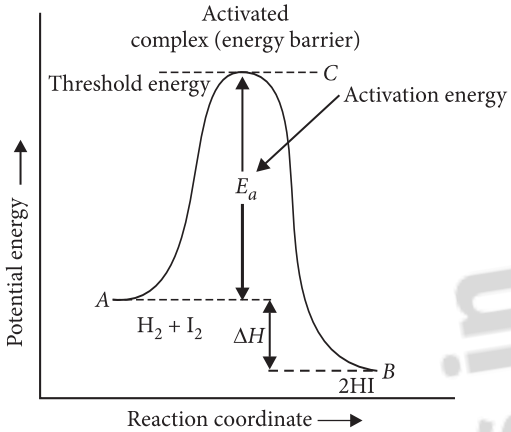

71. Here $k_2 = 2k_1$, $T_1 = 298$ K, $T_2 = 308$ K

$E_a = ?$

$$\therefore \quad \log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

$$\log\frac{2k_1}{k_1} = \frac{E_a}{2.303R}\left[\frac{308-298}{308\times298}\right]$$

$$\log 2 = \frac{E_a}{2.303R}\left[\frac{10}{91784}\right]$$

$$0.3010 = \frac{1.089\times10^{-4}E_a}{2.303\times8.314}$$

$$E_a = \frac{0.3010\times2.303\times8.314}{1.089\times10^{-4}} = 52922.77\,\text{J mol}^{-1}$$

72. $k = (2.5\times10^{14})e^{\frac{-25000}{T}}$

Comparing it with the Arrhenius equation

$k = A.e^{-Ea/RT}$

We get $A = 2.5\times10^{14}\ \text{s}^{-1}$

$$\frac{E_a}{R} = 25000$$

or $E_a = 25000\times8.314\ \text{J} = 207850\ \text{J mol}^{-1}$

For a first order reaction, $t_{1/2} = \dfrac{0.693}{k}$

or, $\quad k = \dfrac{0.693}{t_{1/2}} = \dfrac{0.693}{300} = 2.31\times10^{-3}\ \text{min}^{-1}$

73. Since the rate of a reaction quadruples when temperature changes from 293 K to 313 K,

$\therefore \quad k_2 = 4k_1$

$T_1 = 293$ K and $T_2 = 313$ K

According to Arrhenius equation

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

Putting the values

$$\log\frac{4k_1}{k_1} = \frac{E_a}{2.303\times8.314\ \text{J K}^{-1}\ \text{mol}^{-1}}\left[\frac{(313-293)\text{K}}{293\text{K}\times313\ \text{K}}\right]$$

$$0.6021 = \frac{E_a\times20\ \text{K}}{2.303\times8.314\ \text{JK}^{-1}\ \text{mol}^{-1}\times293\ \text{K}\times313\ \text{K}}$$

$$\therefore\ E_a = \frac{0.6021\times2.303\times8.314\times293\times313}{20}\ \text{J mol}^{-1}$$

$= 52863.3\ \text{J mol}^{-1} = 52.86\ \text{kJ mol}^{-1}$

74. $(t_{1/2})_1 = 40$ min, $(t_{1/2})_2 = 20$ min

$T_1 = 300$ K, $T_2 = 320$ K

$E_a = ?$

For 1st order reaction,

$$k_1 = \frac{0.693}{(t_{1/2})_1} = \frac{0.693}{40} = 0.017\ \text{min}^{-1}$$

$$k_2 = \frac{0.693}{(t_{1/2})_2} = \frac{0.693}{20} = 0.034\ \text{min}^{-1}$$

From Arrhenius equation,

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

$$\log\frac{0.034}{0.017} = \frac{E_a}{2.303\times8.314}\left[\frac{20}{320\times300}\right]$$

$$E_a = \frac{0.3010\times2.303\times8.314\times320\times300}{20}$$

$E_a = 27663.8\ \text{J} = 27.6\ \text{kJ}$

75. For first order reaction, $t_{1/2} = \dfrac{0.693}{k}$

$$t_{1/2} = 200\,\text{min} = 200\times60 = 12000\ \text{s}$$

$$\Rightarrow\ k = \frac{0.693}{12000\ \text{s}} = 5.8\times10^{-5}\ \text{s}^{-1}$$

Also, for first order decomposition of H_2O_2, rate constant is

$$\log k = 14.2 - \frac{1.0\times10^4}{T}\text{K}$$

Comparing the above equation with the Arrhenius equation,

$$\log k = \log A - \frac{E_a}{2.303RT}$$

$$\Rightarrow\ E_a = 2.303\times8.314\times1.0\times10^4 = 1.91\times10^5\,\text{J mol}^{-1}$$

76. Given, $T_1 = 300$ K, $T_2 = 310$ K

$k_1 = 2\times10^{-2}$, $k_2 = 4\times10^{-2}$, $\quad E_a = ?$

Using Arrhenius equation,

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$$\log\frac{k_2}{k_1}=\frac{E_a}{2.303R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$$

or $\log\dfrac{4\times10^{-2}}{2\times10^{-2}}=\dfrac{E_a}{2.303\times8.314}\left[\dfrac{1}{300}-\dfrac{1}{310}\right]$

or $\log 2=\dfrac{E_a}{2.303\times8.314}\times\left[\dfrac{310-300}{310\times300}\right]$

or $E_a=\dfrac{0.301\times2.303\times8.314\times310\times300}{10}$

$= 53598 \text{ J mol}^{-1} = 53.598 \text{ kJ mol}^{-1}$

77. $k_1 = 0.02 \text{ s}^{-1}$, $k_2 = 0.07 \text{ s}^{-1}$

$T_1 = 500$ K, $T_2 = 700$ K

$$\log\frac{k_2}{k_1}=\frac{E_a}{2.303R}\left[\frac{T_2-T_1}{T_2T_1}\right]$$

$$\log\frac{0.07}{0.02}=\frac{E_a}{2.303R}\left[\frac{700-500}{700\times500}\right]$$

$$E_a=\frac{0.544\times2.303\times8.314\times700\times500}{200}$$

$= 18228.07 \text{ J mol}^{-1}$

78. $$\log\frac{k_2}{k_1}=\frac{E_a}{2.303R}\left(\frac{T_2-T_1}{T_1T_2}\right)$$

$$E_a=\left(\frac{2.303\times R\times T_1\times T_2}{T_2-T_1}\right)\log\frac{k_2}{k_1}$$

$$E_a=\frac{2.303\times8.314\text{J mol}^{-1}\text{K}^{-1}\times650\text{K}\times700\text{K}}{700\text{K}-650\text{K}}\times\log\frac{2.39\times10^{-8}}{2.15\times10^{-8}}$$

$$E_a=\frac{19.147\times650\times700}{50}(\log 2.39-\log 2.15)\text{ J mol}^{-1}$$

$= 174237.7\ (0.3783 - 0.3324) \text{ J mol}^{-1}$

$= 174237.7 \times 0.0459 \text{ J mol}^{-1}$

$= 7997.51 \text{ J mol}^{-1}$

79. Given : Slope, $m = -4250$ K,

$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$

$E_a = ?$

Using formula,

$$\log k=\log A-\frac{E_a}{2.303\,R}\left(\frac{1}{T}\right)$$

Comparing with $y = mx + c$ [straight line equation]

$$-\frac{E_a}{2.303\,R}=\frac{-4250}{1}$$

$E_a = 2.303 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 4250 \text{ K}$

$= 81{,}375.35 \text{ J mol}^{-1} = 81.37 \text{ kJ mol}^{-1}$

80. $k_1 = 4.5\times10^3 \text{ s}^{-1}$, $T_1 = 10 + 273$ K, $k_2 = 1.5\times10^4 \text{ s}^{-1}$

$T_2 = ?$, $E_a = 60 \text{ kJ mol}^{-1}$

From Arrhenius equation,

$$\log\frac{k_2}{k_1}=\frac{E_a}{2.303R}\left[\frac{T_2-T_1}{T_1T_2}\right]$$

$$\log\frac{1.5\times10^4}{4.5\times10^3}=\frac{60000\text{ J mol}^{-1}}{2.303\times8.314\text{ J K}^{-1}\text{ mol}^{-1}}\left[\frac{T_2-283}{283T_2}\right]$$

or $\log 3.333 = 3133.63\left[\dfrac{T_2-283}{283T_2}\right]$

or $\dfrac{0.5228}{3133.63}=\dfrac{T_2-283}{283T_2}$

or $0.0472T_2 = T_2 - 283$ or $0.9528T_2 = 283$

or $T_2=\dfrac{283}{0.9528}=297\text{K}$ or $(297-273)^\circ\text{C}=24^\circ\text{C}$

81. $T_1 = 295$ K, $k_1 = k$ (say)

$T_2 = 305$ K, $k_2 = 2k$, $E_a = ?$

Using Arrhenius equation,

$$\log\frac{k_2}{k_1}=\frac{E_a}{2.303\,R}\left[\frac{T_2-T_1}{T_1T_2}\right]$$

$$\log\frac{2k}{k}=\frac{E_a}{2.303\times8.314}\times\frac{305-295}{295\times305}$$

$$E_a=\frac{2.303\times8.314\times295\times305\times\log2}{10}$$

$$=\frac{2.303\times8.314\times295\times305\times0.3010}{10}$$

$= 51855.2 \text{ J mol}^{-1}$

82. (i) *Refer to answer 69.*

(ii) *Refer to answer 81.*

83. It is due to improper orientation. Energy more than threshold energy and proper orientation are the two main factors which are responsible for a reaction to occur.

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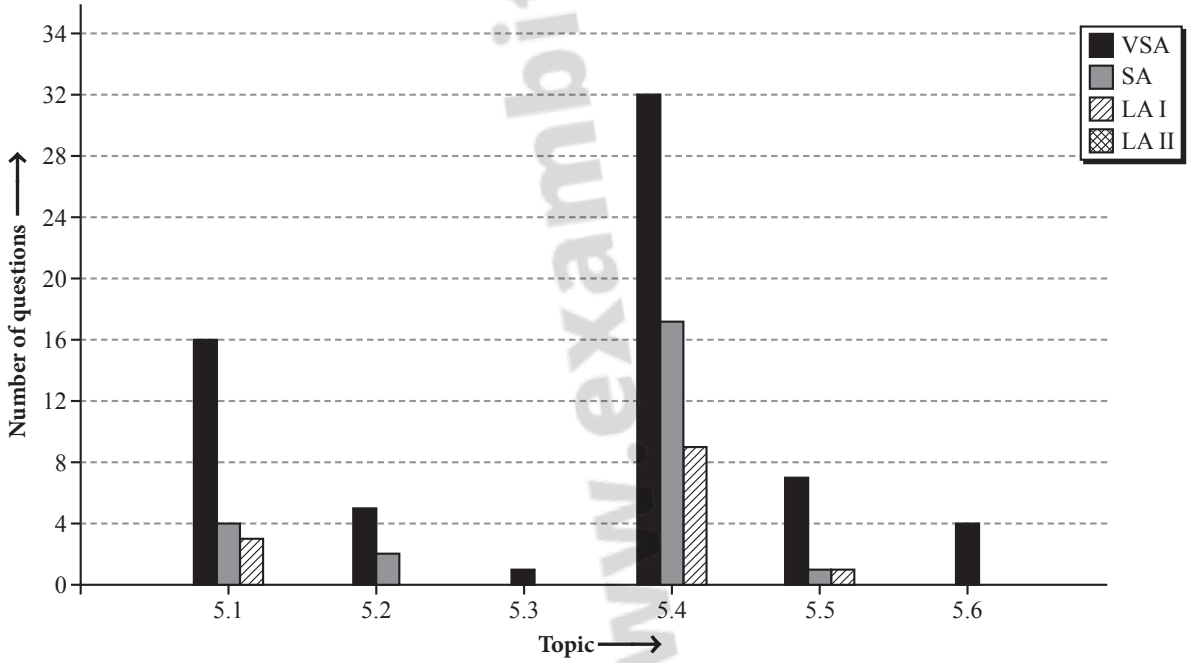
CHAPTER
5

Surface Chemistry

5.1 Adsorption
5.2 Catalysis
5.3 Colloids
5.4 Classification of Colloids
5.5 Emulsions
5.6 Colloids Around Us

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Classification of Colloids.*
- Maximum VSA type questions were asked from *Classification of Colloids.*
- Maximum SA and LA I type questions were asked from *Classification of Colloids.*

QUICK RECAP

- **Surface chemistry :** It deals with phenomena that occur at the surfaces or interfaces.
- **Adsorption :** It is the process of accumulation of molecular species at the surface rather than in the bulk of a solid or liquid.
- **Adsorbate :** The molecular species or substance, which concentrates or accumulates at the surface.
- **Adsorbent :** The material on the surface of which the adsorption takes place.

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Distinction between adsorption and absorption

Adsorption	Absorption
It is a surface phenomenon, *i.e.*, it occurs only at the surface of the adsorbent.	It is a bulk phenomenon, *i.e.*, occurs throughout the body of the material.
In this phenomenon, the concentration on the surface of adsorbent is different from that in the bulk.	In this phenomenon, the concentration is same throughout the material.
Its rate is high in the beginning and then decreases till equilibrium is attained.	Its rate remains same throughout the process.

- **Desorption :** The process of removing an adsorbed substance from the surface.
- **Sorption :** The term used when both absorption and adsorption occur simultaneously.
- ΔG, ΔH and ΔS all are – ve for adsorption.
- **Types of adsorption :** Depending on forces which hold the adsorbate on the surface of adsorbent, adsorption is divided into two classes :
  - **Physical adsorption :** When the particles are held to the surface by the physical forces like van der Waals forces, the adsorption is called *physical adsorption* or *physisorption*.
  - **Chemical adsorption :** When the particles are held to the surface by the chemical forces or by chemical bonds, the adsorption is called *chemical adsorption* or *chemisorption*.

Differences between physisorption and chemisorption

Property	Physisorption	Chemisorption
Enthalpy	Low enthalpy, is the order of 20-40 kJ mol^{-1}	High enthalpy, is the order of 80-240 kJ mol^{-1}
Reversibility	Reversible process	Irreversible process
Effect of temperature	With the increase in temperature, extent of adsorption decreases because adsorption is a exothermic process and kinetic energy of gas molecules increases with temperature.	Chemisorption first increases with temperature upto a certain extent and then decreases. A gas adsorbed at low temperature by physical adsorption may change into chemisorption at high temperature.
Selectivity	Not selective in nature. Does not depend upon the nature or chemical properties of gas and adsorbent.	Highly selective in nature. Depends upon the nature of gas and adsorbent.
Nature and state of adsorbate	The extent of adsorption depends upon the ease of liquefaction of the gas.	The state of adsorbed molecules may be different from that in the bulk.
Activation energy	No appreciable energy needed	High activation energy needed
Pressure	Increase in pressure increases adsorption	Increase in pressure decreases adsorption
Layers	Multimolecular layer	Mono-molecular layer

- **Factors affecting adsorption of gases on solids :**
  - **Nature of adsorbent :** Greater the strained forces on the surface, more is the ease with which adsorption takes place on the surface. The activated adsorbents have high adsorbing power.
  - **Surface area of adsorbent :** Greater the surface area, more is the adsorption.
  - **Nature of gas being adsorbed :** Easily liquefiable gases like NH_3, HCl, Cl_2, SO_2, CO_2, etc. (whose critical temperature is high) are adsorbed to greater extent.

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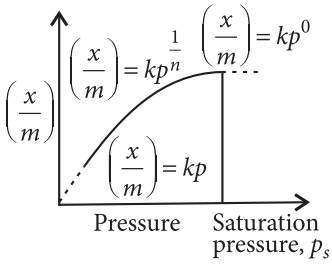
- **Pressure :** At constant temperature, adsorption increases with increase in pressure. The effect of pressure is large at low temperature.
- **Temperature :** Since adsorption is an exothermic process so according to Le-Chatelier's principle adsorption decreases with increase in temperature.

Freundlich adsorption isotherm : The plot of $\frac{x}{m}$ *vs* pressure at constant temperature is called *Freundlich adsorption isotherm*, where, m = mass of the adsorbent, x = mass of the adsorbate

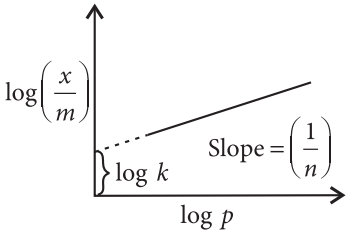
For low pressure, $\frac{x}{m} \propto p$

For high pressure, $\frac{x}{m} \propto p^0$

For intermediate pressures, $\frac{x}{m} \propto p^{1/n}$ $(n > 1)$



$$\log \frac{x}{m} = \log k + \frac{1}{n} \log p$$



Similarly, for adsorption of solutes from solutions, $\frac{x}{m} = k.C^{1/n}$ where, C is the equilibrium concentration, *i.e.*, when adsorption is complete.

Plot of $\log \frac{x}{m}$ *vs* log C is linear.

Promoters : The substances that enhance the activity of a catalyst *e.g.*, Mo acts as a promoter in Haber's process.

Poisons : The substances that decrease the activity of a catalyst.

- **Activity :** It is the capacity of a catalyst to increase the speed of the chemical reaction.
- **Selectivity :** It is the ability of a catalyst to direct a reaction to yield a particular product.

Catalysis : Substances which accelerate the rate of a chemical reaction and themselves remain chemically and quantitatively unchanged after the reaction, are known as *catalysts*, and the phenomenon is known as *catalysis*.

Homogeneous Catalysis

The process in which the reactants and the catalyst are in the same phase.
e.g., oxidation of SO_2 to SO_3 by NO as catalyst (lead chamber process), hydrolysis of methyl acetate by HCl, hydrolysis of sugar by H_2SO_4.

Shape-selective Catalysis

The catalytic reaction that depends upon the pore structure of the catalyst and the size of the reactant and product molecules, *e.g.,* Zeolites (have honey–comb like structures).

Heterogeneous Catalysis

The process in which the reactants and the catalyst are in different phases. *e.g.,* oxidation of SO_2 to SO_3 by Pt, manufacture of NH_3 from N_2 and H_2 by Fe (Haber's process), oxidation of NH_3 to NO by Pt (Ostwald's process), hydrogenation of vegetable oils by Ni.

Enzyme Catalysis

Many biochemical reactions are catalysed by complex nitrogeneous organic compounds (proteins or enzymes) which are also called *biochemical catalysts* and the phenomenon is known as *biochemical catalysis*.

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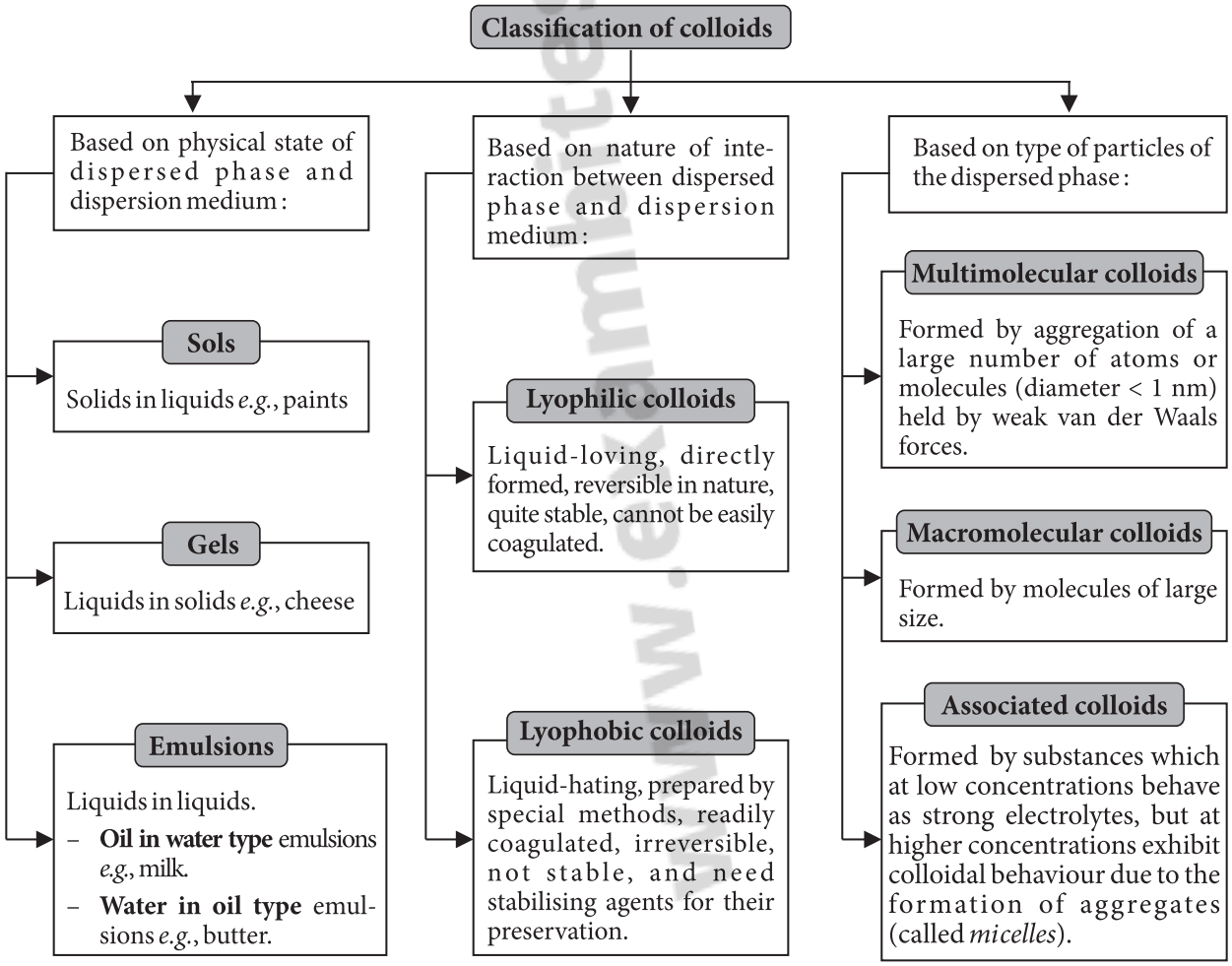
Some Enzymatic Reactions

	Enzyme	Source	Enzymatic reaction
1.	Invertase	Yeast	Sucrose → glucose and fructose
2.	Zymase	Yeast	Glucose → ethyl alcohol and carbon dioxide
3.	Diastase	Malt	Starch → maltose
4.	Maltase	Yeast	Maltose → glucose
5.	Urease	Soyabean	Urea → ammonia and carbon dioxide
6.	Pepsin	Stomach	Proteins → Peptides

Colloid : A colloid is a heterogeneous system in which one substance (*dispersed phase*) is dispersed as very fine particles in another substance called *dispersion medium*. The range of diameters of colloidal particles is between 1 and 1000 nm (10^{-9} to 10^{-6} m).



Properties of colloidal solutions :

- **Colligative properties :** Colloids show colligative properties like relative lowering of vapour pressure, elevation of boiling point, etc. and magnitude of colligative properties of colloids is much less than true solutions due to larger size of colloidal particles.
- **Tyndall effect (Optical property) :** Scattering of light by colloidal particles due to which the path of light beam becomes visible.
- **Brownian movement (Mechanical property):** Zig-zag movement of colloidal particles due to the unbalanced bombardment by the molecules of dispersion medium.
- **Charge on colloidal particles :** Colloidal particles always carry an electric charge and nature of charge (+ve or –ve) is same on all the particles in a given colloidal solution. The charge is due to preferential adsorption of ions from solution.

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Positively charged sols	Negatively charged sols
Hydrated metallic oxides, *e.g.*, $Al_2O_3 \cdot xH_2O$, $Fe_2O_3 \cdot xH_2O$, metal hydroxides, *e.g.*, $Fe(OH)_3$, $Al(OH)_3$, basic dye stuff like Prussian blue, haemoglobin (blood), etc.	Metallic particles, *e.g.*, Cu, Ag, Au Metal sulphides, *e.g.*, As_2S_3, CdS, Acidic dyes like eosin, congo red etc, sols of gelatin, gum, starch, etc.

- **Electrophoresis (Electrical property) :** Movement of colloidal particles towards one of the electrodes on passage of electricity through colloidal solution. The direction depends on the type of charge on colloidal particles.
- **Coagulation of colloids :** Precipitation of colloidal solution by induced aggregation of colloidal particles.
  - **Lyophobic sols:** They can be coagulated by electrophoresis, boiling, persistent dialysis, mixing of oppositely charged sols and addition of electrolytes.
  - **Hardy–Schulze rules :**
    - In case of electrolytes, the ion carrying charge opposite to that of colloidal particles is effective in causing coagulation and greater the valency of the ion causing coagulation, greater is the coagulating power.
    - The minimum concentration of an electrolyte in millimoles per litre required to cause precipitation of a sol in two hours is called *coagulating value*. The smaller the quantity needed, the higher will be the coagulating power of an ion.
  - **Lyophilic sols :** They can be coagulated by addition of electrolytes or addition of a suitable solvent.

Emulsions : Colloids in which both dispersed phase and dispersion medium are in liquid state.

- **Types of emulsions :**
  - **Oil in water :** Dispersed phase is oil, *e.g.*, milk, body lotion.
  - **Water in oil :** Dispersed phase is water, *e.g.*, butter, cold cream.
- **Emulsification :** Process of making an emulsion.
- **Emulsifying agent :** Used to stabilize the emulsion *e.g.*, soaps and detergents.
- **Demulsification :** Separating the two components of an emulsion. Methods used are boiling, freezing, changing pH.

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Previous Years' CBSE Board Questions

5.1 Adsorption

VSA (1 mark)

1. Why are powdered substances more effective adsorbents than their crystalline forms? *(AI 2019)*

2. Why is adsorption always exothermic? *(2018C, AI 2014)*

3. Why is a finely divided substance more effective as an adsorbent? *(1/3, 2018C, AI 2011C)*

4. Write one similarity between physisorption and chemisorption. *(Delhi 2017)*

5. Differentiate between adsorption and absorption. *(1/3, Delhi 2016)*

6. Physisorption is reversible while chemisorption is irreversible. Why ? *(Foreign 2015)*

7. Define the following term :
Adsorption *(1/3, Delhi 2015C, 2014C, 1/2, AI 2013)*

8. In reference to Freundlich adsorption isotherm write the expression for adsorption of gases on solids in the form of an equation. *(1/3, Delhi 2014, 1/3, Foreign 2014)*

9. What is the effect of temperature on chemisorption? *(AI 2014)*

10. What type of forces are responsible for the occurrence of physisorption? *(1/3, Foreign 2014)*

11. Define the following term :
Sorption *(1/3, Delhi 2014C)*

12. Of physisorption or chemisorption, which has a higher enthalpy of adsorption? *(AI 2013)*

13. Out of NH_3 and CO_2, which gas will be adsorbed more readily on the surface of activated charcoal and why? *(Delhi 2012C)*

14. Adsorption of a gas on surface of solid is generally accompanied by a decrease in entropy, still it is a spontaneous process. Explain. *(1/3, Delhi 2012C)*

15. Physisorption is multi-layered, while chemisorption is mono-layered. *(1/3, Delhi 2012C)*

16. Write two applications of adsorption. *(AI 2012C)*

SA (2 marks)

17. Write two differences between physisorption and chemisorption. *(2020)*

18. (i) What is the role of activated charcoal in gas mask?
(ii) How does chemisorption vary with temperature? *(2/3, Delhi 2019)*

19. Give reasons for the following observations :
(i) NH_3 gas absorbs more readily than N_2 gas on the surface of charcoal.
(ii) Powdered substances are more effective adsorbents. *(2/3, Foreign 2015)*

20. Write the differences between physisorption and chemisorption with respect to the following :
(i) Specificity
(ii) Temperature dependence
(iii) Reversibility and
(iv) Enthalpy change *(Delhi 2013)*

LA I (3 marks)

21. Give three points of difference between physisorption and chemisorption. *(2020)*

22. Giving appropriate examples, explain how the two types of processes of adsorption (physisorption and chemisorption) are influenced by the prevailing temperature, the surface area of adsorbent and the activation energy of the process? *(AI 2014C)*

23. What is an adsorption isotherm? Describe Freundlich adsorption isotherm. *(Delhi 2013, AI 2012C)*

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5.2 Catalysis

VSA (1 mark)

24. Define the following term :
Shape-selective catalysis
(1/3, 2020, 1/3, Delhi 2015C, 1/2, AI 2012, 2011)

25. $CO_{(g)}$ and $H_{2(g)}$ react to give different products in the presence of different catalysts. Which ability of the catalyst is shown by these reactions? *(2018)*

26. Write one difference of the following :
Homogeneous catalysis and heterogeneous catalysis *(1/3, Delhi 2017)*

27. Give reason for the following observation :
It is necessary to remove CO when ammonia is prepared by Haber's process.
(1/3, Delhi 2015)

28. What are biocatalysts ? Give an example.
(Foreign 2014)

SA (2 marks)

29. Name the two groups into which phenomenon of catalysis can be divided. Give an example of each group with the chemical equation involved. *(Delhi 2012)*

30. Explain how the phenomenon of adsorption find application in the following processes :
(i) Production of vaccum
(ii) Heterogeneous catalysis *(Foreign 2011)*

5.3 Colloids

VSA (1 mark)

31. Write one difference of the following :
Solution and colloid *(1/3, Delhi 2017)*

5.4 Classification of Colloids

VSA (1 mark)

Read the given passage and answer the questions number (32 - 36) that follow.

Colloidal particles always carry an electric charge which may be either positive or negative. For example, when $AgNO_3$ solution is added to KI solution, a negatively charged colloidal sol is obtained. The presence of equal and similar charges on colloidal particles provide stability to the colloidal sol and if, somehow, charge is removed coagulation of sol occurs. Lyophobic sols are readily coagulated as compare to lyophilic sols. *(2020)*

32. What is the reason for the charge on sol particles?

33. Why the presence of equal and similar charges on colloidal particles provide stability?

34. Why a negatively charged sol is obtained on adding $AgNO_3$ solution to KI solution?

35. Name one method by which coagulation of lyophobic sol can be carried out.

36. Out of KI or K_2SO_4, which electrolyte is better in the coagulation of positive sol?

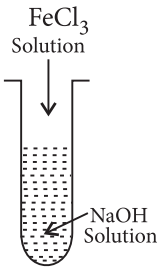
37. Define the following term.
Kraft temperature *(1/3, 2020)*

38. Define the following term.
Peptization *(1/3, 2020, AI 2012)*

39. Define the following term with a suitable example :
Associated colloids *(1/2, 2020)*

40. What is difference between an emulsion and a gel? *(Delhi 2019)*

41. A colloidal sol is prepared by the given method in figure. What is the charge on hydrated ferric oxide colloidal particles formed in the test tube? How is the sol represented?



(1/3, Delhi 2019)

42. Write one difference between the following :
Lyophobic sol and lyophilic sol
(1/3, Delhi 2017, Delhi 2014C)

43. What type of colloid is formed when a liquid is dispersed in a solid? Give an example.
(AI 2017)

44. What type of colloid is formed when a solid is dispersed in a liquid? Give an example.
(AI 2017)

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45. What type of colloid is formed when a gas is dispersed in a liquid? Give an example. *(AI 2017)*

46. Define the term 'Tyndall effect'. *(1/3, AI 2017C)*

47. Write the main reason for the stability of colloidal sols. *(Delhi, AI 2016)*

48. Out of $BaCl_2$ and KCl, which one is more effective in causing coagulation of a negatively charged colloidal sol? Give reason. *(Delhi 2015)*

49. Write the dispersed phase and dispersion medium of butter. *(AI 2015, Foreign 2014)*

50. In reference to surface chemistry, define dialysis. *(Delhi 2015C, 2014C, AI 2014C)*

51. Define the following term :
Electrophoresis *(1/3, Delhi 2015C)*

52. Give one example each of sol and gel. *(Delhi 2014)*

53. Give one example each of lyophobic sol and lyophilic sol. *(Delhi 2014)*

54. What are the dispersed phase and dispersion medium in milk? *(Delhi 2014)*

55. Name the temperature above which the formation of micelles takes place. *(Foreign 2014)*

56. Based on the type of dispersed phase, what type of colloid is micelles? *(Foreign 2014)*

57. Which aerosol depletes ozone layer? *(AI 2013)*

58. To which colloidal system does milk belong? *(AI 2013C)*

59. Which complex ion is formed when undecomposed AgBr is washed with hypo solution in photography ? *(AI 2013C)*

60. How can a colloidal solution and true solution of the same colour be distinguished from each other? *(Delhi 2012C)*

61. How is a sol different from an emulsion ? *(AI 2012C)*

62. What are lyophobic colloids? Give an example for them. *(AI 2011)*

63. Define the following term giving an example :
Hydrosol *(1/3, Foreign 2011)*

SA (2 marks)

64. What happens when
(a) a freshly prepared precipitate of $Fe(OH)_3$ is shaken with a small amount of $FeCl_3$ solution?
(b) persistent dialysis of a colloidal solution is carried out? *(2/3, 2018)*

65. Write one difference between each of the following :
(i) Multimolecular colloid and macromolecular colloid
(ii) Sol and gel *(2/3, Delhi 2017)*

66. Write one difference in each of the following :
(a) Multimolecular colloid and associated colloid
(b) Coagulation and peptization *(2/3, AI 2017)*

67. (a) Write the dispersed phase and dispersion medium of milk.
(b) Write the chemical method by which $Fe(OH)_3$ sol is prepared from $FeCl_3$. *(2/3, AI 2017)*

68. (i) Out of $MgCl_2$ and $AlCl_3$, which one is more effective in causing coagulation of negatively charged sol and why ?
(ii) Out of sulphur sol and proteins, which one forms multimolecular colloids? *(2/3, Delhi 2016)*

69. Give reasons for the following observations :
(i) Leather gets hardened after tanning.
(ii) Lyophilic sol is more stable than lyophobic sol. *(2/3, Delhi 2015)*

70. (i) Based on type of particles of dispersed phase, give one example each of associated colloid and multimolecular colloid
(ii) Write an important characteristic of lyophilic sols. *(2/3, Delhi 2014)*

71. Define the following terms :
(i) Peptization
(ii) Sol *(2/3, Delhi 2014C)*

72. Define the following terms :
(i) Tyndall effect
(ii) Electrophoresis *(2/3, Delhi 2014C)*

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73. Write the dispersed phase and dispersion medium of the following colloidal systems :
(i) Smoke
(ii) Milk *(Delhi 2013)*

74. What is the difference between multimolecular and macromolecular colloids? Give one example of each. *(Delhi 2013)*

75. What are the characteristics of the following colloids? Give one example of each.
(i) Multimolecular colloids
(ii) Lyophobic sols *(2/3, AI 2013)*

76. How are the following colloidal solutions prepared?
(a) Sulphur in water (b) Gold sol *(Delhi 2013C)*

77. Explain the following terms giving one example for each.
(i) Micelles (ii) Aerosol *(Delhi, AI 2012)*

78. Explain the cleaning action of soap. Why do soaps not work in hard water? *(AI 2012)*

79. (i) Same substances can act both as colloids and crystalloids. Explain
(ii) What will be the charge on AgI colloidal particles when it is prepared by adding small amount of $AgNO_3$ solution to KI solution in water? What is responsible for the development of this charge ? *(2/3, Delhi 2012C)*

80. Define the following terms :
(i) Aerosol
(ii) Coagulation of colloids *(Foreign 2011)*

LA I (3 marks)

81. Define lyophobic and lyophilic sol with a suitable example of each. Why is coagulation of lyophilic sol difficult as compared to lyophobic sol? *(2020)*

82. Give reasons for the following :
(a) Brownian movement provides stability to the colloidal solution.
(b) True solution does not show Tyndall effect.
(c) Addition of alum purifies water. *(AI 2019)*

83. Define the following terms :
(i) Lyophilic colloid
(ii) Zeta potential
(iii) Associated colloids *(AI 2016)*

84. Define the following terms :
(i) Brownian movement
(ii) Peptization
(iii) Multimolecular colloids *(AI 2015)*

85. Describe the following processes :
(i) Dialysis
(ii) Electrophoresis
(iii) Tyndall effect *(AI 2015C, 2011C)*

86. Explain what is observed when :
(i) A beam of light is passed through a colloidal solution.
(ii) NaCl solution is added to hydrated ferric oxide sol.
(iii) Electric current is passed through a colloidal solution. *(AI 2013C)*

87. What is meant by coagulation of a colloidal solution? Describe briefly any three methods by which coagulation of lyophobic sols can be carried out. *(Delhi 2012)*

88. Classify colloids where the dispersion medium is water. State their characteristics and write an example of each of these classes. *(AI 2011)*

89. Distinguish between multimolecular, macromolecular and associated colloids. Give one example of each. *(Delhi 2011C)*

5.5 Emulsions

VSA (1 mark)

90. Define the following term with a suitable example :
O/W emulsion *(1/2, 2020)*

91. What happens when an emulsion is centrifuged? *(1/3, 2018)*

92. Write one difference between the following :
O/W emulsion and W/O emulsion *(1/3, Delhi 2017)*

93. What are emulsions? Give an example. *(AI 2017C, Delhi 2015C)*

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94. Give one example each of 'oil in water' and 'water in oil' emulsion. *(Delhi 2014)*

95. What is an emulsion? *(Delhi 2014C, AI 2012, Foreign 2011)*

96. What are emulsions? Name an emulsion in which water is a dispersed phase. *(AI 2014C)*

SA (2 marks)

97. What is the difference between oil/water (O/W) type and water/oil (W/O) type emulsions? Given an example of each type. *(Delhi 2013)*

LA I (3 marks)

98. What are emulsions? What are their different types? Give one example of each type. *(AI 2014, Delhi 2013C)*

5.6 Colloids Around Us

VSA (1 mark)

99. Hardening of leather in tanning industry is based on
(a) electrophoresis
(b) electro-osmosis
(c) mutual coagulation
(d) Tyndall effect. *(2020)*

100. Why are medicines more effective in colloidal state? *(Delhi 2019)*

101. Give reasons for the following observations :
A delta is formed at the meeting point of sea water and river water. *(1/3, Foreign 2015, 2014)*

102. Explain the following :
Artificial rain is caused by spraying salt over clouds. *(1/3, Delhi 2012C)*

Detailed Solutions

1. A finely divided substance is more effective as adsorbent because
(i) it has more surface area so more adsorption occurs.
(ii) the number of active sites (active centres) and the extent of adsorption increases.

2. In adsorption, there is always a decrease in residual unbalanced forces on the surface. This results in decrease in surface energy which appears as heat. Hence, adsorption is unconditionally an exothermic process.

3. *Refer to answer 1.*

4. Physical adsorption and chemical adsorption both increase with increase in surface area of the adsorbent.

5. Adsorption is a surface phenomenon. In this process the adsorbate is concentrated on the surface of the adsorbent and does not penetrate into the bulk whereas, absorption of a substance takes place throughout the bulk of the material. In adsorption, concentration of adsorbate is high on the surface of adsorbent, while during absorption concentration is uniform throughout. *e.g.*, water vapour is adsorbed by silica gel whereas absorbed by anhydrous calcium carbide.

6. Physisorption takes place with the help of non-covalent bonding between an adsorbate and an adsorbent; it makes the process reversible. Chemisorption, on the other hand, takes place with the help of covalent bonding; it makes the process irreversible.

7. Adsorption is the phenomenon of attracting and retaining the molecules of a substance on the surface of a liquid or a solid resulting in higher concentration of the molecules on the surface.

8. $\frac{x}{m} = kp^{1/n} (n > 1)$

$$\log \frac{x}{m} = \log k + \frac{1}{n} \log p$$

where, $\frac{x}{m}$ is the mass of gas adsorbed per gram of the adsorbent and p is the pressure of gas.

9. Effect of temperature : Chemisorption is an exothermic process. Hence, according to Le Chatelier principle, rate of adsorption decreases with rise in temperature. Rate of adsorption is low at lower temperature because of high energy of activation hence, with increasing temperature rate of adsorption increases initially.

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10. The forces operating in these cases are weak van der Waals' forces.

11. The term sorption is used to describe both the processes adsorption and absorption.

12. Chemisorption has higher enthalpy of adsorption.

13. NH_3 gas will be adsorbed more readily on the surface because it has higher critical temperature than CO_2 gas.

Higher the critical temperature, more easily the gas can be liquified, *i.e.*, greater are the attraction of the gas molecules on the surface of the adsorbent, hence greater will be the adsorption.

14. For the process to be spontaneous, ΔG must be negative.

As ΔS is negative, ΔG can be negative only if ΔH is negative and greater than $T\Delta S$.

15. Physical adsorption occurs due to inter-molecular attractive forces between the adsorbate and adsorbent. If the size of the adsorbent pores is close to the size of adsorbate molecules, multilayer adsorption takes place, *i.e.,* adsorption takes place until all the pores are filled with adsorbate molecules, whereas in chemisorption chemical bonds are formed between adsorbate and adsorbent molecules. Therefore, it is monolayered.

16. Applications of adsorption :
(i) Deionisation of water
(ii) In chromatographic analysis.

17.

S. No.	Criteria	Physisorp-tion	Chemisorp-tion
(i)	Specificity	It is not specific in nature.	It is highly specific in nature.
(ii)	Temperature dependence	It decreases with increase in temperature. Thus, low temperature is favourable for physisorption.	It increases with increase in temperature. Thus, high temperature is favourable for chemi-sorption.
(iii)	Reversibility	Reversible in nature.	Irreversible in nature.
(iv)	Enthalpy change	Low enthalpy of adsorption.	High enthalpy of adsorption.

(Any two)

18. (i) Activated charcoal in gas mask adsorb the poisonous gases present in air and thus purify the air for breathing.
(ii) *Refer to answer 9.*

19. (i) Higher the critical temperature of gas, more readily it can get adsorbed on the surface of an adsorbent due to stronger van der Waals' forces.

NH_3 (132°C) has a higher critical temperature than dinitrogen (–147°C). Thus, NH_3 gas adsorbs more readily than N_2 gas on the surface of charcoal.
(ii) *Refer to answer 1.*

20. *Refer to answer 17.*

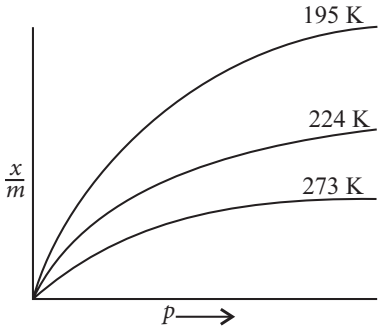
21. *Refer to answer 17.*

22. Effect of temperature : Physisorption decreases with increase of temperature and chemisorption first increases then decreases with increase of temperature.

Surface area : Greater the surface area, greater is the physisorption and chemisorption.

Activation energy : In physisorption, no appreciable activation energy is needed. In chemisorption, sometimes high activation energy is needed.

23. Adsorption isotherm : It is the variation in the amount of gas adsorbed by the adsorbent with pressure at constant temperature.



Adsorption isotherm

These curves indicate that on increasing temperature, physical adsorption decreases at a fixed pressure.

Freundlich adsorption isotherm : It is an empirical relationship between the quantity of gas adsorbed by unit mass of solid adsorbent and pressure at a particular temperature.

$$\frac{x}{m} = kp^{1/n} (n > 1) \quad \ldots(i)$$

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when, $n = 1, \Rightarrow \frac{x}{m} = kp$ or $\frac{x}{m} \propto p$

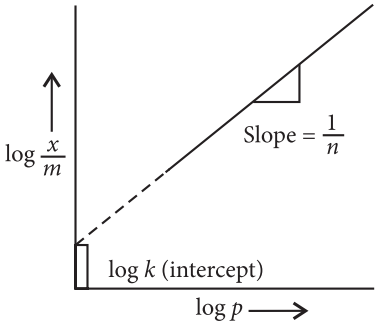
where, x is the mass of gas adsorbed on mass m of the adsorbent at pressure p. k and n are constants which depend on the nature of the adsorbent and the gas at the particular temperature.

Taking log in Eq. (i), gives

$$\log \frac{x}{m} = \log k + \frac{1}{n} \log p$$

The validity of Freundlich isotherm can be verified by plotting $\frac{x}{m}$ on y-axis and log p on x-axis.

If it comes to be a straight line, the Freundlich isotherm is valid.



24. The catalytic reaction that depends upon the pore structure of the catalyst and the size of the reactant and product molecules is called shape selective catalysis.

25. The selectivity of a catalyst is its ability to yield a particular product in the reaction *e.g.*,

$$CO + 3H_2 \xrightarrow{Ni} CH_4 + H_2O$$

$$CO + 2H_2 \xrightarrow{Cu/ZnO\text{-}Cr_2O_3} CH_3OH$$

26. Homogeneous catalysis : Catalyst is present in the same phase as reactants.

Heterogeneous catalysis : Catalyst is present in different phase as that of reactants.

27. CO is a catalytic poison. It reacts with iron to form iron carbonyl thus inhibiting the activity of catalyst.

28. Enzymes are termed as biocatalysts as they help in catalysis of biological reactions.

For example, inversion of cane sugar with the help of invertase enzyme :

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11(aq)}} + H_2O_{(l)} \xrightarrow{\text{Invertase}} \underset{\text{Glucose}}{C_6H_{12}O_{6(aq)}} + \underset{\text{Fructose}}{C_6H_{12}O_{6(aq)}}$$

29. Catalysis is divided into following two groups:

Homogeneous catalysis : When reactants and the catalysts are in the same phase, *i.e.,* liquid or gas, the catalysis is known as homogeneous catalysis.

$$e.g., 2C_2H_5OH_{(l)} \xrightarrow{\text{conc. } H_2SO_{4(l)}} C_2H_5OC_2H_{5(l)} + H_2O_{(l)}$$

$$\text{or } \frac{3}{2}O_{2(g)} \xrightarrow{NO_{(g)}} O_{3(g)}$$

Heterogeneous catalysis : When reactants and the catalysts are in different phases, the catalysis is known as heterogeneous catalysis. In most cases, the catalyst is solid, while reactants are either liquids or gases. Here, the catalyst is usually a metal or an oxide in finely divided form *e.g.,*

$$\text{Vegetable oils}_{(l)} + H_{2(g)} \xrightarrow{Ni_{(s)}} \text{Vegetable ghee}_{(s)}$$

$$4NH_{3(g)} + 5O_{2(g)} \xrightarrow{Pt_{(s)}} 4NO_{(g)} + 6H_2O_{(g)}$$

30. (i) **Production of vacuum :** Adsorption can be applied to create condition of high vacuum. Vessel which has already been exhausted by vacuum pump is connected to a bulb containing charcoal. The remaining traces of air inspite of low pressure are adsorbed by the charcoal almost completely.

(ii) **Role of adsorption in heterogeneous catalysis :** The reactant molecules in gaseous state or in solutions are adsorbed on the surface of the solid catalyst by physisorption or chemisorption. As result, the concentration of the reactant molecules on the surface increases and hence, the rate of reaction increases.

31. Solution : In true solution, the size of the particles is about 10^{-10} m.

Colloid : In a colloid, the size of the particles is in between 10^{-7} to 10^{-9} m.

32. The charge on sol particles is due to preferential adsorption of ions from solution.

33. The main reason for the stability of colloids is the electrostatic stabilisation *i.e.,* equal and same type of charge on the colloidal particles which causes repulsion between them and prevents the coagulation of the sol.

34. When $AgNO_3$ solution is added to aqueous KI solution, a negatively charged sol of AgI is formed. This is due to selective adsorption of I^- ions from the dispersion medium.

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$$AgI + \underset{\text{Dispersion medium}}{I^-} \longrightarrow \underset{\text{Negative sol}}{[AgI]I^-}$$

35. Coagulation is usually done by adding electrolytes. Coagulation also occurs on electrophoresis or prolonged dialysis or mixing two oppositely charged colloidal sols.

36. K_2SO_4 is more effective in causing coagulation of positively charged colloidal sol.
Because greater the valency of the coagulating ion, greater is its power to bring about coagulation.

37. The formation of micelles takes place only above a particular temperature called Kraft temperature (T_k).

38. Peptization is the process of conversion of a precipitate into colloidal state in the presence of some electrolyte.

39. The substances which when dissolved in a medium at low concentrations behave as normal, strong electrolytes but at higher concentrations exhibit colloidal state properties due to formation of aggregated particles are called associated colloids *e.g.,* soap.

40. In an emulsion both the dispersed phase and dispersion medium are liquids while in a gel the dispersed phase is a liquid and dispersion medium is a solid.

41. $FeCl_3 + NaOH \longrightarrow \underset{\text{Negatively charged sol}}{Fe_2O_3 \cdot xH_2O \vdots OH^-}$

42. Difference between lyophilic and lyophobic sol :

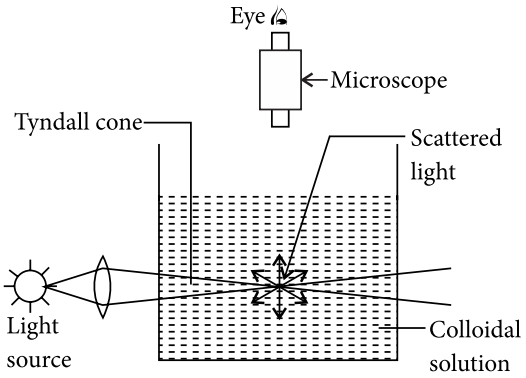
Lyophobic sol	Lyophilic sol
These are solvent repelling.	These are solvent attracting.

43. When a liquid is dispersed in a solid, 'gel' is formed, *e.g.,* jellies.

44. When a solid is dispersed in a liquid, a colloid is formed which is known as 'sol' *e.g.,* paints.

45. When a gas is dispersed in a liquid, foam is formed, *e.g.,* froth.

46. When a beam of light is passed through a colloidal solution and viewed perpendicular to the path of incident light, the path of beam is illuminated by a bluish light. This phenomenon is called Tyndall effect. This is due to the fact that colloidal particles scatter light in all the directions in space.



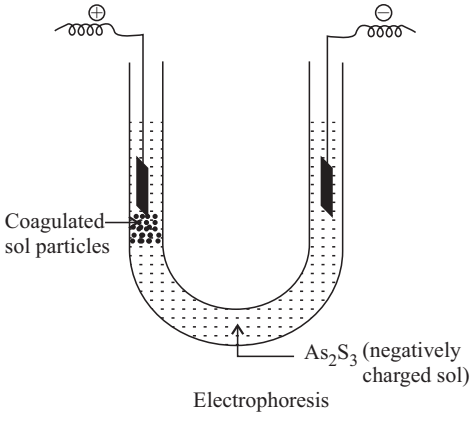
47. The main reason for the stability of colloids is the Brownian movement and electrostatic stabilisation, *i.e.,* equal and same type of charge on the colloidal particles which causes repulsion between them and prevents the coagulation of the sol.

48. $BaCl_2$ is more effective in causing coagulation of negatively charged colloidal sol.
Because greater the valency of the coagulating ion, greater is its power to bring about coagulation.

49. Dispersed phase : Liquid
Dispersion medium : Solid

50. Dialysis : It is the process of removing a dissolved substance from a colloidal solution by means of diffusion through a suitable membrane.
A bag of suitable membrane containing the colloidal solution is suspended in a vessel through which fresh water is continuously flowing.
The molecules and ions diffuse through membrane into the water and pure colloidal solution is left behind.

51. The movement of colloidal particles under an applied electric potential is called electrophoresis.
Positively charged colloidal particles move towards the cathode, while negatively charged particles move towards the anode.



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52.

Type of colloid	Dispersed phase	Dispersion medium	Examples
Sol	Solid	Liquid	Paints or Cell fluids
Gel	Liquid	Solid	Cheese or Jellies

53. A colloidal sol in which dispersed phase and dispersion medium attract each other is called lyophilic colloid. *e.g.*, gum. A colloidal sol in which dispersed phase and dispersion medium repel each other is called lyophobic colloid. *e.g.*, gold solution.

54. Liquid fat is the dispersed phase and water is the dispersion medium.

55. When the formation of micelles takes place only above a particular temperature called Kraft temperature (T_k).

56. Associated colloids

57. CFC (Chlorofluorocarbon)

58. Emulsion

59. When the developed film is immersed in sodium thiosulphate (hypo) solution it removes unchanged silver bromide as a complex ion (sodium argentothisoluphate).
This is known as fixing.

$$AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr$$

After fixing, the film is not sensitive to light.

60. When a powerful beam of light is passed through true and colloidal solutions each kept in a glass vessel then, colloidal solution exhibits tyndall effect whereas true solution does not.

61. Sol is a type of colloid in which the dispersed phase is solid and the dispersion medium is a liquid. Examples include mud, milk of magnesia.
Emulsion is a type of colloid in which the dispersed phase is liquid and dispersion medium is also a liquid. Examples include milk, face cream, etc.

62. *Refer to answer 53.*

63. Hydrosol : It is a colloidal solution of a solid in water as the dispersion medium. *e.g.*, starch solution.

64. (a) On treating a precipitate of iron (III) oxide with a small amount of $FeCl_3$ solution, a reddish brown coloured colloidal solution is formed. In this case, Fe^{3+} ions from ferric chloride are adsorbed by $Fe(OH)_3$ precipitate.

$$\underset{\text{ppt.}}{Fe(OH)_3} + \underset{\text{Electrolyte}}{Fe^{3+}} \longrightarrow \underset{\text{Colloidal sol}}{Fe(OH)_3.\ Fe^{3+}}$$

(b) When dialysis is persistent and prolonged, the traces of electrolyte are also removed. These electrolytes stabilise the colloid and when removed completely, make the colloid unstable and the colloid gets coagulated.

65.

(i)	**Multimolecular Colloids**	**Macromolecular Colloids**
	When a large number of small molecules or atoms (diameter < 1 nm) of a substance combine together in a dispersion medium to form aggregates, having size in the colloidal range, the colloidal solutions thus, formed are known as multimolecular colloids, *e.g.*, gold sol, sulphur sol, etc.	When substances which possess very high molecular masses are dispersed in suitable dispersion medium, the colloidal solutions thus, formed are called macromolecular colloids, *e.g.*, cellulose, starch, etc.
(ii)	**Sol**	**Gel**
	Dispersed phase is solid whereas dispersion medium is liquid.	Dispersed phase is liquid whereas dispersion medium is solid.

66. (a) Difference between multimolecular colloid and associated colloid :

Multimolecular colloid	Associated colloid
The particles of this type of colloids are aggregates of atoms or molecules with diameter less than 1 nm. *e.g.*, sulphur sol consists of colloidal particles which are aggregate of S_8 molecules.	They are substances which at low concentration behave as true solution and at higher concentration exhibit colloidal behaviour due to formation of aggregated particles.

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(b) Difference between coagulation and peptization :

Coagulation	Peptization
It is the process of settling of colloidal particles.	It is the process responsible for the formation of stable dispersion of colloidal particles in dispersion medium.

67. (a) *Refer to answer 54.*

(b) Hydrolysis is the chemical method by which $Fe(OH)_3$ sol is prepared from $FeCl_3$.

$$FeCl_3 + 3H_2O \longrightarrow Fe(OH)_{3(sol)} + 3HCl$$

68. (i) According to Hardy-Schulze rule, for negatively charged sol greater the valency of positive ion added to it, greater is its coagulation power.

In $AlCl_3$, Al has +3 charge which is more than Mg with +2 charge in $MgCl_2$. Thus, $AlCl_3$ is more effective in causing coagulation of negatively charged sol.

(ii) Proteins are macromolecules which cannot form multimolecular colloids while sulphur sol have smaller S_8 molecules which can form multimolecular colloids.

69. (i) Animal hides are colloidal in nature. When a hide, which has positively charged particles is soaked in tannin, containing negatively charged colloidal particles, mutual coagulation takes place. This results in the hardening of leather.

(ii) Lyophilic sol is more stable than lyophobic sol because it is highly hydrated in the solution.

70. (i) Associated colloid : Soap

Multimolecular colloid : Sulphur sol

(ii) Lyophilic sols are reversible sols. These are quite stable and cannot be easily precipitated.

71. (i) *Refer to answer 38.*

(ii) *Refer to answer 61.*

72. (i) *Refer to answer 46.*

(ii) *Refer to answer 51.*

73. (i) Dispersed phase of smoke = Solid

Dispersion medium of smoke = Gas

(ii) Dispersed phase of milk = Fat (liquid)

Dispersion medium of milk = Water (liquid)

74. *Refer to answer 65(i)*

75. (i) *Refer to answer 65(i).*

(ii) *Refer to answer 53.*

76. (a) Sulphur sol is prepared by the oxidation of H_2S with SO_2.

$$SO_2 + 2H_2S \xrightarrow{\text{Oxidation}} \underset{\text{(Sol)}}{3S} + 2H_2O$$

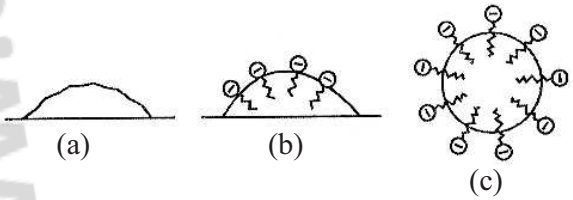
(b) Gold sol is prepared by the reduction of $AuCl_3$ with HCHO.

$$2AuCl_3 + 3HCHO + 3H_2O \xrightarrow{\text{Oxidation}} 2Au\ \text{(Sol)} + 3HCOOH + 6HCl$$

77. (i) Aggregated particles of associated colloids at high concentration are called micelles, *e.g.,* soaps.

(ii) Colloid of a liquid in a gas is called aerosol, *e.g.,* fog, sprays, etc.

78. The cleansing action of soap is due to the fact that soap molecules form micelle around the oil droplet in such a way that hydrophobic part is in the oil droplet and hydrophilic part interact with water, the oil droplet surrounded by stearate ions is now pulled in water and removed from the dirty surface. Thus, soap helps in emulsification and washing away of oils and fats. The negatively charged sheath around the globules prevents them from coming together and forming aggregates.



Hard water contains calcium and magnesium ions. These ions form insoluble calcium and magnesium salts when sodium or potassium soaps are dissolved in hard water. These insoluble soaps separate as scum in water and are useless as cleansing agent.

79. (i) The same substance can act as both colloid and crystalloid. It depends on the size of the particles.

When the size of the particles lies between 1 to 1000 nm, it behaves as a colloid. If particle size is less than 1 nm, it exists as a true solution and behave like a crystalloid.

(ii) When small amount of $AgNO_3$ solution is added to excess aqueous KI solution, a negatively

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charged sol of AgI is formed. This is due to selective adsorption of I^- ions from the dispersion medium.

$$AgI + \underset{\text{Dispersion medium}}{I^-} \longrightarrow$$

80. (i) *Refer to answer 77(ii).*
(ii) **Coagulation :** The process of aggregating together the colloidal particles into large sized particle which ultimately settle down under the force of gravity as a precipitate is called coagulation.

81. A colloidal sol in which dispersed phase and dispersion medium attract each other is called lyophilic colloid. *e.g.*, gum. A colloidal sol in which dispersed phase and dispersion medium repel each other is called lyophobic colloid. *e.g.*, gold solution.
In lyophobic sol, there is hardly and affinity between the particles of dispersion medium and dispersed phase. Therefore, it is unstable and can be easily coagulated. Since there is strong affinity between the particles in case of lyophilic sol, so coagulation is rather difficult.

82. (a) The Brownian movement has a stirring effect which does not permit the particles to settle and thus, it is responsible for the stability of colloidal solutions.
(b) Tyndall effect is not observed in true solution as the diameter of dispersed particles is much smaller as compared to wavelength of the light used.
(c) The water obtained from natural sources often contains suspended impurities. Alum is added to such water to coagulate the suspended impurities and make water fit for drinking purposes.

83. (i) *Refer to answer 53.*
(ii) The difference of potential between fixed layer and diffused layer of a colloidal sol is known as electrokinetic or zeta potential.
(iii) *Refer to answer 39.*

84. (i) **Brownian movement :** When the colloidal particles are observed under the ultramicroscope, the particles are seen to be in constant motion in zig-zag path.
This zig-zag motion of colloidal particles is called Brownian movement.
(ii) *Refer to answer 38.*
(iii) *Refer to answer 65(i).*

85. (i) *Refer to answer 50.*
(ii) *Refer to answer 51.*
(iii) *Refer to answer 46.*

86. (i) Scattering of light by the colloidal particles takes place and the path of light becomes visible (Tyndall effect).
(ii) The positively charged colloidal particles of ferric hydroxide sol get coagulated by the oppositely charged Cl^- ions provided by NaCl.
(iii) On passing electric current through a sol, colloidal particles start moving towards oppositely charged electrodes where they lose their charge and get coagulated (electrophoresis).

87. The process of setting of colloidal particles is called coagulation of the sol. It is also known as precipitation. Following are the three methods by which coagulation of lyophobic sols can be carried out :
(i) **Electrophoresis :** In this process, the colloidal particles move towards oppositely charged electrodes and get discharged resulting in coagulation.
(ii) **Mixing of two oppositely charged sols :** When equal proportions of oppositely charged sols are mixed, they neutralise each other resulting in coagulation.
(iii) **Prolonged dialysis :** By prolonged dialysis, electrolytes present in sol are removed completely and colloid becomes unstable resulting in coagulation.

88. (i) **Hydrosol :** When solids is dispersed in water, it is called hydrosol, *e.g.*, gold sol, starch sol.
(ii) **Emulsion:** When liquid is dispersed in water, it is called emulsion, *e.g.*, milk.
(iii) **Foam :** When gas is dispersed in water, it is called foam or froth, *e.g.*, soap lather, whipped cream.

89. *Refer to answers 65(i) and 66(a).*

90. The emulsion in which oil is the dispersed phase and water is the dispersion medium is known as O/W emulsion *e.g.,* milk.

91. On centrifugation, emulsion is decomposed back into its constituent liquids. This process is called demulsification.

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92.

O/W emulsion	W/O emulsion
Oil is dispersed phase and water is the dispersion medium.	Water is dispersed phase and oil is the dispersion medium.

93. Emulsion : It is a colloidal system when both the dispersed phase and the dispersion medium are in the liquid state, *e.g.*, milk.

94. Oil in water emulsion : Milk
Water in oil emulsion : Butter

95. *Refer to answer 93.*

96. *Refer to answer 93.*
Butter is an emulsion in which water acts a dispersed phase and oil acts as the dispersion medium.

97. The two types of emulsions are :
(i) Oil-in-water type in which small droplets of an oil are dispersed in water.
Example : Milk, cod liver oil.
(ii) Water-in-oil type in which water droplets are dispersed in an oil.
Example : Butter.

98. *Refer to answer 93 and 97.*

99. (c) : Hardening of leather in tanning industry is based on mutual coagulation.

100. Medicines are more effective in colloidal form because in this form, these are more easily assimilated due to large surface area.

101. Sea water contains a lot of electrolytes. River contains colloids of sand and clay. When they meet the electrolytes neutralise the charge on colloidal particles which results in the precipitation of sand, clay etc. thus, resulting in a delta formation.

102. Clouds are colloidal dispersion of water particles in air. These water particles carry some charge over them. On spraying oppositely charged colloidal dust or sand particles over a cloud from an aeroplane, the colloidal water particles present in the cloud will get neutralized and as a result they will come closer and will grow in size to form bigger water drops and ultimately will coagulate or precipitate causing artificial rain.

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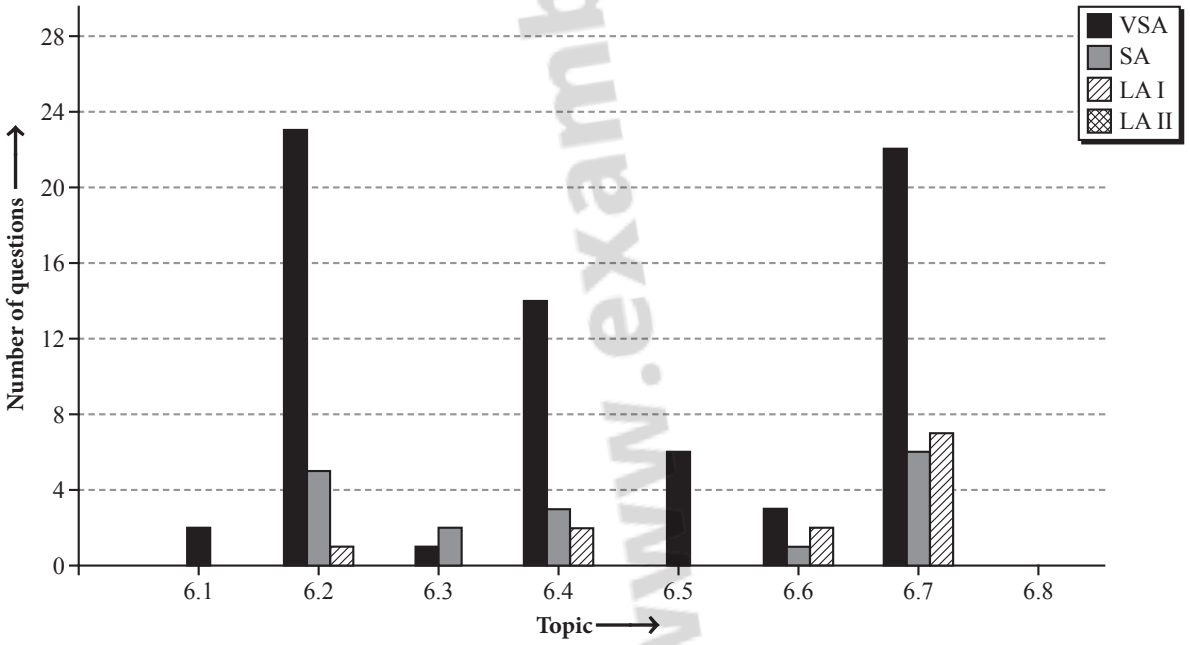
CHAPTER
6

General Principles and Processes of Isolation of Elements

6.1 Occurrence of Metals
6.2 Concentration of Ores
6.3 Extraction of Crude Metal from Concentrated Ore
6.4 Thermodynamic Principles of Metallurgy
6.5 Electrochemical Principles of Metallurgy
6.6 Oxidation Reduction
6.7 Refining
6.8 Uses of Aluminium, Copper, Zinc and Iron

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Refining*.
- Maximum VSA type questions were asked from *Concentration of Ores and Refining*.
- Maximum SA and LA I type questions were asked from *Refining*.

QUICK RECAP

- **Minerals :** Naturally occurring chemical substances in the earth's crust obtainable by mining.
- **Ores :** The minerals from which metal can be extracted profitably and conveniently.
- **Metallurgy :** The entire scientific and technological process used for isolation of the metal from its ore.

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Concentration of ore : Removal of gangue from ore is known as *concentration* or *dressing* of ore.

- Depending upon the nature of the ore and the impurities present, different methods are used for concentration of ore.

Method	Principle and ores specification	Process
Hydraulic washing	It is based on the difference in the specific gravities of the gangue and the ore particles and is mostly used for oxide ores.	The powdered ore is agitated with water. The heavier ore particles settle down while lighter impurities are washed away.
Electro-magnetic separation	It is based on the difference in magnetic properties of the gangue and ore particles and is used for magnetite (Fe_3O_4), haematite (Fe_2O_3), wolframite ($FeWO_4$), chromite ($FeO{\cdot}Cr_2O_3$), etc.	The powdered ore containing impurity is dropped on a belt which rotates around a magnetic roller. The magnetic particles fall nearer to the roller while non-magnetic particles fall farther off.
Froth floatation process	It is based on the difference in wetting properties of gangue and ore particles and is mostly used for sulphide ores.	Powdered ore is mixed with pine oil and water and violently agitated with air. Ore particles rise to the surface in the form of froth and impurities remain in water.
Leaching	It is based on the difference in chemical properties of gangue and ore particles and is used for extraction of Au, Ag and pure alumina from bauxite ore.	*Baeyer's process for* Al *extraction* : $Al_2O_3{\cdot}2H_2O + 2NaOH + H_2O \longrightarrow 2Na[Al(OH)_4]$ $2Na[Al(OH)_4] + 2CO_2 \longrightarrow Al_2O_3{\cdot}xH_2O + 2NaHCO_3$ $Al_2O_3{\cdot}xH_2O \xrightarrow{1200°C} Al_2O_3 + xH_2O$ *Mac Arthur Forest cyanide process for* Ag *extraction* : $4Ag + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na[Ag(CN)_2] + 4NaOH$ $2Na[Ag(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Ag\downarrow$

Extraction and isolation of metals :

- **Calcination :** It is the process of heating the ore (like carbonate and hydrated oxide) strongly below its melting point either in the absence of air or in the limited supply.
- **Roasting :** It is the process of heating the ore (commonly used for sulphide ores) strongly below the melting point in a sufficient supply of air to bring about its oxidation and remove volatile impurities.

Reduction or conversion of ores to metals : The ore obtained after calcination or roasting is reduced to metal and choice of reducing agent depends upon the nature of ore.

- Carbon or carbon monoxide is used for oxides of Fe, Cu, Zn, Mg, Co, etc. and the process is called *smelting*.
- Electropositive metals like Na, Al, Mg or hydrogen are used for reduction of ores of Mn, Cr, Ti, Mo, W, etc.
- Water gas is used for nickel ores.
- Auto-reduction process is used for ores of Pb, Hg, Cu, etc.
- Electrolytic reduction is used for highly electropositive metals.
- Hydrometallurgy or displacement method is used for Ag, Au, etc.

Refining of metals : Purification of crude metals by removing the impurities present in it is called *refining*.

- Different methods are used depending upon the differences in properties of the metal and the impurity.

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Refining/purification of the crude metal

Methods	Metals purified
Liquation	Used for low melting metals like Sn, Pb, Hg, Bi, etc.
Distillation	Used for volatile metals like Zn, Hg, Cd, etc. or metals containing non-volatile impurities.
Poling	Used for metals which contain impurities of their own oxides. *e.g.,* Cu.
Cupellation	Used for metals containing easily oxidisable impurities. *e.g.*, Ag containing Pb impurities.
Electrolytic refining	Used for metals like Cu, Ag, Au, Al which get deposited at cathode and impurities get deposited at anode. Solution of a soluble metal salt acts as the electrolyte.
Mond's process	Used for refining of Ni. $4CO + \underset{\text{Impure}}{Ni} \xrightarrow{60°-80°C} Ni(CO)_4 \xrightarrow{180°C} \underset{\text{Pure}}{Ni} + 4CO$
Zone refining	Used to produce extremely pure metals (semiconductors) like Si, Ge, Ga, B and In.
van Arkel method	Used for ultra-pure metals like Ti, Zr which are used in space technology. $\underset{\text{Impure}}{Ti_{(s)}} + 2I_{2(g)} \xrightarrow{523\ K} TiI_{4(g)} \xrightarrow{1673\ K} \underset{\text{Pure}}{Ti_{(s)}} + 2I_{2(g)}$ $\underset{\text{Impure}}{Zr_{(s)}} + 2I_{2(g)} \xrightarrow{870\ K} ZrI_{4(g)} \xrightarrow{1800\ K} \underset{\text{Pure}}{Zr_{(s)}} + 2I_{2(g)}$
Chromatographic methods	Based on the principle that different components of a mixture are differently adsorbed on an adsorbent *e.g.,* lanthanoids are purified by using ion exchange as adsorbent.

Extraction of Al, Fe, Zn and Cu

Extraction of Al

Ores : Bauxite ($Al_2O_3 \cdot xH_2O$)

Concentration : *(Leaching)*

$Al_2O_3 + 2NaOH + 3H_2O \longrightarrow 2Na[Al(OH)_4]$

$2Na[Al(OH)_4] + CO_2 \longrightarrow Al_2O_3 \cdot xH_2O + 2NaHCO_3$

$Al_2O_3 \cdot xH_2O \xrightarrow{1470K} Al_2O_3 + xH_2O$

Electrolytic reduction :

(*Hall and Heroult process*)

Pure Al_2O_3 + cryolite (electrolyte)

$2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$

At cathode : $Al^{3+}_{(melt)} + 3e^- \longrightarrow Al_{(l)}$

At anode : $C_{(s)} + O^{2-}_{(melt)} \longrightarrow CO_{(g)} + 2e^-$

$C_{(s)} + 2O^{2-}_{(melt)} \longrightarrow CO_{2(g)} + 4e^-$

Purification : Electrolytic method *(Hoope's process)*

Extraction of Fe

Ores : Haematite (Fe_2O_3), Siderite ($FeCO_3$)

Concentration : *(Electromagnetic separation)*

Calcination and roasting : (Removal of H_2O, S, C and As)

$Fe_2O_3 \cdot 3H_2O \xrightarrow{\Delta} Fe_2O_3 + 3H_2O$

$FeCO_3 \longrightarrow FeO + CO_2$

$4FeO + O_2 \longrightarrow 2Fe_2O_3$

$S + O_2 \longrightarrow SO_2$

$4As + 3O_2 \longrightarrow 2As_2O_3$

Smelting : *(Blast furnace)*

$Fe_2O_3 + 3CO \xrightarrow{400\text{-}700°C} 2Fe + 3CO_2$

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$; $CaO + SiO_2 \longrightarrow CaSiO_3$(slag)

$2CO \xrightarrow{\text{Hot iron}} CO_2 + C$; $SiO_2 + 2C \xrightarrow{1200°C} Si + 2CO$

$MnO_2 + 2C \longrightarrow Mn + 2CO$; $P_4O_{10} + 10C \longrightarrow 4P + 10CO$

Pig iron so formed is then remelted and cooled to form cast iron.

Purification : By oxidising impurities in reverberatory furnace lined with haematite.

$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$

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Extraction of Zn	Extraction of Cu
Ores : (i) Zinc blende : ZnS (ii) Calamine : $ZnCO_3$ (iii) Zincite : ZnO → **Concentration :** *(Froth floatation method)* → **Roasting :** $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ $ZnS + 2O_2 \rightarrow ZnSO_4$ $2ZnSO_4 \xrightarrow{\Delta} 2ZnO + 2SO_2 + O_2$ → **Reduction :** $ZnO + C \rightarrow Zn + CO$ → **Purification :** *Electrolytic refining, distillation*	**Ores :** Copper pyrites ($CuFeS_2$) → **Concentration :** *(Froth floatation method)* → **Roasting :** $2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2\uparrow$ $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$ → **Smelting :** Sand + coke $FeO + SiO_2 \rightarrow FeSiO_3$(Slag) → **Bessemerisation :** Sand + hot blast of air $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$ $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$ The copper so obtained upon bessemerisation is called *blister copper*. → **Purification :** Blister copper can be purified by poling or electrolytic refining to get pure copper.

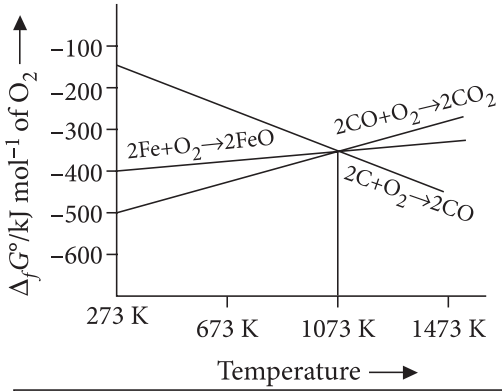
Thermodynamic and electrochemical principles of extraction :

Thermodynamic principles :

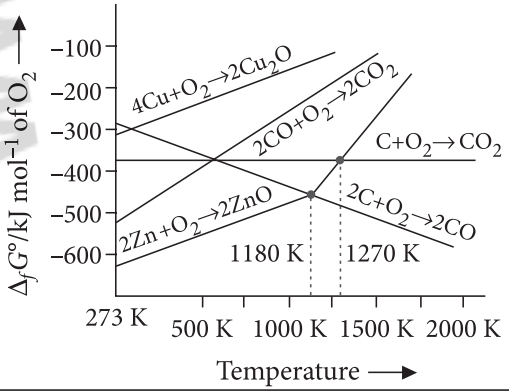
- Thermodynamics help in understanding the conditions of temperature and selecting suitable reducing agent in a metallurgical process.
- **Gibbs' Helmholtz equation** is used to check the feasibility of a reduction process.
- $\Delta G° = \Delta H° - T\Delta S°$, if $\Delta G° < 0$, the reduction is feasible.
- For a reaction, enthalpy change is fixed but temperature factor can be controlled and it can be the deciding factor for feasibility of that reaction.

Coupled reactions : If reactants and products of two reactions are put together in a system and the net ΔG of two possible reactions is –ve, the overall reaction will take place. These reactions are called *coupled reactions*.

Ellingham diagram : It consists of plots of $\Delta_f G°$ *vs.* temperature for the formation of oxides of elements.



Ellingham diagram for formation of FeO from Fe, CO from C and CO_2 from CO



Ellingham diagram showing the formation of Cu_2O from Cu, ZnO from Zn, CO from C and CO_2 from C and CO.

- It provides a sound idea about selecting a reducing agent in reduction of oxides.
- Such diagrams help in predicting the feasibility of thermal reduction of an ore.

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ΔG must be –ve at a given temperature for a reaction to be feasible.

Electrochemical principles :

- Electrolysis is used to carry out the reduction of a molten metal salt.
- The electrochemical principles of this method can be understood through the equation, $\Delta G^\circ = -nFE^\circ$
- More reactive metals have large negative values of the electrode potentials so, their reduction is difficult.
- If the difference of two E° values corresponds to a positive E° and consequently, negative ΔG°, then the less reactive metal will come out of the solution and the more reactive metal will go to the solution.

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Previous Years' CBSE Board Questions

6.1 Occurrence of Metals

VSA (1 mark)

1. Name the chief ores of aluminium and zinc. *(AI 2014 C)*

2. Differentiate between a mineral and an ore. *(AI 2011)*

6.2 Concentration of Ores

VSA (1 mark)

3. Name the depressant which is used to separate PbS and ZnS containing ore in froth floatation process. *(One word, 2020; Foreign 2014)*

4. **Assertion (A) :** Au and Ag are extracted by leaching their ores with a dil. solution of NaCN.

Reason (R) : Impurities associated with these ores dissolve in NaCN.

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is correct, but Reason (R) is incorrect statement.

(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

5. Explain the role of the following :
NaCN in the separation of ZnS and PbS. *(1/3, 2020)*

6. Write chemical equation involved to obtain :
(ii) Ag from $[Ag(CN)_2]^-$ complex *(1/2, 2020)*

7. Write the principle of the following :
Froth floatation process *(1/3, AI 2017; 1/2, Delhi 2014)*

8. In the extraction of Al, impure Al_2O_3 is dissolved in conc. NaOH to form sodium aluminate and leaving impurities behind. What is the name of this process? *(1/3, Delhi 2016)*

9. Describe the role of the following :
NaCN in froth floatation process *(1/2, AI 2014)*

10. What is the function of collectors in the froth floatation process for the concentration of ores? *(Foreign 2014)*

11. What role is played by CO_2 in getting pure alumina (Al_2O_3) in the extraction of aluminium? *(Delhi 2014 C)*

12. Describe the underlying principle of froth floatation process of concentration of ores. *(1/2, Delhi 2014 C)*

13. What are the collectors used in froth floatation process? Name a substance that can be used as such. *(AI 2014 C)*

14. Describe the underlying principle of recovery of silver from the solution obtained by leaching silver ore with a solution of NaCN. *(1/2, AI 2014 C)*

15. What is the role of CO_2 in the extractive metallurgy of aluminium from its ore? *(AI 2014 C; 1/3, Delhi, foreign 2011)*

16. Which solution is used for the leaching of silver metal in the presence of air in the metallurgy of silver? *(1/2, Delhi 2013)*

17. Which of the following ores can be concentrated by froth floatation method and why?
Fe_2O_3, ZnS, Al_2O_3 *(1/2, Delhi 2013)*

18. Name the method used for removing gangue from sulphide ores. *(1/2, AI 2013)*

19. Explain the role of the following :
NaCN in the extraction of silver from silver ore. *(1/2, Delhi 2013 C)*

20. What is the role of collectors in froth floatation process? *(AI 2012)*

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21. What is the role of depressants in the froth floatation process of dressing of ores? *(Delhi 2012 C)*

22. Describe the principle controlling the following process :
Froth floatation method of concentration of a sulphide ore. *(1/2, AI 2011)*

23. Why is it that only sulphide ores are concentrated by froth floatation process? *(AI 2011)*

24. What type of ores can be concentrated by magnetic separation method? *(AI 2011)*

25. Describe the principle controlling the following process :
Preparation of pure alumina (Al_2O_3) from bauxite ore. *(1/2, AI 2011)*

SA (2 marks)

26. Write the chemical equation involved in the leaching of bauxite ore to prepare pure alumina. *(2020)*

27. What is the role of
(a) Depressants in froth floatation?
(b) Concentrated sodium hydroxide in leaching of alumina from bauxite? *(2/3, AI 2019)*

28. (a) Out of PbS and $PbCO_3$ (ores of lead), which one is concentrated by froth floatation process preferably?
(b) What is the significance of leaching in the extraction of aluminium? *(2/3, Delhi 2017)*

29. (a) Write the role of dilute NaCN in the extraction of silver.
(b) What is the role of the collectors in the froth floatation process? Give an example of a collector. *(2/3, AI 2017)*

30. Write the chemical reactions involved in the extraction of silver from silver ore. *(Foreign 2014)*

LA I (3 marks)

31. Write the principle behind the froth floatation process. What is the role of collectors in this process? *(AI 2014)*

6.3 Extraction of Crude Metal from Concentrated Ore

VSA (1 mark)

32. Answer the following :
Differentiate between roasting and calcination. *(1/3, AI 2015 C)*

SA (2 marks)

33. Define the following terms :
(i) Roasting
(ii) Calcination *(Delhi 2014 C)*

34. Giving examples differentiate between calcination and roasting. *(AI 2013 C)*

6.4 Thermodynamic Principles of Metallurgy

VSA (1 mark)

35. Explain the role of the following :
SiO_2 in the metallurgy of Cu containing Fe as impurity. *(1/3, 2020)*

36. Write chemical equations involved to obtain :
Cu from Cu_2S *(1/2, 2020)*

37. What is the role of silica in the extraction of copper? *(1/3, Delhi 2019, 2015, 2014C)*

38. Why does copper obtained in the extraction from copper pyrites have a blistered appearance? *(1/3, AI 2017)*

39. What is the role of coke in the extraction of iron from its oxides? *(1/3, Delhi 2016)*

40. What is the role of limestone in the extraction of iron from its oxides? *(1/3, Delhi 2016)*

41. Which form of the iron is the purest form of commercial iron? *(1/3, Delhi 2015)*

42. What is 'copper matte'? *(1/3, Foreign 2015)*

43. Describe the role of the following :
SiO_2 in the extraction of copper from copper matte *(1/2, AI, Foreign 2014)*

44. Out of C and CO, which is a better reducing agent at the lower temperature range in the blast furnace to extract iron from the oxide ore? *(1/2, Delhi 2013)*

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45. What is the composition of 'copper matte'? *(Delhi 2013)*

46. Give reason for the following :
Zinc oxide can be reduced to metal by heating with carbon but Cr_2O_3 cannot be reduced by heating with carbon.
(1/2, Delhi 2013 C)

47. Although thermodynamically feasible, in practice, magnesium metal is not used for reduction of alumina in the metallurgy of aluminium. Why? *(AI 2012 C)*

48. Describe the principle controlling the following process :
Preparation of cast iron from pig iron.
(1/2, AI 2011)

SA (2 marks)

49. (i) What is the role of silica in the extraction of copper?
(ii) How is 'cast iron' different from 'pig iron'? *(2/3, AI 2015)*

50. How is wrought iron different from steel? *(AI 2013)*

51. Why is copper matte put in silica lined converter? *(Delhi 2012 C)*

LA I (3 marks)

52. Write down the reactions taking place in different zones in the blast furnace during the extraction of iron. How is pig iron different from cast iron? *(Delhi 2015 C)*

53. Write down the reactions which occur in upper, middle and lower zones in the blast furnace during the extraction of iron from iron ore. *(AI 2015C; Delhi 2011 C)*

6.5 Electrochemical Principles of Metallurgy

VSA (1 mark)

54. Write the role of the following :
$NaAlF_4$ in the extraction of aluminium
(1/2, 2020; 1/3, 2018C; AI 2016; 2015 C; Delhi, AI 2013; Delhi 2012)

55. What type of metals are generally extracted by the electrolytic method? *(1/3, Delhi 2019)*

56. Which reducing agent is employed to get copper from the leached low grade copper ore? *(Delhi 2014)*

57. Write the role of graphite rod in the electrometallurgy of aluminium.
(1/2, Delhi, Foreign 2012)

58. How is copper extracted from a low grade ore of it? *(AI 2012)*

59. Which method is employed for extracting copper from low grade ores and scraps?
(AI 2012 C)

6.6 Oxidation Reduction

VSA (1 mark)

60. What is the role of dilute NaCN in the extraction of gold? *(1/3, 2018C, Foreign 2015)*

61. What is the role of zinc metal in the extraction of silver? *(AI 2014)*

62. Write the reaction involved in the following process :
Recovery of gold after gold ore has been leached with NaCN solution.
(1/2, Foreign 2011)

SA (2 marks)

63. The extraction of gold by leaching with NaCN involves both oxidation and reduction. Justify giving chemical equations.
(AI 2012C)

LA I (3 marks)

64. Write chemical reactions taking place in the extraction of aluminium from bauxite ore.
(AI 2019)

65. Write the chemical reactions involved in the process of extraction of Gold. Explain the role of dilute NaCN and Zn in this process. *(2018)*

6.7 Refining

VSA (1 mark)

66. Which of the following is refined by the zone refining by the zone process?
(a) Cu (b) Zn
(c) Ge (d) Sn *(2020)*

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67. Explain the role of the following :
Iodine in the refining of Ti.
(2/3, 2020; 1/2, Delhi 2013C)

68. Write the role of the following :
CO in the refining of Ni *(2/3, 2020)*

69. What is the role of
Carbon monoxide in Mond's process?
(1/3, AI 2019)

70. Write the principle of method used for the refining of germanium. *(1/3, Delhi 2017)*

71. Write the principle of electrolytic refining.
(1/3, AI 2017)

72. Name the method of refining of metals such as germanium.
(1/3, Delhi 2016)

73. Name the method of refining of nickel.
(1/3, AI 2016, 2014)

74. Indicate the principle behind the method used for the refining of zinc.
(1/3, Delhi 2015)

75. What is the principle behind the zone refining of metals?
(1/3, AI 2015, Foreign 2014, Delhi 2013C, AI 2012C)

76. Indicate the principle behind the method used for the refining of nickel.
(1/3, Foreign 2015)

77. What is meant by the term 'chromatography'?
(1/3, AI 2015C)

78. Name the method used for refining of copper metal. *(AI 2014)*

79. Write the role of the following :
Iodine in the refining of zirconium.
(1/2, Foreign 2014)

80. Describe the underlying principle of the following process :
Vapour phase refining of metals.
(1/2, Delhi 2014C)

81. Describe the underlying principle of the following process :
Electrolytic refining of a crude metal.
(1/2, AI 2014C)

82. Name the methods used for the vapour phase refining of impure titanium and nickel metals. *(Delhi 2013 C, 2012C)*

83. Write the chemical reaction which takes place in Mond's process for refining of nickel.
(Delhi 2013C)

84. Describe the following :
The role of carbon monoxide in the refining of crude nickel. *(1/2, Delhi 2012)*

85. What types of metals are usually purified by the method of zone refining? Give an example. *(Delhi 2012C)*

86. Describe the principle controlling the following process :
Vapour phase refining of titanium metal.
(1/2, AI 2011)

87. Write the reactions involved in the following process :
Refining of zirconium by van Arkel method.
(1/2, Foreign 2011)

SA (2 marks)

88. Write the principle of the following refining methods :
(a) vapour phase refining
(b) chromatography *(2020)*

89. Describe the principle controlling each of the following processes :
(i) Zone refining of metals
(ii) Electrolytic refining of metals
(AI 2017C, 2011)

90. Write the principle of the following method :
Vapour phase refining. *(Delhi 2014)*

91. Describe the principle involved in each of the following processes.
(i) Zone refining of a metal
(ii) Vapour phase refining of metals.
(AI 2014, 2013)

92. Give an example of zone refining of metals.
(Delhi 2013)

93. Describe the principle involved in each of the following processes.
(i) Mond process for refining of nickel.
(ii) Column chromatography for purification of rare elements. *(Delhi 2012)*

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LA I (3 marks)

94. Write the principles of the following methods :
(i) Vapour phase refining
(ii) Zone refining
(iii) Chromatography *(Delhi 2017)*

95. Outline the principles of refining of metals by the following methods :
(i) Distillation
(ii) Zone refining
(iii) Electrolysis *(Delhi 2015 C)*

96. Outline the principles behind the refining of metals by the following methods :
(i) Zone refining method
(ii) Chromatographic method
(Delhi 2014)

97. Explain the principle of the method of electrolytic refining of metals. Give one example. *(AI 2014)*

98. Which methods are usually employed for purifying the following metals?
(i) Nickel
(ii) Germanium
Mention the principle behind each one of them. *(AI 2012)*

99. State the principle on which each of the following process operates :
(i) Electrolytic refining of a metal.
(ii) Vapour phase refining of a metal.
(Delhi 2011, Foreign 2010)

100. State the principles of the following methods of refining crude metals :
(i) Zone refining
(ii) Liquation method
(iii) Chromatographic method *(AI 2011 C)*

Detailed Solutions

1. The chief ore of aluminium is bauxite and that of zinc is zinc blende or sphalerite.

2. The naturally occurring chemical substances present in the earth's crust which can be obtained by mining are called minerals. Those minerals from which metals can be extracted economically are called ores.

3. NaCN

4. (c) : Au and Ag dissolve in NaCN solution to form their soluble complexes while the impurities remain unaffected which are filtered off.

5. NaCN is used as a depressant in froth floatation process which selectively prevents ZnS from coming to the froth but allows PbS to come with the froth.

6. $2[Ag(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{2-} + 2Ag$

7. The principle behind this method is that the sulphide ore is preferentially wetted by oils while gangue is preferentially wetted by water. Then on agitation froth is formed that carries mineral particles and can be skimmed off.

8. Leaching

9. *Refer to answer 5.*

10. Collectors enhance non-wettability of the mineral particles. Pine oil can be used as collector.

11. The aluminate in solution is neutralised by passing CO_2 gas and hydrated Al_2O_3 is precipitated.

$$2Na[Al(OH)_4]_{(aq)} + CO_{2(g)} \rightarrow Al_2O_3.xH_2O_{(s)} + 2NaHCO_{3(aq)}$$

12. *Refer to answer 7.*

13. *Refer to answer 10.*

14. During leaching Ag is oxidised to Ag^+ which then combines with CN^- ions (from NaCN) to form soluble complex $[Ag(CN)_2]^-$. Silver is then recovered from this complex by displacement method using more electropositive zinc metal.

$$2[Ag(CN)_2]^-_{(aq)} + Zn_{(s)} \rightarrow 2Ag_{(s)} + [Zn(CN)_4]^{2-}_{(aq)}$$

15. *Refer to answer 11.*

16. Dilute solution of NaCN or KCN in the presence of air (for O_2).

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17. ZnS (Froth floatation method is used to remove gangue from sulphide ores).

18. Froth floatation method

19. *Refer to answer 14.*

20. *Refer to answer 10.*

21. *Refer to answer 5.*

22. *Refer to answer 7.*

23. This is because the sulphide ore particles becomes lighter when preferentially wetted by oil and rise to surface along with the froth, whereas gangue particles are preferentially wetted by water become heavier and settle down.

24. If either the ore or the gangue particles are capable of being attracted by magnetic field, then such ore is concentrated by magnetic separation method.

25. Al_2O_3 present in bauxite is soluble in concentrated NaOH solution whereas impurities are not, Al_2O_3 is reprecipitated from the solution.

26. Leaching of bauxite ore to prepare pure alumina :

$$Al_2O_3 + 2NaOH_{(aq)} + 3H_2O_{(l)} \rightarrow 2Na[Al(OH)_4]_{(aq)}$$

$$2Na[Al(OH)_4]_{(aq)} + CO_{2(g)} \rightarrow Al_2O_3 \cdot xH_2O_{(s)} + 2NaHCO_{3(aq)}$$

$$Al_2O_3 \cdot xH_2O_{(s)} \xrightarrow{1470\ K} Al_2O_{3(s)} + xH_2O_{(g)}$$

27. (a) *Refer to answer 5.*

(b) Bauxite ore contains impurity of Fe_2O_3. Fe_2O_3 is basic in nature and aluminium oxide is amphoteric, using NaOH it dissolves aluminium oxide but iron oxide remains unreacted and insoluble which can be separated by filtration.

28. (a) Sulphide ores are concentrated by froth floatation process preferably, hence, PbS is concentrated more preferably than $PbCO_3$ by this method.

(b) The significance of leaching in the extraction of aluminium is to concentrate pure alumina (Al_2O_3) from bauxite ore.

29. (a) In the extraction of silver, silver metal is leached with a dilute solution of NaCN in the presence of air to form a soluble complex from which the metal is obtained later by replacement.

$$4Ag + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na[Ag(CN)_2] + 4\ NaOH$$

(b) *Refer to answer 10.*

30. $\underset{\text{Argentite}}{Ag_2S} + 4NaCN \rightleftharpoons 2Na[Ag(CN)_2] + Na_2S$

$$4Na_2S + 2H_2O + 5O_2 \rightarrow 2Na_2SO_4 + 4NaOH + 2S$$

$$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag$$

31. Froth floatation method : The method has been in use for removing gangue from sulphide ores. In this process, a suspension of the powdered ore is made with water. To it collectors and froth stabilisers are added.

Collectors (*e.g.*, pine oils, fatty acids, xanthates, etc.) enhance non-wettability of the mineral particles and froth stabilisers (*e.g.*, cresols, aniline) stabilise the froth.

The mineral particles become wet by oils while the gangue particles by water. A rotating paddle agitates the mixture and draws air in it. As a result, froth is formed which carries the mineral particles. The froth is light and skimmed off.

32.

Roasting	Calcination
1. It is the process of heating the ore below its melting point with excess of air. 2. Sulphide ore is converted into oxide form. $2ZnS + 3O_2 \xrightarrow{Heat} 2ZnO + 2SO_2\uparrow$ and volatile impurities are removed as oxides SO_2, As_2O_3, etc.	1. It is the process of heating the ore below its melting point in the absence or limited supply of air. 2. Carbonate ores are converted into oxide. $ZnCO_3 \xrightarrow{Heat} ZnO + CO_2\uparrow$ and water and organic impurities are removed.

33. *Refer to answer 32.*

34. *Refer to answer 32.*

35. During the metallurgy of copper from copper pyrites, $CuFeS_2$, roasting gives FeO besides Cu_2O and SO_2.

$$\underset{\text{(Copper pyrites)}}{4CuFeS_{2(s)}} + 11O_{2(g)} \rightarrow 4FeO_{(s)} + 2Cu_2O_{(s)} + 8SO_{2(g)}$$

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To remove FeO, SiO_2 is added to form slag.

$$\underset{\text{(Basic oxide)}}{FeO} + \underset{\text{(Acidic oxide)}}{SiO_2} \rightarrow \underset{\text{Slag}}{FeSiO_3}$$

36. $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$
$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$

37. *Refer to answer 35.*

38. Copper obtained in the extraction from copper pyrites has a blistered appearance due to the evolution of SO_2.
$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$
$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$

39. Coke reduces iron oxide to iron.
$FeO_{(s)} + C_{(s)} \rightarrow Fe_{(s/l)} + CO_{(g)}$

40. Limestone decomposes to form CaO and CO_2. CaO thus formed acts as a flux and combines with silica (present as impurity) to form fusible calcium silicate slag.
$CaCO_3 \rightarrow CaO + CO_2$
$\underset{\text{(Silica)}}{CaO + SiO_2} \rightarrow \underset{\text{(slag)}}{CaSiO_3}$

41. Wrought iron is the purest form of commercial iron.

42. Copper matte is mixture of Cu_2S and FeS.

43. *Refer to answer 35.*

44. Ellingham diagram for oxides of metals and carbon shows
$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ (i)
$2C_{(s)} + O_{2(g)} \rightarrow 2CO_{(g)}$ (ii)
$2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$ (iii)
that at 673 K, $\Delta_f G°(CO_2) < \Delta_f G°(CO)$ hence formation of CO_2 gas is more favourable than CO gas. Hence, out of C and CO, CO is a better reducing agent.

45. Cu_2S and FeS.

46. Carbon is suitable reducing agent for reduction of zinc oxide. Reduction of Cr_2O_3 by carbon is not thermodynamically favourable.

47. Below the temperature (1623K) corresponding to the point of intersection of Al_2O_3 and MgO curves in Ellingham diagram magnesium can reduce alumina. But magnesium is a much costlier metal than aluminium and hence the process will be uneconomical.

48. Pig iron is melted with scrap iron and coke using hot air blast. Due to this, impurities such as C, S and P present in the pig iron are removed as CO_2, SO_2 and P_2O_5 and carbon content is reduced to about 3%.

49. *(i) Refer to answer 35.*
(ii) Impure iron obtained from blast furnace is known as pig iron and cast into pigs (blocks). It contains 4-5 percent of carbon along with some phosphorus, silicon, manganese and sulphur.
Cast iron contains a less percentage of carbon (1.8 to 3 percent) made by melting pig iron with scrap iron and coke using hot air blast.

50. Wrought iron is the purest form of iron. Steel is an alloy of iron which contains requisite amount of C, Mn, Ni, Cr, etc.

51. The copper matte containing Cu_2S and FeS is put in silica lined converter. Some silica is also added and hot air blast is blown to convert remaining FeS to FeO, which is removed as slag with silica.

52. Reduction of iron oxide in blast furnace :
(i) Lower zone of the blast furnace :
$C + O_2 \longrightarrow CO_2 + \text{heat}$
$C + CO_2 \longrightarrow 2CO$
Coke is burnt to give temperature up to 2200 K at lower part of the blast furnace.
(ii) Middle zone of the blast furnace : CO and heat move up in the furnace. The temperature range in the middle zone of the blast furnace is 900-1500 K.
$FeO + CO \longrightarrow Fe + CO_2$
Lime stone is also decomposed to CaO which removes silicate impurity of the ore as slag.
$CaCO_3 \xrightarrow{1100\ K} CaO + CO_2$
$C + CO_2 \longrightarrow 2CO$
$CaO + SiO_2 \longrightarrow \underset{\text{Slag}}{CaSiO_3}$
(iii) Upper zone of the blast furnace : Temperature range in this zone is 500-800 K.
$3Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2$
$Fe_3O_4 + 4CO \longrightarrow 3Fe + 4CO_2$
$Fe_2O_3 + CO \longrightarrow 2FeO + CO_2$

Refer to answer 49(ii).

53. *Refer to answer 52.*

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54. (i) Purified alumina (Al_2O_3) is mixed with cryolite (Na_3AlF_6) which lowers the melting temperature of electrolyte and also increase its conductivity.

55. Active metals like Na, Mg, Ca, Al, etc. which cannot be reduced by common reducing agents are extracted by electrolytic method.

56. Scrap iron or H_2.

57. Graphite anode is used in the electrometallurgy of aluminium from alumina, Al_2O_3.

Oxygen liberated at high temperature reacts with graphite to form both CO_2 and CO gases and prevent the liberation of O_2 gas at the anode which may react with Al metal to give Al_2O_3 again.

$$\text{At anode : } C_{(s)} + O^{2-}\,(\text{melt}) \rightarrow CO_{(g)} + 2e^-$$

$$C_{(s)} + 2O^{2-}\,(\text{melt}) \rightarrow CO_{2(g)} + 4e^-$$

58. Copper is extracted by hydrometallurgy from low grade ores. It is leached out using acid or bacteria. The solution containing Cu^{2+} is treated with scrap iron or H_2.

$$Cu^{2+}_{(aq)} + H_{2(g)} \rightarrow Cu_{(s)} + 2H^+_{(aq)}$$

59. *Refer to answer 61.*

60. NaCN is used for leaching of gold ore in the presence of air to form soluble gold complex from which metal is displaced by adding more reactive metal.

$$4Au_{(s)} + 8CN^-_{(aq)} + 2H_2O_{(l)} + O_{2(g)} \rightarrow 4[Au(CN)_2]^-_{(aq)} + 4OH^-_{(aq)}$$

61. Zinc in the extraction of silver : Role of zinc is to recover silver from the complex by displacement reaction. Silver ore is leached with dilute solution of NaCN in the presence of air or oxygen to form a soluble complex.

$$4Ag_{(s)} + 8CN^-_{(aq)} + 2H_2O_{(l)} + O_{2(g)} \rightarrow \underset{\text{Soluble complex}}{4[Ag(CN)_2]^-_{(aq)}} + 4OH^-_{(aq)}$$

$$2[Ag(CN)_2]^-_{(aq)} + Zn_{(s)} \rightarrow [Zn(CN)_4]^{2-}_{(aq)} + 2Ag_{(s)}$$

62. Gold is recovered from $[Au(CN)_2]^-$ complex by displacement method using a more electropositive metal zinc.

$$2[Au(CN)_2]^-_{(aq)} + Zn_{(s)} \rightarrow 2Au_{(s)} + [Zn(CN)_4]^{2-}_{(aq)}$$

63. During the leaching process, Au is first oxidised to Au^+ by O_2 of the air which then combines with CN^- ions to form the soluble complex, sodium dicyanoaurate (I).

$$\underset{\text{Gold (Impure)}}{4Au_{(s)}} + 8NaCN_{(aq)} + 2H_2O_{(l)} + O_{2(g)} \rightarrow \underset{\text{Sod. dicyanoaurate (I)}\ \text{(Soluble complex)}}{4Na[Au(CN)_2]_{(aq)}} + 4NaOH_{(aq)}$$

Gold is then extracted from this complex, by displacement method using a more electropositive zinc metal. In this reaction,

Zn acts as a reducing agent. It reduces Au^+ to Au while itself gets oxidised to Zn^{2+} which combines with CN^- ions to form soluble complex, sodium tetracyanozincate (II).

$$2Na[Au(CN)_2]_{(aq)} + Zn_{(s)} \rightarrow \underset{\text{Gold}}{2Au_{(s)}} + \underset{\text{Sod. tetracyanozincate (II)}}{Na_2[Zn(CN)_4]_{(aq)}}$$

Thus, extraction of Au by leaching with NaCN involves both oxidation and reduction.

64. *Refer to answer 26.*

On electrolytic reduction of pure alumina, aluminium gets deposited on cathode which can be purified by Hoope's process.

$$2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$$

$$\text{At cathode : } Al^{3+}_{(melt)} + 3e^- \longrightarrow Al_{(l)}$$

$$\text{At anode : } C_{(s)} + 2O^{2-}_{(melt)} \longrightarrow CO_{2(g)} + 4e^-$$

65. $4Au_{(s)} + 8CN^-_{(aq)} + 2H_2O_{(l)} + O_{2(g)} \rightarrow 4[Au(CN)_2]^-_{(aq)} + 4OH^-_{(aq)}$

$$2[Au(CN)_2]^-_{(aq)} + Zn_{(s)} \rightarrow [Zn(CN)_4]^{2-}_{(aq)} + 2Au_{(s)}$$

NaCN is used for leaching of gold ore in the presence of air to form soluble gold complex from which metal is displaced by adding more reactive metal. Role of zinc is to recover gold from the complex by displacement reaction.

66. (c) : Zone refining method is useful for refining metals of very high purity *e.g.*, germanium, silicon, boron, gallium, indium, etc.

67. Iodine forms a volatile compound with titanium which on further heating decomposes to give pure titanium.

$$\underset{\text{(Impure)}}{Ti} + 2I_2 \xrightarrow{\Delta} TiI_4 \xrightarrow{\Delta} \underset{\text{(Pure)}}{Ti} + 2I_2$$

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68. The role of CO is to convert impure nickel to volatile nickel tetracarbonyl whereas the impurities are left behind in the solid state. The vapour of nickel tetracarbonyl on heating at 450-470 K decomposes to give pure nickel metal and carbon monoxide.

$$\underset{\text{Impure nickel}}{Ni} + 4CO \xrightarrow{330-350K} \underset{\text{Nickel tetracarbonyl}}{Ni(CO)_4}$$

$$Ni(CO)_4 \xrightarrow{450-470\ K} \underset{\text{Pure nickel}}{Ni} + 4CO$$

69. *Refer to answer 68.*

70. Germanium is refined by zone refining method which is based on the principle that the impurities are more soluble in the molten state than in the solid state of the metal.

71. In this process, impure metal is made the anode and a strip of pure metal is made the cathode. A solution of a soluble salt of the same metal is taken as the electrolyte and net result is the transfer of metal in pure form from the anode to the cathode.

72. Zone refining is used for refining of semiconductors or other metals of very high purity.

73. Mond's process : Impure nickel is treated with carbon monoxide at 60-80°C to form volatile nickel carbonyl which decomposes to give pure nickel at 180°C.

$$\underset{\text{Impure}}{Ni} + 4CO \rightarrow Ni(CO)_4 \rightarrow \underset{\text{Pure}}{Ni} + 4CO$$

74. Zinc is refined by electrolytic refining.
In this method, the impure metal is made to act as anode. A strip of the same metal in pure form is used as cathode. They are put in a suitable electrolytic bath containing soluble salt of the same metal. The more basic metal remains in the solution and the less basic ones go to the anode mud.
At anode : $Zn \rightarrow Zn^{2+} + 2e^-$
At cathode : $Zn^{2+} + 2e^- \rightarrow Zn$

75. Zone refining is based on the principle that the impurities are more soluble in the melt than in the solid state of the metal.

76. *Refer to answer 73.*

77. The term chromatography was derived from the Greek word "chroma", meaning colour and "graphy", for writing. It was used for the first time to separate the colour pigments of plants. Chromatography is a technique for analysing or separating mixtures of gases, liquids or dissolved substances.

78. Copper is refined by electrolytic refining.

79. In this method, crude metal is heated in an evacuated vessel with iodine. The metal iodide formed is decomposed on a tungsten filament at high temperature (1800 K) to get pure metal.

$$\underset{\text{Impure}}{Zr} + 2I_2 \rightarrow ZrI_4 \rightarrow \underset{\text{Pure}}{Zr} + 2I_2$$

80. Vapour phase refining : In this method, the metal is converted into its volatile compound by heating it with a specific reagent at lower temperature. It is then decomposed to give pure metal. So, the two requirements are :
(a) The metal should form a volatile compound with an available reagent.
(b) The volatile compound should be easily decomposable, so that the recovery is easy.

81. *Refer to answer 71.*

82. Titanium is refined by van Arkel method and nickel is refined by Mond's process.

83. $\underset{\text{Impure}}{Ni} + 4CO \xrightarrow{330-350\ K} Ni(CO)_4$

$$Ni(CO)_4 \xrightarrow{450-470\ K} \underset{\text{Pure}}{Ni} + 4CO$$

84. *Refer to answer 68.*

85. The metals such as germanium, silicon, gallium, etc. which are used as semiconductors are purified by zone refining which is based upon the principle that impurities are more soluble in the melt than in the solid state of metals.

86. In this method, the metal is converted into its volatile compound. It is then decomposed to give pure metal.

$$\underset{\text{Impure metal}}{Ti} + 2I_2 \xrightarrow{523\ K} TiI_4$$

$$TiI_4 \xrightarrow{1700\ K} \underset{\text{Pure metal}}{Ti} + 2I_2$$

87. Refining of zirconium by van Arkel method.

$$Zr_{(s)} + 2I_{2(s)} \xrightarrow{870\ K} ZnI_{4(g)}$$

$$ZrI_{4(g)} \xrightarrow[\text{Tungsten filament}]{2075\ K} \underset{\text{Pure}}{Zr_{(s)}} + 2I_{2(s)}$$

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88. **(a)** *Refer to answer 80*

(b) Chromatographic method is based on the principle that different components of a mixture are differently adsorbed on an adsorbent. The adsorbed components are removed (eluted) by using suitable eluent. There are several chromatographic techniques such as paper chromatography, column chromatography, gas chromatography, etc.

89. (i) *Refer to answer 75.*

(ii) *Refer to answer 71.*

90. *Refer to answer 80.*

91. (i) *Refer to answer 75.*

(ii) *Refer to answer 80.*

92. *Refer to answer 85.*

93. (i) *Refer to answer 73.*

(ii) Chromatographic method is based on the principle that different components of a mixture are differently adsorbed on an adsorbent. In column chromatography, an adsorbent such as alumina, silica get etc. is packed in a column. This forms the stationary phase. The mixture to be separated is dissolved in a suitable solvent (mobile phase) and applied to the top of the column. The adsorbed components are removed (eluted) by using suitable eluent.

94. (i) *Refer to answer 80.*

(ii) *Refer to answer 75.*

(iii) *Refer to answer 88(b).*

95. (i) **Distillation :** This process is used for volatile metals.

Impure metal is heated in a retort and its vapours are separately condensed in a receiver. The non-volatile impurities are left behind.

e.g., Zn, Cd, Hg, etc. are purified by this method.

(ii) *Refer to answer 75.*

(iii) *Refer to answer 74.*

96. (i) *Refer to answer 75.*

(ii) *Refer to answer 88(b).*

97. *Refer to answer 74.*

98. (i) *Refer to answer 73.*

(ii) *Refer to answer 70.*

99. (i) *Refer to answer 74.*

(ii) *Refer to answer 80.*

100. (i) *Refer to answer 75.*

(ii) **Refining by liquation :** In this method a low melting metal like tin is heated on a sloping surface of a reverberatory furnace. Metal melts and flows down leaving impurities behind.

(iii) *Refer to answer 88(b).*

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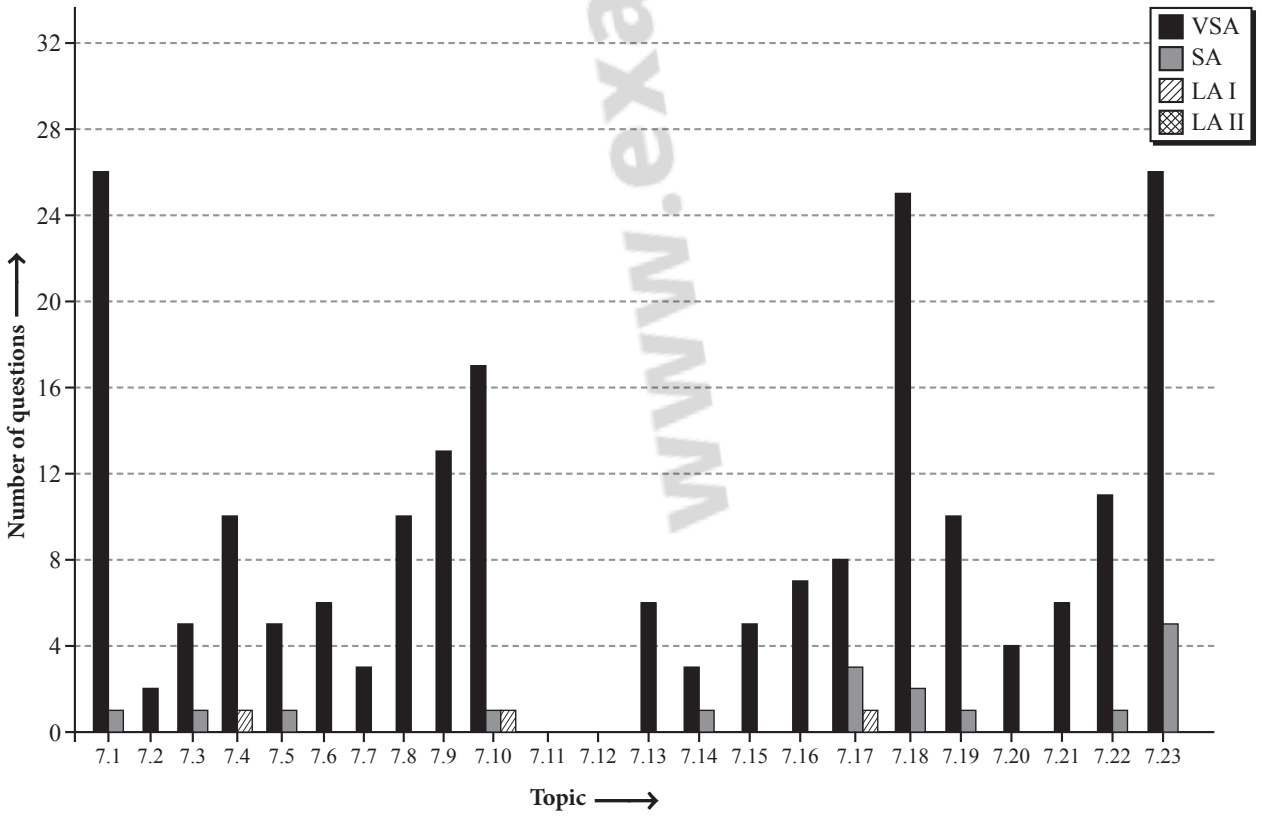
CHAPTER
7

The *p*-Block Elements (Group 15 to 18)

7.1 Group 15 Elements
7.2 Dinitrogen
7.3 Ammonia
7.4 Oxides of Nitrogen
7.5 Nitric Acid
7.6 Phosphorus – Allotropic Forms
7.7 Phosphine
7.8 Phosphorus Halides
7.9 Oxoacids of Phosphorus
7.10 Group 16 Elements
7.11 Dioxygen
7.12 Simple Oxides
7.13 Ozone
7.14 Sulphur - Allotropic Forms
7.15 Sulphur Dioxide
7.16 Oxoacids of Sulphur
7.17 Sulphuric Acid
7.18 Group 17 Elements
7.19 Chlorine
7.20 Hydrogen Chloride
7.21 Oxoacids of Halogens
7.22 Interhalogen Compounds
7.23 Group 18 Elements

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Group 18 Elements.*
- Maximum VSA type questions were asked from *Group 15 Elements, Group 17 elements and Group 18 Elements.*
- Maximum SA type questions were asked from *Group 18 Elements.*

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QUICK RECAP

Elements in which the last electron enters any one of the three *p*-orbitals of their respective outermost shell are called *p-block elements*.

GROUP 15 ELEMENTS (NITROGEN FAMILY)

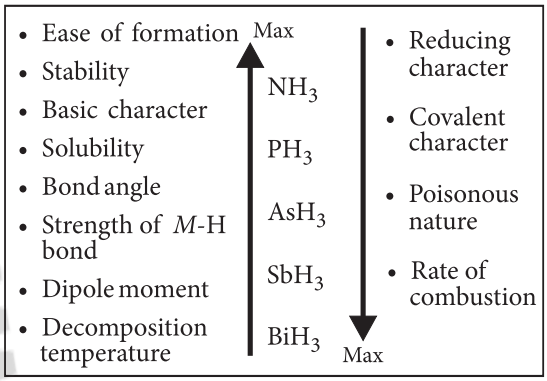
Group 15 elements are collectively called *pnictogens*.

General characteristics :

Electronic configuration	$ns^2\ np^3$
Elements	$_7N$, $_{15}P$, $_{33}As$, $_{51}Sb$, $_{83}Bi$, $_{115}Mc$
Physical state and metallic character	N_2 (unreactive gas), P_4 (solid non-metal), As_4 and Sb_4 (Solid metalloids), Bi (metal)
Atomic radii	Increase down the group, smaller than that of group 14 elements due to increased nuclear charge.
Melting and boiling points	M.pt. increases from N to As and then decreases whereas b. pt. increases from N to Sb and decreases very slightly.
Ionisation enthalpy	Decreases regularly down the group due to increase in size.
Electronegativity	Decreases down the group.
Allotropy	Nitrogen (α and β-Nitrogen), phosphorus (white, red, scarlet, violet, α- black, β-black), arsenic (grey, yellow, black), antimony (metallic, yellow, explosive)

Chemical properties :

- Stability of +3 oxidation state increases and that of +5 decreases down the group due to *inert pair effect*.
- **Halides :** All the elements form trihalides of the type MX_3 and except nitrogen, all form pentahalides of the type MX_5.
  - **Stability :** $NF_3 > NCl_3 > NBr_3$
  - **Lewis acid strength :** $PCl_3 > AsCl_3 > SbCl_3$ and $PF_3 > PBr_3 > PI_3$
  - **Lewis base strength :** $NI_3 > NBr_3 > NCl_3 > NF_3$
  - **Bond angle :** $PF_3 < PCl_3 < PBr_3 < PI_3$ (increasing *b.p.* – *b.p.* repulsions)
- **Hydrides :** All the elements form hydrides of the type MH_3 which are covalent and pyramidal in shape. Their general trends are :



- **Oxides :** All these elements form oxides of the type X_2O_3, X_2O_4 and X_2O_5.

N_2O_3	P_2O_3	As_2O_3	Sb_2O_3	Bi_2O_3	Acidic nature increases ↓
N_2O_4	P_2O_4	As_2O_4	Sb_2O_4	Bi_2O_4	
N_2O_5	P_2O_5	As_2O_5	Sb_2O_5	Bi_2O_5	
Acidic nature decreases →					

Preparation, properties and uses of some important compounds :

Compound	Preparation	Properties	Uses
N_2 $N \equiv N$	$NH_4Cl + NaNO_2 \longrightarrow N_2\uparrow + 2H_2O + NaCl$ $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2\uparrow + 4H_2O + Cr_2O_3$ $Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2\uparrow$	$6Li + N_2 \xrightarrow{Heat} 2Li_3N$ $3Mg + N_2 \xrightarrow{Heat} Mg_3N_2$ $N_{2(g)} + 3H_{2(g)} \underset{}{\overset{773\ K}{\rightleftharpoons}} 2NH_{3(g)}$ $N_{2(g)} + O_{2(g)} \underset{}{\overset{Heat}{\rightleftharpoons}} 2NO_{(g)}$	Used in manufacture of nitric acid, ammonia, calcium cyanamide and other nitrogen compounds. Liquid dinitrogen is used as a refrigerant to preserve biological materials, food items and in cryosurgery.

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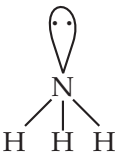
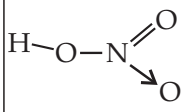
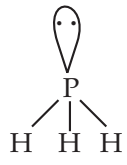
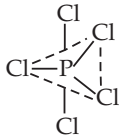
NH_3 	$NH_2CONH_2 + 2H_2O \longrightarrow$ $(NH_4)_2CO_3 \rightleftharpoons 2NH_3 + H_2O + CO_2$ $2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + 2H_2O + CaCl_2$ $(NH_4)_2SO_4 + 2NaOH \longrightarrow 2NH_3 + 2H_2O + Na_2SO_4$ *Haber's process* : $N_{2(g)} + 3H_{2(g)} \overset{773\ K}{\rightleftharpoons} 2NH_{3(g)}$; $\Delta H_f^\circ = -46.1$ kJ/mol	$NH_{3(g)} + H_2O_{(l)} \rightleftharpoons NH_{4(aq)}^+ + OH_{(aq)}^-$ $ZnSO_{4(aq)} + 2NH_4OH_{(aq)} \longrightarrow Zn(OH)_{2(s)} + (NH_4)_2SO_{4(aq)}$ (white ppt.) $FeCl_{3(aq)} + NH_4OH_{(aq)} \longrightarrow Fe_2O_3 . xH_2O_{(s)} + NH_4Cl_{(aq)}$ (brown ppt.) $Cu_{(aq)}^{2+} + 4NH_{3(aq)} \rightleftharpoons [Cu(NH_3)_4]_{(aq)}^{2+}$ (blue) (deep blue) $Ag_{(aq)}^+ + Cl_{(aq)}^- \longrightarrow AgCl_{(s)}$ (colourless) (white ppt.) $AgCl_{(s)} + 2NH_{3(aq)} \longrightarrow [Ag(NH_3)_2]Cl_{(aq)}$ (white ppt.) (colourless)	Used in refrigerators, manufacturing of rayon, HNO_3, $NaHCO_3$, nitrogenous fertilizers.
HNO_3 	$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$ *Ostwald's Process* : $4NH_3 + 5O_2 \xrightarrow[500\ K,\ 9\ bar]{Pt/Rh\ gauge\ catalyst} 4NO + 6H_2O$ $2NO + O_2 \rightleftharpoons 2NO_2$ $3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$	$HNO_{3(aq)} + H_2O_{(l)} \longrightarrow H_3O_{(aq)}^+ + NO_{3(aq)}^-$ $3Cu + 8HNO_3(dilute) \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$ $Cu + 4HNO_3(conc.) \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$ $4Zn + 10HNO_3(dilute) \longrightarrow 4Zn(NO_3)_2 + 5H_2O + N_2O$ $Zn + 4HNO_3(conc.) \longrightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2$ $I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$ $C + 4HNO_3 \longrightarrow CO_2 + 2H_2O + 4NO_2$ $S_8 + 48HNO_3 \longrightarrow 8H_2SO_4 + 48NO_2 + 16H_2O$ $P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$ $NO_3^- + 3Fe^{2+} + 4H^+ \longrightarrow NO + 3Fe^{3+} + 2H_2O$ $[Fe(H_2O)_6]^{2+} + NO \longrightarrow [Fe(H_2O)_5(NO)]^{2+} + H_2O$ (brown)	Used as fertilizers, explosives, perfumes and dyes.
PH_3 	$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$ $Ca_3P_2 + 6HCl \longrightarrow 3CaCl_2 + 2PH_3$ $P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$ (sodium hypophosphite) $PH_4I + KOH \longrightarrow KI + H_2O + PH_3$	$3CuSO_4 + 2PH_3 \longrightarrow Cu_3P_2 + 3H_2SO_4$ $3HgCl_2 + 2PH_3 \longrightarrow Hg_3P_2 + 6HCl$ $PH_3 + HBr \longrightarrow PH_4Br$	The spontaneous combustion of phosphine is technically used in *Holme's signals*. It is also used in *smoke screens*.
PCl_5 	$P_4 + 10Cl_2 \longrightarrow 4PCl_5$ (white or red)	$PCl_5 \xrightarrow[(in\ excess)]{H_2O} H_3PO_4 + HCl$ $PCl_5 \xrightarrow{P_4S_{10}} PSCl_3$ $PCl_5 \xrightarrow{SO_2} SOCl_2 + POCl_3$ $PCl_5 \xrightarrow{P_4O_{10}} POCl_3$	Used as chlorinating and dehydrating agent.

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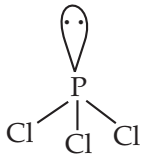
PCl_3 	$P_4 + 6Cl_2 \longrightarrow 4PCl_3$ $P_4 + 8SOCl_2 \longrightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$	$PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$ $3CH_3COOH + PCl_3 \longrightarrow 3CH_3COCl + H_3PO_3$ $3C_2H_5OH + PCl_3 \longrightarrow 3C_2H_5Cl + H_3PO_3$	Used as reagent in organic synthesis and as a precursor of PCl_5, $POCl_3$ and $PSCl_3$. It is used in the synthesis of some organic compounds, *e.g.*, C_2H_5Cl, CH_3COCl.

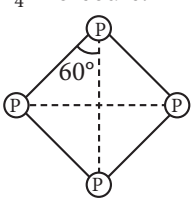
Preparation and properties of oxides of nitrogen :

Formula	O.S.	Preparation	Properties
N_2O	+1	$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$	colourless gas, neutral
NO	+2	$2NaNO_2 + 2FeSO_4 + 3H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + 2NaHSO_4 + 2H_2O + 2NO$	colourless gas, neutral
N_2O_3	+3	$2NO + N_2O_4 \xrightarrow{250\ K} 2N_2O_3$	blue solid, acidic
NO_2	+4	$2Pb(NO_3)_2 \xrightarrow{673\ K} 2PbO + 4NO_2 + O_2$	brown gas, acidic
N_2O_4	+4	$2NO_2 \underset{heat}{\overset{cool}{\rightleftharpoons}} N_2O_4$	colourless solid/liquid, acidic
N_2O_5	+5	$4HNO_3 + P_4O_{10} \longrightarrow 4HPO_3 + 2N_2O_5$	colourless solid, acidic

Allotropes of phosphorus :

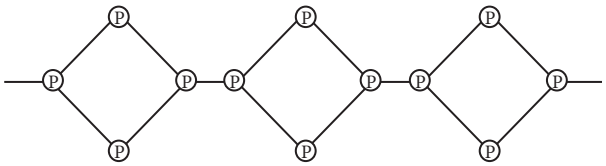
Property	White phosphorus	Red Phosphorus	Black phosphorus
Colour	White, but turns yellow on exposure	Dark red	Black
State	Waxy solid, can be cut with knife	Brittle powder	Crystalline with greasy touch
Smell	Garlic smell	Odourless	—
Density	1.84	2.1	2.69
Ignition temperature	307 K	543 K	673 K
Melting point	317 K	Sublimes in absence of air at 560 K	860 K

- White phosphorus consists of discrete tetrahedral P_4 molecule.



Structure of white phosphorus

- Red phosphorus is polymeric and consists of inter linked P_4 tetrahedra.



Structure of red phosphorus

- Black phosphorus is thermodynamically most stable at room temperature and has two forms : α-black phosphorus and β-black phosphorus.

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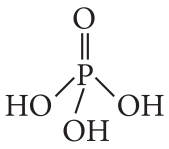
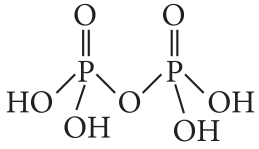
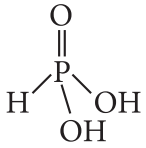
Oxoacids of phosphorus :



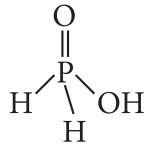
H_3PO_4
Orthophosphoric acid



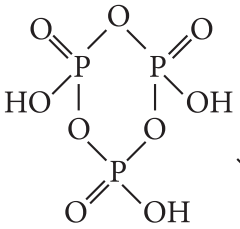
$H_4P_2O_7$
Pyrophosphoric acid



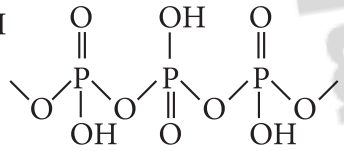
H_3PO_3
Orthophosphorous acid



H_3PO_2
Hypophosphorous acid



$(HPO_3)_3$
Cyclotrimetaphosphoric acid



$(HPO_3)_n$
Polymetaphosphoric acid

GROUP 16 ELEMENTS (OXYGEN FAMILY)

Group 16 elements are collectively called *chalcogens*.

General characteristics :

Electronic configuration	ns^2np^4
Elements	$_8O$, $_{16}S$, $_{34}Se$, $_{52}Te$, $_{84}Po$, $_{116}Lv$
Physical state and metallic characters	O_2 (gas), S_8 (solid non-metal), Se and Te (solid metalloid), Po (radioactive)
Atomic radii	Increase down the group
Ionisation enthalpy	Decreases down the group
Electronegativity	Decreases down the group
Electron gain enthalpy	Increases from oxygen to sulphur and then decreases.
Melting and boiling points	Increase down the group upto Te and then decreases.
Allotropy	All elements show allotropy

Chemical properties :

- Stability of –2 oxidation state decreases down the group, stability of +4 oxidation state increases and that of +6 oxidation state decreases down the group due to *inert pair effect*.
- **Hydrides :** All the elements form stable hydrides of the type H_2M. Their general trends are :
  - **Boiling point :** $H_2O > H_2Te > H_2Se > H_2S$
  - **Volatility** : $H_2S > H_2Se > H_2Te > H_2O$
  - **Bond angle :** $H_2O > H_2S > H_2Se > H_2Te$
  - **Acidic character :** $H_2O < H_2S < H_2Se < H_2Te$
  - **Reducing power :** $H_2Te > H_2Se > H_2S > H_2O$
- **Halides :** All elements form halides of the type EX_6, EX_4 and EX_2.
- **Oxides :**

Simple oxides	MgO, Al_2O_3
Mixed oxides	Pb_3O_4, Fe_3O_4
Acidic oxides	SO_2, Cl_2O_7, CO_2, N_2O_5
Basic oxides	Na_2O, CaO, BaO
Amphoteric oxides	Al_2O_3
Neutral oxides	CO, NO, N_2O

Dioxygen (O_2) :

- **Preparation :**

$$2KClO_3 \xrightarrow[MnO_2]{\Delta} 2KCl + 3O_2$$

$$2Ag_2O_{(s)} \xrightarrow{\Delta} 4Ag_{(s)} + O_{2(g)}$$

$$2Pb_3O_{4(s)} \xrightarrow{\Delta} 6PbO_{(s)} + O_{2(g)}$$

$$2HgO_{(s)} \longrightarrow 2Hg_{(l)} + O_{2(g)}$$

$$2PbO_{2(s)} \longrightarrow 2PbO_{(s)} + O_{2(g)}$$

$$2H_2O_{2(aq)} \longrightarrow 2H_2O_{(l)} + O_{2(g)}$$

- **Properties :**
  - Dioxygen is colourless and odourless gas, soluble in water and paramagnetic in nature.
  - Dioxygen directly reacts with most of the metals (except noble metals like Au, Pt), non-metals (except noble gases).

$2Ca + O_2 \longrightarrow 2CaO$ (Basic oxide)

$4Al + 3O_2 \longrightarrow 2Al_2O_3$ (Amphoteric oxide)

$P_4 + 5O_2 \longrightarrow P_4O_{10}$ (Acidic oxide)

$C + O_2 \longrightarrow CO_2$ (Acidic oxide)

$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$ (Roasting)

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$ (Combustion)

$$2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$$

(Catalytic oxidation)

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$4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$
(Catalytic oxidation)

- **Uses :**
  - For artificial respiration in hospitals and by mountaineers, pilots and divers.
  - In oxy-hydrogen and oxy-acetylene torches which are used for cutting and welding of metals.
  - Liquid dioxygen is used as a rocket fuel.

Ozone (O_3) :

- **Preparation :** $3O_2 \xrightarrow{\text{electric discharge}} 2O_3$
- **Properties :** It is a pale blue gas, dark blue liquid and violet black solid.
- **Oxidising action :**

$O_3 \longrightarrow O_2 + O$

$PbS + 4O_3 \longrightarrow PbSO_4 + 4O_2$

$2I^-_{(aq)} + H_2O_{(l)} + O_{3(g)} \longrightarrow 2OH^-_{(aq)} + I_{2(g)} + O_{2(g)}$

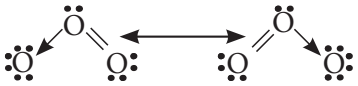
(used for estimation of O_3 by reacting I_2 with hypo.)

- **Reducing action :**

$H_2O_2 + O_3 \longrightarrow H_2O + 2O_2$

$BaO_2 + O_3 \longrightarrow BaO + 2O_2$

- **Structure :**



Oxidation state of O is +1 and –1.

- **Uses :**
  - Used for bleaching ivory, flour, delicate fabrics, etc.
  - As germicide and disinfectant for sterilising water.
  - Manufacture of $KMnO_4$ and artificial silk.

Allotropes of sulphur :

- **Rhombic sulphur (α-sulphur) :** It has S_8 molecules, yellow in colour, m.pt. 385.8 K, specific gravity 2.06 g cm^{-3}, insoluble in water, soluble in CS_2.
- **Monoclinic sulphur (β-sulphur) :** It has S_8 molecules, colourless, needle-shaped crystals, m.pt. 393 K, specific gravity 1.98 g cm^{-3}, soluble in CS_2.

Sulphur dioxide (SO_2) :

- **Preparation :**
  - **By heating sulphur in air :**

$S + O_2 \xrightarrow{\Delta} SO_2$

  - **Lab method :** By heating Cu with conc. H_2SO_4.

$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$

It is also prepared by treating a sulphite with dilute H_2SO_4.

$SO_3^{2-}{}_{(aq)} + 2H^+_{(aq)} \longrightarrow H_2O_{(l)} + SO_{2(g)}$

- **Properties :**
  - **As reducing agent :**

$SO_2 + Cl_2 + 2H_2O \longrightarrow H_2SO_4 + 2HCl$

$2KMnO_4 + 5SO_2 + 2H_2O \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$

  - **As oxidising agent :**

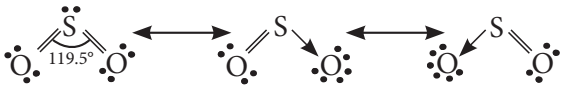
$2H_2S + SO_2 \longrightarrow 2H_2O + 3S\downarrow$

  - **Bleaching action :** Its bleaching action is due to reduction and is temporary.

$SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2[H]$

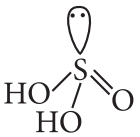
Coloured matter + [H] ⟶ Colourless matter

- **Structure :** SO_2 is a gas having sp^2 hybridisation and *V*-shape.

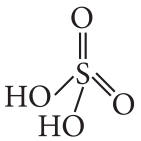


- **Uses :**
  - In the manufacture of sulphuric acid, sulphites and hydrogen sulphide.
  - As a disinfectant, fumigant and preservative.
  - For bleaching delicate articles.

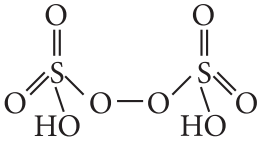
Oxoacids of sulphur :



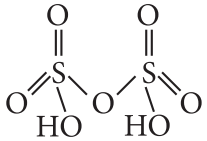
H_2SO_3
Sulphurous acid



H_2SO_4
Sulphuric acid



$H_2S_2O_8$
Peroxodisulphuric acid



$H_2S_2O_7$
Pyrosulphuric acid (Oleum)

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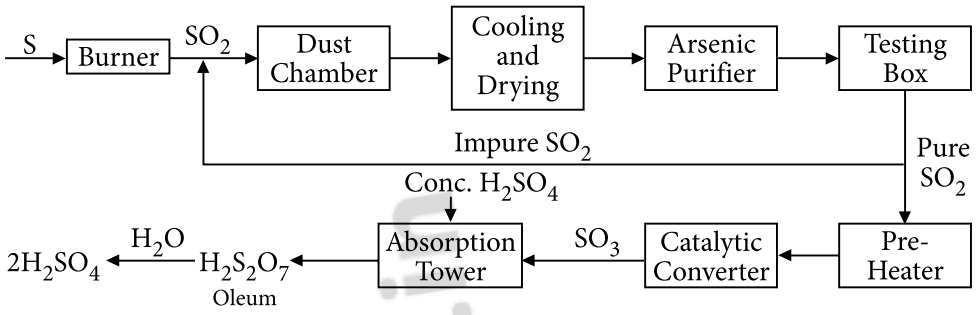
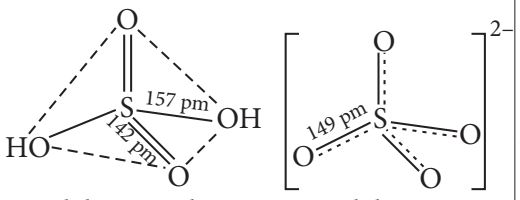
Sulphuric acid (H_2SO_4) :

Preparation (Contact process) :



Structure	Properties	Uses
 Sulphuric acid Sulphate ion	H_2SO_4 + NaOH → $NaHSO_4 + H_2O$ + NaOH → $Na_2SO_4 + 2H_2O$ + Zn → $ZnSO_4 + H_2$ + $C_{12}H_{22}O_{11}$ → $C + H_2O$ + S_8 → $SO_2 + H_2O$ + Na_2S → $Na_2SO_4 + H_2S$ + $BaCl_2$ → $BaSO_4 + 2HCl$ + $K_4[Fe(CN)_6]$ → $K_2SO_4 + FeSO_4 + (NH_4)_2SO_4 + CO\uparrow$	It is used as oxidising agent, dehydrating agent and for the preparation of dyes, drugs, explosives, volatile acids, etc.

GROUP 17 ELEMENTS (HALOGEN FAMILY)

Group 17 elements are collectively called halogens.

General characteristics :

Electronic configuration	ns^2np^5
Elements	$_9F$, $_{17}Cl$, $_{35}Br$, $_{53}I$, $_{85}At$, $_{117}Ts$
Colour and physical state	F_2 (pale yellow gas), Cl_2 (greenish yellow gas), Br_2 (reddish brown liquid), I_2 (purple solid)
Atomic radii	Increase down the group
Ionisation enthalpy	Very high and decreases down the group
Electronegativity	Decreases down the group
Electron gain enthalpy	Cl > F > Br > I
Melting and boiling points	Increase down the group
Bond energy	$Cl_2 > Br_2 > F_2 > I_2$
Heat of hydration	$F^- > Cl^- > Br^- > I^-$

Chemical properties :

- F shows only –1 oxidation state while other elements show –1, +1, +3, +5 and +7 oxidation states.
- **General trends :**
  - **Reactivity :** $F_2 > Cl_2 > Br_2 > I_2$
  - **Boiling points :** HF > HI > HBr > HCl
  - **Melting points :** HI > HF > HBr > HCl
  - **Bond lengths :** HI > HBr > HCl > HF
  - **Bond dissociation enthalpy :** HF > HCl > HBr > HI
  - **Acidic strength :** HI > HBr > HCl > HF
  - **Thermal stability :** HF > HCl > HBr > HI
  - **Reducing power :** HI > HBr > HCl > HF
- **Oxides :**
  - Fluorine forms two oxides OF_2 and O_2F_2 called *oxygen fluorides*, other halogens form oxides in which oxidation states of these halogens range from +1 to +7.
  - The higher oxides of halogens are more stable than the lower ones.

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- **Metal halides :**
  - **Ionic character :** $MF > MCl > MBr > MI$
  - For metals exhibiting more than one oxidation states, the halides in higher oxidation states will be more covalent than the one in lower oxidation states.

Chlorine (Cl_2) :

- **Preparation :**

$PbO_2 + 4HCl \longrightarrow PbCl_2 + 2H_2O + Cl_2$

$2KMnO_4 + 16HCl \longrightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$

$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$

- **Manufacture :**
  - **Deacon's process :** By oxidation of hydrogen chloride gas by atmospheric oxygen in the presence of $CuCl_2$ at 723 K.

$4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$

  - **Electrolytic process :** By the electrolysis of brine solution.
  - **Down's process :** Obtained as by-product during manufacture of sodium by electrolysis of fused NaCl.
- **Properties :** It is a yellowish green gas, poisonous in nature, soluble in water. Its aqueous solution is known as *chlorine water*.
  - **Bleaching action and oxidising property :**

$Cl_2 + H_2O \longrightarrow HOCl + HCl$

$HOCl \longrightarrow HCl + [O]$

Coloured matter + [O] $\longrightarrow$ Colourless matter

The bleaching action of chlorine is permanent and is due to its oxidising nature.

  - **Action of hydrogen :**

$H_2 + Cl_2 \xrightarrow[\text{Charcoal catalyst}]{\text{U.V. light}} 2HCl$

  - **Displacement reactions :**

$2KBr + Cl_2 \longrightarrow 2KCl + Br_2$

$2KI + Cl_2 \longrightarrow 2KCl + I_2$

  - **Action of NaOH :**

$2NaOH + Cl_2 \longrightarrow NaCl + NaOCl + H_2O$ (cold and dil.)

$6NaOH + 3Cl_2 \longrightarrow 5NaCl + NaClO_3 + 3H_2O$ (hot and conc.)

  - **Addition reactions :**

$SO_2 + Cl_2 \longrightarrow SO_2Cl_2$

$CO + Cl_2 \longrightarrow COCl_2$

- **Uses :** It is used as a bleaching agent, disinfectant and in the manufacture of $CHCl_3$, CCl_4, DDT, bleaching powder, poisonous gas phosgene ($COCl_2$), tear gas (CCl_3NO_2) and mustard gas ($ClC_2H_4SC_2H_4Cl$).

Hydrochloric acid (HCl) :

Preparation	Properties	Uses
$NaCl + H_2SO_4 \xrightarrow{420\ K} NaHSO_4 + HCl$ $NaHSO_4 + NaCl \xrightarrow{823\ K} Na_2SO_4 + HCl$	Colourless and pungent smelling gas, easily liquifiable, extremely soluble in water. $NH_3 + HCl \longrightarrow NH_4Cl$ $Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$ $Na_2SO_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2$ 3 parts of conc. HCl and 1 part of conc. HNO_3 is used for dissolving noble metals *e.g.*, gold, platinum $Au + 4H^+ + NO_3^- + 4Cl^- \longrightarrow AuCl_4^- + NO + 2H_2O$ $3Pt + 16H^+ + 4NO_3^- + 18Cl^- \longrightarrow 3PtCl_6^{2-} + 4NO + 8H_2O$	In manufacture of Cl_2, NH_4Cl and glucose, in medicine and as a laboratory reagent. For extracting glue from bones and purifying bone black.

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Oxyacids of halogens :

Variation of the general properties of oxyacids of halogens					
Halogen	**Hypohalous acids (O.S. of halogen = +1)**	**Halous acids (O.S. of halogen = +3)**	**Halic acids (O.S. of halogen = +5)**	**Perhalic acid (O.S. of halogen = +7)**	← Electronegativity decreases ← Thermal stability decreases ← Oxidising power decreases ← Acidic strength decreases
F	HOF	—	—	—	
Cl	HClO	$HClO_2$	$HClO_3$	$HClO_4$	
Br	HBrO	—	$HBrO_3$	$HBrO_4$	
I	HIO	—	HIO_3	HIO_4	
—— Oxidation number of the central atom increases (+1, +3, +5, +7) ⟶ —— Thermal stability increases ⟶ —— Covalent character of *X*—O bond increases ⟶ —— Oxidising power decreases ⟶ —— Acidity increases ⟶ —— Electronegativity of the central atom remains the same ——					

Interhalogens compounds : Halogens combine amongst themselves to form a number of interhalogens of the type XX', XX'_3, XX'_5 and XX'_7, where X is a larger size halogen (more electropositive) and X' is smaller size halogen.

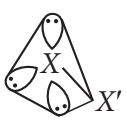
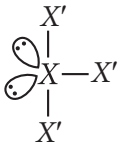
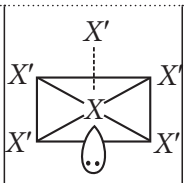
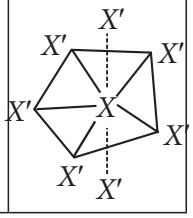
▶ **Preparation :**

$Cl_2 + F_2 \xrightarrow{437\ K} 2ClF$; $I_2 + 3Cl_2 \longrightarrow 2ICl_3$
(equal volume) (excess)

$Cl_2 + 3F_2 \xrightarrow{573\ K} 2ClF_3$; $Br_2 + 3F_2 \longrightarrow 2BrF_3$
(excess)

$I_2 + Cl_2 \longrightarrow 2ICl$; $Br_2 + 5F_2 \longrightarrow 2BrF_5$
(equimolar) (excess)

▶ **Structures :**

Type	Hybridisation	Shape	Structure
XX'	sp^3	Linear	
XX'_3	sp^3d	T-shaped	
XX'_5	sp^3d^2	Square pyramidal	
XX'_7	sp^3d^3	Pentagonal bipyramidal	

GROUP 18 ELEMENTS (NOBLE GASES)

These are monoatomic gases and are also known as *rare gases* or *aerogens*.

General characteristics :

Electronic configuration	ns^2np^6
Elements	$_2$He, $_{10}$Ne, $_{18}$Ar, $_{36}$Kr, $_{54}$Xe, $_{86}$Rn, $_{118}$Og
Physical state	Gases
Atomic radii	Increase down the group.
Electron gain enthalpy	Positive
Melting and boiling points	Very low due to weak dispersion forces

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Chemical properties : Noble gases are least reactive due to high ionization enthalpy and more positive electron gain enthalpy.

- **Xenon-fluorine compounds :**

$$\underset{\text{(excess)}}{Xe} + F_2 \xrightarrow{673\ K,\ 1\ bar} XeF_2$$

$$Xe + \underset{\text{(1 : 5 ratio)}}{2F_2} \xrightarrow{873\ K,\ 7\ bar} XeF_4$$

$$Xe + \underset{\text{(1 : 20 ratio)}}{3F_2} \xrightarrow{573\ K,\ 60-70\ bar} XeF_6$$

- **Xenon-oxygen compounds :**

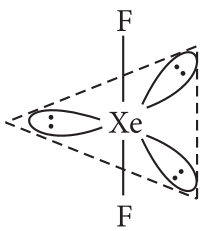
$$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$$

$$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$$

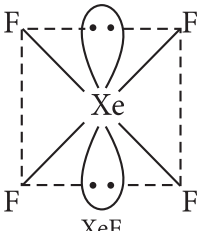
$$XeF_6 + H_2O \longrightarrow XeOF_4 + 2HF$$

$$XeF_6 + 2H_2O \longrightarrow XeO_2F_2 + 4HF$$

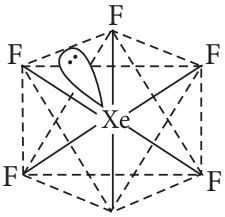
Structures of some compounds of xenon :



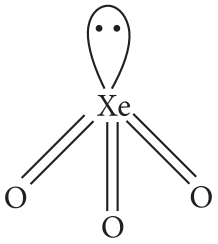
XeF_2
Linear, sp^3d



XeF_4
Square planar, sp^3d^2



XeF_6
Distorted octahedral, sp^3d^3



XeO_3
Pyramidal, sp^3

Uses :

Helium	Neon	Argon	Krypton	Xenon	Radon
– To lift weather balloons and air ships. – As breathing mixture. – For inflating the tyres of aeroplanes.	– For advertising. – For filling sodium vapour lamps. – In becon light.	– To create an inert atmosphere. – In geiger counters. – To date the age of rocks.	– For runway and approach lights in airports. – In high efficiency miners' cap lamps.	– In electric flash bulbs for high speed photography. – Krypton and xenon are more efficient than argon in gas filled lamps.	– In the treatment of cancer. – In X-ray photography for the detection of flaws in metals and other solids.

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Previous Years' CBSE Board Questions

7.1 Group 15 Elements

VSA (1 mark)

1. Why is the single N—N bond weaker than the single P—P bond?
(1/5, Delhi 2019, Foreign 2014)

2. Give reason:
Nitrogen does not form pentahalide.
(1/3, Delhi 2017, 1/3, Delhi AI 2016, 1/3 AI 2011C)

3. Arrange the following in the increasing order of property mentioned :
NH_3, PH_3, AsH_3, SbH_3, BiH_3 (Base strength)
(1/5, Delhi 2016)

4. Account for the following :
BiH_3 is the strongest reducing agent amongst all the hydrides of group 15.
(1/5, Foreign 2015)

5. Why does $R_3P = O$ exist but $R_3N{=}O$ does not? (R = alkyl group) *(1/5, Delhi 2015C)*

6. Why is NH_3 more basic than PH_3?
(1/5, AI 2015C, 2014C)

7. Give reasons for the following :
$(CH_3)_3P{=}O$ exists but $(CH_3)_3N{=}O$ does not. *(1/3, AI 2014)*

8. Account for the following :
Bi is a strong oxidizing agent in the +5 state.
(1/5, Foreign 2014)

9. Account for the following :
PCl_5 is known but NCl_5 is not known.
(1/5, Foreign 2014)

10. Arrange the following in the increasing order of their basic character :
NH_3, PH_3, AsH_3, SbH_3, BiH_3 *(Foreign 2014)*

11. Arrange the following group of substances in the order of the property indicated against each group :
NH_3, PH_3, AsH_3, SbH_3 – increasing order of boiling points. *(1/5, Delhi 2014C)*

12. Assign reasons for the following :
NF_3 is an exothermic compound whereas NCl_3 is not. *(1/5, AI 2014C, 2013C, 2012, 2011, 1/3, Delhi 2011)*

13. Why is nitrogen gas very unreactive?
(AI 2014C)

14. Give reasons for the following : PH_3 has lower boiling point than NH_3.
(1/5, Delhi 2013)

15. Explain the following :
$BiCl_3$ is more stable than $BiCl_5$.
(1/2, Delhi 2013C)

16. Account for the following :
Nitrogen is found in gaseous state.
(1/5, Delhi 2013C)

17. Though nitrogen exhibits +5 oxidation state, it does not form pentahalide. Why?
(1/2, AI 2013C, Delhi 2012C)

18. Account for the following :
NF_3 is an exothermic compound but NCl_3 is an endothermic compound.
(1/3, AI 2013C, 2012, 2011, 1/5, Delhi 2011)

19. Bismuth is a strong oxidising agent in the pentavalent state. Explain.
(1/3, AI 2013C, 1/5, Delhi 2012C)

20. Explain the following observations :
The molecules NH_3 and NF_3 have dipole moments which are of opposite direction.
(1/5, Delhi 2012)

21. Explain the following observation :
Phosphorus has greater tendency for catenation than nitrogen. *(1/5, AI 2012)*

22. Why is BiH_3 the strongest reducing agent amongst all the hydrides of group 15 elements? *(1/5, AI 2012C)*

23. Account for the following :
Tendency to form pentahalides decreases down the group in group 15 of the periodic table. *(1/5, Delhi 2011)*

24. Draw the structure of the following molecule :
NF_3 *(1/3, Foreign 2011)*

25. Explain the following :
+3 oxidation state becomes more and more stable from As to Bi in the group.
(1/5, Delhi 2011C)

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26. Account for the following :
$BiCl_3$ is less covalent than PCl_3.
(1/5, Delhi 2011C)

SA (2 marks)

27. Among the hydrides of group-15 elements, which have the
(a) lowest boiling point
(b) maximum basic character
(c) highest bond angle
(d) maximum reducing character? *(2018)*

7.2 Dinitrogen

VSA (1 mark)

28. What happens when $(NH_4)_2Cr_2O_7$ is heated?
(1/2, Delhi 2017)

29. Give reasons for the following :
N_2 is less reactive at room temperature.
(1/3, AI 2015)

7.3 Ammonia

VSA (1 mark)

30. On adding NaOH to ammonium sulphate, a colourless gas with pungent odour is evolved which forms a blue coloured complex with Cu^{2+} ion. Identify the gas. *(Delhi 2016)*

31. Account for the following :
Bond angle in NH_4^+ is higher than NH_3.
(1/5, Foreign 2015)

32. What happens when ammonium chloride is treated with $Ca(OH)_2$? *(1/5, AI 2015C)*

33. Why does NH_3 act as a Lewis base?
(AI 2014)

34. Mention the optimum conditions for the industrial manufacture of ammonia by Haber's process. *(1/3, Foreign 2011)*

SA (2 marks)

35. How is ammonia prepared on the large scale? Name the process and mention the optimum conditions for the production of ammonia by this process. *(2/5, AI 2014C)*

7.4 Oxides of Nitrogen

VSA (1 mark)

36. Give reasons for the following :
N_2O_5 is more acidic than N_2O_3. *(1/3, AI 2017)*

37. $Pb(NO_3)_2$ on heating gives a brown gas which undergoes dimerisation on cooling? Identify the gas. *(AI 2016)*

38. Draw the structure of the following compound: N_2O_5 *(1/3, Delhi 2014, AI 2012)*

39. Why does NO_2 dimerise? *(AI 2014, 2012C)*

40. Using VSEPR theory predict the probable structure of the following : N_2O_3
(1/5, Delhi 2014C)

41. What is the covalency of nitrogen in N_2O_5?
(Delhi 2013)

42. Explain the following : NO_2 readily forms a dimer. *(1/2, Delhi 2013C)*

43. Explain the following :
The bond angles (O—N—O) are not of the same value in NO_2^- and NO_2^+.
(1/5, Delhi 2012)

44. Give reason : Nitric oxide becomes brown when released in air. *(1/5, Delhi, AI 2012C)*

45. State reasons for the following :
The N—O bond in NO_2^- is shorter than the N—O bond in NO_3^-. *(1/2, AI, Delhi 2011)*

LA I (3 marks)

46. When concentrated sulphuric acid was added to an unknown salt present in a test tube a brown gas (*A*) was evolved. This gas intensified when copper turnings were added to this test tube. On cooling, the gas (*A*) changed into a colourless solid (*B*).
(i) Identify (*A*) and (*B*).
(ii) Write the structures of (*A*) and (*B*).
(iii) Why does gas (*A*) change to solid on cooling? *(3/5, 2018)*

7.5 Nitric Acid

VSA (1 mark)

47. Write the formula of the compound of phosphorus which is obtained when conc.HNO_3 oxidises P_4. *(AI 2017)*

48. Write the formula of the compound of sulphur which is obtained when conc. HNO_3 oxidises S_8. *(AI 2017)*

49. Write the formula of the compound of iodine which is obtained when conc. HNO_3 oxidises I_2. *(AI 2017)*

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50. Complete the following chemical equation ;
$Cu + HNO_{3(dilute)} \longrightarrow$
(1/5, Delhi 2015C, AI 2012)

51. Complete the following chemical equation :
$I_2 + HNO_3 \underset{(Conc.)}{\longrightarrow}$ *(1/5, Delhi 2011)*

SA (2 marks)

52. When dilute ferrous sulphate solution is added to an aqueous solution containing nitrate ion followed by careful addition of concentrated sulphuric acid along the sides of test tube, a brown ring is formed at the interface between the solution and sulphuric acid layers. Which anion is confirmed by the appearance of brown ring? What is the composition of the brown ring? *(AI 2019)*

7.6 Phosphorus – Allotropic Forms

VSA (1 mark)

53. Give reason for the following :
Red phosphorus is less reactive than white phosphorus. *(1/3, AI 2017, 1/3 Foreign 2011)*

54. Which allotrope of phosphorus is more reactive and why? *(1/5, Delhi 2015)*

55. Write the structural difference between white phosphorus and red phosphorus.
(1/3, Delhi 2014)

56. Complete the following equation :
$P_4 + H_2O \longrightarrow$ *(1/2, AI 2014)*

57. Draw the structure of the following :
Red P_4 *(1/5, Foreign 2014)*

58. White phosphorus is more reactive than red phosphorus. Explain *(1/5, Delhi 2012C)*

7.7 Phosphine

VSA (1 mark)

59. Complete the following chemical reaction equation :
$P_4 + NaOH + H_2O \longrightarrow$
(1/5, Delhi 2015C, 2014C)

60. Complete the following chemical equation :
$Ca_3P_2 + H_2O \longrightarrow$ *(1/3, Delhi 2014)*

61. Complete the following equation :
$HgCl_2 + PH_3 \longrightarrow$ *(1/3, AI 2011)*

7.8 Phosphorus Halides

VSA (1 mark)

62. Account for the following :
Solid PCl_5 is ionic in nature.
(1/5, AI 2019, Delhi 2016)

63. What happens when PCl_5 is heated?
Write the equation involved.
(1/2, Delhi 2017, 1/2, Delhi 2013)

64. Account for the following :
PCl_5 is more covalent than PCl_3.
(1/2, Delhi 2014)

65. Complete the following equation :
$Ag + PCl_5 \longrightarrow$ *(1/2, AI 2014)*

66. Draw the structure of the following :
Solid PCl_5 *(1/2, Delhi 2014C, AI 2014C)*

67. Complete the following chemical equation :
$P_4 + SO_2Cl_2 \longrightarrow$ *(1/2, AI 2014C, 1/5, (Delhi 2012, 1/3, Foreign 2011)*

68. Why does PCl_3 fume in moisture?
(Delhi 2013C, 2012C)

69. Which one of PCl_4^+ and PCl_4^- is not likely to exist and why? *(Delhi 2012)*

70. Explain the following observation :
All the bonds in PCl_5 molecule are not equivalent. *(Delhi 2012)*

71. Account for the following :
PCl_5 acts as an oxidising agent.
(1/5, AI 2012C)

7.9 Oxoacids of Phosphorus

VSA (1 mark)

72. Give reasons :
H_3PO_3 undergoes disproportionation reaction but H_3PO_4 does not. *(1/5, 2018)*

73. What happens when H_3PO_3 is heated?
Write the equation.
(1/2 Delhi 2017, Delhi 2013)

74. Draw the structure of the following :
H_3PO_2 *(1/2, Delhi 2017, Delhi 2013C, 2012, AI 2013C)*

75. Draw the structure of the following :
$H_4P_2O_7$ *(1/2, Delhi 2017, 1/5, Foreign 2015)*

76. Arrange the following in the increasing order of property mentioned :

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H_3PO_3, H_3PO_4, H_3PO_2 (Reducing character) *(1/5, Delhi 2016)*

77. What is the basicity of H_3PO_4? *(Delhi 2015)*

78. H_3PO_2 is a stronger reducing agent than H_3PO_3.*(1/3, Delhi, 1/3, AI, 1/5, Foreign 2014)*

79. What is the basicity of H_3PO_3? *(1/3, AI 2014)*

80. Draw the structure of the following molecule : $(HPO_3)_3$ *(Delhi 2013)*

81. Draw the structure of the following molecule: H_3PO_3 *(1/2, Delhi 2013)*

82. Account for the following : H_3PO_2 has reducing nature. *(1/5, Delhi 2013C)*

83. What is the basicity of H_3PO_2 acid and why? *(1/5, 2012, AI 2011)*

84. Write a reaction to show the reducing behaviour of H_3PO_2. *(Delhi 2012C)*

7.10 Group 16 Elements

VSA (1 mark)

85. Give reason :
Sulphur is a polyatomic solid whereas oxygen is a diatomic gas.
(1/5, 2020, 1/5, 2018, Delhi 2015C, AI 2012)

86. Account for the following :
Tendency to show –2 oxidation state decreases from oxygen to tellurium. *(1/5, 2020)*

87. Give reason :
SO_2 is reducing while TeO_2 is an oxidising agent. *(1/5, 2020, 1/3, AI 2016)*

88. Arrange the following in the order of property indicated :
H_2O, H_2S, H_2Se, H_2Te – increasing acidic character *(1/2, Delhi 2019, 1/3, Delhi 2014)*

89. Arrange the following in increasing order of property indicated, giving reason.
Hydrides of group 16 - reducing character *(1/5, AI 2019)*

90. Account for the following : SF_6 is inert towards hydrolysis. *(1/5, 2018C, AI 2014C, Delhi 2013C, 1/2 Delhi, AI 2011)*

91. Give reason :
Thermal stability decreases from H_2O to H_2Te. *(1/3, Delhi 2017)*

92. Account for the following :
There is large difference between the melting and boiling points of oxygen and sulphur. *(1/5, Delhi 2015)*

93. Give reason for the following :
H_2Te is the strongest reducing agent amongst all the hydrides of group 16 elements. *(1/3, AI 2015)*

94. Elements of group 16 generally show lower value of first ionization enthalpy compared to the corresponding elements of group 15. Why? *(1/5, AI 2015C, 1/3, AI 2011C)*

95. Give reason for the following :
Oxygen has less electron gain enthalpy with negative sign than sulphur. *(1/3, AI 2014)*

96. Arrange the following group of substances in the order of the property indicated against the group :
O, S, Se, Te – increasing order of electron gain enthalpy with negative sign. *(1/5, Delhi 2014C)*

97. Assign reason for the following :
H_2S is more acidic than H_2O *(1/5, AI 2014C, 2011)*

98. Assign reason for the following :
Sulphur has a greater tendency for catenation than oxygen. *(1/5, AI 2014C, Delhi 2013, AI 2012)*

99. Account for the following :
H_2S is less acidic than H_2Te. *(1/5, Delhi 2013C)*

100. Account for the following :
Boiling point of water is much higher than that of hydrogen sulphide. *(1/3, AI 2013C)*

101. Account for the following :
Thermal stability of water is much higher than that of H_2S. *(1/5, Delhi 2012C)*

SA (2 marks)

102. Account for the following :
(i) H_2S has lower boiling point than H_2O.
(ii) Reducing character decreases from SO_2 to TeO_2. *(2/3, Foreign 2015)*

LA I (3 marks)

103. Among the hydrides of group 16, write the hydride

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(i) Which is a strong reducing agent?
(ii) Which has maximum bond angle?
(iii) Which is most thermally stable?
Give suitable reason in each. *(3/5, 2020)*

7.13 Ozone

VSA (1 mark)

104. Give reason for the following :
Ozone is thermodynamically less stable than oxygen. *(1/5, Delhi 2019)*

105. Complete the following reaction :
$PbS_{(s)} + O_3 \longrightarrow$ *(1/5, AI 2019)*

106. Account for the following :
Ozone is thermodynamically unstable.
(1/5, Delhi 2016)

107. How the supersonic jet aeroplanes are responsible for the depletion of ozone layers?
(1/5, Delhi 2015)

108. Account for the following :
The two O—O bond lengths in the ozone molecule are equal. *(1/3, Delhi 2014)*

109. Account for the following :
O_3 acts as a powerful oxidising agent.
(1/5, AI 2011C)

7.14 Sulphur – Allotropic Forms

VSA (1 mark)

110. Account for the following :
Sulphur in vapour form exhibits paramagnetic behaviour.
(1/5, Delhi 2019, 1/3, AI 2014, 1/5, Foreign 2014, 2011, 1/5, Delhi 2012, 2011C)

111. Give reason for the following :
Above 1000 K sulphur shows para-magnetism. *(1/5, AI 2019)*

112. Which allotrope of sulphur is thermally stable at room temperature? *(Foreign 2015)*

SA (2 marks)

113. Name the two most important allotropes of sulphur. Which one of the two is stable at room temperature? What happens when the stable form is heated about 370 K?
(Foreign 2014)

7.15 Sulphur Dioxide

VSA (1 mark)

114. Account for the following : Moist SO_2 gas acts as a reducing agent. *(1/5, 2020)*

115. Write a chemical reaction to test sulphur dioxide gas. Write chemical equation involved. *(1/5, Delhi 2019)*

116. Complete the following chemical equation :
$Fe^{3+} + SO_2 + H_2O \longrightarrow$
(1/5, 2018C, Foreign 2011, Delhi 2011C)

117. What happens when :
SO_2 gas is passed through an aqueous solution Fe^{3+} salt?
(1/2, AI 2016, 1/3, AI 2011)

118. What happens when sulphur dioxide reacts with chlorine in the presence of charcoal?
(1/3, Delhi 2015)

7.16 Oxoacids of Sulphur

VSA (1 mark)

119. Account for the following :
Sulphurous acid is a reducing agent.
(1/3, 2020)

120. Draw the structure of an oxoacid of sulphur containing S — O — S linkage. *(1/5, 2020)*

121. Draw the structure of the following :
$H_2S_2O_8$
(1/5, AI 2019, 1/2, AI 2017, 1/5, Foreign 2014, 2011, 1/3, Delhi 2013, 2012)

122. Draw the structure of the following :
$H_2S_2O_7$ *(1/2, Delhi 2017, 1/3, AI 2013, 1/5, Delhi 2012)*

123. Draw the structure of the following :
H_2SO_3 *(1/2, AI 2017, 2015)*

124. Draw the structure of the following :
H_2SO_4 *(1/5, Delhi 2015C, 2014)*

125. Predict the shape and the asked angle (90° or more or less) in the following case :
SO_3^{2-} and the angle O — S — O
(1/5, Delhi 2012)

7.17 Sulphuric Acid

VSA (1 mark)

126. Complete the following equation :
$S + H_2SO_4 \longrightarrow$ (Conc.) *(1/5, 2020)*

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127. Write the conditions to maximize the yield of H_2SO_4 by Contact process. *(1/5, Delhi 2016)*

128. Complete the following equation :
Cu + conc. $H_2SO_4 \longrightarrow$ *(1/3, AI 2014)*

129. Complete the following equation :
$CaF_2 + H_2SO_4 \longrightarrow$ *(1/2, AI 2014)*

130. Complete the following equation :
C + conc. $H_2SO_4 \longrightarrow$
(1/2, AI 2014, Delhi, Foreign 2011)

131. Why is $K_{a_2} << K_{a_1}$ for H_2SO_4 in water? *(1/2, Foreign 2014)*

132. Account for the following :
Concentrated sulphuric acid has charring action on carbohydrates. *(1/3, AI 2013C)*

133. Complete the following equation :
$SO_3 + H_2SO_4 \longrightarrow$ *(1/5, AI 2011)*

SA (2 marks)

134. Write one reaction as an example of each, to show that conc. H_2SO_4 acts as
(i) an oxidising agent, and
(ii) a dehydrating agent. *(2/3, 2020)*

135. What happens when
(i) conc. H_2SO_4 is added to Cu?
(ii) SO_3 is passed through water?
Write the equations. *(Delhi 2017)*

136. With the help of chemical equation explain the principle of Contact process in brief for the manufacture of sulphuric acid. *(2/3, AI 2013C)*

LA I (3 marks)

137. Describe the Contact process for the manufacture of sulphuric acid with special reference to the reaction conditions, catalysts used and yield in the process. (No diagram is required.) *(3/5, Delhi 2015C, 2012C)*

7.18 Group 17 Elements

VSA (1 mark)

138. **Assertion (A) :** F — F bond in F_2 molecule is weak.
Reason (R) : F atom is small in size.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

139. **Assertion (A) :** F_2 has lower bond dissociation enthalpy than Cl_2.
Reason (R) : Fluorine is more electronegative than chlorine.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

140. Complete the following chemical reaction :
$I^-_{(aq)} + H^+_{(aq)} + O_{2(g)} \longrightarrow$ *(1/3, 2020)*

141. Account for the following : Acidic character increases from HF to HI.
(1/5, 2020, Delhi 2015)

142. Account for the following :
Hydration enthalpy of F^- ion is more than Cl^- ion. *(1/5, 2020, 1/3, Delhi 2017)*

143. Arrange the following in order of property indicated :
HF, HCl, HBr, HI – decreasing bond enthalpy
(1/2, Delhi 2019, AI 2011C)

144. Explain the following observation :
Despite lower value of its electron gain enthalpy with negative sign, fluorine (F_2) is a stronger oxidising agent than Cl_2.
(1/5, Delhi 2019, AI 2019, 2012)

145. Arrange the following in increasing order of property indicated, giving reason.
Hydrides of group 17 - acidic strength
(1/5, AI 2019)

146. Arrange the following in the decreasing order of their reducing character :
HF, HCl, HBr, HI *(1/5, 2018)*

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147. Complete the following chemical equation :
$F_2 + 2Cl^- \rightarrow$ *(1/2, Delhi 2017)*

148. Give reason for the following :
Electron gain enthalpies of halogens are largely negative. *(1/3, AI 2017)*

149. F_2 has lower bond dissociation enthalpy than Cl_2. Why? *(1/5, Delhi 2015, 2013)*

150. Why are halogens coloured?
(1/5, Delhi 2015C, AI 2012)

151. Answer the following :
Why are halogens strong oxidising agents?
(1/5, AI 2015C)

152. Arrange the following in the order of property indicated :
HF, HCl, HBr, HI – increasing bond dissociation enthalpy *(1/2, Delhi 2014)*

153. Arrange the following groups of substances in the order of the property indicated :
F_2, Cl_2, Br_2, I_2 – increasing order of bond dissociation enthalpy.
(1/5, Delhi 2014C, 2011C)

154. Why is F_2 a stronger oxidising agent than Cl_2? *(AI 2014C, 1/3, AI 2011C)*

155. Assign reason for the following :
HCl is a stronger acid than HF though fluorine is more electronegative than chlorine. *(1/5, AI 2014C, AI 2013C)*

156. Account for the following :
Fluorine does not exhibit positive oxidation state. *(1/5, Delhi 2013, 2011C, 1/5, AI 2012)*

157. Fluorine exhibits only –1 oxidation state whereas other halogens exhibit +1, +3, +5 and +7 oxidation state also. Why is it so?
(Delhi 2013C)

158. Account for the following :
HF is not stored in glass bottles but is kept in wax-coated bottles. *(1/3, AI 2013C)*

159. Despite having greater polarity, hydrogen fluoride boils at a lower temperature than water. *(1/5, AI 2012)*

160. Explain giving reason for the following situation :
In aqueous medium HCl is stronger acid than HF. *(1/2, Foreign 2011)*

161. Suggest a possible reason for the following observation :
Fluorine forms the largest number of interhalogen compounds amongst the halogens. *(Delhi 2011C)*

162. Account for the following :
Electron gain enthalpy with negative sign for fluorine is less than that for chlorine.
(1/3, AI 2011C)

SA (2 marks)

163. Write the reaction of F_2 with water. Why does I_2 not react with water? *(2/5, 2020)*

164. Compare the oxidizing action of F_2 and Cl_2 by considering parameters such as bond dissociation enthalpy, electron gain enthalpy and hydration enthalpy. *(2/5, Delhi 2016)*

7.19 Chlorine

VSA (1 mark)

165. Complete the following reaction equation :
NaOH (cold and dilute) + $Cl_2 \longrightarrow$
(1/5, 2020, 1/5, Delhi 2018C, 1/3, Delhi 2011C)

166. Complete the following reaction :
$NH_3 + 3Cl_{2(excess)} \longrightarrow$ *(1/2, Delhi 2017)*

167. Complete the following reaction :
$Cl_2 + H_2O \longrightarrow$ *(1/2, Delhi 2017)*

168. Write balanced equation for the following reaction :
Chlorine reacts with dry slaked lime.
(1/3, Delhi 2015C)

169. Name two poisonous gases which can be prepared from chlorine gas. *(AI 2013)*

170. Account for the following :
Bleaching of flowers by Cl_2 is permanent while that of SO_2 is temporary?
(1/3, AI 2013C)

171. Complete the following chemical equation :
NaOH + $Cl_2 \longrightarrow$
(hot and conc.) *(1/5, Delhi 2012)*

172. Account for the following :
Chlorine water loses its yellow colour on standing. *(1/5, AI 2012C)*

173. Complete this reaction :
$6NaOH + 3Cl_2 \longrightarrow$ *(1/3, AI 2011C)*

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174. Complete the following reaction equation :
$I_2 + H_2O + Cl_2 \longrightarrow$ *(AI 2011C)*

SA (2 marks)

175. How can you prepare Cl_2 from HCl and HCl from Cl_2? Write reactions only. *(AI 2019)*

7.20 Hydrogen Chloride

VSA (1 mark)

176. Complete the following chemical reaction :
$MnO_2 + 4HCl \longrightarrow$ *(1/3, 2020)*

177. Write balanced chemical equations for the following process :
MnO_2 is heated with conc. HCl.
(1/2 Delhi 2019)

178. What happens when
HCl is added to MnO_2? *(1/2, Delhi 2017)*

179. Account for the following :
Iron dissolves in HCl to form $FeCl_2$ and not $FeCl_3$. *(1/5, Foreign 2015, 2014)*

7.21 Oxoacids of Halogens

VSA (1 mark)

180. Account for the following :
Fluorine forms only one oxoacid.
(1/3, 2020, 1/5, Delhi 2016)

181. Give reason :
$HClO_4$ is a stronger acid than HOCl.
(1/5, 2020, 1/5, Foreign 2015)

182. Draw the structure of the following :
$HClO_3$ *(1/5, AI 2019, 2018, 1/2 AI 2017)*

183. Draw the structure of the following : $HClO_4$
(1/2, AI 2014, 2011, 1/5 Foreign 2014)

184. Account for the following :
Fluorine does not form oxoacids.
(1/5, AI 2012C)

185. How would you account for the following :
The oxidising power of oxoacids of chlorine follows the order :
$HClO_4 < HClO_3 < HClO_2 < HClO$
(1/2, Foreign 2011)

7.22 Interhalogen Compounds

VSA (1 mark)

186. Draw the shape of the following molecule :
BrF_3 *(1/2, 2020, 1/5, Delhi 2015C, 2014, 2014C, 2013, 2011, AI 2014, 2011)*

187. Give reason :
When Cl_2 reacts with excess of F_2, ClF_3 is formed and not FCl_3. *(1/5, 2018)*

188. Draw the structure of the following :
BrF_5 *(1/2, 2018C, AI 2017, 1/5, Delhi 2016)*

189. Draw the structure of the following :
ClF_3 *(1/2, AI 2017, Delhi 2015, AI 2012)*

190. Give reason : ICl is more reactive than I_2.
(1/3, AI 2016, 2012, AI 2012C, 2011C)

191. Complete the following chemical equation :
$F_2(\text{Excess}) + Cl_2 \xrightarrow{300°C}$ *(1/5, AI 2014C)*

192. F_2 is more reactive than ClF_3 but ClF_3 is more reactive than Cl_2. Give reason.
(1/5, Delhi 2013C)

193. Predict the shape and the asked angle (90° or more or less) in the following cases.
ClF_3 and the angle F — Cl — F
(1/5, Delhi 2012)

194. Account for the following :
$BrCl_3$ is more stable than $BrCl_5$.
(1/5, AI 2012C)

195. Why does fluorine not play the role of a central atom in interhalogen compounds?
(AI 2011)

196. Complete the following chemical equation :
$Br_2 + F_2 \text{ (excess)} \longrightarrow$ *(Foreign 2011)*

SA (2 marks)

197. How are interhalogen compounds formed? What general compositions can be assigned to them? *(AI 2013)*

7.23 Group 18 Elements

VSA (1 mark)

198. Draw the shape of the following molecule :
$XeOF_4$
(1/2, 2020, 1/2, Delhi 2017, 1/5, AI 2017C, 2015,2015C, 2013, 2012, Foreign 2014)

199. Complete the following chemical reaction :
$XeF_6 + KF \longrightarrow$ *(1/3, 2020)*

200. Give reason :
Helium does not form compounds like xenon. *(1/5, 2020)*

201. Boiling point of noble gases increase from He to Rn *(1/3, 2020)*

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202. Complete the following equation :
$XeF_2 + H_2O \rightarrow$ *(1/5, 2020, 1/2, Delhi 2017, AI 2014, 1/5, Delhi 2011, 2011C)*

203. Draw the structure of XeF_2. *(1/5, 2020, 1/2, 2018C, 1/5, Foreign 2015, Delhi 2014, 2012, 2013, 2011,AI 2014)*

204. Draw the structure of the following :
XeF_4 *(1/5, Delhi 2019, 2018, 2016, 2014, 1/2, Delhi 2017, AI 2016, 2014)*

205. Write balanced chemical equations for the following processes :
XeF_2 undergoes hydrolysis.
(1/2, Delhi 2019, 1/5, 2018C)

206. Complete the following reaction :
$XeF_6 + NaF \longrightarrow$ *(1/5, AI 2019)*

207. Complete the following reaction :
$XeF_4 + SbF_5 \longrightarrow$ *(1/5, 2018)*

208. Complete the following equation :
$XeF_4 + O_2F_2 \longrightarrow$ *(1/5, 2018 C, 1/2, Delhi 2014C, 2012, 1/2, AI 2014, 2012)*

209. Complete the following reaction :
$XeF_6 + 2H_2O \longrightarrow$ *(1/2, Delhi 2017)*

210. Draw the structure of the following :
XeF_6 *(1/2, Delhi 2017, AI 2013, 1/5, Delhi 2012)*

211. Complete the following reaction :
$XeF_6 + 3H_2O \longrightarrow$ *(1/2, Delhi 2017)*

212. What happens when XeF_4 reacts with SbF_5?
(AI 2016)

213. Give reason for the following :
Helium is used in diving apparatus as a diluent for oxygen. *(1/3, AI 2015, 2013C)*

214. Draw the structure of the following : XeO_3
(1/5, AI 2015C, 2014)

215. Account for the following :
Unlike xenon, no distinct chemical compound of helium is known. *(1/5, Foreign 2014)*

216. Write the formula and describe the structure of noble gas species which is isostructural with BrO_3^-. *(1/5, Delhi 2014C)*

217. Complete the following chemical reaction equation : $XeF_4 + H_2O \longrightarrow$
(1/2, AI 2014C, 2012, 2011, 2011C, 1/5, Delhi 2012)

218. What inspired N.Bartlett for carrying out reaction between Xe and PtF_6?
(Delhi 2013)

219. Complete the following chemical reaction equation :
$XeF_6 + H_2O \longrightarrow$ *(1/5, Delhi 2012, 1/3, Foreign 2011)*

220. Predict the shape and asked angle (90° or more or less) in the following case.
XeF_2 and the angle F – Xe – F
(1/5, Delhi 2012)

221. Explain the following observations:
Helium forms no real chemical compound.
(1/5, AI 2012)

222. Noble gases have low boiling points. Why?
(Delhi 2012C, AI 2011)

223. Complete the following reaction equation :
$XeF_2 + PF_5 \longrightarrow$
(1/5, Delhi 2011C, 1/2, AI 2011C)

SA (2 marks)

224. Write the products of the following reaction :
$6XeF_4 + 12H_2O \rightarrow$
Is this reaction a disproportionation reaction? Give reasons in support of your answer. *(2020)*

225. (i) Which noble gas is used in filling balloons for meteorological observations?
(ii) Complete the equation :
$XeF_2 + PF_5 \longrightarrow$ *(2/5, Delhi 2015)*

226. Write the balanced chemical equations for obtaining XeO_3 and $XeOF_4$ from XeF_6.
(1/5, Delhi 2013C)

227. Explain the following :
(i) Xenon does not form such fluorides as XeF_3 and XeF_5.
(ii) Out of noble gases, only xenon is known to form real chemical compounds.
(Delhi 2012C)

228. How is XeO_3 obtained? Write the related chemical equations. Draw the structure of XeO_3. *(AI 2012C)*

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Detailed Solutions

1. The single N—N bond is weaker than the single P—P bond because of high interelectronic repulsion of the non-bonding electrons, occurring due to the small bond length.

2. Nitrogen cannot expand its octet due to absence of *d*-orbitals.

3. Increasing (Lewis) base strength order is :
$BiH_3 < SbH_3 < AsH_3 < PH_3 < NH_3$
The reason for this order is that as we move from NH_3 to BiH_3, the size of the central atom increases. Hence, lone pair is not easily available for donation. The electron density on the central atom decreases on moving from NH_3 to BiH_3 and so the basic strength also decreases.

4. Among hydrides of group-15 elements, the bond length increases from N – H to Bi – H with increasing size of element. Bi – H bond is longest and weakest, it can break more easily and evolve H_2 gas which acts as the reducing agent.

5. $R_3N = O$ molecule has five covalent bonds with N atom. The octet in N cannot be extended as it does not have *d*-orbitals for the formation of $p\pi$-$d\pi$ bond.

In the case of $R_3P = O$, P can extend its octet since it has empty *d*-orbitals in its valence shell and form $p\pi$-$d\pi$ bond.

6. Lewis basic nature of NH_3 and PH_3 molecules is due to the presence of lone pairs on N and P atoms, respectively. P atom is much larger than N atom and also has empty *d*-orbitals. Electron density due to lone pair on P gets diffused because of the presence of d-orbitals and so the lone pair is not easily available for donation. Hence, PH_3 is less basic than NH_3.

7. *Refer to answer 5.*

8. On moving down the group, the stability of +5 oxidation state decreases while +3 oxidation state increases due to inert pair effect.
Thus, +5 oxidation state of Bi is less stable and Bi(V) is a stronger oxidising agent.

9. Nitrogen cannot expand its valency beyond 4 due to absence of *d*-orbitals whereas phosphorus show pentavalency due to presence of *d*-orbitals.

10. *Refer to answer 3.*

11. $PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3$
The abnormally high boiling point of NH3 is due to the intramolecular H-bonding. Further as we move from PH_3 to BiH_3 the molecular mass increasing. As a result, the van der waals forces of attraction increase and the boiling points increase regularly from PH_3 to BiH_3.

12. In case of nitrogen, only NF_3 is known to be stable. N–F bond strength is greater than F–F bond strength, therefore, formation of NF_3 is spontaneous. In case of NCl_3, N—Cl bond strength is lesser than Cl—Cl bond strength. Thus, energy has to be supplied during the formation of NCl_3.

13. The bond dissociation enthalpy of triple bond in $N \equiv N$ is very high due to $p\pi$ - $p\pi$ overlap. Hence, N_2 is less reactive at room temperature.

14. Unlike NH_3, PH_3 molecules are not associated through hydrogen bonding in liquid state. Therefore, the boiling point of PH_3 is lower than NH_3.

15. $BiCl_3$ is more stable than $BiCl_5$. On moving down the group, the stability of + 5 oxidation state decreases while + 3 oxidation state increases due to inert pair effect.

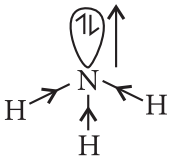
16. Nitrogen exists as a diatomic molecule with a triple bond between two atoms. These N_2 molecules are held together by weak van der Waals force of attraction which can be easily broken by the collision of the molecules at room temperature. Therefore, N_2 is a gas at room temperature.

17. *Refer to answer 2.*

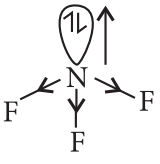
18. *Refer to answer 12.*

19. *Refer to answer 8.*

20.



In NH_3 the bond moment and lone pair moment work in same direction. Hence, net dipole moment is higher.



In NF_3 the bond moment and lone pair moment work in opposite direction. Hence, net dipole moment is lower.

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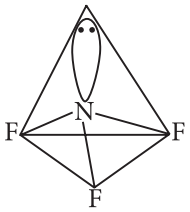
21. The property of catenation depends upon the strength of the element - element bond. Since, P – P (213 kJ mol^{-1}) bond strength is much more than N – N (159 kJ mol^{-1}) bond strength so, phosphorus shows marked catenation properties than nitrogen.

22. *Refer to answer 4.*

23. Due to inert pair effect the stability of +5 oxidation state decreases down the group in group 15. Hence, tendency to form pentahalide decreases down the group 15 of the periodic table.

24. Total no. of electrons around the central N atom = 5
No. of bond pairs = 3
No. of lone pairs = 1
Hybridisation = sp^3
Therefore, according to VSEPR theory; NF_3 should be pyramidal.



25. On moving down the group, the stability of + 5 oxidation state decreases while that of + 3 oxidation state increases due to inert pair effect.

26. $BiCl_3$ is less covalent than PCl_3 because the size of Bi^{3+} is much larger than P^{3+} (According to Fajan's rule)

27. (a) PH_3 (Phosphine)
(b) NH_3 (ammonia)
(c) NH_3 (ammonia)
(d) BiH_3 (Bismuth hydride)

28. When orange red crystals of ammonium dichromate are heated, these crystals decompose violently evolving nitrogen and green coloured chromic oxide is formed.

$$\underset{\text{(Orange red)}}{(NH_4)_2Cr_2O_7} \xrightarrow{\Delta} \underset{\text{(Green)}}{Cr_2O_3} + N_2 + 4H_2O$$

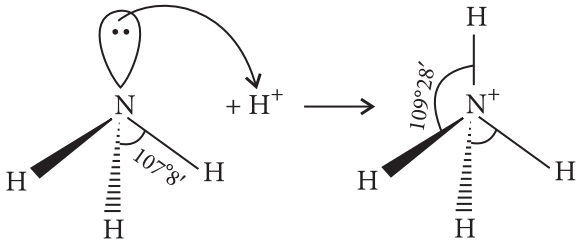
29. In N_2 molecule N atoms are held by triple bonds. It has very high bond dissociation energy (941.4 kJ mol^{-1}). Therefore N_2 is inert at room temperature.

30. $(NH_4)_2SO_4 + 2NaOH \xrightarrow{\Delta}$

$$\underset{\substack{\text{(colourless and}\\ \text{pungent odoured gas)}}}{2NH_3} + Na_2SO_4 + 2H_2O$$

$$Cu^{2+} + 4NH_3 \longrightarrow \underset{\text{(Blue colour)}}{[Cu(NH_3)_4]^{2+}}$$

31. N in NH_3 is sp^3–hybridized. It has three bond pairs and one lone pair around N. Due to stronger lone pair-bond pair repulsions than bond pair-bond pair replusions, the tetrahedral angle decreases from 109°28′ to 107°8′. As a result, NH_3 is pyramidal. However, when it reacts with a proton, it forms NH_4^+ ion which has four bond pairs and no lone pair. Due to the absence of lone pair-bond pair repulsion and presence of four identical bond pair-bond pair interaction, NH_4^+ assumes tetrahedral geometry with a bond angle of 109°28′. This explains why the bond angle in NH_4^+ is higher than in NH_3.



NH_3 molecule (Pyramidal) NH_4^+ ion (Tetrahedal)

32. $2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + 2H_2O + CaCl_2$

33. NH_3 has a lone pair of electrons on the N-atom which it can donate to an electron acceptor. Hence, NH_3 acts as a Lewis base.

34. Optimum conditions for the production of ammonia are :
Temperature = about 700 K
Pressure = about 200 atm (200×10^5 Pa)
Catalyst = iron oxide with small amounts of K_2O and Al_2O_3 (as promoters).

35. Ammonia is manufactured industrially by Haber's process.

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$; $\Delta_r H° = -46.1$ kJ mol^{-1}

This is a reversible exothermic reaction. High pressure about 200 atm, low temperature about 700 K and use of catalyst such as iron oxide with small amounts of Al_2O_3 and K_2O would favour the formation of ammonia according to Le-Chatelier's principle.

36. Oxidation state of N in N_2O_5 = + 5
Oxidation state of N in N_2O_3 = + 3
As the oxidation number of central element increases, acidic strength increases. Hence, N_2O_5 is more acidic than N_2O_3.

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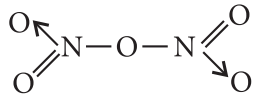
37. $2Pb(NO_3)_2 \xrightarrow{673\ K} \underset{\text{Brown}}{4NO_2} + 2PbO + O_2$

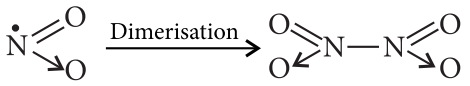
NO_2 dimerises as :

$2NO_2 \underset{\text{Heat}}{\overset{\text{Cool}}{\rightleftharpoons}} N_2O_4$

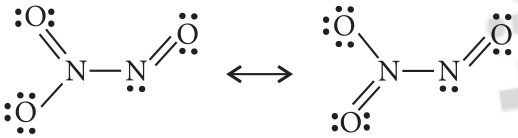
38. The structure of N_2O_5 is



39. Because NO_2 contains odd number of valence electrons and on dimerisation, it is converted to stable N_2O_4 molecule with even number of electrons.

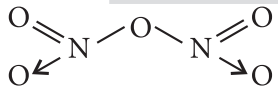


40.



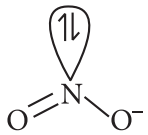
Planar structure

41. In N_2O_5 covalency of nitrogen is four.



42. *Refer to answer 39.*

43. In NO_2^-, 'N' atom has sp^2-hybridisation whereas in NO_2^+, 'N' atom has sp-hybridisation.

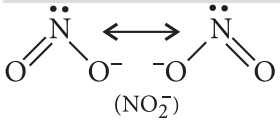


Hence, bond angles are not of the same value.

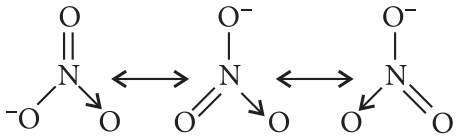
44. Nitric oxide forms brown fumes of nitrogen dioxide (NO_2) instantaneously in the presence of air.

$2NO + O_2 \longrightarrow 2NO_2$

45. In NO_2^-, the average N—O bond order is 1.5 due to two resonating structures whereas in NO_3^-, the average N—O bond order is 1.33 due to three resonating structures. Higher the bond order, shorter is the bond length.



(NO_2^-)



(NO_3^-)

46. (i) Light brown fumes of nitrogen dioxide are evolved on heating the nitrate with concentrated H_2SO_4.

$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$

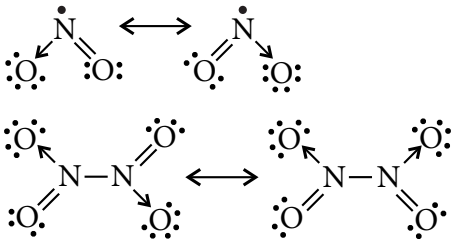
$4HNO_3 \rightarrow 2H_2O + \underset{\text{Brown fumes } (A)}{4NO_2} + O_2$

These fumes intensify when copper turnings are added.

$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

$\underset{\text{Brown gas } (A)}{2NO_2} \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \underset{\text{Colourless solid } (B)}{N_2O_4}$

(ii)



(iii) *Refer to answer 39.*

47. H_3PO_4 (phosphoric acid) is obtained when conc. HNO_3 oxidises P_4.

$P_4 + 20HNO_{3(conc.)} \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$

48. When conc. HNO_3 oxidises S_8, it forms sulphuric acid.

$S_8 + 48HNO_3 \rightarrow 8H_2SO_4 + 48NO_2 + 16H_2O$

49. Iodic acid (HIO_3) is formed when conc. HNO_3 oxidises I_2.

$I_2 + 10HNO_3 \longrightarrow \underset{\text{Iodic acid}}{2HIO_3} + 10NO_2 + 4H_2O$

50. $3\ Cu + 8HNO_{3\ (dilute)} \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

51. $I_2 + 10HNO_{3(conc.)} \rightarrow 2HIO_3 + 10NO_2 + 4H_2O$

52. The brown ring at the interface between the solution and sulphuric acid layers indicates the presence of nitrate ion in solution.

$NO_3^- + 3Fe^{2+} + 4H^+ \longrightarrow NO + 3Fe^{3+} + 2H_2O$

$[Fe(H_2O)_6]^{2+} + NO \longrightarrow \underset{\text{Brown}}{[Fe(H_2O)_5(NO)]^{2+}} + H_2O$

53. White phosphorus is made up of discrete P_4 tetrahedra which are subjected to very high angular strain as the angles are 60°. Red phosphorus is a polymer of P_4 tetrahedra, which has much less angular strain. This high angular strain makes white phosphorus unstable and highly reactive.

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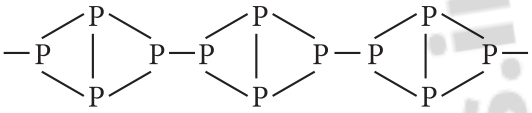
54. White phosphorus is most reactive of all the allotropes because it is unstable due to the angular strain on P_4 molecule with bond angle of 60°.

55. White phosphorus consists of discrete tetrahedral P_4 molecule.
Red phosphorus is polymeric, consisting of chains of P_4 tetrahedra linked together.

56. $P_4 + H_2O \longrightarrow$ No reaction

57.



Red phosphorus

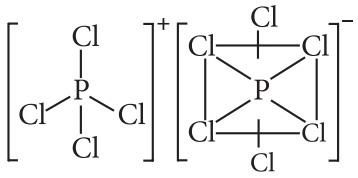
58. *Refer to answer 53.*

59. $P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$

60. $Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

61. $3HgCl_2 + 2PH_3 \longrightarrow Hg_3P_2 + 6HCl$

62. Phosphorus pentachloride is a salt containing the tetrahedral cation $[PCl_4]^+$ and the octahedral anion $[PCl_6]^-$ therefore, it is ionic in solid state.



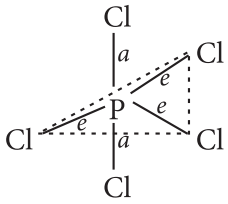
63. PCl_5 on heating, sublimes but it decomposes into phosphorus trichloride and chlorine on stronger heating.

$$PCl_5 \xrightarrow{\Delta} PCl_3 + Cl_2$$

64. In +5 oxidation state charge/radius ratio is higher than that in +3 oxidation state. Hence, +5 oxidation state has more polarising power than that of +3 oxidation state and pentahalides (in O.S. +5) are more covalent than trihalides.

65. $2Ag + PCl_5 \longrightarrow 2AgCl + PCl_3$

66.



In PCl_5, there are three equatorial and two axial bonds present. Since, three equatorial bonds are repelled by two bond pairs and two axial bonds are repelled by three bond pairs so, axial bonds are weaker and longer than the equatorial bonds.

67. $P_4 + 10SO_2Cl_2 \longrightarrow 4PCl_5 + 10SO_2$

68. PCl_3 hydrolyses in the presence of moisture giving fumes of HCl.

$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$

69. PCl_4^- because PCl_3 cannot form bond with Cl^- ions.

70. *Refer to answer 66.*

71. The oxidation state of phosphorus in PCl_5 is +5. As P has five electrons in its valence shell, it cannot increase its oxidation state beyond +5 by donating electrons. It can decrease its oxidation number from +5 to +3 or some lower value. So, PCl_5 acts as an oxidising agent.

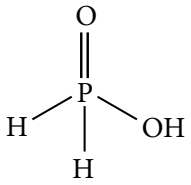
72. The oxoacid of phosphorus containing +3 oxidation state, undergoes disproportionation to yield compounds in higher and lower oxidation states. Hence, H_3PO_3 undergoes disproportionation reaction but H_3PO_4 does not, as in it phosphorus is already in highest oxidation state (+5).

$$4\overset{+3}{H_3}PO_3 \xrightarrow{\Delta} \overset{-3}{PH_3} + 3\overset{+5}{H_3}PO_4$$

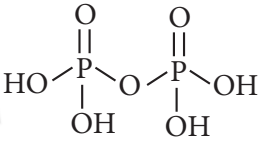
73. When phosphorus acid (H_3PO_3) is heated, it forms orthophosphoric acid and phosphine.

$$4H_3PO_3 \xrightarrow{200\ °C} 3H_3PO_4 + PH_3$$

74. H_3PO_2 :

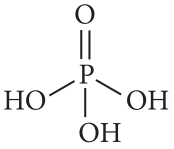


75. $H_4P_2O_7$ (Pyrophosphoric acid) :



76. Reducing character of oxyacids of phosphorus depends on the number of P–H bonds. More the number of P–H bonds in oxyacid, more is the reducing character. H_3PO_2 has two P–H bonds, H_3PO_3 has one P–H bond and H_3PO_4 has no P–H bond. Thus, order of reducing character is $H_3PO_2 > H_3PO_3 > H_3PO_4$.

77. Basicity of oxoacids of P is equal to the number of P—OH bonds in the molecule.



H_3PO_4
Orthophosphoric acid

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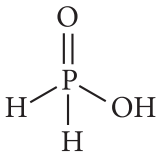
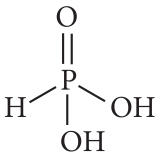
It is tribasic due to the presence of three replaceable hydrogen atoms.

78. The acids which contain more P—H bond, have strong reducing properties. Hypophosphorus acid (H_3PO_2) contains two P—H bonds, whereas orthophosphorus acid (H_3PO_3) has one P—H bond.

Hence, H_3PO_2 is stronger reducing agent than H_3PO_3.

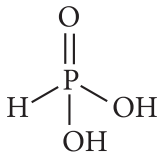


H_3PO_2



H_3PO_3

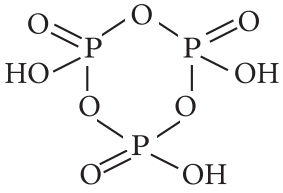
79.



(H_3PO_3)
Orthophosphorus acid

It is dibasic due to the presence of two replaceable hydrogen atoms.

80.

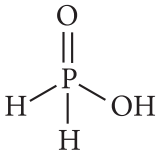


$(HPO_3)_3$

81. *Refer to answer 78.*

82. *Refer to answer 78.*

83.



It is monobasic acid due to the presence of one replaceable hydrogen.

84. $4AgNO_3 + 2H_2O + H_3PO_2 \longrightarrow 4Ag + 4HNO_3 + H_3PO_4$

85. O_2 molecules are held together by weak van der Waals' forces because of the small size and high electronegativity of oxygen. Sulphur shows catenation and the molecule is made up of eight atoms (S_8) with strong intermolecular attractive forces. Hence, sulphur exists as solid at room temperature.

86. Tendency to show –2 oxidation state decreases from oxygen to tellurium because electronegativity decreases on moving down the group.

87. The +6 oxidation state of S is more stable than +4 therefore, SO_2 acts as a reducing agent. Further, since the stability of +6 oxidation state decreases from S to Te therefore, the reducing character of the dioxides decreases while their oxidising character increases.

Thus, TeO_2 acts as an oxidising agent.

88. $H_2O < H_2S < H_2Se < H_2Te$

As the atomic size increases down the group, the bond length increases and hence, the bond strength decreases. Consequently, the cleavage of M — H bond (M = O, S, Se, Te, etc.) becomes easier. As a result, the tendency to release hydrogen as proton increases *i.e.*, acidic strength increases down the group.

89. Reducing character : $H_2S < H_2Se < H_2Te$

Hydrides of all elements of group 16 except that of oxygen, *i.e.*, water are reducing agents. The reducing character, however, increases from H_2S to H_2Te because of increasing size of atoms from S to Te which results in weaker M-H bond.

90. In SF_6, S atom is sterically protected by six F atoms and does not allow any reagent to attack on the S atom. Due to these reasons, SF_6 is kinetically an inert substance.

91. As the size of central atom increases, M — H bond becomes weaker and longer and hence, thermal stability decreases while going from H_2O to H_2Te.

92. Oxygen molecules are held together by weak Van der Waals' forces because of the small size and high electronegativity of oxygen. Sulphur shows catenation and the molecule is made up of 8 atoms with strong intermolecular forces. Hence, there is large difference in the melting and boiling points of oxygen and sulphur.

93. Bond dissociation enthalpy and hence thermal stability of hydrides decreases from H_2O to H_2Te. Therefore, H_2Te releases hydrogen readily. Hence, H_2Te is the strongest reducing agent amongst all the hydrides of group 16.

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94. The first ionization enthalpy of group 16 elements is lower than those of group 15 elements despite their smaller atomic radii and higher nuclear charge. This is due to the relatively symmetrical and more stable configuration of the elements of group 15 as compared to those of the elements of group 16.

95. The electron gain enthalpy of oxygen is less negative than sulphur. This is due to its small size. As a result of which the electron-electron repulsion in the relatively small 2*p*-subshell are comparatively larger and hence the incoming electrons are not accepted with same ease as in case of other (sulphur) elements of this group.

96. Electron gain enthalpy of oxygen is less negative than sulphur due to compact size of oxygen atom (inter-electronic repulsion is more in O). From sulphur onwards enthalpy again becomes less negative upto Po.

	O	< Po <	Te	<	Se	<	S
$\Delta_{eg}H$(kJ mol^{-1})	−141		−174		−190		−195	−200

(−ve sign shows release of energy)

97. *Refer to answer 88.*

98. The property of catenation depends upon *E* – *E* bond strength of the element. As S – S bond is much stronger (213 kJ mol^{-1}) than O – O bond (138 kJ mol^{-1}), sulphur has greater tendency for catenation than oxygen.

99. *Refer to answer 88.*

100.

	H_2O		H_2S
Boiling point	373 K	>	213 K

The abnormally high boiling point of H_2O is due to strong intermolecular H-bonding. Since, all other elements have much lower electronegativity than oxygen, they do not undergo H-bonding.

101. *Refer to answer 91.*

102. (i) *Refer to answer 100.*
(ii) *Refer to answer 87.*

103. (i) Bond dissociation enthalpy and hence thermal stability of hydrides decreases from H_2O to H_2Te. Therefore, H_2Te releases hydrogen readily. Hence, H_2Te is the strongest reducing agent amongst all the hydrides of group 16.
(ii) All the group 16 hydride have angular shape involving sp^3 hybridisation due to stronger lone pair-lone pair repulsions than bond pair-bond pair repulsions, the bond angle in water decreases from the tetrahedral value. As we move down the group from O to Te, the size of central atom increases and its electronegativity decreases. As a result the position of the two bond pairs shifts away from the central atom as we move from H_2O to H_2Te. Consequently the repulsions between the bond pairs decreases from H_2O to H_2Te and therefore bond angle decreases. Thus H_2O has the highest bond angle among group 16 hydrides.
(iii) As the size of the element increases down the group, the E—H bond dissociation energy decreases and hence E—H bond breaks more easily. Thus H_2O has the highest thermal stability among group 16 hydrides.

104. Ozone is thermodynamically unstable and decomposes into oxygen.

$$2O_3 \xrightarrow{\Delta} 3O_2;\ \Delta H = -\text{ve}$$

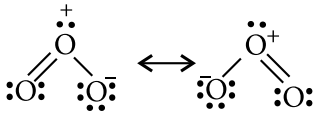
The above conversion is exothermic *i.e.*, ΔH is negative. Also, entropy increases *i.e.*, ΔS = + ve. Thus, ΔG for the decomposition of ozone is negative. Hence, it is thermodynamically unstable.

105. $PbS_{(s)} + 4O_{3(g)} \longrightarrow PbSO_{4(s)} + 4O_{2(s)}$

106. *Refer to answer 104.*

107. Nitrogen oxide emitted from the exhausts of supersonic jet aeroplanes readily combines with ozone to form nitrogen dioxide and diatomic oxygen. Since supersonic jets fly in the stratosphere near the ozone layer, they are responsible for the depletion of ozone layer.

108. The two O — O bond lengths in the ozone molecule are equal as it is a resonance hybrid of the following two forms :



109. Ozone is a powerful oxidising agent because ozone has higher energy content than dioxygen hence, decomposes to give dioxygen and atomic oxygen.

$O_{3(g)}$	$\longrightarrow$	$O_{2(g)}$	+	$O_{(g)}$
Ozone		Dioxygen		Atomic oxygen

The atomic oxygen thus liberated brings about the oxidation while molecular oxygen is set free.

110. At elevated temperature, sulphur vapour exists as S_2 molecule which has two unpaired electrons in the antibonding π^*-orbitals like O_2 hence, exhibits paramagnetism.

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111. *Refer to answer 110.*

112. Rhombic sulphur

113. Sulphur exists in numerous allotropes of which yellow, rhombic (α-sulphur) and monoclinic (β-sulphur) are most important. The stable form is rhombic, which transforms to monoclinic sulphur, when heated around 370 K.

114. In presence of moisture, SO_2, acts as a reducing agent due to the evolution of nascent hydrogen.

115. Sulphur dioxide turns acidified potassium dichromate solution from orange to green.

$$\underset{\text{(orange)}}{K_2Cr_2O_7} + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + \underset{\text{(green)}}{Cr_2(SO_4)_3} + H_2O$$

116. $2Fe^{3+} + SO_2 + 2H_2O \rightarrow 2Fe^{2+} + SO_4^{2-} + 4H^+$

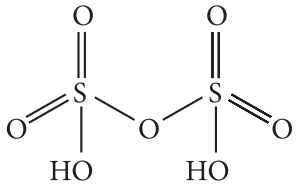
117. *Refer to answer 116.*

118. $SO_2 + Cl_2 \rightarrow SO_2Cl_2$

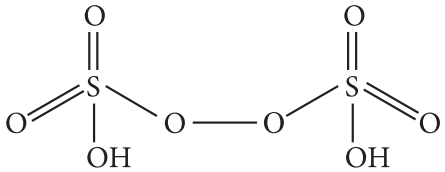
119. Due to the presence of a lone pair of electrons on the sulphur atom, sulphurous acid can be easily oxidised to sulphuric acid therefore, it acts as a reducing agent.

$$H_2O + Br_2 + H_2SO_3 \longrightarrow 2HBr + H_2SO_4$$

120. $H_2S_2O_7$ (Pyrosulphuric acid)

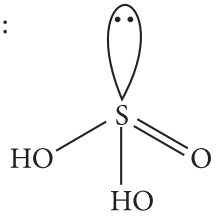


121. $H_2S_2O_8$:

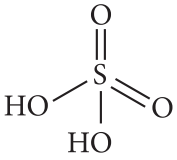


122. *Refer to answer 120.*

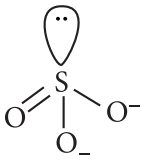
123. H_2SO_3 :



124. H_2SO_4 (Sulphuric acid) :



125.



sp^3 Hybridisation

Shape : Pyramidal; O—S —O angle > 90°

126. $S_8 + \underset{\text{(Conc.)}}{16H_2SO_4} \rightarrow 24SO_2 + 16H_2O$

127. In Contact process, the rate determining step is $2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$, $\Delta H° = -196.6$ kJ mol^{-1} This reaction is reversible and exothermic *i.e.*, ΔH is negative. Thus, according to Le-Chatelier's principle, the conditions to maximise the yield are as follows :

(a) At low temperature: As heat is evolved in the reaction so, at lower temperature the reaction proceeds more in forward direction.

(b) At high pressure: As three moles of gaseous reactants give two moles of gaseous products thus, at higher pressure reaction moves in forward direction.

128. $Cu + \underset{\text{(conc.)}}{2H_2SO_4} \longrightarrow CuSO_4 + SO_2 + 2H_2O$

129. $CaF_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HF$

130. $C + 2H_2SO_4\text{ (conc.)} \longrightarrow CO_{2(g)} + 2H_2O_{(l)} + 2SO_{2(g)}$

131. $H_2SO_{4(aq)} + H_2O_{(l)} \rightarrow H_3O^+_{(aq)} + HSO^-_{4(aq)}$

$K_{a_1} > 10$, very large

$HSO^-_{4(aq)} + H_2O_{(l)} \rightarrow H_3O^+_{(aq)} + SO^{2-}_{4(aq)}$

$K_{a_2} = 1.2 \times 10^{-2}$

K_{a_2} is smaller than K_{a_1} because dissociation of HSO_4^- is less probable due to presence of negative charge on the ion.

132. Concentrated H_2SO_4 removes water from organic compounds hence, it has charring action on carbohydrates.

$$\underset{\text{Sugar}}{C_{12}H_{22}O_{11}} \xrightarrow{\text{conc. } H_2SO_4} \underset{\text{Sugar charcoal}}{12C} + 11H_2O$$

133. $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$

134. (i) $\overset{0}{C} + 2H_2\overset{+6}{S}O_4 \longrightarrow 2\overset{+4}{S}O_2 + \overset{+4}{C}O_2 + 2H_2O$

(ii) $HCOOH \xrightarrow{\text{Conc. } H_2SO_4} H_2O + CO$

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135. (i) When conc. H_2SO_4 reacts with Cu, CuO is formed which gets further converted into $CuSO_4$.

$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$

(ii) When SO_3 is passed through water, it dissolves SO_3 to give H_2SO_4.

$SO_3 + H_2O \longrightarrow H_2SO_4$

136. Contact process : It involves three stages :

(i) Burning of sulphur or sulphide ore in air to generate SO_2.

$S + O_2 \rightarrow SO_2$

$4FeS_2 + 11O_2 \rightarrow 8SO_2 + 2Fe_2O_3$

(ii) Conversion of SO_2 to SO_3 by reaction with oxygen in the presence of V_2O_5 catalyst.

$$2SO_{2(g)} + O_{2\,(g)} \overset{V_2O_5}{\rightleftharpoons} 2SO_{3(g)};$$

$$\Delta_r H° = -196 \text{ kJ mol}^{-1}$$

(iii) The SO_3 gas from the catalytic converter is absorbed in conc. H_2SO_4 to form oleum ($H_2S_2O_7$). Dilution of oleum with water gives H_2SO_4 of desired concentration.

$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$ (Oleum)

$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$

137. *Refer to answers 127 and 136.*

138. (b) : Due to small size, the lone pairs of electrons on F-atoms repel the bond pair between F-atoms. Therefore, F—F bond in F_2 molecule is weak.

139. (b) : Increasing bond dissociation enthalpy order is

$I_2 < F_2 < Br_2 < Cl_2$

Bond dissociation enthalpy of F_2 is less than that of Br_2 and Cl_2 due to the lone pair - lone pair repulsion.

140. $4I^-_{(aq)} + 4H^+_{(aq)} + O_{2(g)} \longrightarrow 2H_2O + 2I_2$

141. The acidic strength of the hydrohalic acids is in the order :

$$HF < HCl < HBr < HI$$

This order is a result of bond dissociation enthalpies of H — *X* bond which decreases from H — F to H — I as the size of halogen atom increases.

142. As the size of halogen increases down the group from F to I, the hydration enthalpy of corresponding halide ion decreases. Hence, F^- ion has higher hydration enthalpy (524 kJ mol^{-1}) than Cl^- ion (378 kJ mol^{-1}).

143. HF > HCl > HBr > HI

As the size of halogen atom increases from F to I, bond dissociation enthalpies of H—*X* bond decreases from H—F to H—I.

144. Fluorine is the strongest oxidising agent because of its low bond dissociation and high heat of hydration as compared to those of chlorine. It oxidise other halide ions in solution or even in solid phase.

$F_2 + 2X^- \longrightarrow 2F^- + X_2$ (X = Cl, Br or I)

$Cl_2 + 2X^- \longrightarrow 2Cl^- + X_2$ (X = Br or I)

$Br_2 + 2I^- \longrightarrow 2Br^- + I_2$

145. *Refer to answer 141.*

146. The decreasing order of reducing character of the given hydrogen halides is

HI > HBr > HCl > HF

147. $F_2 + 2Cl^- \rightarrow 2F^- + Cl_2$

148. Halogens have the smallest size in their respective periods and therefore have high effective nuclear charge. As a result, they readily accept one electron to acquire noble gas configuration. Hence, electron gain enthalpies of halogens are largely negative.

149. F_2 has lower bond dissociation enthalpy than Cl_2 because F atom is very small and hence the electron-electron repulsions between the lone pairs of electrons are very large.

150. Halogens absorb radiations in visible region which results in excitation of outer electrons to higher level resulting in different colours.

151. General electronic configuration of halogens is ns^2np^5. They easily accept one electron to complete their octet. This makes them good oxidising agent.

152. HI < HBr < HCl < HF

153. Increasing bond dissociation enthalpy order is

$I_2 < F_2 < Br_2 < Cl_2$

Bond dissociation enthalpy of F_2 is less than that of Br_2 and Cl_2 due to the lone pair - lone pair repulsion.

154. *Refer to answer 144.*

155. HF is the weakest acid because of its high bond dissociation energy due to small size of fluorine atom.

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156. Since, fluorine is the most electronegative element, it shows only a negative oxidation state of –1, and does not show any positive oxidation state (except +1 in HOF).

157. Fluorine is most electronegative element and it cannot exhibit any positive oxidation state (except +1 in HOF). Other halogens have *d*-orbitals and therefore, can expand their octets and show +1, +3, +5 and +7 oxidation states also.

158. HF acid attacks glass with the formation of fluoro silicate ions. Thus, it is stored in wax-coated glass bottles to prevent the reaction.

159. Higher boiling point of H_2O is due to the extensive H-bonding in H_2O than HF.

160. In aqueous medium HCl is stronger acid than HF because bond dissociation enthalpy of H—Cl is lower than that of HF.

161. Fluorine is the most electronegative element among halogens and it cannot exhibit any positive oxidation state. Therefore, it form largest number of interhalogen compounds.

162. The electron gain enthalpy of fluorine is less negative than that of chlorine due to the small size of fluorine atom.

163. $2F_2 + 2H_2O \longrightarrow 4H^+ + 4F^- + O_2$

$3F_2 + 3H_2O \longrightarrow 6H^+ + 6F^- + O_3$

The electrode potential of F_2/F^- couple is the maximum while that of I_2/I^- is minimum, *i.e.,* F_2 is reduced most easily but I_2 is reduced least easily. F_2 reduces H_2O to O_2 and O_3 but I_2 does not react with water.

164. Oxidising power of a substance depends on the factors like bond dissociation enthalpy, electron gain enthalpy and hydration enthalpy. Due to small size of fluorine, its electron gain enthalpy is less than that of chlorine. However, its low bond dissociation enthalpy and high hydration enthalpy compensate the low electron gain enthalpy. Fluorine because of its small size has higher hydration enthalpy than chlorine. Also, due to repulsion between electrons it has lower bond dissociation energy. Thus, fluorine has better oxidising action than chlorine.

165. $2NaOH_{(dil.)} + Cl_{2(aq)} \rightarrow NaCl_{(aq)} + NaClO_{(aq)} + H_2O_{(l)}$

166. $NH_3 + 3Cl_{2(excess)} \longrightarrow \underset{\text{Nitrogen trichloride (explosive)}}{NCl_3} + 3HCl$

167. $Cl_2 + H_2O \longrightarrow \underset{\text{Hydrochloric acid}}{HCl} + \underset{\text{Hypochlorous acid}}{HOCl} \longrightarrow 2HCl + \underset{\text{Nascent oxygen}}{[O]}$

168. $2Cl_2 + 2Ca(OH)_{2(dil.)} \xrightarrow{\text{Cold}} CaCl_2 + Ca(OCl)_2 + 2H_2O$

169. (i) Phosgene

(ii) Calcium hypochlorite

170. The bleaching action of Cl_2 is due to oxidation of coloured substances to colourless substances by nascent oxygen. Since, the bleaching action of Cl_2 is due to oxidation and that of SO_2 is due to reduction, therefore, bleaching effect of Cl_2 is permanent while that of SO_2 is temporary.

171. $6NaOH + 3Cl_2 \rightarrow 5NaCl + NaClO_3 + H_2O$ (hot and conc.)

172. Chlorine water on standing loses its yellow colour due to the formation of HCl and HClO.

173. *Refer of answer 171.*

174. $I_2 + 6H_2O + 5Cl_2 \rightarrow \underset{\text{Iodic acid}}{2HIO_3} + 10HCl$

175. $MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$

$Cl_2 + H_2S \longrightarrow 2HCl + S$

176. $MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$

177. *Refer to answer 176.*

178. *Refer to answer 176.*

179. Its reaction with iron produces H_2.

$Fe + 2HCl \rightarrow FeCl_2 + H_2$

Liberation of hydrogen prevents the formation of ferric chloride.

180. Fluorine forms only one oxoacid HOF. Because for the formation of other oxoacids *d*-orbitals are required for the multiple $p\pi$ - $d\pi$ bonding between extra oxygen atoms and fluorine.

High electronegativity and small size of fluorine also favours the formation of only one oxoacid.

181. As the number of oxygen bonded to the central atom increases, the oxidation number of the central atom increases causing a weakening of

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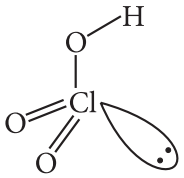
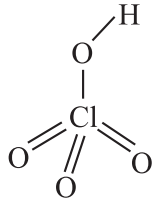
the O—H bond strength and an increase in the acidity. Hence, $HClO_4$ is stronger acid than HClO.

182. $HClO_3$:



183.



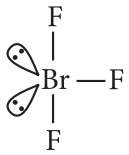
$HClO_4$ (Perchloric acid)

184. *Refer to answer 180.*

185. As the stability of the oxoanion increases, its tendency to decompose to give O_2 decreases and hence its oxidising power decreases. Since, the stability of the oxoanion decreases in the order : $ClO_4^- > ClO_3^- > ClO_2^- > ClO^-$ therefore oxidising power of their oxoacids increases in the reverse order :

$HClO_4 < HClO_3 < HClO_2 < HClO$

186.

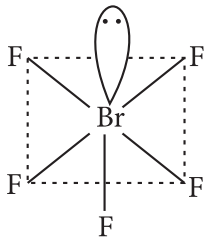


BrF_3

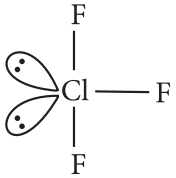
187. Because of smaller size of F and bigger size of Cl, chlorine can accomodate three fluorine atoms around it but fluorine does not. Also Cl has vacant *d*-orbitals and hence can show +3 oxidation state, but F has no *d*-orbitals.

$$\underset{\text{Excess}}{Cl_2 + 3F_2} \xrightarrow{250\ ^\circ C} 2ClF_3$$

188. BrF_5 :



189. ClF_3 :



190. Interhalogen compounds are more reactive than halogens (except flourine) because $X — X'$ bond (I—Cl bond) in interhalogens is weaker than $X—X$ bond (I—I bond) in halogens except F — F bond. In other words, I —Cl bond is weaker than I — I bond. Thus, ICl is more reactive than I_2.

191. $3F_2$ (excess) + $Cl_2 \xrightarrow{300^\circ C} 2ClF_3$

192. Interhalogen compounds are more reactive than halogen because the $X' — X$ bond in interhalogens is weaker than $X — X$ bond in halogens. But in case of flourine, the F—F bond is weaker. This is because of the small size of fluorine atoms, the F—F bond distance is very small due to which there is appreciable inter-electronic repulsion. This repulsion weakens the bond between two fluorine atoms. Hence, F_2 is more reactive than ClF_3 but ClF_3 is more reactive than chlorine.

193. *Refer to answer 189.*

ClF_3 is T-shaped molecule.

Angle F — Cl — F : Less than 90°

194. Oxidation state of Br in $BrCl_5$ is +5, whereas in case of $BrCl_3$ is +3. As Br is more stable in +3 oxidation state than +5, due to inert pair effect. Therefore, it is unstable and readily reduces from +5 to +3 oxidation state.

195. Fluorine does not have *d*-orbitals and it cannot show higher oxidation state. Therefore, it does not play the role of a central atom in interhalogen compounds.

196. $Br_2 + 5F_2$ (excess) $\longrightarrow 2BrF_5$

197. Interhalogen compounds are formulated as XX', XX'_3, XX'_5 and XX'_7 where, X is halogen of larger size and X' of smaller size.

The interhalogen compounds can be prepared by direct combination or by the action of halogen on lower interhalogen compounds. The product formed depends upon some specific conditions, for example,

$$Cl_2 + F_2 \xrightarrow{437\ K} 2ClF$$

(Equal volume)

$$\underset{\text{(Excess)}}{Cl_2 + 3F_2} \xrightarrow{573\ K} 2ClF_3$$

$$I_2 + Cl_2 \rightarrow 2ICl$$

(equimolar)

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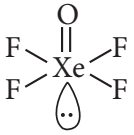
$I_2 + 3Cl_2 \rightarrow 2ICl_3$
Excess

$Br_2 + 3F_2 \rightarrow 2BrF_3$
(diluted with water)

$Br_2 + 5F_2 \rightarrow 2BrF_5$
(excess)

198. $XeOF_4$:



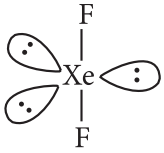
199. $XeF_6 + KF \longrightarrow K^+ [XeF_7]^-$

200. Extremely small size and fully filled outer orbital makes helium very stable and resistant to chemical reactions and hence, it does not form compounds unlike bigger atoms of other elements of noble gas family.

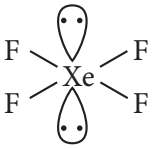
201. As the size of noble gases increases, van der Waals' forces of attraction increase accordingly and hence, the boiling point increases from He to Rn.

202. $2XeF_{2(s)} + 2H_2O_{(l)} \rightarrow 2Xe_{(g)} + 4HF_{(aq)} + O_{2(g)}$

203. XeF_2 :



204.



205. *Refer to answer 202.*

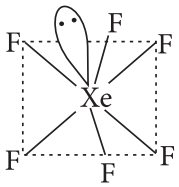
206. $XeF_6 + NaF \longrightarrow Na^+[XeF_7]^-$

207. $XeF_4 + SbF_5 \longrightarrow [XeF_3]^+ [SbF_6]^-$

208. $XeF_4 + O_2F_2 \xrightarrow{143\ K} XeF_6 + O_2$

209. $XeF_6 + 2H_2O \longrightarrow XeO_2F_2 + 4HF$

210. XeF_6 : (Xenon hexafluoride)



211. $XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$

212. *Refer to answer 207.*

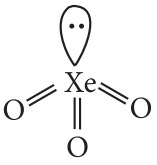
213. Helium is used in diving apparatus as diluent for oxygen because of its low solubility (as compared to N2) in blood, a mixture of oxygen and helium is used in diving apparatus used by deep sea divers.

214. XeO_3 :

Hybridisation = sp^3

Geometry = Tetrahedral

Shape = Pyramidal



215. *Refer to answer 200.*

216. The central atom Br has seven electrons. Four of these electrons form two double bonds or coordinate bonds with two oxygen atoms while the fifth electron forms a single bond with O^- ion. The remaining two electrons form one lone pair. Hence, in all there are three bond pairs and one lone pair around Br atom in BrO_3^-. Therefore, according to VSEPR theory, BrO_3^- should be pyramidal.
Here, BrO_3^- has $26(7 + 3 \times 6 + 1 = 26)$ valence electrons. A noble gas species having 26 valence electrons is $XeO_3(8 + 3 \times 6 = 26)$. Thus, like BrO_3^-, XeO_3 is also pyramidal.

217. $6XeF_4 + 12H_2O \rightarrow 2XeO_3 + 24HF + 3O_2 + 4Xe$

218. Neil Bartlett first prepared a red compound which is formulated as $O_2^+PtF_6^-$. He then realised that the first ionisation enthalpy of molecular oxygen (1175 kJ/mol) is almost identical with Xe (1170 kJ/mol). He made efforts to prepare same type of compound with Xe and was successful in preparing another red compound $Xe^+PtF_6^-$.

219. $XeF_6 + H_2O \longrightarrow XeOF_4 + 2HF$

220. *Refer to answer 203.*
F — Xe — F : 180°

221. Helium has completely filled ns^2 electronic configurations in its valence shell. Due to its small size and high IE, helium is chemically unreactive. Thus, it forms no real chemical compound.

222. Noble gases being monoatomic gases are held together by weak London dispersion forces, therefore they have low boiling points.

223. $XeF_2 + PF_5 \longrightarrow [XeF]^+[PF_6]^-$

224. *Refer to answer 217.*

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It is a disproportionation reaction in which XeF_4 (+4) undergoes disproportionation giving xenon (0) and xenon trioxide (+6).

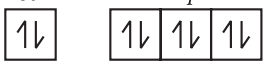
225. (i) Helium is used for filling balloons for meteorological observations because it is non-inflammable.

(ii) *Refer to answer 223.*

226. $XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

$XeF_6 + H_2O \longrightarrow XeOF_4 + 2HF$

227. (i) $5s^2$ $5p^6$



All the orbital of Xe have paired electrons. The promotion of one, two or three electrons from 5*p*-filled orbitals to the 5*d*-vacant orbitals will give rise to two, four and six-half filled orbitals. Therefore, xenon can combine with even number of fluorine atoms, not odd. Thus, it cannot form XeF_3 and XeF_5.

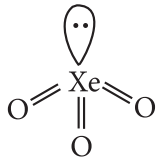
(ii) Except radon which is radioactive, Xe has least ionisation energy among noble gases and hence it readily forms chemical compounds particularly with oxygen and fluorine.

228. XeO_3 can be obtained by hydrolysis of XeF_4 and XeF_6.

$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24\,HF + 3O_2$.

$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

Structure :



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CHAPTER
8

The *d*- and *f*- Block Elements

8.1 Position in the Periodic Table

8.2 Electronic Configurations of the *d*-Block Elements

8.3 General Properties of the Transition Elements (*d*-Block)

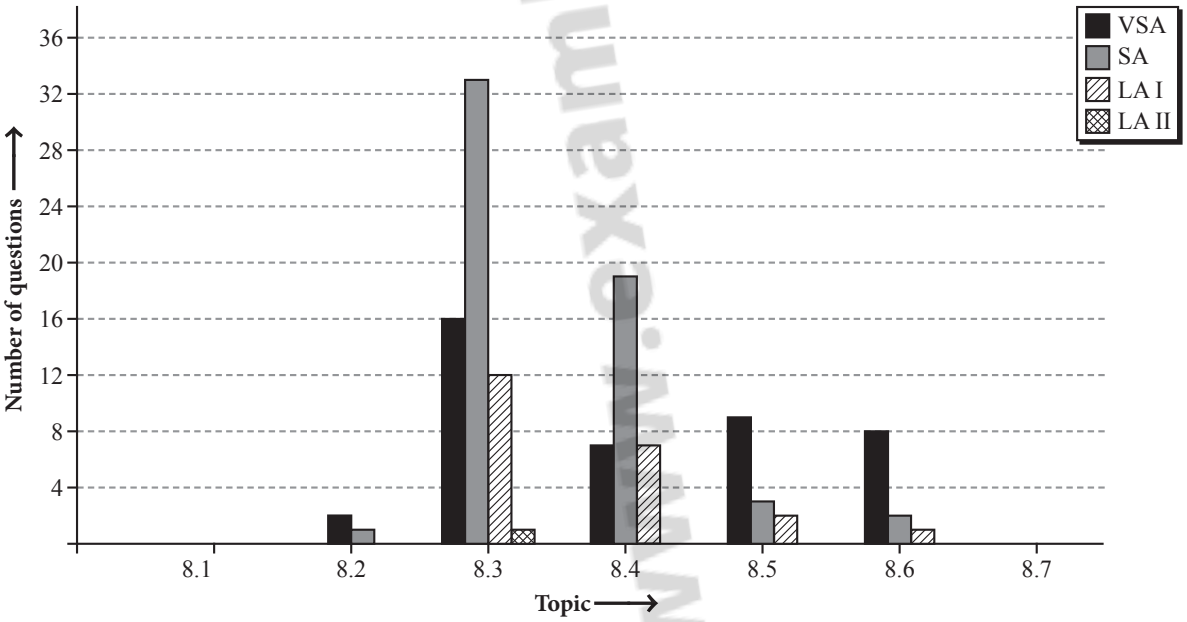
8.4 Some Important Compounds of Transition Elements

8.5 The Lanthanoids

8.6 The Actinoids

8.7 Some Applications of *d*-and *f*-Block Elements

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *General Properties of the Transition Elements* (*d-Block*).
- Maximum VSA type questions were asked from *General Properties of the Transition Elements* (*d-Block*).
- Maximum SA and LA I type questions were asked from *General Properties of the Transition Elements* (*d-Block*).

QUICK RECAP

TRANSITION ELEMENTS (*d*-BLOCK ELEMENTS)

- Elements in which the last electron enters any one of the five *d*-orbitals of their respective penultimate shell are known as *transition elements* or *d-block elements*.
- Their general electronic configuration is $(n-1)d^{1-10}ns^{0-2}$.

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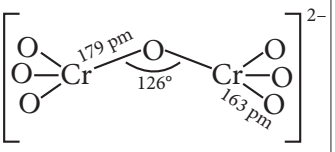
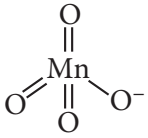
Transition series : *d*-block consists of four transition series,
1st Transition series or 3*d* series $_{21}Sc - _{30}Zn$
2nd Transition series or 4*d* series $_{39}Y - _{48}Cd$
3rd Transition series or 5*d* series $_{57}La$, $_{72}Hf - _{80}Hg$
4th Transition series or 6*d* series $_{89}Ac$, $_{104}Rf - _{112}Cn$

General characteristics :

Melting and boiling points	High due to strong metallic bonding
Enthalpies of atomisation	High due to strong interatomic interactions
Ionisation enthalpies	Generally increases from left to right in a series
Oxidation states	Variable due to participation of *ns* and $(n-1)d$ electrons
Atomic radii	Decrease from left to right but become constant when pairing of electrons takes place
Complex formation	Form complexes due to high nuclear charge and small size and availability of empty *d*-orbitals to accept lone pair of electrons donated by ligands.
Coloured compounds	Form coloured compounds due to *d*-*d* transitions
Magnetic properties	Transition metal ions and their compounds are paramagnetic due to presence of unpaired electrons in the $(n-1)d$-orbitals and it is calculated by using the formula, $\mu = \sqrt{n(n+2)}$ where, *n* is the no. of unpaired electrons.
Catalytic behaviour	Due to variable oxidation states and ability to form complexes
Interstitial compounds	Due to empty spaces in their lattices, small atoms can be easily accommodated
Alloy formation	Due to similar atomic sizes

Some important compounds :

Compounds	Preparation	Properties	Uses
Potassium dichromate ($K_2Cr_2O_7$)  	From sodium dichromate (obtained from chromite ore) $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$ $2Na_2CrO_4 + 2H^+ \longrightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$ $Na_2Cr_2O_7 + 2KCl \longrightarrow K_2Cr_2O_7 + 2NaCl$	Orange red crystalline solid, oxidising agent having melting point 398°C. **Oxidising agent in acidic medium** : $Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$ **Oxidises :** I^- to I_2, H_2S to S, Sn^{2+} to Sn^{4+} and Fe^{2+} to Fe^{3+}	In dyeing, photography and leather industry.
Potassium permanganate ($KMnO_4$)  	From potassium manganate (obtained from pyrolusite) $2MnO_2 + 4KOH + O_2 \longrightarrow 2K_2MnO_4 + 2H_2O$ $2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$	Deep purple crystalline solid, oxidising agent, having melting point 240°C. **Oxidising agent in acidic medium :** $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$ **Oxidises :** I^- to I_2, Fe^{2+} to Fe^{3+}, $C_2O_4^{2-}$ to CO_2, S^{2-} to S, SO_3^{2-} to SO_4^{2-}, NO_2^- to NO_3^- **Oxidising agent in faintly alkaline or neutral medium :** $MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$ **Oxidises :** I^- to IO_3^-, $S_2O_3^{2-}$ to SO_4^{2-}, Mn^{2+} to MnO_2	As a disinfectant, germicide, and Baeyer's reagent (alkaline $KMnO_4$).

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- Volumetric titrations involving $KMnO_4$ are carried out only in the presence of dil. H_2SO_4 but not in the presence of HCl or HNO_3 because HCl is oxidised to Cl_2 and HNO_3 is itself a strong oxidising agent and partly oxidises the reducing agent. H_2SO_4 does not give any oxygen of its own to oxidise the reducing agent.

INNER TRANSITION ELEMENTS (*f*-BLOCK ELEMENTS)

- **Lanthanoids :** Last electron enters one of the 4*f*-orbitals. Cerium (at. no. 58) to lutetium (at. no. 71).
- **Actinoids :** Last electron enters one of the 5*f*-orbitals. Thorium (at. no. 90) to lawrencium (at. no. 103).

General electronic configuration : $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

General characteristics of lanthanoids :

Atomic and ionic radii	Decrease on going from La to Lu.
Oxidation states	Most common oxidation state of lanthanoids is +3. Some elements exhibit +2 and +4 oxidation states due to extra stability of empty, half-filled or fully-filled *f*-subshell, *e.g.,* Ce^{4+} acts as an oxidising agent and gets reduced to Ce^{3+}, Eu^{2+}, Yb^{2+} act as strong reducing agents and get oxidised to Eu^{3+} and Yb^{3+}.
Action of air	All the lanthanoids are silvery white soft metals and tarnish readily in moist air. They burn in oxygen of air and form oxides (Ln_2O_3 type).
Coloured ions	They form coloured trivalent metal ions due to *f-f* transitions of unpaired electrons. La^{3+} and Lu^{3+} are colourless ions due to empty ($4f^0$) or fully ($4f^{14}$) orbitals.
Magnetic properties	La^{3+}, Lu^{3+} are diamagnetic while trivalent ions of the rest of lanthanoids are paramagnetic.
Reducing agents	They readily lose electrons and are good reducing agents.
Electropositive character	Highly electropositive because of low ionisation energies.
Alloy formation	They form alloys easily with other metals especially iron.
Tendency to form complexes	Lanthanoids do not have much tendency to form complexes due to low charge density because of their large size. The tendency to form complexes and their stability increases with increasing atomic number.

Lanthanoid contraction : In lanthanoid series, with increasing atomic number, there is progressive decrease in atomic/ionic radii (M^{3+} ions) from La^{3+} to Lu^{3+}.

- **Reason :** Due to addition of new electrons into *f*-subshell and imperfect shielding of one electron by another in the *f*-orbitals, there is greater effect of increased nuclear charge than screening effect hence contraction in size occurs.
- **Consequences :** Their separation is difficult, they have small differences in properties and 4*d* and 5*d* transition series have almost same atomic radii (Zr and Hf have similar properties due to same size).

Uses of lanthanoids : Used in making mischmetal, an alloy of a lanthanoid metal (~ 95%) with iron (~ 5%) and traces of S, C, Ca and Al. It is used to make tracer bullets, shells and lighter flints.

General characteristics of actinoids :

Ionic radii	Like lanthanoids, ionic radii decrease across the series. Actinoid contraction is greater due to poor sheilding effect of the 5*f*-electrons. Further, 5*f*-orbitals extend in space beyond 6*s* and 6*p*-orbitals whereas 4*f*-orbitals are burried deep inside the atom.

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Oxidation states	Like lanthanoids, most common oxidation state is +3. They also show oxidation state of +4, +5, +6 and +7, *e.g.,* in Th, Pa, U and Np respectively. They show a large number of oxidation states because of very small energy gap between 5*f*, 6*d* and 7*s* subshells.
Action of air, alkalies and acids	Like lanthanoids they are also silvery white metals, tarnish rapidly in air forming oxide coating and are not attacked by alkalies and are less reactive towards acids.
Coloured ions	Coloured due to *f-f* transition except $Ac^{3+}(5f^{0})$, $Cm^{3+}(5f^{7})$ and $Th^{4+}(5f^{0})$ which are colourless.
Magnetic properties	They are strongly paramagnetic.
Density	All actinoids except thorium and americium have high densities.
Melting and boiling points	High melting and boiling points however there is no regular trend with rise in atomic number.
Ionisation energy	They have low ionisation energies than lanthanoides.
Reducing agents	All actinoids are strong reducing agents.
Electropositive character	Highly electropositive metals.

Uses of actinoids : Th is used in atomic reactor and treatment of cancer. U and Pu are used as fuel in nuclear reactor.

Differences between lanthanoids and actinoids :

Lanthanoids	Actinoids
They show +2 and +4 oxidation states in few cases besides +3.	They show higher oxidation states of +4, +5, +6 and +7 besides +3.
Except promethium, they are non-radioactive.	All actinoids are radioactive.
They do not form oxo-cations.	They form oxo-cations like UO_2^{2+}, PuO_2^{2+}, UO_2^{+}, etc.
The compounds of lanthanoids are less basic.	Actinoid compounds are more basic.
They have less tendency of complex formation.	They have greater tendency of complex formation.

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Previous Years' CBSE Board Questions

8.2 Electronic Configurations of the *d*-Block Elements

VSA (1 mark)

1. Account for the following :
Zn, Cd, Hg are considered as *d*-block elements but not as transition elements.
(1/5, 2020)

2. Account for the following :
Zn is not considered as a transition element.
(1/5, AI 2014)

SA (2 marks)

3. What are the transition elements? Write two characteristics of the transition elements.
(Delhi 2015)

8.3 General Properties of the Transition Elements (*d*-Block)

VSA (1 mark)

Read the given passage and answer the questions number (4 to 8) that follow :

The *d*-block of the periodic table contains the elements of groups 3-12 and are known as transition elements. In general, the electronic configuration of these elements is $(n - 1)d^{1-10}$ ns^{1-2}. The *d*-orbitals of the penultimate energy level in their atoms receive electrons giving rise to three rows of the transition metals *i.e.,* 3*d*, 4*d* and 5*d* series. However, Zn, Cd and Hg are not regarded as transition elements. Transition elements exhibit certain characteristic properties like variable oxidation states, complex formation, formation of coloured ions and alloys, catalytic activity, etc. Transition metals are hard (except Zn, Cd and Hg) and have a high melting point.
(2020)

4. Why are Zn, Cd and Hg non-transition elements?

5. Which transition metal of 3*d* series does not show variable oxidation states?

6. Why do transition metals and their compounds show catalytic activity?

7. Why are melting points of transition metals high?

8. Why is Cu^{2+} ion coloured while Zn^{2+} ion is colourless in aqueous solution?

9. Out of zinc and tin, whose coating is better to protect iron objects? *(One word, 2020)*

10. Out of the following transition elements, the maximum number of oxidation states are shown by
(a) Sc ($Z = 21$) (b) Cr ($Z = 24$)
(c) Mn ($Z = 25$) (d) Fe ($Z = 26$).
(2020)

11. **Assertion (A) :** Transition metals have high melting point.
Reason (R) : Transition metals have completely filled *d*-orbitals.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

12. Account for the following :
Copper(I) compounds are white whereas copper(II) compounds are coloured.
(1/5, 2020)

13. Write the formula of an oxoanion of chromium (Cr) in which it shows the oxidation state equal to its group number. *(Delhi 2017)*

14. Write the formula of an oxoanion of manganese (Mn) in which it shows the oxidation state equal to its group number.
(Delhi 2017)

15. How would you account for the following :
Transition metals form coloured compounds?
(1/3, Delhi 2015)

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16. Zn^{2+} salts are white while Cu^{2+} salts are coloured. Why? *(AI 2015)*

17. Why do transition elements show variable oxidation states? *(Delhi 2014C)*

18. Assign reason for the following :
Copper (I) ion is not known in aqueous solution. *(1/2, Delhi 2011)*

19. Explain giving reasons :
Transition metals and their compounds generally exhibit a paramagnetic behaviour. *(1/2, AI 2011)*

SA (2 marks)

20. Calculate the spin-only moment of Co^{2+} (Z = 27) by writing the electronic configuration of Co and Co^{2+}. *(2/5, 2020)*

21. Give reason and select one atom/ion which will exhibit asked property :
(i) Sc^{3+} or Cr^{3+} (exhibit diamagnetic behaviour)
(ii) Cr or Cu (high melting and boiling point) *(2/5, 2020)*

22. Give reasons for the following :
(i) Transition metals form alloys.
(ii) Mn_2O_3 is basic whereas Mn_2O_7 is acidic. *(2/3, Delhi 2019)*

23. Use the data to answer the following and also justify giving reasons :

	Cr	Mn	Fe	Co
$E°_{M^{2+}/M}$	–0.91	–1.18	–0.44	–0.28
$E°_{M^{3+}/M^{2+}}$	–0.41	+1.57	+0.77	+1.97

(i) Which is a stronger reducing agent in aqueous medium, Cr^{2+} or Fe^{2+} and why?
(ii) Which is the most stable ion in +2 oxidation state and why? *(AI 2019)*

24. (i) How is the variability in oxidation states of transition metals different from that of the *p*-block elements?
(ii) Out of Cu^{+} and Cu^{2+}, which ion is unstable in aqueous solution and why? *(2/5, Delhi 2017)*

25. Account for the following :
(i) Transition metals show variable oxidation states.
(ii) Zn, Cd and Hg are soft metals. *(2/5, AI 2017)*

26. Give reasons :
(i) Mn shows the highest oxidation state of +7 with oxygen but with fluorine it shows the highest oxidation state of +4.
(ii) Transition metals show variable oxidation states. *(2/3, Delhi 2016)*

27. Account for the following :
(i) Transition metals show variable oxidation states.
(ii) Cu^{+} ion is unstable in aqueous solution. *(2/3, AI 2015)*

28. Why do transition elements show variable oxidation states? In 3*d* series (Sc to Zn), which element shows the maximum number of oxidation states and why? *(Foreign 2015)*

29. How would you account for the following :
(i) Highest fluoride of Mn is MnF_4 whereas the highest oxide is Mn_2O_7.
(ii) Transition metals and their compounds show catalytic properties. *(2/3, Foreign 2015)*

30. What is meant by 'disproportionation'? Give an example of a disproportionation reaction in aqueous solution. *(Delhi 2015C, AI 2015C)*

31. Account for the following :
(i) Mn^{2+} is more stable than Fe^{2+} towards oxidation to +3 state.
(ii) The enthalpy of atomization is lowest for Zn in 3*d* series of the transition elements. *(Delhi 2014)*

32. (i) Write two characteristic of the transition elements.
(ii) Which of the 3*d*-block elements may not be regarded as the transition elements and why? *(2/3, Foreign 2014)*

33. Assign a reason for each of the following observations :
(i) The transition metals (with the exception of Zn, Cd and Hg) are hard and have high melting and boiling points.
(ii) The ionisation enthalpies (first and second) in the first series of the transition elements are found to vary irregularly. *(Delhi 2014C)*

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34. Assign reasons for the following :
(i) Copper(I) ion is not known to exist in aqueous solutions.
(ii) Both O_2 and F_2 stabilize high oxidation states of transition metals but the ability of oxygen to do so exceeds that of fluorine. *(AI 2014C)*

35. Give reasons for the following :
(i) Transition metals exhibit a wide range oxidation states.
(ii) Cobalt (II) is very stable in aqueous solutions but gets easily oxidised in the presence of strong ligands.
(2/3, AI 2014C)

36. Assign reasons for the following :
(i) Cu(I) ion is not known to exist in aqueous solutions.
(ii) Transition metals are much harder than the alkali metals. *(2/3, AI 2014C)*

37. Assign reasons for the following :
(i) Transition metals and many of their compounds act as good catalysts.
(ii) Transition metals generally form coloured compounds. *(AI 2014C)*

38. (i) Which metal in the first transition series (3*d*-series) exhibits +1 oxidation state most frequently and why?
(ii) Which of following cations are coloured in aqueous solutions and why?
Sc^{3+}, V^{3+}, Ti^{4+}, Mn^{2+}
(At. Nos. Sc = 21, V = 23, Ti = 22, Mn = 25)
(Delhi 2013)

39. How would you account for the following?
(i) Transition metals exhibit variable oxidation states.
(ii) Transition metals and their compounds act as catalyst. *(2/3, Delhi 2013)*

40. How would you account for the following?
(i) Many of the transition elements are known to form interstitial compounds.
(ii) The metallic radii of the third (5*d*) series of transition metals are virtually the same as those of the corresponding group member of the second (4*d*) series.
(2/3, Delhi 2012)

41. How would you account for the following?
(i) With the same *d*-orbital configuration (d^4) Cr^{2+} is a reducing agent while Mn^{3+} is an oxidising agent.
(ii) Most of the transition metal ions exhibit characteristic colours in aqueous solutions. *(2/3, Delhi 2012)*

42. How would you account for the following :
(i) The $E^{\circ}_{M^{2+}/M}$ for copper is positive (+0.34 V). Copper is the only metal in the first series of transition elements showing this behaviour.
(ii) The metallic radii of the third (5*d*) series of transition metals are nearly the same as those of the corresponding members of the second (4*d*) series. *(2/3, AI 2012)*

43. Explain the following observations :
(i) Many of the transition elements are known to form interstitial compounds.
(ii) There is a general increase in density from titanium ($Z = 22$) to copper ($Z = 29$).
(2/3, AI 2012)

44. Explain the following observations.
(i) With the same *d*-orbital configuration (d^4), Cr^{2+} is a reducing agent while Mn^{3+} is an oxidising agent.
(ii) There is hardly any increase in atomic size with increasing atomic numbers in a series of transition metals.
(2/3, AI 2012)

45. Explain the following :
(i) The enthalpies of atomization of transition metals are quite high.
(ii) The transition metals and many of their compounds act as good catalysts.
(Delhi 2012C)

46. Give reason :
(i) Sc (21) is a transition element but Ca (20) is not.
(ii) The Fe^{2+} is much more easily oxidised to Fe^{3+} than Mn^{2+} to Mn^{3+}.
(2/3, AI 2012C)

47. How is the variability in oxidation states of transition elements different from that of non-transition elements? Illustrate with examples. *(2/5, AI 2012C)*

48. Account for the following :
(i) Mn^{2+} compounds are more stable than Fe^{2+} compounds towards oxidation to their +3 state.

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(ii) Cr^{2+} is reducing and Mn^{3+} oxidizing when both have d^4 configuration. *(2/5, AI 2012C)*

49. Assign reasons for each of the following :
(i) Transition metals generally form coloured compounds.
(ii) Manganese exhibits the highest oxidation state of +7 among the 3rd series of transition elements. *(Delhi 2011)*

50. How would you account for the following :
(i) Cr^{2+} is reducing in nature while with the same *d*-orbital configuration (d^4) Mn^{3+} is an oxidising agent.
(ii) In a transition series of metals, the metal which exhibits the greatest number of oxidation states occurs in the middle of the series. *(AI 2011)*

51. State reasons for the following :
(i) Cu (I) ion is not stable in an aqueous solution.
(ii) Unlike Cr^{3+}, Mn^{2+}, Fe^{3+} and the subsequent other V^{2+} ions of the 3*d* series of elements, the 4*d* and the 5*d* series metals generally do not form stable cationic species. *(AI 2011)*

52. Explain giving a suitable reason for each of the following :
(i) Transition metals and their compounds are generally found to be good catalysts.
(ii) Metal-metal bonding is more frequent for the 4*d* and the 5*d* series of transition metals than that for the 3*d* series. *(AI 2011)*

LA I (3 marks)

53. Following ions are given :
Cr^{2+}, Cu^{2+}, Cu^{+}, Fe^{2+}, Fe^{3+}, Mn^{3+}
Identify the ion which is
(i) a strong reducing agent.
(ii) unstable in aqueous solution.
(iii) a strong oxidising agent.
Give suitable reason in each. *(2020)*

54. Give reasons :
(i) E° value for Mn^{3+}/Mn^{2+} couple is much more positive than that for Fe^{3+}/Fe^{2+}.
(ii) Iron has higher enthalpy of atomization than that of copper.
(iii) Sc^{3+} is colourless in aqueous solution whereas Ti^{3+} is coloured. *(2018)*

55. Account the following :
(i) Transition metals form large number of complex compounds.
(ii) The lowest oxide of transition metal is basic whereas the highest oxide is amphoteric or acidic.
(iii) E° value for the Mn^{3+}/Mn^{2+} couple is highly positive (+1.57 V) as compared to Cr^{3+}/Cr^{2+}. *(3/5, Delhi 2017)*

56. Following are the transition metal ions of 3*d* series :
Ti^{4+}, V^{2+}, Mn^{3+}, Cr^{3+}
(Atomic numbers : Ti = 22, V = 23, Mn = 25, Cr = 24)
Answer the following :
(i) Which ion is most stable in aqueous solution and why?
(ii) Which ion is strong oxidising agent and why?
(iii) Which ion is colourless and why? *(3/5, AI 2017)*

57. Account for the following :
(i) Mn shows the highest oxidation state of +7 with oxygen but with fluorine, it shows the highest oxidation state of +4.
(ii) Cr^{2+} is a strong reducing agent.
(iii) Cu^{2+} salts are coloured, while Zn^{2+} salts are white. *(3/5, AI 2016)*

58.

$E^{\circ}_{(M^{2+}/M)}$	Cr	Mn	Fe	Co	Ni	Cu
	–0.91	–1.18	–0.44	–0.28	–0.25	+0.34

From the given data of E° values, answer the following questions :
(i) Why is $E^{\circ}_{(Cu^{2+}/Cu)}$ value exceptionally positive?
(ii) Why is $E^{\circ}_{(Mn^{2+}/Mn)}$ value highly negative as compared to other elements?
(iii) Which is a stronger reducing agent Cr^{2+} or Fe^{2+}? Give reason. *(3/5, AI 2015)*

59. Assign suitable reasons for the following :
(i) The Mn^{2+} compounds are more stable than Fe^{2+} towards oxidation to their +3 state.
(ii) In the 3*d* series from Sc (Z = 21) to Zn (Z = 30), the enthalpy of atomization of Zn is the lowest.

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(iii) Sc^{3+} is colourless in aqueous solution whereas Ti^{3+} is coloured. *(Foreign 2014)*

60. Give reasons for the following :
(i) Mn^{3+} is a good oxidising agent.
(ii) $E°_{M^{2+}/M}$ values are not regular for first row transition metals (3*d*-series).
(iii) Although 'F' is more electronegative than 'O', the highest Mn fluoride is MnF_4, whereas the highest oxide is Mn_2O_7. *(3/5, AI 2013)*

61. Give reasons :
(i) *d*-block elements exhibit more oxidation states than *f*-block elements.
(ii) The enthalpies of atomization of the transition metals are high.
(iii) The variation in oxidation states of transition metals is of different type from that of the non-transition metals. *(3/5, AI 2013C)*

62. Explain the following :
(i) Copper (I) ion is not stable in an aqueous solution.
(ii) With same (d^4) configuration Cr (II) is reducing whereas Mn (III) is oxidising.
(iii) Transition metals in general act as good catalysts. *(3/5, Foreign 2011)*

63. How would you account for the following :
(i) Metal-metal bonding is more extensive in the 4*d* and 5*d* series of transition elements than the 3*d* series.
(ii) Mn (III) undergoes disproportionation reaction easily.
(iii) Co (II) is easily oxidised in the presence of strong ligands. *(3/5, Foreign 2011)*

64. Account for the following :
(i) The transition metals and their compounds act as good catalysts.
(ii) The lowest oxide of transition metal is basic, the highest is amphoteric/acidic.
(iii) A transition metal exhibits higher oxidation states in oxides and fluorides. *(3/5, AI 2011C)*

LA II (5 marks)

65. The elements of 3*d* transition series are given as
Sc Ti V Cr Mn Fe Co Ni Cu Zn
Answer the following :
(i) Write the element which shows maximum number of oxidation states. Given reason.
(ii) Which element has the highest melting point?
(iii) Which element shows only +3 oxidation state?
(iv) Which element is a strong oxidising agent in +3 oxidation state and why? *(AI 2016)*

8.4 Some Important Compounds of Transition Elements

VSA (1 mark)

66. Account for the following :
Chromates change their colour when kept in an acidic solution. *(1/5, 2020)*

67. Orange colour of $Cr_2O_7^{2-}$ ion changes to yellow when treated with an alkali. Why? *(1/5, Delhi 2017)*

68. Complete the following equation :
$2MnO_4^- + 6H^+ + 5NO_2^- \longrightarrow$ *(1/2, Delhi 2015)*

69. Complete the following equation :
$3MnO_4^{2-} + 4H^+ \longrightarrow$ *(1/3, Foreign 2015)*

70. Complete the following equation :
$MnO_4^- + 8H^+ + 5e^- \longrightarrow$ *(1/5, Delhi 2014)*

71. Complete the following chemical equation:
$SO_2 + MnO_4^- + H_2O \longrightarrow$ *(1/2, AI 2014C)*

72. Give reason :
Orange solution of potassium dichromate turns yellow on adding sodium hydroxide to it. *(1/5, AI 2013C)*

SA (2 marks)

73. Write the balanced chemical equations involved in the preparation of $KMnO_4$ from pyrolusite ore (MnO_2). *(2020)*

74. Write the balanced ionic equations showing the oxidising action of acidified dichromate ($Cr_2O_7^{2-}$) solutions with (i) iron (II) ion and (ii) tin (II) ion. *(2020)*

75. When MnO_2 is fused with KOH in the presence of KNO_3 as an oxidizing agent,

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it gives a dark green compound (*A*). Compound (*A*) disproportionates in acidic solution to give purple compound (*B*). An alkaline solution of compound (*B*) oxidises KI to compound (*C*) whereas an acidified solution of compound (*B*) oxidises KI to (*D*). Identify (*A*), (*B*), (*C*) and (*D*). *(Delhi 2019)*

76. Complete and balance the following chemical equations :
(i) $Fe^{2+} + MnO_4^- + H^+ \longrightarrow$
(ii) $MnO_4^- + H_2O + I^- \longrightarrow$ *(2018)*

77. Complete the following equations :
(i) $2MnO_4^- + 16H^+ + 5S^{2-} \longrightarrow$
(ii) $KMnO_4 \xrightarrow{Heat}$ *(2/5, AI 2017)*

78. When chromite ore $FeCr_2O_4$ is fused with NaOH in presence of air, a yellow coloured compound (*A*) is obtained which on acidification with dilute sulphuric acid gives a compound (*B*). Compound (*B*) on reaction with KCl forms an orange coloured crystalline compound (*C*).
(i) Write the formulae of the compounds (*A*), (*B*) and (*C*).
(ii) Write one use of compound (*C*).
(Delhi 2016)

79. Complete the following chemical equations :
(i) $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow$
(ii) $Cr_2O_7^{2-} + 3Sn^{2+} + 14H^+ \rightarrow$
(Delhi 2016)

80. Complete the following equations.
(i) $2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta}$
(ii) $Cr_2O_7^{2-} + 14H^+ + 6I^- \longrightarrow$
(2/5, AI 2016)

81. Describe the preparation of potassium permanganate. How does the acidified permanganate solution react with oxalic acid? Write the ionic equations for the reaction. *(AI 2015C)*

82. Describe the oxidising action of potassium dichromate and write the ionic equations for its reaction with (i) an iodide (ii) H_2S.
(AI 2015C)

83. How do you prepare :
(i) K_2MnO_4 from MnO_2?
(ii) $Na_2Cr_2O_7$ from Na_2CrO_4?
(2/5, Delhi 2014)

84. Complete the following equations :
(i) $Cr_2O_7^{2-} + 2OH^- \longrightarrow$
(ii) $MnO_4^- + 4H^+ + 3e^- \longrightarrow$ *(AI 2014)*

85. Complete the following equations :
(i) $2MnO_4^- + 5S^{2-} + 16H^+ \rightarrow$
(ii) $Cr_2O_7^{2-} + 2OH^- \rightarrow$ *(Foreign 2014)*

86. Complete the following equations :
(i) $2CrO_4^{2-} + 2H^+ \longrightarrow$
(ii) $KMnO_4 \xrightarrow{Heat}$ *(AI 2013)*

87. Complete the following chemical equations :
(i) $Cr_2O_7^{2-} + H^+ + I^- \longrightarrow$
(ii) $MnO_4^- + NO_2^- + H^+ \longrightarrow$ *(Delhi 2012)*

88. Complete the following chemical equations :
(i) $MnO_{4(aq)}^- + S_2O_{3(aq)}^{2-} + H_2O_{(l)} \rightarrow$
(ii) $Cr_2O_{7(aq)}^{2-} + Fe_{(aq)}^{2+} + H_{(aq)}^+ \rightarrow$ *(AI 2011)*

89. Complete the following reactions in an aqueous medium :
(i) $MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow$
(ii) $Cr_2O_7^{2-} + H_2S + H^+ \rightarrow$
(2/5, Foreign 2011)

90. Complete the following chemical equations :
(i) $Fe^{3+} + I^- \rightarrow$
(ii) $CrO_4^{2-} + H^+ \rightarrow$ *(2/5, Foreign 2011)*

91. Describe the reactions involved in the preparation of $K_2Cr_2O_7$ from chromite ore.
(2/5, AI 2011C)

LA I (3 marks)

92. Explain the method of preparation of sodium dichromate from chromite ore. Give the equation representing oxidation of ferrous salts by dichromate ion. *(AI 2019)*

93. Complete the following reactions.
(i) $MnO_2 + KOH + O_2 \longrightarrow$
(ii) $I^- + MnO_4^- + H^+ \longrightarrow$
(iii) $Cr_2O_7^{2-} + Sn^{2+} + H^+ \longrightarrow$ *(AI 2019)*

94. Complete the following chemical equations :
(i) $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow$
(ii) $2CrO_4^{2-} + 2H^+ \rightarrow$
(iii) $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow$
(Delhi 2013)

95. Describe the preparation of potassium permanganate from pyrolusite ore. Write balanced chemical equation for one reaction to show the oxidizing nature of potassium permanganate. *(3/5, AI 2013C)*

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96. Describe the preparation of potassium dichromate from chromite ore. What is the effect of change of pH on dichromate ion? *(3/5, AI 2012C)*

97. Describe the preparation of potassium permanganate from pyrolusite ore. What happens when acidified potassium permanganate solution reacts with ferrous sulphate solution? Write balanced chemical equations. *(3/5, AI 2012C)*

98. Complete the following chemical equations :

(i) $MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow$

(ii) $KMnO_4 \xrightarrow{\text{Heated}}$

(iii) $Cr_2O_7^{2-} + H_2S + H^+ \longrightarrow$ *(Delhi 2011)*

8.5 The Lanthanoids

VSA (1 mark)

99. Give reasons for the following :
Eu^{2+} is a strong reducing agent. *(1/3, Delhi 2019)*

100. Account for the following :
Zr and Hf have almost similar atomic radii. *(1/5, AI 2015)*

101. Name a member of the lanthanoid series which is well known to exhibit +2 oxidation state. *(1/5, Delhi 2014)*

102. Name a member of the lanthanoid series which is well known to exhibit +4 oxidation state. *(1/5, AI 2014)*

103. What are the different oxidation states exhibited by the lanthanoids? *(1/3, Foreign 2014)*

104. How would you account for the following :
Zr (Z = 40) and Hf (Z = 72) have almost identical radii. *(1/3, Delhi 2013, 1/5, AI 2013C)*

105. How would you account for the following :
Among lanthanoids, Ln(III) compounds are predominant. However, occasionally in solutions or in solid compounds, +2 and +4 ions are also obtained. *(1/3, AI 2012)*

106. Give reason :
There is a gradual decrease in the size of atoms with increasing atomic number in the series of lanthanoids. *(1/3, Delhi 2012C)*

107. What is meant by 'lanthanoid contraction'? *(AI 2011)*

SA (2 marks)

108. What is lanthanoid contraction? What are its two consequences? *(Delhi 2013C)*

109. What is lanthanoid contraction and what is it due to? Write two consequences of lanthanoid contraction. *(2/5, Delhi 2013C)*

110. What is lanthanoid contraction? Mention its main consequences. *(2/3, Delhi 2012C)*

LA I (3 marks)

111. What is lanthanoid contraction? What are the consequences of lanthanoid contraction? *(Delhi 2015C)*

112. What is lanthanoid contraction? What is its effect on the chemistry of the elements which follow the lanthanoids? *(3/5, AI 2011C)*

8.6 The Actinoids

VSA (1 mark)

113. Write one similarity and one difference between the chemistry of lanthanoid and actinoid elements. *(1/5, Delhi 2017)*

114. Chemistry of actinoids is complicated as compared to lanthanoids. Give two reasons. *(1/5, Delhi 2017)*

115. Give reasons :
Actinoids show irregularities in their electronic configurations. *(1/3, Delhi 2016)*

116. How would you account for the following :
Actinoid contraction is greater than lanthanoid contraction? *(1/3, Delhi 2015)*

117. Give reasons for the following :
Actinoids exhibit a greater range of oxidation states than lanthanoids. *(1/3, AI 2014C, Delhi, AI 2012, 1/2, Delhi 2011)*

118. Assign reasons for the following :
From element to element actinoid contraction is greater than the lanthanoid contraction. *(1/3, AI 2014C)*

119. How would you account for the following :
Lanthanoids form primarily +3 ions, while the actinoids usually have higher oxidation

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states in their compounds, +4 or even +6 being typical. *(1/3, Delhi 2012)*

120. Explain giving reasons :
The chemistry of actinoids is not as smooth as that of lanthanoids. *(1/2, AI 2011)*

SA (2 marks)

121. Why do actinoids show a wide range of oxidation states? Write one similarity between the chemistry of lanthanoids and actinoids. *(2/5, AI 2015)*

122. With reference to structural variability and chemical reactivity, write the differences between lanthanoids and actinoids. *(2/5, AI 2014)*

LA I (3 marks)

123. Give three points of difference between lanthanoids and actinoids. *(3/5, 2020)*

Detailed Solutions

1. Zn, Cd, Hg are considered as *d*-block elements but not as transition elements because they do not have partly filled d-orbitals in their atomic state or their common oxidation states (*i.e.*, Zn^{2+}, Cd^{2+}, Hg^{2+}).

2. *Refer to answer 1.*

3. Elements which have incompletely filled *d*-orbitals in their ground state or in any one of their oxidation states are called transition elements.
Characteristics of transition elements :
(i) They show variable oxidation states.
(ii) They exhibit catalytic properties.

4. *Refer to answer 1.*

5. Scandium (Sc) exhibits only (+3) oxidation state.

6. The transition metals and their compounds, are known for their catalytic activity. This activity is ascribed to their ability to adopt multiple oxidation states, ability to adsorb the reactant(s) and ability to form complexes. Vanadium (V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in catalytic hydrogenation) are some of the examples. Catalysis involves the formation of bonds between reactant molecules and atoms at the surface of the catalyst.

7. Strong metallic bonds between its atoms of transition elements are responsible for the high melting points.

8. Zn^{2+} ion has completely filled *d*-subshell and no *d-d* transition is possible. So zinc salts are white.

Configuration of Cu^{2+} is [Ar] $3d^9$. It has partly filled *d*-subshell and hence it is coloured due to *d-d* transition.

9. Zinc

10. (c)

11. (c) : Because of strong metallic bonding present in atoms of transition elements, they have high melting points. Transition elements show variable oxidation states.

12. Cu(I) compounds have completely filled *d*-orbitals and there are no vacant *d*-orbitals for promotion of electrons whereas in Cu(II) compounds have one unpaired electron which is responsible for colour formation.

13. Oxoanion of chromium in which it shows +6 oxidation state equal to its group number is $Cr_2O_7^{2-}$ (dichromate ion).

14. Formula of oxoanion of manganese is MnO_4^-.
Oxidation state of Mn in this oxoanion = + 7
Group number of Mn is 7.

15. Due to presence of vacant *d*-orbitals and *d-d* transitions, compounds of the transition metals are generally coloured.
When an electron from a lower energy *d*-orbital is excited to a higher energy *d*-orbital, the energy of excitation corresponds to the frequency which generally lies in the visible region. The colour observed corresponds to the complementary colour of the light absorbed. The frequency of the light absorbed is determined by the nature of the ligand.

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16. *Refer to answer 8.*

17. Transition elements can use their *ns* and $(n-1)d$ orbital electrons for bond formation therefore, they show variable oxidation states.
For example, Sc has $ns^2(n-1)\,d^1$ electronic configuration.
It utilizes two electrons from its *ns* subshell then its oxidation state = +2. When it utilizes both the electrons then its oxidation state = +3.

18. In aqueous solutions, Cu^+ undergoes disproportionation to form a more stable Cu^{2+} ion.
$2Cu^+_{(aq)} \rightarrow Cu^{2+}_{(aq)} + Cu_{(s)}$
Cu^{2+} in aqueous solutions is more stable than Cu^+ ion because hydration enthalpy of Cu^{2+} is higher than that of Cu^+. It compensates the second ionisation enthalpy of Cu involved in the formation of Cu^{2+} ions.

19. Transition metals and most of their compounds contain unpaired electrons in the $(n-1)d$ orbitals hence show paramagnetic behaviour.

20. (b) $Co = [Ar]3d^7 4s^2$
$Co^{2+} = [Ar]3d^7$
i.e., there are three unpaired electrons ($n = 3$).
Hence, $\mu = \sqrt{n(n+2)}$ B.M.
$= \sqrt{3(3+2)} = 3.87$ B.M.

21. (i) Sc^{3+} has $3d^0$ outer electronic configuration, therefore it is diamagnetic in nature whereas Cr^{3+} has $3d^3$ outer electronic configuration. So, it is paramagnetic due to presence of unpaired electrons.
(ii) In a particular series, the metallic strength increases upto middle with increasing number of unpaired electrons, *i.e.*, upto d^5 configuration. After Cr, the number of unpaired electrons goes on decreasing. Accordingly, the m.pt and b.pt. decrease after middle (Cr) because of increasing pairing of electrons.

22. (i) Transition metals form alloys because they have similar atomic radii.
(ii) Basic nature of oxides decreases and acidic nature increases with increase in oxidation state of the metal. Oxidation state of Mn in Mn_2O_3 is +3 while in Mn_2O_7 is +7.

23. (i) Cr^{2+} is a stronger reducing agent than Fe^{2+}.
$E^\circ_{Cr^{3+}/Cr^{2+}}$ is negative (–0.41 V) whereas $E^\circ_{Fe^{3+}/Fe^{2+}}$ is positive (+ 0.77 V). Thus Cr^{2+} is easily oxidized to Cr^{3+} but Fe^{2+} cannot be easily oxidized to Fe^{3+}. Hence, Cr^{2+} is stronger reducing agent than Fe^{2+}.
(ii) More positive is the value of E°, reaction will be more feasible.
As $E^\circ_{Co^{3+}/Co^{2+}}$ is maximum, thus Co^{2+} ion is most stable.

24. (i) Variable oxidation states of transition metals arise due to incomplete filling of *d*-orbitals and it differs from each other by unity *e.g.*, $V^{(V)}$, $V^{(IV)}$, $V^{(III)}$, $V^{(II)}$. In *p*-block elements oxidation states differ generally by a unit of two. *e.g.*, Sn(II), Sn(IV), PCl_3, PCl_5, etc.
(ii) *Refer to answer 18.*

25. (i) *Refer to answer 17.*
(ii) In Zn, Cd and Hg, all the electrons in *d*-subshell are paired. Hence, the metallic bonds are weak. That is why they are soft metals with low melting and boiling points.

26. (i) Manganese can form $p\pi$ - $d\pi$ bond with oxygen by utilising 2*p*-orbital of oxygen and 3*d*-orbital of manganese due to which it can show highest oxidation state of +7. While with fluorine it cannot form such $p\pi$ - $d\pi$ bond thus, it can show a maximum of +4 oxidation state.
(ii) *Refer to answer 17.*

27. (i) *Refer to answer 17.*
(ii) *Refer to answer 18.*

28. *Refer to answer 17.*
Among the 3*d* series manganese (Mn) exhibits the largest number of oxidation states from +2 to +7 because it has maximum number of unpaired electrons.
Mn – [Ar] $3d^5\,4s^2$

29. (i) *Refer to answer 26(i).*
(ii) *Refer to answer 6.*

30. Disproportionation reaction involves the oxidation and reduction of the same substance. The examples of disproportionation reaction are :
(i) Aqueous NH_3 when treated with Hg_2Cl_2 (solid) forms mercury aminochloride disproportionatively.
$Hg_2Cl_2 + 2NH_3 \longrightarrow Hg + Hg(NH_2)Cl + NH_4Cl$
(ii) $2Cu^+ \longrightarrow Cu + Cu^{2+}$

31. (i) Electronic configuration of Mn^{2+} is $3d^5$ which is half filled and hence stable. Therefore,

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third ionization enthalpy is very high, *i.e.*, 3rd electron cannot be lost easily. In case of Fe^{2+}, electronic configuration is $3d^6$. Hence, it can lose one electron easily to give the stable configuration $3d^5$.

(ii) Zinc ($Z = 30$) has completely filled *d*-orbital ($3d^{10}$), so *d*-orbitals do not take part in interatomic bonding. Hence, metallic bonding is weak.

This is why it has very low enthalpy of atomisation (126 kJ mol^{-1}).

32. (i) *Refer to answer 3.*
(ii) *Refer to answer 1.*

33. (i) As we move along transition metal series from left to right (*i.e.,* Ti to Cu), the atomic radii decrease due to increase in nuclear charge. Hence, the atomic volume decreases. At the same time, atomic mass increases. Hence, the density from titanium (Ti) to copper (Cu) increases. The atoms of transition metals have strong metallic bonds between them, thus these have high melting and boiling points.
(ii) Irregular variation of ionisation enthalpies is mainly attributed to varying degree of stability of different 3*d*-configurations (*e.g.*, d^0, d^5, d^{10} are exceptionally stable).

34. (i) *Refer to answer 18.*
(ii) This is due to ability of oxygen to form multiple bonds with metals.

35. (i) *Refer to answer 17.*
(ii) The tendency to form complexes is high for Co(III) as compared to Co(II). Co^{2+} ions are very stable and are difficult to oxidise. Co^{3+} ions are less stable and are reduced by water. In contrast many Co(II) complexes are readily oxidised to Co(III) complexes and Co(III) complexes are very stable, *e.g.*,

$$[Co(NH_3)_6]^{2+} \xrightarrow{\text{Air}} [Co(NH_3)_6]^{3+}$$

This happens because the crystal field stabilisation energy of Co(III) with a $d^6(t_{2g}^6)$ configuration is higher than for Co(II) with a d^7 $(t_{2g}^6 e_g^1)$ arrangement.

36. (i) *Refer to answer 18.*
(ii) This is attributed to the involvement of greater number of electrons from $(n-1)d$ in addition to the *ns* electrons in the interatomic metallic bonding.

37. (i) *Refer to answer 6.*
(ii) *Refer to answer 15.*

38. (i) Copper exhibits +1 oxidation state in its compounds. Electronic configuration of Cu in the ground state is $3d^{10}$ $4s^1$. So, Cu can easily lose $4s^1$ electron to give a stable $3d^{10}$ configuration. Thus, it shows +1 oxidation state.
(ii) Only those ions will be coloured which have partially filled *d*-orbitals facilitating *d-d* transition. Ions with d^0 and d^{10} will be colourless.

From electronic configuration of the ions, $V^{3+}(3d^2)$ and $Mn^{2+}(3d^5)$, are all coloured. $Ti^{4+}(3d^0)$ and $Sc^{3+}(3d^0)$ are colourless.

39. (i) *Refer to answer 17.*
(ii) *Refer to answer 6.*

40. (i) Transition metals form a large number of interstitial compounds because small atoms of certain non metallic elements (H, B, C, N, etc.) get trapped in voids or vacant spaces of lattices of the transition metals. As a result of filling up of the interstitial spaces such interstitial compounds are hard and rigid.
(ii) This is due to lanthanoid contraction.

41. (i) $E°$ values for the Cr^{3+}/Cr^{2+} and Mn^{3+}/Mn^{2+} couples are

$$Cr^{3+}_{(aq)} + e^- \longrightarrow Cr^{2+}_{(aq)};\ E° = -0.41\ V$$

$$Mn^{3+}_{(aq)} + e^- \longrightarrow Mn^{2+}_{(aq)};\ E° = +1.551\ V$$

These $E°$ values indicate that Cr^{2+} is strongly reducing while Mn^{3+} is strongly oxidising agent.
(ii) Since, transition elements contain partially filled *d*-subshells, therefore, electrons in these subshells go from lower *d*-subshells to higher *d*-subshells. This is called *d-d* transition. This transition takes place by absorbing energy from the visible light. The mixture of the wavelength which is not absorbed is transmitted out. This accounts for the colour of transition elements.

42. (i) Electrode potential ($E°$) value is the sum of three factors :
(a) Enthalpy of atomisation $\Delta_a H$ for $Cu_{(s)} \rightarrow Cu_{(g)}$
(b) Ionisation enthalpy $\Delta_i H$ for $Cu_{(g)} \rightarrow Cu^{2+}_{(g)}$
(c) Hydration enthalpy $\Delta_{hyd} H$ for $Cu^{2+}_{(g)} \rightarrow Cu^{2+}_{(aq)}$
In case of copper the sum of enthalpy of atomisation and ionisation enthalpy is greater than enthalpy of hydration.
This is why $E°_{M^{2+}/M}$ for Cu is positive.

(ii) *Refer to answer 40(ii).*

43. (i) *Refer to answer 40(i).*
(ii) As we move along transition metal series from left to right (*i.e.,* Ti to Cu), the atomic radii decrease due to increase in nuclear charge. Hence the atomic volume decreases. At the same time, atomic mass increases. Hence, the density from titanium (Ti) to copper (Cu) increases.

44. (i) *Refer to answer 41(i).*
(ii) As one proceeds along a transition series, the nuclear charge increases which tends to decrease the size but the addition of electrons in the *d*-subshell increases the screening effect which counterbalances the effect of increased nuclear charge. As a result, the atomic radii remain practically same after chromium.

45. (i) As transition metals have a large number of unpaired electrons in the *d*-orbitals of their atoms they have strong interatomic attraction or metallic bonds. Hence, they have high enthalpy of atomization.
(ii) *Refer to answer 6.*

46. (i) Sc(21) is a transition element but Ca(20) is not because Sc has incompletely filled 3*d* orbitals.
(ii) *Refer to answer 31(i).*

47. The variability in oxidation states of transition metals is due to the incomplete filling of *d*-orbitals. Their oxidation states differ from each other by unity.

For example, Fe^{3+} and Fe^{2+}, Cu^{2+} and Cu^{+}, etc.
In case of non transition elements the oxidation states normally differ by units of two. For example, Pb^{2+} and Pb^{4+}, Sn^{2+} and Sn^{4+}, etc. It arises due to expansion of octet and inert pair effect.

48. (i) *Refer to answer 31(i).*
(ii) *Refer to answer 41(i).*

49. (i) *Refer to answer 15.*
(ii) As manganese has maximum number of unpaired electrons (5) in 3*d* subshell in addition to 2 electrons in the 4*s* subshell, it can use the 7 electrons for bonding purpose. Thus, it exhibits +7 oxidation state.

50. (i) *Refer to answer 41(i).*
(ii) Middle of the transition series contains greater number of unpaired electrons in $(n-1)d$ and *ns* orbitals.

51. (i) *Refer to answer 18.*
(ii) This is because due to lanthanoid contraction the expected increase in size does not occur hence they have very high value of ionisation enthalpies.

52. (i) *Refer to answer 6.*
(ii) The metals of 4*d* and 5*d*-series have more frequent metal bonding in their compounds than the 3*d*-metals because 4*d* and 5*d*-orbitals are more exposed in space than the 3*d*-orbitals. So the valence electrons are less tightly held and form metal-metal bonding more frequently.

53. (i) Cr^{2+} is reducing since its configuration is converted to d^3 from d^4. d^3 has half filled t_{2g} configuration with higher stability.
(ii) Cu^{+} is unstable in aqueous solution
In aqueous solutions, Cu^{+} undergoes disproportionation to form a more stable Cu^{2+} ion.

$$2Cu^{+}_{(aq)} \rightarrow Cu^{2+}_{(aq)} + Cu_{(s)}$$

(iii) Mn^{3+} is a strong oxidising agent because electronic configuration of Mn^{2+} is $3d^5$ which is half filled and hence stable. Therefore, third ionization enthalpy is very high, *i.e.,* 3rd electron cannot be lost easily.

54. (i) From the relation, $\Delta G^\circ = -nFE^\circ$
More positive is the value of E°, reaction will be feasible.

$$Mn^{3+} \xrightarrow{+e^-} Mn^{2+}; \qquad Fe^{3+} \xrightarrow{+e^-} Fe^{2+}$$

$3d^4$ — $3d^5$ more stable (half filled); $3d^5$ more stable (half filled) — $3d^6$

Hence, E°_{value} for Mn^{3+}/Mn^{2+} couple is much more positive than that for Fe^{3+}/Fe^{2+}.
(ii) Greater the number of unpaired electrons, stronger is the metallic bond and therefore, higher is the enthalpy of atomisation. Since, iron has greater number of unpaired electrons than copper hence has higher enthalpy of atomisation.
(iii) Only those ions are coloured which have partially filled *d*-orbitals facilitating *d-d* transitions. Sc^{3+} has no unpaired electron but Ti^{3+} has one unpaired electron and hence, Ti^{3+} is coloured.

55. (i) Transition metals form a large number of complex compounds due to following reasons :
– Comparatively smaller size of metal ions.
– High ionic charges.
– Availability of *d*-orbitals for bond formation.

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(ii) Lowest oxidation compounds of transition metals are basic due to their ability to get oxidised to higher oxidation states. Whereas, the higher oxidation state of metal and compounds gets reduced to lower ones and hence are acidic in nature.

e.g., MnO is basic whereas Mn_2O_7 is acidic.

(iii) Much larger third ionisation energy of Mn(where change is d^5 to d^4) is mainly responsible for this. This also explains that +3 state of Mn is of little importance.

56. (i) Ti^{4+} has highest oxidation state among the given ions. Ti^{4+} has stable inert gas configuration and hence, most stable in aqueous solution.

On the other hand, V^{2+}, Mn^{3+}, Cr^{3+} have unstable electronic configuration and hence, are less stable.

(ii) Due to presence of highest oxidation state of Ti, it acts as the strongest oxidising agent among the given ions.

(iii) Due to absence of unpaired electron in Ti^{4+}, it is a colourless ion.

E.C. of Ti^{4+} = $[Ar]3d^0 4s^0$

57. (i) *Refer to answer 26(i).*

(ii) Cr^{2+} is reducing since its configuration is converted to d^3 from d^4. d^3 has half filled t_{2g} configuration with higher stability.

(iii) *Refer to answer 8.*

58. (i) *Refer to answer 42(i).*

(ii) Mn^{2+} ion has stable half-filled ($3d^5$) electronic configuration. Its ionisation enthalpy value is lower in comparison to hydration enthalpy. Hence $E^\circ_{Mn^{2+}/Mn}$ is more negative.

(iii) *Refer to answer 23(i).*

59. (i) *Refer to answer 31(i).*

(ii) *Refer to answer 31(ii).*

(iii) Only those ions are coloured which have partially filled *d*-orbitals facilitating *d-d* transitions.

Sc^{3+} with $3d^0$ configuration is colourless while Ti^{3+} ($3d^1$) is coloured.

60. (i) Mn^{2+} is more stable due to half filled d^5 configuration and Mn^{3+} easily changes to Mn^{2+} hence, it is a good oxidising agent.

(ii) The $E^\circ_{M^{2+}/M}$ values are not regular which can be explained from the irregular variation of ionisation enthalpes *i.e.*, IE_1 + IE_2 and also the sublimation enthalpies which are relatively much less for manganese and vanadium.

(iii) *Refer to answer 26(i).*

61. (i) All transition elements except the first and the last member in each series show a large number of variable oxidation states. This is because difference of energy in the $(n-1)d$ and ns orbitals is very little.

Hence, electrons from both the energy levels can be used for bond formation.

(ii) *Refer to answer 45(i).*

(iii) *Refer to answer 47.*

62. (i) *Refer to answer 18.*

(ii) *Refer to answer 41(i).*

(iii) *Refer to answer 6.*

63. (i) *Refer to answer 52(ii).*

(ii) Mn^{3+} is less stable and changes to Mn^{2+} which is more stable due to half-filled *d*-orbital configuration. That is why, Mn^{3+} undergoes disproportionation reaction.

(iii) *Refer to answer 35(ii).*

64. (i) *Refer to answer 6.*

(ii) *Refer to answer 55(ii).*

(iii) A transition metal exhibits higher oxidation states in oxides and fluorides because oxygen and fluorine are highly electronegative elements, small in size and strongest oxidising agents. For example, vanadium shows as oxidation state of +5 in V_2O_5. Manganese shows +7 in Mn_2O_7.

65. (i) Mn shows maximum no. of oxidation states from +2 to +7 because Mn has maximum number of unpaired electrons in 3*d* sub-shell.

(ii) Cr has maximum melting point, because it has 6 unpaired electrons in the valence shell, hence it has strong interatomic interaction.

(iii) Sc shows only +3 oxidation state because after losing 3 electrons, it has noble gas electronic configuration.

(iv) Mn is strong oxidising agent in +3 oxidation state because change of Mn^{3+} to Mn^{2+} give stable half filled (d^5) electronic configuration,

$E^\circ_{(Mn^{3+}/Mn^{2+})}$ = 1.5 V.

66. In aqueous solution,

$$\underset{\substack{\text{Dichromate ion}\\\text{(Orange red)}}}{Cr_2O_7^{2-}} + H_2O \rightleftharpoons \underset{\substack{\text{Chromate ion}\\\text{(Yellow)}}}{2CrO_4^{2-}} + 2H^+$$

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When an acid is added (*i.e.*, pH of solution decreased), the concentration of H^+ ions is increased and the reaction proceeds in the backward direction producing an orange red dichromate solution.

67. Orange colour of $Cr_2O_7^{2-}$ ion changes to yellow when an alkali such as NaOH is added because on addition of an alkali, the concentration of H^+ ions decreases and hence, the reaction proceeds in the forward direction producing yellow solution containing CrO_4^{2-} ions.

$$\underset{\text{orange}}{Cr_2O_7^{2-}} + 2OH^- \longrightarrow \underset{\text{yellow}}{2CrO_4^{2-}} + H_2O$$

68. $2MnO_4^- + 6H^+ + 5NO_2^- \longrightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$

69. $3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O$

70. $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

71. $2MnO_4^- + 5SO_2 + 2H_2O \longrightarrow 2Mn^{2+} + 5SO_4^{2-} + 4H^+$

72. *Refer to answer 67.*

73. Preparation of potassium permanganate : Potassium permanganate is prepared by the fusion of MnO_2 (pyrolusite) with potassium hydroxide and an oxidising agent like KNO_3 to form potassium manganate which disproportionates in a neutral or acidic solution to form permanganate.

$2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O$

$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

or, $3K_2MnO_4 + 4HCl \rightarrow 2KMnO_4 + MnO_2 + 2H_2O + 4KCl$

74. $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \longrightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

$Cr_2O_7^{2-} + 3Sn^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 3Sn^{4+} + 7H_2O.$

75. (*A*) is K_2MnO_4, (*B*) is $KMnO_4$, (*C*) is KIO_3 and (*D*) is I_2.

$MnO_2 + 2KOH + KNO_3 \longrightarrow \underset{(A)}{K_2MnO_4} + KNO_2 + H_2O$

$3K_2MnO_4 + 2H_2SO_4 \longrightarrow \underset{(B)}{2KMnO_4} + MnO_2 + 2K_2SO_4 + 2H_2O$

$2KMnO_4 + H_2O + KI \longrightarrow 2MnO_2 + 2KOH + \underset{(C)}{KIO_3}$

$2KMnO_4 + 8H_2SO_4 + 10KI \longrightarrow 6K_2SO_4 + 2MnSO_4 + 8H_2O + \underset{(D)}{5I_2}$

76. (i) $5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$

(ii) $2MnO_4^- + H_2O + I^- \rightarrow 2OH^- + 2MnO_2 + IO_3^-$

77. (i) $2MnO_4^- + 16H^+ + 5S^{2-} \longrightarrow 2Mn^{2+} + 8H_2O + 5S$

(ii) $2KMnO_4 \xrightarrow{\text{heat}} K_2MnO_4 + MnO_2 + O_2$

78. (i) $4FeCr_2O_4 + 16NaOH + 7O_2 \longrightarrow \underset{(A)}{8Na_2CrO_4} + 2Fe_2O_3 + 8H_2O$

$2Na_2CrO_4 + H_2SO_4 \longrightarrow \underset{(B)}{Na_2Cr_2O_7} + Na_2SO_4 + H_2O$

$Na_2Cr_2O_7 + 2KCl \longrightarrow \underset{(C)}{K_2Cr_2O_7} + 2NaCl$

(ii) Potassium dichromate is used as a powerful oxidising agent in industries and for staining and tanning of leather.

79. (i) $8MnO_{4\,(aq)}^- + 3S_2O_{3\,(aq)}^{2-} + H_2O_{(l)} \rightarrow 8MnO_{2(aq)} + 6\,SO_{4\,(aq)}^{2-} + 2OH^-_{(aq)}$

(ii) $Cr_2O_7^{2-} + 3Sn^{2+} + 14H^+ \longrightarrow 2Cr^{3+} + 3Sn^{4+} + 7H_2O$

80. (i) $2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta} 2K_2MnO_4 + 2H_2O$

(ii) $Cr_2O_7^{2-} + 14H^+ + 6I^- \longrightarrow 2Cr^{3+} + 3I_2 + 7H_2O$

81. *Refer to answer 73.*

Ionic equation for the reaction between oxalic acid and acidified permanganate solution :

$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

82. Potassium dichromate is a strong oxidising agent. In the presence of dilute sulphuric acid, one molecule of $K_2Cr_2O_7$ gives 3 atoms of available oxygen.

(i) *Refer to answer 80(ii).*

(ii) $K_2Cr_2O_7 + 3H_2S + 4H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 3S + K_2SO_4 + 7H_2O$

83. (i) *Refer to answer 80(i).*

(ii) Sodium dichromate can be crystallised out from sodium chromate solution by acidifying it with sulphuric acid.

$2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$

84. (i) $Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$

(ii) $MnO_4^- + 4H^+ + 3e^- \longrightarrow MnO_2 + 2H_2O$

85. (i) $H_2S \longrightarrow 2H^+ + S^{2-}$

$5S^{2-} + 2MnO_4^- + 16H^+ \longrightarrow 2Mn^{2+} + 8H_2O + 5S$

(ii) $Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$

86. (i) $2CrO_4^{2-} + 2H^+ \longrightarrow Cr_2O_7^{2-} + H_2O$

(ii) $2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$

87. (i) *Refer to answer 80(ii).*

(ii) *Refer to answer 68.*

88. (i) *Refer to answer 79(i).*

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(ii) $Cr_2O_{7(aq)}^{2-} + 14H_{(aq)}^{+} + 6Fe_{(aq)}^{2+} \longrightarrow 2Cr_{(aq)}^{3+} + 6Fe_{(aq)}^{3+} + 7H_2O_{(l)}$

89. (i) $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \longrightarrow 2Mn^{2+} + 8H_2O + 10CO_2$

(ii) $Cr_2O_7^{2-} + 3H_2S + 8H^+ \longrightarrow 2Cr^{3+} + 7H_2O + 3S$

90. (i) $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$

(ii) $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$

91. $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$

$2Na_2CrO_4 + 2H^+ \longrightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$

$Na_2Cr_2O_7 + 2KCl \longrightarrow K_2Cr_2O_7 + 2NaCl$

Potassium dichromate is converted to chromate if pH is increased.

$$Cr_2O_7^{2-} \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} CrO_4^{2-}$$

92. The yellow solution of sodium chromate is acidified with sulphuric acid to give a orange solution of sodium dichromate $Na_2Cr_2O_7$ which is crystallised.

$$\underset{\text{Sodium chromate}}{2Na_2CrO_4} + H_2SO_4 \rightarrow \underset{\text{Sodium dichromate}}{Na_2Cr_2O_7} + Na_2SO_4 + H_2O$$

The solution of sodium dichromate is treated with potassium chloride to obtain potassium dichromate.

$$Na_2Cr_2O_7 + 2KCl \rightarrow \underset{\text{Potassium dichromate}}{K_2Cr_2O_7} + 2NaCl$$

Refer to answer 88 (ii).

93. (i) *Refer to answer 80(i).*

(ii) $2MnO_4^- + 10I^- + 16H^+ \longrightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

(iii) *Refer to answer 79(ii).*

94. (i) *Refer to answer 88(ii).*

(ii) *Refer to answer 90(ii).*

(iii) *Refer to answer 89(i).*

95. *Refer to answer 73.*

Oxidising nature of $KMnO_4$

$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

96. *Refer to answer 91.*

97. *Refer to answer 73.*

$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$

98. (i) *Refer to answer 89(i).*

(ii) *Refer to answer 86(ii).*

(iii) *Refer to answer 89(ii).*

99. Eu^{2+} has a strong tendency to loose electrons to attain the more stable +3 oxidation state of lanthanoids hence, it is a strong reducing agent.

100. Due to lanthanoid contraction the elements of 4*d* and 5*d*-series have similar atomic radii *e.g.*, Zr = 145 pm and Hf = 144 pm.

101. Europium (Eu) is well known to exhibit +2 oxidation state due to its half-filled *f* orbital in +2 oxidation state.

102. Lanthanoids showing +4 oxidation state are $_{58}Ce$, $_{59}Pr$, $_{65}Tb$ and $_{66}Dy$.

103. Lanthanum and all the lanthanoids predominantly show +3 oxidation state. However, some of the lanthanoids also show +2 and +4 oxidation states in solution or in solid compounds. This irregularity arises mainly due to attainment of stable empty ($4f^0$), half-filled ($4f^7$) and fully filled ($4f^{14}$) sub shell.

e.g., $Ce^{4+} : 4f^0$, $Eu^{2+} : 4f^7$

$Tb^{4+} : 4f^7$, $Yb^{2+} : 4f^{14}$

104. *Refer to answer 100.*

105. *Refer to answer 103.*

106. As the atomic number increases, each succeeding element contains one more electron in the 4*f* orbital and one extra proton in the nucleus. The 4*f* electrons are rather ineffective in screening the outer electrons from the nucleus. As a result, there is gradual increase in the nuclear attraction for the outer electrons. Consequently, the atomic size gradually decreases. This is called lanthanoid contraction.

107. The steady decrease in the atomic and ionic radii (having the same charge) with increase in atomic number across the series from lanthanum to lutetium is known as lanthanoid contraction.

108. Lanthanoid contraction : The steady decrease in the atomic and ionic radii of lanthanoid elements with increase in atomic number is called lanthanoid contraction. It is caused due to imperfect shielding of nuclear charge by 4*f*-electrons.

Consequences of lanthanoid contraction :

(i) The basic strength of oxides and hydroxides of lanthanoids decrease with increasing atomic number.

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(ii) Atomic and ionic sizes of 4*d* transition series elements and 5*d* series elements are similar. *e.g.,* atomic radii of zirconium(Zr) is same as that of hafnium Hf.

109. *Refer to answer 108.*

110. *Refer to answer 108.*

111. *Refer to answer 108.*

112. *Refer to answer 108.*

113. Similarity : The elements of both the series are electropositive in nature. They are reactive metals and act as strong reducing agents.
Difference : Lanthanoids except promethium are non-radioactive elements, while all actinoids are radioactive elements.

114. Chemistry of actinoids is more complicated than lanthanoids because
(i) actinoids show greater number of oxidation states due to the comparable energies of 5*f*, 6*d* and 7*s* orbitals.
(ii) most of the actinoids are radioactive and the study of their chemistry in the laboratory is difficult.

115. The irregularities in the electronic configurations of actinoids are due to extra stabilities of the f^0, f^7 and f^{14} orbitals.

116. The actinoid contraction is more than lanthanoid contraction because 5*f*-electrons are more poorly shielded than 4*f*-electrons.

117. Actinoids exhibit greater range of oxidation states than lanthanoids. This is because there is less energy difference between 5*f* and 6*d* orbitals in actinoids than the energy difference between 4*f* and 5*d* orbitals in case of lanthanoids.

118. *Refer to answer 116.*

119. *Refer to answer 117.*

120. The chemistry of actinoids is not as smooth as lanthanoid because they show greater number of oxidation states due to comparable energies of 5*f*, 6*d* and 7*s* orbitals.

121. *Refer to answer 117.*
Refer to answer 113.

122. Structure : All the lanthanoids are silvery white soft metals. Hardness of Lanthanoids increases with increasing atomic number.
The actinoid metals are all silvery in appearance but display a variety of structures. The structural variability is due to irregularities in metallic radii which are greater than that of lanthanoids.
Chemical reactivity : Earlier members of lanthanoid series are quite reactive similar to calcium but with increasing atomic number they behave more like aluminium.
The actinoids are highly reactive in finely divided state.

123. Difference between lanthanoids and actinoids are following :
(i) Electronic configuration : The general electronic configuration of lanthanoids is $[Xe]4f^{\,1-14}\,5d^{0-1}\,6s^2$ whereas, that of actinoids is $[Rn]\,5f^{\,1-14}\,6d^{0-1}\,7s^2$. Thus, lanthanoids involve the filling of 4*f*-orbitals whereas, actinoids involve the filling of 5*f*-orbitals.
(ii) Oxidation states : Lanthanoids have principal oxidation state of +3. In addition, the lanthanoids show limited oxidation states such as +2, +3 and +4 because of large energy gap between 4*f* and 5*d* subshells. On the other hand, actinoids show a large number of oxidation states because of small energy gap between 5*f* and 6*d* subshells.
(iii) Chemical reactivity : First few members of lanthanoids are quite reactive almost like calcium, whereas, actinoids are highly reactive metals especially in the finely divided state. Lanthanoids react with dilute acids to liberate H_2 gas whereas actinoids react with boiling water to give a mixture of oxide and hydride.

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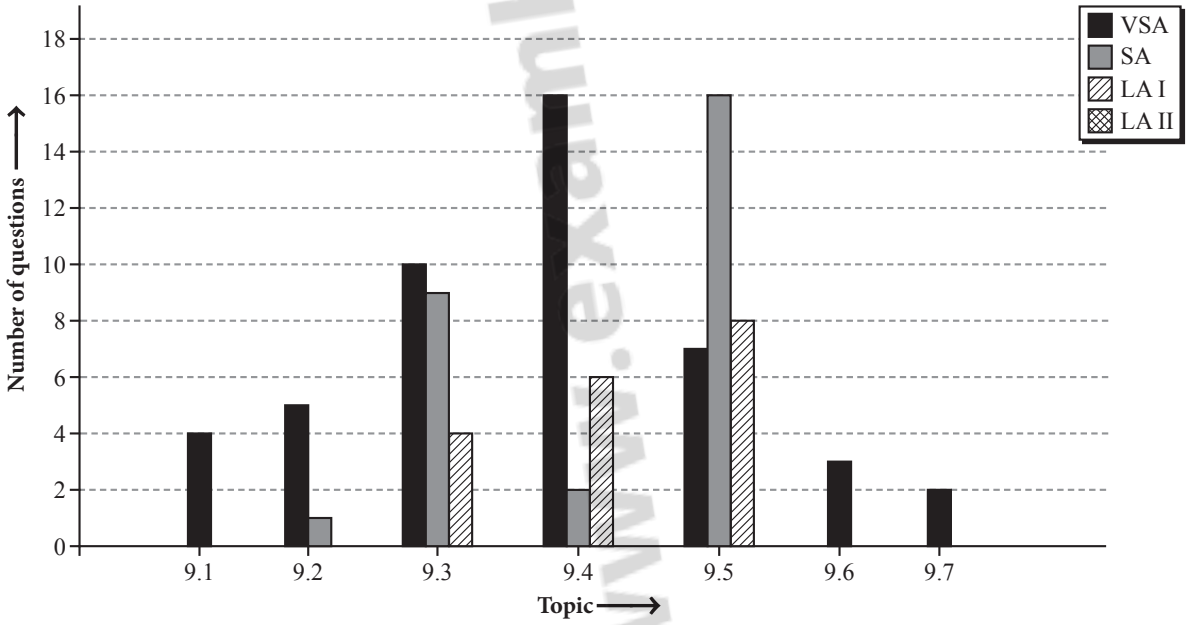
CHAPTER
9

Coordination Compounds

9.1 Werner's Theory of Coordination Compounds
9.2 Definition of Some Important Terms Pertaining to Coordination Compounds
9.3 Nomenclature of Coordination Compounds
9.4 Isomerism in Coordination Compounds
9.5 Bonding in Coordination Compounds
9.6 Bonding in Metal Carbonyls
9.7 Importance and Applications of Coordination Compounds

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Bonding in Coordination Compounds.*
- Maximum VSA type questions were asked from *Isomerism in Coordination Compounds.*
- Maximum SA and LA I type questions were asked from *Bonding in Coordination Compounds.*

QUICK RECAP

Werner's coordination theory :

- It explains the nature of bonding in complexes. Metals show two different kinds of valencies:
  - **Primary valency :** Non directional and ionisable. It is equal to the oxidation state of the central metal ion.

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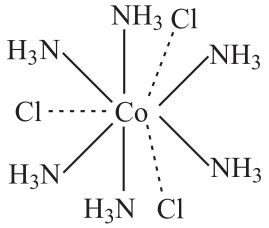
– **Secondary valency :** Directional and non-ionisable. It is equal to the coordination number of the metal. It is commonly satisfied by neutral and negatively charged or sometimes by positively charged ligands.

▶ The ionisation of the coordination compound is written as :

$[Co(NH_3)_6]Cl_3 \rightleftharpoons [Co(NH_3)_6]^{3+} + 3Cl^-$



$[Co(NH_3)_6]Cl_3$

Representation of $CoCl_3 \cdot 6NH_3$ complex according to Werner's theory

Addition compounds : The compounds formed by combination of two or more simple compounds are called *addition compounds*. They are of two types :

▶ **Double salt :** A compound formed by combination of two or more simple compounds, which is stable in solid state only is called *double salt*. In solution it breaks into component ions.

e.g., $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$; Potash alum

$FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$; Mohr's salt

$KCl \cdot MgCl_2 \cdot 6H_2O$; Carnallite

▶ **Classification of ligands :**

Ligands

On the basis of charge

→ **Negative ligands** CN^-, F^-, Cl^-, NO_2^-, NO_3^-, OH^-, O^{2-}

→ **Positive ligands** NO_2^+, NO^+, $N_2H_5^+$

→ **Neutral ligands** $H_2\ddot{O}$, $\ddot{N}H_3$, CO, NH_2OH, $CH_3\ddot{N}H_2$

On the basis of number of donor sites

→ **Monodentate :** Only one donor site *e.g.*, H_2O, NH_3

→ **Bidentate :** Two donor sites *e.g.*, $(COO^-)_2$ (Oxalato), $CH_2—NH_2$ | $CH_2—NH_2$ (Ethylenediamine)

→ **Polydentate :** More than two donor sites *e.g.*, EDTA (Hexadentate)

On the basis of bonding

→ **Chelating ligands :** A bidentate or polydentate ligand which forms more than one coordinate bonds in such a way that a ring is formed.

$CH_2—NH_2$ | $CH_2—NH_2$ (Ethylenediamine)

→ **Ambidentate ligands :** Monodentate ligand which contains more than one coordinating atom (or donor atom).

$M \leftarrow O^-—N=O$, $M \leftarrow SCN$, $M \leftarrow CN$

or or or

$M \leftarrow N(=O)(-O^-)$ $M \leftarrow NCS$ $M \leftarrow NC$

▶ **Complex compound :** A compound formed by combination of two or more simple compounds which retain its identity both in solid and solution states is called *complex compound*.

e.g., $K_4[Fe(CN)_6]$, Potassium ferrocyanide

$[Cu(NH_3)_4]SO_4$, Cupramine sulphate

Some important terms pertaining coordination compound :

▶ **Coordination entity :** The central metal atom or ion and ligand taken together is called *coordination entity*. It may be positive, negative or neutral.

e.g., $[Cu(NH_3)_4]^{2+}$, $[Fe(CN)_6]^{4-}$, $[Ni(CO)_4]$

▶ **Central atom :** The atom or ion with which definite number of ligands are attached in a definite geometry is called *central atom/ion*. Any atom/ion which has high positive charge density or vacant orbitals of suitable energy may be central atom or ion, *e.g.,* transition metals, lanthanoids. It is Lewis acid (electron acceptor).

▶ **Ligands :** Molecules or ions which are bound to the central atom/ion in the coordination entity are called *ligands*. A molecule or ion which has high negative charge or dipole or lone pair of electrons may be ligands. It is Lewis base (electron donor).

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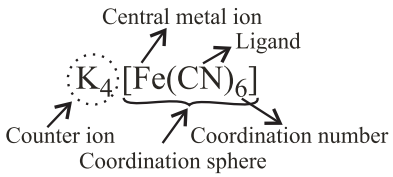
- **Coordination number (C.N.) :** The total number of coordinate bonds through which the central metal atom or ion is attached with ligands is known as coordination number. Examples : $[Ag(CN)_2]^-$: C.N. = 2, $[Cu(NH_3)_4]^{2+}$: C.N. = 4, $[Cr(H_2O)_6]^{3+}$: C.N. = 6
- **Coordination sphere :** The central atom and the ligands which are directly attached are collectively known as coordination sphere. It is non-ionisable and written enclosed in square brackets. The ionisable groups are written outside the brackets.
  Example :

$$[Cu(NH_3)_4]\ SO_4 \rightleftharpoons [Cu(NH_3)_4]^{2+} + SO_4^{2-}$$

Coordination sphere — Ionisable group



- **Coordination polyhedron :** The spatial arrangement of the ligand atoms which are directly attached to the central atom defines a coordination polyhedron about the central atom, *e.g.*, $[(Co(NH_3)_6)]^{2+}$ is octahedral $[Ni(CO)_4]$ is tetrahedral. Octahedral is most common coordination polyhedron.

Homoleptic and heteroleptic complexes :

- **Homoleptic complexes :** Complexes in which a metal is bound to only one kind of ligands are called *homoleptic complexes*.
  e.g., $[Co(NH_3)_6]^{3+}$, $[Ti(H_2O)_6]^{3+}$, $[Cu(CN)_4]^{3-}$
- **Heteroleptic complexes :** Complexes in which the central atom is bound to different type of ligands are called *heteroleptic complexes.*
  e.g., $[Co(NH_3)_4Cl_2]$, $K_2[Fe(CN)_5NO]$, $[Fe(H_2O)_5NO]SO_4$

Nomenclature of coordination compounds :

- **Rules for writing the formula of coordination compounds :**
  - Formula of the cation whether simple or complex must be written first, followed by anion.
  - The coordination sphere is written in square brackets.
  - Within the coordination sphere the sequence of symbols is, first the metal atom followed by anionic ligand then neutral ligand finally cationic ligand. Ligands of same type are arranged alphabetically.
  - Polyatomic ligands are enclosed in parentheses.
  - The number of cations or anions to be written in the formula is calculated on the basis that total positive charge must be equal to the total negative charge, as the complex as a whole is electrically neutral.
- **Rules for naming coordination compounds :**
  - The cation is named first then the anion.
  - In naming coordination sphere, ligands are named first in alphabetical order followed by metal atom and then oxidation state of metal by a roman numeral in parentheses.
  - The complex part is written as one word.
  - When the coordination sphere is anionic, name of central metal ends in *–ate*.
- **Naming of ligands :**
  - Name of anionic ligands end in *–o*.
    e.g., Cl^- : Chlorido
  - Neutral ligands (with a few exceptions) retain their names *e.g.,* NH_3 : Ammine
  - Name of cationic ligands end in – *ium*.
    e.g., NO_2^+ : Nitronium
  - Certain ligands are represented by abbreviations in parentheses instead of their complex structural formulae.
    e.g., ethylenediamine(*en*).
  - Ambidentate ligands are named by using different names of ligands or by placing the symbol of donor atom.
    e.g., —SCN^- (Thiocyanato-S or Thiocyanato), —NCS^- (Thiocyanato-N or Isothiocyanato), —ONO^- (Nitrito-O or Nitrito), —NO_2^- (Nitrito-N or Nitro)
  - The prefixes *di-*, *tri-*, *tetra-*, *penta-* and *hexa-* are used to indicate the number of each ligand. If the ligand name includes such a prefix, the ligand name should be placed in parentheses and preceded by *bis-*(2), *tris-*(3), *tetrakis-*(4), *pentakis-*(5) and *hexakis-*(6).

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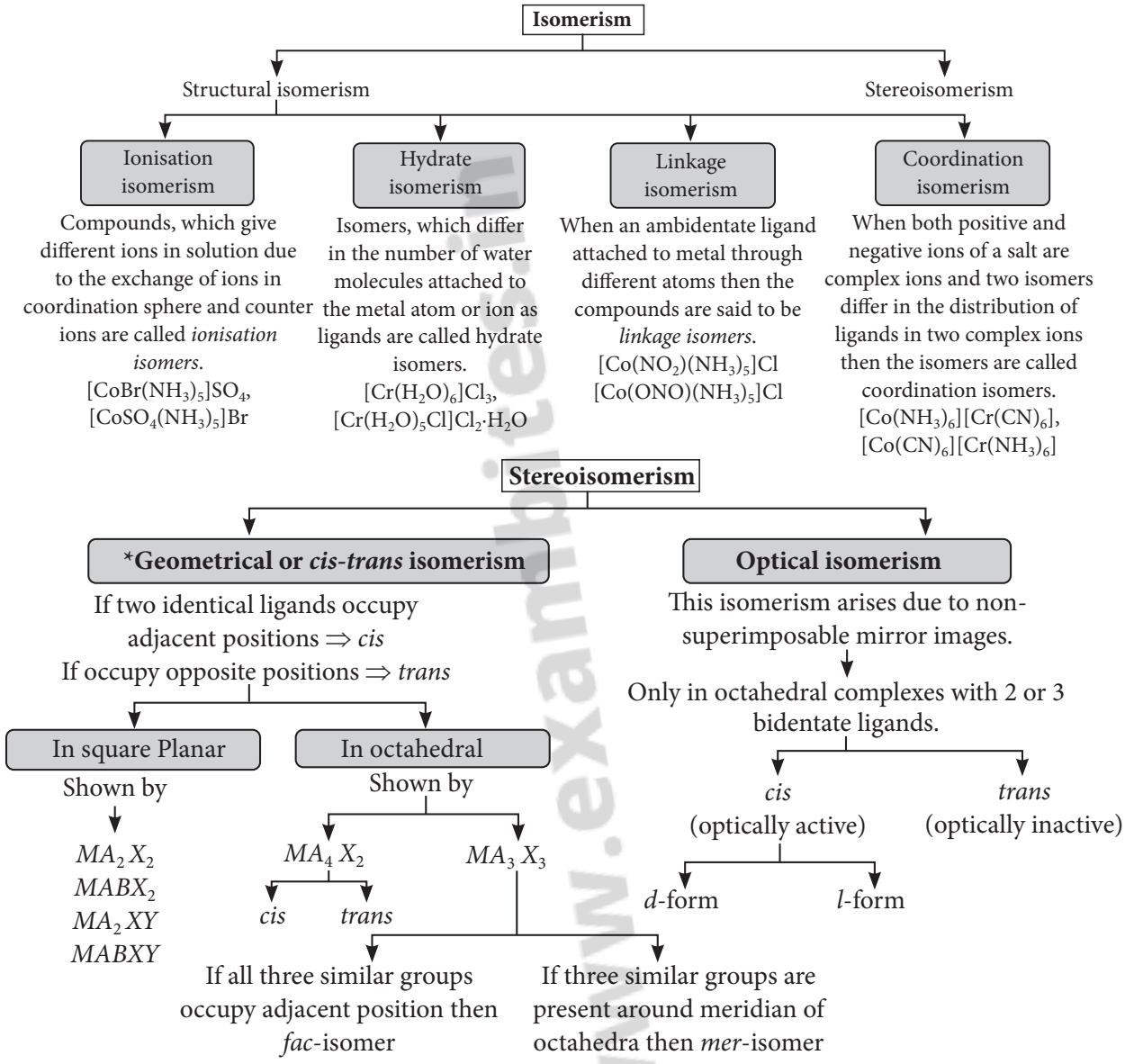
Isomerism : Two or more substances having the same molecular formula but different structural or spatial arrangement are called *isomers* and phenomenon is called *isomerism*.



(* Geometrical isomerism is not shown by tetrahedral complexes.)

Bonding in coordination compounds :

▶ **Valence bond theory :** It was developed by Pauling.

- A suitable number of vacant orbitals must be present in the central metal atom or ion for the formation of coordinate bonds with the ligands.
- Central metal ion can use appropriate number of *s*, *p* or *d*-orbitals for hybridisation depending upon the total number of ligands.
- The outer orbital (high spin) or inner orbital (low spin) complexes are formed depending upon whether outer *d*-orbitals or inner *d*-orbitals are used.

C. No.	Type of hybridisation	Geometry	Examples
2	*sp*	Linear	$[Ag(NH_3)_2]^+$, $[Ag(CN)_2]^-$
3	sp^2	Trigonal planar	$[HgI_3]^-$

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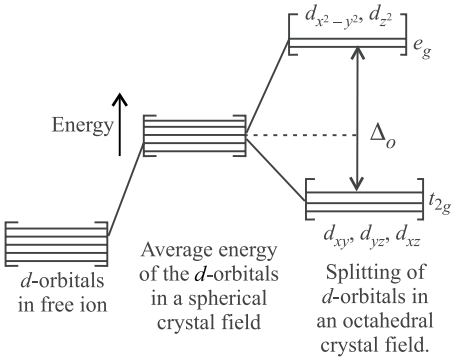
4	sp^3	Tetrahedral	$Ni(CO)_4$, $[NiX_4]^{2-}$, $[ZnCl_4]^{2-}$, $[CuX_4]^{2-}$, where, $X = Cl^-, Br^-, I^-$
	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$, $[Cu(NH_3)_4]^{2+}$, $[Ni(NH_3)_4]^{2+}$
5	dsp^3	Trigonal bipyramidal	$[Fe(CO)_5]$, $[CuCl_5]^{3-}$
	sp^3d	Square pyramidal	$[SbF_5]^{2-}$
6	d^2sp^3	Octahedral (Inner orbital)	$[Cr(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{3-}$
	sp^3d^2	Octahedral (Outer orbital)	$[FeF_6]^{3-}$, $[Fe(H_2O)_6]^{2+}$, $[Ni(NH_3)_6]^{2+}$

Inner orbital complexes	Outer orbital complexes
Involves inner *d*-orbitals *i.e.*, $(n-1)d$-orbitals.	Involves outer *d*-orbitals *i.e.*, *nd*-orbitals.
Low spin complexes	High spin complexes
Have less or no unpaired electrons. *e.g.*, $[Co(NH_3)_6]^{3+}$, $[Co(CN)_6]^{4-}$	Have large number of unpaired electrons. *e.g.*, $[MnF_6]^{3-}$, $[CoF_6]^{3-}$

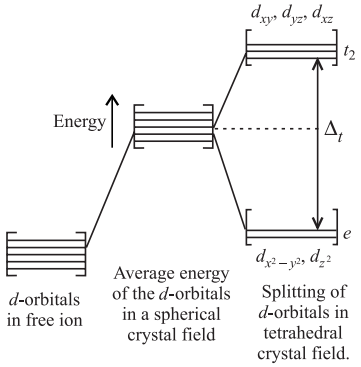
- Low spin complexes are generally diamagnetic and high spin complexes are paramagnetic.
- Paramagnetism ∝ No. of unpaired electrons.
- Magnetic moment $= \sqrt{n(n+2)}$ B.M. where n = number of unpaired electrons.

▶ **Crystal field theory :** It assumes the ligands to be point charges and there is electrostatic force of attraction between ligands and metal atom or ion. When ligands approach the central metal ion, then the five degenerate orbitals do not possess equal energy any more and results in splitting, which depends upon nature of ligand field strength.

- Greater the ease with which the ligand can approach the metal ion, the greater will be the crystal field splitting caused by it.
- Crystal field splitting in octahedral coordination complexes is shown as :



- If $\Delta_o < P$ (where 'P' is energy required for forced pairing of electrons) then the electrons will remain unpaired and a high spin complex is formed.
- If $\Delta_o > P$, then pairing of electrons takes place and a low spin complex is formed.
- Crystal field splitting in tetrahedral complexes is shown as :



- Difference in energy between e and t_2 level is less in tetrahedral complexes.

$$\Delta_t = \frac{4}{9}\Delta_o$$

- **Spectrochemical series :** Arrangement of ligands in the order of increasing field strength.

Weak field ligands → Increasing order of CFSE (Δ_o) → Strong field ligands

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < NO_2^- < CN^- < CO$

⏩ **Colour of coordination compounds :** The magnitude of CFSE (Δ_o) for most of the complexes is of the same order as the energy of a photon of visible light. Hence, whenever

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d-d transition takes place, it imparts colour to the complex. The colour of the complex is the colour complementary to the wavelength absorbed.

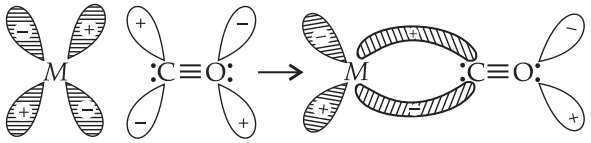
Bonding in metal carbonyls :

- Metal–carbon bond in metal carbonyls possesses both σ- and π-characters.
- Its formation involves the following steps :
  - Overlap of filled π_{2p} orbital of CO with suitable empty metal *d*-orbital resulting in the formation of sigma bond.

Vacant metal orbital | Orbital containing lone pair

$M \leftarrow$ C σ-bond

  - π-overlap involving filled metal *d*-orbital with an empty antibonding π^*2p orbital of same CO. This results in formation of $M \rightarrow$ C π-bond. This is also called *back bonding*.



Filled metal *d*-orbital | Empty π^*2p orbital of CO | $M \rightarrow C$ π-bond

Applications of coordination compounds :

- Coordination compounds are of great importance in biological system. *e.g.*, chlorophyll, haemoglobin, vitamin B_{12}, etc. are coordinate compounds of Mg, Fe and Co respectively.
- Coordination compounds are used for qualitative and quantitative analysis, extraction of metals, electroplating, photography and as dyes.
- *cis*-Platin is used in cancer treatment, EDTA is often used for treatment of lead poisoning.
- Coordination compounds are also used as catalyst.

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Previous Years' CBSE Board Questions

9.1 Werner's Theory of Coordination Compounds

VSA (1 mark)

1. One mole of $CrCl_3 \cdot 6H_2O$ compound reacts with excess $AgNO_3$ solution to yield two moles of $AgCl_{(s)}$. The structural formula of the compound is
(a) $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$
(b) $[Cr(H_2O)_3Cl_3] \cdot 3H_2O$
(c) $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$
(d) $[Cr(H_2O)_6Cl_3]$ *(2020)*

2. When a coordination compound $CrCl_3 \cdot 6H_2O$ is mixed with $AgNO_3$, 2 moles of AgCl are precipitated per mole of the compound. Write structural formula of the complex. *(AI 2019, 1/2, Delhi 2016)*

3. What is the difference between a complex and a double salt? *(AI 2019)*

4. When a coordination compound $CoCl_3 \cdot 6NH_3$ is mixed with $AgNO_3$, 3 moles of AgCl are precipitated per mole of the compound. Write structural formula of the complex. *(1/2, AI 2016)*

9.2 Definition of Some Important Terms Pertaining to Coordination Compounds

VSA (1 mark)

5. The coordination number of 'Co' in the complex $[Co(en)_3]^{3+}$ is
(a) 3 (b) 6
(c) 4 (d) 5 *(2020)*

6. **Assertion (A) :** $[Pt(en)_2Cl_2]^{2+}$ complex is less stable than $[Pt(NH_3)_4Cl_2]^{2+}$ complex.
Reason (R) : $[Pt(en)_2Cl_2]^{2+}$ complex shows chelate effect.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

7. Write the coordination number and oxidation state of platinum in the complex $[Pt(en)_2Cl_2]$. *(2018)*

8. Which of the following is more stable complex and why?
$[Co(NH_3)_6]^{3+}$ and $[Co(en)_3]^{3+}$ *(Delhi 2014)*

9. What do you understand by 'denticity of a ligand'? *(Foreign 2011)*

SA (2 marks)

10. Explain the following terms giving a suitable example in each case :
(i) Ambidenate ligand
(ii) Denticity of a ligand *(2/3, AI 2011)*

9.3 Nomenclature of Coordination Compounds

VSA (1 mark)

11. Write the formula of the following coordination compound :
Iron (III) hexacyanidoferrate(II) *(1/3, 2018)*

12. Write the IUPAC name of the following complex : $[Cr(NH_3)_3Cl_3]$. *(1/3, Delhi 2017)*

13. Write the IUPAC name of the following complex : $[Co(NH_3)_5(CO_3)]Cl$. *(1/3, Delhi 2017)*

14. When a coordination compound $CrCl_3 \cdot 6H_2O$ is mixed with $AgNO_3$, 2 moles of AgCl are precipitated per mole of the compound. Write IUPAC name of the complex. *(1/2, Delhi 2016)*

15. Write the IUPAC name of the complex $[Cr(NH_3)_4Cl_2]Cl$. *(1/3, AI 2014)*

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16. Write down the formula of :
Tetrammineaquachloridocobalt(III)chloride.
(AI 2012C)

17. Name the following coordination compound :
$K_3[CrF_6]$. *(Foreign 2011)*

18. Write the IUPAC name of
$[PtCl(NH_2CH_3)(NH_3)_2]Cl$. *(Delhi 2011C)*

19. Write the IUPAC name of $[Pt(NH_3)_4Cl_2]Cl_2$.
(AI 2011C)

20. Write the IUPAC name of
$[Cr(NH_3)_6][Co(CN)_6]$. *(AI 2011C)*

SA (2 marks)

21. Give the formulae of the following compounds :
(a) Potassium tetrahydroxidozincate(II)
(b) Hexaammineplatinum(IV) chloride
(2020)

22. Using IUPAC norms write the formulae for the following :
(i) Pentaamminenitrito–O–cobalt(III) chloride
(ii) Potassium tetracyanidonickelate(II)
(Delhi 2019)

23. Using IUPAC norms write the formulae for the following :
(a) Sodium dicyanidoaurate(I)
(b) Tetraamminechloridonitrito-N-platinum(IV) sulphate *(AI 2017)*

24. Using IUPAC norms write the formulae for the following :
(a) *Tris*(ethane-1,2,diamine)chromium(III) chloride
(b) Potassium tetrahydroxozincate(II)
(AI 2017)

25. Using IUPAC norms write the formulae for the following :
(a) Potassium trioxalatoaluminate(III)
(b) Dichlorido*bis*(ethane-1,2-diamine) cobalt(III) *(AI 2017)*

26. (i) Write down the IUPAC name of the following complex :
$[Cr(NH_3)_2Cl_2(en)]Cl$ (*en* = ethylenediamine)
(ii) Write the formula for the following complex: Pentaamminenitrito-O-cobalt (III).
(Delhi 2015)

27. Using IUPAC norms write the formulae for the following coordination compounds :
(i) Hexaamminecobalt(III) chloride
(ii) Potassium tetrachloridonickelate(II)
(AI 2015)

28. (i) Write down the IUPAC name of the following complex : $[Cr\ (en)_3]Cl_3$.
(ii) Write the formula for the following complex : Potassium trioxalatochromate (III).
(Foreign 2015)

29. Name the following coordination compounds and draw their structures :
(i) $[CoCl_2(en)_2]Cl$
(ii) $[Pt(NH_3)_2Cl(NO_2)]$
(At. No. Co = 27, Pt = 78)
(Foreign 2011)

LA I (3 marks)

30. Write the IUPAC name of the following :
(i) $[Co(NH_3)_6]Cl_3$ (ii) $[NiCl_4]^{2-}$
(iii) $K_3[Fe(CN)_6]$ *(AI 2015C)*

31. Write down the IUPAC name for each of the following complexes :
(i) $[Co(NH_3)_5Cl]Cl_2$ (ii) $K_3[Fe(CN)_6]$
(iii) $[NiCl_4]^{2-}$ *(Delhi 2014C)*

32. Write the IUPAC name and draw the structure of each of the following complex entities :
(i) $\left[Co\begin{pmatrix}COO\\|\\COO\end{pmatrix}_3\right]^{3-}$ (ii) $[Cr(CO)_6]$
(iii) $[PtCl_3(C_2H_4)]^-$
(At. nos. Cr = 25, Co = 27, Pt = 78)
(AI 2014C)

33. Write the IUPAC names of the following coordination compounds :
(i) $[Cr(NH_3)_3Cl_3]$
(ii) $K_3[Fe(CN)_6]$
(iii) $[CoBr_2(en)_2]^+$, (*en* = ethylenediamine)
(Delhi 2013)

9.4 Isomerism in Coordination Compounds

VSA (1 mark)

34. **Assertion (A) :** Linkage isomerism arises in coordination compounds because of ambidentate ligand.

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Reason (R) : Ambidentate ligand like NO_2 has two different donor atoms *i.e.*, N and O.

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is correct, but Reason (R) is incorrect statement.

(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

35. What type of isomerism is shown by the complex $[Co(en)_2 Cl_2]^{2+}$? Name the structure of an isomer of this complex which is optically active. *(1/2, 2020)*

36. What type of isomerism is exhibited by the complex $[Co(NH_3)_5Cl]SO_4$? *(1/3, 2018)*

37. What type of isomerism is shown by the complex $[Co(NH_3)_6][Cr(CN)_6]$? *(1/3, Delhi 2017, Foreign 2014)*

38. What type of isomerism is shown by the complex $[Co(en)_3]Cl_3$? *(1/3, Delhi 2017)*

39. What type of isomerism is shown by the complex $[Co(NH_3)_5(SCN)]^{2+}$? *(1/3, AI 2017)*

40. Draw one of the geometrical isomers of the complex $[Pt(en)_2Cl_2]^{2+}$ which is optically active. *(1/3, Delhi 2016)*

41. Draw one of the geometrical isomers of the complex $[Pt(en)_2Cl_2]^{2+}$ which is optically inactive. *(AI 2016)*

42. Draw the geometrical isomers of complex $[Pt(NH_3)_2Cl_2]$. *(1/3, Delhi 2015)*

43. Write down the IUPAC name of the complex $[Pt(en)_2Cl_2]^{2+}$. What type of isomerism is shown by this complex? *(AI 2015)*

44. Draw the geometrical isomers of complex $[Pt(en)_2Cl_2]^{2+}$. *(1/3, Foreign 2015)*

45. What type of isomerism is exhibited by the complex $[Co(en)_3]^{3+}$? (*en* = ethane-1,2-diamine) *(1/3, AI 2014)*

46. What type of isomerism is exhibited by the following complex :
$[Co(NH_3)_5SO_4]Cl$ *(Foreign 2014)*

47. What type of isomerism is exhibited by the complex $[Co(NH_3)_5NO_2]^{2+}$? *(Foreign 2014)*

48. Indicate the types of isomerism exhibited by the complex $[Co(NH_3)_5(NO_2)](NO_3)_2$. (At no. Co = 27) *(AI 2012C)*

49. Give IUPAC name of ionisation isomer of $[Ni(NH_3)_3NO_3]Cl$. *(AI 2012C)*

SA (2 marks)

50. Write IUPAC name of the complex $[Cr(NH_3)_4Cl_2]^+$. Draw structures of geometrical isomers for this complex. *(Delhi 2019)*

51. Draw the structure of isomers, if any and write the name of the following complexes :
(i) $[Cr(NH_3)_4Cl_2]^+$
(ii) $[Co(en)_3]^{3+}$ *(Foreign 2011)*

LA I (3 marks)

52. Indicate the types of isomerism exhibited by the following complexes :
(i) $[Co(NH_3)_5(NO_2)]^{2+}$
(ii) $[Co(en)_3]Cl_3$ (*en* = ethylene diamine)
(iii) $[Pt(NH_3)_2Cl_2]$ *(Delhi 2015C)*

53. Write the IUPAC name of the complex $[Cr(NH_3)_4Cl_2]^+$. What type of isomerism does it exhibit? *(Delhi 2014)*

54. Draw the structures of optical isomers of each of the following complex ions :
$[Cr(C_2O_4)]^{3-}$, $[PtCl_2(en)_2]^{2+}$, $[Cr(NH_3)_2Cl_2(en)]^+$ *(Delhi 2014C)*

55. Write the types of isomerism exhibited by the following complexes :
(i) $[Co(NH_3)_5Cl]SO_4$
(ii) $[Co(en)_3]^{3+}$
(iii) $[Co(NH_3)_6]$ $[Cr(CN)_6]$ *(Delhi 2013)*

56. Name the following coordination entities and draw the structures of their stereoisomers :
(i) $[Co(en)_2Cl_2]^+$ (*en* = ethane-1,2-diamine)
(ii) $[Cr(C_2O_4)_3]^{3-}$
(iii) $[Co(NH_3)_3Cl_3]$
(Atomic numbers Cr = 24, Co = 27) *(AI 2012)*

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57. Write the structures and names of all the stereoisomers of the following compounds :
(i) $[Co(en)_3]Cl_3$ (ii) $[Pt(NH_3)_2Cl_2]$
(iii) $[Fe(NH_3)_4Cl_2]Cl$ *(AI 2011)*

9.5 Bonding in Coordination Compounds

VSA (1 mark)

58. The crystal field splitting energy for octahedral (Δ_0) and tetrahedral (Δ_t) complexes is related as
(a) $\Delta_t = \frac{2}{9}\Delta_0$ (b) $\Delta_t = \frac{5}{9}\Delta_0$
(c) $\Delta_t = \frac{4}{9}\Delta_0$ (d) $\Delta_t = 2\,\Delta_0$ *(2020)*

59. Write the IUPAC name and hybridisation of the complex $[CoF_6]^{3-}$ (Given : Atomic number of Co = 27) *(1/2, 2020)*

60. Write the hybridisation and number of unpaired electrons in the complex $[CoF_6]^{3-}$. (Atomic no. of Co = 27) *(1/3, 2018)*

61. Why a solution of $[Ni(H_2O)_6]^{2+}$ is green while a solution of $[Ni(CN)_4]^{2-}$ is colourless? (At. no. of Ni = 28) *(1/3, Delhi 2017)*

62. Write the hybridisation and magnetic character of $[Co(C_2O_4)_3]^{3-}$. (At. no. of Co = 27) *(1/3, Delhi 2017)*

63. Why is $[NiCl_4]^{2-}$ paramagnetic but $[Ni(CO)_4]$ is diamagnetic? (At. no. : Cr = 24, Co = 27, Ni = 28) *(1/3, AI 2014)*

64. Explain the following term giving a suitable example :
Crystal field splitting in an octahedral field. *(1/3, AI 2011)*

SA (2 marks)

65. Write IUPAC name and hybridization of the following complexes :
(i) $[Ni(CO)_4]$ (ii) $[CoF_6]^{3-}$
(Atomic number Ni = 28, Co = 27) *(2020)*

66. Out of $[CoF_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$, which one complex is
(i) diamagnetic (ii) more stable
(iii) outer orbital complex and
(iv) low spin complex ?
(Atomic no. of Co = 27) *(Delhi 2019)*

67. (a) Why is $[NiCl_4]^{2-}$ paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic?
(Atomic number of Ni = 28)
(b) Why are low spin tetrahedral complexes rarely observed? *(2/3, AI 2017)*

68. For the complex $[Fe(CN)_6]^{3-}$, write the hybridization type, magnetic character and spin nature of the complex.
(At. number : Fe = 26). *(2/3, Delhi 2016)*

69. For the complex $[Fe(H_2O)_6]^{3+}$, write the hybridization, magnetic character and spin of the complex. (At. number Fe = 26) *(2/3,AI 2016)*

70. (i) On the basis of crystal field theory, write the electronic configuration of d^4 ion if $\Delta_o < P$.
(ii) Write the hybridization and magnetic behaviour of the complex $[Ni(CO)_4]$.
(At. no. of Ni =28) *(2/3, Delhi 2015)*

71. Write the hybridization and shape of the following complexes :
(i) $[CoF_6]^{3-}$ (ii) $[Ni(CN)_4]^{2-}$
(Atomic number : Co = 27, Ni = 28) *(AI 2015)*

72. Write the state of hybridization, shape and IUPAC name of the complex $[CoF_6]^{3-}$. (Atomic no. of Co = 27) *(Foreign 2014)*

73. Write the state of hybridization, shape and IUPAC name of the complex $[Ni(CN)_4]^{2-}$. (Atomic no. of Ni = 28) *(Foreign 2014)*

74. Write the state of hybridization, shape and IUPAC name of the complex $[Co(NH_3)_6]^{3+}$. (Atomic no. of Co = 27) *(Foreign 2014)*

75. Give the formula of each of the following coordination entities :
(i) Co^{3+} ion is bound to one Cl^-, one NH_3 molecule and two bidentate ethylene diamine (*en*) molecules.
(ii) Ni^{2+} ion is bound to two water molecules and two oxalate ions.
Write the name and magnetic behaviour of each of the above coordination entities.
(At. nos. Co = 27, Ni = 28) *(2/3, Delhi 2012)*

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76. State a reason for each of the following situations :
(i) Co^{2+} is easily oxidised to Co^{3+} in presence of a strong ligand.
(ii) The molecular shape of $Ni(CO)_4$ is not the same as that of $[Ni(CN)_4]^{2-}$.
(2/3, Delhi 2012)

77. Using valence bond theory, explain the geometry and magnetic behaviour of $[Co(NH_3)_6]^{3+}$.
(At. no. of Co = 27) *(2/3, Delhi 2012C)*

78. Using valence bond theory of complexes, explain the geometry and magnetic nature of $[Ni(CN)_4]^{2-}$. (At. no. of Ni = 28)
(2/3, Delhi 2012C)

79. $[Fe(H_2O)_6]^{3+}$ is strongly paramagnetic whereas $[Fe(CN)_6]^{3-}$ is weakly paramagnetic. Explain (At. no. Fe = 26] *(Delhi 2012C)*

80. Give the name, the stereochemistry and the magnetic behaviour of the following complexes :
(i) $[Co(NH_3)_5Cl]Cl_2$
(ii) $K_2[Ni(CN)_4]$ *(Foreign 2011)*

LA I (3 marks)

81. Write the hybridisation and magnetic character of the following complexes :
(i) $[Fe(H_2O)_6]^{2+}$ (ii) $[Ni(CN)_4]^{2-}$
[Atomic number : Fe = 26, Ni = 28]
(AI 2019)

82. For the complex $[NiCl_4]^{2-}$, write
(i) the IUPAC name
(ii) the hybridization type
(iii) the shape of the complex.
(Atomic no. of Ni = 28) *(AI 2013)*

83. What is meant by crystal field splitting energy? On the basis of crystal field theory, write the electronic configuration of d^4 in terms of t_{2g} and e_g in an octahedral field when
(i) $\Delta_o > P$
(ii) $\Delta_o < P$ *(AI 2013)*

84. Write the name, the structure and the magnetic behaviour of each one of the following complexes :
(i) $[Pt(NH_3)_2Cl(NO_2)]$
(ii) $[Co(NH_3)_4Cl_2]Cl$ (iii) $Ni(CO)_4$
(At. nos. Co = 27, Ni = 28, Pt = 78)
(Delhi 2012)

85. Name the following coordination entities and describe their structures.
(i) $[Fe(CN)_6]^{4-}$
(ii) $[Cr(NH_3)_4Cl_2]^{+}$
(iii) $[Ni(CN)_4]^{2-}$ *(AI 2012)*

86. Write the name, stereochemistry and magnetic behaviour of the following :
(At. nos. Mn = 25, Co = 27, Ni = 28)
(i) $K_4[Mn(CN)_6]$
(ii) $[Co(NH_3)_5Cl]Cl_2$
(iii) $K_2[Ni(CN)_4]$ *(Delhi 2011)*

87. For the complex $[Fe(en)_2Cl_2]Cl$, identify the following :
(i) Oxidation number of iron
(ii) Hybrid orbitals and shape of the complex
(iii) Magnetic behaviour of the complex
(iv) Number of its geometrical isomers
(v) Whether there may be optical isomer also.
(vi) Name of the complex. *(Delhi 2011)*

88. Write the state of hybridization, the shape and the magnetic behaviour of the following complex entities :
(i) $[Cr(NH_3)_4Cl_2]Cl$
(ii) $[Co(en)_3]Cl_3$
(iii) $K_2[Ni(CN)_4]$ *(AI 2011)*

9.6 Bonding in Metal Carbonyls

VSA (1 mark)

89. Out of NH_3 and CO, which ligand forms a more stable complex with a transition metal and why? *(1/3, AI 2015)*

90. State a reason for the following situation :
CO is a stronger complexing reagent than NH_3. *(1/3, Delhi 2012)*

91. Why is CO a stronger ligand than Cl^-?
(Foreign 2011)

9.7 Importance and Applications of Coordination Compounds

VSA (1 mark)

92. Give two examples of coordination compounds used in industries.
(1/3, Delhi 2012C)

93. Give names of two complexes which are used in medicines. *(1/3, Delhi 2012C)*

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Detailed Solutions

1. For one mole of the compound, two moles of AgCl are precipitated which indicates two ionisable chloride ions in the complex. Hence, its structural formula is $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$

2. *Refer to answer 1.*

3. Double salts dissociate into ions completely when dissolved in water. On the other hand, in complexes, the complex ion does not dissociate.

4. Structural formula : $[Co(NH_3)_6]Cl_3$

5. **(b) :** Ethane-1, 2-diamine (*en*) is a bidendate ligand.

6. **(d) :** Formation of chelate rings increases the stability of the complex.

7. Coordination number and oxidation state of Pt in the complex $[Pt(en)_2Cl_2]$ are 6 and +2 because *en* is a bidentate and neutral ligand.

8. $[Co(en)_3]^{3+}$ is more stable complex than $[Co(NH_3)_6]^{3+}$ due to chelate effect as it forms rings.

9. **Denticity :** The number of coordinating groups present in a ligand is called the denticity of ligand. For example, bidentate ligand ethane-1, 2-diamine has two donor nitrogen atoms which can link to central metal atom.

$$H_2\ddot{N}-CH_2-CH_2-\ddot{N}H_2$$

Ethane-1, 2-diamine

10. (i) **Ambidentate ligand :** A unidentate ligand which can coordinate to central metal atom through two different atoms is called ambidentate ligand. For example, NO_2^- ion can coordinate either through nitrogen or through oxygen to the central metal atom/ion.

(ii) *Refer to answer 9.*

11. $Fe_4[Fe(CN)_6]_3$

12. IUPAC name of the complex $[Cr(NH_3)_3Cl_3]$ is triamminetrichloridochromium(III).

13. Pentaamminecarbonatocobalt(III) chloride

14. IUPAC name of the complex $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$ is : pentaaquachloridochromium (III) chloride.

15. IUPAC name of the complex $[Cr(NH_3)_4Cl_2]Cl$ is tetraamminedichloridochromium(III) chloride.

16. $[Co(NH_3)_4(H_2O)(Cl)]Cl_2$

17. Potassium hexafluoridochromate(III).

18. Diamminechlorido(methylamine)platinum(II) chloride.

19. Tetraamminedichloridoplatinum(IV) chloride.

20. Hexaamminechromium(III) hexacyanidocobaltate(III).

21. (a) $K_2[Zn(OH)_4]$
Potassium tetrahydroxidozincate(II)

(b) $[Pt(NH_3)_6]Cl_4$
Hexaammineplatinum(IV) chloride

22. (i) $[Co(NH_3)_5(ONO)]Cl_2$
(ii) $K_2[Ni(CN)_4]$

23. (a) $Na[Au(CN)_2]$
Sodium dicyanidoaurate(I)

(b) $[Pt(NH_3)_4(Cl)(NO_2)]SO_4$
Tetraamminechloridonitrito-N-platinum(IV) sulphate

24. (a) $[Cr(en)_3]Cl_3$
Tris(ethane-1,2-diamine)chromium(III) chloride

(b) $K_2[Zn(OH)_4]$
Potassium tetrahydroxidozincate(II)

25. (a) $K_3[Al(C_2O_4)_3]$
Potassium trioxalatoaluminate(III)

(b) $[CoCl_2(en)_2]^+$
Dichlorido*bis*(ethane–1,2-diamine)cobalt(III) ion

26. (i) Diamminedichlorido(ethane-1,2-diamine) chromium(III) chloride.

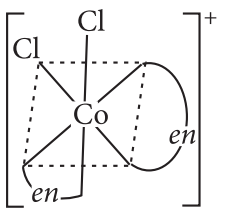
(ii) $[Co(NH_3)_5(ONO)]^{2+}$

27. (i) $[Co(NH_3)_6]Cl_3$ (ii) $K_2[NiCl_4]$

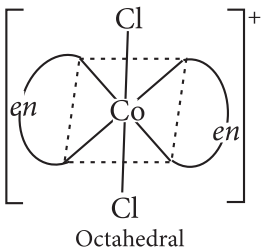
28. (i) *Tris*(ethylenediammine)chromium(III) chloride

(ii) $K_3[Cr(ox)_3]$

29. (i) Dichlorido*bis* (ethane-1, 2-diamine) cobalt(III) chloride.



Octahedral
cis-$[CoCl_2(en)_2]^+$



Octahedral
trans-$[CoCl_2(en)_2]^+$

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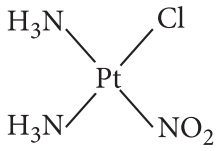
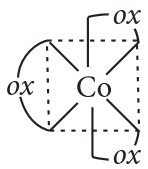
(ii) Diamminechloridonitrito-N-platinum(II)



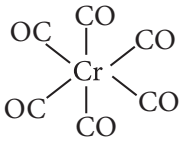
30. (i) Hexaamminecobalt(III) chloride
(ii) Tetrachloridonickelate(II) ion
(iii) Potassium hexacyanidoferrate(III)

31. (i) Pentaamminechloridocobalt(III) chloride
(ii) *Refer to answer 30(iii).*
(iii) *Refer to answer 30(ii).*

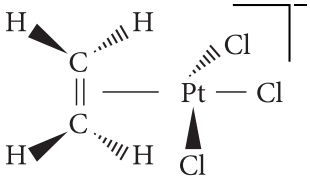
32. (i) Trioxalatocobaltate(III)



(ii) Hexacarbonylchromium(0)



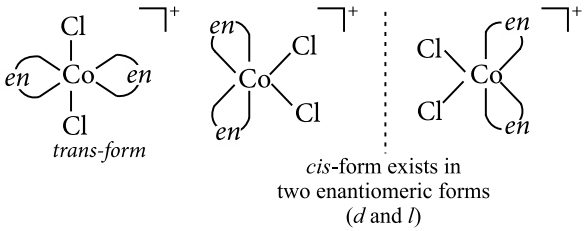
(iii) Trichloridoetheneplatinum(III)



33. (i) Triamminetrichloridochromium(III)
(ii) *Refer to answser 30(iii)*
(iii) Dibromido*bis*(ethane-1,2-diamine) cobalt(III) ion

34. (a)

35. $[CoCl_2(en)_2]^+$: Dichlorido*bis*(ethane-1, 2-diamine)cobalt(III) ion
This complex exhibit both optical and geometrical isomerism and *cis* isomer is optically active.



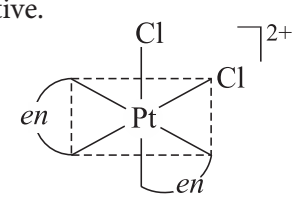
36. Ionisation isomerism : $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]Cl$

37. Coordination isomerism

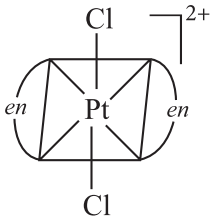
38. The complex, $[Co(en)_3]Cl_3$ shows optical isomerism.

39. The complex $[Co(NH_3)_5(SCN)]^{2+}$ shows linkage isomerism as SCN^- is an ambidentate ligand.

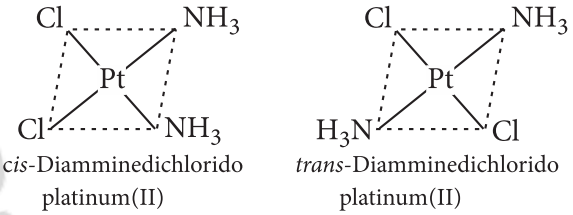
40. *cis*-isomer of the complex $[Pt(en)_2Cl_2]^{2+}$ is optically active.



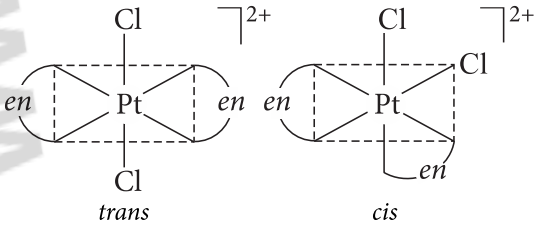
41. *Trans*-isomer is optically inactive due to the presence of plane of symmetry.



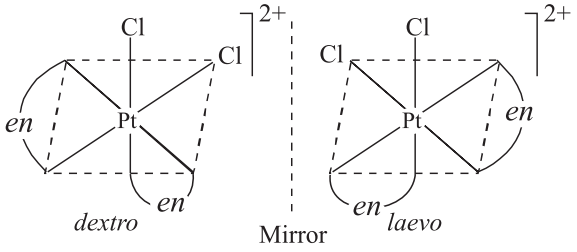
42. Geometrical isomers



43. IUPAC name - Dichlorido*bis*(ethane-1,2-diamine)platinum(IV) ion. This complex shows geometrical and optical isomerism.



cis-isomer is optically active.



44. *Refer to answer 43.*

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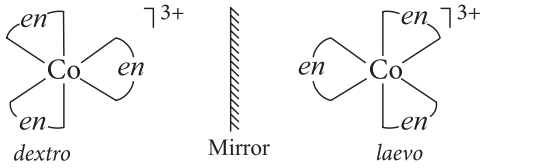
45. Optical isomerism :



Tris(ethane-1, 2-diamine) cobalt(III) ion — *Tris*(ethane-1, 2-diamine) cobalt(III) ion

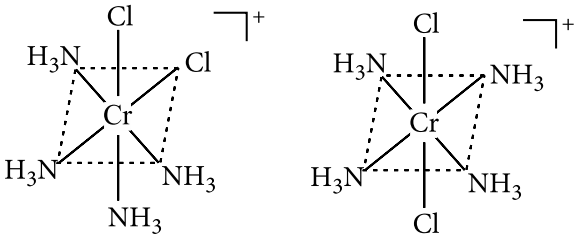
46. *Refer to answer 36.*

47. Linkage isomerism : $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(ONO)(NH_3)_5]^{2+}$

48. (i) Ionisation isomers :
$[Co(NH_3)_5(NO_2)](NO_3)_2$ and
$[Co(NH_3)_5(NO_3)](NO_2)(NO_3)$
(ii) Linkage isomers : $[Co(NH_3)_5(NO_2)](NO_3)_2$ and $[Co(NH_3)_5(ONO)](NO_3)_2$

49. Ionisation isomer of $[Ni(NH_3)_3NO_3]Cl$ is $[Ni(NH_3)_3Cl]NO_3$. The IUPAC name of this complex is triamminechloridonickel(II) nitrate.

50. *Tetra*amminedichloridochromium(III) ion
Geometrical isomers of $[Cr(NH_3)_4Cl_2]^+$;



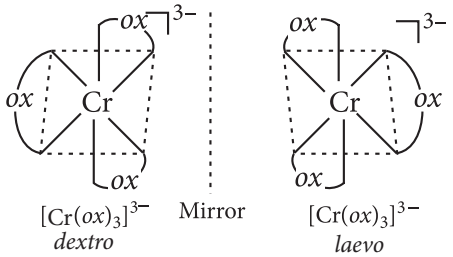
cis-Tetraamminedichlorido chromium(III) ion — *trans*-Tetraamminedichlorido chromium(III) ion

51. (i) *Refer to answer 50.*
(ii) *Refer to answer 45.*

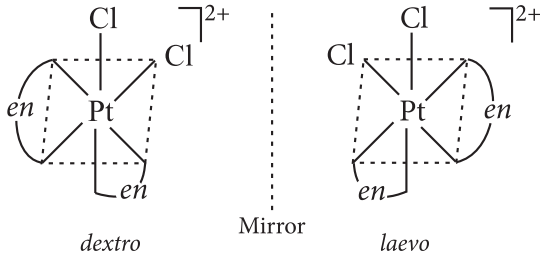
52. (i) *Refer to answer 47.*
(ii) *Refer to answer 45.*
(iii) *Refer to answer 42.*

53. *Refer to answer 50.*

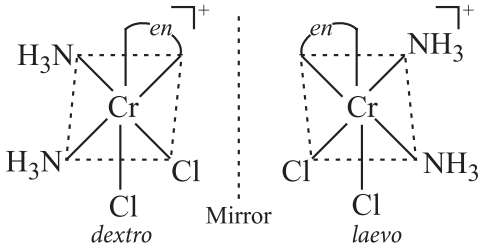
54. $[Cr(C_2O_4)]^{3-}$:



$[PtCl_2(en)_2]^{2+}$:

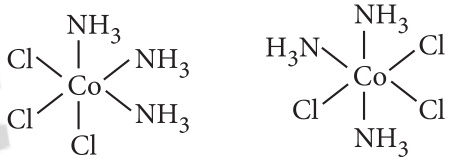


$[Cr(NH_3)_2Cl_2(en)]^+$:

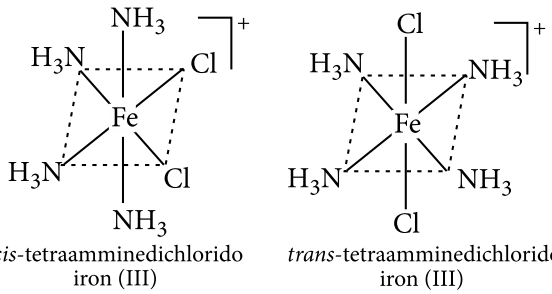


55. (i) *Refer to answer 36.*
(ii) *Refer to answer 45.*
(iii) *Refer to answer 37.*

56. (i) *Refer to answer 35.*
(ii) $[Cr(C_2O_4)_3]^{3-}$: Trioxalatochromate(III) ion
It will show optical isomerism.
Refer to answer 54.
(iii) $[Co(NH_3)_3Cl_3]$: Triamminetrichloridocobalt(III)
It will exist in two geometrical isomeric forms :



57. (i) *tris*(ethane-1,2-diamine) cobalt(III) chloride : *Refer to answer 45.*
(ii) *Refer to answer 42.*
(iii) $[Fe(NH_3)_4Cl_2]Cl$



58. **(c)**

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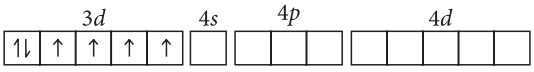
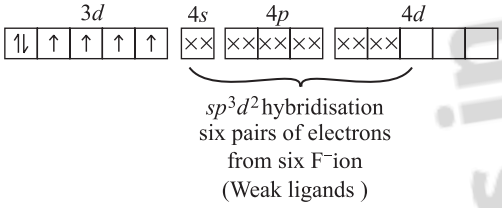
59. The IUPAC name of $[CoF_6]^{3-}$ is hexafluoridocobaltate(III).
Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.

Co^{3+} :



Co^{3+} in $[CoF_6]^{3-}$:



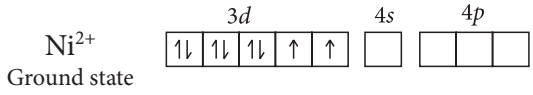
No. of unpaired electrons = 4

60. *Refer to answer 59.*

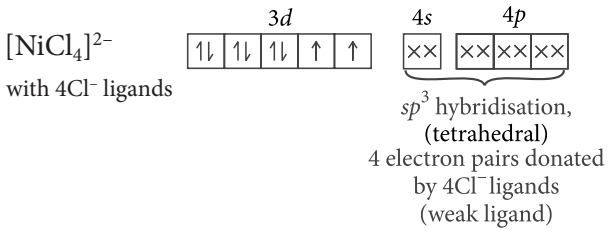
61. $[Ni(H_2O)_6]^{2+}$ is a high spin complex (Δ_o small) while $[Ni(CN)_4]^{2-}$ is a low spin square planar complex.
In $[Ni(H_2O)_6]^{2+}$ complex, *d-d* transitions are taking place on absorbing low energy radiation (red component of spectrum) from visible region showing green as the complementary colour.
In $[Ni(CN)_4]^{2-}$ complex, *d-d* transitions do not take place because there is no unpaired electron present hence, complex is colourless.

62. The complex $[Co(C_2O_4)_3]^{3-}$ has d^2sp^3-hybridisation and it is diamagnetic in nature due to the absence of unpaired electrons.

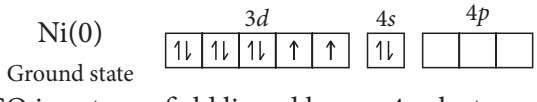
63. $[NiCl_4]^{2-}$ contains Ni^{2+} ion with $3d^8$ configuration.



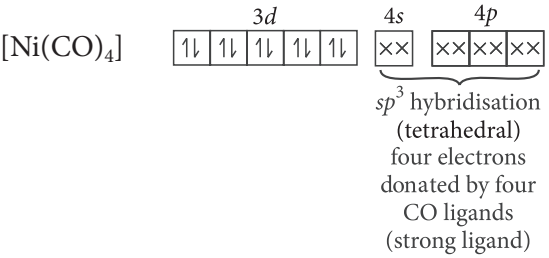
Cl^- is a weak field ligand. Hence, outer 4*s* and 4*p*-orbitals are used in hybridisation.



It has two unpaired electrons hence, it is paramagnetic.
$[Ni(CO)_4]$ contains Ni(0) – $3d^84s^2$ configuration.

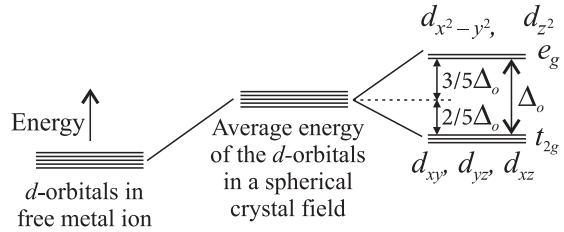


CO is a strong field ligand hence, 4*s*-electrons will shift to 3*d*-orbital making 4*s*-orbital vacant.



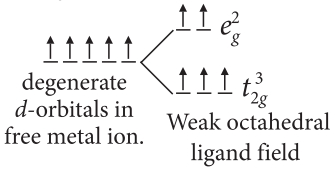
The complex has all paired electrons hence, it is diamagnetic.

64. The splitting of the degenerate *d*-orbitals into three orbitals of lower energy, t_{2g} set and two orbitals of higher energy e_g set due to the interaction of ligand in an octahedral crystal field is known as crystal field splitting in an octahedral field.

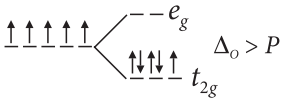


d-orbital splitting in an octahedral crystal field

$[Mn(H_2O)_6]^{2+}$ contains : Mn^{2+} – $3d^5$ configuration, C.N. = 6, octahedral structure, H_2O (weak field ligand) hence $\Delta_o < P$.



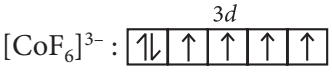
Electrons enter into e_g orbital before pairing. It has five unpaired electrons.
$[Mn(CN)_6]^{4-}$ contains : Mn^{2+} ion-$3d^5$ configuration, C.N. = 6, octahedral geometry, CN^- (strong field ligand) hence, $\Delta_o > P$.



Hence, pairing of electrons in t_{2g} orbitals takes place. It has only one unpaired electron.

65. (i) Name; Tetracarbonylnickel (0)
For hybridisation; *Refer to answer 63.*
(ii) *Refer to answer 59.*

66. Formation of $[CoF_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$ can be represented as :



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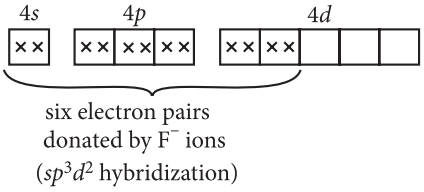
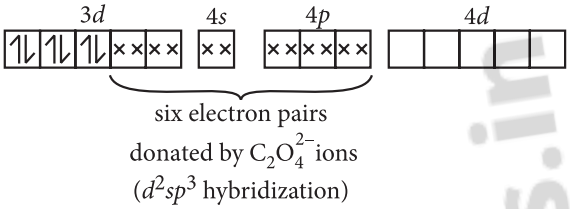


$[Co(C_2O_4)_3]^{3-}$:



(i) $[Co(C_2O_4)_3]^{3-}$ is diamagnetic as all electrons are paired.

(ii) $[Co(C_2O_4)_3]^{3-}$ is more stable as $C_2O_4^{2-}$ is a chelating ligand and forms chelate rings.

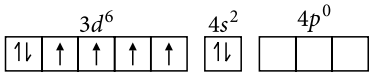
(iii) $[CoF_6]^{3-}$ is outer orbital complex as it undergoes sp^3d^2 hybridization using the outer 4*d*-orbital.

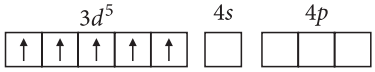
(iv) $[Co(C_2O_4)_3]^{3-}$ is low spin complex due to absence of any unpaired electron.

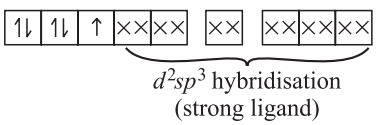
67. (a) Due to the presence of weak field ligand, *i.e.*, Cl^- in the complex $[NiCl_4]^{2-}$ two unpaired electrons are present in 3*d*-orbitals of Ni-atom hence, this complex is paramagnetic in nature. On the other hand, due to the presence of strong field ligand *i.e.*, CN^- in the complex $[Ni(CN)_4]^{2-}$, no unpaired electron is present in 3*d*-orbitals of Ni-atom (as strong field ligand causes pairing of electrons), hence, it is diamagnetic in nature.

(b) In tetrahedral coordination entity formation, the *d*-orbital splitting is smaller. Consequently, the orbital splitting energies are not sufficiently large to force pairing and therefore, low spin configurations are rarely observed.

68. Fe atom ($Z = 26$)

Ground state : 

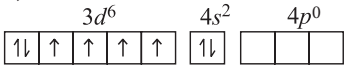
Fe^{3+} ion : 

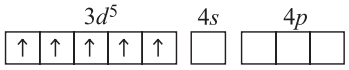
$[Fe(CN)_6]^{3-}$ ion 

The complex ion has inner orbital octahedral geometry (low spin) and is paramagnetic due to the presence of one unpaired electron.

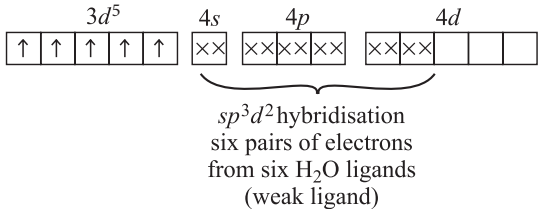
69. $[Fe(H_2O)_6]^{3+}$:

Fe atom ($Z = 26$)

Ground state : 

Fe^{3+} ion : 

$[Fe(H_2O)_6]^{3+}$:

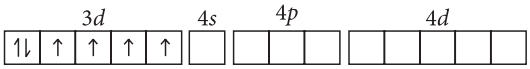


The complex ion has outer orbital octahedral geometry (high spin) and is paramagnetic due to the presence of five unpaired electrons.

70. (i) For d^4 ion, if $\Delta_o < P$, the fourth electron enters one of the e_g orbitals giving the configuration $t_{2g}^3\ e_g^1$. Ligands for which $\Delta_o < P$ are known as weak field ligands and form high spin complexes.

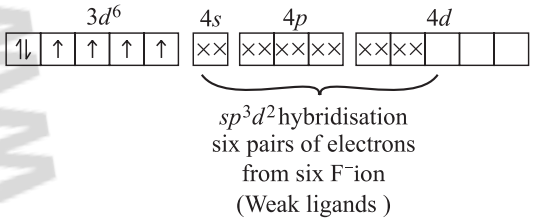
(ii) *Refer to answer 63.*

71. (i) Oxidation state of Co ion in $[CoF_6]^{3-}$ is +3.

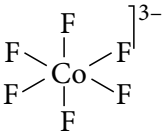
Co^{3+} : 

(In ground state)

$[CoF_6]^{3-}$:

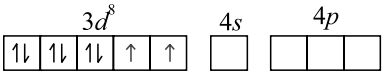


Structure – Outer orbital octahedral complex



Nature – Paramagnetic

(ii) In $[Ni(CN)_4]^{2-}$: Ni is present as Ni(II) with $3d^8$ configuration.



(In ground state)

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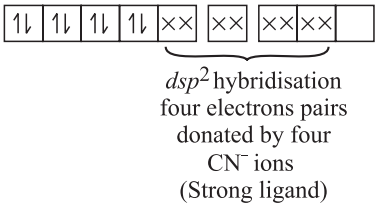
$[Ni(CN)_4]^{2-}$:



The complex ion has square planar geometry and is diamagnetic in nature.

72. *Refer to answer 71(i).*

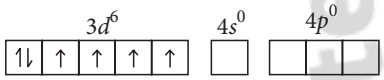
IUPAC Name : Hexafluoridocobaltate(III) ion

73. *Refer to answer 71(ii).*

IUPAC name : Tetracyanidonickelate(II) ion

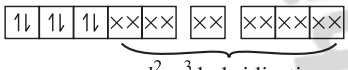
74. Oxidation of cobalt in $[Co(NH_3)_6]^{3+}$ is +3.

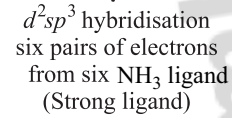
Co^{3+} :



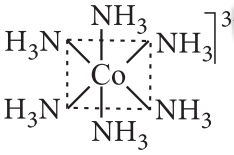
Ground State

$[Co(NH_3)_6]^{3+}$ ion :





Hybridisation – d^2sp^3
Structure – Octahedral
(low spin)
Nature – Diamagnetic



IUPAC name : Hexaamminecobalt(III) ion

75. (i) $[Co(en)_2Cl(NH_3)]^{2+}$

Amminechlorido*bis*(ethane-1,2-diamine)cobalt(III) ion

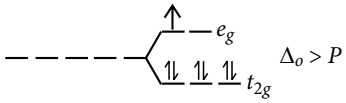
In presence of strong NH_3 and *en* ligand, Co^{3+} ($3d^6$) forms low spin complex. Hence, complex is diamagnetic.

(ii) $[Ni(C_2O_4)_2(H_2O)_2]^{2-}$:

Diaquadioxalatonickelate(II) ion

In the presence of weak H_2O and *ox* ligand, Ni(II) forms high spin complex (sp^3d^2 hybridisation). It is paramagnetic.

76. (i) In presence of strong field ligand, Co(II) has electronic configuration $t_{2g}^6 e_g^1$.



It can easily lose one electron present in e_g orbital to give stable t_{2g}^6 configuration. This is why Co^{2+} is easily oxidised to Co^{3+} in the presence of strong field ligand.

(ii) *Refer to answers 71(ii) and 63.*

77. *Refer to answer 74.*

78. *Refer to answer 71(ii).*

79. In presence of CN^-, the $3d$ electrons pair up leaving only one unpaired electron, the hybridisation is d^2sp^3 forming an inner orbital octahedral complex whereas in the presence of H_2O, $3d$ electrons do not pair up. The hybridisation is sp^3d^2 forming an outer orbital octahedral complex containing five unpaired electrons. Hence, it is strongly paramagnetic.

80. (i) $[Co(NH_3)_5Cl]Cl_2$: Pentaamminechloridocobalt (III) chloride.

Co(III) : $[Ar]3d^6 4s^0$: d^2sp^3 hybridisation leads to octahedral shape.

Magnetic behaviour : Diamagnetic.

Stereochemistry : Complex $[Co(NH_3)_5Cl]Cl_2$ do not exhibit geometrical and optical isomerism.

(ii) $K_2[Ni(CN)_4]$: Potassium tetracyanidonickelate(II)

Ni(II) : $[Ar]\ 3d^8 4s^0$ – dsp^2 hybridisation leads to square planar shape.

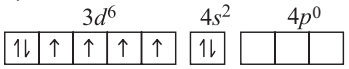
Magnetic behaviour : Diamagnetic.

Stereochemistry : Complex $[Ni(CN)_4]$ do not exhibit geometrical isomerism and optical isomerism because all the possible arrangement are equivalent.

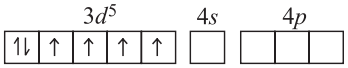
81. (i) $[Fe(H_2O)_6]^{2+}$:

Fe atom ($Z = 26$)

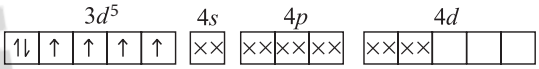
Ground state :

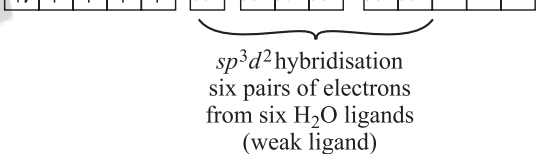


Fe^{2+} ion :



$[Fe(H_2O)_6]^{2+}$:





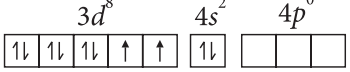
The complex ion has outer orbital octahedral geometry (high spin) and is paramagnetic due to the presence of four unpaired electrons.

(ii) *Refer to answer 71(ii).*

82. Tetrachloridonickelate(II) ion

Ni atom ($Z = 28$)

Ground state :



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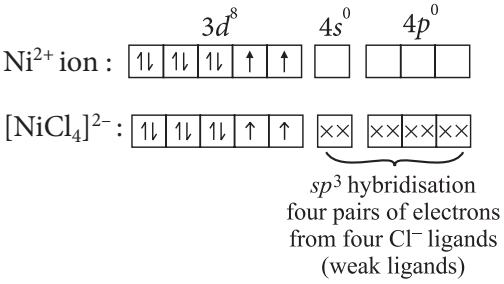
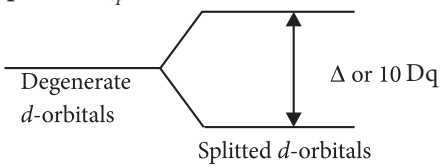


The complex ion has tetrahedral geometry and is paramagnetic due to the presence of unpaired electrons.

83. The difference of energy between two splitted levels of *d*-orbitals is called crystal field splitting energy. It is denoted by Δ or 10 Dq.

For octahedral Δ_o, for tetrahedral it is Δ_t and for square planar Δ_{sp}.



(i) When $\Delta_o > P$, $t_{2g}^4 e_g^0$

(ii) When $\Delta_o < P$, $t_{2g}^3 e_g^1$

84. (i) $[Pt(NH_3)_2Cl(NO_2)]$:

Diamminechloridonitrito-N-platinum(II)

It is square planar and diamagnetic.

(ii) $[Co(NH_3)_4Cl_2]Cl$:

Tetraamminedichloridocobalt (III) chloride

It is octahedral and diamagnetic.

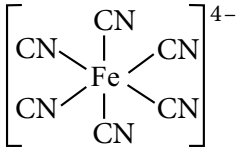
(iii) $Ni(CO)_4$: Tetracarbonylnickel(0)

It is tetrahedral and diamagnetic.

85. (i) $[Fe(CN)_6]^{4-}$: Hexacyanidoferrate(II) ion

Hybridisation - d^2sp^3

Structure : Inner orbital octahedral complex



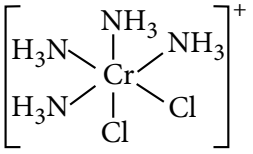
Octahedral

(ii) $[Cr(NH_3)_4Cl_2]^+$:

Tetraamminedichloridochromium(III) ion

Hybridisation - d^2sp^3

Structure : Inner orbital octahedral complex

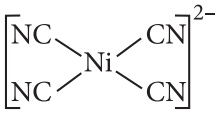


Octahedral

(iii) $[Ni(CN)_4]^{2-}$: Tetracyanidonickelate(II) ion

Hybridisation - dsp^2

Structure - Square planar



Square planar

86.

	Formulae	Name of the Complex	Hybridisation of metal ion involved	Geometry of complex (stereochemistry)	Magnetic behaviour
(i)	$K_4[Mn(CN)_6]$	Potassium hexacyanidomanganate(II)	d^2sp^3	Octahedral	Paramagnetic
(ii)	$[Co(NH_3)_5Cl]Cl_2$	Pentaamminechloridocobalt(III) chloride	d^2sp^3	Octahedral	Diamagnetic
(iii)	$K_2[Ni(CN)_4]$	Potassiumtetracyanidonikelate (II)	dsp^2	Square planar	Diamagnetic

None of these complexes will show stereoisomerism.

87. (i) $[Fe(en)_2Cl_2]Cl$

$x + 0 \times 2 + (-1) \times 2 + (-1) \times 1 = 0$

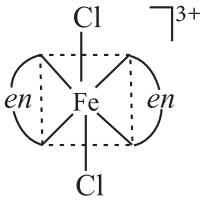
$\therefore \quad x = +3$

Oxidation number of iron = +3

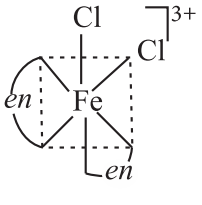
(ii) d^2sp^3 hybridisation and octahedral shape.

(iii) Paramagnetic due to presence of one unpaired electron.

(iv) 2, *cis*- and *trans*-isomers.



trans-bis(ethane-1,2-diamine) dichloridoiron(III) chloride



cis-bis(ethane-1,2-diamine) dichloridoiron(III) chloride

(v) *cis*-$[Fe(en)_2Cl_2]$ has optical isomer.

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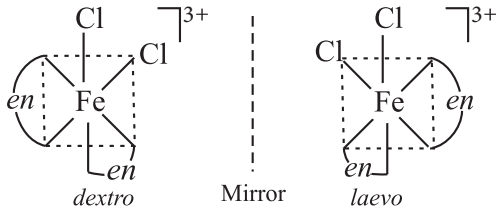

(vi) dichlorido*bis*(ethane-1,2-diamine)iron(III) chloride

88.

	Complex	**Central metal ion/ atom**	**Hybridisation of metal ion involved**	**Geometry of complex**	**Magnetic behaviour**
(i)	$[Cr(NH_3)_4Cl_2]Cl$	Cr^{3+}	d^2sp^3	Octahedral	Paramagnetic
(ii)	$[Co(en)_3]Cl_3$	Co^{3+}	d^2sp^3	Octahedral	Diamagnetic
(iii)	$K_2[Ni(CN)_4]$	Ni^{2+}	dsp^2	Square planar	Diamagnetic

89. In CO both lone pair of electrons and vacant π^* orbitals are present. Hence, it acts as electron pair σ donor as well as π acceptor by back bonding. Hence, *M*—CO bond is stronger.

$$M \underset{\sigma}{\overset{\pi}{\rightleftharpoons}} CO$$

NH_3 is electron pair donor only. Accumulation of negative charge on the metal ion takes place, hence *M*—NH_3 bond is weaker.

$M \leftarrow NH_3$

90. *Refer to answer 89.*

91. Because CO has vacant molecular orbitals with which it can form π-bond with metal through back donation.

92. (i) Na_2EDTA is used for estimation of hardness of water.

(ii) Wilkinson's catalyst is used as catalyst for hydrogenation.

93. *cis*-Platin is used in the treatment of cancer. EDTA is used in lead poisoning.

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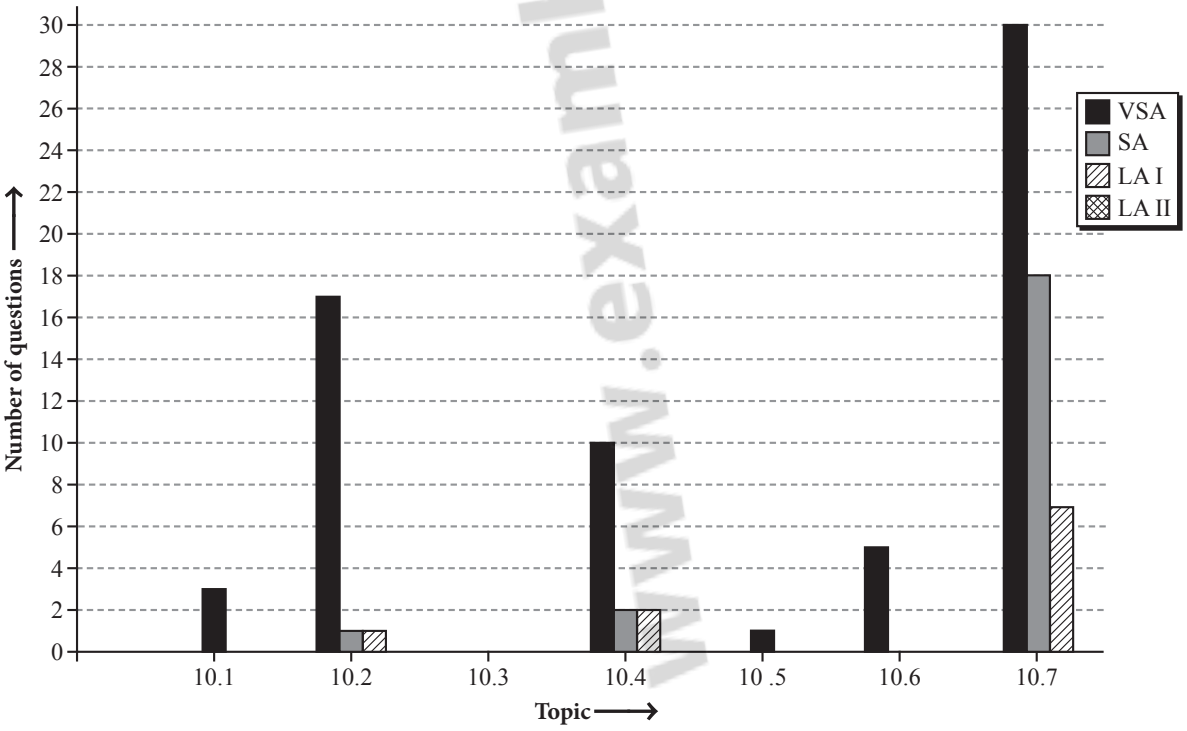
CHAPTER
10

Haloalkanes and Haloarenes

10.1 Classification
10.2 Nomenclature
10.3 Nature of C—*X* Bond
10.4 Methods of Preparation of Haloalkones
10.5 Methods of Preparation of Haloarenes
10.6 Physical Properties
10.7 Chemical Reactions

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Chemical Reactions.*
- Maximum VSA type questions were asked from *Chemical Reactions.*
- Maximum SA and LA I type questions were asked from *Chemical Reactions.*

QUICK RECAP

Haloalkanes and haloarenes : The replacement at hydrogen atom(s) in a hydrocarbon, aliphatic or aromatic, by halogen atom(s) results in the formation of alkyl halide (haloalkane) and aryl halide (haloarene) respectively.

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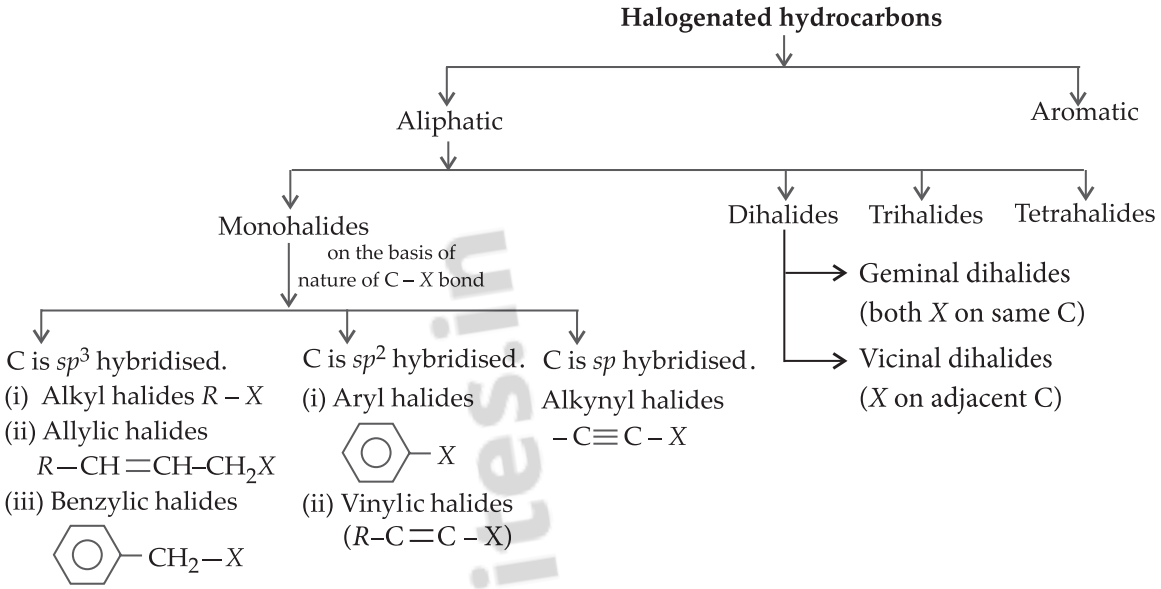
Classification :



HALOALKANES

Alkyl halides : General formula is *RX*, where *R* = alkyl group.

General methods of preparation :

▶ **From alcohols :** Alkyl halides can be prepared from alcohols in a number of ways :

– **Action of hydrogen halides :**

$$R-OH \xrightarrow[\text{HCl/anhyd. } ZnCl_2]{} R\text{Cl} + H_2O$$

$$R-OH \xrightarrow[\text{or KBr + conc. } H_2SO_4 \text{ reflux}]{\text{HBr/}H_2SO_4 \text{ reflux}} R-\text{Br} + H_2O$$

$$R-OH \xrightarrow[\text{or KI + } H_3PO_4 \text{ reflux}]{\text{HI reflux}} R-\text{I} + H_2O$$

- Order of reactivity of halogen acids towards the above reaction is :
  HI > HBr > HCl
- Order of reactivity of different alcohols towards the above reaction is :
  tertiary > secondary > primary

– **Action of phosphorus halides :**

$$R-OH \xrightarrow{PCl_5} R-\text{Cl} + POCl_3 + HCl$$

$$R-OH \xrightarrow{PCl_3} R-\text{Cl} + H_3PO_3$$

$$R-OH \xrightarrow[(P + Br_2)]{PBr_3} R-\text{Br} + H_3PO_3$$

$$R-OH \xrightarrow[(P + I_2)]{PI_3} R-\text{I} + H_3PO_3$$

– **Action of thionyl chloride (Darzen's process) :**

$$R—OH + SOCl_2 \xrightarrow{\text{Pyridine}} R—Cl + SO_2\uparrow + HCl\uparrow$$

▶ **Halogenation of alkanes :**

$$R—H + X_2 \xrightarrow[\text{or } 250-400\ ^\circ C]{h\upsilon} RX + HX$$

– Abstraction of hydrogen for a particular halogen follows the order
  allylic > 3° > 2° > 1° > CH_4
– Reactivity of halogens for the above reaction follows the order :
  $F_2 > Cl_2 > Br_2 > I_2$

▶ **Halide exchange method :**

$$R—Cl + NaI \xrightarrow{\text{dry acetone}} R—I + NaCl$$

This is known as *Finkelstein reaction*.

– Fluoroalkanes can also be prepared by halide exchange method by reacting alkyl chloride with inorganic fluorides. This is known as *Swart's reaction*.

$$2CH_3CH_2Cl + Hg_2F_2 \longrightarrow \underset{\text{Mercurous fluoride}}{} 2CH_3CH_2F + \underset{\text{Mercurous chloride}}{Hg_2Cl_2}$$

▶ **Action of hydrogen halides on ethers :**

$$R—O—R + 2HX \longrightarrow 2R - X + H_2O$$

▶ **Addition of hydrogen halides to alkene :**

Alkenes add on a molecule of hydrogen halide to form alkyl halide.

$$>C=C< + HX \longrightarrow -\underset{H}{\overset{|}{C}}-\underset{X}{\overset{|}{C}}-$$

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$$CH_3-CH=CH_2 + HCl \longrightarrow CH_3-\underset{\displaystyle Cl}{\underset{|}{CH}}-CH_3$$

– Reaction follows electrophilic addition mechanism and takes place as per Markownikoff's rule. However, in presence of peroxide, addition of HBr takes place as per anti-Markownikoff's rule.

$$CH_3-CH=CH_2 + HBr$$

Absence of Peroxide | Presence of Peroxide

$CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$ (Markownikoff's addition)

$CH_3-CH_2-CH_2Br$ (Anti-Markownikoff's addition)

Anti-Markownikoff's rule is also known as *Peroxide effect* or *Kharasch effect*. HCl and HI do not show peroxide effect.

Physical properties :

- Alkyl halides being polar in nature are insoluble in water as they cannot break H-bonding already existing in water.
- They have higher melting and boiling points, than those of the hydrocarbons of comparable molecular mass. For the same alkyl group boiling point follows the order as :

  R—I > R—Br > R—Cl > R—F
- Decreasing order of boiling points among the isomeric alkyl halides follows the order : 1° > 2° > 3° alkyl halides
- Decreasing order of density among the alkyl halides is RI > RBr > RCl > RF.

  For alkyl iodide decreasing order of density is as follows :

  $CH_3I > CH_3CH_2I > CH_3CH_2CH_2I$.
- Bond strength of C—X bond follows the order

  CH_3—F > CH_3—Cl > CH_3—Br > CH_3—I

  i.e., bond strength of C—X bond decreases as the size of halogen atoms increases.
- Correct stability order of RX is as follows :

  R—F > R—Cl > R—Br > R—I

Chemical properties :

- **Nucleophilic substitution reactions :** The reactions in which a stronger nucleophile displaces a weaker nucleophile are called nucleophilic substitution reactions. Better the leaving group, more facile is the nucleophilic substitution reaction.
- **S_N1 and S_N2 mechanisms :** The nucleophilic substitution can proceed via S_N1 mechanism or S_N2 mechanism.

Unimolecular (S_N1)	Bimolecular (S_N2)
It is first order reaction.	It is second order reaction.
Generally carried out in polar protic solvents like water, alcohol and acetic acid.	Carried out in polar aprotic solvents like acetone, DMSO, acetonitrile or DMF.
Takes place in two steps through carbocation as the intermediate.	Takes place in one step through transition state.
Rate of reaction : $3° > 2° > 1° > CH_3$ halides (fastest) (slowest)	Rate of reaction : $CH_3 > 1° > 2° > 3°$ halides (fastest) (slowest)
Greater the stability of carbocation, faster will be the reaction.	Less the steric hindrance in transition state, faster will be the reaction.
Tends to proceed with weak nucleophiles, *e.g.*, CH_3OH, H_2O, CH_3CH_2OH, etc.	Tends to proceed with strong nucleophiles, *e.g.*, CH_3O^-, CN^-, OH^-, etc.
Both retention and inversion of configuration takes place.	Inversion of configuration takes place (Walden inversion).

- Primary allylic and primary benzylic halides show higher reactivity in S_N1 reactions than other primary alkyl halides due to the greater stabilisation of allylic and benzylic carbocation intermediates by resonance.
- Vinylic and aryl halides are unreactive in nucleophilic substitution reactions. This is because of double bond character of C — X bond due to resonance.

Optical isomerism/Enantiomerism :

- Optical isomer is known as *dextrorotatory isomer* (latin : *dexter* means right) (*d*-form or +ve) if it rotates the plane polarised light to the right (clockwise) and *laevorotatory*

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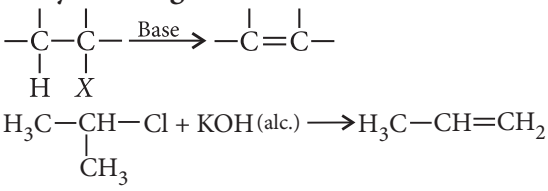
isomer (latin : *laevo* means left) (*l*-form or –ve) if it rotates the plane polarised light to the left (anticlockwise).

- An equimolar mixture of the *d*-form and *l*-form will be optically inactive and is called *racemic mixture* (or *dl- form or* (±)*-mixture*).
- The process of conversion of an enantiomer into racemic mixture is known as *racemisation*.
- **Chirality :** The compound is said to have chirality if the central carbon atom is attached to four different groups and this centre is called *chiral* (*asymmetric*) *centre* or *stereogenic centre* or *stereocentre*.
- **Achirality :** The compound is said to have achirality if the central carbon atom have atleast two identical groups and this centre is called *achiral* (*symmetric*).
- If molecule has a plane of symmetry it is *achiral* (*not chiral*) and if molecule has no plane of symmetry it is *chiral*.
- **Enantiomers/*d*- and *l*-isomers :** They are the optical isomers which are non-superimposable mirror images (or dissymmetric).
- **Diastereomers :** They are the optical isomers which are not mirror images of each other. They have different physical properties and magnitude of specific rotation.
- **Meso compounds :** These compounds have two or more even number of chiral carbon atoms and have an internal plane of symmetry. They are optically inactive due to internal compensation.

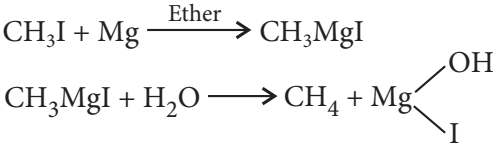
Elimination reactions:

- **Dehydrohalogenation :**

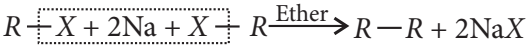


Reaction with metals:

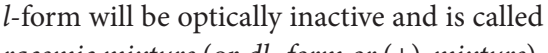
- **Reaction with Mg metal :**



- **Wurtz reaction :**



Reduction of alkyl halides :



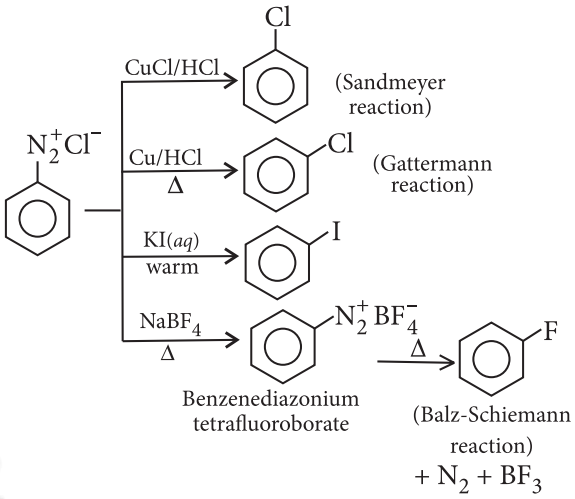
HALOARENES

Aryl halides : General formula is Ar*X*, where Ar = aryl group.

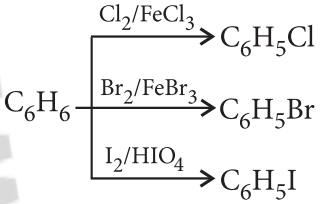
General methods of preparation :

- **By Raschig process :**

- **From benzenediazonium salt :**



- **By direct halogenation of benzene :**



Physical properties :

- Aryl halides are colourless stable liquids with pleasant odour.
- These are insoluble in water but readily miscible with organic solvents.
- Most of them are steam volatile, heavier than water.
- Their boiling points are higher than corresponding alkyl halides. The boiling points rise gradually from fluoro to iodo compounds.

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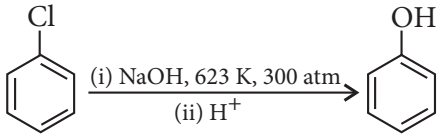
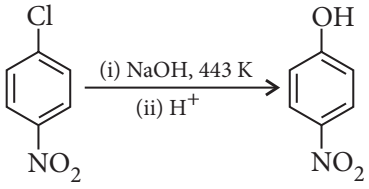
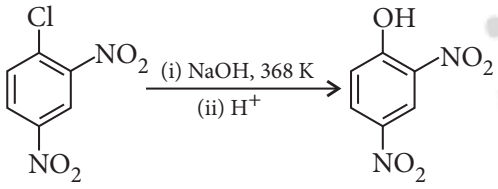
Chemical properties :

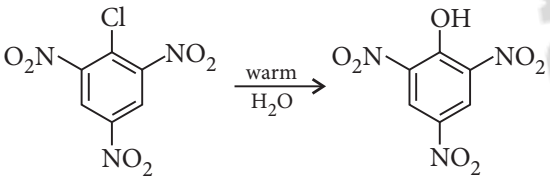
▶ Nucleophilic substitution reactions :



The presence of an electron withdrawing group (—NO_2) at *ortho*- and *para*-positions increases the reactivity of haloarenes.

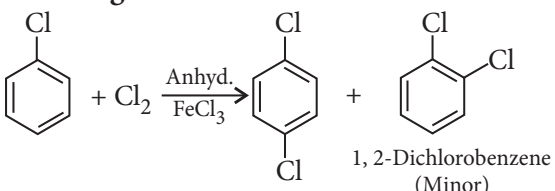






▶ Electrophilic substitution reactions :

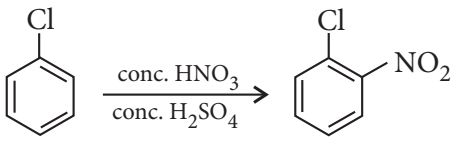
– **Halogenation :**



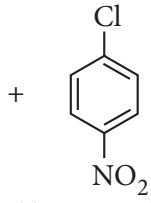
1, 4-Dichlorobenzene (Major)

1, 2-Dichlorobenzene (Minor)

– **Nitration :**

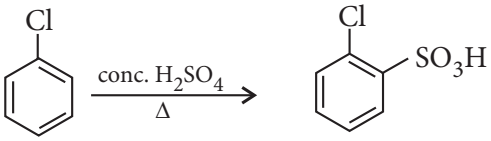


1-Chloro-2-nitrobenzene (Minor)

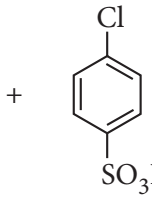


1-Chloro-4-nitrobenzene (Major)

– **Sulphonation :**

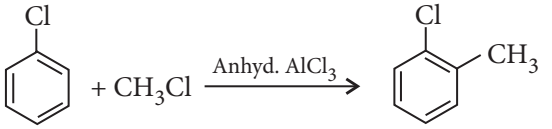


2-Chlorobenzene-sulphonic acid (Minor)

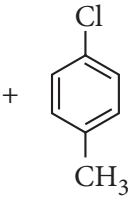


4-Chlorobenzene-sulphonic acid (Major)

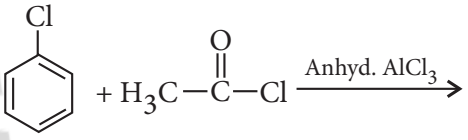
– **Friedal–Crafts reaction :**

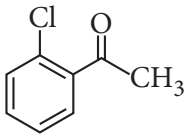


1-Chloro-2-methylbenzene (Minor)

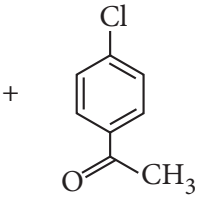


1-Chloro-4-methylbenzene (Major)



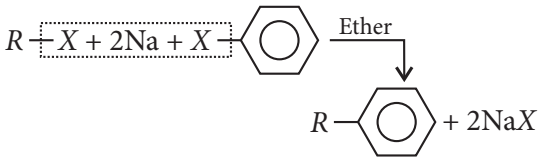


2-Chloroacetophenone (Minor)

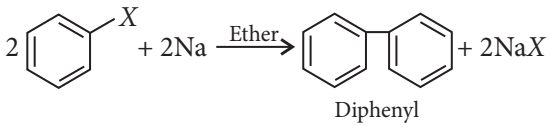


4-Chloroacetophenone (Major)

▶ Wurtz Fittig reaction :



▶ Fittig reaction :



Diphenyl

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Uses and environmental effects of some important compounds :

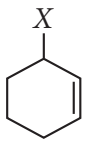
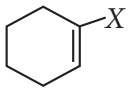
Compounds	Uses	Effects
Chloroform ($CHCl_3$)	– Its major use is in the production of Freon refrigerant, R-22. – It is used as a solvent for resins, rubbers, oils and fats, alkaloids, iodine and many other substances. – In the past, it was extensively used as anaesthetic for surgery but now it is rarely used as it causes liver damage. – It is used in preparation of chloretone (drug) and chloropicrin (insecticide). – It is used to preserve anatomical species.	– It is oxidised to poisonous gas, carbonyl chloride, known as *phosgene*. $2CHCl_3 + O_2 \xrightarrow{\text{Light}} 2COCl_2 + 2HCl$ Phosgene gas causes liver and kidney damage. – Inhaling chloroform vapours depresses the central nervous system, causes dizziness, fatigue and headache.
Iodoform (CHI_3)	– It was earlier used as an antiseptic in dressing of wounds due to liberation of iodine. Due to its objectionable smell, it has been replaced by other formulations containing iodine. – It is used as methylating agent in organic synthesis.	– It has strong smell.
Carbon tetrachloride (CCl_4)	– It is used in the manufacture of refrigerants and propellants for aerosol cans. – It is used as feed stock in the synthesis of CFC's and other chemicals.	– Exposure CCl_4 causes liver cancer. – It causes Diziness, light headness, nausea and vomiting, which can cause permanent damage to nerve cells. – It deplets the ozone layer.
Freons	– They are used as refrigerants, blowing agents, propellants in medical applications and degreasing solvent.	– Freons cause disruption of ozone layer by initiating radical chain reactions in stratosphere. – This anthropogenic compound is a green-house gas and its effect is more than CO_2.
DDT	– In 1940, it was used as a pesticide.	– It is a persistent organic pollutant, strongly absorbed by soil. – It is lipophilic so has a high potential to bioaccumulate. – It may be directly genotoxic but may also induce enzymes to produce other genotoxic intermediates and DNA adducts.

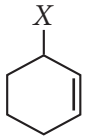
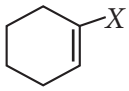
www.pdfkeyboard.com

Previous Years' CBSE Board Questions

10.1 Classification

VSA (1 mark)

1. Out of  and  , which is an example of allylic halide? *(AI 2017)*

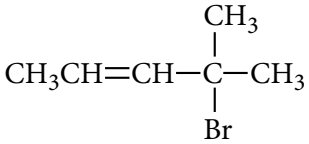
2. Out of  and  , which is an example of vinylic halide? *(AI 2017)*

3. Out of and , which is an example of a benzylic halide? *(AI 2017)*

10.2 Nomenclature

VSA (1 mark)

4. Write the IUPAC name of



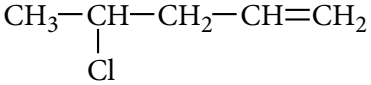
(AI 2019, Delhi 2013)

5. Write the structure of 1-bromo-4-chlorobut-2-ene. *(Delhi 2017)*

6. Write the structure of 3-bromo-2-methylprop-1-ene. *(Delhi 2017)*

7. Draw the structure of 2-bromopentane. *(Delhi 2014C)*

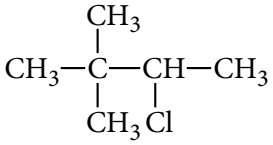
8. Write the IUPAC name of



(Delhi 2013)

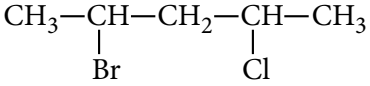
9. Write the IUPAC name of $(CH_3)_2CHCH(Cl)CH_3$. *(Delhi 2013)*

10. Write the IUPAC name of the following compound:



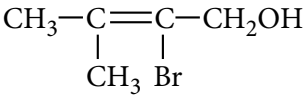
(AI 2013)

11. Write the IUPAC name of the following compound:



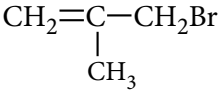
(AI 2013)

12. Write the IUPAC name of the following:



(AI 2013C, 2012C, Foreign 2011)

13. Give the IUPAC name of the following compound:



(Delhi 2012, AI 2011)

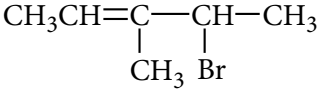
14. Write the IUPAC name of the following compound:
$(CH_3)_3CCH_2Br$ *(Delhi 2011)*

15. Write the IUPAC name of the following compound:
$CH_2{=}CHCH_2Br$ *(AI 2011)*

16. Write the structure of the following compound:
1, 4-dibromobut-2-ene *(Delhi 2011C)*

17. Write the structure of the following compound:
2-(2-Bromophenyl)butane *(Delhi 2011C)*

18. Give IUPAC name of the following organic compound:



(Delhi 2011C)

19. Write the structure of the following compound:
2-(2-chlorophenyl)-1-iodooctane *(AI 2011C)*

20. Write the structure of the following compound:
1-bromo-4-*sec*-butyl-2-methylbenzene *(AI 2011C)*

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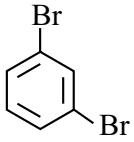
SA (2 marks)

21. Write the IUPAC names of the following compounds :
(i) $CH_2=CHCH_2Br$ (ii) $(CCl_3)_3CCl$
(AI 2014C)

LA I (3 marks)

22. Give the IUPAC names of the following compounds :
(i) $CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-CH_3$
(ii) 
(iii) $CH_2=CH-CH_2-Cl$ *(AI 2015C)*

10.4 Methods of Preparation of Haloalkanes

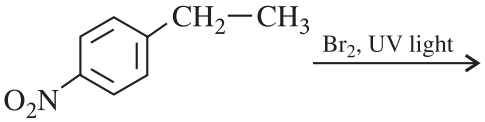
VSA (1 mark)

23. How can you convert the following?
But-1-ene to 1-iodobutane *(1/3, 2020)*

24. Give reason for the following :
Thionyl chloride method is preferred for preparing alkyl chloride from alcohols.
(1/3, AI 2019)

25. Write equation for preparation of 1-iodobutane from 1-chlorobutane. *(1/3, AI 2019)*

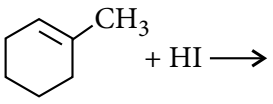
26. How do you convert :
Propene to 1-iodopropane? *(1/3, AI 2016)*

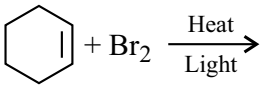
27. Write the major product in the following :

(1/3, AI 2016)

28. Write the structure of the major product in the following reaction :
$CH_3-CH=\underset{\displaystyle CH_3}{\underset{|}{C}}-CH_3 + HBr \longrightarrow$
(1/3, AI 2015)

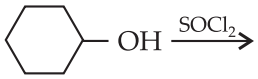
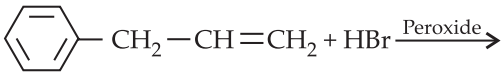
29. A hydrocarbon C_5H_{12} gives only one monochlorination product. Identify the hydrocarbon. *(Delhi 2013C)*

30. What happens when bromine attacks $CH_2=CH-CH_2-C\equiv CH$? *(AI 2012)*

31. Draw the structure of major monohalogen product formed in the following reaction :

(Delhi 2012C)

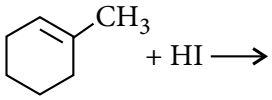
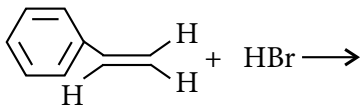
32. Draw the structure of major monohalogen product in the following reaction :

(Delhi 2012C)

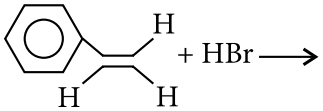
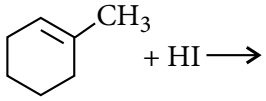
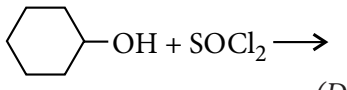
SA (2 marks)

33. Draw the structure of major monohalo product in each of the following reactions :
(i) 
(ii) 
(Delhi 2014)

34. Write the mechanism of the following reaction :
$CH_3CH_2OH \xrightarrow{HBr} CH_3CH_2Br + H_2O$
(AI 2014)

LA I (3 marks)

35. Complete the following reaction equations :
(i) 
(ii) 
(iii) $CH_3CH_2CH=CH_2 + HBr \longrightarrow$
(Foreign 2011)

36. Complete the equations for the following reactions :
(i) 
(ii) 
(iii) 
(Delhi 2011C)

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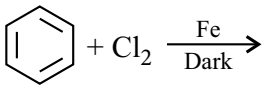
10.5 Methods of Preparation of Haloarenes

VSA (1 mark)

37. Draw the structure of major monohalo product in the following reaction :



(Delhi 2012C)

10.6 Physical Properties

VSA (1 mark)

38. Give reason for the following :
p-dichlorobenzene has higher melting point than that of *ortho* or *meta* isomer.
(1/3, AI 2019, 1/2, Delhi 2013)

39. Explain the following :
Alkyl halides, though polar, are immiscible with water.
(1/2, AI 2017C, 1/3, Foreign 2015, Delhi 2013C, AI 2013C, 2012C)

40. Explain why the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.
(1/3, Delhi 2016, 2013C, 2011C)

41. Give reason :
n-Butyl bromide has higher boiling point than *t*-butyl bromide. *(1/3, Delhi 2015)*

42. Answer the following :
Haloalkanes easily dissolve in organic solvents, why? *(1/3, Delhi 2011)*

10.7 Chemical Reactions

VSA (1 mark)

Read the given passage and answer the questions number (43 - 47) that follow.

The substitution reaction of alkyl halide mainly occurs by S_N1 or S_N2 mechanism. Whatever mechanism alkyl halides follow for the substitution reaction to occur, the polarity of the carbon halogen bond is responsible for these substitution reactions. The rate of S_N1 reactions are governed by the stability of carbocation whereas for S_N2 reactions steric factor is the deciding factor. If the starting material is a chiral compound, we may end up with an inverted product or racemic mixture depending upon the type of mechanism followed by alkyl halide. Cleavage of ethers with HI is also governed by steric factor and stability of carbocation, which indicates that in organic chemistry, these two major factors help us in deciding the kind of product formed. *(2020)*

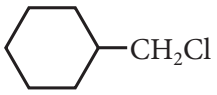
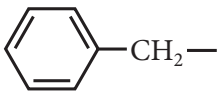
43. Predict the stereochemistry of the product formed if an optically active alkyl halide undergoes substitution reaction by S_N1 mechanism.

44. Name the instrument used for measuring the angle by which the plane polarised light is rotated.

45. Predict the major product formed when 2-bromopentane reacts with alcoholic KOH.

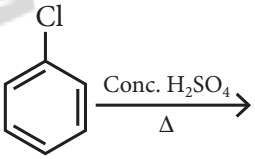
46. Give one use of CHI_3.

47. Write the structures of the products formed when anisole is treated with HI.

48. Out of  and  which will react faster in S_N1 reaction with OH^-? *(One word, 2020)*

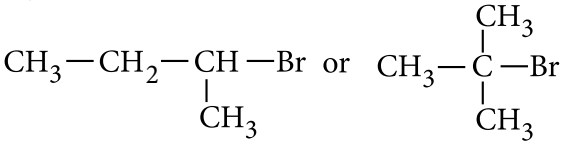
49. Racemisation occurs in
(a) S_N2 reaction
(b) S_N1 reaction
(c) Neither S_N2 nor S_N1 reactions
(d) S_N2 reaction as well as S_N1 reaction.
(2020)

50. Write the major product(s) of the following reaction :



(1/3, 2020)

51. Which alkyl halide from the following pair would you expect to react more rapidly by an S_N2 mechanism?



(AI 2019)

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52. Give reason for the following :
The presence of $-NO_2$ group at *ortho* and *para* position increases the reactivity of haloarenes towards nucleophilic substitution reactions. *(1/3, AI 2019)*

53. Out of 2-bromopentane, 2-bromo-2-methyl butane, and 1-bromopentane, which compound is most reactive towards elimination reaction and why? *(1/3, AI 2019)*

54. Out of chlorobenzene and benzyl chloride, which one gets easily hydrolysed by aqueous NaOH and why? *(2018)*

55. Out of $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-Cl$ and $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-Cl$, which is more reactive towards S_N1 reaction and why?
(Delhi 2016)

56. Write the structure of an isomer of compound C_4H_9Br which is most reactive towards S_N1 reaction. *(AI 2016)*

57. Which would undergo S_N2 reaction faster in the following pair and why?

CH_3-CH_2-Br and $CH_3-\underset{\displaystyle Br}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-CH_3$

(Delhi 2015)

58. Which would undergo S_N1 reaction faster in the following pair :

CH_3-CH_2-Br and $CH_3-\underset{\displaystyle Br}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-CH_3$ *(AI 2015)*

59. Which would undergo S_N2 reaction faster in the following pair and why?
CH_3-CH_2-Br and CH_3-CH_2-I
(Foreign 2015)

60. Identify the chiral molecule in the following pair :

(AI 2014)

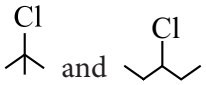
61. Which halogen compound in each of the following pairs will react faster in S_N2 reaction :
(i) CH_3Br or CH_3I
(ii) $(CH_3)_3CCl$ or CH_3Cl
(AI 2014, Delhi 2014C)

62. What happens when CH_3-Br is treated with KCN? *(Delhi 2013)*

63. What happens when ethyl chloride is treated with aqueous KOH? *(Delhi 2013)*

64. Why is (±)-butan-2-ol is optically inactive?
(1/2, Delhi 2013)

65. Which compound in the following pair undergoes faster S_N1 reaction?



(Delhi 2013, 2013C, 2012C)

66. How may methyl bromide be preferentially converted to methyl isocyanide?
(Delhi 2013C)

67. Account for the following :
Grignard's reagents should be prepared under anhydrous conditions.
(1/3, Delhi 2013C, 1/3, AI 2012C)

68. Predict the order of reactivity of four isomeric bromobutanes in S_N1 reaction.
(Delhi 2012C)

69. Predict the order of reactivity of the following compounds in S_N1 reaction.
$C_6H_5CH_2Br$, $C_6H_5C(CH_3)(C_6H_5)Br$,
$C_6H_5CH(C_6H_5)Br$, $C_6H_5CH(CH_3)Br$
(Delhi 2012C)

70. Give a chemical test to distinguish between C_2H_5Br and C_6H_5Br. *(AI 2012C)*

71. Which will react faster in S_N2 displacement, 1-bromopentane or 2-bromopentane and why? *(Foreign 2011)*

72. Which will react faster in S_N1 displacement reaction :
1-Bromobutane or 2-bromobutane and why?
(Foreign 2011)

SA (2 marks)

73. Write the mechanism of the following S_N1 reaction

$(CH_3)_3C-Br \xrightarrow{\text{Aq. NaOH}} (CH_3)_3C-OH + NaBr$

(2/3, 2020)

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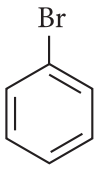
74. Give reasons :
(i) C—Cl bond length in chlorobenzene is shorter than C—Cl bond length in CH_3—Cl.
(ii) S_N1 reactions are accompanied by racemization in optically active alkyl halides. *(2/3, Delhi 2016)*

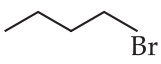
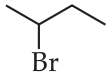
75. How do you convert?
(i) Chlorobenzene to biphenyl
(ii) 2-Bromobutane to but-2-ene
(2/3, AI 2016)

76. Write the major product(s) in the following :
(i) $2CH_3-\underset{\underset{Cl}{|}}{CH}-CH_3 \xrightarrow[\text{Dry ether}]{Na}$
(ii) $CH_3-CH_2-Br \xrightarrow{AgCN}$ *(2/3, AI 2016)*

77. Give reasons:
(i) Racemic mixture is optically inactive.
(ii) The presence of nitro group ($-NO_2$) at *o*/*p* positions increases the reactivity of haloarenes towards nucleophilic substitution reactions.
(2/3, Delhi 2015)

78. Write the structure of the major product in each of the following reactions :
(i) $CH_3-CH_2-CH_2-\underset{\underset{Br}{|}}{CH}-CH_3 + KOH \xrightarrow[\text{Heat}]{\text{Ethanol}}$
(ii)  $+ CH_3Cl \xrightarrow{\text{anhyd. } AlCl_3}$
(2/3, AI 2015)

79. (i) Why is butan-1-ol optically inactive but butan-2-ol is optically active?
(ii) Although chlorine is an electron withdrawing group, yet it is *ortho*-, *para*-directing in electrophilic aromatic substitution reactions. Why?
(2/3, Foreign 2015, Delhi 2012)

80. (i) Which alkyl halide from the following pair is chiral and undergoes faster S_N2 reaction?
(a)  (b) 
(ii) Out of S_N1 and S_N2, which reaction occurs with
(a) inversion of configuration
(b) racemisation? *(2/3, Delhi 2014)*

81. Write chemical equations when
(i) ethyl chloride is treated with aqueous KOH.
(ii) chlorobenzene is treated with CH_3COCl in presence of anhydrous $AlCl_3$.
(Foreign 2014)

82. (i) Which alkyl halide from the following pairs would you expect to react more rapidly by an S_N2 mechanism and why?

$CH_3-CH_2-\underset{\underset{Br}{|}}{CH}-CH_3$

$CH_3-CH_2-CH_2-CH_2-Br$
(ii) Racemisation occurs in S_N1 reactions. Why? *(Foreign 2014)*

83. Write chemical equations when
(i) methyl chloride is treated with $AgNO_2$.
(ii) bromobenzene is treated with CH_3Cl in the presence of anhydrous $AlCl_3$.
(Foreign 2014)

84. What are ambident nucleophiles? Explain with an example. *(2/3, AI 2014C)*

85. Chlorobenzene is extremely less reactive towards a nucleophilic substitution reaction. Give two reasons for the same. *(Delhi 2013)*

86. Account for the following:
(i) The C—Cl bond length in chlorobenzene is shorter than that in CH_3—Cl.
(ii) Chloroform is stored in closed dark brown bottles. *(Delhi 2013)*

87. Give reasons for the following :
(i) Ethyl iodide undergoes S_N2 reaction faster than ethyl bromide.
(ii) C—*X* bond length in halobenzene is smaller than C—*X* bond length in CH_3—*X*. *(2/3, AI 2013)*

88. Haloalkanes undergo nucleophilic substitution whereas haloarenes undergo electrophilic substitution. Explain. *(2/3, Delhi 2012C)*

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89. Answer the following :
(i) What is known as a racemic mixture? Give an example.
(ii) Of the two bromo derivatives, $C_6H_5CH(CH_3)Br$ and $C_6H_5CH(C_6H_5)Br$, which one is more reactive in S_N1 substitution reaction and why?
(2/3, Delhi 2011)

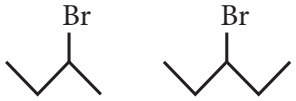
90. Write the mechanism of the following reaction :

$$n\text{-BuBr} + KCN \xrightarrow{EtOH,\ H_2O} n\text{-BuCN}$$

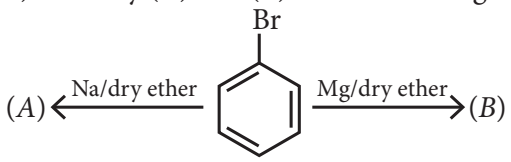
(1/3, Delhi 2011C)

LA I (3 marks)

91. (i) Write the structure of major alkene formed by β-elimination of 2, 2, 3-trimethyl-3-bromopentane with sodium ethoxide in ethanol.
(ii) Which one of the compounds in the following pairs is chiral?



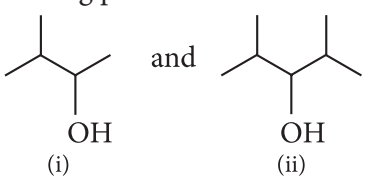
(iii) Identify (*A*) and (*B*) in the following :



(2020)

92. (i) Out of $(CH_3)_3C-Br$ and $(CH_3)_3C-I$, which one is more reactive towards S_N1 and why?
(ii) Write the product formed when *p*-nitrochlorobenzene is heated with aqueous NaOH at 443 K followed by acidification.
(iii) Why *dextro* and *laevo*-rotatory isomers of Butan-2-ol are difficult to separate by fractional distillation? *(Delhi 2019)*

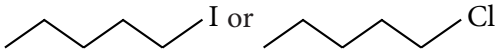
93. (a) Identify the chiral molecule in the following pair :



(b) Write the structure of the product when chlorobenzene is treated with methyl chloride in the presence of sodium metal and dry ether.
(c) Write the structure of the alkene formed by dehydrohalogenation of 1-bromo-1-methylcyclohexane with alcoholic KOH. *(2018)*

94. Following compounds are given to you :
2-Bromopentane, 2-Bromo-2-methylbutane, 1-Bromopentane
(i) Write the compound which is most reactive towards S_N2 reaction.
(ii) Write the compound which is optically active.
(iii) Write the compound which is most reactive towards β-elimination reaction.
(Delhi 2017)

95. Answer the following questions:
(i) What is meant by chirality of a compound? Give an example.
(ii) Which one of the following compounds is more easily hydrolysed by KOH and why?
$CH_3CHClCH_2CH_3$ or $CH_3CH_2CH_2Cl$
(iii) Which one undergoes S_N2 substitution reaction faster and why?



(AI 2012)

96. Rearrange the compounds of each of the following sets in order of reactivity towards S_N2 displacement:
(i) 2-Bromo-2-methylbutane, 1-Bromopentane, 2-Bromopentane
(ii) 1-Bromo-3-methylbutane, 2-Bromo-2-methylbutane,3-bromo-2-methylbutane
(iii) 1-Bromobutane, 1-Bromo-2, 2-dimethyl propane, 1-Bromo-2-methylbutane
(AI 2011)

97. (a) Write a chemical test to distinguish between :
(i) Chlorobenzene and benzyl chloride
(ii) Chloroform and carbon tetrachloride
(b) Why is methyl chloride hydrolysed more easily than chlorobenzene?
(Delhi 2011C)

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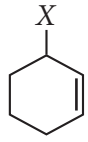
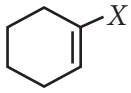
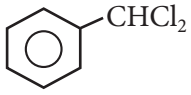
Detailed Solutions

1.  is an allylic halide as the halogen is attached to sp^3-hybridised carbon atom which is next to carbon-carbon double bond.

2.  is a vinylic halide as the halogen is attached to sp^2-hybridised carbon.

3.  is a benzylic halide as the halide group is attached to sp^3-hybridised carbon atom next to aromatic ring.

4. 4-Bromo-4-methylpent-2-ene

5. $Cl-\overset{4}{C}H_2-\overset{3}{C}H=\overset{2}{C}H-\overset{1}{C}H_2-Br$

1-Bromo-4-chlorobut-2-ene

6. $H_2C(Br)-C(CH_3)=CH_2$

3-Bromo-2-methylprop-1-ene

7. $H_3C-CH(Br)-CH_2-CH_2-CH_3$

8. 4-Chloropent-1-ene

9. 2-Chloro-3-methylbutane

10. 3-Chloro-2,2-dimethylbutane

11. 2-Bromo-4-chloropentane

12. $\overset{4}{C}H_3-\overset{3}{C}(CH_3)(Br)=\overset{2}{C}H-\overset{1}{C}H_2OH$

2-Bromo-3-methylbut-2-en-1-ol

13. $\overset{1}{C}H_2=\overset{2}{C}(CH_3)-\overset{3}{C}H_2Br$

3-Bromo-2-methylpropene

14. $H_3C-\overset{2}{C}(CH_3)_2-\overset{1}{C}H_2-Br$

1-Bromo-2,2-dimethylpropane

15. $\overset{1}{C}H_2=\overset{2}{C}H\overset{3}{C}H_2Br$

3-Bromoprop-1-ene

16. $H_2C(Br)-CH=CH-CH_2(Br)$

17. $CH_3-CH(Br)-CH_2-CH_3$

18. $\overset{1}{C}H_3\overset{2}{C}H=\overset{3}{C}(CH_3)(Br)-\overset{4}{C}H(Br)-\overset{5}{C}H_3$

4-Bromo-3-methylpent-2-ene

19. $IH_2C-CH(Cl)-(CH_2)_5-CH_3$

20. $CH_3-CH_2-CH(Br)-CH_3$ (with CH_3 on the ring)

21. (i) *Refer to answer 15.*

(ii) 2-(Trichloromethyl)-1, 1,1, 2, 3, 3, 3-heptachloropropane

22. (i) 2-Bromobutane

(ii) 1,3-Dibromobenzene

(iii) 3-Chloropropene

23.

$$\underset{\text{But-1-ene}}{CH_3-CH_2-CH=CH_2} + HBr \xrightarrow{\text{Peroxide}} CH_3CH_2CH_2CH_2Br \xrightarrow[\text{NaI}]{\text{Acetone}} \underset{\text{1-Iodobutane}}{CH_3CH_2CH_2CH_2I}$$

24. Thionyl chloride is preferred because in this reaction alkyl halide is formed along with gases SO_2 and HCl. These two gaseous products are escapable, hence the reaction gives pure alkyl halides.

25. Alkyl iodides are often prepared by the reaction of alkyl chlorides/bromides with NaI in dry acetone. This is known as Finkelstein reaction.

$$CH_3CH_2CH_2CH_2Cl \xrightarrow[\text{Acetone}]{\text{NaI}} CH_3CH_2CH_2CH_2I$$

26.

$$\underset{\text{Propene}}{CH_3CH=CH_2} + HBr \xrightarrow{\text{Peroxide}} CH_3CH_2CH_2Br \xrightarrow[\text{Acetone}]{\text{NaI}} \underset{\text{1-Iodopropane}}{CH_3-CH_2-CH_2I}$$

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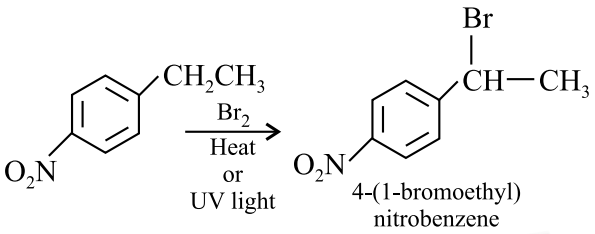
27.



4-(1-bromoethyl) nitrobenzene

28. $CH_3-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle Br}{\underset{|}{\overset{|}{C}}}}-CH_3$

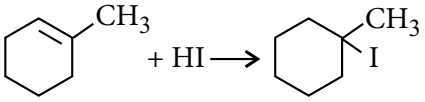
2-Bromo-2-methylbutane

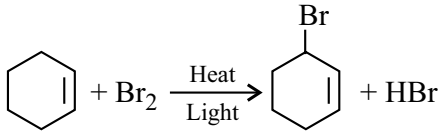
(Markovnikov's addition)

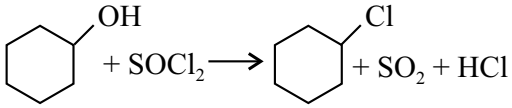
29. $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}-CH_3$

2,2-Dimethylpropane

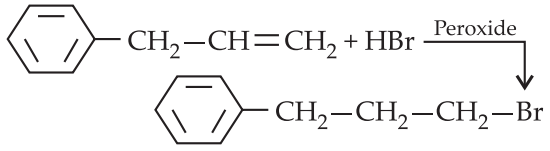
30. $CH_2(Br)-CH(Br)-CH_2-C\equiv CH$

31. 

32. 

33. (i) 

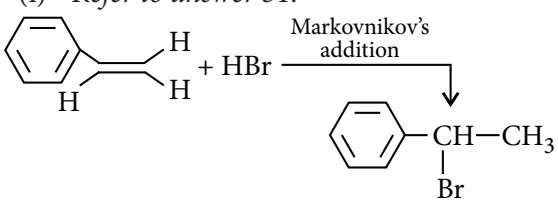
Chlorocyclohexane

(ii) 

34. $CH_3CH_2\ddot{O}H + H^+ \longrightarrow CH_3CH_2-\overset{+}{\ddot{O}}H_2$

$Br^- + CH_3CH_2-\overset{+}{\ddot{O}}H_2 \longrightarrow CH_3CH_2-Br + H_2O$

35. (i) *Refer to answer 31.*

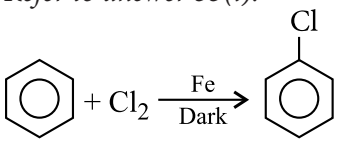
(ii) 

(iii) $CH_3CH_2CH=CH_2 + HBr \longrightarrow CH_3CH_2-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$

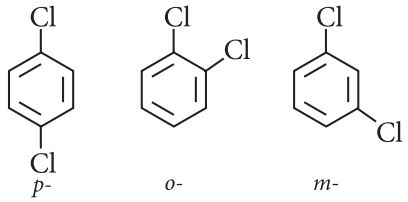
36. (i) *Refer to answer 35(ii).*

(ii) *Refer to answer 31.*

(iii) *Refer to answer 33(i).*

37. 

38. *p*-Dichlorobenzene has higher melting point than those of *o*-and *m*-isomers because it is more symmetrical and packing is better in solid form. Hence, it has stronger intermolecular forces of attraction than *o*-and *m*-isomers.



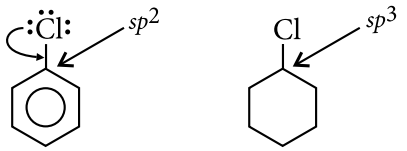
39. Alkyl halides are polar but are insoluble in water because energy required to break the intermolecular H – bond among water molecules is much higher than energy released by water halide interaction.

40. There are two reasons :

(i) In case of chlorobenzene, carbon to which chlorine is attached is sp^2-hybridised and is more electronegative than the corresponding carbon in cyclohexyl chloride which is sp^3-hybridised. So the net dipole moment is lower in chlorobenzene.

(ii) In chlorobenzene C—Cl bond has some double bond character so its bond length is smaller.

Hence dipole moment is smaller than cyclohexyl chloride which has a longer C—Cl single bond.



Chlorobenzene Cyclohexyl chloride

41. *n*-Butyl bromide, being a straight chain molecule have strong intermolecular forces whereas *t*-butyl bromide being a branched chain molecule have weaker intermolecular forces due to smaller surface area.

Hence, boiling point of *n*-butyl bromide is higher than that of *t*-butyl bromide.

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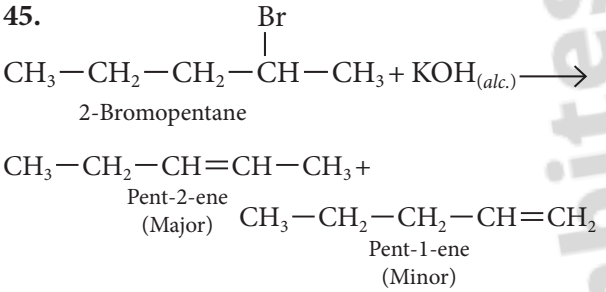
42. Haloalkanes dissolve in organic solvents because the intermolecular attractions between haloalkanes and organic solvent molecules have the same strength as in the separate haloalkanes and solvent molecules.

43. Stereochemical aspects of nucleophilic substitution reaction in S_N1 proceeds with racemisation.

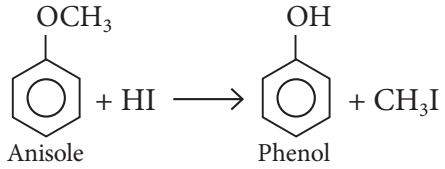
44. Polarimeter is used for measuring the angle by which the plane polarised light is rotated.

45.

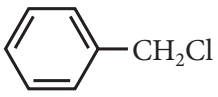


46. Iodoform (CHI_3) is used as a mild antiseptic.

47.

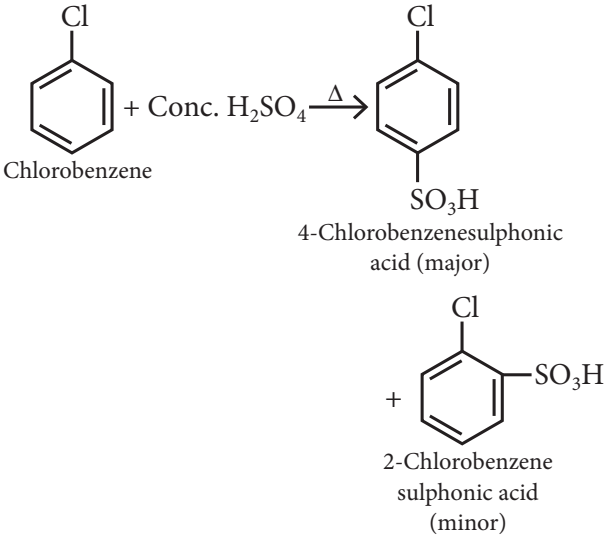


48.



49. **(b)** : S_N1 reactions are accompanied by racemisation.

50.



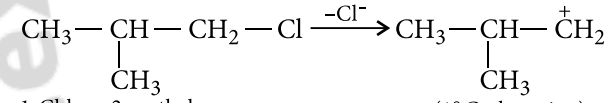
51. 2° alkyl halide *i.e.*, $CH_3-CH_2-\underset{\displaystyle CH_3}{\overset{|}{CH}}-Br$ would undergo S_N2 reaction faster than compared to 3° alkyl halide *i.e.*, $(CH_3)_3CBr$ due to formation of less sterically hindered transition state.

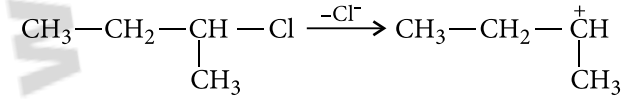
52. The presence of nitro group at *o*-and *p*-positions withdraws electrons from the benzene ring and thus, facilitates the attack of the nucleophile on haloarenes. The carbanion thus formed is further stabilised by resonance.

53. 2-Bromo-2-methylbutane will give fastest elimination reaction because it is a *tert*-halide and *tert*-halides prefer elimination reaction.

54. Benzyl chloride gets easily hydrolysed by aq. NaOH due to formation of stable benzyl carbocation. But due to partial double bond character of C — Cl bond in chlorobenzene, it does not hydrolyse.

55. The S_N1 reaction proceeds through carbocation formation thus, the compound which forms more stable carbocation will be more reactive.





As, 2° carbocation is more stable than 1° carbocation thus, 2-chlorobutane is more reactive towards S_N1 reaction.

56. $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-Br$

tert-butyl bromide

57. $CH_3—CH_2—Br$ would undergo S_N2 reaction faster due to formation of less sterically hindered transition state.

58. $CH_3-\underset{\displaystyle Br}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_3$ will undergo S_N1 reaction faster due to the formation of stable carbocation.

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59. Since I^- is a better leaving group than Br^-, thus, CH_3CH_2I undergoes S_N2 reaction faster than CH_3CH_2Br.

60. is a chiral molecule.

61. (i) CH_3I will give faster S_N2 reaction.
(ii) CH_3Cl will give faster S_N2 reaction.

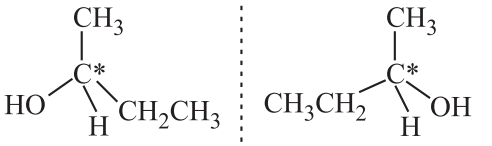
62. CH_3CN is formed by nucleophilic substitution reaction.

$$CH_3Br + KCN \rightarrow CH_3CN + KBr$$

63. When ethyl chloride is treated with aqueous KOH, ethanol is formed,

$$CH_3CH_2Cl + KOH_{(aq)} \longrightarrow CH_3CH_2OH + KCl$$

64. (±)-Butan-2-ol is optically inactive because the two enantiomeric forms are present in equal amounts therefore, it does not rotate the plane of polarized light and is optically inactive.



50% – (+) Butan-2-ol 50% – (–) Butan-2-ol

65. Tertiary halide reacts faster than the secondary halide because of the greater stability of *tert*-carbocation.

66. AgCN is mainly covalent in nature and nitrogen is free to donate electron pair forming isocyanide as main product.

$$\underset{\text{Methyl bromide}}{CH_3Br} + AgCN \longrightarrow \underset{\text{Methyl isocyanide}}{CH_3N{\rightleftharpoons}C} + AgBr$$

67. Grignard reagents react with water to form alkanes.

$$R{-}Mg{-}X + H_2O \longrightarrow R{-}H + Mg\begin{matrix}X\\OH\end{matrix}$$

So, they must be prepared under anhydrous conditions.

68. $(CH_3)_3CBr > CH_3CH_2\underset{\displaystyle Br}{\underset{|}{C}}H{-}CH_3 > (CH_3)_2CHCH_2Br > CH_3CH_2CH_2CH_2Br$

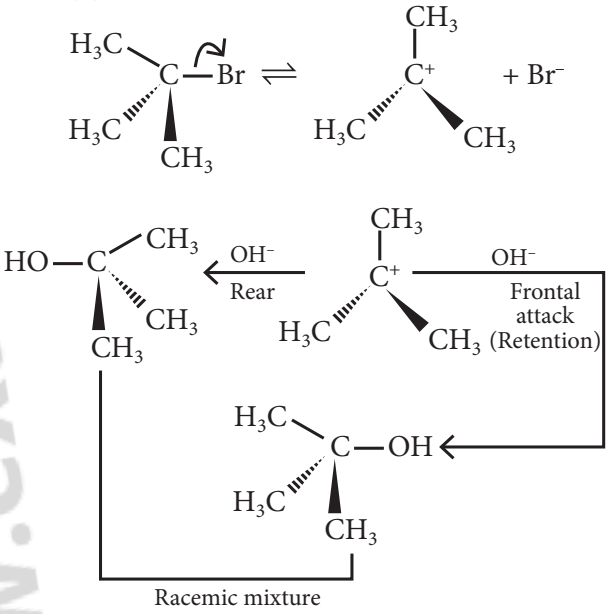
69. $C_6H_5C(CH_3)(C_6H_5)Br > C_6H_5CH(C_6H_5)Br > C_6H_5CH(CH_3)Br > C_6H_5CH_2Br$

70. C_2H_5Br reacts with $AgNO_3$ to give yellow precipitate of AgBr while C_6H_5Br will not.

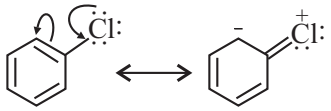
71. 1-Bromopentane is a primary alkyl halide, hence reacts faster in S_N2 displacement than secondary halide 2-bromopentane due to less steric hindrance in 1° alkyl halide than 2° alkyl halide.

72. 2-Bromobutane will react faster in S_N1 displacement reaction because it will form more stable secondary carbocation intermediate.

73. **(a)**

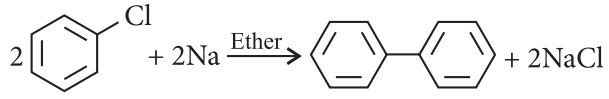


74. (i) In halobenzene C—X bond has partial double bond character due to resonance while CH_3—*X* bond is single bond.
Thus, bond length of C—*X* bond in halobenzene is smaller than that in CH_3—*X*.



(ii) In S_N1 reaction carbocation intermediate is formed which is a planar molecule so, an incoming nucleophile can attack from either side and a equimolar mixture of two components are formed and resulting mixture is optically inactive.

75. (i)



Chlorobenzene Biphenyl

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(ii) $CH_3-\underset{\text{2-Bromobutane}}{\overset{\overset{Br}{|}}{CH}}-CH_2-CH_3 \xrightarrow[\Delta]{\text{Ethanolic KOH}} \underset{\text{But-2-ene}}{CH_3-CH=CH-CH_3} + HBr$

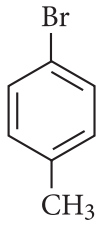
76. (i) $CH_3-\overset{\overset{CH_3}{|}}{CH}-\overset{\overset{CH_3}{|}}{CH}-CH_3$

(ii) CH_3CH_2NC

77. (i) Racemic mixture contains equal amount of *d*- and *l*-forms, hence rotation due to one enantiomer is cancelled by another.

(ii) *Refer to answer 52.*

78. (i) $\underset{\text{Pent-2-ene}}{CH_3-CH_2-CH=CH-CH_3}$ (Saytzeff rule)

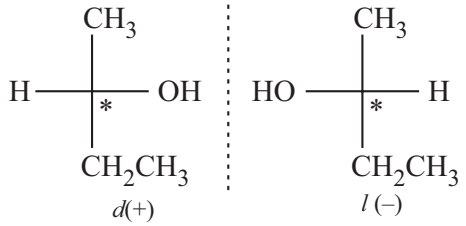
(ii)  (Friedel—Crafts alkylation)

4-Bromotoluene

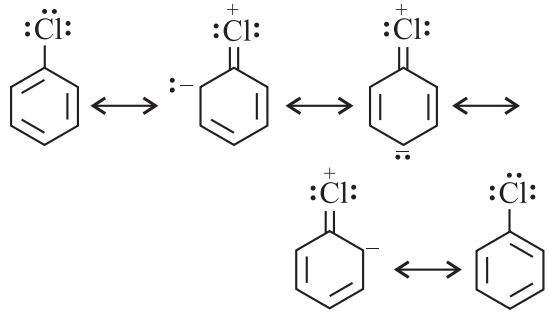
79. (i) Butan-1-ol is achrial, *i.e.*, does not have chiral 'C' atom which is attached to four different groups, therefore, it is optically inactive.

$\underset{\substack{\text{Butan-1-ol}\\\text{(Optically inactive)}\\\text{(No chiral carbon)}}}{CH_3-CH_2-CH_2-CH_2OH}$

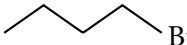
Butan-2-ol is chiral, *i.e.*, has chiral 'C' atom, attached to four different groups.

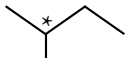


d(+) *l* (–)

(ii) 

Although Cl is electron withdrawing (-*I* effect) but still *o*- and *p*-directing as due to +*R* effect, electron density is maximum at *o*- and *p*-positions.

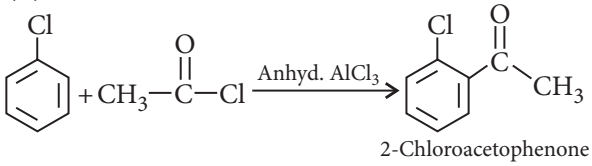
80. (i) (a)  undergoes faster S_N2 reaction.

(b)  is chiral.

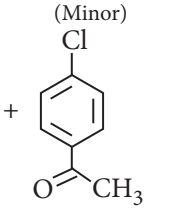
(ii) (a) S_N2 reaction occurs with inversion of configuration.

(b) S_N1 reaction occurs with racemisation.

81. (i) *Refer to answer 63.*

(ii) 

2-Chloroacetophenone (Minor)

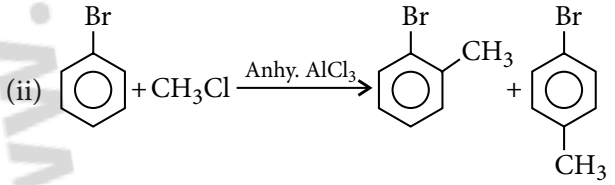


4-Chloroacetophenone (Major)

82. (i) 1-Bromobutane is 1° alkyl halide while 2-bromobutane is 2° alkyl halide. Due to steric hindrance in 2° alkyl halides, 1° alkyl halide will react faster than 2° alkyl halide in S_N2 reaction.

(ii) *Refer to answer 74(ii).*

83. (i) $CH_3Cl + AgNO_2 \longrightarrow CH_3NO_2 + AgCl$

(ii) 

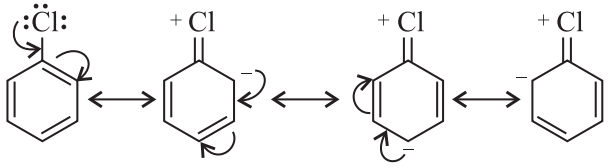
84. A nucleophile which can attack from more than one centres, is known as ambident nucleophile, *e.g.*, $\ddot{\bar{C}}\equiv N:$ cyanide ion

$R - X + AgCN \longrightarrow R - NC + AgX$

$R - X + KCN \longrightarrow R - CN + KX$

85. Haloarenes are much less reactive than haloalkanes towards nucleophilic substitution reactions due to the following reasons.

(i) Resonance effect : In haloarenes the electron pairs on halogen atom are in conjugation with π-electrons of the ring and the following resonating structures are possible.



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C—Cl bond acquires a partial double bond character due to resonance. As a result, the bond cleavage in haloarene is difficult than haloalkane and therefore, are less reactive towards nucleophilic substitution reaction.

(ii) In halorenes, halogen is attached to sp^2-hybridised carbon while in haloalkanes, halogen is attached to sp^3-hybridised carbon.

86. (i) *Refer to answer 74(i).*

(ii) Chloroform when exposed to air and sunlight changes to phosgene which is a poisonous gas.

$$CHCl_3 + \frac{1}{2}O_2 \rightarrow \underset{\text{Phosgene}}{COCl_2} + HCl$$

Thus, it is kept in dark coloured bottles to prevent the oxidation.

87. (i) *Refer to answer 59.*

(ii) *Refer to answer 74(i).*

88. In haloarenes –ve charge gets localised on arenes using resonance, therefore they undergo electrophilic substitution.

Haloalkanes have electrophilic carbon centre due to polarity of $C \rightarrow X$ bond.

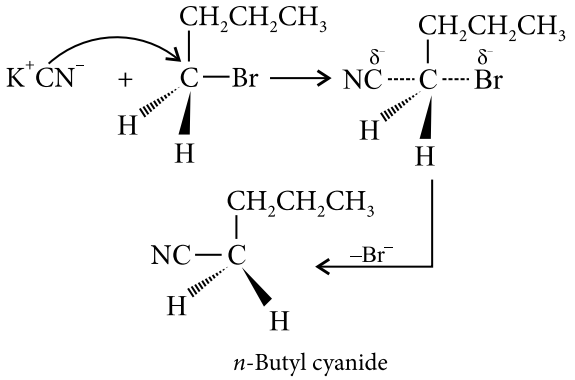
89. (i) An equimolar mixture of a pair of enantiomers is called racemic mixture. A racemic mixture is optically inactive due to external compensation.

Example :

Butan-2-ol (50% mixture of *d*- and *l*-form)

(ii) Of the two bromo derivatives, $C_6H_5CH(C_6H_5)Br$ is more reactive than $C_6H_5CH(CH_3)Br$ for S_N1 reaction because its carbocation is resonance stabilised by two phenyl groups.

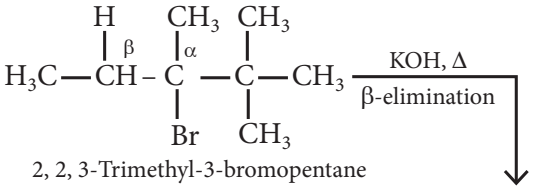
90. Normal butyl bromide will give S_N2 reaction :



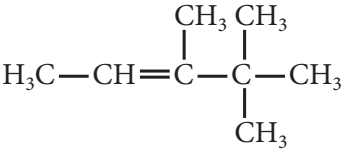
n-Butyl cyanide

91.

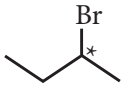
(i)



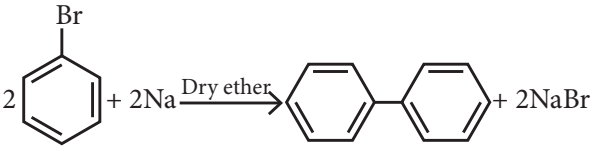
2, 2, 3-Trimethyl-3-bromopentane



(ii)

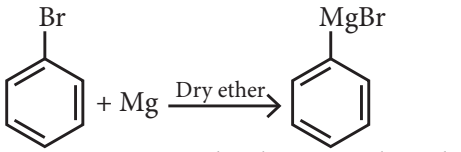


(iii)



Biphenyl

(*A*)

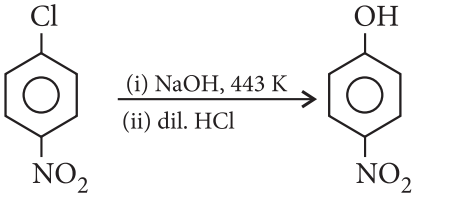


Phenylmagnesium bromide

(*B*)

92. (i) $(CH_3)_3C—I$ is more reactive because C—I bond has the minimum bond dissociation enthalpy.

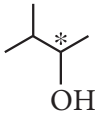
(ii)



4-Nitrophenol

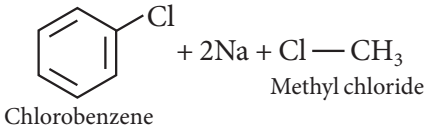
(iii) *Dextro* and *laevo*-rotatory isomers of butan-2-ol are difficult to separate by fractional distillation because they have almost same boiling point.

93. (a)



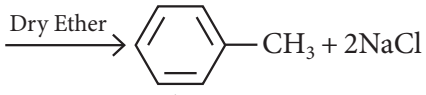
(i)

(b)



Chlorobenzene

Methyl chloride



Toluene

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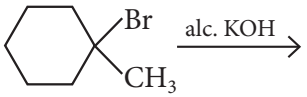
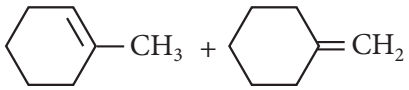
(c) 

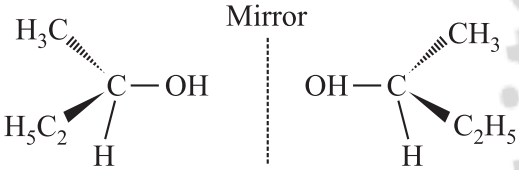
1-Bromo-1-methylcyclohexane



(major) (minor)

94. (i)1-Bromopentane
(ii) 2-Bromopentane
(iii) 2-Bromo-2-methylbutane

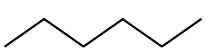
95. (i) The molecules which are not superimposable on their mirror images are called chiral molecules. The property of non-superimposability of a structure on its mirror image is called chirality.



Non superimposable mirror image of 2-butanol

(ii) $CH_3CH_2\underset{\underset{Cl}{|}}{C}HCH_3$ hydrolyses easily with KOH because it is secondary halide.

(iii) As iodide is a better leaving group because of its large size, therefore,

undergoes S_N2 reaction faster than 

96. (i) 1-Bromopentane > 2-Bromopentane > 2-Bromo-2-methylbutane.
(ii) 1-Bromo-3-methylbutane > 3-Bromo-2-methylbutane > 2-Bromo-2-methylbutane
(iii) 1-Bromobutane > 1-Bromo-2-methylbutane > 1-Bromo-2,2-dimethylpropane

97. (a) (i) Benzyl chloride gives white precipiate with $AgNO_3$ solution while chlorobenzene does not.
(ii) $CHCl_3$ with aniline in presence of alc. KOH gives foul smelling isocyanides whereas CCl_4 does not.
(b) CH_3Cl is hydrolysed easily than C_6H_5Cl as chlorobenzene has partial double bond character between C—Cl bond which is difficult to break.

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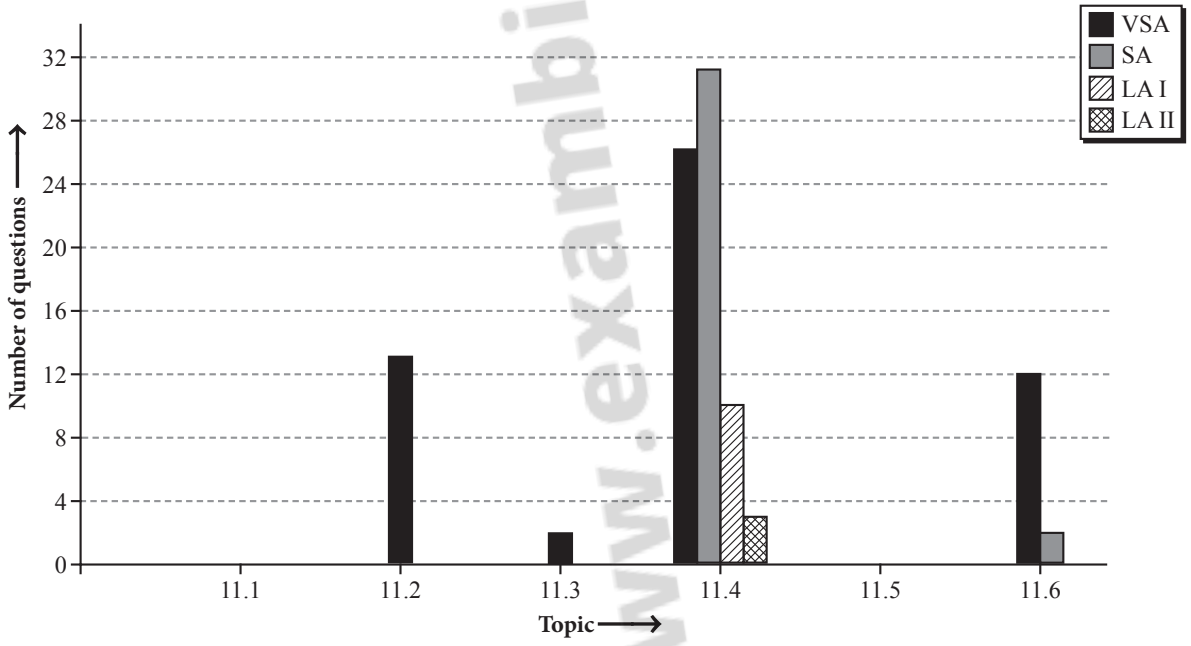
CHAPTER
11

Alcohols, Phenols and Ethers

11.1 Classification
11.2 Nomenclature
11.3 Structures of Functional Groups
11.4 Alcohols and Phenols
11.5 Some Commercially Important Alcohols
11.6 Ethers

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Alcohols and Phenols.*
- Maximum VSA type questions were asked from *Alcohols and Phenols.*
- Maximum SA and LA I type questions were asked from *Alcohols and Phenols.*

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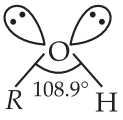
QUICK RECAP

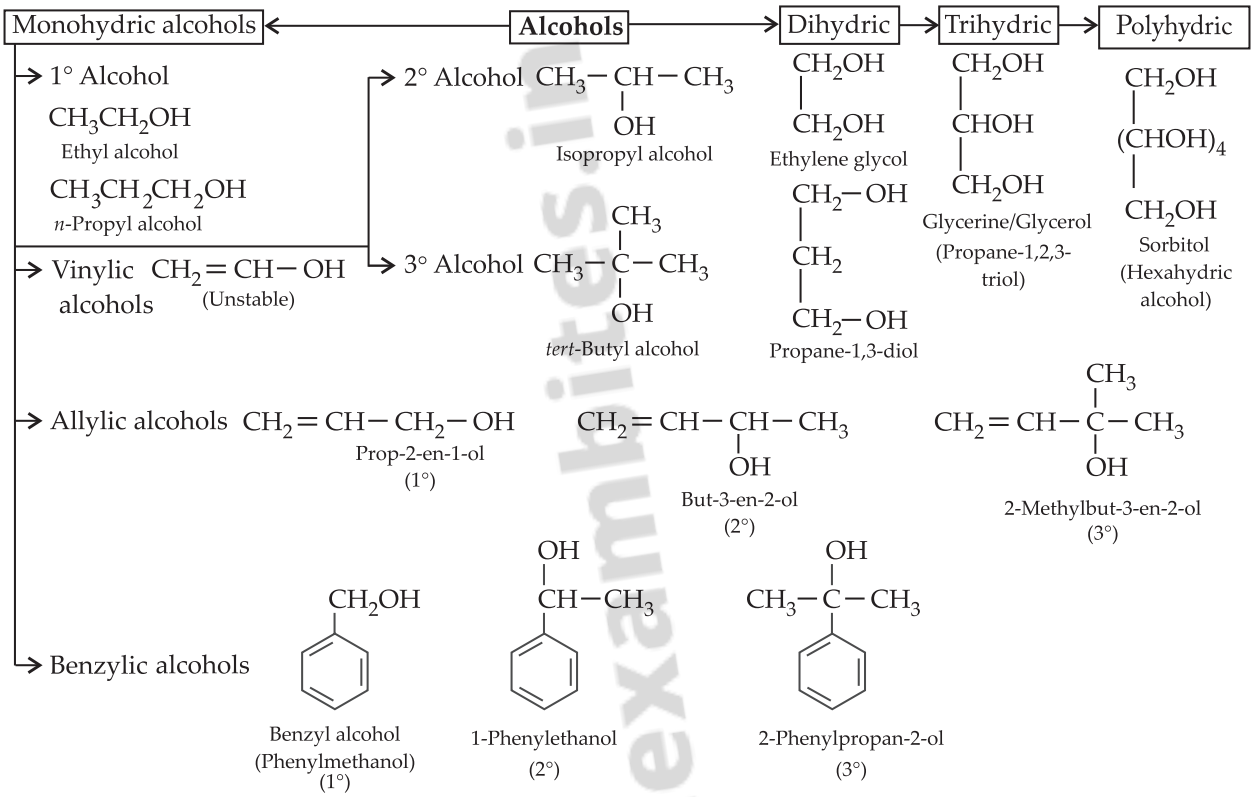
ALCOHOLS

- **General formula :** Alcohols are the hydroxy derivatives of alkanes having general formula $C_nH_{2n+1}OH$.
- **Nomenclature :** In common system, alcohols are named as *alkyl alcohol*. According to IUPAC system, alcohols are called '*alkanols*', by replacing '*-e*' of alkane by '*-ol*'.
- **Structure :** In alcohols, $R-OH$, the O-atom of hydroxyl group is attached to C-atom

by a sigma (σ) bond formed by the overlap of sp^3 hybridised orbital of C-atom with sp^3 hybridised orbital of O-atom.



Classification : Alcohols are classified as monohydric, dihydric, trihydric and polyhydric alcohols depending upon the number of hydroxyl (—OH) groups present in the molecule.



Preparation :

▶ **From alkenes :**

– **By acid catalysed hydration :**

$$CH_3CH{=}CH_2 + H_2O \overset{H^+}{\rightleftharpoons} CH_3{-}\underset{OH}{CH}{-}CH_3$$

– **By hydroboration-oxidation :**

$$CH_3CH{=}CH_2 + (H{-}BH_2)_2 \longrightarrow CH_3{-}\underset{H}{CH}{-}\underset{BH_2}{CH_2}$$

$$(CH_3CH_2CH_2)_2BH \xleftarrow{CH_3CH=CH_2}$$

$$(CH_3CH_2CH_2)_3B \xrightarrow{CH_3CH=CH_2}$$

$$3CH_3CH_2CH_2OH + B(OH)_3 \xleftarrow[OH^-]{H_2O_2}$$

Propan-1-ol

▶ **From carbonyl compounds :**

– **By reduction of aldehydes and ketones :**

$$RCHO + H_2 \xrightarrow{Pd} RCH_2OH$$

$$RCOR' \xrightarrow{NaBH_4} R{-}\underset{OH}{CH}{-}R'$$

– **By reduction of carboxylic acids and esters :**

$$RCOOH \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} RCH_2OH$$

$$RCOOR' \xrightarrow{H^+/H_2O} RCOOH + R'OH$$

– **From Grignard reagents :**

$$RCHO + R'MgX \xrightarrow{H_2O} R{-}\underset{\displaystyle\xrightarrow{H_2O}}{\overset{R'}{\overset{|}{CH}}}{-}OMgX$$

$$Mg(OH)X + R{-}\underset{R'}{CH}{-}OH$$

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$$RCOR + R'MgX \longrightarrow R-\underset{\displaystyle R}{\overset{\displaystyle R'}{\overset{|}{\underset{|}{C}}}}-OMgX \xrightarrow{H_2O} R-\underset{\displaystyle R}{\overset{\displaystyle R'}{\overset{|}{\underset{|}{C}}}}-OH + Mg(OH)X$$

Physical properties :

- **Physical state :** Lower alcohols are colourless liquids with characteristic smell while higher alcohols are colourless, odourless waxy solids.
- **Solubility :** Solubility of alcohols in water is due to their ability to form hydrogen bonds with water molecules. The solubility decreases with increase in size of alkyl groups and solubility increases with increase in branching and the order is $1° < 2° < 3°$.
- **Boiling points :** The boiling points of alcohols increase with increase in the number of carbon atoms as van der Waals' forces increase and the boiling points decrease with increase of branching in carbon chain due to decrease in van der Waals' forces with decrease in surface area and the order is $1° > 2° > 3°$.

Chemical properties : Alcohols react both as nucleophiles (when the bond between O – H is broken) and electrophiles (when the bond between C – O is broken).

- **Reactions involving cleavage of O—H bond :**
  - **Acidity of alcohols (reaction with metals) :**

$$2ROH + 2Na \longrightarrow \underset{\text{Sodium alkoxide}}{2RONa} + H_2$$

  - The above reaction shows that alcohols are acidic in nature.
  - The acid strength of alcohols decreases in the following order :

$$R\rightarrow CH_2OH > \underset{\text{Secondary}}{\overset{R}{\overset{R'}{\overset{\diagdown\ \diagup}{CH}}}OH} >> \underset{\text{Tertiary}}{R\rightarrow\overset{R}{\overset{R}{\overset{\diagdown\ \diagup}{C}}}-OH}$$

Primary

  - **Esterification :**

$$R'OH + (RCO)_2O \text{ or } R'COCl \longrightarrow RCOOR'$$

- **Reactions involving cleavage of C—O bond :**
  - **Reaction with hydrogen halides :**

$$ROH + HX \xrightarrow[\Delta]{\text{Anhyd, } ZnCl_2} RX + H_2O$$

  - **Lucas test :** Lucas reagent is a solution of conc. HCl with anhyd. $ZnCl_2$.
  - With Lucas reagent,

Primary alcohols – No cloudiness
Secondary alcohols – Cloudiness in 5 minutes
Tertiary alcohols – Cloudiness immediately

  - **Reaction with phosphorus trihalides :**

$$ROH + PCl_3 \longrightarrow RCl + H_3PO_3$$

  - **Dehydration :**

$$C_2H_5OH \xrightarrow[443\ K]{H_2SO_4} CH_2{=}CH_2 + H_2O$$

$$CH_3\underset{\displaystyle OH}{\overset{|}{C}}HCH_3 \xrightarrow[440\ K]{85\%\ H_3PO_4} CH_3CH{=}CH_2 + H_2O$$

$$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH \xrightarrow[358\ K]{20\%H_3PO_4} CH_3-\overset{\displaystyle CH_2}{\overset{\|}{C}}-CH_3 + H_2O$$

Thus, the relative ease of dehydration of alcohols follows the order :

Tertiary > Secondary > Primary

  - **Oxidation :**

$$RCH_2OH \xrightarrow{[O]} \underset{\text{Aldehyde}}{R-\overset{\displaystyle H}{\overset{|}{C}}=O} \xrightarrow{[O]} \underset{\text{Carboxylic acid}}{R-\overset{\displaystyle OH}{\overset{|}{C}}=O}$$

$$RCH_2OH \xrightarrow{CrO_3 \text{ or } Cu/573\ K} RCHO$$

$$CH_3CH{=}CHCH_2OH \xrightarrow{PCC} CH_3CH{=}CHCHO$$

$$\underset{\text{sec-alcohol}}{R-\underset{\displaystyle OH}{\overset{|}{CH}}-R'} \xrightarrow{CrO_3 \text{ or } Cu/573\ K} \underset{\text{Ketone}}{R-\overset{\displaystyle O}{\overset{\|}{C}}-R'}$$

$$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH \xrightarrow[573\ K]{Cu} CH_3-\overset{\displaystyle CH_3}{\overset{|}{C}}=CH_2$$

Some commercially important alcohols :

- **Methanol (wood spirit) :** It is produced by catalytic hydrogenation of CO in the presence of ZnO – Cr_2O_3 as catalyst at high temperature and pressure.

$$CO + 2H_2 \xrightarrow[200\text{-}300\ atm,\ 573\text{-}673\ K]{ZnO-Cr_2O_3} CH_3OH$$

  - Methanol is a colourless liquid and is highly poisonous in nature.

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– Ingestion of small quantities can cause blindness while large quantities can cause death.
– It is used as a solvent in paints, varnishes and for making formaldehyde.

▶ **Ethanol :** It is commercially manufactured by using fermentation of sugar present in molasses, sugarcane or fruits such as grapes.

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{Invertase}} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Zymase}} \underset{\text{Ethanol}}{2C_2H_5OH} + 2CO_2$$

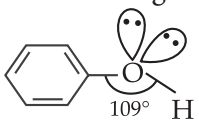
– Ethanol is colourless liquid and is used as a solvent in paint industry and in preparation of a number of carbon compounds.
– Commercial alcohol made unfit for drinking by mixing it with copper sulphate and pyridine, it is called *denatured alcohol*.

PHENOLS

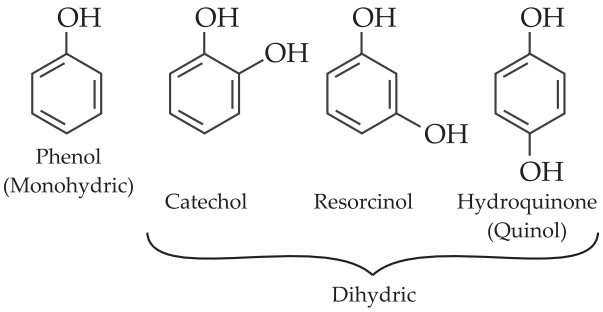
General formula : Phenols are the compounds in which hydroxy (—OH) group is directly linked to aromatic ring having formula, C_6H_5OH.

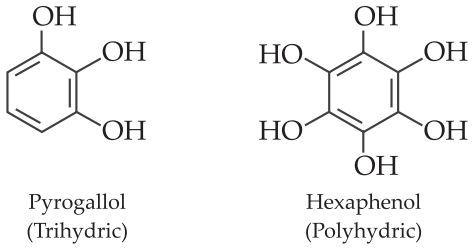
Nomenclature : The simplest hydroxy derivative of benzene is phenol, also called *carbolic acid*. It is its common name and also an accepted IUPAC name.

Structure : In phenols, the —OH group is attached to sp^2 hybridised C-atom of an aromatic ring.



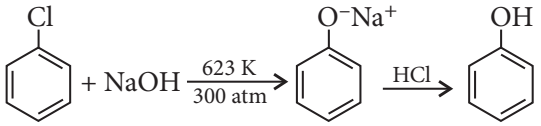
Classification : Like alcohols, phenols are also classified as mono and polyhydric phenols.



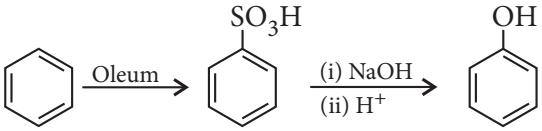


Preparation :

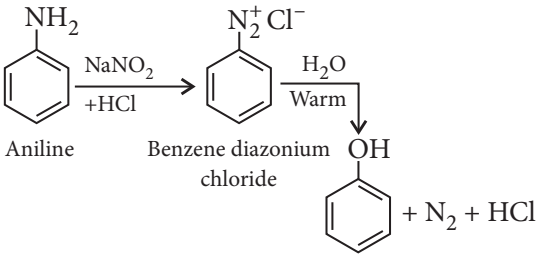
▶ **From haloarenes :**



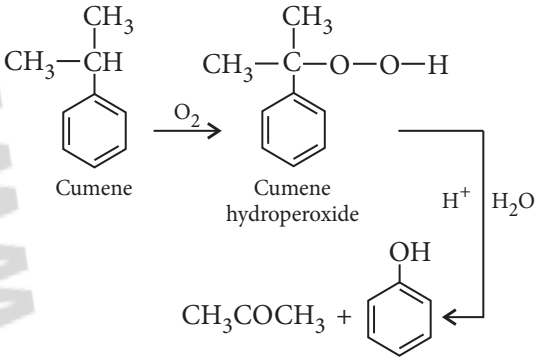
▶ **From benzenesulphonic acid :**



▶ **From diazonium salts :**



▶ **From cumene :**



Physical properties :

▶ **Physical state :** Phenols are colourless crystalline solids or liquids.

▶ **Odour :** They have characteristic phenolic odours.

▶ **Solubility :** Like alcohols, phenols are soluble in water due to the formation of hydrogen bonding with water.

– Phenols are less soluble than alcohols due to large hydrocarbon (benzene ring) part.

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– Phenols are soluble in alcohols, ethers and also in NaOH.

▶ **Boiling points :** The boiling points of phenols are much higher than the corresponding aromatic hydrocarbons and haloarenes due to intermolecular hydrogen bonding.

Chemical properties :

▶ **Acidity of phenols :** Phenols are weakly acidic in nature due to polar O—H bond directly attached to sp^2-hybridised C-atom.

– They turn blue litmus red and react with alkali metals and alkalies to form their salts.

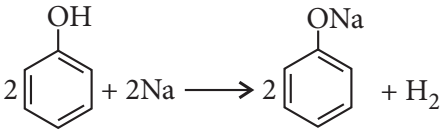
– Phenol is weaker acid than carboxylic acid. It does not react with sodium carbonate (Na_2CO_3) and sodium bicarbonate ($NaHCO_3$).

– Phenols are more acidic than alcohols which can be explained on the basis of resonance.

– Electron withdrawing groups increase the acidic strength of phenols.

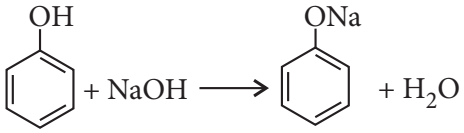
– Electron releasing groups decrease the acidic strength of phenols.

– **Reaction with metals :**



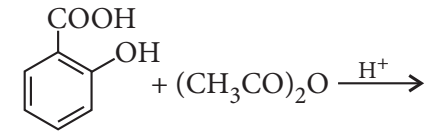
Phenol Sodium phenoxide

In addition to this, phenols react with aqueous sodium hydroxide to form sodium phenoxides.

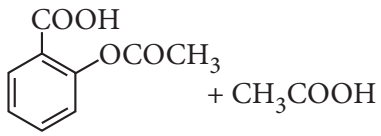


Sodium phenoxide

▶ **Esterification :**



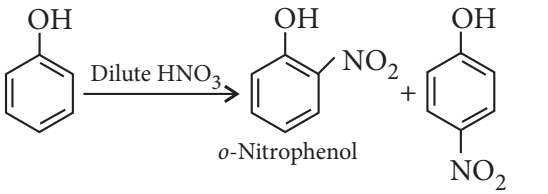
Salicylic acid



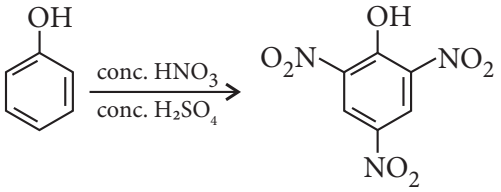
Acetylsalicylic acid
(Aspirin)

▶ **Electrophilic aromatic substitution :**

– **Nitration :**

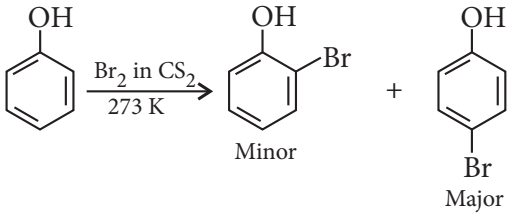


o-Nitrophenol *p*-Nitrophenol

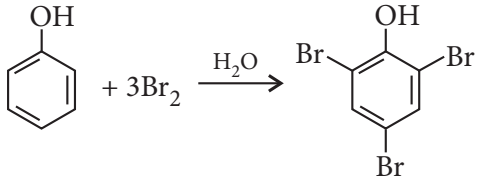


2,4,6-Trinitrophenol
(Picric acid)

– **Halogenation :**

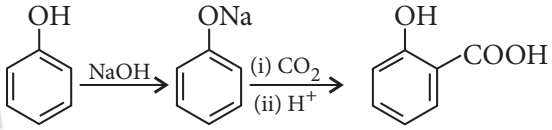


Minor Major



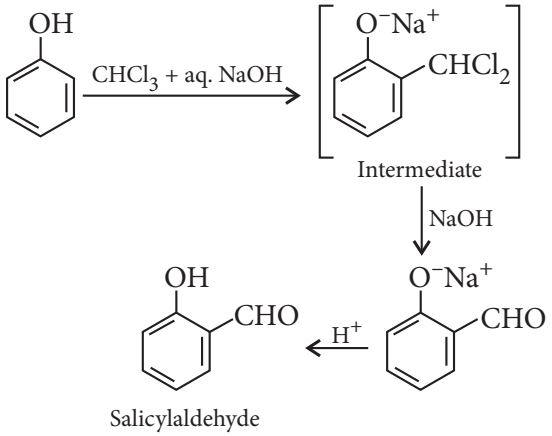
2,4,6-Tribromophenol

▶ **Kolbe's reaction :**



2-Hydroxybenzoic acid
(Salicylic acid)

▶ **Reimer-Tiemann reaction :**



Salicylaldehyde

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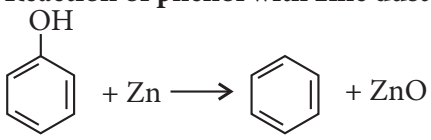
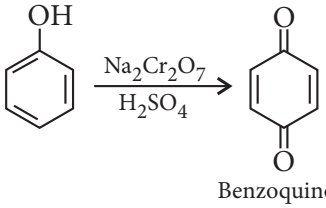
▶ **Reaction of phenol with zinc dust :**



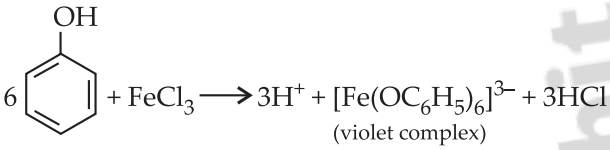
▶ **Oxidation**



Benzoquinone

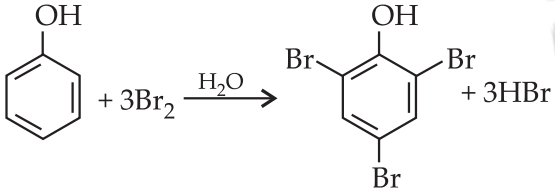
▶ **Test for phenols :**

– **Ferric chloride test :** Phenol gives violet colour with neutral $FeCl_3$ solution.



(violet complex)

– **Bromine water test :** Phenol gives white ppt. with Br_2-water due to the formation of 2,4,6-tribromophenol.



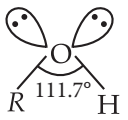
(white ppt.)

ETHERS

General formula : Ethers are the compounds having general formula, $C_nH_{2n+2}O$; where, n is always greater than 1.

Nomenclature : Common names of ethers are derived from the names of alkyl/ aryl groups written as separate words in alphabetical order and adding the word '*ether*' at the end. If both the alkyl groups are the same, the prefix '*di*' is added before the alkyl group. According to IUPAC system of nomenclature, ethers are regarded as hydrocarbon derivatives in which a hydrogen atom is replaced by an $-OR$ or $-OAr$ group, where R and Ar represent alkyl and aryl groups, respectively. The larger R group is chosen as the parent hydrocarbon.

Structure : In ethers, the four electron pairs, *i.e.*, the two bond pairs and two lone pairs of electrons on O-atom are arranged approximately in a tetrahedral arrangement.



Classification : Ethers can be classified as symmetrical or simple ethers having formula, $R-O-R$ and unsymmetrical or mixed ethers having formula, $R-O-R'$.

Preparation :

▶ **From alcohols by dehydration :**

$$2ROH \xrightarrow{H_2SO_4,\ 413\ K} ROR$$

▶ **From alkyl halide :**

$$R{-}X + RONa \xrightarrow[-NaX]{\Delta} ROR$$

(Williamson synthesis)

– Williamson synthesis can be used to prepare ethers containing 2° or 3° alkyl groups through S_N2 mechanism. In the case of 2° and 3° alkyl halides, elimination takes place. It cannot be used to prepare diaryl ethers.

– Dehydration of alcohols for the formation of ethers follows the order : 1° > 2° > 3°.

Physical properties :

▶ **Physical state and odour :** Dimethyl ether and ethyl methyl ether are exceptionally gases at room temperature while all other ethers are colourless liquids with characteristic ethereal smell.

▶ **Solubility :** Ethers are soluble in water to a certain extent due to hydrogen bonding.

– Solubility decreases with increase in molecular mass.

– Ethers are fairly soluble in all organic solvents such as alcohol, chloroform, benzene, etc.

▶ **Boiling points :** Ethers have lower boiling points than isomeric alcohols due to their inability to form hydrogen bonds and get associated.

– But lower ethers have slightly higher boiling points than *n*-alkanes of comparable molecular masses due to dipole-dipole interactions.

– Higher ethers (containing carbon atom more than four) have slightly

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lower boiling points than *n*-alkanes of comparable molecular masses due to weak van der Waals' forces of attraction.

– **Polarity :** Ethers are polar in nature.

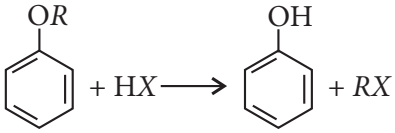
– **Density :** Ethers have low density. All ethers are lighter than water.

Chemical properties :

▶ **Cleavage of C—O bond in ethers :**

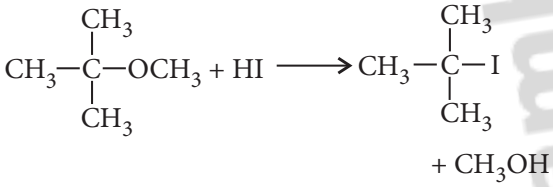
$ROR + HX \longrightarrow RX + ROH$

$ROH + HX \longrightarrow R{-}X + H_2O$



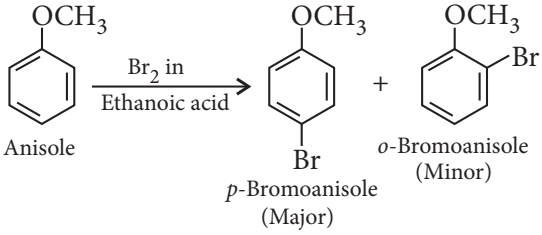
$ROR' + HX \longrightarrow RX + R'OH$

The order of reactivity of hydrogen halides is as follows : HI > HBr > HCl.

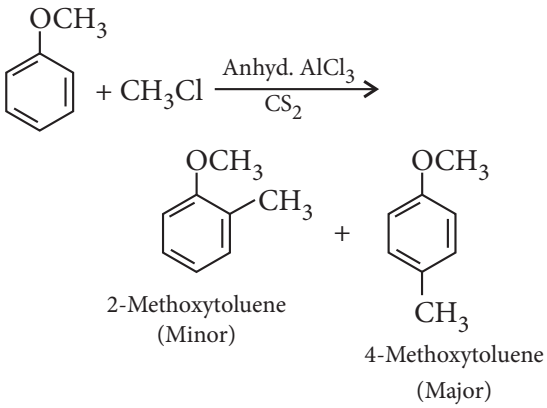


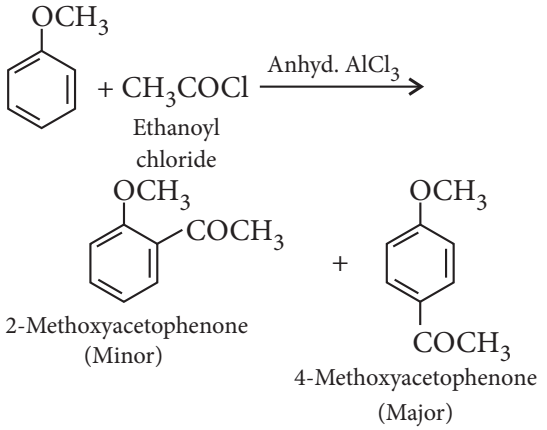
▶ **Electrophilic substitution :**

– **Halogenation :**

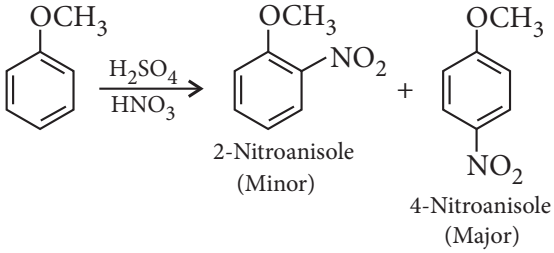


– **Friedel-Crafts reaction :**





– **Nitration :**



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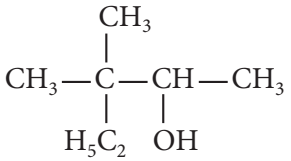
Previous Years' CBSE Board Questions

11.2 Nomenclature

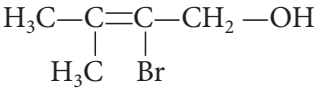
VSA (1 mark)

1. Write the IUPAC name of the following :



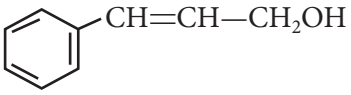
(2018)

2. Write the IUPAC name of the following compound :



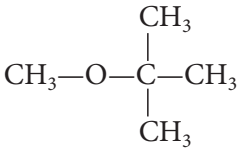
(AI 2017, AI 2012C)

3. Write the IUPAC name of the following compound :



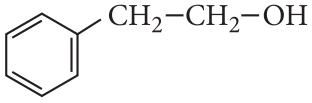
(AI 2017)

4. Write the IUPAC name of the following compound :



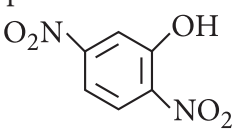
(AI 2017)

5. Write the IUPAC name of the given compound.



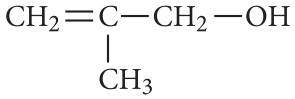
(AI 2016)

6. Write the IUPAC name of the given compound :



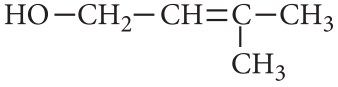
(Delhi 2015)

7. Write the IUPAC name of the given compound :



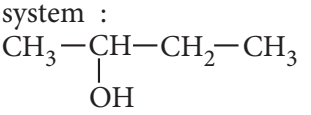
(AI 2015)

8. Write the IUPAC name of the given compound:



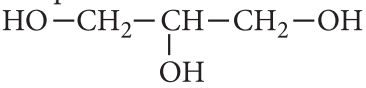
(Foreign 2015)

9. Name the following according to IUPAC system :



(1/2, Delhi 2015C)

10. Write IUPAC name of the following compound :

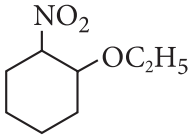


(Foreign 2014)

11. Draw the structural formula of 2-Methylpropan-2-ol molecule. *(Delhi 2012)*

12. Draw the structure of hex-1-en-3-ol compound. *(Delhi 2012)*

13. Write IUPAC name of the following :



(1/3, AI 2012C)

11.3 Structures of Functional Groups

VSA (1 mark)

14. **Assertion (A) :** The C—O—H bond angle in alcohols is slightly less than the tetrahedral angle.
Reason (R) : This is due to the repulsive interaction between the two lone electron pairs on oxygen.

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

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15. The C—O bond is much shorter in phenol than in ethanol. Give reason. *(Delhi 2012C)*

11.4 Alcohols and Phenols

VSA (1 mark)

16. Carry out the following conversion :
Phenol to salicylaldehyde *(1/5, 2020)*

17. Carry out the following conversion :
Propene to propanol *(1/5, 2020)*

18. Predict the reagent for carrying out the following conversion :
Phenol to benzoquinone *(1/5, 2020)*

19. Predict the reagent for carrying out the following conversion :
Phenol to 2,4,6-tribromophenol *(1/5, 2020)*

20. Write the preparation of phenol from cumene. *(1/3, 2020)*

21. How can you convert the following :
Phenol to chlorobenzene. *(1/3, 2020)*

22. How can you convert the following :
Sodium phenoxide to *o*-hydroxybenzoic acid. *(1/3, 2020)*

23. How can you convert the following?
Ethanol to propanenitrile *(1/3, 2020)*

24. Write the equation involved in the following reaction : Kolbe's reaction
(1/2, AI 2019, Delhi 2014C, 2013C)

25. Give reason for the following :
Phenol is more acidic than ethanol.
(1/3, AI 2017C, 2015)

26. Write the equation involved in the acetylation of salicylic acid. *(Delhi 2015)*

27. Which of the following isomers is more volatile :
o-nitrophenol or *p*-nitrophenol?
(Delhi 2014)

28. Write the equation involved in the following reaction :
Reimer –Tiemann reaction.
(1/2, AI 2014, 2013)

29. How is toluene obtained from phenol?
(1/3, Delhi 2013C)

30. Give a chemical test to distinguish between 2-pentanol and 3-pentanol.
(1/2, Delhi 2013C)

31. Of the two hydroxy organic compounds *R*OH and *R*′OH, the first one is basic and other is acidic in behaviour. How is *R* different from *R*′? *(Delhi 2013C)*

32. How would you obtain ethane-1, 2-diol from ethanol ? *(AI 2013C)*

33. How would you obtain acetophenone from phenol? *(AI 2013C)*

34. *Ortho*-nitrophenol has lower boiling point than *p*-nitrophenol. Why? *(Delhi 2012C)*

35. Give a chemical test to distinguish between benzoic acid and phenol. *(Delhi 2012C)*

36. Illustrate the following name reaction
Reimer-Tiemann Reaction. *(Delhi 2012C)*

37. Give a chemical test to distinguish between 2-propanol and 2-methyl-2-propanol.
(Delhi 2012C)

38. *Ortho*-nitrophenol is more acidic than *ortho*-methoxyphenol. Why? *(1/2, Delhi 2012C)*

39. Give the structure and IUPAC name of the product formed when propanone is reacted with methylmagnesium bromide followed by hydrolysis. *(AI 2012C)*

40. Explain the following giving one example :
Reimer-Tiemann reaction. *(1/3, Delhi 2011)*

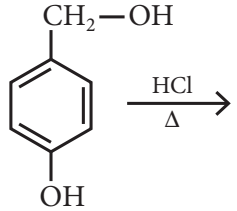
41. How would you convert ethanol to ethene?
(AI 2011)

SA (2 marks)

42. Out of *t*-butyl alcohol and *n*-butanol, which one will undergo acid catalysed dehydration faster and why? *(2/5, 2020)*

43. Give the mechanism for the formation of ethanol from ethene. *(2/5, 2020)*

44. Write the major product(s) of the following reactions :

(i) 

(ii) $(CH_3)_3COH \xrightarrow[573\ K]{Cu}$ *(2/3, 2020)*

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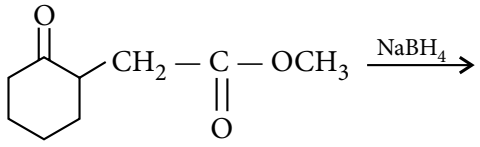
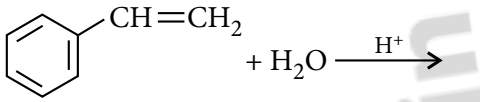
45. Write the structures of the main products in the following reactions :

(i) 

(ii) 

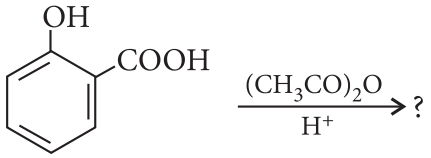
(2/3, 2018)

46. Explain the mechanism of dehydration steps of ethanol.

$$CH_3CH_2OH \xrightarrow[443\ K]{H^+} CH_2{=}CH_2 + H_2O$$

(2/3, 2018C, Delhi 2015C)

47. Write the product(s) in the following reactions :

(i) 

(ii) $CH_3-CH=CH-CH_2-OH \xrightarrow{PCC} ?$

(2/5, Delhi 2017)

48. Give simple chemical tests to distinguish between the following pairs of compounds :

(i) Ethanol and phenol

(ii) Propanol and 2-methylpropan-2-ol.

(2/5, Delhi 2017)

49. (a) Arrange the following compounds in the increasing order of their acid strength :

p-cresol, *p*-nitrophenol, phenol

(b) Write the mechanism (using curved arrow notation) of the following reaction :

$$CH_2{=}CH_2 \xrightarrow{H_3O^+} CH_3-CH_2^+ + H_2O$$

(AI 2017)

50. Write the structure of the products when Butan-2-ol reacts with the following :

(a) CrO_3 (b) $SOCl_2$ *(AI 2017)*

51. Write the main product(s) in each of the following reactions :

(i) $CH_3-CH=CH_2 \xrightarrow[(ii)\ 3H_2O_2/OH^-]{(i)\ B_2H_6}$

(ii) $C_6H_5-OH \xrightarrow[(ii)\ CO_2,\ H^+]{(i)\ aq.\ NaOH}$ *(Delhi 2016)*

52. Write the final product(s) in each of the following reactions :

(i) $CH_3CH_2-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow{Cu/573\ K}$

(ii) $C_6H_5-OH \xrightarrow[(ii)\ H^+]{(i)\ CHCl_3 + aq.\ NaOH}$

(Delhi 2016)

53. How are the following conversions carried out?

(i) Propene to propane-2-ol

(ii) Benzyl chloride to Benzyl alcohol

(2/3, Delhi 2015C)

54. Write the mechanism of acid dehydration of ethanol to yield ethene. *(AI 2015C)*

55. Write the mechanism of the following reaction :

$$CH_3CH_2OH \xrightarrow{HBr} CH_3CH_2Br + H_2O$$

(AI 2014, 2/3, Foreign 2014)

56. Name the reagents used in the following reactions :

(i) Bromination of phenol to 2, 4, 6-tribromophenol

(ii) Butan-2-one to Butan-2-o1

(iii) Friedel–Crafts alkylation of anisole

(iv) Oxidation of primary alcohol to carboxylic acid *(Foreign 2014)*

57. Name the different reagents needed to perform the following reactions :

(i) Phenol to Benzene

(ii) Dehydration of propan-2-ol to propene

(iii) Friedel-Crafts alkylation of anisole

(iv) Dehydrogenation of ethanol to ethanal

(Foreign 2014)

58. How are the following conversions carried out?

(i) Propene to Propan-2-ol

(ii) Ethyl chloride to Ethanal *(Delhi 2014C)*

59. Explain the following with an example for each :

(i) Kolbe's reaction,

(ii) Reimer-Tiemann reaction

(2/3, AI 2014C)

60. How will you convert :

(i) Propene to propan-2-ol,

(ii) Phenol to 2,4,6-trinitrophenol?

(Delhi 2013)

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61. How will you convert the following :
(i) Propan-2-ol to propanone
(ii) Phenol to 2,4,6-tribromophenol.
(Delhi 2013)

62. Explain the mechanism of the following reaction :

$$CH_3—CH_2—OH \xrightarrow[443\ K]{H^+} CH_2=CH_2 + H_2O$$

(AI 2013)

63. Explain the mechanism of acid catalysed hydration of an alkene to form corresponding alcohol. *(AI 2012)*

64. Explain the following behaviours :
(i) Alcohols are more soluble in water than the hydrocarbons of comparable molecular masses.
(ii) *Ortho*-nitrophenol is more acidic than *ortho*-methoxyphenol. *(AI 2012)*

65. Give a separate chemical test to distinguish between the following pairs of compounds :
(i) Ethanol and Phenol
(ii) 2-Pentanol and 3-Pentanol
(Delhi 2012C)

66. How would you obtain the following :
(i) 2-methylpentan-2-ol from 2-methyl-1-pentene
(ii) Acetophenone from phenol.
(2/3, AI 2012C)

67. How would you obtain
(i) Picric acid (2, 4, 6-trinitrophenol) from phenol.
(ii) 2-Methylpropene from 2-methylpropanol?
(Delhi 2011)

68. How would you obtain the following :
(i) Benzoquinone from phenol
(ii) 2-Methylpropan-2-ol from methyl magnesium bromide? *(AI 2011)*

69. Give the names of the reagents of bringing about the following transformations :
(i) Hexan-1-ol to hexanal
(ii) But-2-ene to ethanol *(Delhi 2011C)*

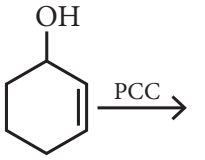
70. Account for the following :
(i) Propanol has higher boiling point than butane.
(ii) *Ortho*-nitrophenol is more acidic than *ortho*-methoxyphenol.
(2/3, Delhi 2011C)

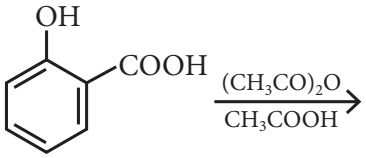
71. Account for the following :
(i) The boiling point of ethanol is higher than that of methanol.
(ii) Phenol is a stronger acid than an alcohol. *(Delhi 2011C)*

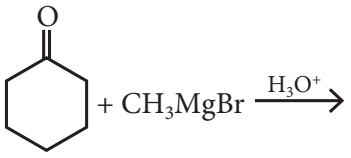
72. Write Reimer–Tiemann reaction giving an example. *(AI 2011C)*

LA I (3 marks)

73. Write the product(s) of the following reactions :

(i) 

(ii) 

(iii) 

(2020)

74. (a) Show how will you synthesise the following alcohol prepared by the reaction of a suitable Grignard reagent on methanal?

$$CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—OH$$

(b) Write the mechanism of the following reaction :

$$CH_2=CH_2 + H_2O \xrightarrow{H^+} CH_3CH_2OH$$

(AI 2019)

75. How do you convert the following?
(i) Phenol to anisole
(ii) Propan-2-ol to 2-methylpropan-2-ol
(iii) Aniline to phenol *(Delhi 2015)*

76. Predict the products of the following reactions :

(i) $CH_3—CH=CH_2 \xrightarrow[(ii)\ 3H_2O_2/OH^-]{(i)\ B_2H_6} ?$

(ii) $C_6H_5OH \xrightarrow{Br_2(aq)} ?$

(iii) $CH_3CH_2OH \xrightarrow{Cu/573\ K} ?$ *(Foreign 2015)*

77. How are the following conversions carried out?

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(i) Benzyl chloride to benzyl alcohol
(ii) Ethyl magnesium chloride to propan-1-ol
(iii) Propene to propan-2-ol.
(AI 2015C, 2014C)

78. (a) Write the mechanism of the following reaction :

$$CH_3CH_2OH \xrightarrow{HBr} CH_3CH_2Br + H_2O$$

(b) Write the equation involved in Reimer–Tiemann reaction. *(Delhi 2014)*

79 (a) Give chemical tests to distinguish between the following pairs of compounds :
(i) Pentan-2-ol and Pentan-3-ol
(ii) Methanol and Phenol
(b) *o*-Nitrophenol is more acidic than *o*-methoxyphenol. Explain why.
(AI 2013C)

80. Draw the structure and name of the product formed if the following alcohols are oxidized. Assume that an excess of oxidising agent is used.
(i) $CH_3CH_2CH_2CH_2OH$
(ii) 2-butenol
(iii) 2-methyl-1-propanol *(Delhi 2012)*

81. (a) Describe the mechanism of hydration of ethene to yield ethanol.
(b) Write Kolbe's reaction with an example.
(AI 2011C)

82. Acid catalysed dehydration of *tert*-butanol is faster than that of *n*-butanol. Explain.
(AI 2011C)

LA II (5 marks)

83. (a) How do you convert the following :
(i) Phenol to Anisole
(ii) Ethanol to Propan-2-ol
(b) Write mechanism of the following reaction :

$$C_2H_5OH \xrightarrow[443\,K]{H_2SO_4} CH_2 = CH_2 + H_2O$$

(c) Why phenol undergoes electrophilic substitution more easily than benzene?
(Delhi 2019)

84. (a) Account for the following :
(i) *o*-nitrophenol is more steam volatile than *p*-nitrophenol.
(ii) *t*-butyl chloride on heating with sodium methoxide gives 2-methyl-propene instead of *t*-butylmethylether.
(b) Write the reaction involved in the following :
(i) Reimer—Tiemann reaction
(ii) Friedal—Crafts alkylation of phenol
(c) Give simple chemical test to distinguish between ethanol and phenol.
(Delhi 2019)

85. (a) Write the formula of reagents used in the following reactions :
(i) Bromination of phenol to 2,4,6- tribromophenol
(ii) Hydroboration of propene and then oxidation to propanol.
(b) Arrange the following compound groups in the increasing order of their property indicated :
(i) *p*-nitrophenol, ethanol, phenol (acidic character)
(ii) propanol, propane, propanal (boiling point)
(c) Write the mechanism (using curved arrow notation) of the following reaction:

$$CH_3-CH_2-\overset{+}{O}H_2 \xrightarrow{CH_3CH_2OH} CH_3-CH_2-\overset{+}{\underset{\underset{H}{|}}{O}}-CH_2-CH_3 + H_2O$$

(Delhi 2017)

11.6 Ethers

VSA (1 mark)

86. **Assertion (A) :** $(CH_3)_3C-O-CH_3$ gives $(CH_3)_3C-I$ and CH_3OH on treatment with HI.
Reason (R) : The reaction occurs by S_N1 mechanism.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. (2020)

87. Carry out the following conversion :
t-Butyl chloride to *t*-butyl ethyl ether
(1/3, 2020)

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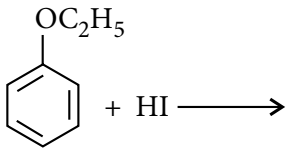
88. Predict the reagent for carrying out the following conversions :
Anisole to *p*-bromoanisole *(1/3, 2020)*

89. Write the equation for the preparation of 2-methyl-2-methoxypropane by Williamson synthesis. *(1/3, 2020)*

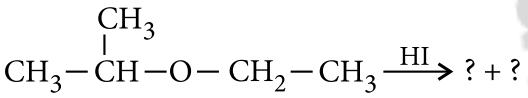
90. Write the equations involved in the following reaction :
Friedel—Crafts alkylation of anisole.
(1/2, AI 2019, 1/3, Delhi 2011)

91. Write the structure of the main products in the following reaction :



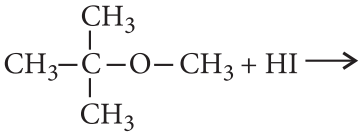
(1/3, 2018)

92. Write the product in the following reaction :



(1/5, Delhi 2017)

93. Write the main product(s) in the following reaction :



(1/3, Delhi 2016, 1/2, AI 2016)

94. How is the following conversion carried out?
Anisole to *p*-bromoanisole *(1/3, Delhi 2015C)*

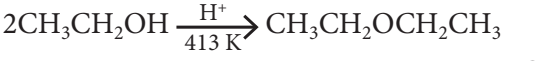
95. Write the equation involved in the following reaction :
Williamson synthesis
(1/2, AI 2014, 2013, Delhi 2014C 1/3, AI 2014C, Delhi 2012C)

96. Account for the following :
Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. *(1/3, Delhi 2011C)*

97. Account for the following :
The boiling points of ethers are lower than isomeric alcohols. *(1/3, AI 2011C)*

SA (2 marks)

98. Write the mechanism of the following reaction :

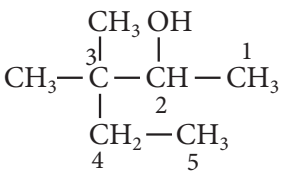


(2/3, 2020, 2/3, AI 2017, Delhi 2016, 2015, 2013, 2013C)

99. Give reasons for the following :
(i) Boiling point of ethanol is higher in comparison to methoxymethane.
(ii) $(CH_3)_3C—O—CH_3$ on reaction with HI gives CH_3OH and $(CH_3)_3C—I$ as the main products and not $(CH_3)_3C—OH$ and CH_3I. *(2/3, AI 2015)*

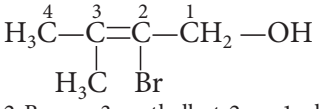
Detailed Solutions

1.



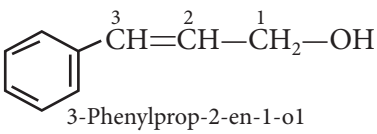
3,3 – Dimethylpentan-2-ol

2.



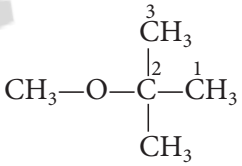
2-Bromo-3-methylbut-2-en-1-ol

3.



3-Phenylprop-2-en-1-o1

4.



2-Methoxy-2-methylpropane

5. 2-Phenylethanol

6. 2,5-Dinitrophenol

7. 2-Methylprop-2-en-1-ol

8. 3-Methylbut-2-en-1-ol

9. Butan-2-ol

10. Propane-1,2,3-triol

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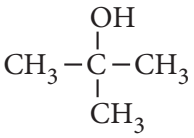
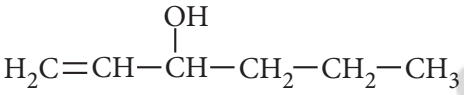
11. 
2-Methylpropan-2-ol

12. Hex-1-en-3-ol



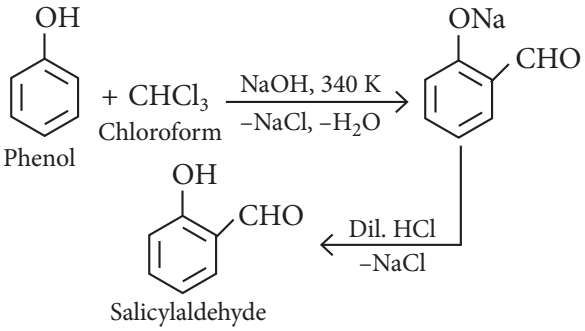
13. 1-Ethoxy-2-nitrocyclohexane.

14. **(a)** : The C—O—H bond angle in alcohols is slightly less than the tetrahedral angle due to greater lone pair-lone pair repulsions than lone pair-bond pair repulsions.

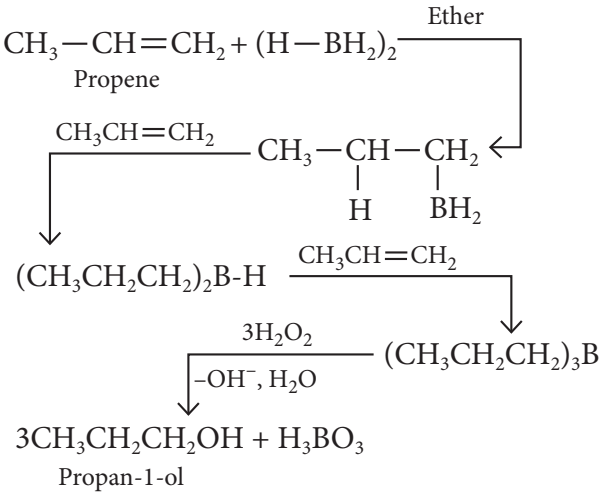
15. Due to resonance C—O bond acquires some partial double bond character.

So, in phenol C—O bond length is smaller than ethanol.

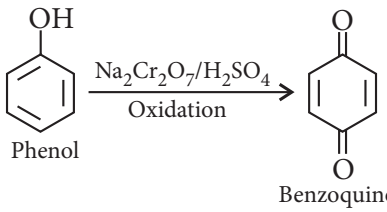
16.



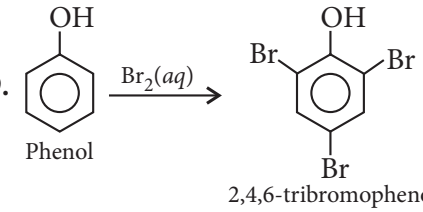
17.



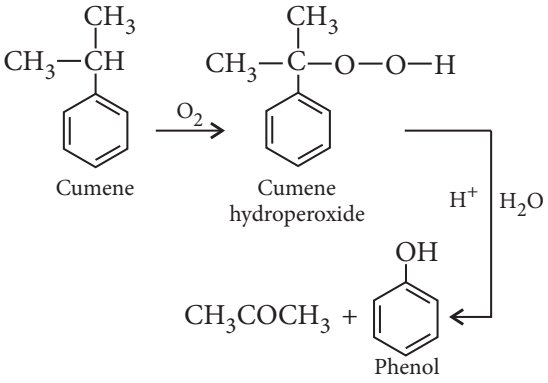
18.



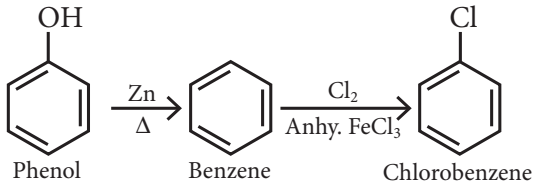
19.



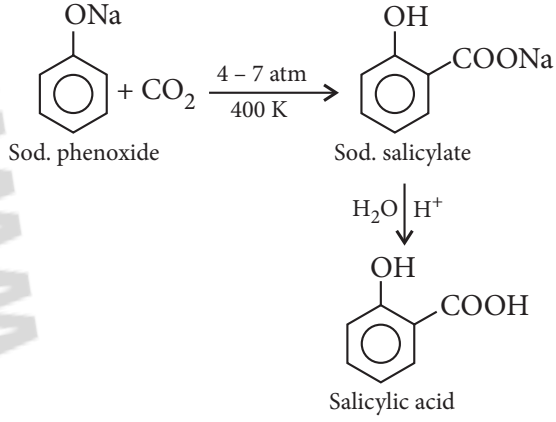
20.



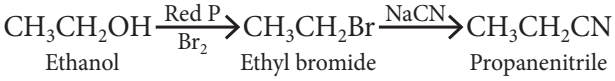
21.



22. **Kolbe's reaction :** When sodium phenoxide is heated with carbon dioxide under pressure, it gives salicylic acid.



23.



24. *Refer to answer 22.*

25. Phenols are more acidic than alcohols. It can be explained on the basis of that alcohol on losing H^+ ions form alkoxide ion and phenol forms phenoxide ion.

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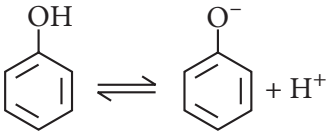
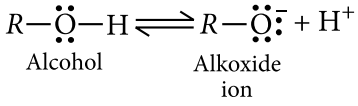
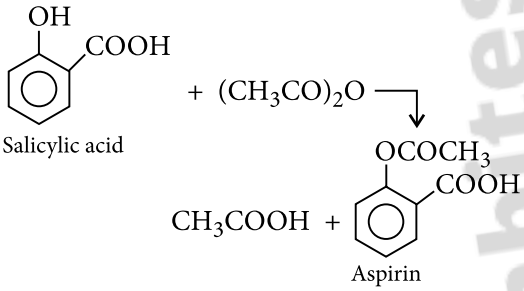
The greater acidity of phenol is due to the stability of the phenoxide ion which is resonance stabilized.



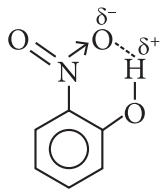
On the other hand, alkoxide ion shows no such resonance stabilisation and is unstable.



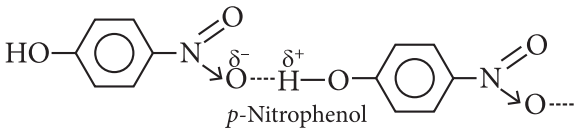
26.



27. *o*-Nitrophenol is more steam volatile than *p*-nitrophenol due to the presence of intramolecular H-bonding. *p*-Nitrophenol shows intermolecular H–bonding.



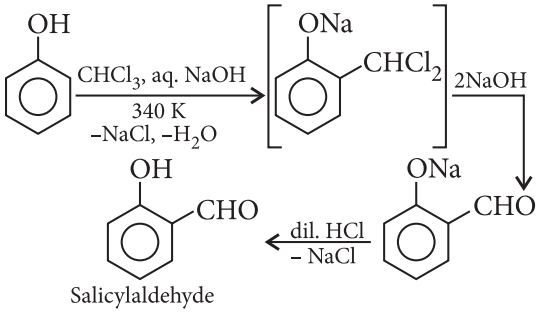
o-Nitrophenol
(intramolecular H-bonding)



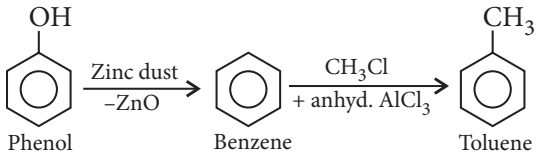
p-Nitrophenol
(intermolecular H-bonding)

That's why *o*-nitrophenol has lower boiling point than *p*-nitrophenol.

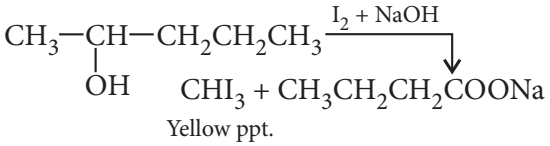
28. Reimer–Tiemann reaction :



29.

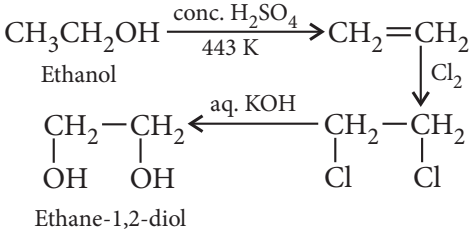


30. On adding I_2 and NaOH, 2-pentanol will give yellow precipitate of iodoform whereas 3-pentanol will not give yellow precipitate.

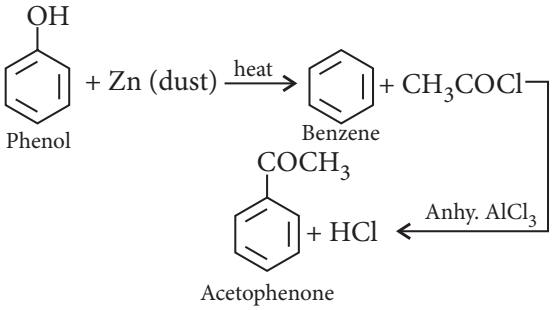


31. *R* is alkyl group and *R*′ is aryl group. *R* must be a group having more electron density than H. *i.e.*, having +*I* effect whereas *R*′ must be having +*R* effect.

32.



33.



34. *Refer to answer 27.*

35. Benzoic acid gives brisk effervescence of CO_2 on addition of $NaHCO_3$ while phenol does not.

36. *Refer to answer 28.*

37. 2-Propanol will give yellow precipitate of iodoform on addition of I_2 and NaOH while 2-methyl-2-propanol will not.

38. As we know that the electron withdrawing groups enhance the acidic character of phenols because they help in the stabilisation of phenoxide ion by dispersing negative charge. Nitro group is an electron withdrawing group whereas methoxy group destabilise the phenoxide ion by intensifying the negative charge. Thus, *o*-nitrophenol is more acidic than *o*-methoxyphenol.

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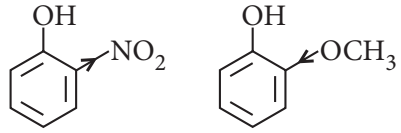


o-Nitrophenol *o*-Methoxyphenol

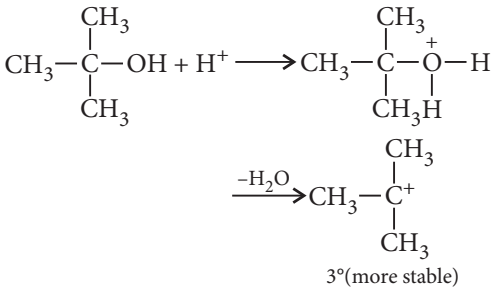
39. $CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + CH_3MgBr \xrightarrow{H_2O}$ $CH_3-\overset{\overset{OH}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$

2-Methylpropan-2-ol

40. *Refer to answer 28.*

41. $\underset{\text{Ethanol}}{CH_3CH_2OH} \xrightarrow[443\ K]{\text{conc. } H_2SO_4} \underset{\text{Ethene}}{CH_2=CH_2}$

42. Acid catalysed dehydration of alcohols follows carbocation mechanism.



3°(more stable)

$CH_3CH_2CH_2CH_2OH + H^+ \longrightarrow$

$CH_3CH_2CH_2CH_2\overset{+}{O}H_2 \xrightarrow[-H_2O]{} CH_3CH_2CH_2\overset{+}{C}H_2$

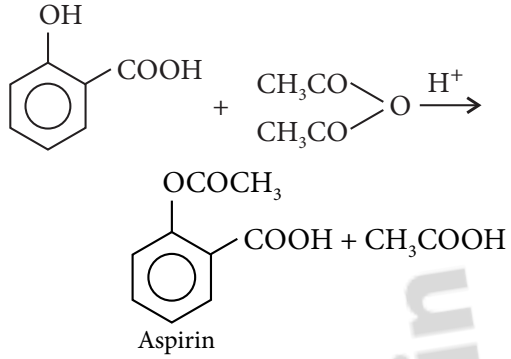
1°(less stable)

Hence, dehydration of *tert*-butanol which forms 3° carbocation is faster than *n*-butanol which forms primary carbocation.

43. $\underset{\text{Ethene}}{H_2C=CH_2} + H^+ \rightleftharpoons H_3C-\overset{+}{C}H_2 \xrightarrow{H_2O}$

$H_3C-CH_2-OH \xleftarrow{-H^+} H_3C-CH_2$

$\underset{\text{Ethanol}}{}$ $\quad\quad\quad

47. (i) 

Aspirin

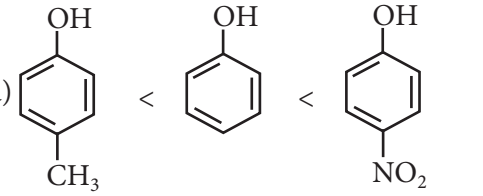
(ii) $CH_3-CH=CH-CH_2OH \xrightarrow{PCC}$ $CH_3-CH=CH-CHO$
But-2-en-1-ol → But-2-enal

48. (i) Distinction between ethanol and phenol.
$FeCl_3$ test : Phenol gives a violet colouration with $FeCl_3$ solution while ethanol does not.

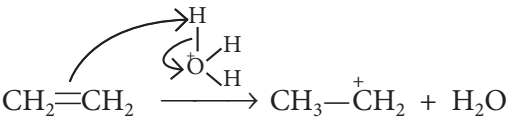
$3C_6H_5OH + FeCl_3 \longrightarrow (C_6H_5O)_3Fe + 3HCl$
Phenol → Violet colouration

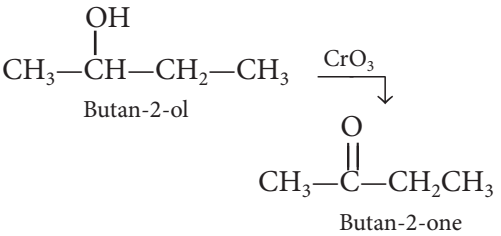
$C_2H_5OH + FeCl_3 \longrightarrow$ No violet colouration

(ii) On oxidation in presence of acidic $KMnO_4$, 1° alcohol (propanol) will give aldehyde while 3° alcohol (2-methylpropan-2-ol) will give a ketone.

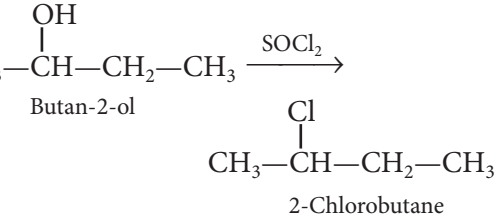
49. (a) 

p-Cresol, Phenol, *p*-Nitrophenol

(b) 

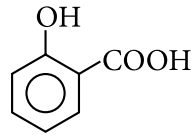
50. (a) 

Butan-2-ol → Butan-2-one

(b) 

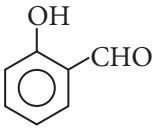
Butan-2-ol → 2-Chlorobutane

51. (i) $CH_3-CH=CH_2 \xrightarrow[(ii)\ 3H_2O_2/OH^-]{(i)\ B_2H_6}$ $CH_3CH_2CH_2OH$
Propanol

(ii) $C_6H_5OH \xrightarrow[(ii)\ CO_2,H^+]{(i)\ aq.\ NaOH}$ 

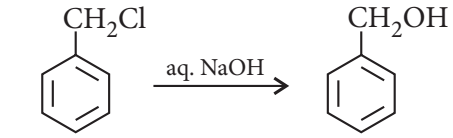
Salicylic acid

52.
(i) $CH_3CH_2-CH(OH)-CH_3 \xrightarrow{Cu/573\ K}$ $CH_3CH_2-C(=O)-CH_3$
Butan-2-one

(ii) $C_6H_5-OH \xrightarrow[(ii)\ H^+]{(i)\ CHCl_3 + aq.\ NaOH}$ 

Salicylaldehyde

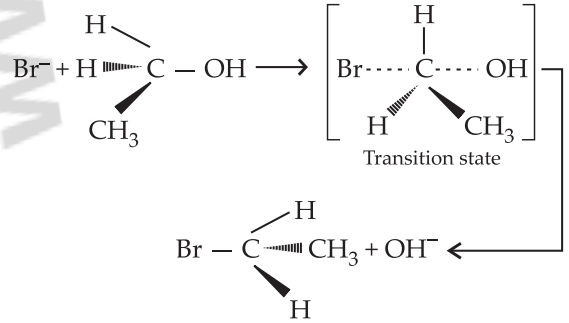
53. (i) $CH_3CH=CH_2 \xrightarrow[H_2O]{H_2SO_4} CH_3-CH(OH)-CH_3$
Propene → Propan-2-ol

(ii) 

Benzyl chloride → Benzyl alcohol

54. *Refer to answer 46.*

55. The reaction proceeds through nucleophilic substitution bimolecular (S_N2) mechanism, as shown below :



Inversion of configuration takes place during the reaction.

56. (i) Bromine water, ($Br_{2(aq)}$)
(ii) Lithium aluminium hydride, ($LiAlH_4$) or H_2/Ni
(iii) Alkyl halide in the presence of anhydrous aluminium chloride, CH_3Cl and $AlCl_3$ (anhy.)

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(iv) Acidified potassium permangante, $KMnO_4$, H_3O^+

57. (i) Zinc dust

(ii) Concentrated H_2SO_4

(iii) *Refer to answer 56(iii).*

(iv) Cu/573 K

58. (i) *Refer to answer 53(i).*

(ii) $CH_3CH_2Cl + KOH_{(aq)} \longrightarrow CH_3CH_2OH$

Ethyl chloride — Ethanol

573 K, Cu

$CH_3-\overset{O}{\overset{\|}{C}}-H$

Ethanal

59. (i) *Refer to answer 22.*

(ii) *Refer to answer 28.*

60. (i) *Refer to answer 53(i).*

(ii) Phenol $\xrightarrow{\text{conc. } HNO_3}$ Picric acid (2,4,6-Trinitrophenol)

61. (i) $CH_3-\underset{OH}{CH}-CH_3 \xrightarrow{Cu/573\ K} CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

Propan-2-ol — Propanone

(ii) Phenol $\xrightarrow{Br_2(aq)}$ 2,4,6-tribromophenol

62. *Refer to answer 46.*

63. *Refer to answer 43.*

64. (i) The solubility of alcohols in water is due to their ability to form hydrogen bonds with water molecules. Hydrocarbons cannot form such hydrogen bonds, hence they are insoluble in water.

(ii) *Refer to answer 38.*

65. (i) *Refer to answer 48 (i).*

(ii) *Refer to answer 30.*

66.

(i) $H_2C=\underset{\text{2-methyl-1-pentene}}{\overset{CH_3}{\overset{|}{C}}-CH_2-CH_2-CH_3} \xrightarrow{H_2O/H_2SO_4} H_3C-\overset{CH_3}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CH_2CH_2CH_3$

2-methylpentan-2-ol

(ii) *Refer to answer 33.*

67. (i) *Refer to answer 60(ii).*

(ii) $CH_3-\underset{\text{2-Methylpropanol}}{\overset{CH_3}{\overset{|}{CH}}-CH_2OH} \xrightarrow{SOCl_2} CH_3-\overset{CH_3}{\overset{|}{CH}}-CH_2-Cl \xrightarrow{\text{alc. KOH}} \underset{\text{2-Methylpropene}}{\overset{CH_3}{\overset{|}{CH_3-C=CH_2}}}$

68. (i) Phenol $\xrightarrow[\text{Oxidation}]{Na_2Cr_2O_7/H_2SO_4}$ Benzoquinone

(ii) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + CH_3MgBr \longrightarrow \left[CH_3-\overset{OMgBr}{\overset{|}{C}}-CH_3\right] \xrightarrow{H_2O} \underset{\text{2-Methylpropan-2-ol}}{CH_3-\overset{OH}{\overset{|}{C}}-CH_3}$

69. (i) Cu at 573 K

(ii) Ozonolysis followed by catalytic hydrogenation

$H_3C-CH=CH-CH_3 \xrightarrow{\text{Ozonolysis}}$

But-2-ene

$2CH_3CHO \xrightarrow{H_2/Ni} 2CH_3CH_2OH$

Ethanol

70. (i) The molecules of butane are held together by weak van der Waals' forces of attraction while those of propanol are held together by stronger intermolecular hydrogen bonding.

$H-\overset{\delta^-}{O}\cdots\overset{\delta^+}{H}-O\cdots$

$C_3H_7 \quad C_3H_7$

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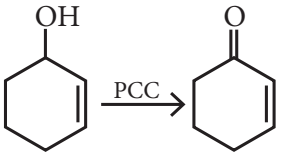
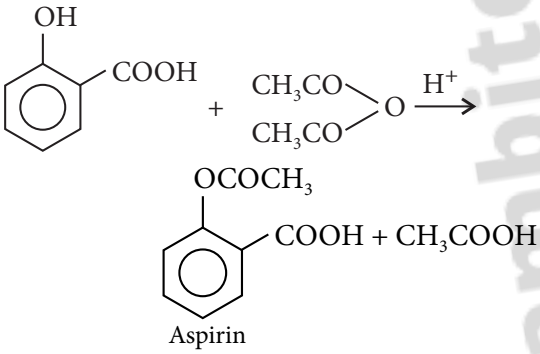
(ii) *Refer to answer 38.*

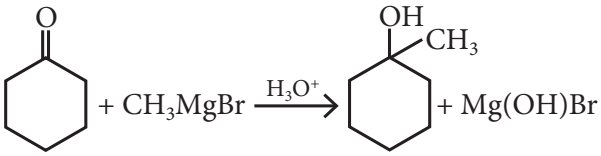
71. (i) It is due to higher molecular weight, more surface area, more van der Waals' forces of attraction in C_2H_5OH than CH_3OH.

(ii) *Refer to answer 25.*

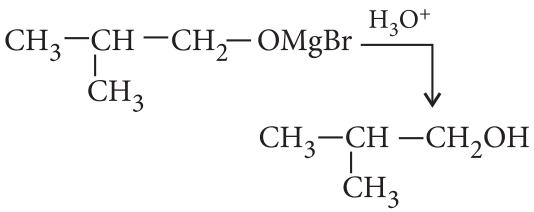
72. *Refer to answer 28.*

73. (i) 

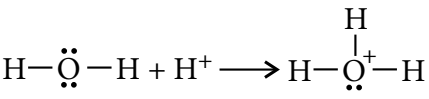
(ii) 

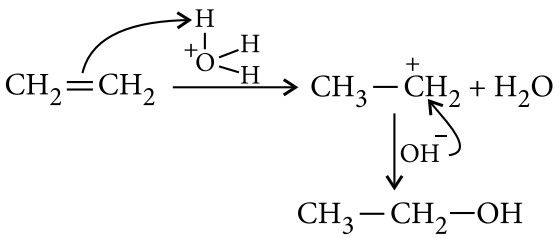
(iii) 

74. (a) $HCHO + (CH_3)_2CHMgBr \longrightarrow$

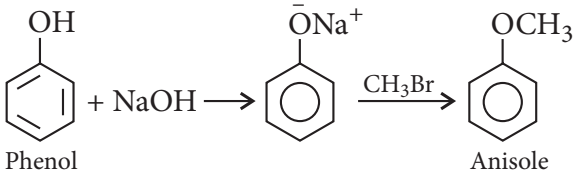


(b) $H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$

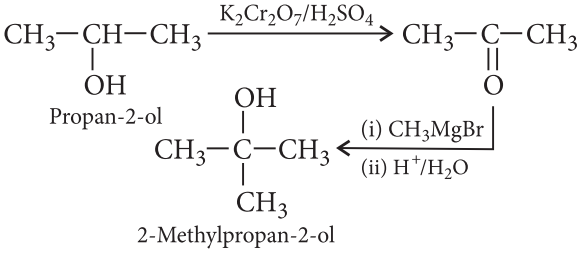




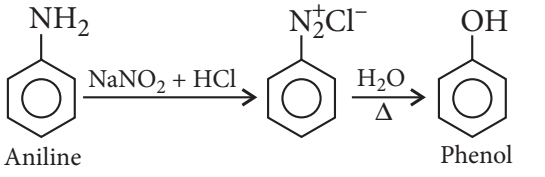
75. (i) Phenol to anisole



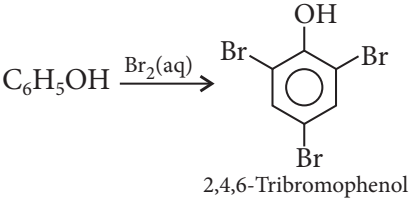
(ii) Propan-2-ol to 2-methylpropan-2-ol



(iii) Aniline to phenol



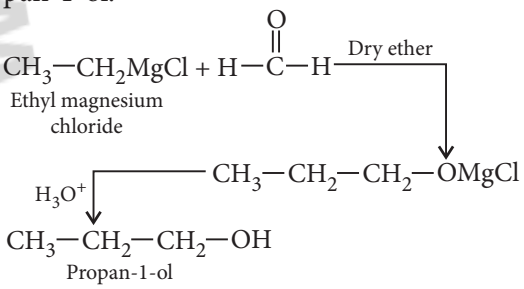
76. (i) $CH_3—CH=CH_2 \xrightarrow[(ii)\ 3H_2O_2/OH^-]{(i)\ B_2H_6} CH_3CH_2CH_2OH$

(ii) 

(iii) $CH_3CH_2OH \xrightarrow{Cu/573\ K} CH_3CHO$

77. (i) *Refer to answer 53(ii).*

(ii) Ethyl magnesium chloride on addition to formaldehyde followed by hydrolysis gives propan-1-ol.



(iii) *Refer to answer 53(i).*

78. (a) *Refer to answer 55.*

(b) *Refer to answer 28.*

79. (a) (i) *Refer to answer 30.*

(ii) *Refer to answer 48(i).*

(b) *Refer to answer 38.*

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80. (i) $CH_3CH_2CH_2CH_2OH \xrightarrow{\text{Oxidation}} CH_3-CH_2CH_2COOH$
Butanoic acid

(ii) $CH_3-CH=CH-CH_2OH \longrightarrow$ $CH_3CH=CH-COOH$
2-Butenol — But-2-enoic acid

(iii) $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2OH \longrightarrow CH_3-\underset{CH_3}{\underset{|}{CH}}-COOH$
2-methyl-1-propanol — 2-methylpropanoic acid

81. (a) *Refer to answer 43.*

(b) *Refer to answer 22.*

82. *Refer to answer 42.*

83. (a) (i) *Refer to answer 75(i).*

(ii) $CH_3CH_2OH \xrightarrow{PCC} CH_3CHO \xrightarrow[H_2O]{CH_3MgBr} CH_3\underset{OH}{\underset{|}{CH}}-CH_3$
Propan-2-ol

(b) *Refer to answer 46.*

(c) Phenols undergo electrophilic substitution reaction more easily than benzene due to strong activating effect of –OH group attached to benzene ring.

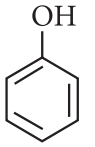
84. (a) (i) *Refer to answer 27.*

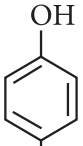
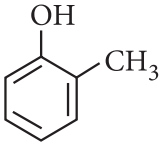
(ii) This is because *tert*-alkyl halides undergo elimination instead of substitution.

$$CH_3-\underset{CH_3}{\overset{CH_2-H}{\overset{|}{\underset{|}{C}}}}-Cl + Na^+\bar{O}C_2H_5 \xrightarrow{\Delta} \underset{\text{Elimination}}{}$$

$$CH_3-\overset{CH_3}{\overset{|}{C}}=CH_2 + C_2H_5OH + NaBr$$
2-methylpropene

(b) (i) *Refer to answer 28.*

(ii)  $+ CH_3-Cl \xrightarrow{\text{Anhy. } AlCl_3}$

 + 
p-Cresol (Major) — *o*-Cresol (Minor)

(c) *Refer to answer 48(i).*

85. (a) (i) Br_2/H_2O

(ii) BH_3 in THF/H_2O_2

(b) (i) *p*-nitrophenol > phenol > ethanol

(ii) Propanol > propanal > propane

(c) $CH_3-CH_2-\overset{+}{O}H_2 \xrightarrow[S_N2 \text{ mechanism}]{CH_3-CH_2-\ddot{O}H}$

$$CH_3-CH_2-\overset{+}{\underset{H}{O}}-CH_2-CH_3 + H_2O$$

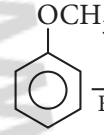
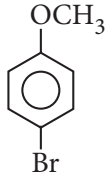
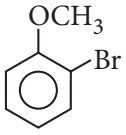
86. **(a) :** $CH_3-\overset{CH_3}{\overset{|}{\underset{|}{C}}}-OCH_3 + HI \longrightarrow$

$$CH_3-\overset{CH_3}{\overset{|}{\underset{|}{C}}}-I + CH_3OH$$

87. The given conversion is not possible by treating sodium ethoxide with *t*-butyl chloride or bromide since, under these conditions an alkene *i.e.*, isobutylene (2-methylpropene) is the main product. If however, t-butyl bromide is heated with a weak nucleophile such as pure ethanol both substitution and elimination occur in which substitution product predominates. Thus,

$$CH_3-\overset{CH_3}{\overset{|}{\underset{|}{C}}}-Br \xrightarrow[\text{Heat}]{CH_3CH_2OH}$$

$(CH_3)_3-OCH_2CH_3 + CH_3-\overset{CH_3}{\overset{|}{C}}=CH_2$
tert-Butyl ethylether (80%) — 2-Methylpropene (20%)

88.  $\xrightarrow[\text{Ethanoic acid}]{Br_2 \text{ in}}$  + 
Anisole — *p*-Bromoanisole — *o*-Bromoanisole

89. $H_3C-Br + Na-O-\overset{CH_3}{\overset{|}{\underset{|}{C}}}-CH_3$ (S_N2, Δ)

$$H_3C-O-\overset{CH_3}{\overset{|}{\underset{|}{C}}}-CH_3 + NaBr$$
2-methyl-2-methoxy propane

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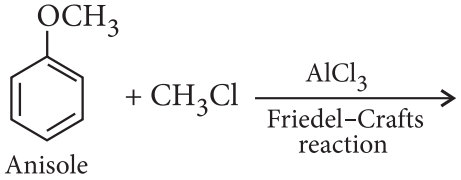
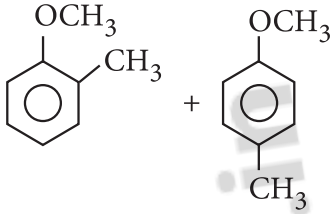
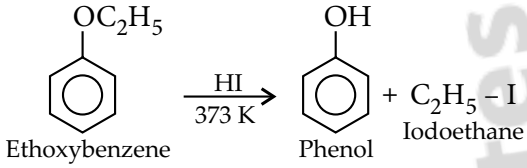
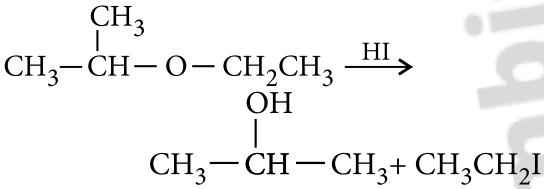
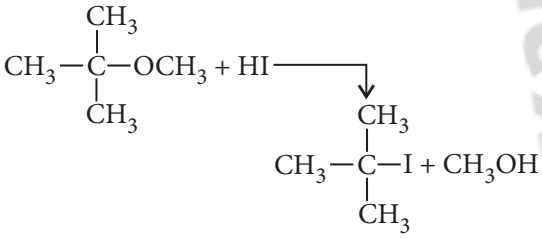
90. 



91. 

92. 

93. 

94. *Refer to answer 88.*

95. Williamson ether synthesis : Alkyl halide when treated with sodium alkoxide gives dialkyl ether.

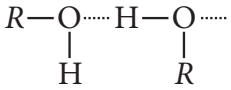
$C_2H_5ONa + C_2H_5Cl \longrightarrow C_2H_5—O—C_2H_5 + NaCl$

96. Acid dehydration of 2° and 3° alcohols give alkenes rather than ethers.

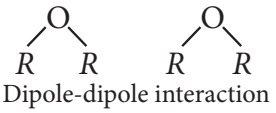
Due to steric hindrance the nucleophilic attack by the alcohol molecule on the protonated alcohol molecule does not occur.

The protonated 2° and 3° alcohols lose water molecules to form stable 2° and 3° carbocations.

97. The boiling points of ethers are much lower than, those of alcohol of comparable molar masses because like alcohols they cannot form intermolecular hydrogen bonds.

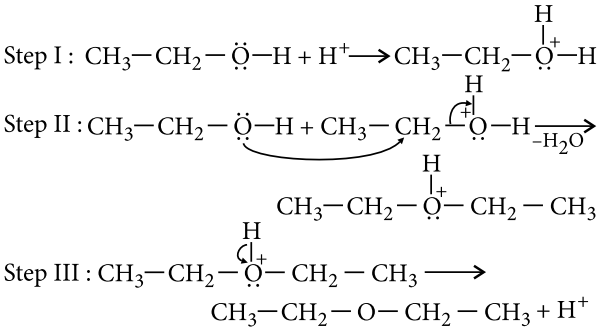


Intermolecular hydrogen bond in alcohol



Dipole-dipole interaction

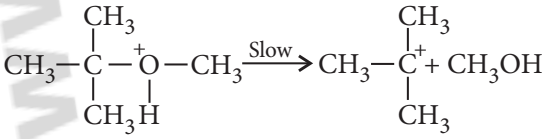
98. Mechanism : The formation of ether is nucleophilic bimolecular reaction.

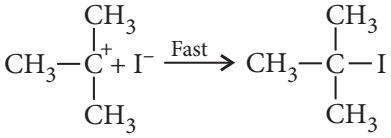


99. (i) Ethanol has higher boiling point because of strong intermolecular hydrogen bonding whereas in methoxymethane, molecules are held by dipole-dipole interaction.

(ii) When one alkyl group is a tertiary group the halide formed is tertiary halide.

In step II, the departure of leaving group (CH_3—OH) creates a more stable carbocation (3°) and the reaction follows S_N1 mechanism.





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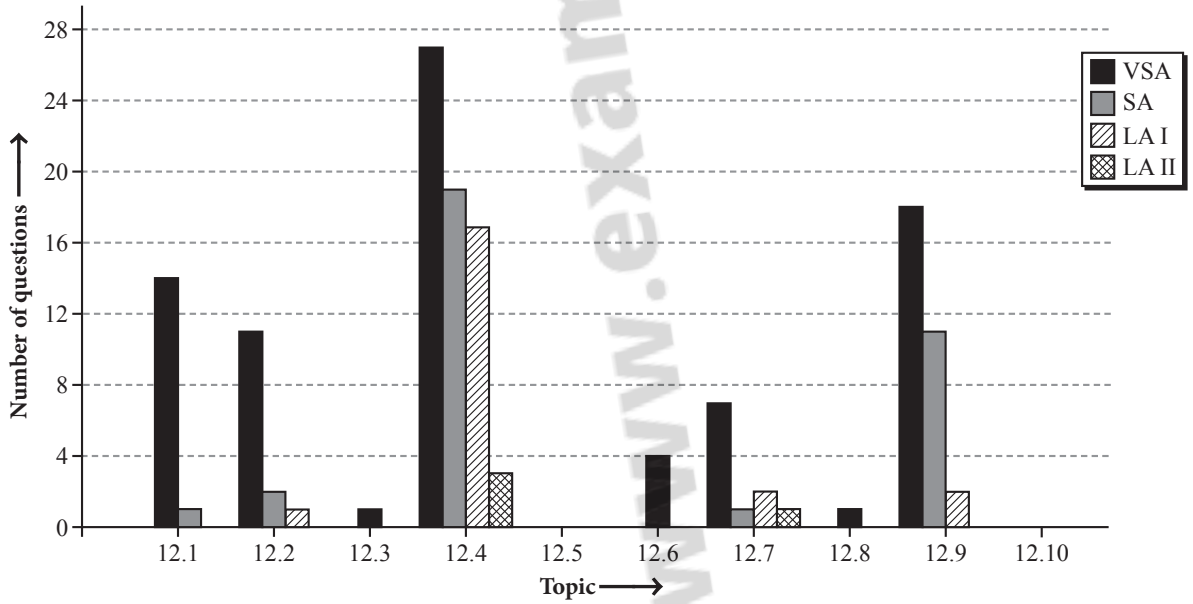
CHAPTER
12

Aldehydes, Ketones and Carboxylic Acids

12.1 Nomenclature and Structure of Carbonyl Group
12.2 Preparation of Aldehydes and Ketones
12.3 Physical Properties
12.4 Chemical Reactions
12.5 Uses of Aldehydes and Ketones
12.6 Nomenclature and Structure of Carboxyl Group
12.7 Methods of Preparation of Carboxylic Acids
12.8 Physical Properties
12.9 Chemical Reactions
12.10 Uses of Carboxylic Acids

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Chemical Reactions of Aldehydes and Ketones.*
- Maximum VSA type questions were asked from *Chemical Reactions of Aldehydes and Ketones.*
- Maximum SA and LA I type questions were asked from *Chemical Reactions of Aldehydes and Ketones.*

QUICK RECAP

ALDEHYDES AND KETONES

- **General formula :** $C_nH_{2n}O$ having $>C=O$ group.
- **Aldehydes :** $\begin{matrix} R \\ H \end{matrix}\!\!>\!C=O$; where R = H, alkyl or aryl.

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▶ **Ketones :** ; where *R* = alkyl or aryl.

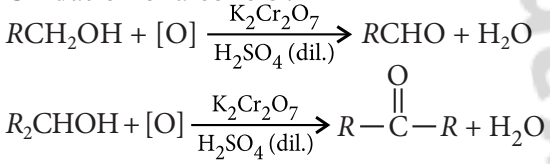
Nomenclature : The common names of most aldehydes are derived from the common names of the corresponding carboxylic acids by replacing the ending –'*ic*' of acid with aldehyde.

– The IUPAC names of open chain aliphatic aldehydes and ketones are derived from the names of the corresponding alkanes by replacing the ending –'*e*' with –'*al*' and –'*one*' respectively.

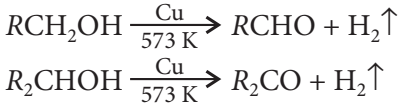
Structure : The C-atom of carbonyl group is sp^2 hybridised and forms three σ-bonds and one π-bond with O atom. Carbonyl carbon with three atoms attached to it lie in a same plane with bond angle 120° (trigonal coplanar structure) and π-electron cloud lies above and below of this plane.

Preparation :

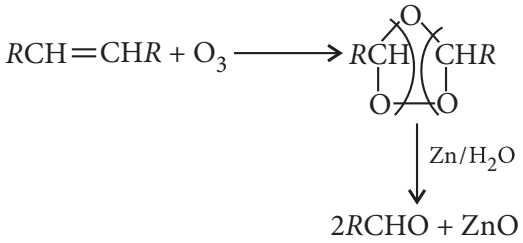
▶ **Oxidation of alcohols :**



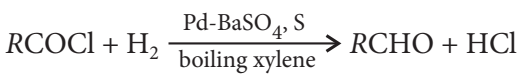
▶ **Catalytic dehydrogenation of alcohols :**



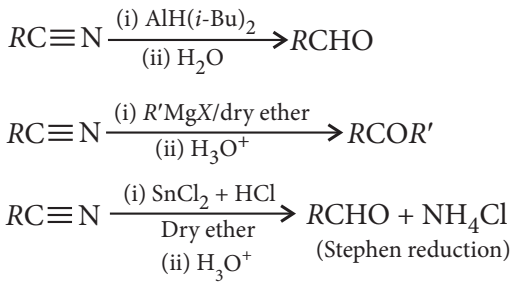
▶ **Reductive ozonolysis of alkenes :**



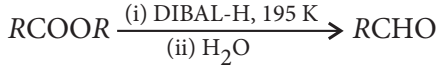
▶ **Rosenmund reduction :**



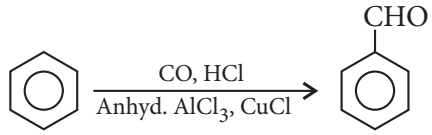
▶ **Reduction of nitriles :**



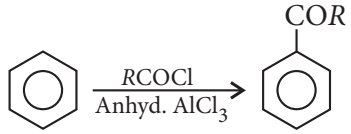
▶ **From esters :**



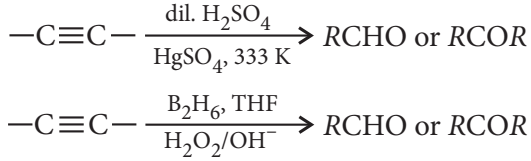
▶ **Gatterman-Koch reaction :**



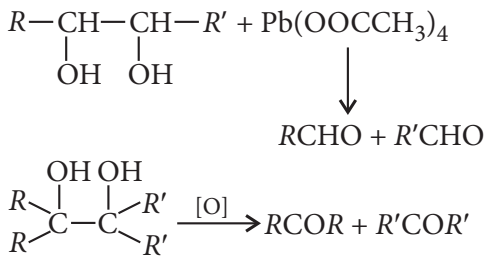
▶ **Friedel-Crafts acylation :**



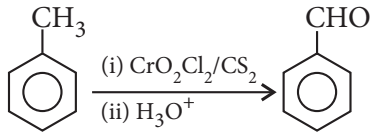
▶ **From alkynes :**



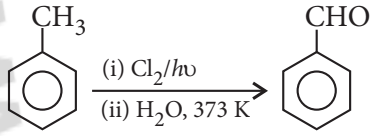
▶ **Oxidation of 1,2-glycols :**



▶ **Etard reaction :**



▶ **Side chain chlorination :**



Physical properties :

▶ **Physical state :** Lower members of aldehydes and ketones (upto C_{10}) are colourless, volatile liquids except formaldehyde which is gas at ordinary temperature.

– Higher members of aldehydes and ketones are solids with fruity odour.

– Lower aldehydes have unpleasant odour but ketones possess pleasant smell.

▶ **Boiling points :** The boiling points of aldehydes and ketones are higher than hydrocarbons and ethers of comparable molecular masses due to weak dipole-dipole interactions.

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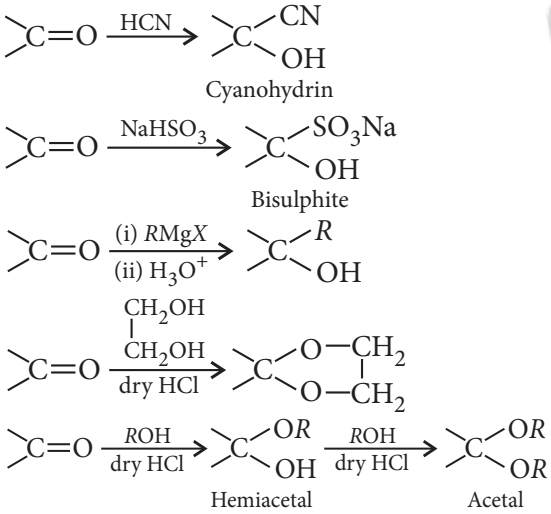
– Their boiling points are lower than those of alcohols of similar molecular masses due to absence of intermolecular hydrogen bonding.
– Among isomeric aldehydes and ketones, ketones have slightly higher boiling points due to the presence of two electron releasing alkyl groups which make carbonyl group more polar.

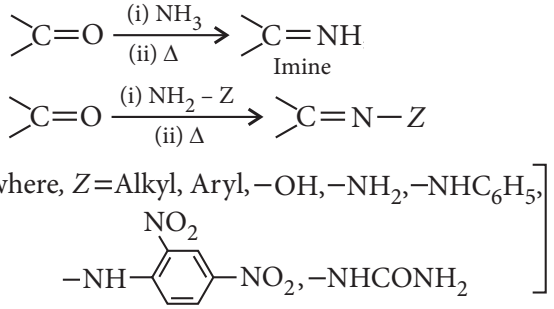
▶ **Solubility :** Lower members of aldehydes and ketones (upto C_4) are soluble in water due to H-bonding between polar carbonyl group and water. However, solubility decreases with increase in molecular weight.
– Aromatic aldehydes and ketones are much less soluble than corresponding aliphatic aldehydes and ketones due to larger benzene ring.
– All carbonyl compounds are fairly soluble in organic solvents.

Chemical properties :

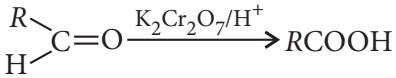
▶ **Nucleophilic addition reactions :**

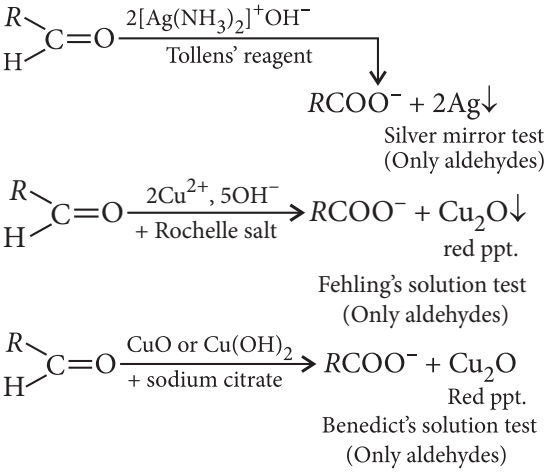


▶ **Nucleophilic addition-elimination reactions :**

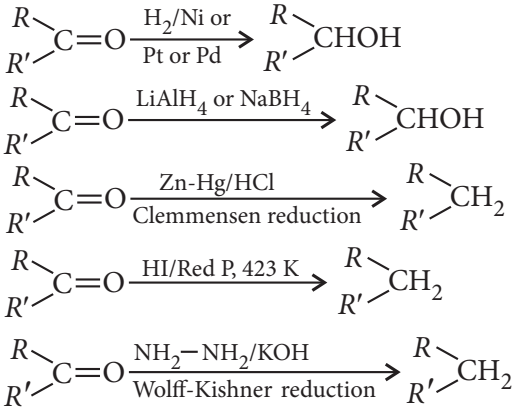


▶ **Oxidation :**





▶ **Reduction :**



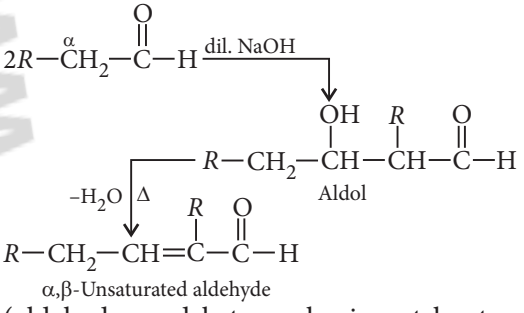
▶ **Haloform reaction :**

$2NaOH + I_2 \longrightarrow NaI + NaOI + H_2O$

$RCOCH_3 + 3NaOI \longrightarrow RCOONa + \underset{\text{Iodoform (yellow ppt.)}}{CHI_3\downarrow} + 2NaOH$

(Given by compounds having CH_3CO— group or $CH_3CH(OH)$— group).

▶ **Aldol condensation :**



(aldehydes and ketones having at least one α-hydrogen)

– **Intramolecular aldol condensation :** It takes place in diketones and gives rise to cyclic products.
– **Crossed aldol condensation :** Aldol condensation is carried out between two different aldehydes and/or ketones and if

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both of them contain α-hydrogen atoms, it gives a mixture of four products.

▶ **Cannizzaro reaction :**

$$\underset{\text{Formaldehyde}}{HCHO} + HCHO \xrightarrow{\text{conc. KOH}} \underset{\text{Methanol}}{CH_3OH} + \underset{\text{Potassium formate}}{HCOO^-K^+}$$

(aldehydes which do not have an α-hydrogen atom)

– **Crossed Cannizzaro reaction :**

$$C_6H_5-\overset{\overset{O}{\|}}{C}-H + H-\overset{\overset{O}{\|}}{C}-H \xrightarrow[\Delta]{OH^-} \underset{\text{Benzyl alcohol}}{C_6H_5CH_2OH} + \underset{\text{Sod. formate}}{HCOO^-Na^+}$$

– **Intramolecular Cannizzaro reaction :** It is given by dialdehydes having no α-hydrogen atoms.

▶ **Electrophilic substitution reactions :** Aromatic aldehydes and ketones undergo electrophilic substitution at the ring in which the carbonyl group acts as a deactivating and *meta* directing group.

▶ **Distinction between aldehydes and ketones :**

Tests with	Aldehydes	Ketones
Schiff's reagent	Pink colour	No colour
Fehling's solution	Red precipitate	No precipitate
Tollens' reagent	Silver mirror	No silver mirror
2,4-dinitrophenyl-hydrazine	Orange-yellow or red well defined crystals with melting points characteristic of individual aldehydes.	Orange-yellow or red well defined crystals with melting points characteristic of individual ketones.

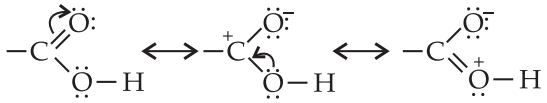
CARBOXYLIC ACIDS

General Formula : $C_nH_{2n}O_2$ having —COOH group.
*R*COOH where, *R*=H or alkyl or aryl.

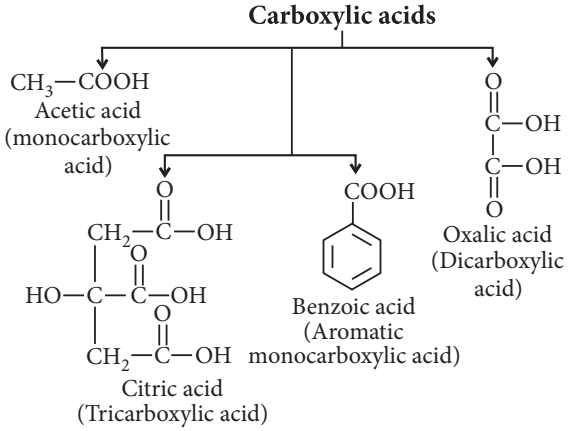
Nomenclature : The common names end with the suffix –'*ic acid*' and have been derived from Latin or Greek names of their natural sources.

– In the IUPAC system, aliphatic carboxylic acids are named by replacing the ending –'*e*' in the name of the corresponding alkane with –'*oic* acid'. In numbering the carbon chain, the carboxylic carbon is numbered one.

Structure : In carboxylic acids, the bonds to the carboxyl carbon lie in one plane and are separated by about 120°. The carboxylic carbon is less electrophilic than carbonyl carbon because of the possible resonance structure.



Classification : They are classified as mono, di, tri and polycarboxylic acids depending upon the number of carboxyl groups present in a molecule.



Aliphatic monocarboxylic acids and aliphatic esters are known as *functional isomers*. Some higher aliphatic monocarboxylic acids (C_{12}—C_{18}) are known as *fatty acids* because they occur in natural fats as esters of glycerol, *e.g.,* palmitic acid and stearic acid are obtained on hydrolysis of fats.

Preparation :

$$\underset{\substack{\text{Primary}\\\text{alcohol}}}{RCH_2OH} \xrightarrow[K_2Cr_2O_7 + \text{dil. } H_2SO_4]{\text{Oxidation, alk. } KMnO_4,\ H_3O^+ \text{ or } CrO_3\text{–}H_2SO_4} RCOOH$$

$$\underset{\text{Aldehyde}}{RCHO} \xrightarrow[\text{or Fehling's solution or Tollens' reagent}]{\text{Oxidation}\ K_2Cr_2O_7 + \text{dil. } H_2SO_4} RCOOH$$

$$\underset{\substack{\text{Alkyl}\\\text{Cyanide}}}{RCN} \xrightarrow[\substack{\text{Mineral acid}\\HCl,\ H_2SO_4,\ \text{etc.}}]{\text{Hydrolysis}} RCOOH$$

$$\underset{\substack{\text{Alkyl}\\\text{Cyanide}}}{RCN} \xrightarrow[\substack{\text{Alkali}\\\text{NaOH or KOH}}]{\text{Hydrolysis}} RCOONa \xrightarrow{HCl} RCOOH$$

$$\underset{\substack{\text{Grignard}\\\text{reagent}}}{RMgX} \xrightarrow[\text{dry ether}]{CO_2\ (\text{dry ice})} RCOOH$$

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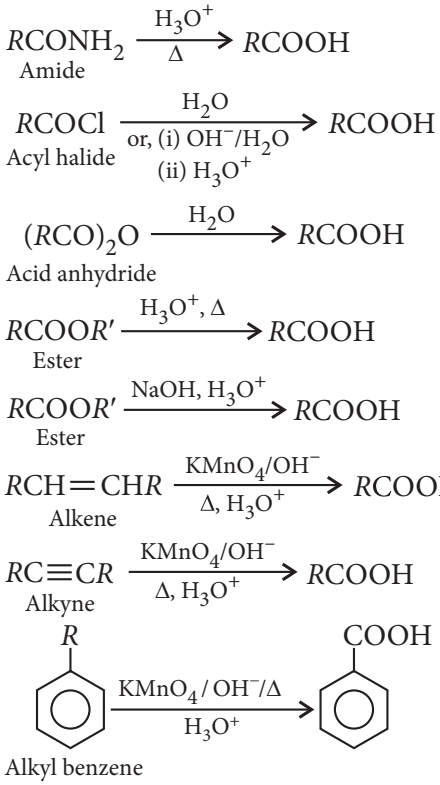
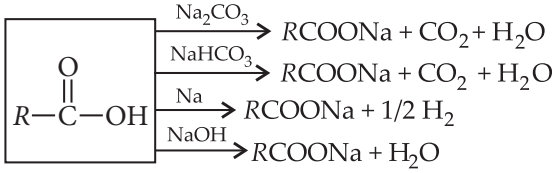


Physical Properties :

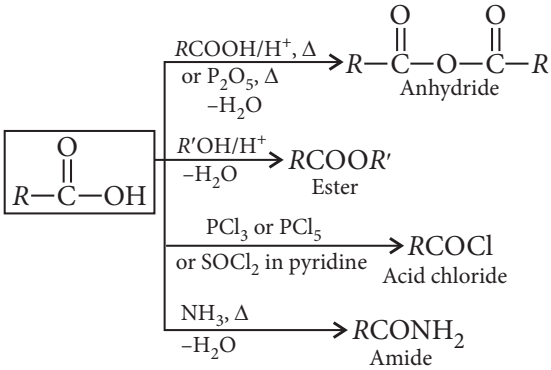
- **Physical state :** The lower fatty acids upto C_9 are colourless liquids. The higher ones are colourless waxy solids.
- **Odour :** The first three members have a sharp pungent odour. The middle ones, C_4 to C_9, have an odour of rancid butter. The higher members do not possess any smell.
- **Solubility :** Simple aliphatic carboxylic acids having upto four carbon atoms are miscible in water due to the formation of hydrogen bonds with water.
  - The solubility decreases with increasing number of carbon atoms. Higher carboxylic acids are practically insoluble in water due to the increased hydrophobic interaction of hydrocarbon part.
  - Benzoic acid, the simplest aromatic carboxylic acid is nearly insoluble in cold water.
  - Carboxylic acids are also soluble in less polar organic solvents like benzene, ether, alcohol, chloroform, etc.
- **Boiling points :** Carboxylic acids are higher boiling liquids than aldehydes, ketones and even alcohols of comparable molecular masses due to more extensive association of their molecules through intermolecular hydrogen bonding. The H-bonds are not broken completely even in the vapour phase.

Chemical reactions :

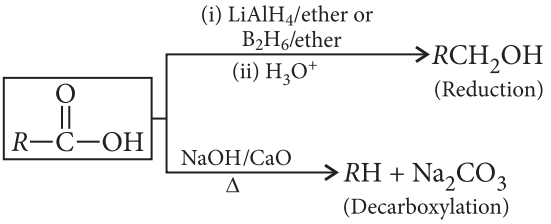
- **Reactions involving cleavage of O—H bond :**



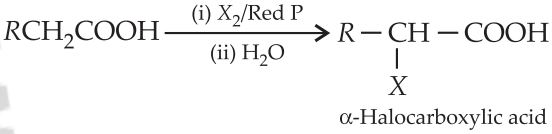
- **Reactions involving cleavage of C—OH bond :**



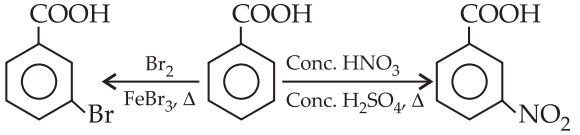
- **Reactions involving —COOH group :**



- **Hell-Volhard-Zelinsky reaction :**



- **Ring substitution in aromatic acids :** Aromatic carboxylic acids undergo electrophilic substitution reactions in which the carboxyl group acts as a *deactivating* and *meta* directing group.



- **Distinction test between phenol and carboxylic acid :**

Test	Phenol	Carboxylic acid
$NaHCO_3$ test	No reaction	Brisk effervescence of CO_2 gas.
$FeCl_3$ test	Violet colour	Buff coloured ppt.

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Previous Years' CBSE Board Questions

12.1 Nomenclature and Structure of Carbonyl Group

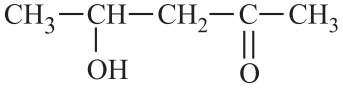
VSA (1 mark)

1. Write the structure of 2-methylbutanal. *(1/5, AI 2015)*

2. Draw the structure of 3-methylpentanal. *(Delhi 2015C)*

3. Write the IUPAC name of the following :
$CH_3—CH_2—CHO$ *(AI 2015C)*

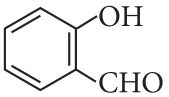
4. Write the IUPAC name of the compound :



(Delhi 2014)

5. Write the structure of 4-chloropentan-2-one. *(AI 2014, Delhi 2013)*

6. Write the IUPAC name of the following compound :



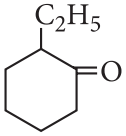
(Foregin 2014)

7. Write the structure of 3-methylbutanal. *(Delhi 2013)*

8. Write the structure of *p*-methylbenzaldehyde molecule. *(Delhi 2013)*

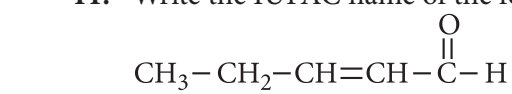
9. Draw the structure of the compound named 4-methylpent-3-en-2-one. *(Delhi 2013C)*

10. Write the IUPAC name of the following :



(AI 2013C)

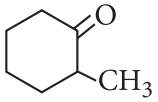
11. Write the IUPAC name of the following :



(AI 2012)

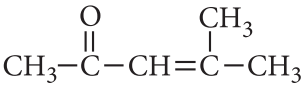
12. Write the IUPAC name of
$Ph—CH=CH—CHO$ *(AI 2012)*

13. Write the IUPAC name of the following :



(AI 2012C)

14. Write the IUPAC name of



(AI 2011C)

SA (2 marks)

15. Draw the structures of the following :
(i) *p*-Methylbenzaldehyde
(ii) 4-Methylpent-3-en-2-one
(2/5, AI 2015C)

12.2 Preparation of Aldehydes and Ketones

VSA (1 mark)

16. What happens when benzene is treated with CH_3COCl in presence of anhydrous $AlCl_3$? *(1/2, 2020)*

17. How can you convert the following : Benzene to acetophenone? *(1/3, 2020)*

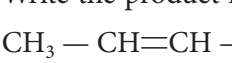
18. Complete the following reaction :
$(C_6H_5CH_2)_2Cd + 2CH_3COCl \longrightarrow$
(1/3, Delhi 2019)

19. Write chemical equations for the following reaction :
Benzoyl chloride is hydrogenated in presence of $Pd/BaSO_4$. *(1/3, Delhi 2019, 1/5, 2018C)*

20. Write the equation involved in the following reaction :
Etard reaction. *(1/2, Delhi 2017)*

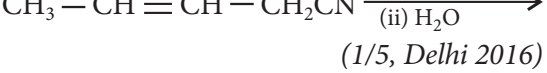
21. Write the reaction involved in the following :
Stephen reduction. *(1/5, AI 2017)*

22. Write the product in the following reaction :



(1/5, AI 2017)

23. Write the product in the following reaction :



(1/5, Delhi 2016)

24. How do you convert the following :
Ethyne to ethanal *(1/3, Foregin 2015)*

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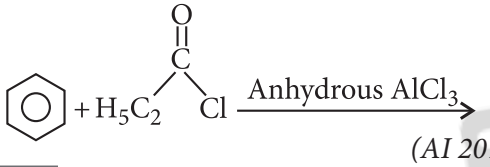
25. How will you obtain the following :
Benzaldehyde from phenol *(1/5, AI 2013C)*

26. Write the structure of the product formed in the following reaction :



(AI 2012C)

SA (2 marks)

27. Write the major product(s) in the following :

(i) $CH_3—CH=CH—CH_2—CN \xrightarrow[(ii)\ H_3O^+]{(i)\ DIBAL–H}$

(ii) $CH_3—CH_2—OH \xrightarrow{CrO_3}$

(2/5, 2020)

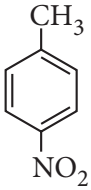
28. Write the equations involved in the following reactions :
(i) Stephen reaction
(ii) Etard reaction *(2/3, Foregin 2015)*

LA I (3 marks)

29. Write the structures of the main products of the following reactions :

(i) $+ C_6H_5COCl \xrightarrow[CS_2]{\text{anhydrous } AlCl_3}$

(ii) $H_3C—C\equiv C—H \xrightarrow{Hg^{2+},\ H_2SO_4}$

(iii)  $\xrightarrow[2.\ H_2O]{1.\ CrO_2Cl_2}$ *(Delhi 2012)*

12.3 Physical Properties

VSA (1 mark)

30. Aldehydes and ketones have lower boiling points than corresponding alcohols. Why?
(AI 2012C)

12.4 Chemical Reactions

VSA (1 mark)

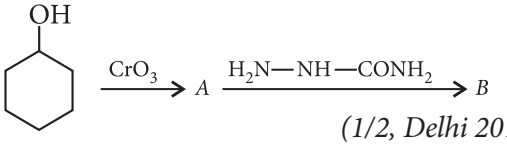
31. Iodoform test is not given by
(a) ethanol (b) ethanal
(c) pentan-2-one (d) pentan-3-one
(2020)

32. **Assertion (A) :** Reactivity of ketones is more than aldehydes.
Reason (R) : The carbonyl carbon of ketones is less electrophilic as compared to aldehydes.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

33. What happens when,
propanone is treated with methyl magnesium iodide and then hydrolysed? *(1/2, 2020)*

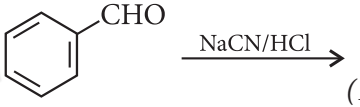
34. How can you convert the following : Acetone to propene ? *(1/3, 2020)*

35. Write structures of compounds *A* and *B* in the following reaction :



(1/2, Delhi 2019)

36. Complete the following reaction :



(1/3, Delhi 2019)

37. How do you convert the following :
Ethanal to propanone *(1/2, 2018)*

38. Give simple chemical tests to distinguish between the following pair of compounds :
Ethanal and propanal
(1/5, 2018C, Delhi 2013, Delhi 2012C)

39. Write the equation involved in the following reaction :
Wolff–Kishner reduction *(1/2, Delhi 2017)*

40. Do the following conversion in not more than two steps :
Propanone to propene *(1/3, Delhi 2017)*

41. Give chemical tests to distinguish between the following pairs of compunds :
Benzaldehyde and acetophenone.
(1/5, AI 2017C, 1/5, Delhi 2015C, AI 2012C)

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42. Write the structure of *A* and *B* in the following reaction :

$$CH_3COCl \xrightarrow{H_2,\ Pd-BaSO_4} A \xrightarrow{H_2N-OH} B$$

(1/5, AI 2016)

43. Distinguish between the following :
$C_6H_5—COCH_3$ and $C_6H_5—CHO$
(1/2, AI 2016, 1/5, AI 2015)

44. Give a simple chemical test to distinguish between the following pair of compounds :
CH_3CH_2CHO and $CH_3CH_2COCH_3$
(1/2, AI 2016, 1/5, AI 2015)

45. Name the reagent used in the following reaction :

$$CH_3—CO—CH_3 \xrightarrow{?} CH_3—\underset{\underset{OH}{|}}{CH}—CH_3$$

(Delhi 2015, 1/2, Foreign 2015)

46. Give simple chemical tests to distinguish between the following pairs of compounds :
(i) Benzaldehyde and benzoic acid
(ii) Propanal and propanone.
(1/5, Delhi 2014)

47. Account for the following :
CH_3CHO is more reactive than CH_3COCH_3 towards reaction with HCN. *(Delhi 2014)*

48. Give simple chemical tests to distinguish between the following pair of compounds :
Propanal and propanone *(1/5, AI 2014)*

49. Give simple chemical tests to distinguish between the following pair of compounds :
Propanal and butan-2-one
(1/5, Foreign 2014)

50. How will you convert the following :
Ethanal to 2-hydroxy propanoic acid
(1/5, AI 2013)

51. Give reason :
pH of reaction should be carefully controlled while preparing ammonia derivatives of carbonyl compounds. *(1/5, Delhi 2013C)*

52. Give reason :
Aldehydes are more reactive than ketones towards nucleophilic reagents.
(1/5, AI 2013C)

53. Arrange the following compounds in an increasing order of their reactivity in nucleophilic addition reactions : ethanal, propanal, propanone, butanone. *(Delhi 2012)*

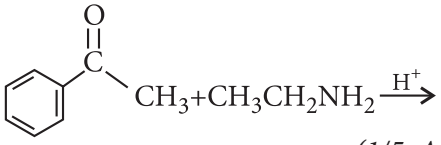
54. Give chemical tests to distinguish between Benzophenone and acetophenone
(1/5, Delhi 2012)

55. Arrange the following compounds in an increasing order of their property as indicated :
Acetaldehyde, acetone, methyl *tert*-butyl ketone (reactivity towards HCN)
(1/5, AI 2012)

56. Predict the organic product of the following reaction :

(1/2, AI 2011C)

57. Predict the product of the following reaction :



(1/5, AI 2011C)

SA (2 marks)

58. Write chemical equations for the following reactions :
(i) Propanone is treated with dilute $Ba(OH)_2$.
(ii) Acetophenone is treated with Zn(Hg)/Conc. HCl *(2/3, Delhi 2019)*

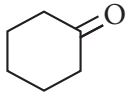
59. Write the structure of major product(s) in the following :

(i) $CH_3 — CH_2 — \underset{\underset{O}{||}}{C} — H \xrightarrow[\text{(b) KOH, glycol/heat}]{\text{(a) } H_2N-NH_2}$

(ii) $CH_3 — \underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}} — CHO \xrightarrow{\text{conc. NaOH}}$

(2/5, AI 2019)

60. (a) Write the product in the following reaction :



(b) Give simple chemical test to distinguish between the following pair of compounds :
Butanal and Butan-2-one
(2/5, AI 2017)

61. Write the equations involved in the following reactions :

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(i) Clemmensen reduction
(ii) Cannizzaro reaction *(Delhi 2017)*

62. Predict the products of the following reactions :

(i) $CH_3-\underset{\underset{CH_3}{|}}{C}=O \xrightarrow[\text{(ii) KOH/Glycol, }\Delta]{\text{(i) } H_2N-NH_2} ?$

(ii) $C_6H_5-CO-CH_3 \xrightarrow{NaOH/I_2} ? + ?$

(2/3, Delhi 2015)

63. A compound '*A*' of molecular formula C_2H_3OCl undergoes a series of reactions as shown below. Write the structure of *A*, *B*, *C* and *D* in the following reactions :

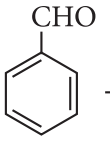
$(C_2H_3OCl)A \xrightarrow{H_2/Pd\text{-}BaSO_4} B \xrightarrow{\text{dil. NaOH}} C \xrightarrow{Heat} D$

(AI 2015)

64. Describe the following reactions :
(i) Acetylation
(ii) Aldol condensation *(2/5, Delhi 2015C)*

65. Write the main product in the following equations :

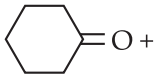
(i) $CH_3-\underset{\underset{O}{||}}{C}-CH_3 \xrightarrow{LiAlH_4} ?$

(ii)  $\xrightarrow[273-283K]{HNO_3/H_2SO_4} ?$

(2/5, Delhi 2015C)

66. Draw the structure of the following derivatives :
(i) Propanone oxime
(ii) Semicarbazone of the CH_3CHO

(2/5, AI 2015C)

67. (i)  $+ H_2N-OH \xrightarrow{H^+}$

(ii) $2C_6H_5CHO + \text{conc. NaOH} \longrightarrow$

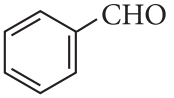
(2/5, Delhi 2014)

68. Account for the following :
(i) CH_3CHO is more reactive than CH_3COCH_3 towards reaction with HCN.
(ii) There are two $-NH_2$ groups in semicarbazide ($H_2NNHCONH_2$).
However, only one is involved in the formation of semicarbazone. *(2/5, Foregin 2014)*

69. Write the chemical equation to illustrate each of the following name reactions :
(i) Rosenmund reduction
(ii) Cannizzaro reaction *(2/5, Foregin 2014)*

70. Complete the following reactions :

(i) $2H-\underset{\underset{O}{||}}{C}-H \xrightarrow{\text{Conc. KOH}}$

(ii)  $\xrightarrow[273-283\ K]{HNO_3/H_2SO_4}$

(2/5, Delhi 2013)

71. How will you convert the following :
(i) Propanone to Propan-2-ol
(ii) Ethanal to 2-hydroxypropanoic acid

(2/5, AI 2013)

72. Give simple chemical test to distinguish between :
(a) Pentan-2-one and pentan-3-one
(b) Ethanal and propanal *(2/5, AI 2013)*

73. Write the products of the following reactions :

(i) $CH_3-\underset{\underset{O}{||}}{C}-CH_3 \xrightarrow[\text{conc. HCl}]{Zn-Hg}$

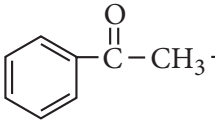
(ii) $CH_3-\underset{\underset{O}{||}}{C}-Cl + H_2 \xrightarrow{Pd-BaSO_4}$

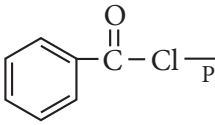
(2/5, AI 2013)

74. Give chemical tests to distinguish between
(i) Propanal and propanone
(ii) Benzaldehyde and acetophenone

(2/5, AI 2012)

75. Predict the products of the following reactions :

(i)  $\xrightarrow{H_2CrO_4}$

(ii)  $\xrightarrow[Pd-BaSO_4]{H_2}$

(2/5, AI 2011C)

76. An organic compound *A*, having the formula, C_3H_8O, on treatment with copper at 573 K, gives *B*. *B* does not reduce Fehling's solution but gives a yellow precipitate of the compound *C* with I_2/NaOH. Deduce the structure of *A*, *B* and *C*. *(2/5, Delhi 2011C)*

LA I (3 marks)

77. Write the products formed when $(CH_3)_3C-CHO$ reacts with the following reagents :

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(i) CH_3COCH_3 in the presence of dilute NaOH
(ii) HCN
(iii) Conc. NaOH *(2020)*

78. (a) How can you distinguish between propanal and propanone?
(b) Draw structures of the following derivatives :
(i) Cyanohydrin of cyclobutanone
(ii) Hemiacetal of ethanal *(3/5, 2020)*

79. An alkene *A* with molecular formula C_5H_{10} on ozonolysis gives a mixture of two compounds, *B* and *C*. Compound *B* gives positive Fehling's test and also reacts with iodine and NaOH solution. Compound *C* does not give Fehling solution test but forms iodoform. Identify the compounds *A*, *B* and *C*. *(3/5, AI 2019)*

80. (*A*), (*B*) and (*C*) are three non-cyclic functional isomers of a carbonyl compound with molecular formula C_4H_8O. Isomers (*A*) and (*C*) give positive Tollens' test whereas isomer (*B*) does not give Tollens' test but gives positive iodoform test. Isomers (*A*) and (*B*) on reduction with Zn(Hg)/conc. HCl give the same product (*D*).
(a) Write the structures of (*A*), (*B*), (*C*) and (*D*).
(b) Out of (*A*), (*B*) and (*C*) isomers, which one is least reactive towards addition of HCN? *(2018)*

81. (a) Write the chemical equation for the reaction involved in Cannizzaro reaction.
(b) Draw the structure of the semicarbazone of ethanal.
(c) How can you distinguish between propanal and propanone? *(3/5, Delhi 2016)*

82. (a) Write the chemical reaction involved in Wolff-Kishner reduction.
(b) Arrange the following in the increasing order of their reactivity towards nucleophilic addition reaction.
$C_6H_5COCH_3$, CH_3-CHO, CH_3COCH_3
(c) *A* and *B* are two functional isomers of compound C_3H_6O. On heating with NaOH and I_2, isomer *B* forms yellow precipitate of iodoform whereas isomer *A* does not form any precipitate. Write the formulae of *A* and *B*. *(3/5, AI 2016)*

83. Write the structures of the main products when acetone ($CH_3-CO-CH_3$) reacts with the following reagents :
(i) Zn – Hg/conc. HCl
(ii) $H_2N-NHCONH_2/H^+$
(iii) CH_3MgBr and then H_3O^+ *(3/5, AI 2015)*

84. How will you convert ethanal into the following compounds? Give the chemical equations involved.
(i) CH_3-CH_3
(ii) $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CHO$
(iii) CH_3CH_2OH *(3/5, Delhi 2015C)*

85. Write the chemical equations to illustrate the following name reactions :
(i) Wolff–Kishner reduction
(ii) Aldol condensation
(iii) Cannizzaro reaction *(3/5, Delhi 2014)*

86. Write the products formed when CH_3CHO reacts with the following reagents :
(i) HCN (ii) H_2N-OH
(iii) CH_3CHO in the presence of dilute NaOH *(3/5, AI 2014)*

87. (a) Write the chemical equations to illustrate the following name reactions :
(i) Rosenmund reduction
(ii) Cannizzaro reaction
(b) Out of $CH_3CH_2-CO-CH_2-CH_3$ and $CH_3CH_2-CH_2-CO-CH_3$, which gives iodoform test? *(3/5, AI 2014)*

88. Write the products formed when ethanal reacts with the following reagents :
(i) CH_3MgBr and then H_3O^+
(ii) Zn-Hg/conc. HCl
(iii) C_6H_5CHO in the presence of dilute NaOH *(3/5, Foregin 2014)*

89. How will you bring about the following conversions?
(i) Propanone to propane
(ii) Benzoyl chloride to benzaldehyde
(iii) Ethanal to but-2-enal *(3/5, Delhi 2013)*

90. An organic compound (*A*) which has characteristic odour, on treatment with NaOH forms two compounds (*B*) and (*C*). Compound (*B*) has the molecular formula C_7H_8O which on oxidation with CrO_3 gives

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back compound (*A*). Compound (*C*) is the sodium salt of the acid. Compound (*C*) when heated with soda lime yields an aromatic hydrocarbon (*D*). Deduce the structures of (*A*), (*B*), (*C*) and (*D*). Write chemical equations for all reactions taking place.

(3/5, AI 2013C, 2012C)

91. An organic compound with molecular formula $C_9H_{10}O$ forms 2,4-DNP derivative, reduces Tollen's reagent and undergoes Cannizzaro reaction. On vigorous oxidation it gives 1,2-benzenedicarboxylic acid. Identify the compound.

(3/5, AI 2012, 2012C)

92. An organic compound (*A*) with molecular formula C_8H_8O forms an orange red precipitate with 2, 4-DNP reagent and gives yellow precipitate on heating with I_2 and NaOH. It neither reduces Tollens' reagent nor Fehling's reagent, nor does it decolourise bromine water or Baeyer's reagent. On drastic oxidation with chromic acid, it gives a carboxylic acid (*B*) having molecular formula $C_7H_6O_2$. Identify the compounds (*A*) and (*B*) and explain the reactions involved.

(3/5, Delhi 2012C)

93. An organic compound with molecular formula $C_5H_{10}O$ does not reduce Tollens' reagent but forms an addition compound with sodium hydrogen sulphite and gives a positive iodoform test. On vigorous oxidation, it gives ethanoic acid and propanoic acid. Identify the compound and write all chemical equations for the reactions.

(3/5, AI 2012C)

LA II (5 marks)

94. (a) An organic compound (*A*) having molecular formula C_4H_8O gives orange red precipitate with 2, 4-DNP reagent. It does not reduce Tollens' reagent but gives yellow precipitate of iodoform on heating with NaOH and I_2. Compound (*A*) on reduction with $NaBH_4$ gives compound (*B*) which undergoes dehydration reaction on heating with conc. H_2SO_4 to form compound (*C*). Compound (*C*) on ozonolysis gives two molecules of ethanal.

Identify (*A*), (*B*) and (*C*) and write their structures. Write the reactions of compound (*A*) with (i) $NaOH/I_2$ and (ii) $NaBH_4$.

(b) Give reason :

(i) Oxidation of propanal is easier than propanone.

(ii) α-Hydrogen of aldehydes and ketones is acidic in nature. *(2020)*

95. (a) Give a plausible explanation for each one of the following :

(i) There are two $-NH_2$ groups in semicarbazide. However, only one such group is involved in the formation of semicarbazones.

(ii) Cyclohexanone forms cyanohydrin in good yield but 2,4,6-trimethylcyclohexanone does not.

(b) An organic compound with molecular formula $C_9H_{10}O$ forms 2,4-DNP derivative, reduces Tollens' reagent and undergoes Cannizzaro reaction. On vigorous oxidation it gives 1,2-benzenedicarboxylic acid. Identify the compound.

(Delhi 2012)

96. (a) Identify *A*, *B* and *C* in the following sequence of reactions :

$$CH_3CHO \xrightarrow[(ii)\ H_2O]{(i)\ C_2H_5MgCl} A \xrightarrow{conc.\ H_2SO_4} B \xrightarrow{HBr + Peroxide} C$$

(b) Predict the structures of the products formed when benzaldehyde is treated with

(i) conc. NaOH

(ii) HNO_3/H_2SO_4 (at 273– 383 K)

(Delhi 2011C)

12.6 Nomenclature and Structure of Carboxyl Group

VSA (1 mark)

97. What is the correct IUPAC name of the given compound?

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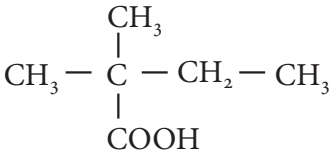
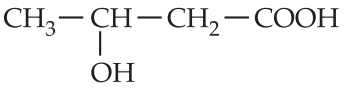


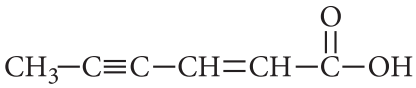
(a) 2,2-Dimethylbutanoic acid
(b) 2-Carboxyl-2-methylbutane
(c) 2-Ethyl-2-methylpropanoic acid
(d) 3-Methylbutanecarboxylic acid *(2020)*

98. Write the IUPAC name of the compound :



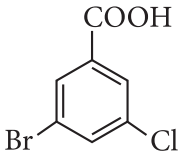
(Delhi 2014)

99. Write the IUPAC name of the following :



(Delhi 2011C)

100. Write the IUPAC name of

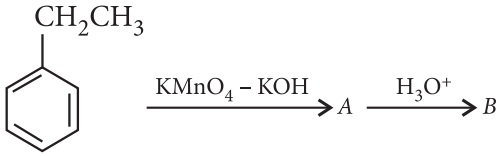


(AI 2011C)

12.7 Methods of Preparation of Carboxylic Acids

VSA (1 mark)

101. Write structures of compounds *A* and *B* in each of the following reaction.



(1/2, Delhi 2019)

102. How do you convert the following :
Toluene to benzoic acid? *(1/2, 2018)*

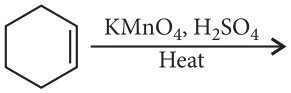
103. Do the following conversion in not more than two steps :
Ethyl benzene to benzoic acid
(1/3, Delhi 2017)

104. Name the reagent used in the following reaction :

(1/2, Delhi 2015)

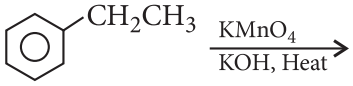
105. How will you carry out the following conversions?
(i) Acetylene to acetic acid
(1/5, Delhi 2013C)

106. Predict the product of the following reaction :



(1/5, AI 2011C)

107. Predict the organic product of the following reaction :



(1/3, AI 2011C)

SA (2 marks)

108. Write a suitable chemical equation to complete each of the following transformations :
(i) Butan-1-ol to butanoic acid
(ii) 4-Methylacetophenone to benzene-1,4-dicarboxylic acid *(2/5, AI 2012)*

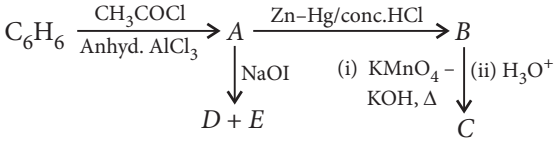
LA I (3 marks)

109. Two moles of organic compound '*A*' on treatment with a strong base gives two compound '*B*' and '*C*'. Compound '*B*' on dehydrogenation with Cu gives '*A*' while acidification of '*C*' yields carboxylic acid '*D*' with molecular formula of CH_2O_2. Identify the compounds *A*, *B*, *C* and *D* and write all chemical reactions involved.
(3/5, Delhi 2013C)

110. How are the following conversions carried out?
(i) Ethyl cyanide to ethanoic acid.
(ii) Butan-1-ol to butanoic acid.
(iii) Benzoic acid to *m*-bromobenzoic acid.
(Delhi 2012)

LA II (5 marks)

111. Write the structures of *A*, *B*, *C*, *D* and *E* in the following reactions :



(Delhi 2016)

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12.8 Physical Properties

VSA (1 mark)

112. Arrange the following in the increasing order of their boiling points.
CH_3CHO, CH_3COOH, CH_3CH_2OH
(1/5, AI 2016, 2015)

12.9 Chemical Reactions

VSA (1 mark)

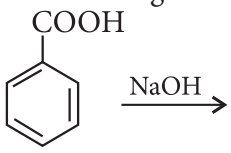
113. **Assertion (A) :** Benzoic acid does not undergo Friedal-Crafts reaction.
Reason (R) : The carboxyl group is activating and undergo electrophilic substitution reaction.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

114. Complete the following reaction :

$$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-COOH \xrightarrow[\text{(ii) } H_2O]{\text{(i) } Br_2/\text{Red } P_4}$$

(1/3, Delhi 2019)

115. Write the structure of major product(s) in the following:



(1/5, AI 2019)

116. Carry out the following conversions:
Propanoic acid to acetic acid. *(1/5, AI 2019)*

117. Do the following conversion in not more than two steps:
Benzoic acid to benzaldehyde.
(1/3, Delhi 2017, 1/5 Delhi 2015C)

118. Why carboxylic acid does not give reactions of carbonyl group?
(1/5, AI 2017C, 1/5, AI 2016)

119. Give simple chemical tests to distinguish between the following pairs of compounds :
Benzoic acid and ethyl benzoate
(1/5, AI 2017C, 1/3, Foreign 2014)

120. Distinguish between
CH_3COOH and HCOOH *(1/2, AI 2016)*

121. Predict the products of the following reaction :

$$CH_3COONa \xrightarrow[\Delta]{NaOH/CaO} ?$$ *(1/3, Delhi 2015)*

122. Name the reagent used in the following reaction :

$$CH_3—COOH \xrightarrow{?} CH_3—COCl$$

(1/2, Foreign 2015)

123. Write the main product in the following equation :

$$CH_3—COOH \xrightarrow{PCl_5}$$ *(1/5, Delhi 2015C)*

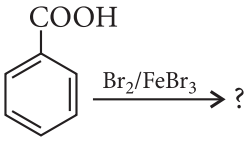
124. Describe the following giving chemical equation :
Decarboxylation reaction
(1/5, Delhi 2015C, 2012)

125. Account for the following :
Carboxylic acids do not give reactions of carbonyl group. *(1/5, AI 2014)*

126. Give simple chemical test to distinguish between the following pair of compounds :
Benzoic acid and phenol
(1/5, AI 2014, Delhi 2013C, 2012, 2012C)

127. Write the chemical equation to illustrate the following name reaction :
Hell-Volhard-Zelinsky reaction
(1/5, Foregin 2014, Delhi 2012)

128. Write the products of the following reaction :



(1/5, AI 2013)

129. Give reasons :
Chloroacetic acid is stronger than acetic acid. *(1/5, Delhi 2013C)*

130. How will you carry out the following conversion :
Toluene to *m*-nitrobenzoic acid
(1/5, Delhi 2013C)

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SA (2 marks)

131. Account for the following :

(a) Aromatic carboxylic acids do not undergo Friedel–Crafts reaction.

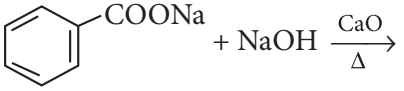
(b) pK_a value of 4-nitrobenzoic acid is lower than that of benzoic acid. *(2018)*

132. Write the reactions involved in the following :

(i) Hell—Volhard—Zelinsky reaction

(ii) Decarboxylation reaction *(Delhi 2017)*

133. (a) Write the product in the following reaction :



(b) Give simple chemical test to distinguish between the following pair of compounds :

Benzoic acid and phenol *(2/5, AI 2017)*

134. How will you convert the following in not more than two steps :

(i) Acetophenone to benzoic acid

(ii) Ethanoic acid to 2-hydroxyethanoic acid *(2/5, AI 2017)*

135. (a) Write the product of the following reaction :

$CH_3COOH \xrightarrow{Cl_2/P}$

(b) Give simple chemical test to distinguish between the following pair of compounds :

Benzaldehyde and benzoic acid

(2/5, Delhi 2014)

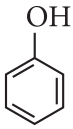
136. Account for the following :

$Cl—CH_2COOH$ is a stronger acid than CH_3COOH. *(2/5, AI 2014)*

137. Although phenoxide ion has more number of resonating structures than carboxylate ion, carboxylic acid is a stronger acid than phenol. Give two reasons. *(2/5, Delhi 2013)*

138. Which acid of each pair shown here would you expect to be stronger?

(i) $F—CH_2—COOH$ or $Cl—CH_2—COOH$

(ii)  or CH_3COOH *(2/5, AI 2013)*

139. Give reasons :

(i) Electrophilic substitution in benzoic acid takes place at *meta*-position.

(ii) Carboxylic acids do not give the characteristic reactions of carbonyl group. *(2/5, AI 2013C, Delhi 2012C)*

140. Arrange the following compounds in an increasing order of their property as indicated :

(i) Benzoic acid, 3,4-dinitrobenzoic acid, 4-methoxybenzoic acid (acid strength)

(ii) $CH_3CH_2CH(Br)COOH$, $CH_3CH(Br)CH_2COOH$, $(CH_3)_2CHCOOH$(acid strength)

(2/5, AI 2012)

141. Write the mechanism of esterification of carboxylic acids. *(2/5, Delhi 2012C)*

LA I (3 marks)

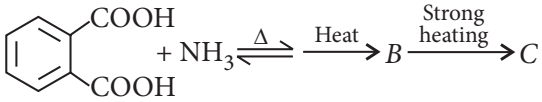
142. Write the structures of compounds *A*, *B* and *C* in each of the following reactions :

(i) $C_6H_5Br \xrightarrow{Mg/dry\ ether} A \xrightarrow[(b)\ H_3O^+]{(a)\ CO_{2(g)}} B \xrightarrow{PCl_5} C$

(ii) $CH_3CN \xrightarrow[(b)\ H_3O^+]{(a)\ SnCl_2/HCl} A \xrightarrow{dil.\ NaOH} B \xrightarrow{\Delta} C$

(Delhi 2017)

143. Identify *B* and *C* in the following sequence of reaction :



(3/5, AI 2011C)

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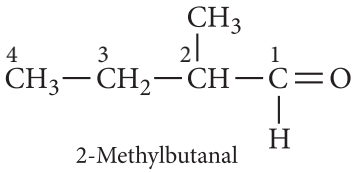
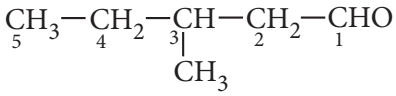
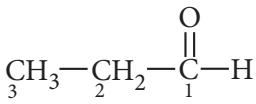
Detailed Solutions

1. 

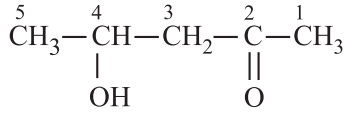
2-Methylbutanal

2. 

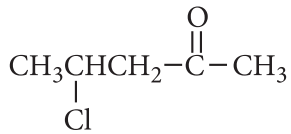
3-Methylpentanal

3. 

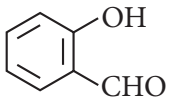
1-Propanal

4. 

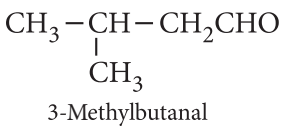
4-Hydroxypentan-2-one

5. 

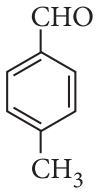
4-Chloropentan-2-one

6. 

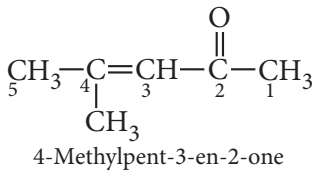
2-Hydroxybenzaldehyde

7. 

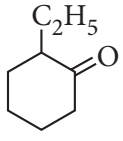
3-Methylbutanal

8. 

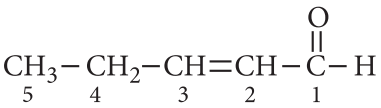
p-Methylbenzaldehyde

9. 

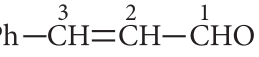
4-Methylpent-3-en-2-one

10. 

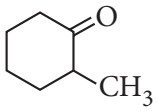
2-Ethylcyclohexanone

11. 

Pent-2-enal

12. 

3-Phenylprop-2-enal

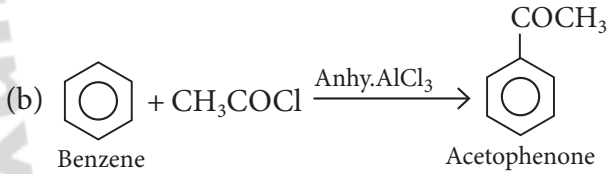
13. 

2-Methylcyclohexanone

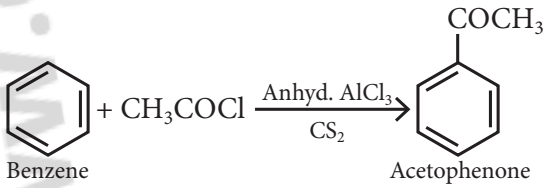
14. *Refer to answer 9.*

15. (i) *Refer to answer 8.*

(ii) *Refer to answer 9.*

16. 

Benzene Acetophenone

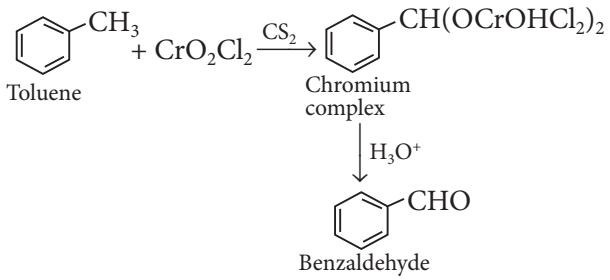
17. 

Benzene Acetophenone

18. $(C_6H_5CH_2)_2Cd + 2CH_3COCl \longrightarrow 2\,CH_3\overset{O}{\overset{\|}{C}}-CH_2C_6H_5 + CdCl_2$

19. $C_6H_5COCl \xrightarrow[Pd\text{-}BaSO_4]{H_2} \underset{\text{Benzaldehyde}}{C_6H_5CHO}$

20. Etard reaction :



Toluene Chromium complex Benzaldehyde

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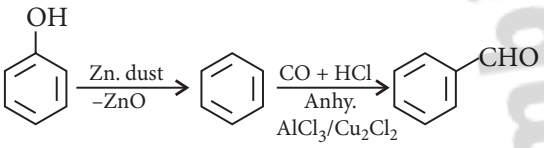
21. Stephen reduction :

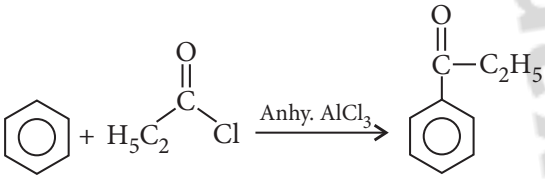
$$R\text{—CN} + SnCl_2 + HCl \xrightarrow{\text{Dry ether}} R - CH = NH \xrightarrow{H_3O^+} R - CHO$$

22. $CH_3 - CH{=}CH - CN \xrightarrow{\text{(a) DIBAL–H}} CH_3{-}CH{=}CH{-}CH{=}NH \xrightarrow{\text{(b) } H_2O} CH_3{-}CH{=}CH{-}CHO$

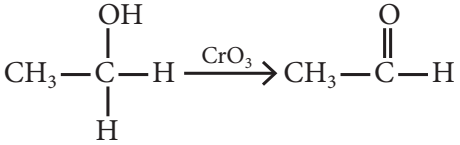
23. $CH_3 - CH = CH - CH_2CN \xrightarrow[\text{(ii) } H_2O]{\text{(i) DIBAL-H}} CH_3 - CH = CH - CH_2CHO$

24. $CH{\equiv}CH \xrightarrow[HgSO_4]{\text{dil. } H_2SO_4} CH_3{-}CHO$
Ethyne — Ethanal

25. 

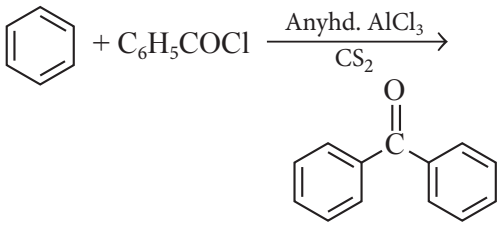
26. 

27. (i) *Refer to answer 23.*

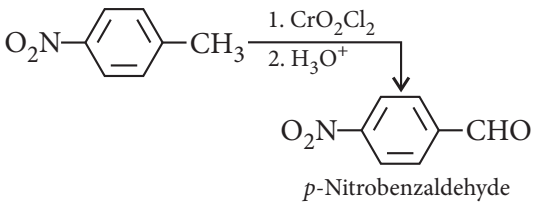
(ii) 

28. (i) *Refer to answer 21.*

(ii) *Refer to answer 20.*

29. (i) 

(ii) $CH_3{-}C{\equiv}CH \xrightarrow[H_2SO_4]{Hg^{2+},} CH_3{-}\overset{O}{\overset{\|}{C}}{-}CH_3$
Propanone

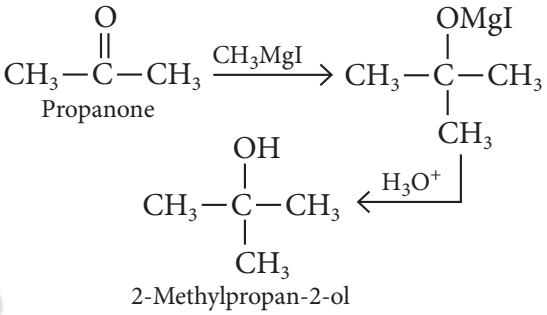
(iii) 

p-Nitrobenzaldehyde

30. The boiling points of aldehydes and ketones are lower than that of corresponding alcohols and acids due to absence of intermolecular H–bonding in aldehydes and ketones.

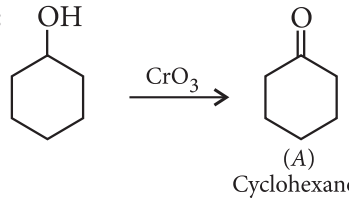
31. (d) : Iodoform test is given by the compound containing $CH_3{-}\overset{O}{\overset{\|}{C}}{-}$ or $CH_3{-}\underset{OH}{\underset{|}{CH}}{-}$ group thus it cannot be given by pentan-3-one, *i.e.*, $CH_3{-}CH_2{-}\overset{O}{\overset{\|}{C}}{-}CH_2{-}CH_3$.

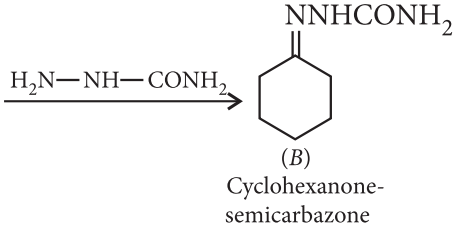
32. (d) : Reactivity of ketones is less than aldehydes.

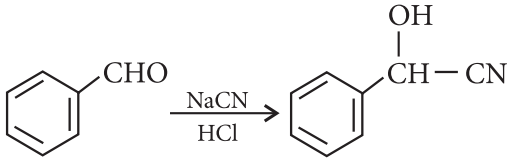
33. 

34. $CH_3{-}\overset{O}{\overset{\|}{C}}{-}CH_3 \xrightarrow{LiAlH_4} CH_3{-}\underset{OH}{\underset{|}{CH}}{-}CH_3$
Propanone (Acetone) — Propan-2-ol

$\xrightarrow[443\ K]{\text{Conc. } H_2SO_4} CH_3{-}CH{=}CH_2$
Propene

35. (c) : 



36. 

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37. $CH_3-\overset{\overset{O}{\|}}{C}-H \xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3MgBr} CH_3-\overset{\overset{OH}{|}}{\underset{\underset{H}{|}}{C}}-CH_3$

Ethanal

Propan-2-ol

$\xrightarrow{Cu/\Delta}$

$CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

Propanone

38. Ethanal and propanal can be distinguished by iodoform test.

Yellow precipitate of iodoform will be formed from ethanal on heating with iodine and sodium hydroxide solution whereas propanol will not give iodoform test.

39. Wolff-Kishner reduction :

$$\begin{matrix} R \\ R' \end{matrix}\!\!>\!C=O \xrightarrow[-H_2O]{NH_2NH_2} \begin{matrix} R \\ R' \end{matrix}\!\!>\!C=NNH_2$$

$$\xrightarrow[\Delta]{KOH/\text{ethylene glycol}} \begin{matrix} R \\ R' \end{matrix}\!\!>\!CH_2 + N_2$$

40. *Refer to answer 34.*

41. Benzaldehyde and acetophenone can be distinguished by Tollens' test.

Benzaldehyde will form silver mirror, on treatment with Tollens' reagent whereas acetophenone will not show Tollens' test.

42. $CH_3COCl \xrightarrow{H_2,Pd\text{-}BaSO_4} CH_3-\underset{\underset{H}{|}}{C}=O$ (*A* (Ethanal)) $\xrightarrow{H_2N-OH} CH_3-\underset{\underset{H}{|}}{C}=N-OH$ (*B* (Ethanal oxime))

43. *Refer to answer 41.*

44. Propanal and butan-2-one can be distinguished by their reactions with Tollens' reagent.

Propanal will form the silver mirror, but butan-2-one does not react.

45. Lithium aluminium hydride ($LiAlH_4$).

46. (i) Benzaldehyde and benzoic acid can be distinguished by sodium bicarbonate test.

Benzoic acid will give effervescence with $NaHCO_3$ but benzaldehyde will not react.

(ii) Propanal and propanone can be distinguished by their reactions with Tollens' reagent.

Propanal will form the silver mirror, but propanone does not react.

47. It is a nucleophilic addition reaction, in which CN^- acts as a nucleophile. CH_3CHO undergoes nucleophilic addition reactions faster than CH_3COCH_3 as in CH_3COCH_3 there are two electron releasing methyl groups attached to the carbonyl carbon that hinders the approach of nucleophile to carbonyl carbon and reduce the electrophilicity of the carbonyl group while in CH_3CHO, there is only one methyl group attached to carbonyl carbon.

48. *Refer to answer 46(ii).*

49. *Refer to answer 44.*

50. $CH_3CHO \xrightarrow{HCN} CH_3-\underset{\underset{OH}{|}}{CH}-CN \xrightarrow{Hydrolysis}$

$CH_3-\overset{\overset{H}{|}}{\underset{\underset{OH}{|}}{C}}-COOH$

2- Hydroxypropanoic acid

51. In strongly acidic medium, ammonia derivatives being basic will react with acids and will not react with carbonyl compound. In basic medium, OH^- will attack carbonyl group.

Therefore, pH of the reaction should be carefully controlled.

52. Ketones are less reactive than aldehydes towards nucleophilic addition reactions because the two electron releasing alkyl groups decrease the magnitude of positive charge on carbonyl carbon and make it less susceptible to nucleophilic attack.

$\begin{matrix} R \\ R \end{matrix}\!\!>\!C=O$ Ketone $\qquad$ $\begin{matrix} R \\ H \end{matrix}\!\!>\!C=O$ Aldehyde

The two bulkier alkyl groups hinder the approach of the nucleophile to the carbonyl carbon. This is called steric factor.

53. Butanone < propanone < propanal < ethanal.

54. Acetophenone and benzophenone can be distinguished by iodoform test.

Acetophenone will give yellow precipitate of iodoform, but benzophenone will not react.

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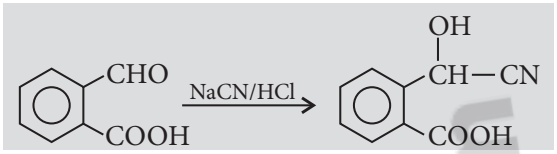
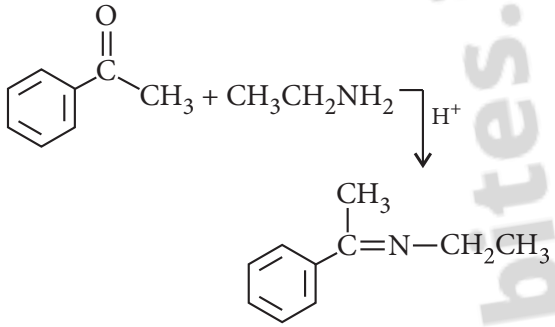
55. $(CH_3)_3C-\underset{CH_3}{\underset{|}{C}}=O < CH_3-\underset{CH_3}{\underset{|}{C}}=O < CH_3-\underset{H}{\underset{|}{C}}=O$

Methyl *tert*-butyl ketone — Acetone — Acetaldehyde

56.



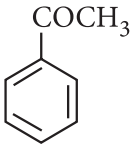
57.



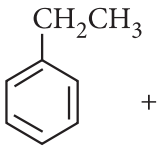
58.

(i) $2\,CH_3COCH_3 \xrightarrow{Ba(OH)_2} CH_3-\underset{CH_3}{\underset{|}{C}}(OH)-CH_2COCH_3$

4-Hydroxy-4-methyl-pentane-2-one

(ii)  $+ 4[H] \xrightarrow{Zn(Hg)/Conc.HCl}$

Acetophenone

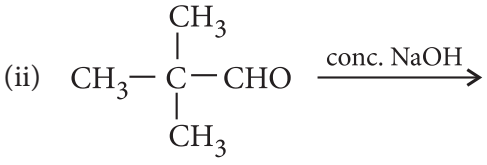
 $+ H_2O$

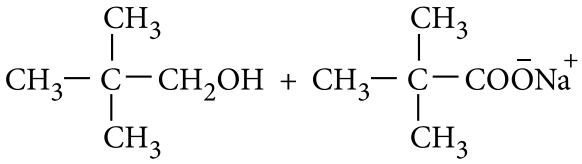
Ethylbenzene

59. (i) $CH_3-CH_2-\underset{O}{\underset{\|}{C}}-H \xrightarrow{NH_2-NH_2}$

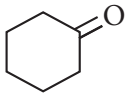
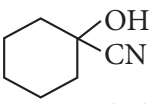
$CH_3-CH_2-\underset{H}{\underset{|}{C}}=NNH_2 \xrightarrow{KOH/ethylene\ glycol/heat} CH_3-CH_2-CH_3 + N_2$

(Wolff-Kishner reduction)

(ii) 



(Cannizzaro reaction)

60. (a)  $+ HCN \longrightarrow$ 

Cyclohexanone — Cyclohexane cyanohydrin

(b) Butanal respond to Fehling test and silver mirror test but butan-2-one does not respond.

61. (i) **Clemmensen reduction :**

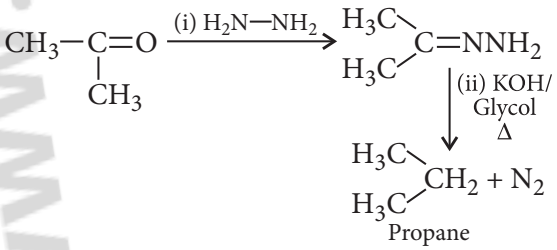
$\underset{H}{\overset{R}{\diagdown}}C=O \xrightarrow[conc.HCl]{Zn-Hg} \underset{H}{\overset{R}{\diagdown}}CH_2 + H_2O$

(ii) **Cannizzaro reaction :**

$2H-\underset{O}{\underset{\|}{C}}-H + KOH_{(conc.)} \xrightarrow{\Delta} CH_3OH + HCOO^-K^+$

Formaldehyde — Methanol — Potassium formate

62. (i)

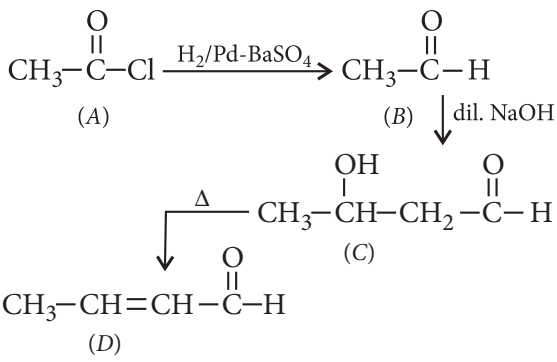


Propane

(ii) $C_6H_5-CO-CH_3 \xrightarrow{NaOH/I_2} CHI_3\downarrow + C_6H_5COO^-Na^+$

Iodoform

63.



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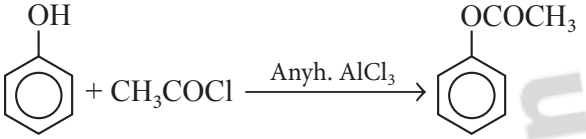
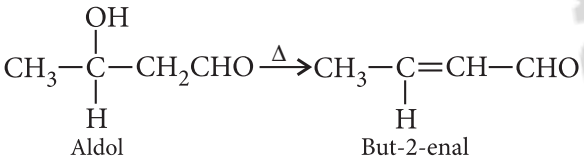
64. (i) **Acetylation :** Introduction of acetyl group $\left(-\overset{\overset{O}{\|}}{C}-CH_3\right)$ in alcohols, phenols or amines is called acetylation.



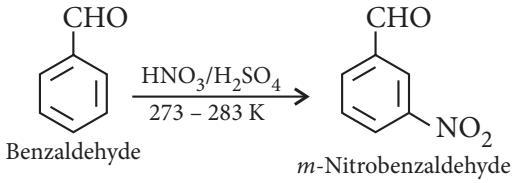
(ii) **Aldol condensation :** Two molecules of an aldehyde or ketones having at least one α-hydrogen atom condense in the presence of a dilute alkali to give β-hydroxyaldehyde or β-hydroxyketone which upon heating give α,β-unsaturated aldehyde or ketone.



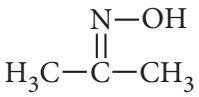
65.

(i) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{LiAlH_4} CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

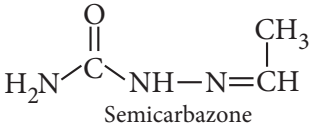
(ii)



66. (i)

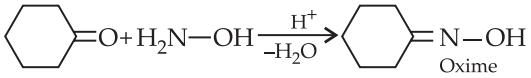


(ii)



67.

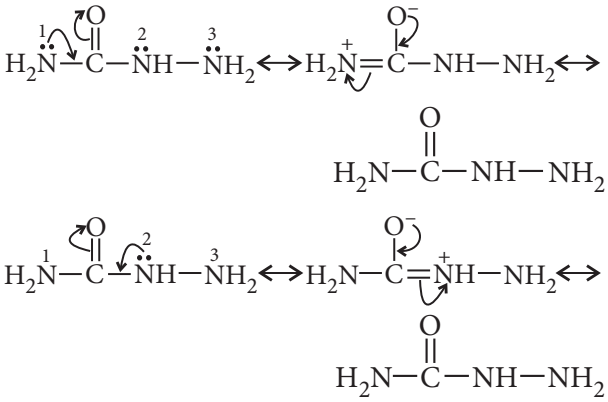
(i)



(ii) $2C_6H_5CHO + \text{conc. } NaOH \longrightarrow C_6H_5COONa + C_6H_5CH_2OH$

68. (i) *Refer to answer 47.*

(ii) Semicarbazide has the following resonance structures arising due to the electron withdrawing nature of the O atom.



Lone pairs of N-1 and N-2 are involved in conjugation with $\diagdown C=O$ group while that of N-3 is not involved in resonance thus, it is involved in the formation of semicarbazone.

69. (i) **Rosenmund's reaction :**

$$\underset{\text{Ethanoyl chloride (Acetyl chloride)}}{CH_3COCl} + H_2 \xrightarrow[\text{Boiling xylene}]{Pd/BaSO_4,\ S} \underset{\text{Ethanal (Acetaldehyde)}}{CH_3CHO} + HCl$$

(ii) *Refer to answer 61(ii).*

70. (i) *Refer to answer 61(ii).*

(ii) *Refer to answer 65(ii).*

71.

(i) $\underset{\text{Propanone}}{CH_3-CO-CH_3} + H_2 \xrightarrow{Ni \text{ or } Pt} \underset{\text{Propan-2-ol}}{CH_3-CHOH-CH_3}$

(ii) *Refer to answer 50.*

72.

(a) NaOI	$CH_3CH_2\overset{\overset{O}{\|}}{C}CH_2CH_3$ Pentan-3-one → No reaction	$CH_3-\overset{\overset{O}{\|}}{C}-(CH_2)_2CH_3$ Pentan-2-one gives iodoform test.
(b) NaOI	CH_3CHO Ethanal gives iodoform test.	$CH_3CH_2CHO \longrightarrow$ Propanal No reaction

73. (i) $\underset{\text{Acetone}}{CH_3-\overset{\overset{O}{\|}}{C}-CH_3} \xrightarrow[\text{conc. HCl}]{Zn-Hg} \underset{\text{Propane}}{CH_3-CH_2-CH_3}$

(ii) $CH_3\overset{\overset{O}{\|}}{C}-Cl \xrightarrow[Pd\text{-}BaSO_4]{H_2} CH_3CHO$

74. (i) *Refer to answer 46(ii).*

(ii) *Refer to answer 41.*

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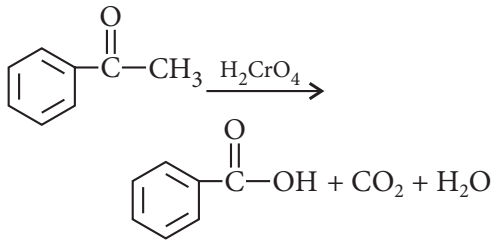
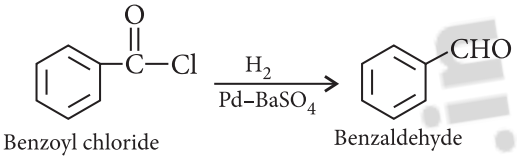
75. (i) 

(ii) 

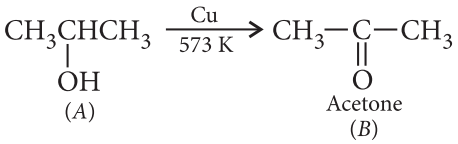
Benzoyl chloride Benzaldehyde

76. Compound *B* gives positive iodoform test, it means it contains $—COCH_3$ (methyl ketone) group *i.e.,* it is a ketone. Moreover, *B* is obtained by the oxidation of *A*, thus *A* must be a 2° alcohol. (As only 2° alcohol give ketones on oxidation with Cu at 573 K. Hence, the structure of compound *A* is $RCHCH_3$ with OH)

Comparing with the given molecular formula gives $R = CH_3$.

Thus, compound *A* is CH_3CHCH_3 (OH) (Propan-2-ol)

The reaction is as follows :



(*A*) Acetone (*B*)

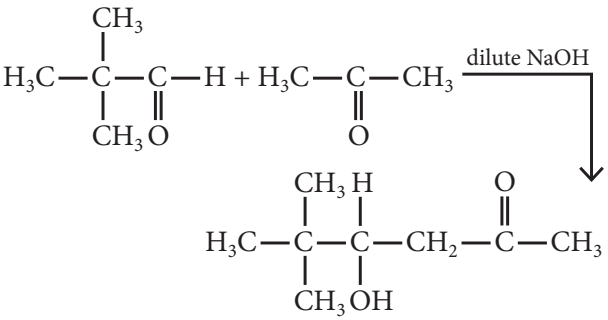
Acetone $\xrightarrow[\text{solution}]{\text{Fehling}}$ No reaction

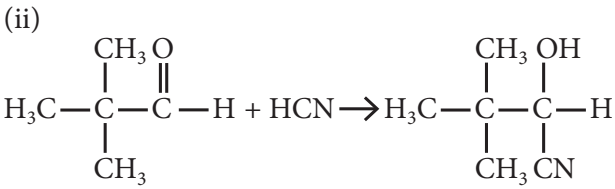
$$CH_3COCH_3 + 3I_2 + 4NaOH \longrightarrow CH_3COONa + 3NaI + \underset{\text{Iodoform}\ (C)}{CHI_3\downarrow} + 3H_2O$$

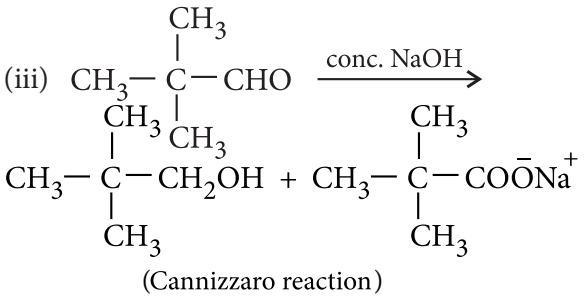
Thus, $A = CH_3CH(OH)CH_3$

$B = CH_3COCH_3$

$C = CHI_3$

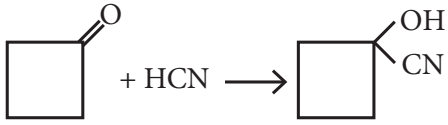
77. (i) 

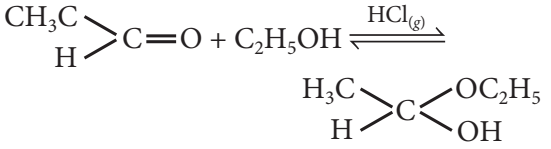
(ii) 

(iii) 

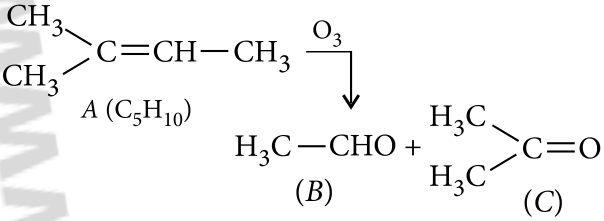
(Cannizzaro reaction)

78. (a) *Refer to answer 46(ii).*

(b) (i) 

(ii) 

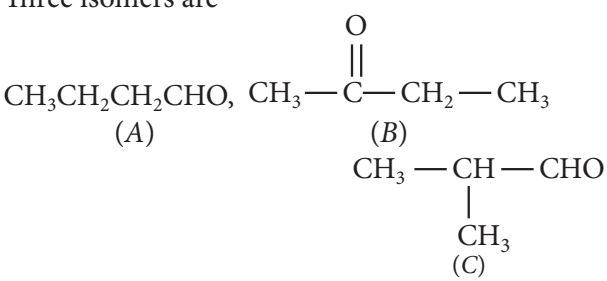
79. As compound *B* gives positive Fehling's test so it is an aldehyde and as it gives positive iodoform test so it is acetaldehyde. Compound *C* does not give Fehling's solution test. So it is a ketone which could be acetone because acetone gives positive iodoform test.



A (C_5H_{10}) (*B*) (*C*)

80. (a) As (*A*) and (*C*) give positive Tollens' test thus these two should be aldehyde while (*B*) should be a ketone (does not give Tollens' test) with $—\overset{\overset{O}{\|}}{C}—CH_3$ group (as it gives positve iodoform test).

Three isomers are



(*A*) (*B*) (*C*)

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$CH_3CH_2CH_2CHO \xrightarrow{Zn(Hg)/conc.\ HCl} CH_3CH_2CH_2CH_3$
(*A*) (*D*)

$CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_3 \xrightarrow{Zn(Hg)/conc.\ HCl} CH_3CH_2CH_2CH_3$
(*B*) (*D*)

(b) Out of (*A*), (*B*) and (*C*) isomers, (*B*) is least reactive towards addition of HCN.

81. (a) *Refer to answer 61(ii).*

(b) $H_3CC(H)=O + H_2N-C(=O)-NH-NH_2 \rightarrow H_3C-C(H)=N-NH-C(=O)-NH_2$
Ethanal, Semicarbazide → (Semicarbazone of ethanal)

(c) *Refer to answer 46(ii).*

82. (a) **Wolff-Kishner reduction :** The carbonyl group of aldehydes and ketones is reduced to $>CH_2$ group on treatment with hydrazine followed by heating with potassium hydroxide in a high boiling solvent such as ethylene glycol.

$(CH_3)_2C=O \xrightarrow[-H_2O]{NH_2NH_2} (CH_3)_2C=NNH_2 \xrightarrow[heat]{KOH/ethylene\ glycol} (CH_3)_2CH_2 + N_2$
Acetone → Propane

(b) Increasing order of reactivity towards nucleophilic addition reaction :

$C_6H_5COCH_3 < CH_3COCH_3 < CH_3CHO$

(c) Formula of compounds *A* and *B* = C_3H_6O
B forms yellow precipitate of iodoform. Hence, *B* must contain $-COCH_3$ group. Therefore, compound '*B*' must be $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$.

A does not give iodoform test and it is functional isomer of *B* thus, it may be CH_3CH_2CHO.

83. (i) $(CH_3)_2C=O \xrightarrow{Zn-Hg/Conc.\ HCl} CH_3CH_2CH_3$
Acetone → Propane

(ii) $(CH_3)_2C=O + H_2N-NHCONH_2 \xrightarrow{H^+} (CH_3)_2C=N-NHCONH_2$
Acetone semicarbazone

(iii) $(CH_3)_2C=O + CH_3MgBr \rightarrow (CH_3)_2C(OH)CH_3 \xleftarrow{H_3O^+} (CH_3)_2C(OMgBr)CH_3$
2-Methylpropan-2-ol

84. (i) $CH_3CHO \xrightarrow{Zn-Hg/conc.\ HCl} CH_3-CH_3$

(ii) $2CH_3CHO \xrightarrow{dil.\ NaOH} CH_3-CH(OH)-CH_2-CHO$

(iii) $CH_3CHO \xrightarrow{H_2/Ni} CH_3-CH_2-OH$

85. (i) *Refer to answer 82(a).*

(ii) *Refer to answer 64(ii).*

(iii) *Refer to answer 61(ii).*

86. (i) $CH_3CHO + HCN \longrightarrow CH_3-C(OH)(H)-CN$

(ii) $CH_3CHO + H_2NOH \longrightarrow CH_3-CH=N-OH$

(iii) $2CH_3CHO + dil.\ NaOH \longrightarrow CH_3-CH(OH)-CH_2-CHO$

87. (a) (i) *Refer to answer 69(i).*

(ii) *Refer to answer 61(ii).*

(b) $CH_3CH_2CH_2-CO-CH_3$ will give iodoform test because it contains acetyl group.

88. (i) $CH_3-CHO + CH_3MgBr \xrightarrow{H_3O^+} CH_3-CH(OH)-CH_3$

(ii) $CH_3CHO \xrightarrow[Conc.\ HCl]{Zn-Hg} CH_3-CH_3$

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(iii) $CH_3-CHO + C_6H_5CHO \xrightarrow{\text{dil. NaOH}} C_6H_5-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2CHO$

89. (i) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}=O \xrightarrow[\text{Clemmensen reduction}]{\text{Zn–Hg + conc. HCl}} CH_3-CH_2-CH_3$

(ii) *Refer to answer 75(ii).*

(iii) *Refer to answer 64(ii).*

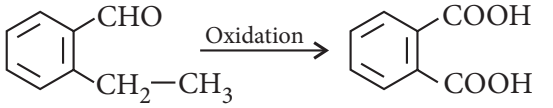
90. Benzaldehyde has a characteristic odour. Reaction of '*A*' with NaOH appears to be Cannizzaro reaction which gives '*B*' (benzyl alcohol) and '*C*' (Sodium salt of benzoic acid). Oxidation of alcohols gives aldehydes. Sodium salt of benzoic acid on heating with soda lime gives benzene (*D*).

$\underset{\text{'}A\text{'}}{C_6H_5CHO} \xrightarrow{NaOH} \underset{\text{'}B\text{'}\ (C_7H_8O)}{C_6H_5CH_2OH} + \underset{\text{'}C\text{' (Sodium salt of benzoic acid)}}{C_6H_5COONa}$

$\underset{\text{'}B\text{'}}{C_6H_5CH_2OH} \xrightarrow{CrO_3} \underset{\text{'}A\text{'}}{C_6H_5CHO}$

$\underset{\text{'}C\text{'}}{C_6H_5COONa} \xrightarrow[\Delta]{NaOH/CaO} \underset{\text{'}D\text{' (Benzene) (Aromatic)}}{C_6H_6} + Na_2CO_3$

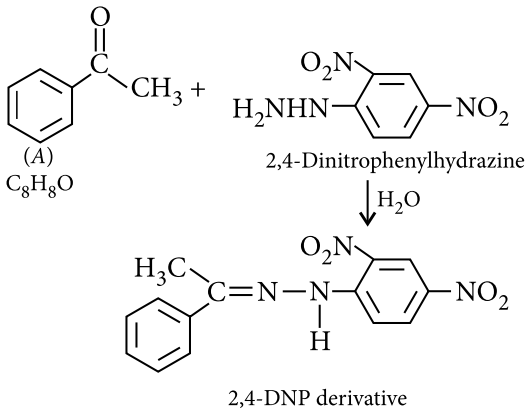
91. The compound forms 2,4-DNP derivative. It shows that it is a carbonyl compound. Further it reduces Tollens' reagent which shows that it contains aldehydic group. It undergoes Cannizzaro reaction indicating that aldehyde group is without any α-hydrogen. On vigorous oxidation, it gives 1,2-benzenedicarboxylic acid which shows that there are two carbon residues on benzene ring. Since the molecular formula is $C_9H_{10}O$, it fits into the structure, 2-ethylbenzaldehyde.

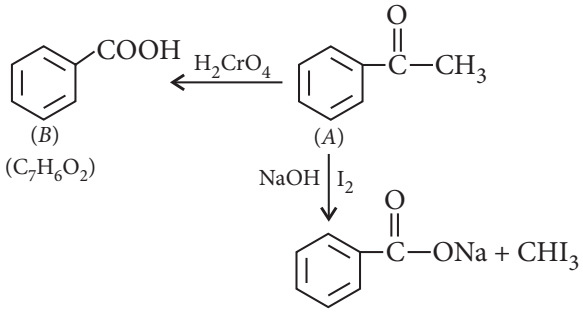


2-Ethylbenzaldehyde — 1,2-Benzenedicarboxylic acid

92. (*A*) forms, 2, 4-DNP derivative. Therefore, it is an aldehyde or a ketone. Since it does not reduce Tollens' or Fehling reagent, (*A*) must be a ketone. (*A*) responds to iodoform test.

Compound (*B*), being an oxidation product of a ketone should be a carboxylic acid. The molecular formula of (*B*) indicates that it should be benzoic acid and compound (*A*) should, therefore, be a mono-substituted aromatic methyl ketone.





93. The given compound does not reduce Tollens' reagent, so it is not an aldehyde but the formation of addition compound with sodium hydrogen sulphite indicates it to be a carbonyl compound. Since this compound gives positive iodoform test, so it should contain $-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3$ group.

On the basis of this information, two possible structures are written as under :

$CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_2-CH_2-CH_3$ and

I

$CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

II

On oxidation, this compound gives ethanoic and propanoic acids which confirm its structure to be I.

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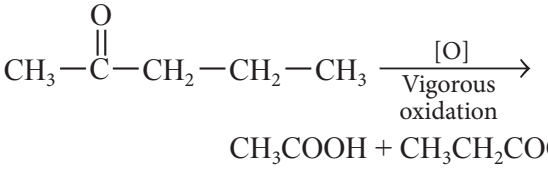
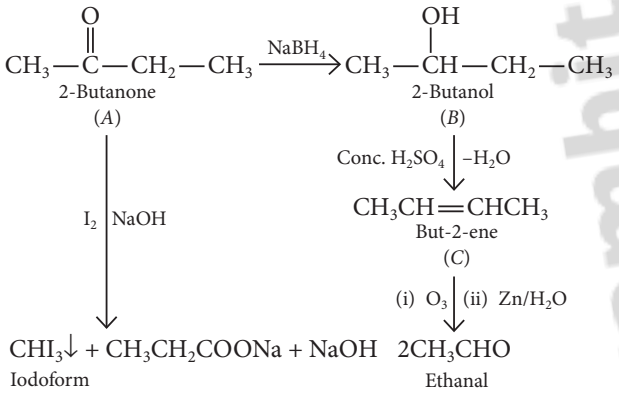


$CH_3COOH + CH_3CH_2COOH$

94. (a) As (*A*) forms 2, 4-DNP derivative, hence it is an aldehyde or ketone. Since it does not reduce Tollens' reagent, hence it is a ketone (*A*). As it gives iodoform reaction, hence it is a methyl ketone, *i.e.*, contains $CH_3-\overset{\overset{O}{\|}}{C}-$ group.

Therefore, (*A*) must be $CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$ *i.e.*, 2-butanone.

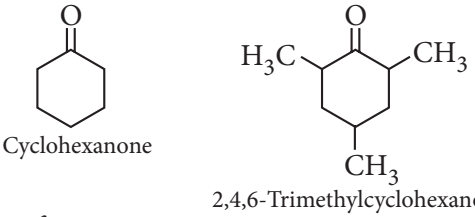


(b) (i) Unlike aldehydes, ketones do not contain any hydrogen atom attached to $>C=O$ group and hence they cannot be oxidised without the cleavage of some carbon-carbon bonds. Thus oxidation of propanal is easier than propanone.

(ii) Due to *–I* effect of the carbonyl group of aldehydes and ketones, it withdraw electrons from the adjacent carbon-carbon bond. This makes α-carbon electron deficient. The α-carbon in turn, withdraws electrons from the $C_\alpha-H$ bonds. As a result, the electron density in $C_\alpha-H$ bond decreases. Hence the α-hydrogens are weakly held and can be easily abstracted by strong bases giving enolate anions which are stabilised by resonance.

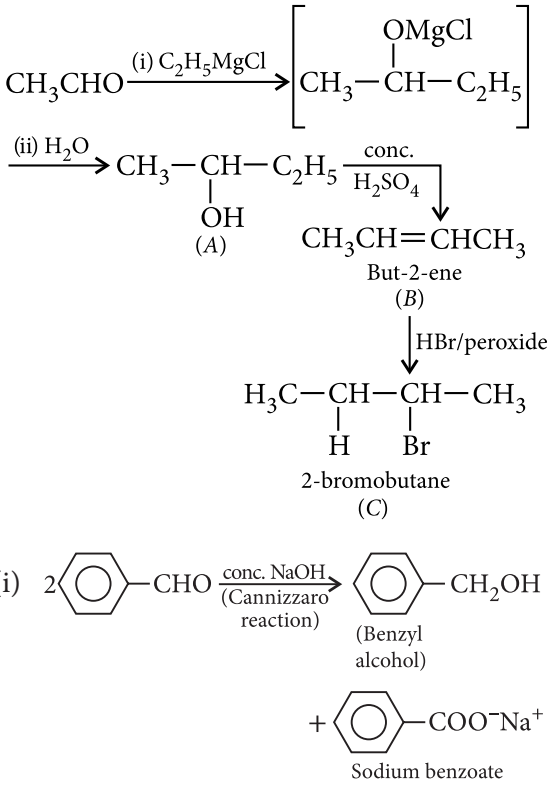
95. (a) (i) *Refer to answer 68(ii).*

(ii) Formation of cyanohydrin involves the nucleophilic attack of cyanide ions (CN^-) at the carbonyl carbon. In cyclohexanone, reaction proceeds but in 2,4,6-trimethylcyclohexanone, the methyl groups cause steric hindrance and yields are poor.



(b) *Refer to answer 91.*

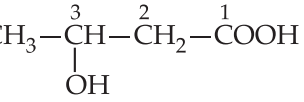
96.

(a) 

(b) (i)

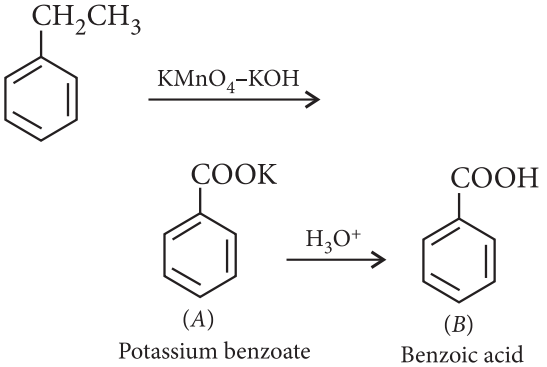
(ii) *Refer to answer 65(ii).*

97. **(a)**

98. 

99. Hex-2-en-4-yn-oic acid

100. 3-Bromo-5-chlorobenzoic acid

101. 

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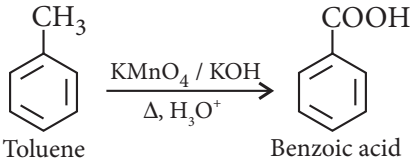
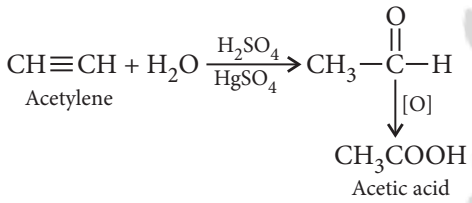
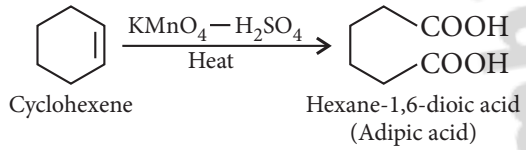
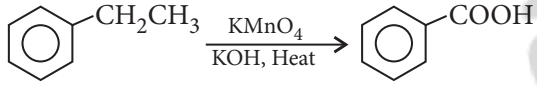
102. 

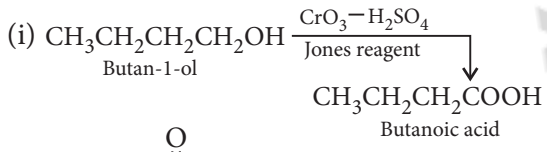
103.

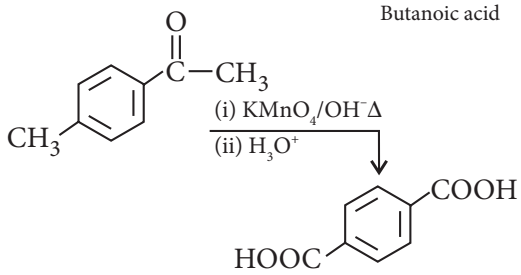
104. Alkaline potassium permanganate ($KMnO_4$, KOH)

105. 

106. 

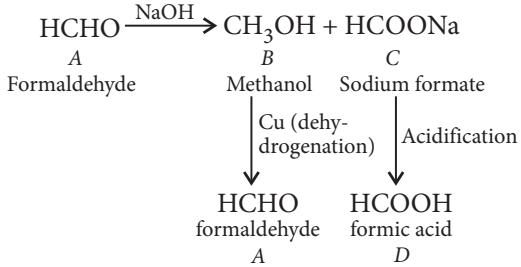
107. 

108. (i) 

(ii) 

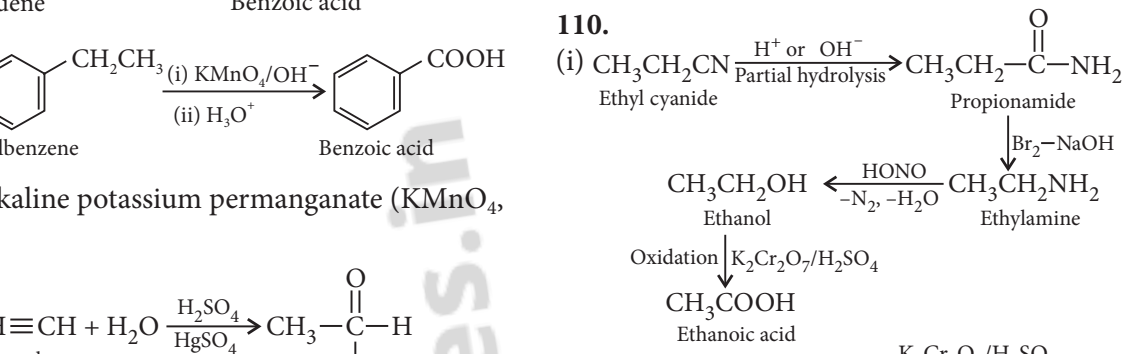
109. Since the molecular formula of *D* is CH_2O_2, thus, *D* is HCOOH (formic acid). *D* is obtained by the acidification of *C*, so, *C* is sodium formate (HCOONa).

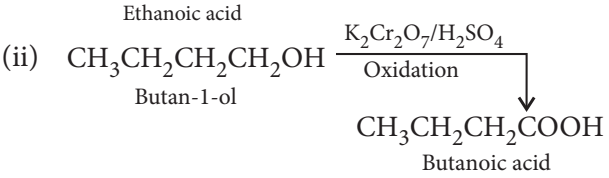
Thus, *A* must be formaldehyde (as it undergoes Cannizzaro reaction with a strong base).

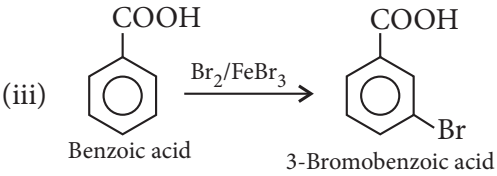


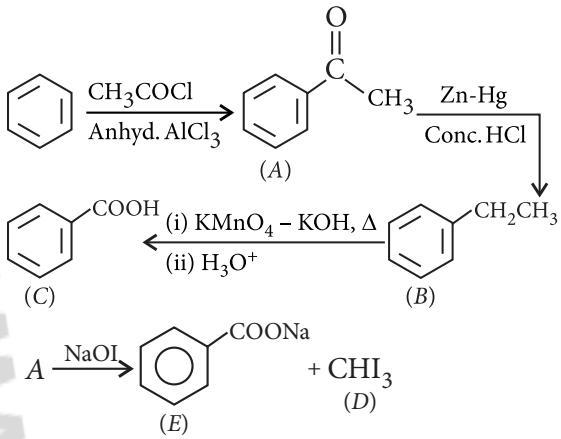
Thus, *A* = Formaldehyde (HCHO)
B = Methanol (CH_3OH)
C = Sodium formate (HCOONa)
D = Formic acid (HCOOH)

110.

(i) 

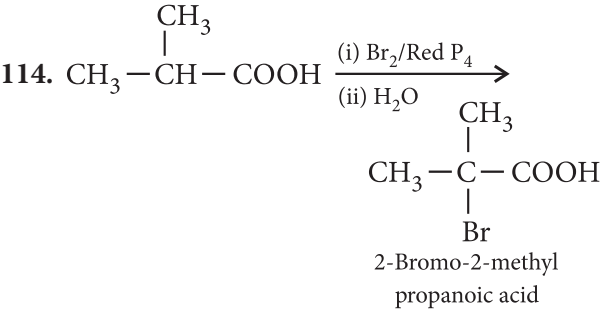
(ii) 

(iii) 

111. 

112. Increasing order of boiling point :
$CH_3—CHO < C_2H_5OH < CH_3—COOH$

113. (c) : Due to presence of electron withdrawing group (—COOH) in aromatic carboxylic acids, they do not undergo Friedel-Crafts reaction.

114. 

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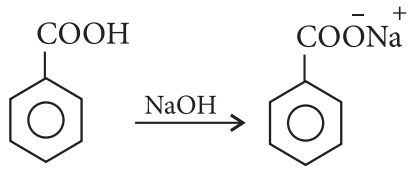
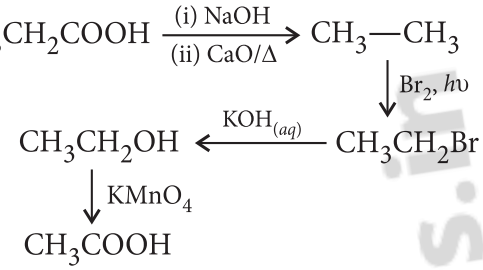
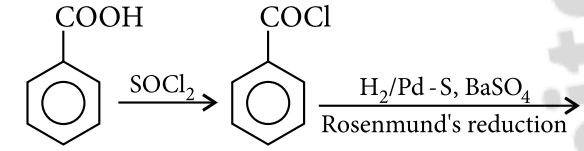
115.

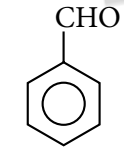


116.

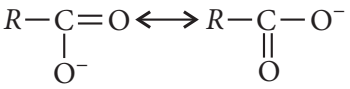


117.





118. The carbonyl group in —COOH is inert and does not show nucleophilic addition reaction like carbonyl compound. It is due to resonance stabilisation of carbyoxylate ion :



119. Benzoic acid and ethyl benzoate can be distinguished by their reactions with sodium bicarbonate solution. Benzoic acid will give effervescence with $NaHCO_3$ whereas ethyl benzoate does not react.

120. Add Tollens' reagent to formic acid and warm. Silver mirror is formed.

$$\underset{\text{Formic acid}}{HCOOH} + 2[Ag(NH_3)_2]^+ + 2OH^- \xrightarrow{\text{Warm}} \underset{\text{Silver mirror}}{2Ag} + CO_2 + 2NH_3 + 2NH_4OH$$

Acetic acid does not give this test.

121. $CH_3COONa \xrightarrow[\Delta]{NaOH/CaO} \underset{\text{Methane}}{CH_4} + Na_2CO_3$

122. $CH_3—COOH \xrightarrow{PCl_5} CH_3—COCl + POCl_3 + HCl$

123. *Refer to answer 122.*

124. Decarboxylation : Sodium or potassium salt of carboxylic acids on heating with soda lime (NaOH and CaO), loses a molecule of carbon dioxide and alkanes are obtained as products.

$$R—\overset{\overset{O}{\|}}{C}—O^-Na^+ + NaOH \xrightarrow{CaO} R—H + Na_2CO_3$$

125. *Refer to answer 118.*

126. Phenol and benzoic acid can be distinguished by their reactions with sodium bicarbonate solution. Benzoic acid will give effervescence with $NaHCO_3$ but phenol will not react.

127. Hell-Volhard-Zelinsky reaction : Carboxylic acids react with chlorine or bromine in the presence of phosphorous to give compounds in which α-hydrogen atom is replaced by halogen atom.

$$\underset{\text{Acetic acid}}{CH_3COOH} + Cl_2 \xrightarrow{\text{Red P}} \underset{\text{Chloroacetic acid}}{ClCH_2—COOH} + HCl$$

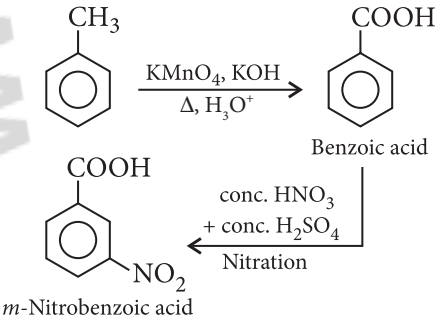
$$ClCH_2—COOH + Cl_2 \xrightarrow{\text{Red P}} \underset{\text{Dichloroacetic acid}}{Cl_2CHCOOH} + HCl$$

$$Cl_2CHCOOH + Cl_2 \xrightarrow{\text{Red P}} \underset{\text{Trichloroacetic acid}}{Cl_3CCOOH} + HCl$$

128. *Refer to answer 110(iii).*

129. Chloroacetic acid has lower pK_a value than acetic acid; 'Cl' in chloroacetic acid shows $-I$ effect, it creates less electron density on oxygen of carboxylic acid. Thus, release of proton becomes easier. Hence, chloroacetic acid is stronger than acetic acid.

130.



131. (a) Due to presence of electron withdrawing group (—COOH) in aromatic carboxylic acids, they do not undergo Friedel-Crafts reaction.

(b) Due to presence of strong electron withdrawing group (—NO_2), 4-nitrobenzoic acid is more acidic than benzoic acid and therefore, pK_a value is lower.

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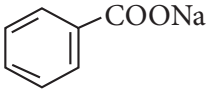
132. (i) *Refer to answer 127.*

(ii) *Refer to answer 124.*

133. (a)  + NaOH $\xrightarrow[\Delta]{CaO}$

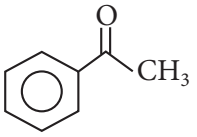
Sodium benzoate

+ Na_2CO_3

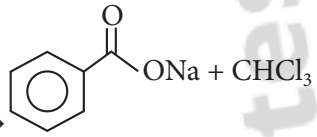
Benzene

(b) *Refer to answer 126.*

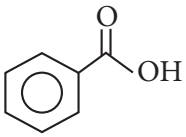
134. (i) Acetophenone to benzoic acid :



Acetophenone

$\xrightarrow{NaOCl}$  + $CHCl_3$

$\downarrow H_3O^+$



Benzoic acid

(ii) Ethanoic acid to 2-hydroxyethanoic acid.

$CH_3COOH \xrightarrow[H_2O]{Cl_2/Red\ P} Cl—CH_2—COOH \xrightarrow{NaOH} HO—CH_2—COOH$

Ethanoic acid — 2-Chloroethanoic acid — 2-Hydroxyethanoic acid

135. (a) $CH_3COOH \xrightarrow{Cl_2/P} \underset{Cl}{CH_2COOH}$

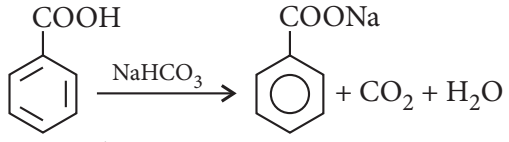
(b) Benzaldehyde when treated with ammoniacal silver nitrate gives silver mirror.

$$C_6H_5CHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow C_6H_5COO^- + 2Ag + 4NH_3 + 2H_2O$$

Silver mirror

Benzoic acid does not give this test.

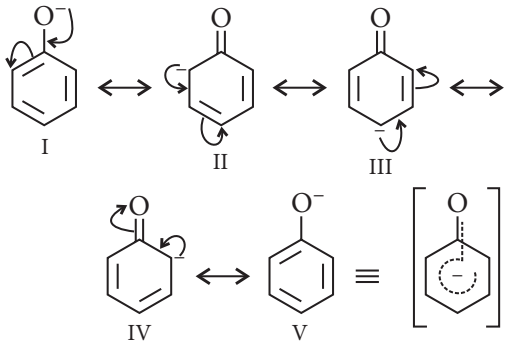
Benzoic acid reacts with sodium bicarbonate to liberate CO_2.



Benzoic acid

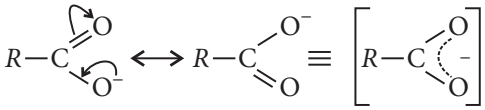
136. *Refer to answer 129.*

137. Phenoxide ion has the following resonating structures :



I II III IV V

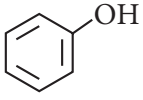
Carboxylate ion has the following resonating structures.



(i) Phenoxide ion is a resonance hybrid of structures I to V, where each structure has a contribution of 20% in the resonance hybrid. On the other hand, each of the two contributing structures of carboxylate ion contribute 50% towards the resonance hybrid. Therefore, the carboxylate ion tends to be more stable than the phenoxide ion and hence has higher acidity.

(ii) The negative charge rests on the electronegative O atom in carboxylate ion. The presence of negative charge on an electronegative atom makes the ion more stable. For the same reason $RCOO^-$ is more stable than the phenoxide ion where the carbon has negative charge on it. For the above two reasons carboxylate ion is more stable and hence carboxylic acid has higher acidity than phenol.

138. (i) $F—CH_2COOH > Cl—CH_2COOH$

(ii) CH_3COOH is stronger than 

139. (i) Electrophilic substitution in benzoic acid takes place at *meta*-position. Due to resonance in benzoic acid, there is high electron density at *meta*-position. Therefore, electrophilic substitution in benzoic acid takes place at *meta*-position.

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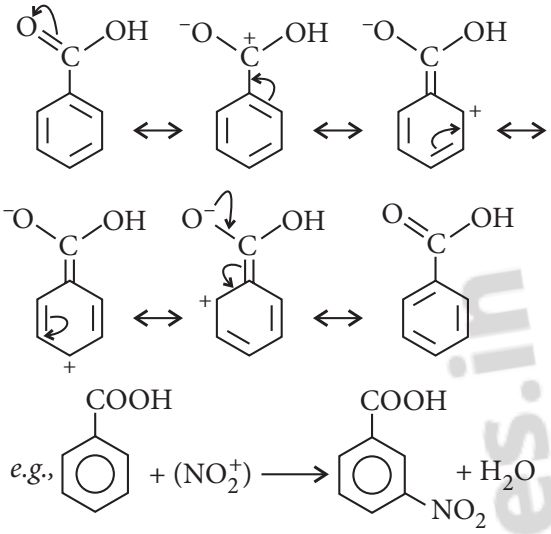


(ii) *Refer to answer 118.*

140. (i) The overall acid strength increases in the order.

4-Methoxybenzoic acid < benzoic acid < 3,4-dinitrobenzoic acid.

(ii) We know that + *I*-effect decreases while –*I*-effect increases the acid strength of carboxylic acids. The overall acid strength increases in the order :

$(CH_3)_2CHCOOH < CH_3CH(Br)CH_2COOH < CH_3CH_2CH(Br)COOH.$

141. Esterification : Carboxylic acids react with alcohols or phenols in the presence of a mineral acid like concentrated H_2SO_4 or HCl gas as catalyst and give ester.

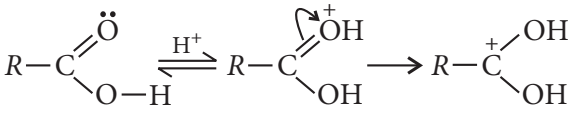
$$R—COOH + R'OH \rightleftharpoons R—COOR' + H_2O$$

$$\text{e.g., } CH_3COOH + C_2H_5OH \overset{H^+}{\rightleftharpoons} CH_3COOC_2H_5 + H_2O$$

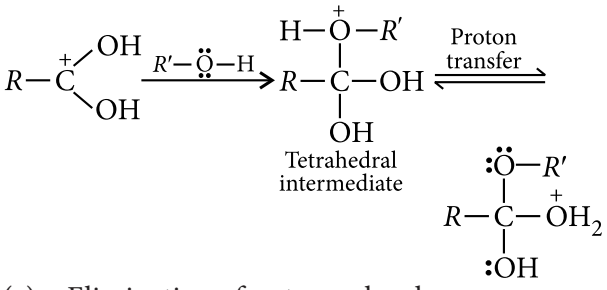
Ethanoic acid Ethanol Ethyl ethanoate

Mechanism of esterification : It is a nucleophilic acyl substitution.

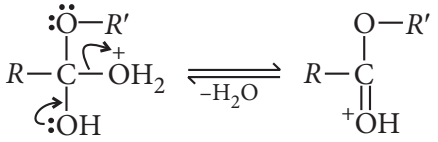
(a) Protonation of carboxyl oxygen :



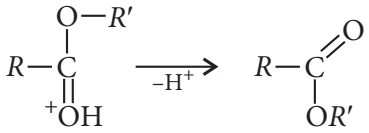
(b) Nucleophilic addition of alcohol :



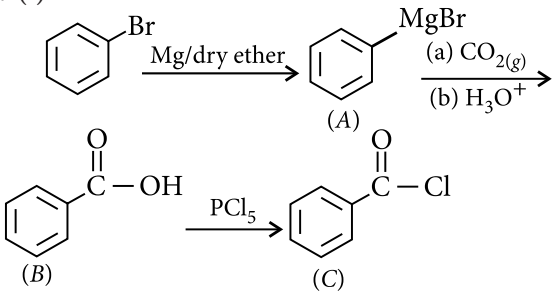
(c) Elimination of water molecule :



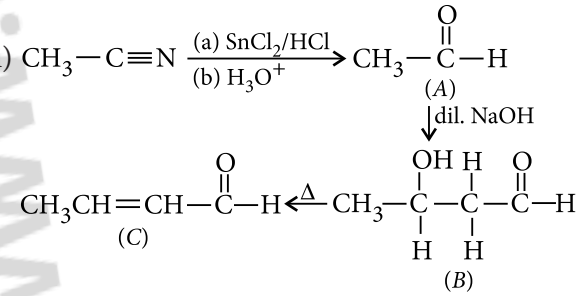
(d) Protonated ester loses a proton to give ester :



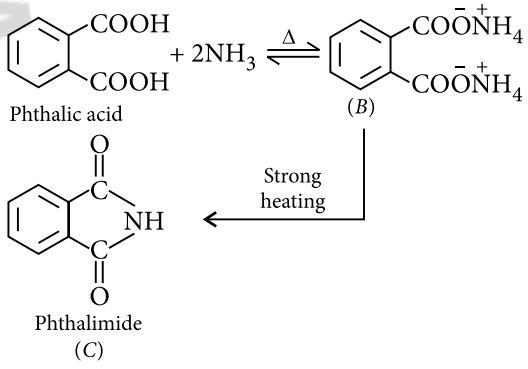
142. (i)



(ii)



143.

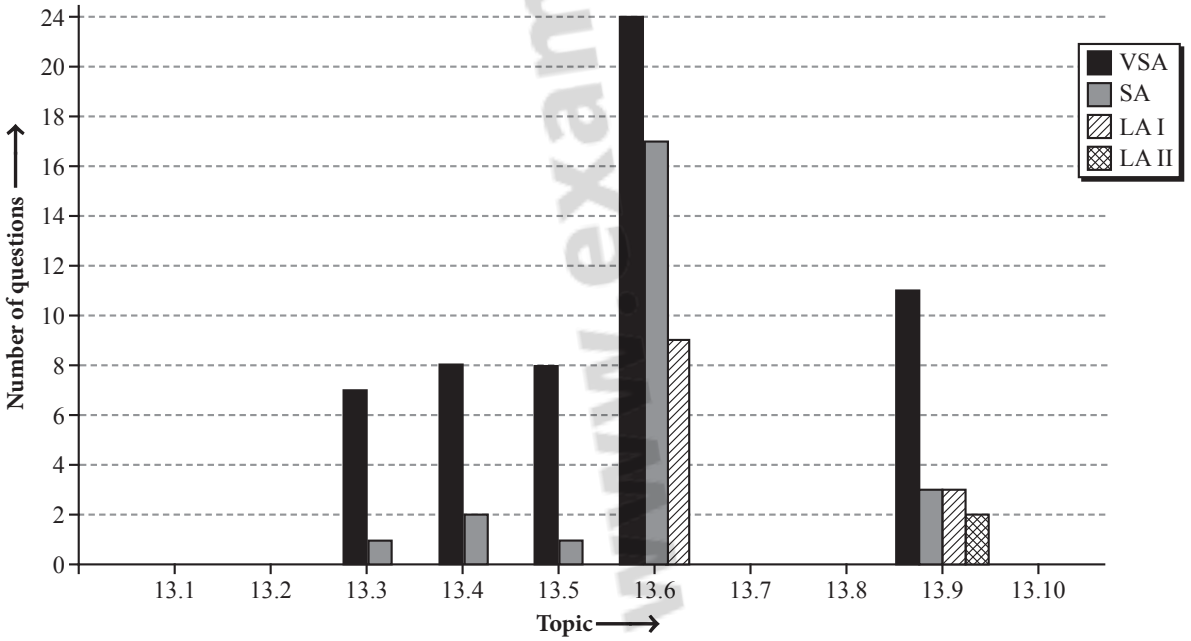


CHAPTER
13

Amines

13.1 Structure of Amines
13.2 Classification
13.3 Nomenclature
13.4 Preparation of Amines
13.5 Physical Properties
13.6 Chemical Reactions
13.7 Method of Preparation of Diazonium Salts
13.8 Physical Properties
13.9 Chemical Reactions
13.10 Importance of Diazonium Salts in Synthesis of Aromatic Compounds

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Chemical Reactions of Amines.*
- Maximum VSA type questions were asked from *Chemical Reactions of Amines.*
- Maximum SA and LA I type questions were asked from *Chemical Reactions of Amines.*

QUICK RECAP

AMINES

Amines : These are alkyl or aryl derivatives of ammonia and are obtained by replacing one, two or three hydrogen atoms by alkyl/aryl groups.

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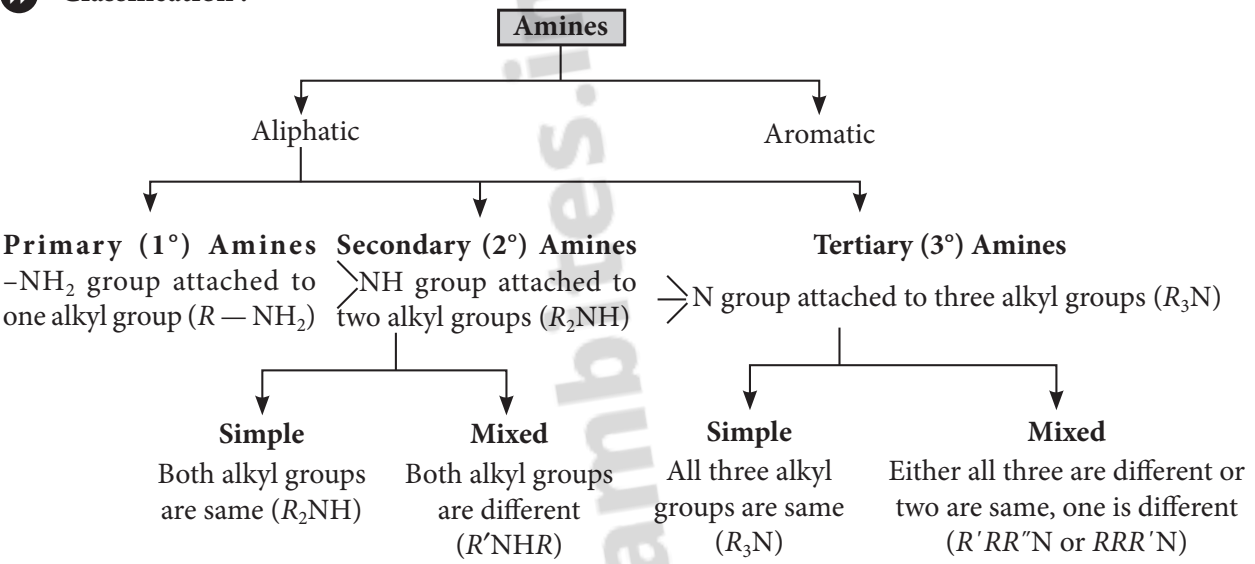
Nomenclature : In common system, an aliphatic amine is named by prefixing alkyl group to amine, *i.e.*, *alkylamine*. In IUPAC system, amines are named as *alkanamines*. In secondary and tertiary amines, when two or more groups are the same, the prefix *di* or *tri* is appended before the name of alkyl group.

Structure : Nitrogen orbitals in amines are sp^3-hybridised and the geometry of amines is pyramidal. Due to the presence of unshared pair of electrons, the angle C — N — *E*, (where *E* is C or H) is less than 109.5°.

Classification :



Preparation :

Reduction of nitro compounds :

$$RNO_2 \xrightarrow[\text{or } H_2/Pd \text{ or } LiAlH_4/\text{ether}]{\text{Sn/HCl or Fe/HCl}}$$

Hofmann's ammonolysis method :

$$RX \xrightarrow[373\ K]{NH_3}$$

Mendius reduction :

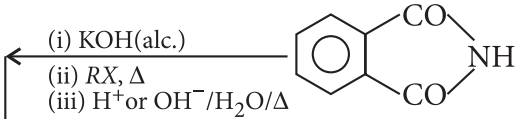
$$RCN \xrightarrow[\text{or } Na(Hg)/C_2H_5OH]{H_2/Ni \text{ or } LiAlH_4/\text{ether}}$$

Reduction of amides :

$$RCONH_2 \xrightarrow[\text{or } Na/C_2H_5OH]{LiAlH_4/\text{ether}}$$

RNH_2

Gabriel phthalimide synthesis :



Hofmann bromamide degradation :

$$RCONH_2 \xrightarrow[KOH,\ \Delta]{Br_2}$$

- **Limitations of Gabriel phthalimide synthesis :**
  - It is used for the preparation of only 1° amines. *tert*-Butylamine is a 1° amine, but cannot be prepared by this method. In this case, elimination takes place.
  - Aromatic 1° amines cannot be prepared by this method because aryl halides do not undergo nucleophilic substitution reaction with potassium phthalimide under mild conditions.

Physical properties :

- Lower amines are gases and liquids but higher amines are solids.
- Primary and secondary amines have higher boiling points than other organic compounds due to hydrogen bonding.
- Primary and secondary amines are soluble in water due to hydrogen bonding between >NH_2 and H_2O molecules.

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Chemical properties :

▶ **Basic character of amines :**

– Amines are basic in nature due to the presence of lone pair of electrons on nitrogen atom.

– Aliphatic amines are stronger bases than ammonia due to *+I* effect of alkyl groups present in amines.

– Aromatic amines are weaker bases than ammonia due to *–I* effect of aryl group.

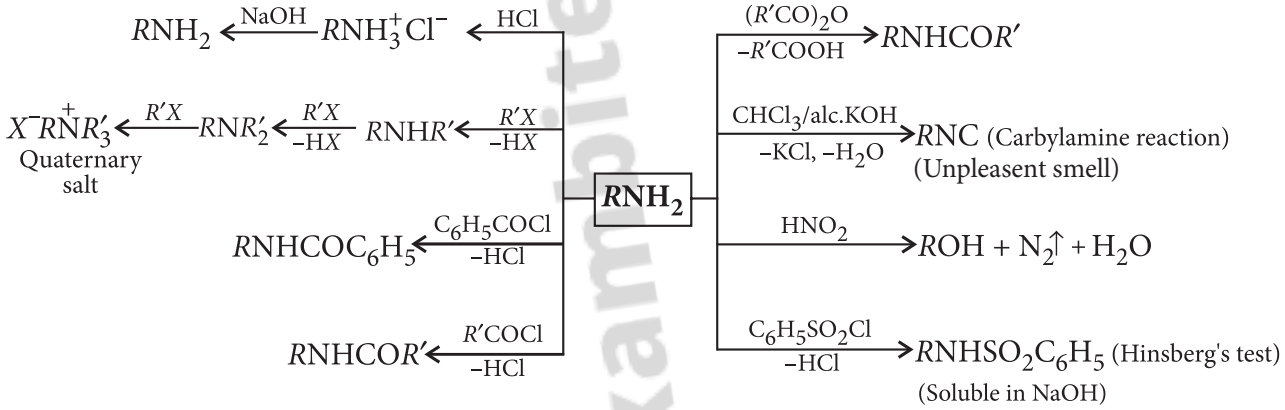
– Beside inductive effect, there are other effects like steric effect, solvation effect, resonance effect which affect the basic strength of amines.

– In gaseous phase, the order of basicity of amines is 3° amine > 2° amine > 1° amine > NH_3.

– In aqueous phase, despite of inductive effect, solvation effect and steric hindrance also play an important role. Thus, the order of basicity of amines is

$(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$
and $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

▶ **Chemical reactions :**



▶ **Identification of primary, secondary and tertiary amines :**

	Test	Primary amine	Secondary amine	Tertiary amine
1.	Reaction with nitrous acid.	Gives alcohol with effervescence of N_2 gas.	Gives oily nitrosoamine which gives Liebermann's nitrosoamine test.	Forms nitrite in cold which is soluble in water and on heating gives nitrosoamine.
2.	Reaction with benzene sulphonyl chloride (*Hinsberg's reagent*)	Gives *N*-alkylbenzene-sulphonamide which is soluble in alkali.	Gives *N*,*N*-dialkylbenzene sulphonamide which is insoluble in alkali.	No reaction
3.	*Carbylamine test* : Reaction with chloroform and alcoholic KOH	Forms carbylamine or isocyanide (*R*NC) with characteristic unpleasant odour.	No reaction	No reaction
4.	*Hofmann's mustard oil reaction* : Reaction with CS_2 and $HgCl_2$.	Forms *N*-substituted isothiocyante with characteristic unpleasant smell of mustard oil.	No reaction	No reaction

▶ **Electrophilic substitution reactions of arylamines :** Aniline undergoes electrophilic substitution reactions. —NH_2 group is *ortho*- and *para*-directing and a powerful activating group.

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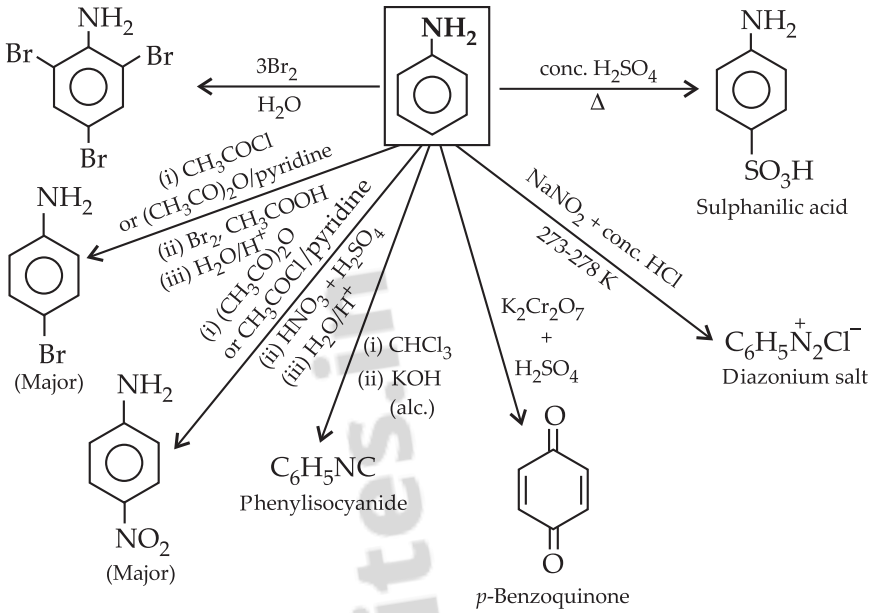


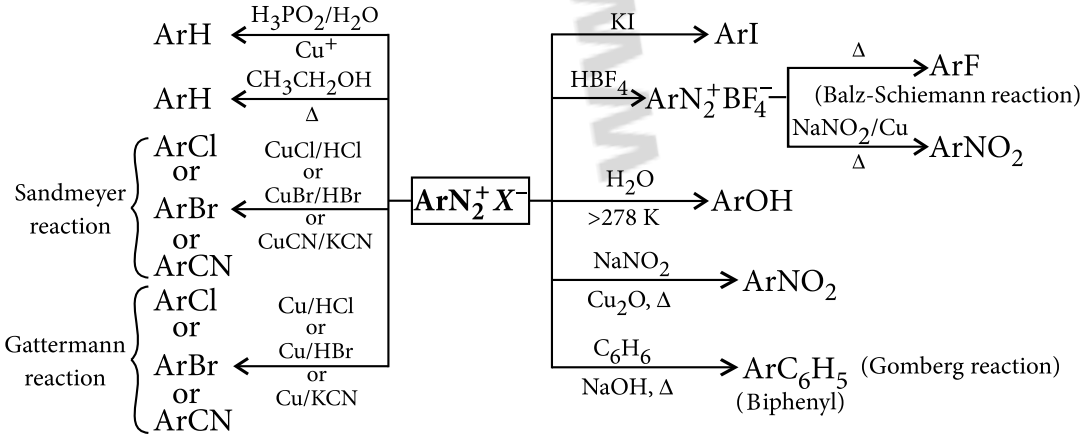
DIAZONIUM SALTS

- **Arenediazonium salts :** They have the general formula $ArN_2^+X^-$, Ar stands for the aryl ($-C_6H_5$) group and X^- is Cl^-, Br^-, NO_3^-, HSO_4^-, BF_4^-.
- **Nomenclature :** They are named by suffixing diazonium to the name of parent hydrocarbon followed by anion.
- **Preparation :** Benzenediazonium chloride is prepared by the reaction of aniline with nitrous acid (sodium nitrite and hydrochloric acid) at 0-5 °C.

$$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{0-5^\circ C} C_6H_5N_2^+Cl^- + NaCl + 2H_2O$$

- **Stability :** Arenediazonium salts are more stable (for short time) than alkanediazonium salts due to dispersal of positive charge over the benzene ring.
- **Physical properties :** These are generally colourless, crystalline solids, which are soluble in water. They are unstable and explode in dry state.
- **Chemical properties :**

▶ **Reactions involving displacement of nitrogen (diazo group) :**



▶ **Reactions involving retention of diazo group (coupling reactions) :**

$$ArN{=}NC_6H_4NH_2 \xleftarrow[pH = 4\text{-}5,\ 0\text{-}5^\circ C]{C_6H_5NH_2/H^+} \boxed{ArN_2^+X^-} \xrightarrow[pH = 9\text{-}10,\ 0\text{-}5^\circ C]{C_6H_5OH/OH^-} ArN{=}NC_6H_4OH$$

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Previous Years' CBSE Board Questions

13.3 Nomenclature

VSA (1 mark)

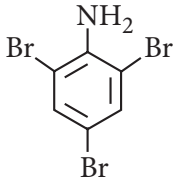
1. Write IUPAC name of the following compound : $(CH_3CH_2)_2NCH_3$ *(Delhi 2017)*

2. Write the structure of 2, 4-dinitrochlorobenzene. *(Delhi 2017)*

3. Write the IUPAC name of the following compound :
$CH_3NHCH(CH_3)_2$ *(Delhi 2017)*

4. Write IUPAC name of the following compound:
$(CH_3)_2N — CH_2CH_3$ *(Delhi 2017)*

5. Write the IUPAC name of the given compound :



(Delhi 2016)

6. Write the structure of *N*-methylethanamine. *(AI 2013)*

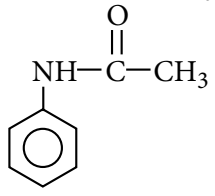
7. Write the structure of 2-aminotoluene. *(AI 2013)*

SA (2 marks)

8. Give IUPAC names of the following compounds :

(a) $CH_2=CH-\underset{\displaystyle CH_3}{\underset{|}{CH}}-NH_2$

(b)



(Delhi 2012C)

13.4 Preparation of Amines

VSA (1 mark)

9. Ammonolysis of alkyl halides is not a good method to prepare pure primary amines. *(1/5, 2020)*

10. Write the structures of *A* and *B* in the following reaction :

$$CH_3COOH \xrightarrow[\Delta]{NH_3} A \xrightarrow{NaOBr} B$$

(1/5, 2020)

11. Carry out the following conversion :
Benzoic acid to aniline. *(1/5, AI 2019)*

12. Write the chemical equation involved in the following reaction :
Hofmann bromamide degradation reaction
(AI 2019, 1/5, 2018, 1/2, AI 2016, 2012, 1/3, Foreign 2011)

13. State and illustrate the following :
Gabriel synthesis *(1/5, 2018, 1/2, AI 2013C, 1/3, Foreign 2011)*

14. How do you convert the following :
Ethanenitrile to ethanamine *(1/3, AI 2015)*

15. Write chemical equation for the following conversion :
Benzyl chloride to 2-phenylethanamine.
(1/3, Delhi 2012)

16. Why cannot primary aromatic amines be prepared by Gabriel phthalimide synthesis? *(1/2, AI 2011C)*

SA (2 marks)

17. How will you convert the following :
(i) Nitrobenzene into aniline
(ii) Ethanoic acid into methanamine
(2/3, Delhi 2014)

18. How are the following conversions carried out?
(i) CH_3CH_2Cl to $CH_3CH_2CH_2NH_2$
(ii) Benzene to aniline *(Delhi 2012C)*

13.5 Physical Properties

VSA (1 mark)

19. Out of CH_3NH_2 and CH_3OH, which has higher boiling point? *(2020)*

20. Arrange the following compounds in decreasing order of their boiling points :
Butanol, Butanamine, Butane *(1/3, 2020)*

21. Propanamine and *N*,*N*-dimethylmethanamine contain the same number of carbon atoms,

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even though propanamine has higher boiling point than *N*,*N*-dimethylmethanamine. Why? *(AI 2019)*

22. Give reasons for the following :
Primary amines have higher boiling point than tertiary amines. *(1/3, AI 2016, 2014, 2011)*

23. Arrange the following in the increasing order of their boiling points :
$C_2H_5NH_2$, C_2H_5OH, $(CH_3)_3N$ *(1/5, Delhi 2015)*

24. Arrange the following compounds in increasing order of solubility in water :
$C_6H_5NH_2$, $(C_2H_5)_2NH$, $C_2H_5NH_2$ *(Delhi 2014, AI 2011C)*

25. Out of CH_3NH_2 and $(CH_3)_3N$, which one has higher boiling point? *(Delhi 2014C)*

26. Account for the following :
Ethylamine is soluble in water whereas aniline is not. *(1/3, Delhi 2014C, AI 2011)*

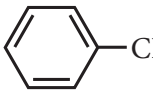
SA (2 marks)

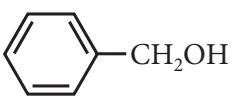
27. Arrange the following compounds as directed :
(i) In increasing order of solubility in water :
$(CH_3)_2NH$, CH_3NH_2, $C_6H_5NH_2$
(ii) In increasing order of boiling point :
$(C_2H_5)_2NH$, $(C_2H_5)_3N$, $C_2H_5NH_2$ *(2/3, 2020)*

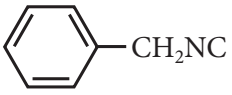
13.6 Chemical Reactions (Amines)

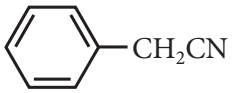
VSA (1 mark)

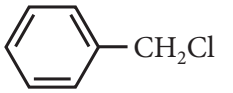
28. Out of the following, the strongest base in aqueous solution is
(a) methylamine (b) dimethylamine
(c) trimethylamine (d) aniline. *(2020)*

29.  on heating with $CHCl_3$ and alcoholic KOH gives foul smell of

(a) 

(b) 

(c) 

(d)  *(2020)*

30. Arrange the following compounds as directed :
In decreasing order of basic strength in aqueous solution :
$(CH_3)_3N$, $(CH_3)_2NH$, CH_3NH_2 *(1/3, 2020)*

31. Give the structures of *A* and *B* in the following sequence of reactions :

$$C_6H_5NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[0°-5°C]{NaNO_2+HCl} B$$

(2/3, 2020)

32. Arrange the following in increasing order of base strength in gas phase.
$(C_2H_5)_3N$, $C_2H_5NH_2$, $(C_2H_5)_2NH$ *(Delhi 2019)*

33. Carry out the following conversion :
Bromomethane to ethanol. *(1/5, AI 2019)*

34. Give reason :
$(CH_3)_2NH$ is more basic than $(CH_3)_3N$ in an aqueous solution. *(1/5, 2018)*

35. Arrange the following in the decreasing order of their basic strength in aqueous solutions :
CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$ and NH_3 *(1/3, 2018C, Delhi 2012)*

36. Describe the following by giving the relevant chemical equation :
Carbylamine reaction *(1/3, 2018C, 1/2, AI 2012)*

37. Give a simple chemical test to distinguish between the following pair of compounds :
$(CH_3)_2NH$ and $(CH_3)_3N$ *(1/5, Delhi 2015)*

38. Arrange the following in increasing order of basic strength :
Aniline, *p*-nitroaniline and *p*-toluidine *(1/2, AI 2015, AI 2015C)*

39. Arrange the following in increasing order of basic strength :
$C_6H_5NH_2$, $C_6H_5NHCH_3$, $C_6H_5CH_2NH_2$ *(1/2, AI 2015, Delhi 2014)*

40. How will you convert the following :
Aniline into *N*-phenylethanamide
(Write the chemical equations involved.) *(1/3, Delhi 2014)*

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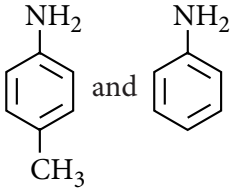
41. Arrange the following in increasing order of basic strength :
$C_6H_5NH_2$, $C_6H_5NHCH_3$, $C_6H_5N(CH_3)_2$
(Delhi 2014)

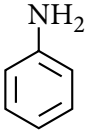
42. Which of the two is more basic and why?
CH_3NH_2 or NH_3 *(Foreign 2014)*

43. Which of the two is more basic and why?



(Foreign 2014)

44. Which of the two is more basic and why?

CH_3NH_2 or  *(Foreign 2014)*

45. Arrange the following in increasing order of their basic strength in aqueous solution :
CH_3NH_2, $(CH_3)_3N$, $(CH_3)_2NH$ *(Delhi 2013)*

46. Complete the following reaction equation :
$C_6H_5NH_2 + Br_{2(aq)} \rightarrow$ *(1/2, AI 2012)*

47. Rearrange the following in an increasing order of their basic strengths :
$C_6H_5NH_2$, $C_6H_5N(CH_3)_2$, $(C_6H_5)_2NH$ and CH_3NH_2 *(AI 2011)*

48. State reasons for the following :
pK_b value for aniline is more than that for ethylamine. *(1/3, AI 2011)*

49. How will you differentiate between aniline and ethylamine? *(AI 2011)*

50. Why is an alkylamine more basic than ammonia? *(Foreign 2011)*

51. How will you bring about the following conversion :
Ethanamine to ethanoic acid *(Delhi 2011C)*

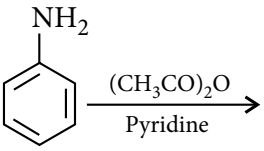
SA (2 marks)

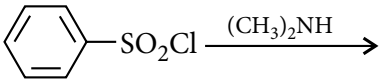
52. How will you distinguish between the following pairs of compounds :
(i) Aniline and ethanamine
(ii) Aniline and *N*-methylaniline
(2/3, 2020)

53. Distinguish between the following :
(i) $CH_3CH_2NH_2$ and $(CH_3CH_2)_2NH$
(ii) Aniline and CH_3NH_2 *(2/5, 2020)*

54. Give reasons :
(i) Although $-NH_2$ group is *o*/*p* directing in electrophilic substitution reactions, yet aniline, on nitration gives good yield of *m*-nitroaniline.
(ii) $(CH_3)_2NH$ is more basic than $(CH_3)_3N$ in an aqueous solution. *(2/5, 2020)*

55. Write the structures of the main products of the following reactions :

(i) 

(ii) 

(2/5, 2018)

56. (a) Give a simple chemical test to distinguish between aniline and *N*, *N*-dimethylaniline.
(b) Arrange the following in the increasing order of their pK_b values :
$C_6H_5NH_2$, $C_2H_5NH_2$, $C_6H_5NHCH_3$
(2/5, 2018)

57. Give reasons for the following :
(i) Aniline does not undergo Friedel–Crafts reaction.
(ii) $(CH_3)_2NH$ is more basic than $(CH_3)_3N$ in an aqueous solution.
(2/3, AI 2016, 2014)

58. How do you convert the following :
(i) $C_6H_5CONH_2$ to $C_6H_5NH_2$
(ii) Aniline to phenol
(2/3, AI 2015)

59. Illustrate the following reactions giving suitable example in each case :
(i) Ammonolysis
(ii) Acetylation of amines
(2/5, Foreign 2015)

60. Give the structures of *A*, *B* and *C* in the following reactions :

(i) $C_6H_5NO_2 \xrightarrow{Sn + HCl} A \xrightarrow[273\ K]{NaNO_2 + HCl} B \xrightarrow{H_2O} C$

(ii) $CH_3CN \xrightarrow{H_2O/H^+} A \xrightarrow[\Delta]{NH_3} B \xrightarrow{Br_2 + KOH} C$

(2/3, Delhi, AI 2014)

61. Account for the following :
(i) Aniline does not give Friedel–Crafts reaction.
(ii) pK_b of methylamine is less than that of aniline. *(2/3, Delhi 2014C)*

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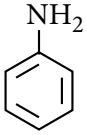
62. (i) Arrange the following compounds in an increasing order of basic strength :
$C_6H_5NH_2$, $C_6H_5N(CH_3)_2$, $(C_2H_5)_2NH$ and CH_3NH_2

(ii) Arrange the following compounds in a decreasing order of pK_b values :
$C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_2H_5)_2NH$ and $C_6H_5NH_2$

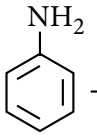
(Delhi 2014C)

63. Complete the following reactions :

(i) $CH_3CH_2NH_2 + CHCl_3 + \text{alc. KOH} \longrightarrow$

(ii)  $+ HCl_{(aq)} \longrightarrow$ *(2/3, AI 2013)*

64. Write the main products of the following reactions :

(i)  $\xrightarrow{Br_{2(aq)}}$?

(ii) $CH_3-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow{Br_2 + NaOH}$?

(2/3, AI 2013)

65. Give chemical tests to distinguish between the following pairs of compounds :

(i) Aniline and ethylamine

(ii) Ethylamine and dimethylamine

(Delhi 2013C)

66. Give reasons :

(i) Aniline is a weaker base than cyclohexyl amine.

(ii) It is difficult to prepare pure amines by ammonolysis of alkyl halides.

(AI 2013C)

67. Give reasons :

(i) Electrophilic substitution in aromatic amines takes place more readily than benzene.

(ii) CH_3CONH_2 is weaker base than $CH_3CH_2NH_2$. *(AI 2013C)*

68. How would you account for the following :

(i) Aniline is a weaker base than cyclohexylamine.

(ii) Methylamine in aqueous medium gives reddish-brown precipitate with $FeCl_3$.

(AI 2012C)

LA I (3 marks)

69. (a) Write the chemical reaction of methyl amine with benzoyl chloride and write the IUPAC name of the product obtained.

(b) Arrange the following in the increasing order of their pK_b values :
$C_6H_5NH_2$, NH_3, $C_2H_5NH_2$, $(C_2H_5)_2NH$

(3/5, 2020)

70. (a) Give one chemical test to distinguish between the compounds of the following pairs :

(i) CH_3NH_2 and $(CH_3)_2NH$

(ii) Aniline and ethanamine

(b) Why aniline does not undergo Friedel–Crafts reaction? *(AI 2019)*

71. Give reasons :

(i) Acetylation of aniline reduces its activation effect.

(ii) CH_3NH_2 is more basic than $C_6H_5NH_2$.

(iii) Although $-NH_2$ is *o*/*p* directing group, yet aniline on nitration gives a significant amount of *m*-nitroaniline.

(Delhi 2017)

72. Write the structures of *A*, *B* and *C* in the following :

(i) $C_6H_5-CONH_2 \xrightarrow{Br_2/aq.\ KOH} A \xrightarrow{NaNO_2 + HCl,\ 0\text{-}5°C} B \xrightarrow{KI} C$

(ii) $CH_3-Cl \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[\Delta]{CHCl_3 + alc.\ KOH} C$

(Delhi 2016)

73. Write the structures of main products when aniline reacts with the following reagents :

(i) Br_2 water (ii) HCl

(iii) $(CH_3CO)_2O$/pyridine *(3/5, Delhi 2015)*

74. Write the chemical equations involved when aniline is treated with the following reagents :

(i) Br_2 water (ii) $CHCl_3$ + KOH

(iii) HCl *(AI 2015)*

75. Write the structures of *A*, *B* and *C* in the following reactions :

(i) $C_6H_5NO_2 \xrightarrow{Sn + HCl} A \xrightarrow[273\ K]{NaNO_2 + HCl} B \xrightarrow[\Delta]{H_2O} C$

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(ii) $CH_3Cl \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[273\ K]{HNO_2} C$

(3/5, Foreign 2015)

76. An aromatic compound '*A*' on treatment with aqueous ammonia and heating forms compound '*B*' which on heating with Br_2 and KOH forms a compound '*C*' of molecular formula C_6H_7N. Write the structures and IUPAC names of compounds *A*, *B* and *C*. *(Delhi 2015C)*

77. Give the structures of products *A*, *B* and *C* in the following reactions :

(i) $CH_3CH_2Br \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[0°C]{HNO_2} C$

(ii) $CH_3COOH \xrightarrow[\Delta]{NH_3} A \xrightarrow{NaOH + Br_2} B \xrightarrow[\text{Alc. NaOH}]{CHCl_3 +} C$

(Delhi 2014, 2013)

13.9 Chemical Reactions (Diazonium salts)

VSA (1 mark)

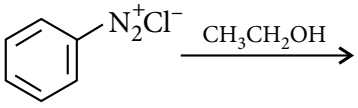
78. Write the structures of *A* and *B* in the following reactions :

$C_6H_6N_2^+Cl^- \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B$

(1/5, 2020)

79. Carry out the following conversion :
p-nitrotoluene to 2-bromobenzoic acid.
(1/5, AI 2019)

80. Write the structure of the main product of the following reaction :



(1/5, 2018)

81. Complete the following reaction equation :

$C_6H_5N_2Cl + H_3PO_2 + H_2O \longrightarrow$

(Delhi 2015C, AI 2013, 2012)

82. The conversion of primary aromatic amines into diazonium salts is known as _____. *(AI 2014)*

83. Complete the following reaction :

$C_6H_5N_2^+Cl^- \xrightarrow[\text{(Room temp.)}]{H_2O}$ *(1/3, AI 2013)*

84. State and illustrate the following :
Coupling reaction
(1/2, AI 2013C, Foreign 2011)

85. How is the following conversion carried out :
Aniline to *p*-hydroxyazobenzene.
(1/2, Delhi 2012C)

86. How will you bring about the following conversion :
Nitrobenzene to phenol *(1/2, Delhi 2011C)*

87. How will you bring about the following conversion : Aniline to chlorobenzene
Write the chemical equation involved.
(1/2, Delhi 2011C)

88. How will you bring about the following conversion :
Aniline to benzonitrile. *(Delhi 2011C)*

SA (2 marks)

89. (a) Write the reaction involved in the following : Diazotisation
(b) Give reason :
Aromatic diazonium salts are more stable than aliphatic diazonium salts.
(2/5, 2018)

90. Write chemical equations for the following conversions :
(i) Nitrobenzene to benzoic acid.
(ii) Aniline to benzyl alcohol.
(2/3, Delhi 2012)

91. Illustrate the following with an example of reaction in each case :
(i) Sandmeyer reaction
(ii) Coupling reaction.
(AI 2012C, Delhi 2011C)

LA I (3 marks)

92. Write the structures of main products when benzene diazonium chloride reacts with the following reagents :
(i) CuCN (ii) CH_3CH_2OH
(iii) KI *(Delhi 2019)*

93. Write the structures of compounds *A*, *B* and *C* in the following reactions :

(a) $CH_3—COOH \xrightarrow{NH_3/\Delta} A \xrightarrow{Br_2/KOH_{(aq)}} B \xrightarrow{CHCl_3 + \text{alc. KOH}} C$

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(b) $C_6H_5N_2^+BF_4^- \xrightarrow[\Delta]{NaNO_2/Cu} A \xrightarrow{Fe/HCl} B \xrightarrow{CH_3COCl/pyridine} C$

(AI 2017)

94. Give the structure of *A*, *B* and *C* in the following reactions :

(i) $C_6H_5N_2^+Cl^- \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B \xrightarrow[\Delta]{NH_3} C$

(ii) $C_6H_5NO_2 \xrightarrow{Sn + HCl} A \xrightarrow[273\ K]{NaNO_2 + HCl} B \xrightarrow[\Delta]{H_2O/H^+} C$

(Delhi 2013)

LA II (5 marks)

95. An aromatic compound '*A*' of molecular formula C_7H_7ON undergoes a series of reactions as shown below. Write the structures of *A*, *B*, *C*, *D* and *E* in the following reactions :

$(C_7H_7ON)\ A \xrightarrow{Br_2 + KOH} C_6H_5NH_2$

$A \xrightarrow{CH_3CH_2OH} C$

$C_6H_5NH_2 \xrightarrow[273\ K]{NaNO_2 + HCl} B$

$B \xrightarrow{KI} E$

$C_6H_5NH_2 \xrightarrow{CHCl_3 + NaOH} D$

(Delhi 2015)

96. (i) Write the structures of main products when benzenediazonium chloride ($C_6H_5N_2^+Cl^-$) reacts with the following reagents :

(a) HBF_4/Δ (b) Cu/HBr

(ii) Write the structures of *A*, *B* and *C* in the following reactions :

(a) $C_6H_5NO_2 \xrightarrow{Sn + HCl} A \xrightarrow[273\ K]{NaNO_2 + HCl} B \xrightarrow[\Delta]{H_2O} C$

(b) $CH_3Cl \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[273\ K]{HNO_2} C$

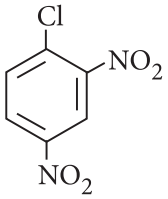
(Foreign 2015)

Detailed Solutions

1. $(H_3CH_2C)_2N—CH_3$

IUPAC name : *N*-Ethyl-*N*-methylethanamine

2.



2, 4-Dinitrochlorobenzene

3. *N*-Methylpropan-2-amine (2° amine)

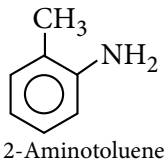
4. $(H_3C)_2N—CH_2CH_3$

N,*N*-Dimethylethanamine

5. 2, 4, 6-Tribromoaniline

6. $CH_3CH_2NHCH_3$

7.



2-Aminotoluene

8. (a) But-3-*en*-2-amine

(b) *N*-Phenylethanamide

9. The ammonolysis of alkyl halides with ammonia is a nucleophilic substitution reaction in which ammonia acts as a nucleophile by donating the electron pair on nitrogen atom to form primary amine as the initial product. Now, the primary amine can act as a nucleophile and combine with alkyl halide (if available) to give secondary amine and the reaction continues in the same way to form tertiary amine and finally quaternary ammonium salt. Thus, a mixture of products is formed and it is not possible to separate individual amines from the mixture.

$R — X \xrightarrow[-HX]{NH_3} R — NH_2 \xrightarrow[-HX]{R-X} R_2NH \xrightarrow[-HX]{R-X} R_3N \xrightarrow{R-X} R_4N^+X^-$

10.

$CH_3COOH \xrightarrow[\Delta]{NH_3} \underset{(A)}{CH_3CONH_2} \xrightarrow{NaOBr} \underset{(B)}{CH_3NH_2}$

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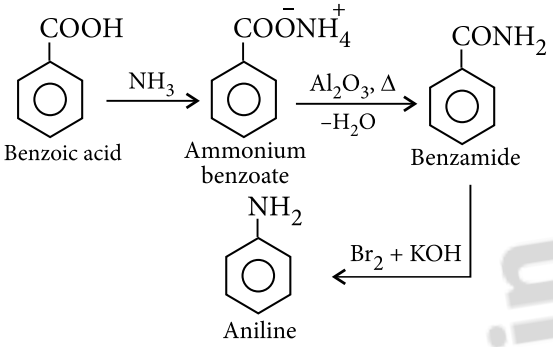
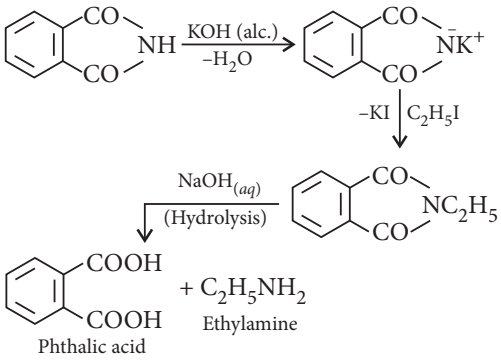
11.



12. $R—CONH_2 + Br_2 + 4NaOH \longrightarrow$
Acid amide

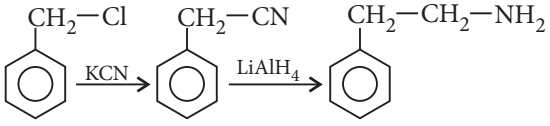
$R—NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$
1° amine

13. Gabriel phthalimide synthesis : In this reaction phthalimide is converted into its potassium salt by treating it with alcoholic potassium hydroxide. Then potassium phthalimide is heated with an alkyl halide to yield an *N*-alkylpthalimide which is hydrolysed to phthalic acid and primary amine by alkaline hydrolysis.



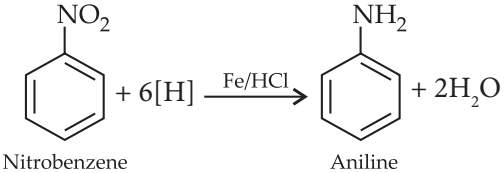
14. $CH_3CN + 4[H] \xrightarrow{Na + C_2H_5OH} CH_3CH_2NH_2$
Ethanenitrile — Ethanamine

15.



16. Primary aromatic amines cannot be prepared by Gabriel phthalimide synthesis because aryl halides do not undergo nucleophilic substitution with the anion formed by phthalimide.

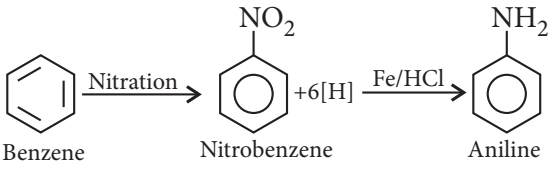
17. (i)



(ii) $CH_3COOH \xrightarrow{PCl_5} CH_3—\overset{O}{\overset{\|}{C}}—Cl \xrightarrow{NH_3} CH_3—\overset{O}{\overset{\|}{C}}—NH_2 \xrightarrow{Br_2/KOH} CH_3—NH_2$
Ethanoic acid (Acetic acid) — Methanamine

18. (i) $CH_3CH_2Cl \xrightarrow{alc.\ NaCN} CH_3CH_2C \equiv N \xrightarrow[Ni/H_2]{Reduction} CH_3CH_2CH_2NH_2$
Chloroethane — Propane nitrile — 1-Propanamine

(ii)



19. CH_3OH has higher boiling point than CH_3NH_2. The hydrogen bonding in alcohols is stronger than that of amines because oxygen is more electronegative than nitrogen.

20. Decreasing order of boiling points of given compounds :
Butanol > butanamine > butane

21. Primary amines (R – NH_2) have two hydrogen atoms on nitrogen which can undergo intermolecular hydrogen bonding whereas no such hydrogen bonding is present in tertiary amines (R_3N). So, primary amines boil at a higher temperature than tertiary amines.

22. *Refer to answer 21.*

23. Increasing order of boiling points :
$(CH_3)_3N < C_2H_5NH_2 < C_2H_5OH$
Tertiary amine does not have hydrogen to form hydrogen bonding and hydrogen bonding in alcohol is stronger than that of primary amines because oxygen is more electronegative than nitrogen.

24. $C_6H_5NH_2 < (C_2H_5)_2NH < C_2H_5NH_2$
1° amines are more soluble in water than 2° amines. Aniline due to large hydrophobic benzene ring is least soluble.

25. *Refer to answer 21.*

26. Ethylamine is soluble in water due to formation of intermolecular hydrogen bonds with water molecules. However, in aniline due to large

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hydrophobic aryl group the extent of hydrogen bonding decreases considerably and hence aniline is insoluble in water.

27. (i) $C_6H_5NH_2 < (CH_3)_2NH < CH_3NH_2$
1° amines are more soluble in water than 2° amines. Aniline due to large hydrophobic benzene ring is least soluble.
(ii) 1° amines have two, 2° amines have one while 3° amines have no hydrogen linked to nitrogen. The degree of association due to hydrogen bonding and hence the boiling point increase as $(C_2H_5)_3N < (C_2H_5)_2NH < C_2H_5NH_2$

28. (b) : The order of basicity of the given compounds is
$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > C_6H_5NH_2$.
There is a subtle interplay of the inductive effect, solvation effect and steric hindrance of alkyl groups which decides the basic strength of alkyl amines in the aqueous state.

29. (**b**) **:** Primary amines when warmed with chloroform and alcoholic solution of KOH produce isocyanides or carbylamines which have foul smell.

30. In case of small alkyl groups like $—CH_3$ the order of basicity is secondary amine > primary amine > tertiary amine due to solvation effect and $+I$ effect of $—CH_3$ group.

	$(CH_3)_2NH$	>	CH_3NH_2	>	$(CH_3)_3N$
pK_b	3.27		3.38		4.22

31. $C_6H_5NO_2 \xrightarrow{Fe + HCl} \underset{(A)}{C_6H_5NH_2} \xrightarrow[0\text{-}5°C]{NaNO_2 + HCl} \underset{(B)}{C_6H_5N_2^+Cl^-}$

32. $C_2H_5NH_2 < (C_2H_5)_2NH < (C_2H_5)_3N$

33. $\underset{\text{Bromomethane}}{CH_3—Br} \xrightarrow{HCN} CH_3—CN \xrightarrow{Ni/H_2}$

$$CH_3CH_2NH_2 \xrightarrow{HNO_2} \underset{\text{Ethanol}}{CH_3CH_2OH}$$

34. In aqueous solution, 2°amine is more basic than 3°amine due to the combination of inductive effect, solvation effect and steric hindrance.

35. $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

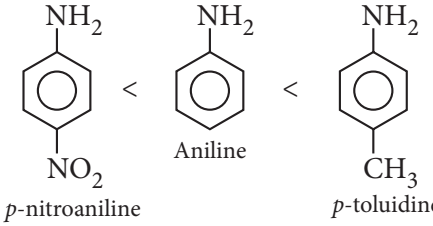
36. Carbylamine reaction is the reaction in which 1° amines produce a bad smelling compound when treated with chloroform in the presence of alkali.

$$RNH_2 + CHCl_3 + 3KOH\ (alc.) \longrightarrow R—N \rightleftarrows C + 3KCl + 3H_2O$$

It is the test for primary amines.

37. When treated with benzenesulphonyl chloride (Hinsberg's reagent), $(CH_3)_2NH$ forms insoluble *N*, *N*-dialkylbenzene sulphonamide which is insoluble in KOH whereas tertiary amine does not react at all.

38.

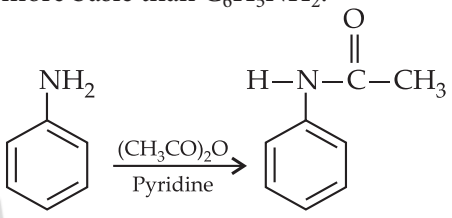


p-nitroaniline Aniline *p*-toluidine

Electron withdrawing group ($–NO_2$) on benzene ring decreases the basicity and electron donating group ($–CH_3$) on benzene ring increases the basicity of compound.

39. $C_6H_5NH_2 < C_6H_5NHCH_3 < C_6H_5CH_2NH_2$
$C_6H_5NH_2$ and $C_6H_5NHCH_3$ are less basic than aliphatic amine $C_6H_5CH_2NH_2$ due to lone pair of nitrogen is in conjugation with benzene ring. But due to $+I$ effect of $—CH_3$ group in $C_6H_5NHCH_3$, it is more basic than $C_6H_5NH_2$.

40.



Aniline *N*-Phenylethanamide (Acetanilide)

41. Increasing order of basic strength in gaseous state is as follows :
$C_6H_5NH_2 < C_6H_5NHCH_3 < C_6H_5N(CH_3)_2$
As the number of $—CH_3$ groups ($+I$ effect) attached to nitrogen increases, its basicity will increase.

42. Methyl amine is more basic than ammonia because of the presence of electron donating methyl group ($+I$ effect), which increases the electron density on nitrogen atom.

43. $CH_3(C_6H_4)\ NH_2$ is more basic than $C_6H_5NH_2$ due to electron releasing nature of methyl group which pushes electrons towards nitrogen.

44. CH_3NH_2 is more basic than $C_6H_5NH_2$ because in aniline the lone pair of electrons on nitrogen are involved in resonance.

45. In case of small alkyl groups like $—CH_3$ the order of basicity is secondary amine > primary

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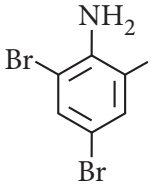
amine > tertiary amine due to solvation effect and +*I* effect of —CH_3 group.

$$(CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$$

46. $C_6H_5NH_2 + Br_{2(aq)} \longrightarrow$  $+ 3HBr$

47. $(C_6H_5)_2NH < C_6H_5NH_2 < C_6H_5N(CH_3)_2 < CH_3NH_2$

48. In aniline, the lone pair of electrons on N-atom is delocalised over benzene ring due to resonance. As a result, electron density on the nitrogen atom decreases. In contrast, in ethylamine, +*I*-effect of —C_2H_5 group increases electron density on the nitrogen atom. Therefore, aniline is a weaker base than ethylamine hence, its pK_b value is more than that for ethylamine.

49. Aniline being an aromatic primary amine on treatment with HNO_2[$NaNO_2$ + HCl (*dil.*)] at 273–278 K followed by treatment with an alkaline solution of β-naphthol gives an orange coloured azo dye. Ethylamine does not give this test.

50. Electron density of N-atom increases due to the +*I* effect of the alkyl group. Hence, alkylamines are stronger bases than ammonia.

$$\ddot{N}H_3, \quad R \rightarrow \ddot{N}H_2$$

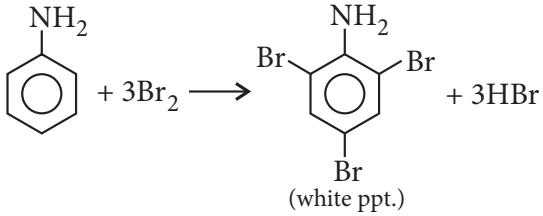
51. $CH_3CH_2NH_2 \xrightarrow[\text{(ii) } H_2O,\ 273\text{-}278\ K]{\text{(i) } NaNO_2/HCl} CH_3CH_2OH \xrightarrow{K_2Cr_2O_7/H^+} CH_3COOH$

52. (i) *Refer to answer 49.*

(ii) Aniline gives carbylamine test, *i.e.,* on treatment with alc. KOH and chloroform followed by heating it gives offensive odour of phenylisocyanide but *N*-methylaniline being secondary amine, does not show this test.

53. (i) When heated with an alcoholic solution of KOH and $CHCl_3$, ethylamine gives foul smelling ethyl isocyanide. Diethylamine does not give this test.

(ii) Aniline gives white or brown precipitate with bromine water.



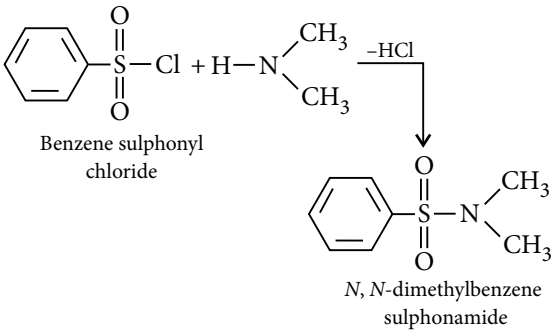
(white ppt.)

Methylamine does not react with bromine water.

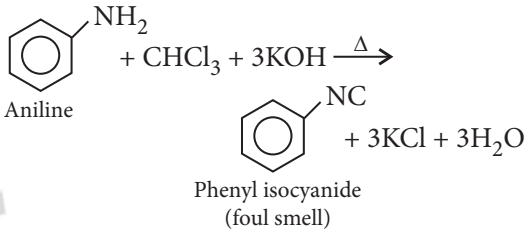
54. (i) Nitration is carried out with conc. HNO_3 in the presence of conc. H_2SO_4. In the presence of these acids, the $-NH_2$ group of aniline gets protonated and is converted into $-\overset{+}{N}H_3$ group. This positively charged group acts as a strong electron withdrawing and *meta*-directing group. Hence, the incoming electrophile goes to *m*-position.

(ii) In aqueous solution, 2°amine is more basic than 3°amine due to the combination of inductive effect, solvation effect and steric reasons.

55. (i) *Refer to answer 40.*

(ii) 

56. (a) Aniline undergoes isocyanide test (carbylamine reaction) whereas, *N*, *N* – dimethylaniline does not.

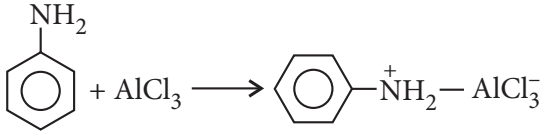


(b) Stronger the base, lower will be the pK_b value.

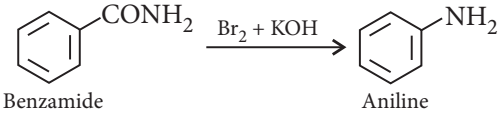
$$C_2H_5NH_2 < C_6H_5NHCH_3 < C_6H_5NH_2$$

pK_b	3.29	4.63	9.38

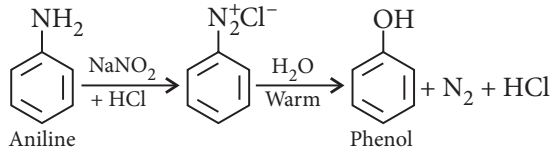
57. (i) In Friedel — Crafts reaction, $AlCl_3$ is added as a catalyst which is a Lewis acid. It forms a salt with aniline due to which the nitrogen of aniline acquires positive charge. This positively charged nitrogen acts as a strong deactivating group, hence aniline does not undergo Friedel — Crafts reaction.



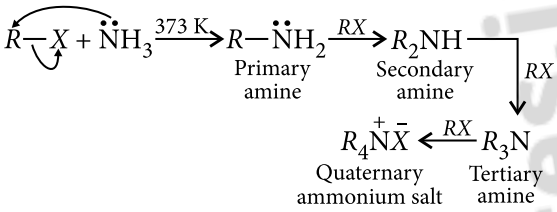
(ii) *Refer to answer 34.*

58. (i) 

Benzamide Aniline

(ii) 

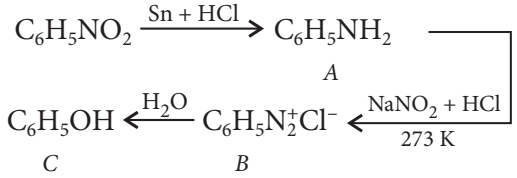
59. (i) **Ammonolysis :** Alkyl halides when treated with ethanolic solution of ammonia give a mixture of primary, secondary, tertiary amines and quaternary ammonium salt.



(ii) **Acetylation of amines :** The process of introducing an acetyl group ($CH_3—\overset{O}{\overset{\|}{C}}—$) into a molecule is called acetylation.

$$CH_3CH_2NH_2 + CH_3—\overset{O}{\overset{\|}{C}}—Cl \xrightarrow{\text{Base}} CH_3—\overset{O}{\overset{\|}{C}}—NHC_2H_5 + HCl$$

N-Ethyl acetamide

60. (i) 

(ii) $CH_3CN \xrightarrow{H_2O/H^+} \underset{A}{CH_3COOH} \xrightarrow{NH_3} \underset{B}{CH_3CONH_2} \xrightarrow{Br_2+KOH} \underset{C}{CH_3NH_2}$

61. (i) *Refer to answer 57(i).*

(ii) In aniline, the lone pair of electrons of N-atom are delocalised over the benzene ring. As a result, electron density on the nitrogen decreases. In contrast, in $CH_3—NH_2$, $+I$ effect of $—CH_3$ group increases the electron density on the N-atom. Therefore, aniline is a weaker base than methylamine and hence, its pK_b value is higher than that of methylamine.

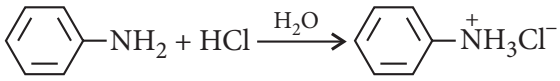
62. (i) Increasing order of basic strength is

$C_6H_5NH_2 < C_6H_5N(CH_3)_2 < CH_3NH_2 < (C_2H_5)_2NH$

(ii) Stronger the base lower will be its pK_b value hence, the decreasing order of pK_b values :

$C_6H_5NH_2 > C_6H_5NHCH_3 > C_2H_5NH_2 > (C_2H_5)_2NH$

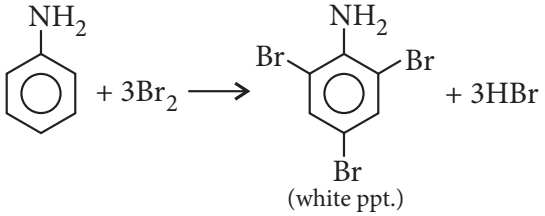
63. (i) $CH_3—CH_2—NH_2 + CHCl_3 + 3KOH \longrightarrow CH_3—CH_2—NC + 3KCl + 3H_2O$

(ii) 

64. (i) *Refer to answer 46.*

(ii) $CH_3CONH_2 \xrightarrow{Br_2 + NaOH} \underset{\text{Methanamine}}{CH_3—NH_2}$

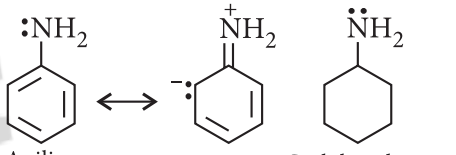
65. (i) Aniline gives white or brown precipitate with bromine water.



Ethylamine does not react with bromine water.

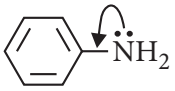
(ii) When heated with an alcoholic solution of KOH and $CHCl_3$, ethylamine gives foul smelling ethyl isocyanide. Dimethylamine does not give this test.

66. (i) Aniline is weaker base than cyclohexylamine because of resonance. Due to electromeric effect, the lone pair on nitrogen is attracted by benzene ring. Hence, donor tendency of $—\ddot{N}H_2$ group decreases. There is no resonance in cyclohexylamine. Electron repelling nature of cyclohexyl group further increases the donor property of NH_2 group. So, cyclohexylamine is a stronger base.



(ii) *Refer to answer 9.*

67. (i) Benzene ring in aromatic amines is highly activated due to the displacement of lone pair of nitrogen towards the ring. It results, increase in the electron density on the ring. This facilitates the electrophilic attack on the ring.



(ii) In CH_3CONH_2, the lone pair of electrons on nitrogen atom is involved in resonance with the

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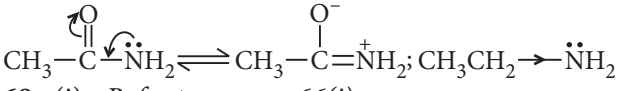
carbonyl group. So, the electron pair of nitrogen is not easily available for protonation. Hence, CH_3CONH_2 is a weaker base than $CH_3CH_2NH_2$.



68. (i) *Refer to answer 66(i).*

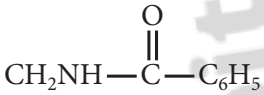
(ii) Methylamine forms hydroxide ions when dissolved in water due to the following acid - base equilibrium.

$$CH_3—NH_2 + H_2O \rightleftharpoons CH_3—\overset{+}{N}H_3 + OH^-$$

These OH^- ions react with Fe^{3+} ions to form ferric hydroxide.

$$2Fe + 6OH^- \longrightarrow 2Fe(OH)_3$$

69. (a) $CH_3NH_2 + C_6H_5COCl \longrightarrow$



N-Methylbenzamide

(b) Stronger the base, lower will be its pK_b value. Hence increasing order of pK_b values is,

$(C_2H_5)_2NH < C_2H_5NH_2 < NH_3 < C_6H_5NH_2$

70. (a) (i) Methylamine gives carbylamine test, *i.e.,* on treatment with alc. KOH and chloroform, followed by heating it gives offensive odour of methyl isocyanide. Dimethylamine does not give this test.

(ii) *Refer to answer 49.*

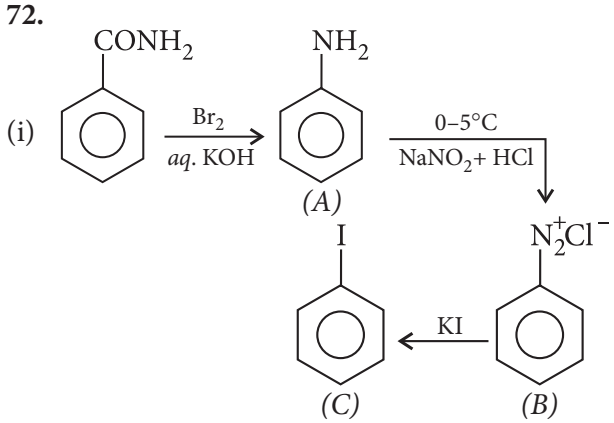
(b) *Refer to answer 57 (i).*

71. (i) After acetylation of aniline, acetanilide is formed in which due to the presence of $—\overset{O}{\overset{\|}{C}}—CH_3$ group having $-I$ effect, electron density on N-atom decreases and hence, activation effect of aniline gets reduced.

(ii) *Refer to answer 44.*

(iii) *Refer to answer 54(i).*

72.



(A) (B) (C)

(ii) $CH_3—Cl \xrightarrow{KCN} \underset{(A)}{CH_3CN} \xrightarrow{LiAlH_4} \underset{(B)}{CH_3CH_2NH_2}$

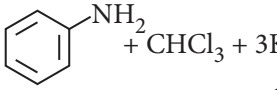
$$\underset{(C)}{CH_3CH_2NC} \xleftarrow[\Delta]{CHCl_3 + alc.\ KOH} CH_3CH_2NH_2$$

73. (i) *Refer to answer 46.*

(ii) *Refer to answer 63(ii).*

(iii) *Refer to answer 40.*

74. (i) *Refer to answer 46.*

(ii)  $+ CHCl_3 + 3KOH \longrightarrow$

Aniline

$+ 3KCl + 3H_2O$

Phenyl isocyanide

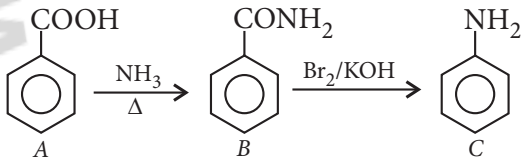
(iii) *Refer to answer 63(ii).*

75. (i) *Refer to answer 60(i).*

(ii) $CH_3Cl \xrightarrow{KCN} \underset{A}{CH_3CN} \xrightarrow{LiAlH_4} \underset{B}{CH_3CH_2NH_2}$

$$\underset{C}{CH_3CH_2OH} \xleftarrow[273\ K]{HNO_2} B$$

76. Formula of the compound '*C*' indicates it is an amine. Since it is obtained by the reaction of Br_2 and KOH with the compound '*B*' so compound '*B*' can be an amide. As '*B*' is obtained from compound '*A*' by reaction with ammonia followed by heating so, compound '*A*' could be an aromatic acid. Formula of compound '*C*' shows it to be aniline, then '*B*' is benzamide and compound '*A*' is benzoic acid. The sequence of reactions can be written as follows :



A B C

77. (i) $CH_3CH_2Br \xrightarrow{KCN} \underset{(A)}{CH_3CH_2CN} \xrightarrow{LiAlH_4}$

$$\underset{(B)}{CH_3CH_2CH_2NH_2} \xleftarrow[HNO_2]{273\ K} \underset{(C)}{CH_3CH_2CH_2OH}$$

(ii) $CH_3COOH \xrightarrow[\Delta]{NH_3} \underset{(A)}{CH_3CONH_2} \xrightarrow{Br_2 + NaOH}$

$$\underset{(B)}{CH_3NH_2} \xleftarrow{CHCl_3 + NaOH} \underset{(C)}{CH_3NC}$$

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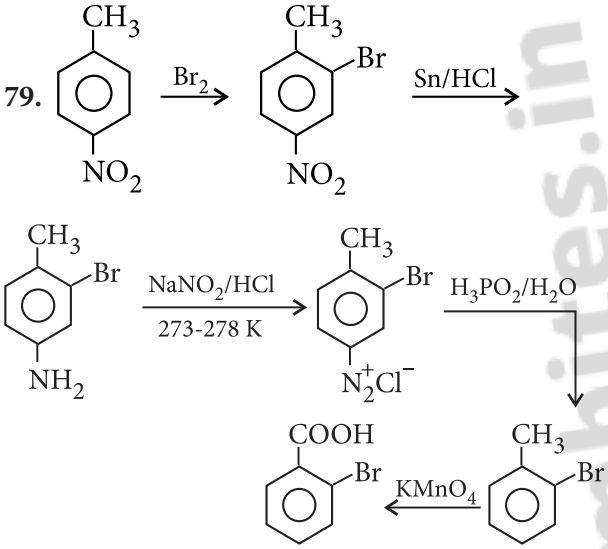
78. $C_6H_5N_2^+Cl^- \xrightarrow{CuCN} C_6H_5CN \xrightarrow{H_2O/H^+} C_6H_5COOH$

Benzonitrile (*A*) — Benzoic acid (*B*)

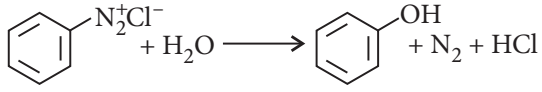
79. 

80. $C_6H_5N_2^+Cl^- + C_2H_5OH \xrightarrow{\text{Reduction}} C_6H_6 + CH_3CHO + N_2 + HCl$

Benzenediazonium chloride — Benzene — Ethanal

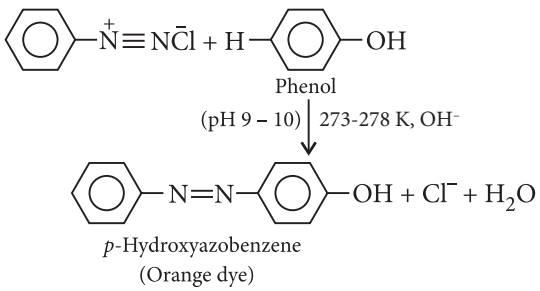
81. $C_6H_5N_2Cl + H_3PO_2 + H_2O \longrightarrow C_6H_6 + N_2 + H_3PO_3 + HCl$

82. Diazotisation reaction.

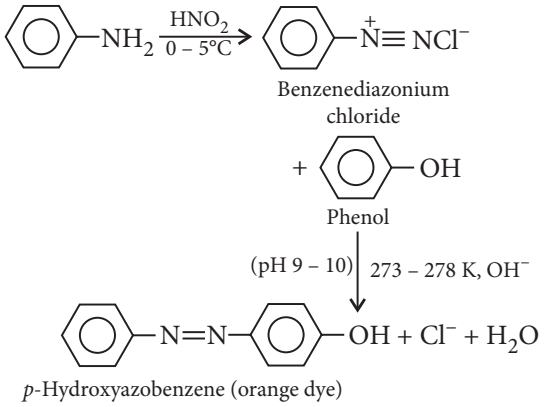
83. 

Phenol

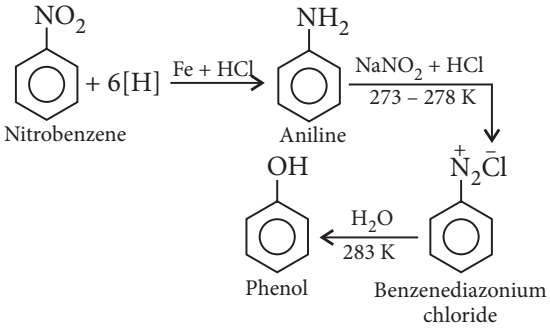
84. Diazonium salts react with aromatic amines in weakly acidic medium and phenols in weakly alkaline medium to form coloured compounds called azo dyes by coupling at *p*-position of amines or phenols.

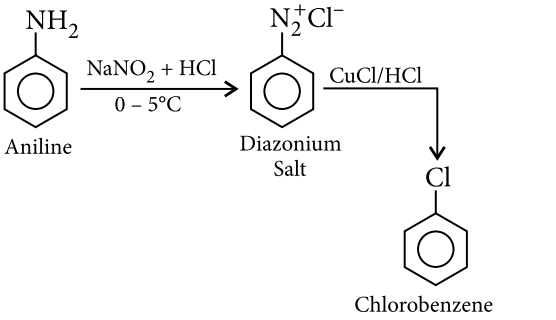


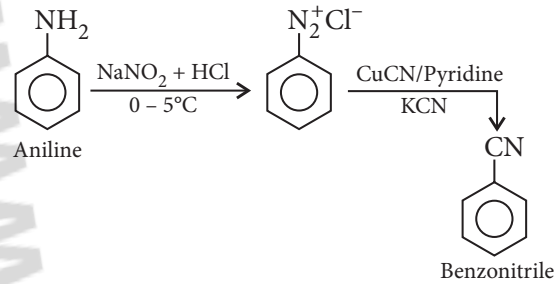
p-Hydroxyazobenzene (Orange dye)

85. 

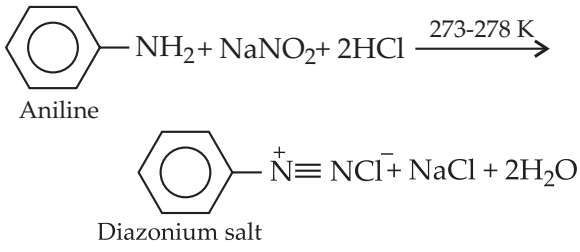
p-Hydroxyazobenzene (orange dye)

86. 

87. 

88. 

89. (a) Diazotisation :



(b) Diazonium salts carry a nitrogen atom with a positive charge. This positive charge is well dispersed in aromatic diazonium salts through resonance as shown below :

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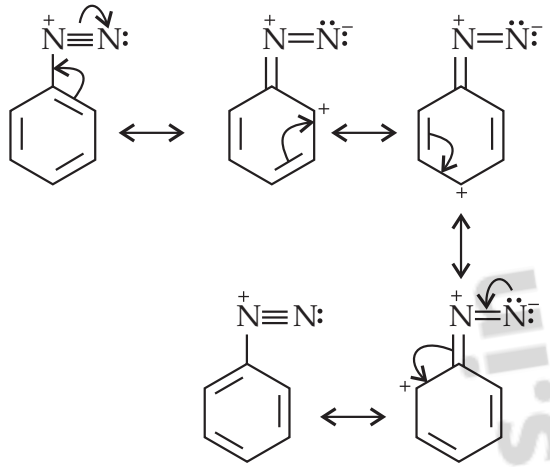
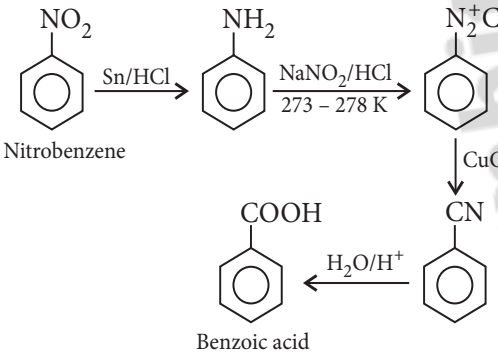
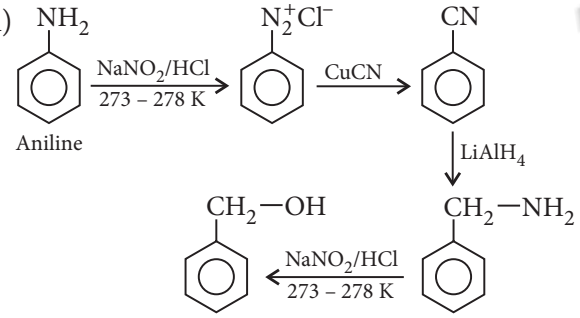


Such a charge delocalisation is not possible in aliphatic diazonium salts and hence they are less stable than aromatic diazonium salts.

90. (i) 

Nitrobenzene

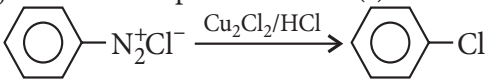
Benzoic acid

(ii) 

Aniline

Benzyl alcohol

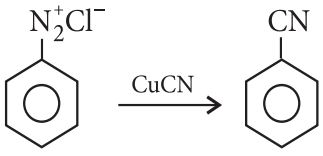
91. (i) **Sandmeyer reaction :** By this reaction nucleophiles like Cl^-, Br^-, CN^- etc. can easily be introduced in the benzene ring by treating the diazonium salt solution with corresponding halogen acid in the presence of Cu(I) ion.



Benzenediazonium chloride

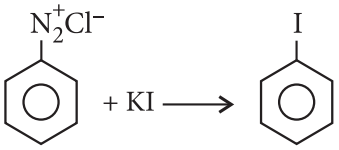
Chlorobenzene

(ii) *Refer to answer 84.*

92. (i) 

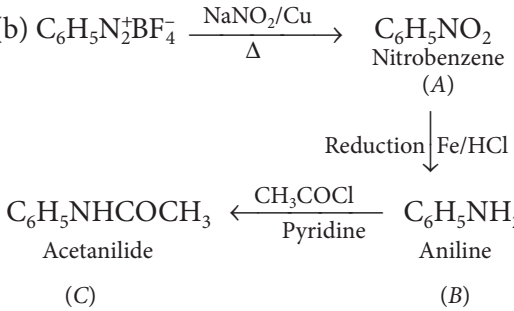
Benzonitrile

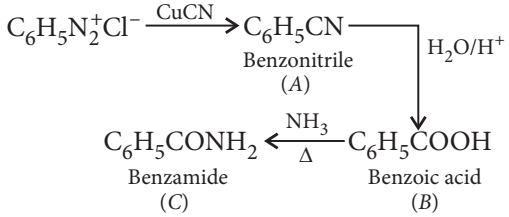
(ii) *Refer to answer 80.*

(iii) 

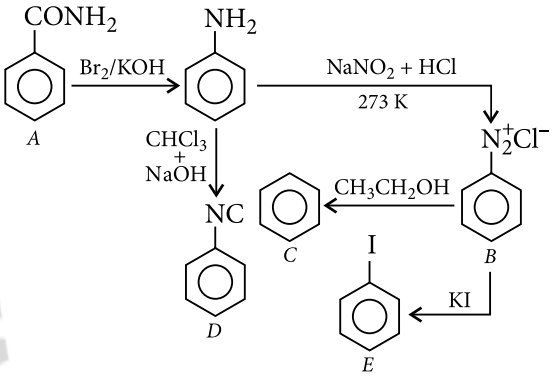
Iodobenzene

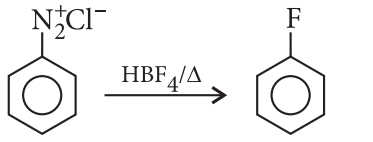
93. (a) *Refer to answer 77 (ii).*

(b) 

94. (i) 

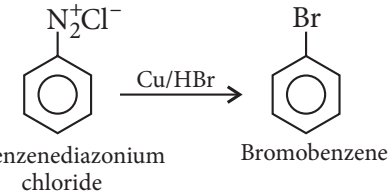
(ii) *Refer to answer 60(i).*

95. 

96. (i) (a) 

Benzenediazonium chloride

Fluorobenzene

(b) 

Benzenediazonium chloride

Bromobenzene

(ii) (a) *Refer to answer 60(i).*

(b) *Refer to answer 75(ii).*

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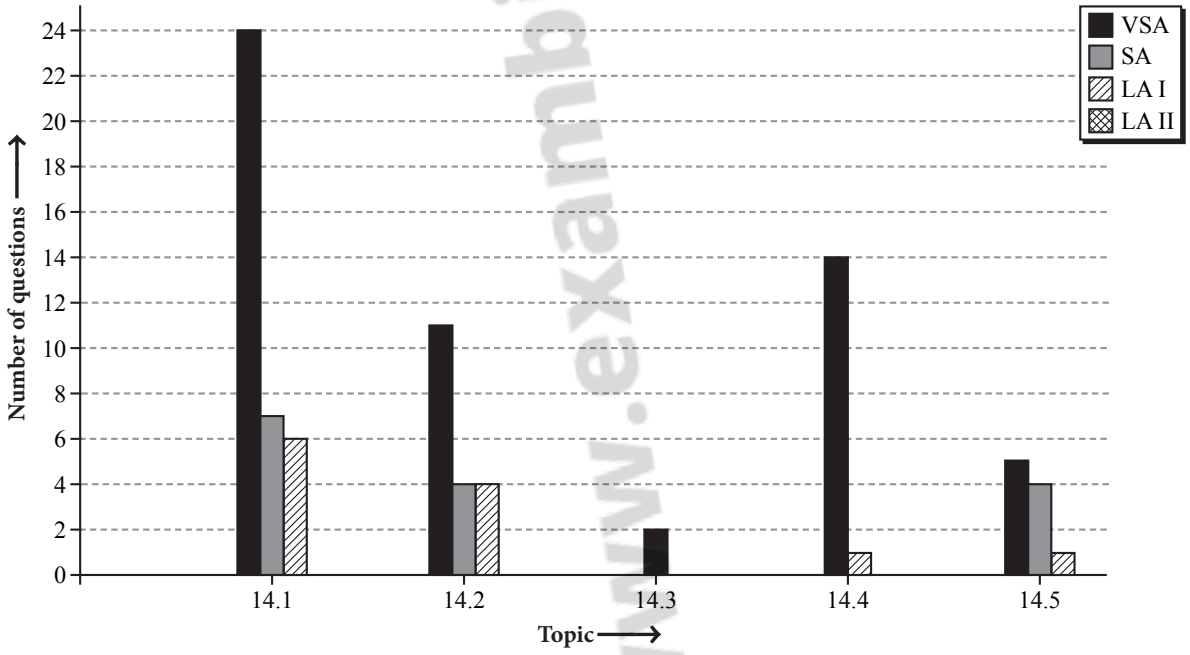
CHAPTER
14

Biomolecules

14.1 Carbohydrates
14.2 Proteins
14.3 Enzymes
14.4 Vitamins
14.5 Nucleic Acids

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Carbohydrates.*
- Maximum VSA type questions were asked from *Carbohydrates.*
- Maximum SA and LA I type questions were asked from *Carbohydrates and Proteins respectively.*

QUICK RECAP

CARBOHYDRATES

- **General formula :** $C_x(H_2O)_y$
- **Sugars or Saccharides :** They are optically active polyhydroxy aldehydes or ketones.

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Classification :

Carbohydrates are classified as either reducing or non-reducing sugars :

Reducing sugars

- Free aldehydic or ketonic group.
- Reduce Fehling's solution and Tollens' reagent
- *e.g.*, maltose and lactose.

Non-reducing sugars

- Do not have free aldehydic or ketonic group.
- Do not reduce Fehling's solution and Tollens' reagent.
- *e.g.*, Sucrose.

▶ On the basis of their behaviour towards hydrolysis :

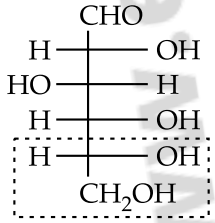
Carbohydrates

Monosaccharides	**Oligosaccharides**	**Polysaccharides**
Cannot be hydrolysed further. $(CH_2O)_n$, where, $n = 3\text{-}7$ *e.g.*, glucose, fructose, ribose, etc.	Yield 2-10 monosaccharides on hydrolysis. $(C_6H_{10}O_5)_n$, where, $n = 2\text{-}10$ *e.g.*, sucrose, maltose, etc.	Yield large number of monosaccharides on hydrolysis. $(C_6H_{10}O_5)_n$, where, $n = 100\text{-}3000$ *e.g.*, starch, cellulose, etc.

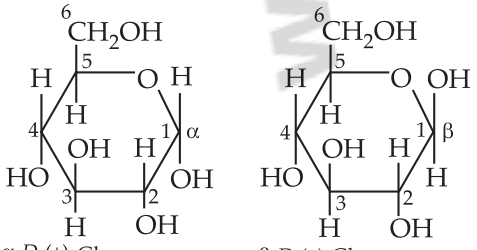
Monosaccharides :

▶ **Glucose ($C_6H_{12}O_6$) :** An aldohexose as it contains six carbon atoms and aldehydic group.

– **Structure :**



Fischer projection of
D-(+)-Glucose



α-*D*-(+)-Glucopyranose β-*D*-(+)-Glucopyranose

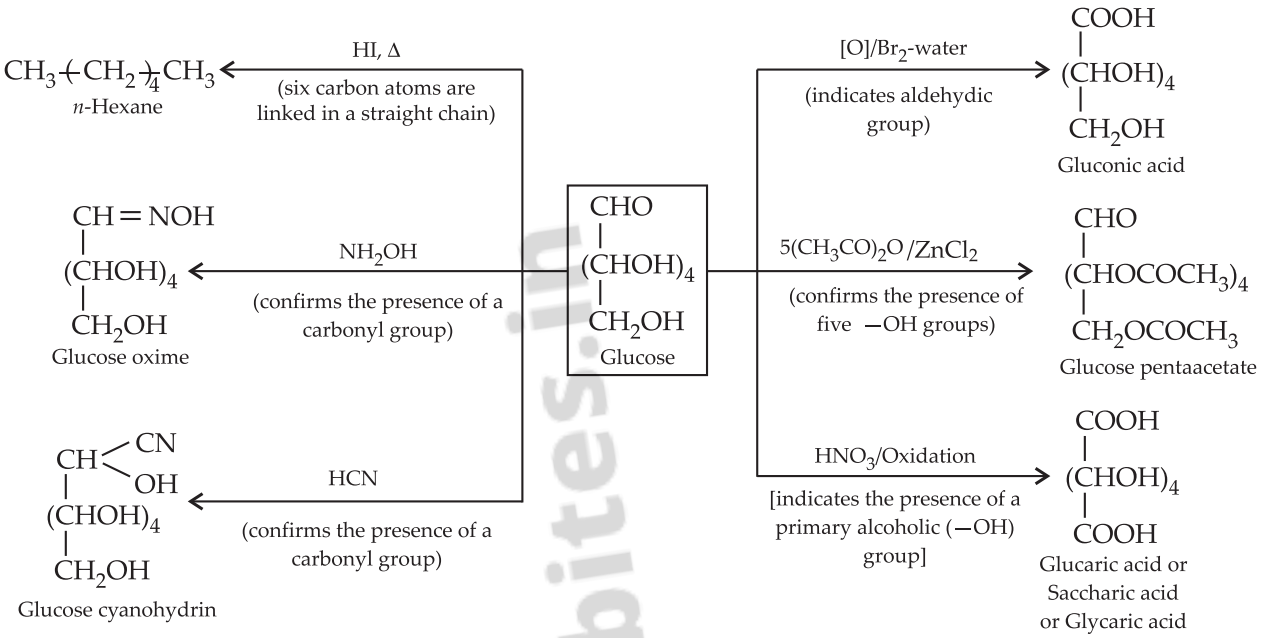
Haworth structures :
(Anomers)

– **Preparation :**

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6} \text{ (\textit{Laboratory method})}$$
Sucrose

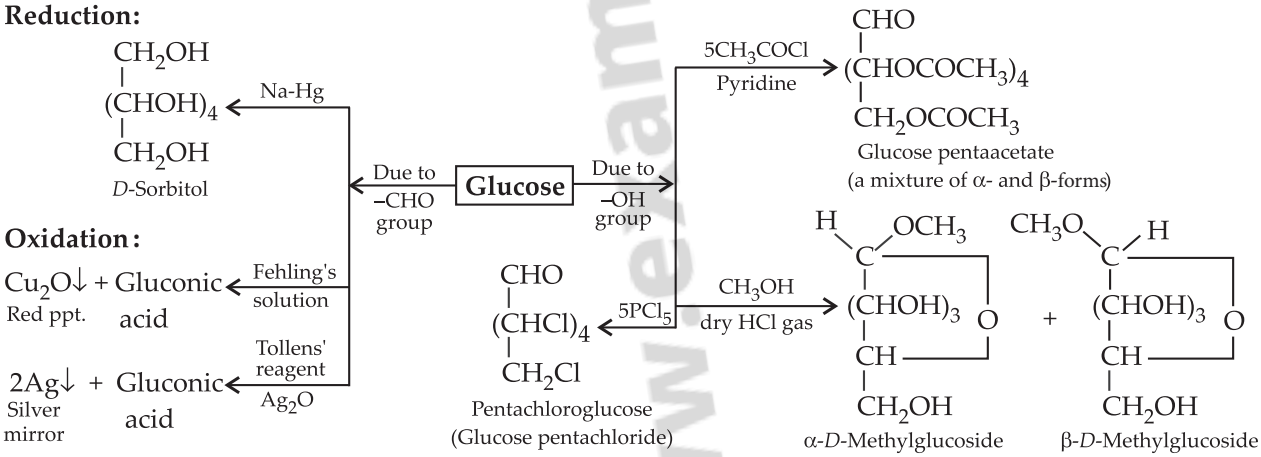
$$\underset{\text{Starch or Cellulose}}{(C_6H_{10}O_5)_n} + nH_2O \xrightarrow[393\text{ K, 2-3 atm}]{H^+/\Delta} \underset{\text{Glucose}}{nC_6H_{12}O_6} \text{ (\textit{Commercial method})}$$

– Glucose was assigned open chain structure on the basis of following evidences :



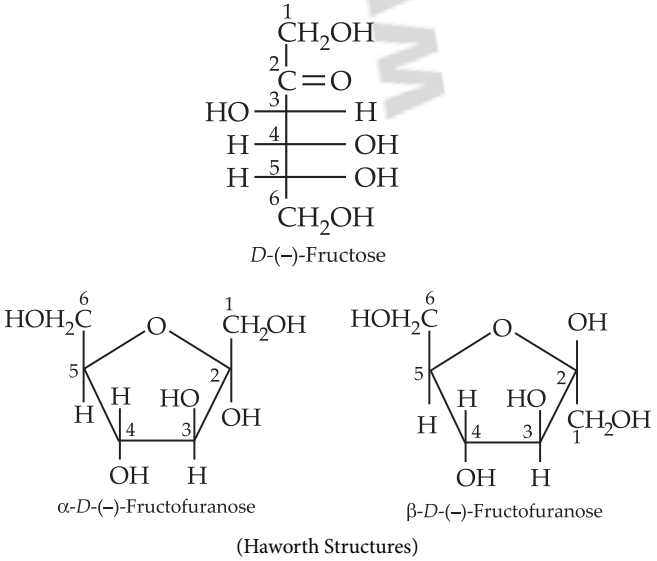
– **Chemical reactions :**

Reduction:



▶ **Fructose ($C_6H_{12}O_6$) :** A ketohexose as it contains six carbon atoms and a ketonic group.

– **Structure :**



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▶ **Mutarotation :** The change in specific rotation of an optically active compound with time to an equilibrium value is called *mutarotation*.

$$\alpha\text{-}D\text{-Glucose} \rightleftharpoons \underset{\text{mixture}}{\text{Equilibrium}} \rightleftharpoons \beta\text{-}D\text{-Glucose}$$

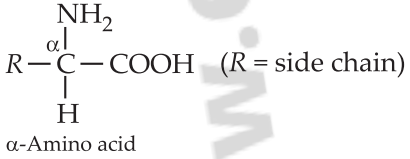
$$[\alpha]_D = +112^\circ \qquad [\alpha]_D = +52.7^\circ \qquad [\alpha]_D = +19^\circ$$

Disaccharides and Polysaccharides

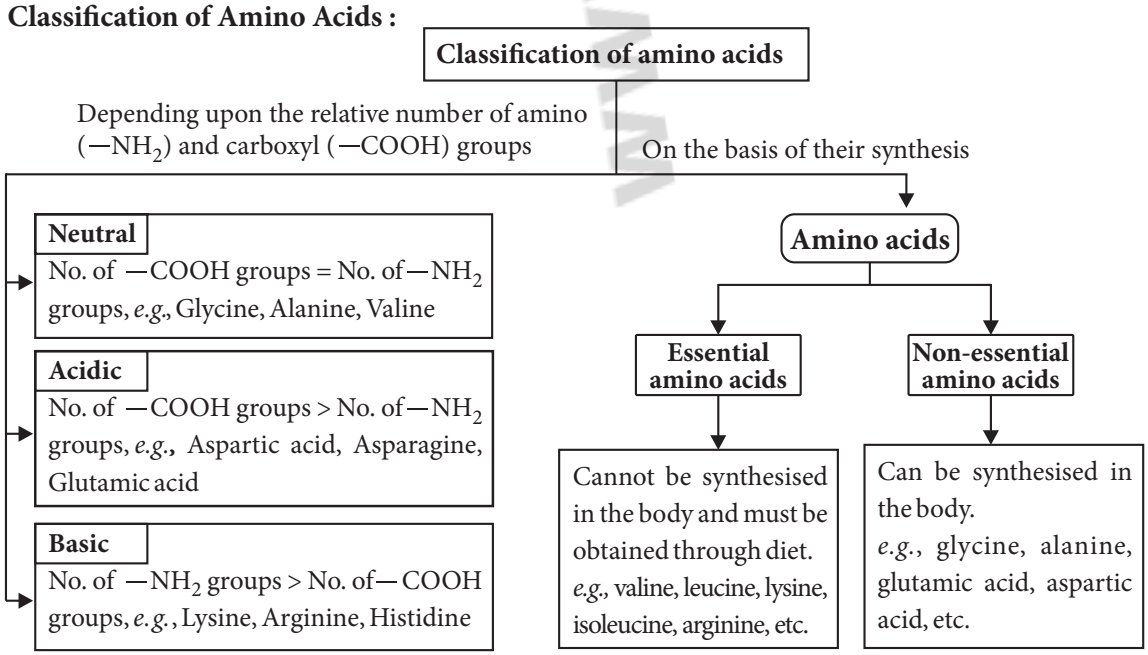
Carbohydrate	Hydrolysis products	Linkage	Reducing property
Sucrose (Disaccharide)	α-*D*-Glucose and β-*D*-Fructose	C-1 (Glucose) and C-2(Fructose)	Non-reducing
Maltose (Disaccharide)	α-*D*-Glucose	C-1 (Glucose) and C-4 (glucose)	Reducing
Lactose (Disaccharide)	β-*D*-Galactose and β-*D*-Glucose	C-1 (Galactose) and C-4 (Glucose)	Reducing
Starch (Polysaccharide)	Amylose and amylopectin	Amylose (C-1 and C-4 glycosidic linkage betweeen α-D-Glucose) Amylopectin (C-1 and C-4 linkage between α-D-Glucose and branching occurs by C-1 and C-6 linkage)	Non-reducing
Cellulose (Polysaccharide)	β-*D*-Glucose	C-1(glucose) and C-4(glucose)	Non-reducing
Glycogen (Polysaccharide)	α-*D*-Glucose	C-1 (glucose) and C-4(glucose)	Non-reducing

PROTEINS

Proteins : They are the biomolecules of the living system made up of nitrogenous organic compounds by condensation polymerisation of α-amino acids.



Classification of Amino Acids :



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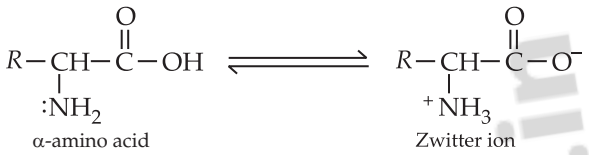
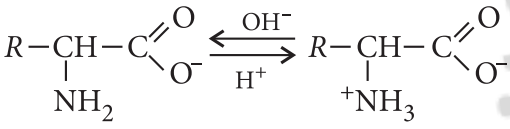
Properties :

- In aqueous solution, the carboxylic group can lose a proton and amino group can accept a proton giving rise to a dipolar ion known as *zwitter ion*. This is neutral but contains both positive and negative charges.

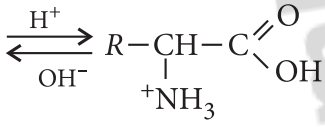


α-amino acid Zwitter ion

- Since these form salts with acids as well as with bases, their chemical reactions are similar to primary amines and carboxylic acids.

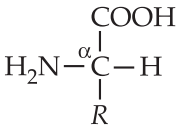


(in alkaline solution) (Neutral form)



(in acidic solution)

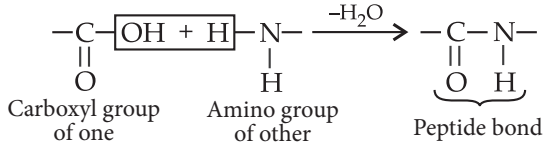
- **Isoelectric point :** The pH at which dipolar ion (zwitter ion) exists as neutral ion, *i.e.*, +ve and –ve charge is equal and it does not migrate to either electrode, is called *isoelectric point*. The amino acids have least solubility in water at isoelectric point which helps in their separation.
- Except glycine, all other naturally occurring α-amino acids are optically active because they contain chiral, asymmetric carbon atom.
- They exist in both *D*- and *L*-forms. Most naturally occurring α-amino acids have *L*-configuration.



L-Amino acid

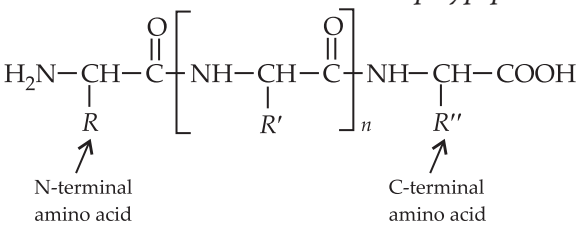
Peptides and their classification :

- **Peptide bond :** The bond formed between two amino acids by the elimination of a water molecule is called a *peptide linkage or bond*.



Carboxyl group of one amino acid — Amino group of other amino acid — Peptide bond

- The products formed by the linking of amino acids by peptide linkage are known as *peptides*.
- Peptides are further divided into *di*, *tri*, *tetra* depending upon the number of amino acids combined.
- **Oligopeptide :** It contains anywhere between 2-10 amino acids.
- **Polypeptides :** Structures with more than ten amino acids are known as *polypeptides*.



N-terminal amino acid — C-terminal amino acid

where, *R*, *R*′, *R*″ may be same or different.

- A polypeptide with more than hundred amino acid residues, having molecular mass higher than 10,000 u is called a *protein*.

Classification of proteins : On the basis of molecular structure, proteins are classified as :

Fibrous proteins

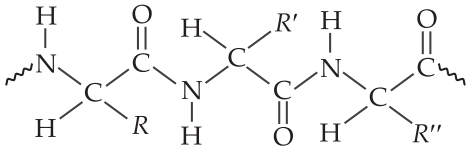
In fibrous proteins, polypeptide chains are parallel and are held together by hydrogen and disulphide bonds. These are insoluble in water, *e.g.*, keratin and myosin.

Globular proteins

Globular proteins result when the polypeptide chains coil around to give three dimensional spherical shape. These are soluble in water, *e.g.*, insulin and albumins.

Structure :

- **Primary structure :** It refers to the number and linear sequence of amino acids held together by peptide bonds.



Primary structure

- **Secondary structure:** It is due to folding or coiling of the peptide chain. It is mainly of two types :
  - **α-helix :** These coils are stabilized by hydrogen bonds between carbonyl oxygen of first amino acid to amide nitrogen of fourth amino acid.

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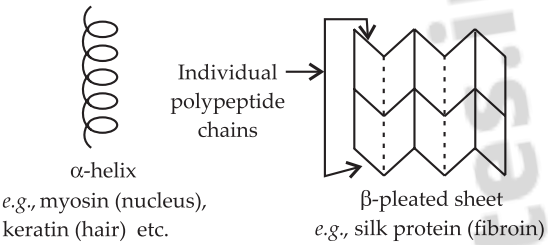
– **β-pleated sheet structure :** β-pleated sheet structure is formed when hydrogen bonds are formed between the carbonyl oxygens and amide hydrogens of two or more adjacent polypeptide chains. The bonding in β-pleated sheet structure is intermolecular H-bonding. The structure is not planar but is slightly pleated. Silk fibroin is rich in β-pleated sheets.



- **Tertiary structure :** It represents overall folding of the polypeptide chains *i.e.*, further folding of the secondary structure and the bonds responsible for such interaction are hydrophobic interactions, hydrogen bonds, ionic interactions, van der Waals' forces and disulphide bonds.
- **Quaternary structure :** The spatial arrangement of the subunits (two or more polypeptide chains) with respect to each other.

Denaturation of proteins :

- When a protein in its native form, is subjected to physical changes like change in temperature or chemical changes like change in pH, the hydrogen bonds are disturbed. Due to this, globules unfold and helix gets uncoiled and protein loses its biological activity. This is called *denaturation of protein*.
- The denaturation causes change in secondary and tertiary structures but primary structure remains intact, *e.g.*, coagulation of egg white on boiling, curdling of milk, formation of cheese, when an acid is added to milk.

ENZYMES

The enzymes are *biocatalysts* produced by living cells which catalyse biochemical reactions in living organisms. Chemically, enzymes are naturally occurring simple or conjugated proteins. Some enzymes may be non-proteins also.

HORMONES

Hormones : They are the molecules that act as intercellular messengers and are poured directly in the blood stream by endocrine glands.

Types of hormones :

- **Steroids :** Estrogens and androgens
- **Polypeptides :** Insulin and endorphins
- **Amino acid derivatives :** Epinephrine and norepinephrine.

VITAMINS

Vitamins: These are complex organic molecules which cannot be produced by the body and must be supplied in small amounts in diet to carry out essential metabolic reactions which are required for normal growth and maintenance of the body.

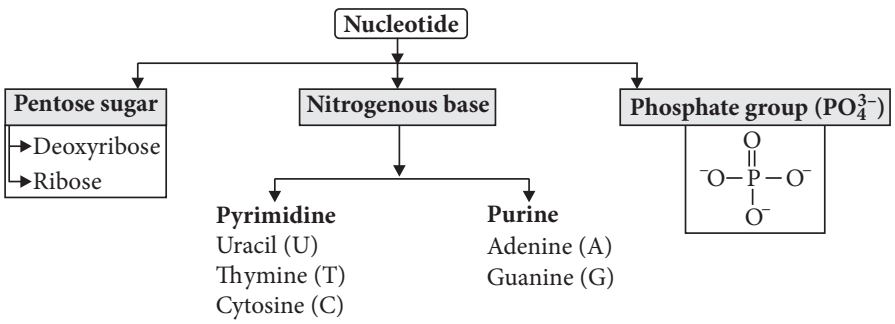
Classification :

- **Water soluble vitamins :** Soluble in water. Must be supplied regularly in diet as they are regularly excreted in urine (except vitamin B_{12}), *e.g.*, Vitamin– B_1, B_2, B_6, B_{12} and C.
- **Fat soluble vitamins :** Soluble in fat and oils. Stored in liver and adipose tissues, *e.g.*, Vitamin – A, D, E and K.

Deficiency of more than one vitamin in the body causes *avitaminosis* while excess intake of vitamins (A and D) may cause *hypervitaminoses*.

NUCLEIC ACIDS

Nucleic acids are the polymers of nucleotides present in nucleus of all living cells and play an important role in transmission of the hereditary characteristics and biosynthesis of proteins.



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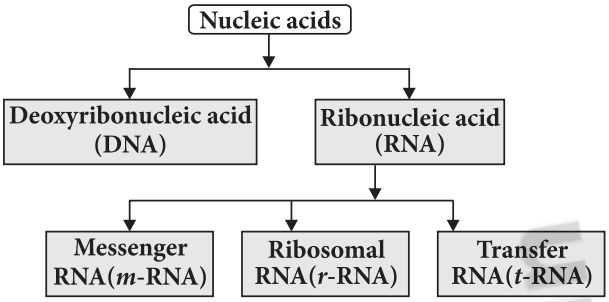
Types of nucleic acids :



Chargaff's rule : Amount of purine bases is always equal to that of pyrimidine bases. Purine base of one strand of DNA molecule pairs with pyrimidine base of the other strand. Adenine (A) pairs with thymine (T) through two H-bonds (A = T) and guanine (G) pairs with cytosine (C) through three H-bonds (G ≡ C). In case of RNA, adenine (A) pairs with uracil (U), (A = U).

Replication : It is the process by which a single DNA molecule produces two identical copies of itself.

Protein synthesis : It occurs in two steps :

- **Transcription :** It is the process of synthesis of RNA.
- **Translation :** The synthesis of proteins occur in the cytoplasm of the cell. The *m*-RNA directs protein synthesis with the help of *r*-RNA and *t*-RNA.

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Previous Years' CBSE Board Questions

14.1 Carbohydrates

VSA (1 mark)

1. Name the disaccharide which on hydrolysis gives two molecules of glucose. *(One word, 2020)*

2. Write the name of linkage joining two monosaccharides. *(One word, 2020)*

3. α–D(+) glucose and β–D(+) glucose are
(a) geometrical isomers
(b) enantiomers
(c) anomers
(d) optical isomers *(2020)*

4. **Assertion (A) :** Sucrose is a non-reducing sugar.
Reason (R) : Sucrose has glycosidic linkage.
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement. *(2020)*

5. What is the basic structural difference between glucose and fructose? *(Delhi 2019)*

6. Write the products obtained after hydrolysis of lactose. *(Delhi 2019)*

7. Define the following with an example of :
Polysaccharides *(1/3, 2018)*

8. Write the product when *D*-glucose reacts with conc. HNO_3. *(1/3, 2018, AI 2012)*

9. Write the name of two monosaccharides obtained on hydrolysis of lactose sugar. *(Delhi 2016)*

10. Write the structural difference between starch and cellulose. *(AI 2016)*

11. Which one of the following is a disaccharide : Starch, Maltose, Fructose, Glucose? *(Delhi 2015)*

12. Write the product obtained when *D*-glucose reacts with $H_2N—OH$. *(AI 2015)*

13. Which one of the following is a monosaccharide :
starch, maltose, fructose, cellulose *(Foreign 2015)*

14. Which of the two components of starch is water soluble? *(Delhi 2014)*

15. Write the product formed on reaction of *D*-glucose with Br_2 water. *(Delhi 2014)*

16. Write the product formed when glucose is treated with HI. *(Delhi 2014)*

17. Define the following term :
Anomers *(AI 2014, Foreign 2014)*

18. Define the following term :
Polysaccharides *(Foreign 2014)*

19. Define the following term : Invert sugar *(Foreign 2014)*

20. What is a glycosidic linkage? *(Delhi 2013)*

21. Name two components of starch. *(Delhi 2013C)*

22. Write a reaction which shows that all the carbon atoms in glucose are linked in a straight chain. *(AI 2012)*

23. State two functions of carbohydrates. *(AI 2012C)*

24. Explain what is meant by the following :
Pyranose structure of glucose? *(AI, Foreign 2011)*

SA (2 marks)

25. Define the following terms :
(i) Oligosaccharides
(ii) Invert sugar *(2020)*

26. Write the reactions showing the presence of following in the open structure of glucose :
(i) an aldehyde group
(ii) a primary alcohol *(2020)*

27. Enumerate the reactions of glucose which cannot be explained by its open chain structures. *(Delhi 2014C, AI 2011)*

28. Write any two reactions of glucose which cannot be explained by the open chain structure of glucose molecule. *(Delhi 2012)*

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29. Write down the structures and names of the products formed when *D*-glucose is treated with (i) Hydroxylamine (ii) Acetic anhydride. *(AI 2012C)*

30. Write down the structures and names of the products formed when *D*-glucose is treated with
(i) Bromine water
(ii) Hydrogen iodide (Prolonged heating).
(AI 2012C)

31. What is essentially the difference between α-form and β-form of glucose? Explain.
(Delhi 2011)

LA I (3 marks)

32. Give the plausible explanation for the following :
(a) Glucose doesn't give 2,4-DNP test.
(b) The two strands in DNA are not identical but are complementary.
(c) Starch and cellulose both contain glucose unit as monomer, yet they are structurally different. *(2020)*

33. Write chemical reactions to show that open structure of *D*-glucose contains the following :
(i) Straight chain
(ii) Five alcohol groups
(iii) Aldehyde as carbonyl group. *(Delhi 2019)*

34. What happens when *D*-glucose is treated with the following reagents?
(a) Br_2 water
(b) HCN
(c) $(CH_3CO)_2O$ *(AI 2019)*

35. Define the following terms :
(i) Glycosidic linkage
(ii) Invert sugar
(iii) Oligosaccharides *(AI 2014)*

36. What is essentially the difference between α-glucose and β-glucose? What is meant by pyranose structure of glucose? *(AI 2012)*

37. Mention the structural feature characterising reducing sugar. *(Delhi 2011C)*

14.2 Proteins

VSA (1 mark)

38. Peptide linkage is present in
(a) carbohydrates (b) vitamins
(c) protein (d) rubber. *(2020)*

39. What is the difference between fibrous protein and globular protein?
(1/3, AI, 2017C, Delhi 2015)

40. Give one example each for fibrous protein and globular protein.
(AI 2016, Delhi 2014, AI 2013C)

41. Amino acids show amphoteric behaviour. Why? *(AI 2015)*

42. What is the difference between acidic amino acids and basic amino acids? *(Foreign 2015)*

43. What type of linkage is responsible for the formation of proteins? *(Delhi, Foreign 2014)*

44. Define the following term :
Essential amino acids *(AI 2014)*

45. Define the following term :
Denaturation of proteins *(Foreign 2014)*

46. Define the following term :
Amino acids *(Foreign 2014)*

47. Define a 'peptide linkage'.
(AI 2014C, 2011, Foreign 2011)

48. Where does the water present in the egg go after boiling the egg? *(Delhi 2012C)*

SA (2 marks)

49. Define the following with an example of each :
(i) Denatured protein
(ii) Essential amino acids *(2/3, 2018)*

50. (a) Amino acids show amphoteric behaviour. Why?
(b) Write one difference between α-helix and β-pleated structures of proteins.
(2/3, 2018)

51. Describe what you understand by primary structure and secondary structure of proteins? *(Delhi 2011, Foreign 2011)*

52. Explain what is meant by a peptide linkage.
(Delhi 2011)

LA I (3 marks)

53. Differentiate between the following :
(i) Amylose and Amylopectin
(ii) Peptide linkage and Glycosidic linkage
(iii) Fibrous proteins and Globular proteins.
(Delhi 2019)

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54. Define the following terms as related to proteins :
(i) Peptide linkage
(ii) Primary structure
(iii) Denaturation *(AI 2015, 2014)*

55. What are essential and non-essential amino acids? Give two examples of each. *(AI 2014C, Delhi 2012C)*

56. (a) Give two differences between globular and fibrous proteins.
(b) What change occurs in the nature of egg protein on boiling? *(Delhi 2013C)*

14.3 Enzymes

VSA (1 mark)

57. Define the following term :
Enzymes *(Foreign 2014, AI 2014 C)*

58. What is meant by biocatalysts? *(Delhi 2012)*

14.4 Vitamins

VSA (1 mark)

59. Why vitamin C cannot be stored in our body? *(Delhi 2016)*

60. Write the name of vitamin whose deficiency causes bone deformities in children. *(Delhi 2015)*

61. Write the name of the vitamin whose deficiency causes bleeding of gums. *(Foreign 2015)*

62. Deficiency of which vitamin causes night-blindness? *(Delhi 2014)*

63. Deficiency of which vitamin causes rickets? *(Delhi 2014)*

64. Deficiency of which vitamin causes scurvy? *(Delhi 2014)*

65. Define the following term :
Vitamins *(Foregin 2014)*

66. Why are vitamin A and vitamin C essential for us? *(Delhi 2014C)*

67. Name the deficiency diseases resulting from lack of vitamins A and E in the diet. *(Delhi 2013C)*

68. Name one of the water soluble vitamin which is powerful antioxidant. Give its one natural source. *(Delhi 2013C, AI 2012C)*

69. Name the only vitamin which can be synthesized in our body. Name the disease caused due to the deficiency of this vitamin. *(Delhi 2013C)*

70. How are hormones and vitamins different in respect of their source and functions? *(AI 2013C)*

71. Name the deficiency disease resulting from lack of vitamin A in the diet. *(Delhi 2011C)*

72. The deficiency of which vitamin causes the disease, 'pernicious anaemia'? *(AI 2011C)*

LA I (3 marks)

73. How are vitamins classified? Name the vitamin responsible for the coagulation of blood. *(Delhi 2015C)*

14.5 Nucleic Acids

VSA (1 mark)

74. What is difference between a nucleoside and nucleotide? *(Delhi 2016, 2014C)*

75. What type of linkage is present in nucleic acids? *(AI 2016)*

76. Name of the base that is found in nucleotide of RNA only. *(Delhi 2014)*

77. Define the following term :
Nucleoside *(Foregin 2014)*

78. Mention one important function of nucleic acids in our body. *(AI 2013C)*

SA (2 marks)

79. Write the structural and functional difference between DNA and RNA. *(Delhi 2013C)*

80. Write the main structural difference between DNA and RNA. Of the two bases, thymine and uracil, which one is present in DNA? *(Delhi 2012)*

81. Name the bases present in RNA. Which one of these is not present in DNA? *(Delhi 2011)*

82. Write the main structural difference between DNA and RNA. Of the four bases, name those which are common to both DNA and RNA. *(AI 2011)*

LA I (3 marks)

83. What are the different types of RNA found in cells of organisms? State the functions of each type. *(Delhi 2012C)*

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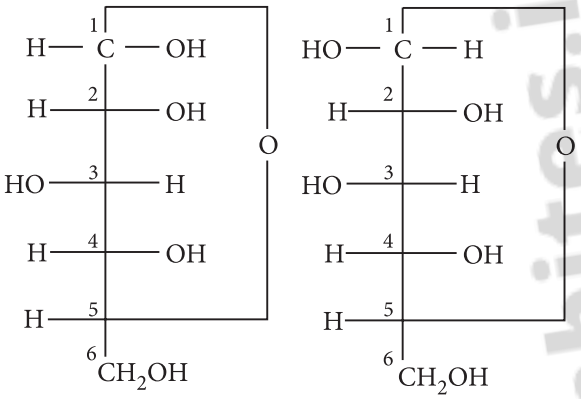
Detailed Solutions

1. Maltose

2. Glycosidic linkage.

3. **(c) :**The pair of stereoisomers which differ only in the configuration of the hydroxyl group at C_1 are called anomers.



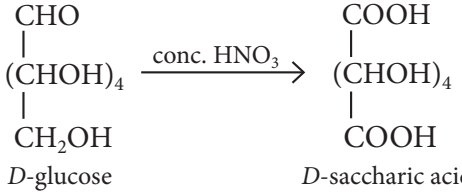
4. **(a) :** Sucrose is disaccharide and its two monosaccharides are held together by a glycosidic linkage. Since the reducing groups of glucose and fructose are involved in glycosidic bond formation, therefore, sucrose is a non-reducing sugar.

5. Glucose contains an aldehydic group while fructose contains a ketonic group.

6. Lactose on hydrolysis gives β-*D*-glucose and β-*D*-galactose.

7. Carbohydrates which yield a large number of monosaccharide units on hydrolysis are called polysaccharides, *e.g.*, cellulose.

8. On oxidation with nitric acid, *D*-glucose yields saccharic acid.

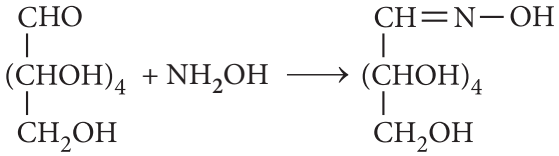


9. *Refer to answer 6.*

10. The basic structural difference between starch and cellulose is of linkage between the glucose units. In starch, there is α-*D*-glycosidic linkage. Both the components of starch-amylose and amylopectin are polymer of α-*D*-glucose. On the other hand, cellulose is a linear polymer of β-*D*-glucose in which C_1 of one glucose unit is connected to C_4 of the other through β-*D*-glycosidic linkage.

11. Maltose is a disaccharide as it consists of two α-*D*-glucose units.

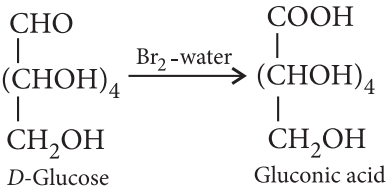
12. *D*-Glucose reacts with $H_2N—OH$ to give glucose oxime.



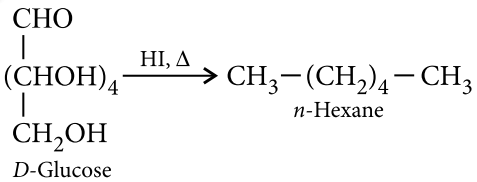
13. Fructose is a monosaccharide because it cannot be hydrolysed to simpler polyhydroxy aldehydes or ketones.

14. Amylose is water soluble and amylopectin is insoluble in water.

15. *D*-Glucose gets oxidised to carboxylic acid (gluconic acid) on reaction with bromine water.



16. On prolonged heating with HI, *D*-glucose forms *n*-hexane.



17. *Refer to answer 3.*

18. *Refer to answer 7.*

19. An equimolar mixture of glucose and fructose, obtained by hydrolysis of sucrose in presence of an acid or the enzyme invertase is called invert sugar.

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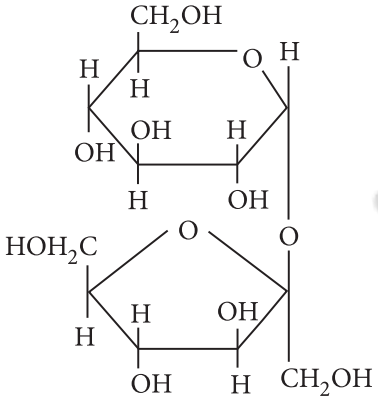
20. The two monosaccharides are joined together by an oxide linkage formed by the loss of water molecule. Such linkage is called glycosidic linkage.



21. Amylose and amylopectin are the two components of starch.

22. Glucose when heated with red P and HI gives *n*-hexane.

$$C_6H_{12}O_6 \xrightarrow[\Delta]{\text{Red P-HI}} CH_3CH_2CH_2CH_2CH_2CH_3$$

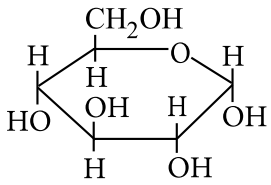
n-Hexane

It indicates the presence of straight chain of six carbon atoms in glucose.

23. (i) Carbohydrates act as storage molecules as starch in plants and glycogen in animals.

(ii) They act as constituent of cell membrane.

24. The six membered cyclic structure of glucose is called pyranose structure (α- or β –), in analogy with heterocyclic compound pyran.

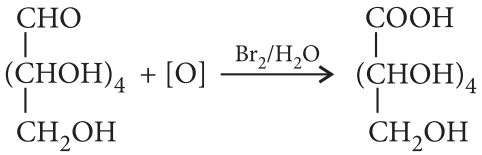


α-*D*-(+)-Glucopyranose

25. (i) These are the carbohydrates which on hydrolysis give 2 - 10 monosaccharides. For example, sucrose, lactose, maltose, etc.

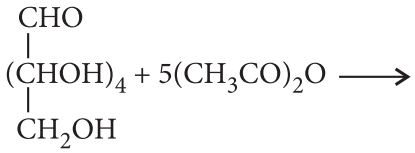
(ii) *Refer to answer 19.*

26. (i) Glucose on oxidation with a mild oxidising agent like bromine water gives gluconic acid containing the same six carbon atoms as present in glucose. This indicates presence of aldehyde group.

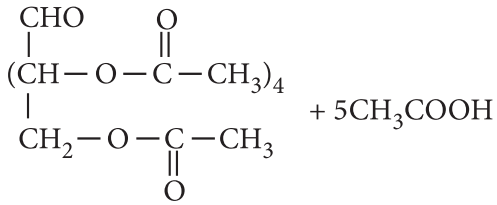


Gluconic acid

(ii) On acetylation with acetic anhydride, glucose gives a pentaacetate. This confirms that glucose contains five – OH groups.



Glucose



Glucose pentaacetate

27. The following reactions of *D*-glucose cannot be explained on the basis of its open chain structure :

(i) *D*-Glucose does not react with sodium bisulphite ($NaHSO_3$).

(ii) It does not give 2, 4-DNP test and Schiff's test.

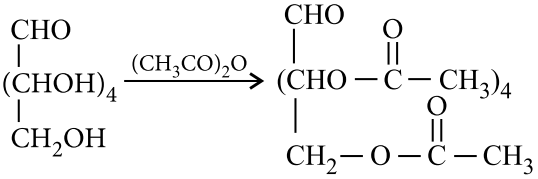
(iii) The pentaacetate of *D*-glucose does not react with hydroxylamine.

(iv) *D*-Glucose shows the phenomenon of mutarotation *i.e.*, when its aqueous solution is kept for sometime its optical activity changes.

(v) On reaction with 1 mole of methanol, it yield two monomethyl derivatives which are known as methyl α-*D*-glucoside and methyl-β-*D*-glucoside.

28. *Refer to answer 27.*

29. (i) *Refer to answer 12.*

(ii) 

Glucose pentaacetate

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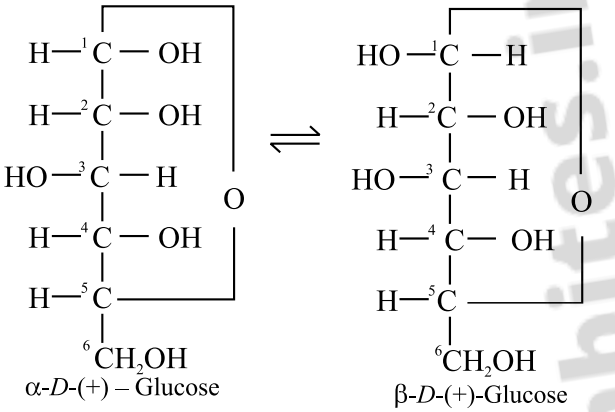
30. (i) *Refer to answer 15.*
(ii) *Refer to answer 16.*

31. In α-*D* Glucose, the –OH group at C_1 is towards right whereas in β-glucose, the –OH group at C_1 is towards left. Such a pair of stereoisomers which differ in the configuration only at C_1 are called anomers.



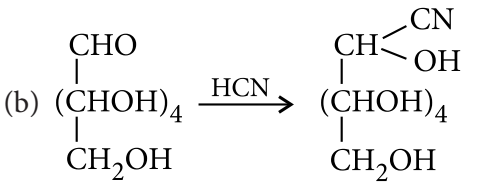
32. (a) Actually, glucose exists in the cyclic hemiacetal form with only a small amount (< 0.05%) of the open chain form. Since, the concentration of the open chain form is low and its reaction with 2,4-DNP is reversible, therefore, formation of 2,4-DNP derivative cannot disturb the equilibrium to regenerate more of the open chain form from the cyclic hemiacetal form and hence, does not give this test.

(b) The two strands in DNA molecule are held together by the hydrogen bonds between purine base of one strand and pyrimidine base of the other and *vice versa*. Because of different sizes and geometries of the bases, the only possible pairing in DNA are G (guanine) and C (cytosine) through three H-bonds, *i.e.*, (C ≡ G) and between A (adenine) and T (thymine) through two H-bonds (*i.e.*, A = T) . Due to this base-pairing principle, the sequence of bases in one strand automatically fixes the sequence of bases in the other strand. Thus, the two strands are complementary and not identical.

(c) The basic structural difference between starch and cellulose is of linkage between the glucose units. In starch, there is α-*D*-glycosidic linkage. Both the components of starch-amylose and amylopectin are polymers of α-*D*-glucose. On the other hand, cellulose is a linear polymer of β-*D*-glucose in which C1 of one glucose unit is connected to C4 of the other through β-*D*-glycosidic linkage.

33. (i) *Refer to answer 22.*
(ii) *Refer to answer 26(ii).*
(iii) *Refer to answer 26(i).*

34. (a) *Refer to answer 15.*

(b) 

Glucose cyanohydrin

(c) *Refer to answer 26(ii).*

35. (i) *Refer to answer 20.*
(ii) *Refer to answer 19.*
(iii) *Refer to answer 25(i).*

36. *Refer to answers 31 and 24.*

37. The sugars which reduce Fehling's solution and Tollens' reagent are called reducing sugars. For example, all monosaccharides containing free $-CHO$ or $-\overset{|}{C}=O$ group are reducing sugars.

38. (c) Peptide linkage is present in proteins.

39. Characteristic differences between globular and fibrous proteins can be given as :

S. No.	Globular proteins	Fibrous proteins
1.	These are cross-linked proteins and are condensation product of acidic and basic amino acids.	These are linear condensation polymer.
2.	These are soluble in water, mineral acids and bases.	These are insoluble in water but soluble in strong acids and bases.

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3.	These proteins have three dimensional folded structure. These are stabilised by internal hydrogen bonding, *e.g.*, egg albumin, enzymes.	These are linear polymers held together by intermolecular hydrogen bonds, *e.g.*, hair, silk.

40. Globular protein – Insulin
Fibrous protein – Keratin

41. As amino acids have both acidic (carboxyl group) and basic groups (amino group) in the same molecule, they react with both acids and bases. Hence, they show amphoteric behaviour.

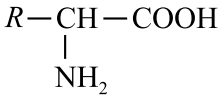
42. Acidic amino acids are those which contain more number of carboxyl groups as compared to amino groups whereas basic amino acids are those which contains more number of amino groups than carboxyl groups.

43. Peptide linkage.

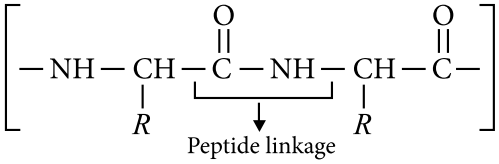
44. Essential amino acids : Amino acids which cannot be synthesized in the body and must be obtained through diet are known as essential amino acids, *e.g.*, valine, leucine, etc.

45. Denaturation : The loss of biological activity of a protein by changing the pH, temperature or by adding some salt due to disruption of the native structure of protein is called denaturation. During denaturation, secondary and tertiary structure of protein is destroyed but primary structure remains intact.

46. Organic compounds containing both amino ($-NH_2$) and carboxyl ($-COOH$) functional groups are called amino acids.



47. Proteins are the polymers of α-amino acids linked by amide formation between carboxyl and amino group. This is called peptide linkage or peptide bond, *e.g.*,



48. An egg contains a soluble globular protein called albumin which is present in the white part. On boiling, denaturation (loss of biological activity) of this protein takes place which results in the formation of insoluble fibrous proteins. The water molecules are utilized in this process.

49. (i) *Refer to answer 45.*
(ii) *Refer to answer 44.*

50. (a) *Refer to answer 41.*
(b) In α-helix structure, intramolecular H-bonding takes place whereas in β-pleated structure, intermolecular H-bonding takes place.

51. Primary structure : The specific sequence in which the various amino acids present in a protein are linked to one another is called its primary structure. Any change in the primary structure creates a different protein.
Secondary structure : The conformation of the polypeptide chain is known as secondary structure. The two types of secondary structure are α-helix and β-pleated sheet structure.
In α-helix structure, the polypeptide chain forms all the possible hydrogen bonds by twisting into a right handed screw (helix) with the — NH groups of each amino acid residue hydrogen bonded to the $\backslash C=O$ group of an adjacent turn of the helix.
In β-pleated sheet structure, all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together by intermolecular hydrogen bonds.

52. *Refer to answer 47.*

53. (i) Amylose is a linear condensation polymer of α-*D*-glucose in which C_1 of one glucose unit is attached to C_4 of the other through α-glycosidic linkage while amylopectin is a highly branched polymer in which C_1 of terminal glucose unit in each chain is further linked to C_6 of some other

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glucose unit in the next chain through C_1-C_6 α-glycosidic linkage.

(ii) Peptide linkage is an amide linkage formed between – COOH group of one α-amino acid and $–NH_2$ group of the other amino acid by loss of a molecule of water whereas a linkage between two monosaccharides units through oxygen atom is called glycosidic linkage.

(iii) *Refer to answer 39.*

54. (i) *Refer to answer 47.*

(ii) *Refer to answer 51.*

(iii) *Refer to answer 45.*

55. Amino acids which cannot be synthesised in the body and must be obtained through diet are known as essential amino acids, *e.g.*, valine and leucine. There are ten essential amino acids. Amino acids which can be synthesised in the body are known as non-essential amino acids, *e.g.*, alanine and glutamic acids.

56. (a) *Refer to answer 39.*

(b) Protein is denatured and its biological activity is lost.

57. Most of the chemical reactions which occur in living systems process at very slow rates under mild condition of temperature and pH. These reactions are catalysed by a group of biomolecules called enzymes.

58. Substances which catalyse chemical reactions taking place in living organisms are called biocatalysts, *e.g.*, enzymes.

59. Vitamin C is soluble in water and regularly excreted in urine and hence cannot be stored in body.

60. Vitamin D

61. Vitamin C

62. Vitamin A

63. Vitamin D

64. Vitamin C

65. Organic compounds required in the diet in small amounts to perform specific biological functions for normal maintenance of optimum growth and health of the organism are called vitamins.

66. The deficiency of vitamin A leads to xerophthalmia and night blindness. The deficiency of vitamin C leads to scurvy.

67. Vitamin – A : Night blindness

Vitamin E : Muscular weakness.

68. Vitamin C is water soluble and powerful antioxidant. Natural source of vitamin C is amla.

69. Vitamin D

Disease caused due to deficiency of Vitamin D is rickets.

70.

	Hormones	Vitamins
(i)	The biomolecules which transfer information from one group of cell to distant tissue or organ.	These are essential dietary factors required by an organism in minute quantities.
(ii)	They are produced in the body in ductless glands.	They are supplied to the body from the food eaten.

71. *Refer to answer 67.*

72. Vitamin B_{12}

73. Vitamins are classified into two groups depending upon their solubility in water or fat.

(i) Fat soluble vitamins, (*e.g.,* vitamin A and D)

(ii) Water soluble vitamins, (*e.g.,* vitamin B and C) vitamin K is responsible for the coagulation of blood.

74. Nucleoside contains pentose sugar, and base whereas nucleotide contains pentose sugar, base as well as phosphate group.

Nucleoside = Base + Sugar

Nucleotide = Base + Sugar + Phosphate

75. Ester linkage

76. Uracil

77. *Refer to answer 74.*

78. DNA is reserve of genetic information and responsible for heredity transmission.

79. Structural differences between DNA and RNA

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(i) The sugar in DNA is deoxyribose while that in RNA is ribose.
(ii) DNA has a double-stranded helical structure, while RNA has a single-stranded helical structure.
Functional differences between DNA and RNA
(i) DNA is the chemical basis of heredity and is responsible for maintaining the identity of different species.
(ii) RNA molecules are responsible for protein synthesis but the message for the synthesis of a particular protein is present in DNA.

80. *Refer to answer 79.*
In DNA, thymine is present.

81. The bases present in RNA are adenine (A), guanine (G) cytosine (C) and Uracil (U).
Uracil is not present in DNA.

82. *Refer to answer 79.*
Common bases in DNA and RNA are adenine, guanine and cytosine.

83. RNA are of three types :
(i) Messenger RNA (*m*-RNA) : Function as messenger carrying the information in a gene to the protein synthesizing machinery.
(ii) Transfer RNA (*t*-RNA) : They transfer the amino acids from cytoplasm to the protein synthesizing machinery.
(iii) Ribosomal RNA (*r*-RNA) : They associates with a set of proteins to form ribosomes. These complex structures, which physically move among an *m*-RNA molecule, catalyze the assembly of amino acids into protein chains. They also bind *t*-RNAs and various molecules necessary for protein synthesis.

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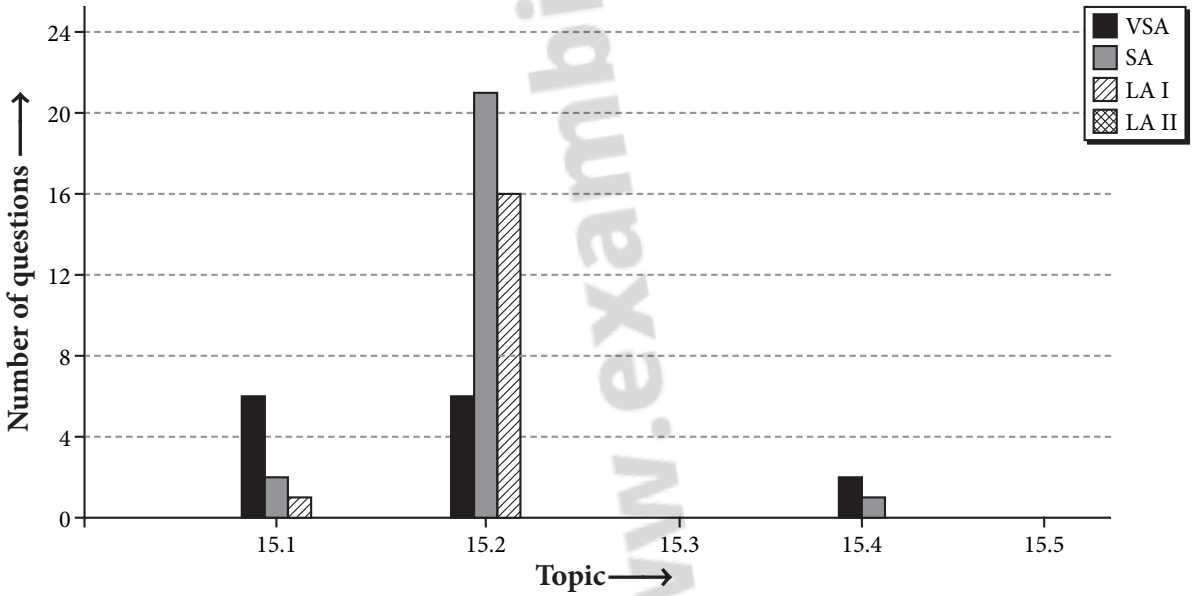
CHAPTER
15

Polymers

15.1 Classification of Polymers
15.2 Types of Polymerisation Reactions
15.3 Molecular Mass of Polymers
15.4 Biodegradable Polymers
15.5 Polymers of Commercial Importance

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Types of Polymerisation Reactions.*
- Maximum VSA type questions were asked from *Classification of Polymers* and *types of Polymerisation Reactions.*
- Maximum SA and LA I type questions were asked from *Types of Polymerisation Reactions.*

QUICK RECAP

- **Polymers :** They are macro-sized, high molecular mass compounds, formed by the combination of a large number of simple molecules or repeating units.
- **Monomers :** The simple molecules or repeating units which combine to give polymers are called *monomers*.
- The process of joining together of a large number of the monomers is termed as *polymerisation*.
- A polymer formed from one type of monomers is called *homopolymer*, *e.g.*, polyethene, PVC, polyacrylonitrile, etc.

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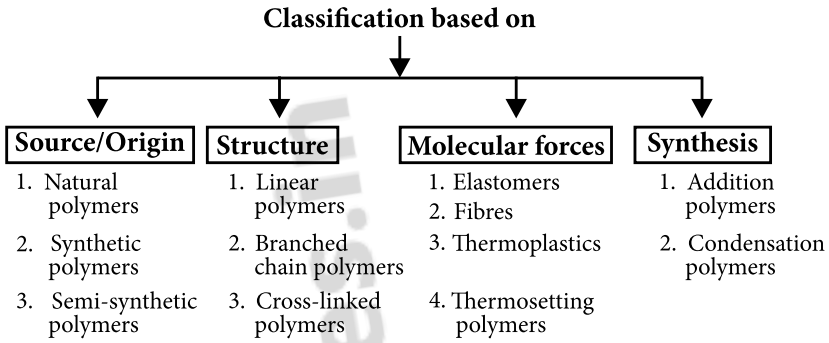
- A polymer formed from two or more different monomers is called *copolymer, e.g.,* nylon-6,6, polyester, bakelite, etc.
- The number of times a monomer unit is repeated in a polymer, is called its *degree of polymerisation*.
- **Classification of polymers :**



- **Natural polymers :** These are substances of natural origin and are mainly found in plants and animals, *e.g.*, starch, cellulose, proteins, etc.
- **Synthetic polymers :** These polymers are prepared in the laboratories, they are also called *man-made polymers*, *e.g.*, teflon, terylene, synthetic rubber, etc.
- **Semi-synthetic polymers :** They are mostly derived from naturally occurring polymers by chemical modifications, *e.g.*, vulcanised rubber, cellulose nitrate, etc.
- **Linear polymers :** In these polymers, the monomers are linked together to form linear chains, *e.g.*, polyethene, polyester, nylon, etc.
- **Branched chain polymers :** In these polymers, the monomers are joined to form long chains or branches of different lengths, *e.g.*, glycogen, starch, etc.
- **Cross-linked polymers :** In these polymers, the monomer units are cross-linked together to form a three-dimensional network polymers, *e.g.*, bakelite, melamine, etc.
- **Elastomers :** These are the polymers having very weak intermolecular forces between the polymer chains. The weak forces permit the polymer to be stretched. Elastomers, thus, possess elastic character, *e.g.*, vulcanised rubber.
- **Fibres :** These are the polymers which have strong intermolecular forces between the chains. These forces are either hydrogen bonds or dipole-dipole interactions, *e.g.*, nylon-6,6.
- **Thermoplastics :** These are the polymers in which the intermolecular forces of attraction are intermediate between those of elastomers and fibres. These polymers do not have any cross-links between the chains, they can be easily moulded on heating *i.e.*, thermoplastics soften on heating and become hard on cooling, *e.g.*, polythene, polystyrene, PVC, etc.
- **Thermosetting polymers :** They have extensive cross-links formed between polymer chains on heating. They undergo a permanent change on heating, *e.g.*, bakelite, melamine, etc.
- **Addition polymerisation :** A polymer formed by direct addition of repeated monomers without the elimination of by-product molecules is called *addition polymer* and the phenomenon is known as *addition polymerisation*, *e.g.*, polythene.
- **Condensation polymerisation :** A polymer formed by the condensation of two or more than two monomers with the elimination of simple molecules like water, ammonia, alcohol, etc. is called *condensation polymer* and the phenomenon is known as *condensation polymerisation*, *e.g.*, terylene.
- **Types of polymerisation reactions :**
- **Addition or chain growth polymerisation :** Depending upon the reactive particles formed, it is of three types :
  - **Free radical polymerisation :** It takes place in three steps :

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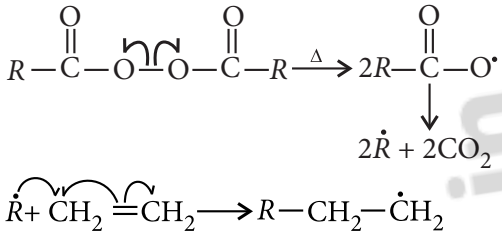
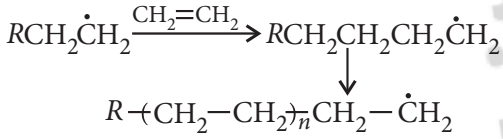
- **Chain initiating step :** This step requires initiators which are free radicals produced on heating, *e.g.*, benzoyl peroxide, acetyl peroxide, dioxygen, etc.

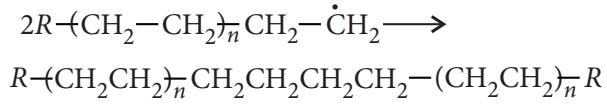


- **Chain propagating step :** The free radical thus formed adds to the double bond of the monomer to form larger free radical.

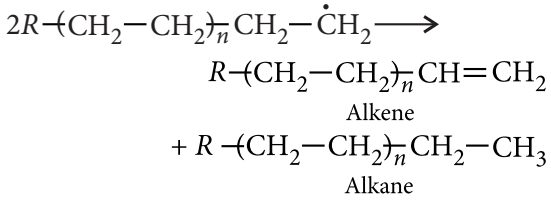


- **Chain terminating step :** The growing free radical chain consumes free radicals either by combination or by disproportionation to get polymer.

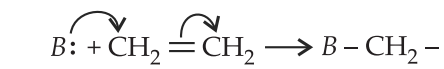
Combination :



Disproportionation :



– **Cationic polymerisation :** Initiated by the use of strong Lewis acids such as HF, $AlCl_3$, H_2SO_4, etc.



e.g., polyvinylether, polyisobutylene, polystyrene, etc.

– **Anionic polymerisation :** Initiated by strong bases such as $NaNH_2$, C_4H_9Li and Grignard reagent, etc.

e.g., polyacrylonitrile, polyvinyl chloride and polymethylmethacrylate, etc.

▶ **Condensation or step growth polymerisation :** In this type of polymerisation reactions, monomers are bifunctional and form bond with the loss of simple molecule of water, alcohol, ammonia, etc.

Differences between LDPE and HDPE :

Polyethylene

Low density polyethylene (LDPE)	High density polyethylene (HDPE)
– Addition or chain growth homopolymer.	– Linear addition or chain growth homopolymer.
– By heating ethylene under high pressure (1000-2000 atm) at temperature of 350-570 K in presence of traces of oxygen or peroxide.	– By heating ethylene at 333-343 K and 6-7 atm in presence of Ziegler-Natta catalyst.
– Free radical addition polymerisation.	– Coordination polymerisation.
– Highly branched polymer.	– Linear molecules, closely packed.
– Low density (0.92 g/cm^3), low melting point (384 K)	– High density (0.97 g/ cm^3), high melting point (403 K).
– Transparent, moderate tensile strength.	– Translucent.
– Chemically inert, tough but flexible.	– Chemically inert, quite harder, greater tensile strength.
– Used for packing, insulation and manufacturing squeeze bottles, pipes, toys, etc.	– Used for manufacturing containers, housewares and pipes, etc.

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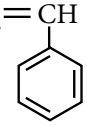
Preparation of some important addition homopolymers :

Polyolefins

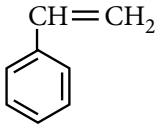
Polymer	Structure of monomers
Polythene	$CH_2=CH_2$ Ethylene
Polypropylene	$CH_3-CH=CH_2$ Propylene
Polytetrafluoroethene (Teflon) (PTFE)	$CF_2=CF_2$ Tetrafluoroethene
Polyacrylonitrile (Orlon) (PAN)	$CH_2=CHCN$ Acrylonitrile (Vinyl cyanide)
Polyvinyl chloride (PVC)	$CH_2=CHCl$ Chloroethene (Vinyl chloride)
Polystyrene (Styron)	$CH_2=CH$ 
Polymethylmethacrylate (PMMA)	$CH_2=C(CH_3)-COOCH_3$ Methylmethacrylate
Polymonochlorotrifluoroethylene (PCTFE)	$Cl-C(F)=CF_2$ Chlorotrifluoroethylene

Polydienes

Polymer	Structure of monomers
Natural rubber (*cis*-1, 4-polyisoprene)	$CH_2=C(CH_3)-CH=CH_2$ Isoprene (2-Methyl-1, 3-butadiene)
Neoprene (synthetic rubber)	$CH_2=C(Cl)-CH=CH_2$ Chloroprene (2-chloro-1, 3-butadiene)
Gutta-percha (*trans*-polyisoprene)	$CH_2=C(CH_3)-CH=CH_2$ Isoprene (2-Methyl-1, 3-butadiene)

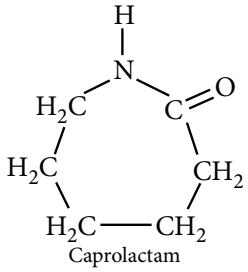
Preparation of some important addition copolymers :

Polydienes

Polymer	Structure of monomers
Buna-S (Styrene Butadiene Rubber) (SBR)	$CH_2=CH-CH=CH_2$ 1,3-Butadiene  Styrene
Buna-N (Nitrile rubber)	$CH_2=CH-CH=CH_2$ 1,3-Butadiene $CH_2=CH-CN$ Acrylonitrile

Preparation of some important condensation homopolymers :

Polyamides

Polymer	Structure of monomers
Nylon-6 (Perlon-L)	 Caprolactam

Polyesters

Polymer	Structure of monomers
Polyhydroxy butyrate (PHB)	$CH_3-CH(OH)-CH_2-COOH$ 3-Hydroxybutanoic acid

Preparation of some important condensation copolymers :

Polyamides

Polymer	Structure of monomers
Nylon-6, 6	$HOOC-(CH_2)_4-COOH$ Adipic acid $H_2N-(CH_2)_6-NH_2$ Hexamethylenediamine
Nylon-6, 10	$HOOC-(CH_2)_8-COOH$ Sebacic acid $H_2N-(CH_2)_6-NH_2$ Hexamethylenediamine
Nylon-2, 6	H_2N-CH_2-COOH Glycine $H_2N-(CH_2)_5-COOH$ 6-Aminohexanoic acid

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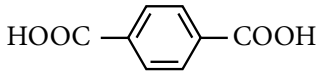
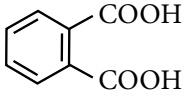
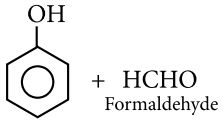
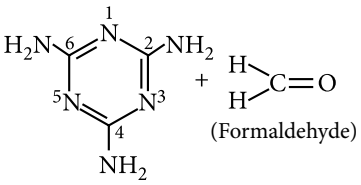
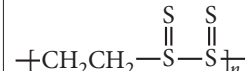
Polyesters

Polymer	Structure of monomers
Terylene (Dacron)	$HOCH_2—CH_2OH$ Ethylene glycol (Ethane-1,2,-diol)  Terephthalic acid (Benzene-1, 4-dicarboxylic acid)
Glyptal (Alkyd resin)	$HOCH_2—CH_2OH$ Ethylene glycol  Phthalic acid (Benzene-1, 2-dicarboxylic acid)
Poly-β-hydroxy butyrate-co-β-hydroxy valerate (PHBV)	$CH_3—\underset{OH}{CH}—CH_2—COOH$ 3-Hydroxybutanoic acid $CH_3—CH_2—\underset{OH}{CH}—CH_2—COOH$ 3-Hydroxypentanoic acid
Poly (glycolic acid) poly(lactic acid) (Dextron)	$HO—CH_2—COOH$ Glycolic acid $HO—\underset{CH_3}{CH}—COOH$ Lactic acid

Formaldehyde resins

Polymer	Structure of monomers
Phenol-formaldehyde resin (Bakelite)	 Phenol + HCHO Formaldehyde
Melamine-formaldehyde resin (Melmac)	 (Formaldehyde) Melamine (2, 4, 6-Triamino-1,3,5-triazine)
Urea-formaldehyde resin	$NH_2CONH_2 + HCHO$ Urea Formaldehyde

Polysulphide

Polymer	Structure of monomers
Thiokol 	$Cl—CH_2—CH_2—Cl$ Ethylene dichloride Na_2S_4 Sodium tetrasulphide

Differences between natural rubber and vulcanized rubber :

	Natural rubber	Vulcanized rubber
1.	Natural rubber is soft and sticky.	Vulcanized rubber is hard and non-sticky.
2.	It has low tensile strength.	It has high tensile strength.
3.	It has low elasticity.	It has high elasticity.
4.	It can be used over a wide narrow range of temperature (from 10°C to 60°C).	It can be used over a wide range of temperature (– 40°C to 100°C).
5.	It has low wear and tear resistance.	It has high wear and tear resistance.
6.	It is soluble in solvents like ether, carbon tetrachloride, petrol, etc.	It is insoluble in all the common solvents.

Molecular mass of polymers

There are two ways of expressing molecular mass of a polymer :

(i) Number average molecular mass ($\bar{M}_n$)

(ii) Weight average molecular mass ($\bar{M}_w$)

(i) **Number average molecular mass ($\bar{M}_n$)** : $\bar{M}_n$ is generally determined by osmotic pressure method. It is obtained by the expression.

$$\bar{M}_n = \frac{\Sigma N_i M_i}{\Sigma N_i}$$

where N_i = number of molecules and M_i = molecules mass

(ii) **Weight average molecular mass ($\bar{M}_w$) :** $\bar{M}_w$ is generally determined by using methods such as light scattering, ultracentrifuge, etc. It is obtained by the expression.

$$\bar{M}_w = \frac{\Sigma N_i M_i^2}{\Sigma N_i M_i}$$

where N_i = number of monomer molecules and M_i = molecular mass.

Biodegradable polymers : Aliphatic polyesters are one of the most important class of biodegradable polymers. Some important examples are given below :

▶ **Poly β-hydroxybutyrate-co-β-hydroxy valerate (PHBV) :** PHBV is used in speciality packaging, orthopaedic devices and in controlled release of drugs. PHBV undergoes bacterial degradation in the environment.

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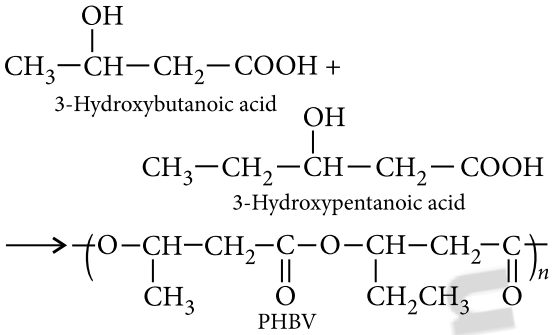
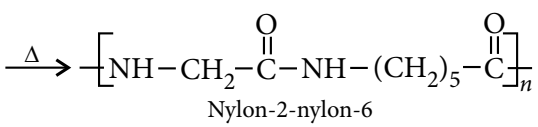


- **Nylon-2-nylon-6 :** It is an alternating polyamide copolymer of glycine (H_2NCH_2COOH) and amino caproic acid [$H_2N(CH_2)_5COOH$] and is biodegradable.



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Previous Years' CBSE Board Questions

15.1 Classification of Polymers

VSA (1 mark)

1. Arrange the following polymers in the increasing order of their intermolecular forces : Polystyrene, Terylene, Buna-S. *(1/3, Delhi 2016)*

2. Arrange the following polymers in the increasing order of their intermolecular forces : Terylene, Polythene, Neoprene. *(1/3, AI 2016)*

3. Which of the following is a natural polymer? Buna-S, Proteins, PVC *(AI 2014)*

4. Based on molecular forces what type of polymer is neoprene? *(AI 2014)*

5. Which of the following is a fibre? Nylon, Neoprene, PVC *(AI 2014)*

6. Explain the following term giving a suitable example : Elastomers *(1/3, AI 2012)*

SA (2 marks)

7. Define thermoplastic and thermosetting polymers. Give one example of each. *(2/3, AI 2013)*

8. How are thermosetting polymers different from thermoplastic polymers? *(2/3, Delhi 2013C, 2012C)*

LA I (3 marks)

9. Differentiate between thermoplastic and thermosetting polymers. Give one example of each. *(AI 2012)*

15.2 Types of Polymerisation Reactions

VSA (1 mark)

10. Give the structure of the monomer of PVC. *(One word , 2020)*

11. Write the structure of the monomers used for getting the following polymer : Nylon-6, 6 *(1/3, AI 2017)*

12. Write the name and structure of the monomers of the following polymer : Buna-S *(1/3, Delhi 2015C, 2013)*

13. Is $+CH_2-\underset{\displaystyle Cl}{\underset{|}{CH}}+_n$ a homopolymer or a copolymer? *(AI 2013)*

14. Name the polymer which is used for making non-stick cooking utensils. *(1/3, Delhi 2013C)*

15. Define the term, 'homopolymerisation' giving an example. *(Delhi 2012)*

SA (2 marks)

16. Write the names and structures of monomers in the following polymers : (a) Bakelite (b) Neoprene *(2020)*

17. Write the name and structures of monomers in the following polymers : (i) Nylon 6, 6 (ii) Terylene *(2/3, 2020)*

18. Classify the following as addition and condensation polymers giving reason : (a) Teflon (b) PHBV *(AI 2019)*

19. Write the structures of monomers of the following polymers. (a) Terylene (b) Buna-N *(AI 2019)*

20. Write the structures of the monomers used for getting the following polymers : (i) Dacron (ii) Buna-N *(2/3, 2018C, Delhi 2017)*

21. Write the structures of the monomers used for getting the following polymers : (i) Nylon-6 (ii) Teflon *(2/3, Delhi 2017)*

22. (i) What is the role of *t*-butyl peroxide in the polymerisation of ethene?
(ii) Identify the monomers in the following polymer :

$$+NH-(CH_2)_6-NH-\overset{\displaystyle O}{\overset{\|}{C}}-(CH_2)_4-\overset{\displaystyle O}{\overset{\|}{C}}+_n$$

(2/3, Delhi 2016)

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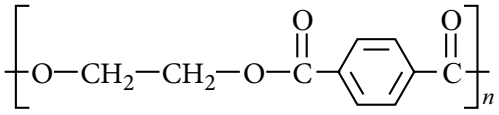
23. (i) What is the role of sulphur in the vulcanisation of rubber?
(ii) Identify the monomers in the following polymer :



(2/3, AI 2016)

24. Write the names and structures of the monomers of the following polymers :
(i) Nylon-6, 6
(ii) Neoprene *(2/3, Delhi 2015, 2013)*

25. Write the names and structures of the monomers of the following polymers :
(i) Neoprene
(ii) Teflon *(2/3, Delhi 2015C)*

26. Explain the term 'copolymerisation' and give two examples of copolymerisation. *(2/3, AI 2015C)*

27. Write the name of monomers used for getting the following polymers :
(i) Bakelite
(ii) Neoprene *(AI 2014)*

28. Write the name of monomers used for getting the following polymers :
(i) Terylene
(ii) Nylon-6,6 *(AI 2014)*

29. Write the name of monomers used for getting the following polymers :
(i) Teflon (ii) Buna-N *(AI 2014)*

30. Give names of the monomers of the following polymers :
(i) Neoprene (ii) Polystyrene
(iii) Polypropene *(Delhi 2014C)*

31. Write the names and structures of monomers used for getting the following polymers :
(i) Buna-S
(ii) Nylon-6, 6 *(AI 2014C)*

32. Distinguish between homopolymers and copolymers. Give one example of each. *(AI 2013)*

33. Draw the structure of the monomer for each of the following polymers.
(i) Nylon-6
(ii) Polypropene *(Delhi 2012)*

34. Explain the following terms giving a suitable example for each :
(i) Condensation polymers
(ii) Addition polymers *(AI 2012)*

35. Write down the structure of monomer and one use of the polymer polystyrene. *(2/3, Delhi 2012C)*

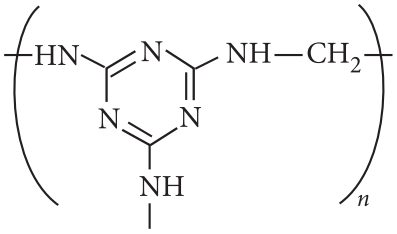
36. Mention two important uses of each of the following :
(i) Bakelite
(ii) Nylon-6 *(2/3, Delhi 2011)*

LA I (3 marks)

37. Write the names and structures of the monomers in the following polymers :
(i) Buna-S
(ii) Nylon-6, 6
(iii) Bakelite *(2020)*

38. Write the structures of monomers used for getting the following polymers :
(i) Nylon-6,6 (ii) Glyptal
(iii) Buna-S. *(Delhi 2019)*

39. (i) Is $+CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}+_n$ a homopolymer or copolymer? Give reason.
(ii) Write the monomers of the following polymer :



(iii) What is the role of sulphur in vulcanization of rubber? *(Delhi 2019)*

40. Write the structures of the monomers used for getting the following polymers :
(i) Neoprene
(ii) Melamine-formaldehyde polymer
(iii) Buna-S *(Delhi 2017)*

41. Write the mechanism of free radical polymerisation. *(Delhi 2016)*

42. Write the names and structures of the monomers of the following polymers :
(i) Buna-S (ii) Glyptal
(iii) Polyvinyl chloride *(AI 2015)*

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43. Write the names and structures of the monomers of the following polymers :
(i) Terylene (ii) Bakelite
(iii) Buna-S *(Foreign 2015)*

44. Write the name of the monomers of the following polymers :
(i) Polythene
(ii) Polyvinyl chloride
(iii) Bakelite *(Delhi 2014C)*

45. Write the names and structures of the monomers of the following polymers :
(i) Bakelite
(ii) Nylon-6
(iii) Polythene *(Delhi 2013)*

46. (a) Differentiate between copolymerisation and homopolymerisation. Give one example of each.
(b) What is the role of benzoyl peroxide in preparation of polythene? *(Delhi 2013C)*

47. Write the monomers of the following polymers and classify them as addition or condensation polymers :
Teflon, Bakelite and Natural rubber *(AI 2013C)*

48. Explain the term copolymerization and give two examples of copolymers and the reactions for their preparations. *(AI 2012C)*

49. Draw the structure of the monomers of the following polymers :
(i) Polythene (ii) PVC
(iii) Teflon *(AI 2011)*

50. What are addition polymers? How are the two types of addition polymers different from each other? Give one example of each type. *(Foreign 2011)*

51. Mention two important uses for each of the following polymers :
(i) Bakelite
(ii) Nylon 6,6
(iii) PVC *(Foreign 2011)*

52. Distinguish between addition polymers and condensation polymers. Classify the following into addition and condensation polymers :
(i) Polythene
(ii) PTFE
(iii) Polybutadiene
(iv) Bakelite *(AI 2011C)*

15.4 Biodegradable Polymers

VSA (1 mark)

53. Write the name and structures of monomers in the following polymers :
PHBV *(1/3, 2020, Delhi 2015)*

54. What are biodegradable polymers? Give one example. *(1/3, AI 2013C, Delhi 2011)*

SA (2 marks)

55. What is a biodegradable polymer? Give an example of a biodegradable aliphatic polyester. *(2/3, AI 2013)*

Detailed Solutions

1. The intermolecular forces are least in case of elastomers like Buna-S while strongest in case of fibres like terylene and in case of thermoplastics like polystyrene the intermolecular forces are intermediate in between elastomers and fibres.

Thus, the increasing order of their intermolecular forces is Buna-S < Polystyrene < Terylene

2. Neoprene < Polythene < Terylene

3. Proteins are natural polymers.

4. Elastomer

5. Nylon is a fibre.

6. Elastomers are the polymers in which polymer chains are held by weakest intermolecular forces. *e.g.,* Buna-S, Buna-N.

7. **Thermoplastics :** Thermoplastics are linear or slightly branched polymers which can be

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repeatedly softened on heating and hardened on cooling and hence can be used again and again without any change in chemical composition and mechanical strength.

Example : Polythene and polystyrene.

Thermosetting Polymers : Thermosetting polymers on heating in a mould get hardened and set and cannot be softned again. This hardening on heating is due to cross-linking between different polymer chains to give a three-dimensional network solid.

Example : Bakelite.

8. *Refer to answer 7.*

9. *Refer to answer 7.*

10. $H_2C{=}CHCl$
(Vinyl chloride)

11. Monomers used for getting nylon–6,6, are $HOOC(CH_2)_4COOH$ (adipic acid) and $H_2N(CH_2)_6NH_2$ (hexamethylenediamine).

12. Buna-S :

$CH_2{=}CH-CH{=}CH_2$ and $C_6H_5CH{=}CH_2$
1,3–Butadiene Styrene

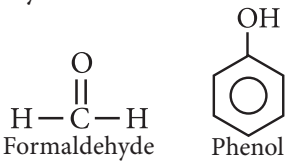
13. Homopolymer

14. Teflon $+CF_2-CF_2+_n$

15. A polymer made by polymerisation of a single monomer is known as homopolymer and the process is known as homopolymerisation *e.g.*, polythene made by polymerisation of ethene molecules.

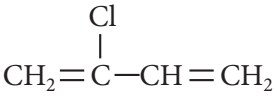
$$n CH_2 = CH_2 \longrightarrow -(CH_2-CH_2)-_n$$
Ethene Polythene

16. (a) **Bakelite :** Monomers are phenol and formaldehyde.



Formaldehyde Phenol

(b) **Neoprene :** Neoprene or polychloroprene is formed by the polymerisation of chloroprene.

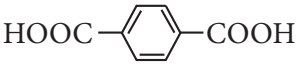


Chloroprene
(2-Chloro-1,3-butadiene)

17. (i) *Refer to answer 11.*

(ii) Monomers of terylene :

$HO-CH_2-CH_2-OH$ and
Ethylene glycol



Terephthalic acid

18. (a) Teflon is an addition polymer as it is made by addition of large number of tetrafluoroethylene molecules.

$$n CF_2{=}CF_2 \longrightarrow +CF_2-CF_2+_n$$
Teflon

(b) PHBV (poly-β-hydroxybutyrate-co-β-hydroxy valerate) is a condensation polymer because it is formed by repeated condensation reaction between two bifunctional (3-hydroxybutanoic acid and 3-hydroxypentanoic acid) monomer units with the elimination of water molecules.

19. (a) *Refer to answer 17(ii).*

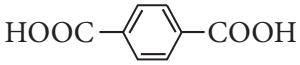
(b) **Monomers of Buna–N :**

$CH_2{=}CH-CH{=}CH_2$
1,3-Butadiene

$H_2C{=}CH-CN$
Acrylonitrile

20. (i) **Monomers of of terylene or dacron :**

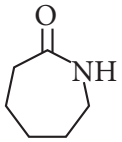
$HOCH_2-CH_2-OH$ and
Ethylene glycol



Terephthalic acid

(ii) *Refer to answer 19(b).*

21. (i) **Monomer of nylon-6 :** Caprolactam;



(ii) **Monomer of teflon :** Tetrafluoroethene;
$(CF_2{=}CF_2)$

22. (i) Polymerisation of ethene requires an initiator to start the polymerisation with free radical mechanism. Thus, peroxide like *t*-butyl peroxide decomposes to give free radical that initiates the reaction.

(ii) Hexamethylenediamine and adipic acid.

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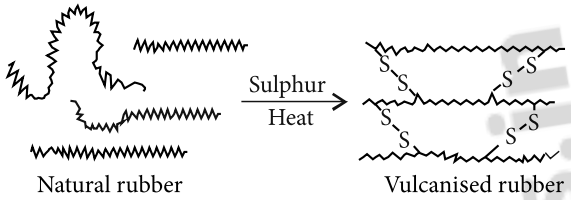
23. (i) Vulcanisation is a process of heating natural rubber with sulphur and an appropriate additive to modify its properties.

It gives greater elasticity and ductility. Sulphur forms cross linked network which gives mechanical strength to the rubber.



(ii) Terephthalic acid and ethylene glycol.

24. (i) **Nylon-6, 6 :** $\underset{\text{Adipic acid}}{HOOC-(CH_2)_4-COOH}$ and $\underset{\text{Hexamethylenediamine}}{H_2N-(CH_2)_6-NH_2}$

(ii) **Neoprene :** $\underset{\text{Chloroprene}}{CH_2=\overset{\overset{Cl}{|}}{C}-CH=CH_2}$

25. (i) *Refer to answer 24(ii).*

(ii) *Refer to answer 21(ii)*

26. Copolymerisation : When the polymers are synthesised by polymerization of two or more than two different monomers then this process is called as copolymerisation. Example,

(i) **Styrene butadiene rubber (SBR) :**

$$\underset{\text{1,3-Butadiene}}{nCH_2=CH-CH=CH_2} + n\underset{\text{Styrene}}{\text{C}_6\text{H}_5\text{CH}=\text{CH}_2} \longrightarrow \underset{\text{Butadiene-styrene copolymer}}{\left[CH_2-CH=CH-CH_2-\underset{C_6H_5}{CH}-CH_2\right]_n}$$

(ii) **Buna–N :**

$$\underset{\text{1,3-Butadiene}}{nCH_2=CH-CH=CH_2} + \underset{\text{Acrylonitrile}}{nCH_2=CHCN} \longrightarrow \underset{\text{Buna - N}}{+CH_2-CH=CH-CH_2-CH_2-\underset{CN}{CH}+_n}$$

27. (i) *Refer to answer 16.*

28. (i) *Refer to answer 17(ii).*

(ii) *Refer to answer 24(i).*

29. (i) *Refer to answer 25(ii).*

(ii) *Refer to answer 19(b).*

30.

Name of polymer	Monomer
(i) Neoprene	Chloroprene
(ii) Polystyrene	Styrene
(iii) Polypropene	Propene

31. (i) **Buna–S :**

$\underset{\text{1,3-Butadiene}}{CH_2=CH-CH=CH_2}$ and $\underset{\text{Styrene}}{C_6H_5CH=CH_2}$

(ii) *Refer to answer 24(i).*

32. Homopolymer : A polymer made by polymerisation of a single monomer is known as homopolymer and the reaction is called homopolymerisation. *e.g.,* Polythene made by ethene molecules.

$$\underset{\text{Ethene}}{nCH_2=CH_2} \rightarrow \underset{\text{Polythene (homopolymer)}}{+CH_2-CH_2+_n}$$

Copolymer : A polymer made by polymerisation of two or more different monomers is called copolymer and the reaction is called copolymerisation. When styrene and butadiene are polymerised together, a polymer called styrene-butadiene rubber is formed.

$$\underset{\text{1,3-Butadiene}}{nCH_2=CH-CH=CH_2} + n\underset{\text{Styrene}}{\text{C}_6\text{H}_5\text{CH}=\text{CH}_2} \longrightarrow \underset{\text{Styrene-butadiene rubber}}{+CH_2-CH=CH-CH_2-\underset{C_6H_5}{CH}-CH_2+_n}$$

33. (i) *Refer to answer 21(i)*

(ii) **Polypropene :** $\underset{\text{Propene}}{CH_3-CH=CH_2}$

34. (i) **Condensation polymers :** The polymers formed by the condensation of two or more bifunctional monomers are called condensation polymers.

For example : Nylon 6,6, Bakelite.

(ii) **Addition polymers :** The polymers formed by the addition reaction of a large number of unsaturated monomers are called addition polymers.

For example : Polythene, polystyrene.

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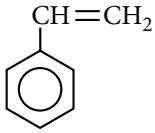
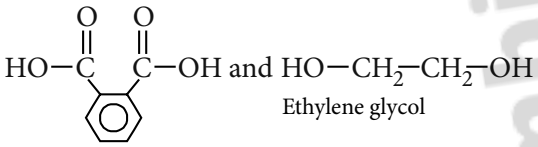
35. Monomer of polystyrene : 
Styrene

Used to make transparent drinking cups.

36. (i) Bakelite is used for making combs, electrical switches, handles of various utensils and phonograph records.
(ii) Nylon–6 is used for making tyre cords, ropes and fabrics.

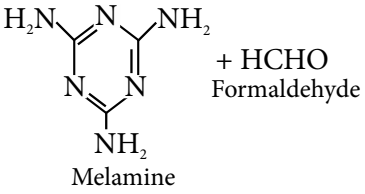
37. (i) *Refer to answer 31(i).*
(ii) *Refer to answer 24(i).*
(iii) *Refer to answer 16(a).*

38. (i) *Refer to answer 24(i).*
(ii) **Glyptal :**


Phthalic acid Ethylene glycol

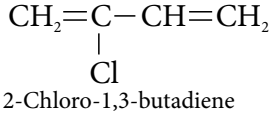
(iii) *Refer to answer 31(i).*

39. (i) It is a homopolymer because it is made up of one type of monomer propylene.
(ii) Melamine-formaldehyde polymer is formed by condensation polymerisation of melamine and formaldehyde.


Melamine Formaldehyde

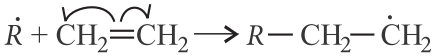
(iii) *Refer to answer 23(i).*

40. (i) Neoprene is formed by the free radical polymerisation of chloroprene.

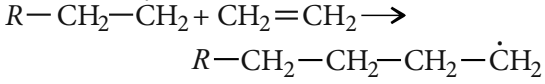

2-Chloro-1,3-butadiene

(ii) *Refer to answer 39(ii).*
(iii) *Refer to answer 31(i).*

41. Chain initiation :

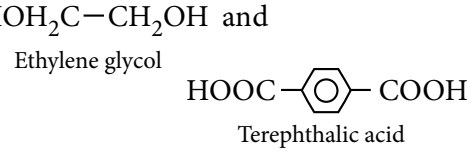


Chain propagation :



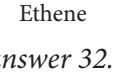
Chain termination :

42. (i) *Refer to answer 31(i).*
(ii) *Refer to answer 38(ii).*
(iii) *Refer to answer 10.*

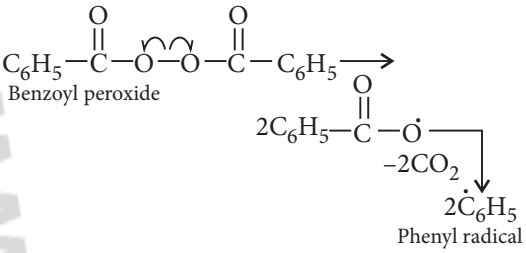
43. (i) 
Ethylene glycol Terephthalic acid

(ii) *Refer to answer 16(a).*
(iii) *Refer to answer 31(i).*

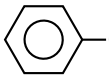
44. (i) Monomer of polythene is ethene.
(ii) Monomer of PVC is vinyl chloride.
(iii) Monomers of bakelite are formaldehyde and phenol.

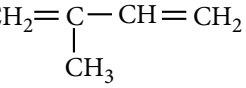
45. (i) *Refer to answer 16(a).*
(ii) *Refer to answer 21(i).*
(iii) **Polythene :** 
Ethene

46. (a) *Refer to answer 32.*
(b) In the preparation of polythene from ethene, benzoyl peroxide acts as an initiator or free radical generator.


Benzoyl peroxide Phenyl radical

47. Teflon : $CF_2 = CF_2$: Addition polymer

Bakelite : HCHO +  : Condensation polymer

Natural rubber :  : Addition polymer.

48. *Refer to answer 26.*

49. (i) *Refer to answer 45(iii).*
(ii) *Refer to answer 10.*
(iii) *Refer to answer 25(ii).*

50. Polymers which are formed by the repeated addition reaction of unsaturated monomer molecules are called the addition polymers.

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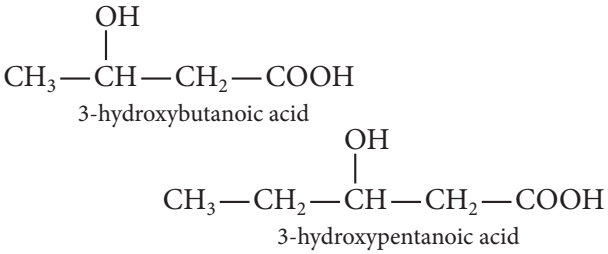
Refer to answer 32.

51. (i) **Bakelite** – (1) In electrical switches.
(2) In making handles of various utensils.
(ii) **Nylon-6,6** – (1) In Fabrics.
(2) In tyre cords.
(iii) **PVC** – (1) In handbags.
(2) In water pipes.

52. *Refer to answer 34.*
(i) Polythene – addition polymer
(ii) PTFE – addition polymer
(iii) Polybutadiene – addition polymer
(iv) Bakelite – condensation polymer

53. Monomers of PHBV (poly-β-hydroxybutyrate-co-β-hydroxy valerate) are



54. Biodegradable polymers : The natural polymer, which disintegrates by itself or by micro-organisms within certain period of time is called biodegradable polymer, *e.g.*, PHBV (poly–β–hydroxybutyrate–co–β–hydroxyvalerate), Nylon 2–nylon 6.

55. *Refer to answer 54.*

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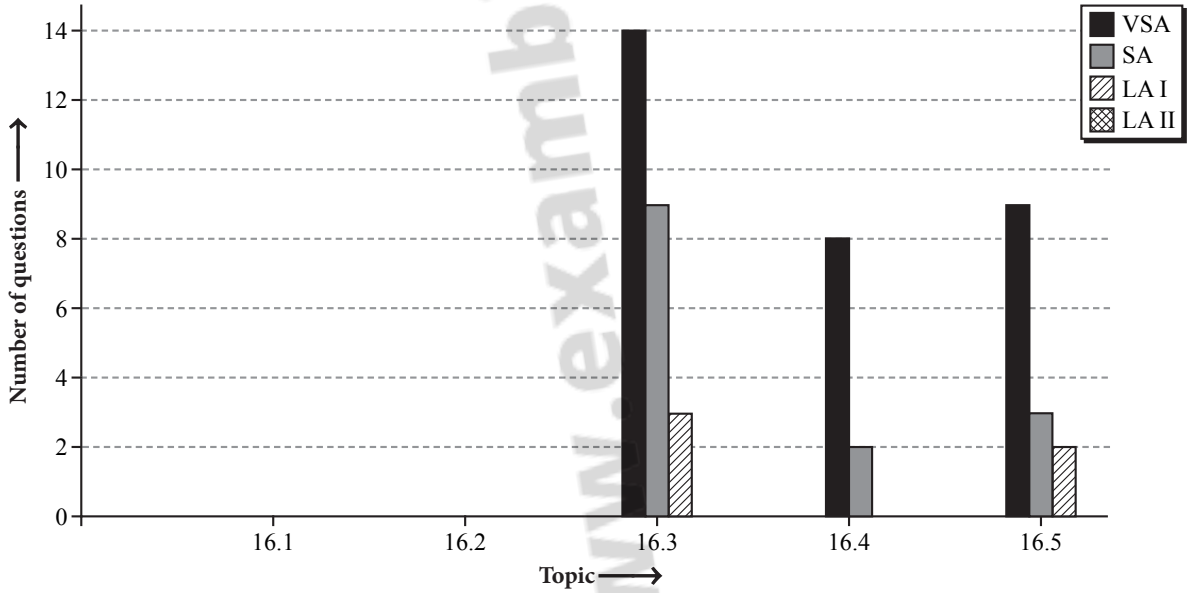
CHAPTER
16

Chemistry in Everyday Life

16.1 Drugs and their Classification
16.2 Drug-Target Interaction
16.3 Therapeutic Action of Different Classes of Drugs
16.4 Chemicals in Food
16.5 Cleansing Agents

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum total weightage is of *Therapeutic Action of Different Classes of Drugs*.
- Maximum VSA type questions were asked from *Therapeutic Action of Different Classes of Drugs*.
- Maximum SA and LA I type questions were asked from *Therapeutic Action of Different Classes of Drugs*.

QUICK RECAP

CHEMICALS IN MEDICINES

- **Drugs :** These are the chemicals of low molecular masses (~ 100-500 u) which interact with macromolecular targets and produce a biological response.
- **Medicines :** These are the drugs which are therapeutic and used for diagnosis, prevention and treatment of diseases.
- Use of chemicals for therapeutic effect is called *chemotherapy*.

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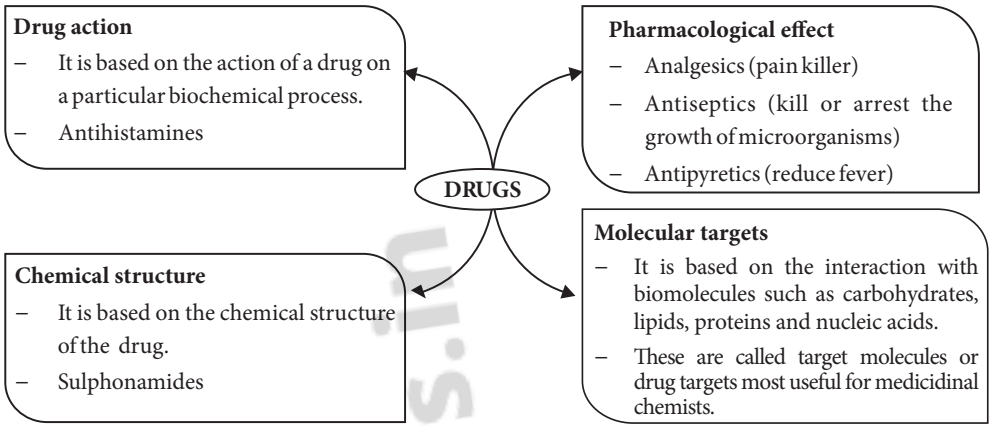
Classification of drugs : Drugs are classified on the basis of



Drug Target Interaction :

- **Enzymes as drug target :** Drugs inhibit any of the two activities of the enzymes, they can block the binding site of the enzyme and prevent the binding of substrate or they can inhibit the catalytic activity of enzyme.
- **Receptors as drug target :** Proteins which transmit communication to the different parts of the body are called *receptors*. Receptor proteins are embedded in the cell membrane and receptor changes its shape to accommodate a chemical messenger which brings about transfer of message into the cell.
- **Drug interact with receptors in two ways :**
  - Drugs bind to their receptor sites and inhibit its natural function (antagonists). These are useful when blocking of message is required.
  - Some drugs mimic the natural messenger by switching on the receptor (agonists). These are useful when there is lack of natural chemical messenger.

Therapeutic action of different classes of drugs :

- **Antacids :** These are chemicals which neutralise excess acid in the gastric juices and give relief from acid indigestion, acidity, heart burns and gastric ulcers, *e.g.*, magnesium hydroxide, calcium carbonate, etc.
- **Antihistamines :** They diminish or abolish the main action of histamine released in the body and hence, prevent allergic reactions, they are also called *anti-allergic drugs*, *e.g.*, diphenylhydramine (benadryl), pheniramine maleate (avil), etc.
- **Tranquilizers :** These are chemical substances used for the treatment of stress, mild and severe mental diseases. They are neurologically active drugs and are also called *psychotherapeutic drugs, e.g.,* veronal, amytal, seconal, equanil, chlorodiazepoxide, etc.

  Veronal, amytal and seconal are called barbiturates. Barbiturates are hypnotic, *i.e.,* sleep producing agents.
- **Analgesics :** These are chemical substances which reduce pain. They are classified as :
  - **Non-narcotic analgesics :** Aspirin and paracetamol belong to this class of drugs. They are effective in relieving skeletal pain such as that due to arthritis. They have many other effects such as reducing fever and preventing platelets coagulation. Aspirin finds use in prevention of heart attacks because of its anti-blood clotting action.
  - **Narcotic analgesics :** Morphine and many of its homologues, when administered in medicinal doses, relieve pain and produce sleep. In poisonous doses, they cause convulsions and ultimately death. These are mainly used for the relief of post operative pain, cardiac pains and pains of terminal cancer.

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- **Antimicrobials :** These are chemical substances used to cure infections due to micro-organisms, *e.g.*, sulphadiazine, sulphadoxine, etc. Antibiotics, antiseptics and disinfectants are antimicrobial drugs.
  - **Antibiotics :** These are chemical substances produced wholly or partly by chemical synthesis which in low concentration inhibit the growth or destroy micro-organisms by intervening in their metabolic processes.
    - The antibiotics may be either *bacteriocidal* (kill the organisms in the body) *e.g.*, penicillin, ofloxacin, etc., or *bacteriostatic* (inhibit the growth of organisms), *e.g.*, erythromycin, chloramphenicol, etc.
    - Antibiotics which kill or inhibit a wide range of Gram-positive and Gram-negative bacteria are said to be *broad spectrum antibiotics*, *e.g.*, tetracycline, chloromycetin and chloramphenicol.
    - Antibiotics which are effective mainly against Gram-positive or Gram-negative bacteria are *narrow spectrum antibiotics*, *e.g.*, penicillin-G.
  - **Antiseptics :** These are chemicals which either kill or prevent the growth of micro-organisms and are applied to the living tissues such as wounds, cuts, ulcers and diseased skin.
    - Dettol is a commonly used antiseptic and it is a mixture of chloroxylenol and terpineol
    - Bithionol is added to soaps.
    - Tincture of iodine, *i.e.*, 2-3% solution in alcohol-water mixture is applied on wounds.
  - **Disinfectants :** These are also used to kill microorganisms, but they are applied to inanimate objects.
    - Some substances can act as antiseptic as well as disinfectant by varying the concentration.
    - 0.2% phenol is an antiseptic, whereas its 1% solution is disinfectant.
    - 0.2-0.4 ppm chlorine in aqueous solution acts as disinfectant.
- **Antifertility drugs :** Chemical substances used to prevent conception or fertilization are called *antifertility drugs*. These are essentially a mixture of estrogen and progesterone derivatives which are more potent than the natural hormones, *e.g.*, mifepristone, ormeloxifene, etc.

CHEMICALS IN FOOD

- **Preservatives :** These are the chemical substances which are added to the food materials to prevent their spoilage and to retain their nutritive value for long periods. These preservatives prevent the rancidity of food and inhibit the growth or kill the microorganisms.
- The preservation of food by adding sufficient amount of salt to it is called *salting*. Salt prevents the water from being available for microbial growth.
- The microbial growth in food materials can also be prevented by adding certain chemical substances. The most common preservative used is sodium benzoate (C_6H_5COONa). It is metabolised by conversion to hippuric acid, $C_6H_5CONHCH_2COOH$ which is ultimately excreted through urine.
- Certain food preservatives such as BHA and BHT are used for edible oils, also act as antioxidants.
- **Artificial sweetening agents :** These are chemical compounds which give sweetening effect to the food and enhance its odour and flavour.

Artificial sweetener	Sweetness value in comparision to cane sugar
Aspartame	100
Saccharin	550
Alitame	2000

- **Antioxidants :** These are the chemical substances which prevent oxidation and subsequent spoilage of the food. These act as sacrificial materials, *i.e.,* they are more

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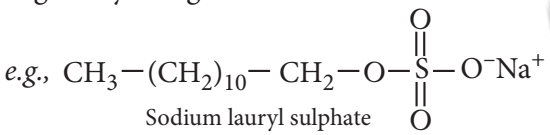
reactive towards oxygen than the materials they are protecting. They also reduce the rate of involvement of free radicals in the ageing process.

CLEANSING AGENTS

- **Soaps :** These are sodium or potassium salts of long chain fatty acids, *e.g.,* stearic, oleic and palmitic acids. Soaps containing sodium salts are prepared by heating glyceryl ester of fatty acid with aqueous NaOH solution and the reaction is known as *saponification*.
- Only sodium or potassium soaps are soluble in water and are used for cleaning purposes. Generally potassium soaps are soft to the skin than sodium soaps.
- **Detergents :** These are the materials which are used for cleaning purposes. They are also called *soapless soaps*.
  - Anionic detergents : Their polar head is negatively charged.



e.g., $CH_3-(CH_2)_{10}-CH_2-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-O^-Na^+$

Sodium lauryl sulphate

Such detergents are used to wash clothes.

  - Cationic detergents : Their polar head is positively charged *e.g.,*

$[CH_3(CH_2)_{15} - N(CH_3)_3]^+Cl^-$

Trimethylhexadecylammonium chloride

These are used as fabric softener and hair conditioner.

  - Non-ionic detergents : Their polar head is neutral. *e.g.,*

$CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$

Polyethylene glycol stearate

Such detergents are used in dish washers.

- **Action of soap in hard water :** Hard water contains Ca^{2+} or Mg^{2+} ions which react with sodium or potassium salt of fatty acid (soap) to form calcium or magnesium salt of fatty acids called *scum*.

$$2C_{17}H_{35}COONa + Ca^{2+} \rightarrow 2Na^+ + (C_{17}H_{35}COO)_2Ca$$

Insoluble soap or scum

$$2C_{17}H_{35}COONa + Mg^{2+} \rightarrow 2Na^+ + (C_{17}H_{35}COO)_2Mg$$

- **Advantages of synthetic detergents over soaps :**
  - They can be used in hard water, in acidic medium while soaps get precipitated.
  - They are more soluble in water and thus, form lather more easily.
  - They are stronger cleansing agents than soaps as they decrease the surface tension to greater extent.

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Previous Years' CBSE Board Questions

16.3 Therapeutic Action of Different Classes of Drugs

VSA (1 mark)

1. Which one of the following is a narcotic analgesic?
Penicillin, Codeine, Ranitidine
(One word, 2020)

2. Give reason :
Antacids should not be used for longer time.
(1/2, 2020)

3. Define the following terms with a suitable example :
Tranquilizers *(1/2, 2020, 1/3, Delhi 2017, AI 2013C, 2012, 2012C)*

4. Why magnesium hydroxide is a better antacid than sodium bicarbonate?
(1/3, Delhi 2019)

5. Define the following :
Antacids *(1/3, AI 2019, 2017, Delhi 2011)*

6. What is tincture of iodine? Write its one use. *(1/3, 2018)*

7. What is meant by a 'broad spectrum antibiotic'? *(1/3, Delhi 2017, Foreign 2011)*

8. What are antiseptics? Give one example.
(1/3, Delhi 2017, AI 2014, Foreign 2011)

9. What are limited spectrum antibiotics ? Give one example. *(1/3, Delhi 2017, 2013C)*

10. What is the following? Give one example.
Antibiotics *(1/3, Delhi 2014)*

11. Which class of drugs is used in sleeping pills?
(1/3, Delhi 2013)

12. What is the cause of a feeling of depression in human beings? Name a drug which can be useful in treating this depression. *(AI 2012)*

13. Differentiate between disinfectants and antiseptics. *(AI 2012, 1/3, Delhi 2011)*

14. Name the main constituents of dettol.
(1/3, Delhi 2011)

SA (2 marks)

15. Define the following terms :
(a) Tranquilizers
(b) Antiseptic *(2020)*

16. Define the following terms with a suitable example in each :
(i) Antibiotics
(ii) Analgesics. *(2/3, Delhi 2019, AI 2012C)*

17. Define the following :
(i) Narrow spectrum antibiotics
(ii) Disinfectants *(2/3, Delhi 2017)*

18. (i) Define antihistamine with an example.
(ii) Which one of the following drugs is an antibiotic?
Morphine, Equanil, Chloramphenicol, Aspirin. *(2/3, Foreign 2014)*

19. Explain the following terms with a suitable example for each :
(i) Disinfectants
(ii) Antacids *(2/3, Delhi 2014C)*

20. (i) What class of drug is ranitidine?
(ii) Which of the following is an antiseptic?
0.2% phenol, 1% phenol *(2/3, AI 2013)*

21. How do antiseptics differ from disinfectants? Give one example of each. (Give two differences). *(2/3, AI 2013C, Delhi 2012C)*

22. What are the following substances ? Give one example of each :
(i) Broad spectrum antibiotics
(ii) Narcotic analgesics *(2/3, AI 2012C)*

23. Explain the following terms with one suitable example for each :
(i) Enzymes
(ii) Analgesics *(2/3, Delhi 2011)*

LA I (3 marks)

24. Answer the following questions :
(i) Why should medicines not be taken without consulting a doctor?

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(ii) What is meant by 'broad spectrum antibiotics'?
(iii) What are the main constituents of dettol? *(Delhi 2014)*

25. What are the following substances ? Give one example of each :
(i) Antihistamines (ii) Tranquilizers
(iii) Broad spectrum antibiotics *(AI 2012C)*

26. Mention the action of the following on the human body in bringing relief from a disease :
(i) Brompheniramine
(ii) Aspirin (iii) Equanil *(Delhi 2011)*

16.4 Chemicals in Food

VSA (1 mark)

27. Define the following term with a suitable example :
Artificial sweeteners
(1/3, Delhi 2019, AI 2019)

28. Among the following, which one acts as a food preservative?
Aspartame, Aspirin, Sodium benzoate, Paracetamol *(1/3, 2018)*

29. Why is use of aspartame limited to cold foods and soft drinks?
(1/3, Delhi 2014, 2011, Foreign 2014)

30. Name sweetening agent used in the preparation of sweets for a diabetic patient.
(1/3, Delhi 2014)

31. What are the following substance? Give one example?
Food preservatives
(1/3, Delhi 2014, 2012, 2011)

32. Which one of the following is a food preservative?
Equanil, Morphine, Sodium benzoate.
(1/3, Delhi 2013)

33. Why do we require artificial sweetening agents? *(Delhi 2013C)*

34. Explain the following terms with one suitable example.
A sweetening agent for diabetic patients
(1/3, Delhi 2011)

SA (2 marks)

35. What are the following? Give one example.
(i) Sweetening agents
(ii) Food preservatives *(2/3, Delhi 2014C)*

36. What are the food preservatives? Name two such substances. *(AI 2012)*

16.5 Cleansing Agents

VSA (1 mark)

37. Which structural unit present in a detergent makes it non-biodegradable?
(One word, 2020)

38. Name the class of the synthetic detergent which is used in toothpaste.
(One word, 2020)

39. Give reason :
Shaving soaps contain glycerol. *(1/2, 2020)*

40. Define the following term with a suitable example :
Anionic detergents.
(1/2, 2020, 1/3, AI 2019, 1/2, Delhi 2013C)

41. Why is bithional added to soap? *(1/3, 2018)*

42. Define the following :
Cationic detergents
(1/3, Delhi 2017, AI 2017C, 1/2 Dehli 2013C)

43. If water contains dissolved Ca^{2+} ions, out of soaps and synthetic detergents, which will you use for cleaning clothes? *(1/3, AI 2013)*

44. What is the following substance? Give one example?
Synthetic detergents
(1/3, Delhi 2012C, 2011)

45. Why do soaps not work in hard water?
(1/3, Delhi 2012C, 2011)

SA (2 marks)

46. Explain the cleansing action of soaps. *(2020)*

47. (i) Why bithional is added in soap?
(iii) Why soaps are biodegradable whereas detergents are non-biodegradable?
(2/3, Delhi 2019)

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48. State the reason in each of the following cases :
(i) Soaps do not work well in hard water.
(ii) Synthetic detergents are better than soaps. *(Foreign 2011)*

LA I (3 marks)

49. (i) What are disinfectants? Give an example.
(ii) What are anionic detergents? Give an example.
(Delhi 2014)

50. What are biodegradable and non-biodegradable detergents ? Give one example of each. *(Delhi 2011C)*

Detailed Solutions

1. Codeine

2. Antacids only stop the symptoms of acidity. They are not the cure for acidity. Excessive use of antacids like bicarbonates can make the stomach alkaline and trigger the production of even more acid. If the acidity advances, it leads to formation of ulcers. In advance stages ulcers become life threatening and its only treatment is removal of the affected part of the stomach. Hence it is not advisable to take antacids for a long period of time.

3. **Tranquilizers :** Those medicines which act on central nervous system and help in reducing anxiety and relieve tension on the nerves are called tranquilizers. For example, valium.

4. Magnesium hydroxide is better antacid than sodium bicarbonate because being insoluble it does not increase the pH above neutrality whereas excess of sodium bicarbonate can make the stomach alkaline and trigger the production of even more acid.

5. **Antacids :** The compounds which reduce or neutralise acidity and raise the pH to appropriate level in stomach are known as antacids. *e.g.*, Ranitidine.

6. 2-3% solution of iodine in alcohol-water mixture is known as tincture of iodine. It is a powerful antiseptic.

7. The antibiotic which is effective against a wide range of Gram positive and Gram negative bacteria is known as broad spectrum antibiotic. For example, chloramphenicol.

8. **Antiseptics :** These are the chemical substances used to prevent the growth of microorganisms or to kill them and are safe to be applied on living tissues. These are applied to wounds and cuts. For examples, (i) iodoform (ii) tincture of iodine.

9. Antibiotics which are effective against a single organism or disease are called limited spectrum antibiotics. *e.g.*, Dysidazirine which is toxic towards certain strains of cancer cells.

10. Antibiotics are chemical substances (prepared wholly or partially by chemical synthesis) which in low concentration either kill or inhibit the growth of microorganisms by intervening in their metabolic processes. *e.g.*,
Dysidazirine - Toxic towards cancer cells.

11. Tranquilizers are used in sleeping pills.

12. Low level of noradrenaline is the cause of depression of human being.
Equanil is used for treatment.

13. **Antiseptics :** *Refer to answer 8.*
Disinfectants : These are chemical substances used to kill micro-organisms in inanimate

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materials and are not safe to be applied to the living tissues. Examples,
(i) 1% solution of phenol.
(ii) Solution of *o*-, *m*- and *p*-methylphenols (cresols) in soapy water is called lysol and is used as a disinfectant.

14. Chloroxylenol and terpineol.

15. (a) *Refer to answer 3.*
(b) *Refer to answer 8.*

16. (i) *Refer to answer 10.*
(ii) Analgesics are drugs that reduce pain without causing impairment of consciousness, mental confusion, incoordination or paralysis or some other disturbances of nervous system.
Examples : Aspirin, Paracetamol, Morphine etc.

17. (i) **Narrow spectrum antibiotics :** The antibiotics that are specifically effective against a limited group of microorganisms are known as narrow spectrum antibiotics.
(ii) *Refer to answer 13.*

18. (i) The drugs which prevents the interaction of histamine with the receptors where histamine exerts its effect are called antihistamines. *e.g.*, Synthetic drug brompheniramine (Dimetapp) and terfenadine (Seldane) acts as antihistamines.
(ii) Chloramphenicol is an antibiotic.

19. (i) *Refer to answer 13.*
(ii) *Refer to answer 5.*

20. (i) It is an antacid as it helps in removing acidity of stomach.
(ii) 0.2% solution of phenol acts as an antiseptic whereas 1% solution of phenol acts as a disinfectant.

21. *Refer to answer 13.*

22. (i) *Refer to answer 7.*
(ii) Narcotic analgesics are pain killers and induce sleep. If taken in high amount they produce coma, convulsions and ultimately, death. *e.g.,* Morphine.

23. (i) Proteins which play an important role of biological catalyst in the body are called enzymes. *e.g.*, Invertase, maltase etc.
(ii) *Refer to answer 16(ii).*

24. (i) We should not take medicine without consulting doctor because we don't know about that particular medicine that what would be its reaction and side effect. May be a single medicine can cause death. It could be harmful for our life.
(ii) *Refer to answer 7.*
(iii) *Refer to answer 14.*

25. (i) *Refer to answer 18(i).*
(ii) *Refer to answer 3.*
(iii) *Refer to answer 7.*

26. (i) Brompheniramine acts as antihistamine (antiallergic). It interferes with the natural action of histamine by competing with histamine for binding sites of receptor where histamine exerts its effect.
(ii) Aspirin acts as an analgesic *i.e.*, used to reduce pain and antipyretic, *i.e.*, brings down the body temperature. Aspirin inhibits the synthesis of compounds which stimulate inflammation in the tissue and cause pain.
(iii) Equanil is used for controlling depression and hypertension.

27. Artificial sweeteners : The substances which provide sweetness to the food without increasing the calories to the body are known as artificial sweetening agents. These are inert and harmless thus excreted from the body in urine unchanged. Saccharin and aspartame are artificial sweetening agents.

28. Sodium benzoate acts as a food preservative.

29. Use of aspartame is limited to cold food and drinks because it is unstable at cooking temperature.

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30. Saccharin is the sweetener used in the preparation of sweets for a diabetic person.

31. Chemicals which prevent the spoilage of food due to microbial growth are called food preservatives. Table salt, sugar, vegetable oils are some commonly used food preservatives.

32. Sodium benzoate-food preservative.
Morphine -narcotic-analgesic, Equanil-tranquilizer.

33. Artificial sweetening agents are required by diabetic persons and people who need to control intake of calories.

34. *Refer to answer 27.*

35. (i) *Refer to answer 27.*
(ii) *Refer to answer 31.*

36. *Refer to answer 31.*

37. Branched hydrocarbon chain

38. Anionic detergents

39. Shaving soaps contain glycerol to prevent rapid drying.

40. Anionic detergent : A detergent whose large part of the molecule is anion is called anionic detergent. These are sodium salts of sulphonated long chain alcohols or hydrocarbons.
e.g., Sodium laurylsulphate.

$CH_3-(CH_2)_{10}CH_2OSO_3^-Na^+$

41. Bithional added to soap to impart antiseptic properties and to reduce the odour produced by bacterial decomposition of organic matter on skin.

42. Cationic detergents : These are quaternary ammonium salts of amines with acetate, chlorides or bromides as anions. Cationic part has long hydrocarbon chain and positive charge lies on nitrogen atom.
e.g., cetyltrimethyl ammonium bromide.

43. As the water contains dissolved Ca^{2+} ions, therefore it is hard water. Hence, synthetic detergents are preferred over the soaps for cleaning the clothes because calcium salts of detergents are soluble in water while calcium salts of soaps are insoluble. As a result, lot of soap is wasted.

44. Synthetic detergents : These are cleansing agents which have all the properties of soaps but actually do not contain any soap. These can be used both in soft and hard water. *e.g.,* sodium *p*-dodecylbenzene-sulphonate.

45. Soaps do not work well with hard water because Mg^{2+} and Ca^{2+} ions of hard water react with soap to form magnesium and calcium salts of fatty acids which are insoluble in water and form scum.

46. Soap has a hydrophobic part (hydrocarbon) that attracts dirt, grease oil, etc. whereas hydrophilic part (— COONa) attracts water that take oil, dirt and grease. Soaps do not work well with hard water because Mg^{2+} and Ca^{2+} ions of hard water react with soap to form magnesium and calcium salts of fatty acids which are insoluble in water and form scum.

$$\underset{\text{Soap}}{2C_{17}H_{35}COONa} + CaCl_2 \rightarrow 2NaCl + \underset{\substack{\text{Insoluble calcium}\\ \text{stearate (soap)}}}{(C_{17}H_{35}COO)_2Ca}$$

47. (i) *Refer to answer 41.*
(ii) Soaps are sodium or potassium salts of long chain fatty acids which can be decomposed by microorganisms while detergents are sodium salts of long chain alkyl hydrogen sulphate or long chain benzene sulphonic acids which cannot be decomposed by bacteria due to branching.

48. (i) *Refer to answer 45.*
(ii) Synthetic detergents are better than soaps because they form foam even in hard water whereas soaps do not. Some of the detergents give foam even in ice cold water.

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49. (i) *Refer to answer 13.*

(ii) *Refer to answer 40.*

50. Biodegradable detergents : Detergents having straight chain hydrocarbons are easily decomposed by micro-organisms. These are called biodegradable detergents.

Example : Sodium lauryl sulphate.

Non-biodegradable detergents : Detergents having highly branched hydrocarbon chains are not degraded by bacteria easily. These are called non-biodegradable detergents.

Example : Sodium 4- (1, 3, 5, 7 - Tetramethyloctyl) benzene sulphonate.

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PRACTICE PAPER 1
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

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SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

The resistance of a conductivity cell when filled with 0.05 M solution of an electrolyte *X* is 100 ohms at 40°C. The same conductivity cell filled with 0.01 M solution of an electrolyte *Y* has a resistance of 50 ohms. The conductivity of 0.05 M solution of electrolyte *X* is 1.0×10^{-4} S cm^{-1}.

1. Calculate the cell constant.

2. What will be the conductivity of 0.01 M *Y* solution?

3. Determine the molar conductivity of 0.01 M *Y* solution.

4. What is conductivity cell?

5. Conductivity of a solution decreases with increasing dilution of solution. Why?

Questions 6 to 10 are one word answers :

6. Which substance is used in gas mask?

7. Name the product when methyl bromide is treated with silver cyanide.

8. Which type of drug cimetidine is?

9. Name an alloy of transition metal with non-transition metal.

10. Which reaction is used for the test of 1°-aromatic amine?

Questions 11 to 15 are multiple choice questions :

11. The compounds $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2.H_2O$ represent

(a) linkage isomerism

(b) ionisation isomerism

(c) coordination isomerism

(d) hydrate isomerism.

12. Carbohydrates are stored in human body as the polysaccharide

(a) starch (b) glycogen

(c) cellulose (d) amylose.

13. Which of the following reactions shows the process of smelting?

(a) $2PbO + PbS \rightarrow 3Pb + SO_2$

(b) $2Na[Au(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Au$

(c) $PbO + C \rightarrow Pb + CO$

(d) $2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$

14. Which group contains coloured ions out of the following?

1. Cu^+ 2. Ti^{4+} 3. Co^{2+} 4. Fe^{2+}

(a) 1, 2, 3, 4 (b) 3, 4

(c) 2, 3 (d) 1, 2

15. A first order reaction takes 40 min for 30% decomposition. What will be $t_{1/2}$?
(a) 77.7 min. (b) 52.5 min.
(c) 46.2 min. (d) 22.7 min.

Questions 16 to 20 :
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : For the coagulation of sols carrying positive charge, PO_4^{3-} ions are more efficient than SO_4^{2-} or Cl^- ions.
Reason (R) : This follows Hardy – Schulze rule.

17. Assertion (A) : Ethyl xanthate is used as a collector in froth floatation process.
Reason (R) : Collectors depress the floatation property of one of the components of the ore and thus help in the separation of different minerals present in the same ore.

18. Assertion (A) : Hydroxylation of ethylene with Baeyer's reagent gives ethylene glycol.
Reason (R) : Baeyer's reagent is $H_2O_2 + FeSO_4$.

19. Assertion (A) : PHBV is an example of biodegradable polymer.
Reason (R) : PHBV is used in speciality packaging, orthopaedic devices, etc.

20. Assertion (A) : Among chalcogens, tendency of catenation is maximum for sulphur.
Reason (R) : S–S bond dissociation energy is higher than O–O bond dissociation energy.

SECTION - B

21. For the cell $Zn|Zn^{2+}_{(aq)} \,||\, Cu^{2+}_{(aq)}|Cu$, derive the relation between E°_{cell} and K_c at 298 K.

22. What factors are responsible for the stability of α-helix structure of proteins? Why is it named as 3.6_{13} helix?

23. How will you bring about the following conversions?
(i) Propanone to propene
(ii) Benzoic acid to *m*-nitrobenzyl alcohol

OR

State one reaction as an example in each case.
(i) Rosenmund's reaction.
(ii) Cannizzaro's reaction.

24. What type of synthetic detergents are biodegradable and why? Give one example of biodegradable detergents.

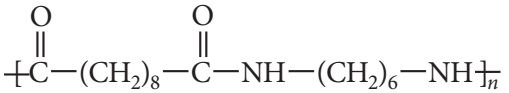
25. How is gold extracted from its ore?

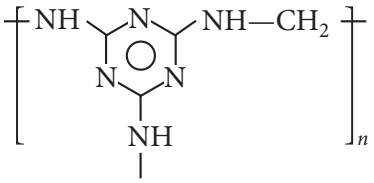
26. Physical adsorption decreases with increase in temperature while chemical adsorption first increases and then decreases with increase in temperature. Explain.

27. Could a copolymer be formed by both addition and condensation polymerisation? Explain with examples.

OR

Identify the monomers in the following polymer structures.

(i) 

(ii) 

SECTION - C

28. Write short notes on :
(i) Ionisation isomerism
(ii) Linkage isomerism
(iii) Coordination isomerism.

29. The density of lead is 11.35 g cm^{-3} and crystallise with *fcc* unit cell. Estimate the radius of lead atom.
(At. mass of lead = 207 g mol^{-1} and $N_A = 6.02 \times 10^{23}$ mol^{-1})

OR

Calculate the number of unit cells in 8.1 g of aluminium if it crystallizes in a *fcc* structure.
(Atomic mass of Al = 27 g mol^{-1})

30. Give explanation for the following statements :
(i) Chloroacetic acid has lower pK_a value than acetic acid.
(ii) Carboxylic acids have higher boiling points than alcohols of comparable molecular masses.

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(iii) Sodium bisulphite is used for the purification of aldehydes and ketones.

31. Account for the following statements :

(i) Cerium (atomic number = 58) forms tetrapositive ion, Ce^{4+} in aqueous solution.

(ii) The second and third members in each group of transition elements have similar atomic radii.

(iii) Write the structures of

(a) MnO_4^- (b) $Cr_2O_7^{2-}$

32. In a pseudo first order hydrolysis of ester in water, the following results are obtained :

***t* in seconds**	0	30	60	90
[Ester] in M	0.55	0.31	0.17	0.085

(i) Calculate the average rate of reaction between the time interval 30 to 60 seconds.

(ii) Calculate the pseudo first order rate constant for the hydrolysis of ester.

33. Explain what happens when

(i) ethanol vapours are passed over alumina at 600 K.

(ii) excess of ethanol is heated with conc. H_2SO_4 at 413 K.

(iii) phenol is treated with acetyl chloride.

OR

Account for the following :

(i) Phenol has a smaller dipole moment than methanol.

(ii) Phenols do not give protonation reactions readily.

(iii) Phenols undergo electrophilic substitution reactions more readily than benzene.

34. Give the structures of *A*, *B* and *C* in the following reactions :

(i) $C_6H_5N_2^+Cl^- \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B \xrightarrow[\Delta]{NH_3} C$

(ii) $C_6H_5NO_2 \xrightarrow{Sn + HCl} A \xrightarrow[273\text{-}278\ K]{NaNO_2 + HCl} B \xrightarrow[\Delta]{H_2O/H^+} C$

SECTION - D

35. (i) Arrange the following in decreasing order of property indicated :

(a) H_2O, H_2S, H_2Se, H_2Te (Boiling point)

(b) *M*F, *M*Cl, *M*Br, *M*I (Ionic character of bond)

(ii) How are XeO_3 and $XeOF_4$ prepared?

(iii) Explain why red phosphorus is less reactive than white phosphorus.

OR

(i) Account for the following statements :

(a) Decomposition of O_3 molecule is a spontaneous process.

(b) SF_6 is inert towards hydrolysis.

(c) H_2S is less acidic than H_2Te.

(d) Nitrogen is relatively inert as compared to phosphorus.

(ii) Name the oxide of chlorine used in paper pulp.

36. (i) Give the order of depression in freezing point of water for the same molar concentrations of acetic acid, trichloroacetic acid and trifluoroacetic acid. Give reason also.

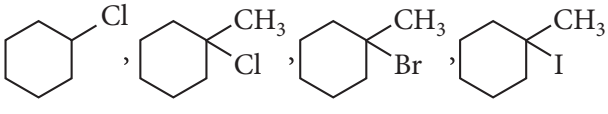
(ii) Calculate the depression in freezing point of water when 20.0 g of $CH_3CH_2CHClCOOH$ is added to 500 g of water.
[Given : $K_a = 1.4 \times 10^{-3}$, $K_f = 1.86\ K\ kg\ mol^{-1}$]

OR

(i) Assuming complete dissociation, calculate the expected freezing point of a solution prepared by dissolving 6.00 g of Glauber's salt, $Na_2SO_4 \cdot 10H_2O$ in 0.100 kg of water. (K_f for water = 1.86 K kg mol^{-1}, atomic masses of Na = 23, S = 32, O = 16, H = 1)

(ii) Phenol associates in benzene to a certain extent to form a dimer. A solution containing 20 g of phenol in 1.0 kg of benzene has its freezing point lowered by 0.69 K. Calculate the fraction of phenol that has dimerised.
(Given : K_f for benzene = 5.1 K m^{-1})

37. (i) Predict the order of reactivity of the following compounds towards S_N1 reaction.



(I) (II) (III) (IV)

(ii) What is Wurtz-Fittig reaction?

(iii) Racemisation occurs in S_N1 reactions. Why?

(iv) State one use each of DDT and iodoform.

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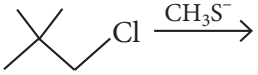
(v) Racemic mixture is optically inactive. Give reason.

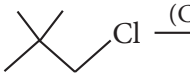
OR

(i) Which reaction in each pair shown below will show the faster rate of disappearance of starting material?

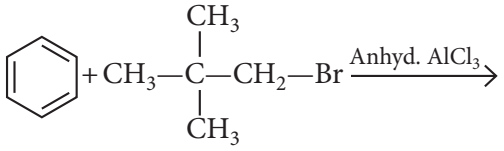
(a) (I) $(CH_3)_3CBr \xrightarrow[H_2O]{EtOH}$

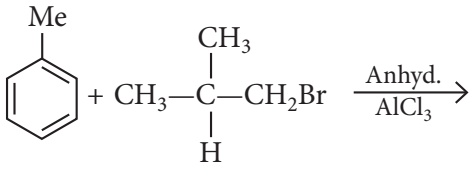
(II) $(CH_3)_3CBr \xrightarrow{EtOH}$

(b) (I)  $\xrightarrow{CH_3S^-}$

(II)  $\xrightarrow{(CH_3)_2CHS^-}$

(ii) Write the major product of the following reactions :

(a) 

(b) 

Detailed Solutions

1. Cell constant, $G = \kappa \times R$

or $G = 1.0 \times 10^{-4}\ S\ cm^{-1} \times 100\ ohm$

$= 1.0 \times 10^{-2}\ cm^{-1}$

2. For electrolyte *Y* :

$$\kappa = G \cdot \frac{1}{R}$$

or, $\kappa = 1.0 \times 10^{-2}\ cm^{-1} \times \frac{1}{50\ ohm}$

$= 2 \times 10^{-4}\ ohm^{-1}\ cm^{-1}$

3. Molar conductance, $\Lambda_m = \frac{\kappa \times 1000}{M}$

$$\Lambda_m = \frac{2 \times 10^{-4}\ ohm^{-1}\ cm^{-1} \times 10^3\ cm^3\ L^{-1}}{0.01\ mol\ L^{-1}}$$

or, $\Lambda_m = 20\ ohm^{-1}\ cm^2\ mol^{-1}$

4. In measurement of the conductivity of ionic solution, a solution cannot be connected to the bridge like a metallic wire or other solid conductor. It is kept in a specially designed vessel which is called conductivity cell.

5. Conductivity always decrease with decrease in concentration, because the number of ions per unit volume that carry the current in a solution decreases on dilution.

6. Activated charcoal

7. Methyl isocyanide

8. Antihistamine

9. Bronze (copper-tin)

10. Diazo test

11. (d)

12. (b) : Glycogen is the reserve carbohydrate occurring mainly in the liver.

13. (c)

14. (b) : Colour of transition elements is due to presence of unpaired electrons.

15. (a) : 30% decomposition means X = 30% of R_0

or $R = R_0 - 0.3R_0 = 0.7R_0$

For first order, $k = \frac{2.303}{t} \log \frac{[R_0]}{[R]}$

$$= \frac{2.303}{40} \log \frac{10}{7} min^{-1} = 8.918 \times 10^{-3}\ min.^{-1}$$

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{8.918 \times 10^{-3}\ min.^{-1}} = 77.7\ min.$$

16. (a) : According to Hardy–Schulze rule, greater the valency of the flocculating ion added, greater is its power to cause precipitation.

17. (c) : Collector enhances the non-wettability of mineral particles.

18. (c) : Baeyer's reagent is 1% alkaline solution of $KMnO_4$.

19. (b) : PHBV undergoes bacterial degradation in the environment hence, it is a biodegradable polymer.

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20. (b) : Sulphur shows maximum property of catenation among chalcogens; actually second among all element (carbon is the first). S—S bond is much stronger than O—O bond and hence sulphur has a much stronger tendency for catenation than oxygen.

21. In this cell, electrons will flow from Zn electrode to Cu electrode.

$$Zn_{(s)} + Cu^{2+}_{(aq)} \rightleftharpoons Zn^{2+}_{(aq)} + Cu_{(s)}$$

Apply the law of mass action,

$$K_c = \frac{[Zn^{2+}]}{[Cu^{2+}]} \qquad (\because [Cu_{(s)}] = [Zn_{(s)}] = 1)$$

According to Nernst equation for cell potential,

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

But at equilibrium E_{cell} is zero.

$$\therefore \quad E^\circ_{cell} = \frac{2.303RT}{nF} \log \frac{[Zn^{2+}]}{[Cu^{2+}]} = \frac{2.303RT}{nF} \log K_c$$

$$\text{or} \quad \log K_c = \frac{n \times 96500}{2.303 \times 8.314 \times 298} \; E^\circ_{cell}$$

$$\log K_c = \frac{n}{0.059} \; E^\circ_{cell}$$

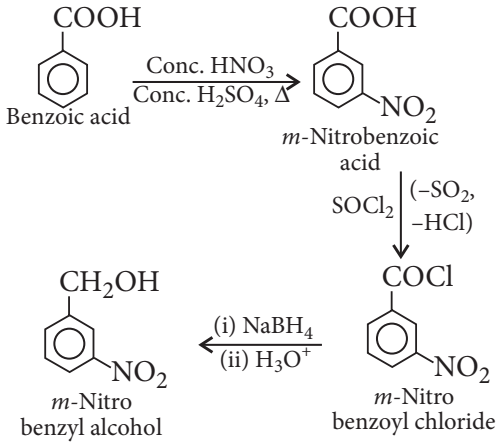
$$E^\circ_{cell} = \frac{0.059}{n} \log K_c \quad \text{or} \quad E^\circ_{cell} = \frac{0.059}{2} \log K_c$$

[$\because n = 2$ for Zn^{2+} and Cu^{2+} cell]

22. The stability of α-helix structure of proteins is due to intramolecular hydrogen bonding between —NH and —CO groups of the same peptide chain. The α-helix is termed as 3.6_{13} helix since each turn of the helix has approximately 3.6 amino acids and hydrogen bonding leads to the formation of 13-membered ring.

23. (i) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow[\text{(Reduction)}]{NaBH_4,\ CH_3OH}$
Propanone

$CH_3-\underset{\underset{\text{Propan -2-ol}}{|}}{\overset{\overset{OH}{|}}{CH}}-CH_3 \xrightarrow[\text{Heat } (-H_2O)]{\text{Conc. } H_2SO_4} CH_3CH{=}CH_2$
Propene

(ii) 

OR

(i) Rosenmund's reaction :

$CH_3COCl + H_2 \xrightarrow[\text{Boiling xylene}]{Pd/BaSO_4,\ S} CH_3CHO + HCl$
Ethanoyl chloride (Acetyl chloride) — Ethanal (Acetaldehyde)

(ii) Cannizzaro's reaction :

$2HCHO + NaOH \longrightarrow CH_3OH + HCOONa$
Methanal (Formaldehyde) — Methanol (Methyl alcohol) — Sodium methanoate (Sod. formate)

24. Those detergents which have linear alkyl chains are biodegradable as they are decomposed by microorganisms easily.

Sodium 4– (1-dodecyl)benzene sulphonate is an example of biodegradable detergent.

25. Gold is extracted by hydrometallurgy.

(i) Gold is extracted from quartz stone. Quartz is broken into chips and powdered.

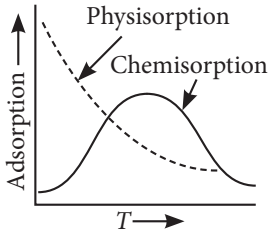
(ii) Powdered ore is treated with dilute aqueous solution of sodium cyanide and filtered.

$Au + 2NaCN \xrightarrow{H_2O} Na[Au(CN)_2] + NaOH$

(iii) Filtrate is treated with more active metal to give gold.

$2Na[Au(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Au$

26. Physisorption is a type of adsorption in which adsorbate molecules are held to the solid surface (adsorbent) by weak van der Waals' forces.



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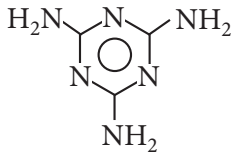
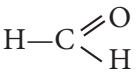
When temperature increases, these forces are weakened resulting in decrease in extent of physisorption. In case of chemisorption, adsorbate molecules are held with adsorbent by chemical bonds with high activation energy. Initial increase in chemisoption is due to heat supplied which acts as activation energy required in chemisorption. But later it decreases due to the exothermic nature of adsorption at equilibrium.

27. Copolymers can be formed both by addition and condensation polymerisation, *e.g.*, styrene butadiene rubber, a copolymer is obtained by addition polymerisation while nylon-6,6 is obtained by condensation polymerisation.

OR

(i) $HOOC—(CH_2)_8—COOH$,
Sebacic acid
$H_2N—(CH_2)_6—NH_2$
Hexamethylenediamine

(ii)  and 
Melamine Methanal

28. (i) **Ionisation isomerism :** When two complexes have same molecular formula but give different ions in solution, they are said to be ionisation isomers and the phenomenon is known as ionisation isomerism. This isomerism arises due to the exchange of ions between coordination sphere and ionisation sphere. For example,

$[Co(NH_3)_5Br]SO_4$ and
Pentaamminebromidocobalt(III) sulphate
$[Co(NH_3)_5SO_4]Br$
Pentaamminesulphatocobalt(III) bromide

(ii) **Linkage isomerism :** When any monodentate ligand in a complex has more than one donor atoms, it may be bonded to metal ion through either of the atoms and gives different isomers. For example, ligand SCN^- may be bonded to the metal either through sulphur or through nitrogen to give two different isomers.

$[Co(NH_3)_5SCN]Cl_2$ and $[Co(NH_3)_5NCS]Cl_2$
Pentaamminethiocyanato cobalt(III) chloride — Pentaammineisothiocyanato cobalt(III) chloride

(iii) **Coordination isomerism :** This type of isomerism arises when both positive and negative ions of a compound are complex ions. There may be an exchange of ligand molecules between the two coordination spheres of two ions to give two coordination isomers.

$[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$
Tetraamminecopper(II) tetrachloridoplatinate(II) — Tetraammineplatinum(II) tetrachloridocuprate(II)

29. $d = \dfrac{Z \times M}{a^3 \times N_A} \Rightarrow a^3 = \dfrac{Z \times M}{d \times N_A}$ ...(i)

For *fcc* unit cell, $Z = 4$

Given, $M = 207$ g mol^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}

$d = 11.35$ g cm^{-3}

Substituting these values in equation (i), we get

$$a^3 = \frac{4 \times 207 \text{ g mol}^{-1}}{11.35 \text{ g cm}^{-3} \times 6.02 \times 10^{23} \text{ mol}^{-1}}$$

$$a^3 = \frac{4 \times 207 \times 10}{11.35 \times 6.02 \times 10^{24}} \text{ cm}^3$$

$$\Rightarrow \quad a = \left(\frac{8280}{11.35 \times 6.02}\right)^{1/3} \times 10^{-8} \text{ cm}$$

$\therefore \quad a = 4.949 \times 10^{-8}$ cm $\Rightarrow$ $a = 494.9$ pm

For *fcc*, $r = \dfrac{a}{2\sqrt{2}}$

$$\therefore \quad r = \frac{494.9}{2\sqrt{2}} \text{ pm} = \frac{494.9\sqrt{2}}{4} \text{ pm} = \frac{494.9 \times 1.414}{4} \text{ pm}$$

$\therefore \quad r = 174.95$ pm

OR

As 27 g of Al will contain 6.023×10^{23} atoms

$\therefore$ 8.1 g Al will contain

$$= \frac{6.023 \times 10^{23}}{27} \times 8.1 \text{ atoms}$$

$= 1.8069 \times 10^{23}$ atoms

In *fcc*, 4 atoms are present in one unit cell

$\therefore$ 1.8069×10^{23} atoms will be present in

$$\frac{1.8069 \times 10^{23}}{4} = 4.517 \times 10^{22} \text{ unit cells}$$

30. (i) 'Cl' in chloroacetic acid shows $-I$ effect which creates less electron density on oxygen of carboxylic acid. Thus, release of proton becomes easier. In case of acetic acid, $+I$ effect of $-CH_3$ group makes the release of proton difficult. Hence, chloroacetic acid is stronger acid than acetic acid. Thus, chloroacetic acid has lower pK_a value than that of acetic acid.

(ii) Carboxylic acids and alcohols both undergo hydrogen bonding among their molecules. Hydrogen bonding among acid molecules is far

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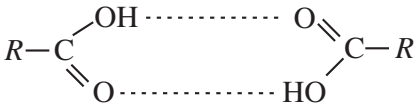
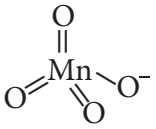
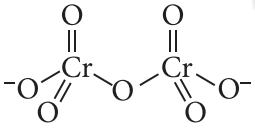
stronger than among alcohol molecules. Hence, carboxylic acids have higher boiling points than alcohols.



Strong hydrogen bonding among carboxylic acid molecules (Dimer formation).

(iii) Sodium bisulphite reacts with aldehydes and ketones to form insoluble, crystalline addition products. These products can be easily separated in pure state. These products can be decomposed by dilute mineral acid or alkali to give back original aldehyde or ketone. Hence, it can be used for purification of aldehydes and ketones.

31. (i) The electronic configuration of Ce ($Z = 58$) is $_{58}Ce = [Xe]\ 4f^1 5d^1 6s^2$.
Cerium can lose four electrons ($4f^1 5d^1 6s^2$) in aqueous solution to acquire stable configuration of rare gas xenon. Due to small size and high charge, Ce^{4+} ion has high hydration energy.
(ii) The second and third members in each group of transition elements have very similar atomic radii due to lanthanoid contraction which arises due to poor shielding effect of *f*-electrons.

(iii) (a)  (b) 

32. (i) Average rate of reaction between the time interval 30 to 60 seconds is

$$r_{av} = \frac{-[0.17 - 0.31]}{60 - 30} = \frac{0.14}{30} = 4.67 \times 10^{-3}\ s^{-1}$$

(ii) $k = \dfrac{2.303}{t} \log \dfrac{[R]_0}{[R]}$

At $t = 30$ s,

$$k = \frac{2.303}{30} \log \frac{0.55}{0.31}$$

$$= \frac{2.303}{30} \times 0.249 = 1.91 \times 10^{-2}\ s^{-1}$$

At $t = 60$ s,

$$k = \frac{2.303}{60} \log \frac{0.55}{0.17}$$

$$= \frac{2.303}{60} \times 0.5099 = 1.96 \times 10^{-2}\ s^{-1}$$

At $t = 90$ s

$$k = \frac{2.303}{90} \log \frac{0.55}{0.085} = \frac{2.303}{90} \times 0.8109$$

$$= 2.07 \times 10^{-2}\ s^{-1}$$

$\therefore$ Average value of k

$$= \frac{1.91 \times 10^{-2} + 1.96 \times 10^{-2} + 2.07 \times 10^{-2}}{3}$$

$$= 1.98 \times 10^{-2}\ s^{-1}$$

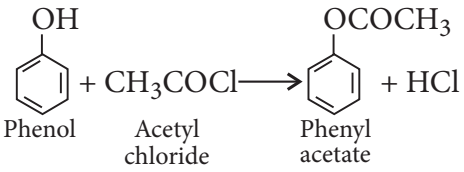
33. (i) Dehydration will take place and ethene will be formed.

$$\underset{\text{Ethanol}}{CH_3CH_2OH} \xrightarrow[600\ K]{Al_2O_3} \underset{\text{Ethene}}{CH_2 = CH_2} + H_2O$$

(ii) Diethyl ether is formed.

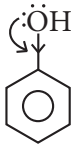
$$\underset{\text{Ethanol}}{2C_2H_5OH} \xrightarrow[413\ K]{\text{Conc. } H_2SO_4} \underset{\text{Ethoxyethane}}{C_2H_5OC_2H_5} + H_2O$$

(iii) Phenyl ethanoate is formed.

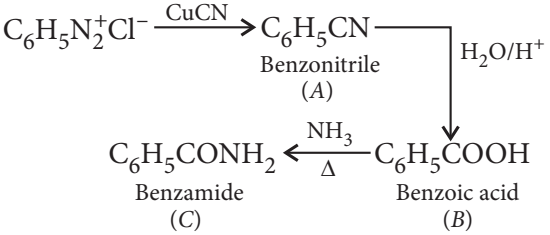


OR

(i) Due to electron withdrawing effect of phenyl group, the C – O bond in phenol is less polar, whereas in case of methanol, the methyl group has electron releasing effect and hence C – O bond in it, is more polar. Hence, dipole moment of methanol is higher than that of phenol.
(ii) Due to electron withdrawing effect of phenyl group the electron density on the oxygen atom of –OH group in phenol is less. Hence, phenols do not undergo protonation readily.
(iii) Phenols undergo electrophilic substitution reactions more readily than benzene because electron releasing –OH group activates the benzene by increasing electron density on the ring and facilitates attack of electrophile.



34. (i)



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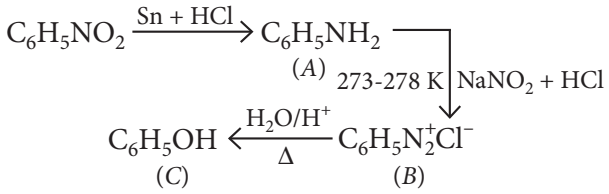
(ii)



35. (i) (a) $H_2O > H_2Te > H_2Se > H_2S$ - Boiling point. Because the boiling points of hydride increase with the increase in molecular mass of the elements. H_2O has exceptionally high boiling point because of intermolecular hydrogen bonding.

(b) $MF > MCl > MBr > MI$ - Ionic character

As the electronegativity of halogens decreases, the ionic character of $M - X$ decreases.

(ii) XeO_3 can be obtained by the hydrolysis of either XeF_4 or XeF_6.

$$6XeF_{4(s)} + 12H_2O_{(l)} \rightarrow 4Xe_{(g)} + 2XeO_{3(s)} + 24\ HF_{(aq)} + 3O_{2(g)}$$

$$XeF_{6(s)} + 3H_2O_{(l)} \rightarrow XeO_{3(s)} + 6HF_{(aq)}$$

$XeOF_4$ is obtained when XeF_6 reacts with small amount of water.

$$XeF_{6(s)} + H_2O_{(l)} \rightarrow XeOF_{4(l)} + 2HF_{(aq)}$$

(iii) White phosphorus is made up of discrete P_4 tetrahedra which are subjected to very high angular strain as the angles are 60°. Red phosphorus is a polymer of P_4 tetrahedra, which has much less angular strain. This high angular strain makes white phosphorus unstable and highly reactive.

OR

(i) (a) For spontaneity of a reaction, ΔG must be negative. Decomposition of O_3 is an exothermic process (ΔH = –ve) and occurs with increase in entropy (ΔS = +ve). These two effects reinforce each other which results in large negative Gibbs energy change according to the relation,

$\Delta G = \Delta H - T\Delta S$

Thus, decomposition of O_3 is a spontaneous reaction.

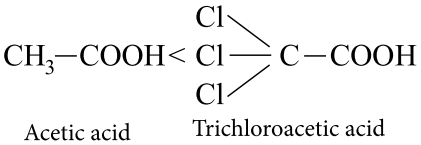
(b) In SF_6, S is surrounded by 6 F^- octahedrally. Therefore, attack of water molecule on S is sterically hindered. Hence, it is an inert substance.

(c) In a group, on moving down the group there is decrease in E–H bond dissociation enthalpy which increases acidic character. Thus, H_2S is less acidic than H_2Te.

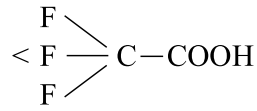
(d) Because P — P single bond is much weaker than N ≡ N triple bond and the bond length of nitrogen is small and bond dissociation energy is very large which makes it inert and unreactive and thus phosphorus becomes more reactive.

(ii) ClO_2 is used as a bleaching agent for paper pulp and textiles and in water treatment.

36. (i) The depression in freezing points are in the order :



Acetic acid Trichloroacetic acid



Trifluoroacetic acid

Fluorine being most electronegative, has the highest electron withdrawing inductive effect. Consequently, trifluoroacetic acid is the strongest acid while acetic acid is the weakest acid. Hence, trifluoroacetic acid ionises to the largest extent while acetic acid ionises to the minimum extent to give ions in their solutions in water. Greater the ions produced, greater is the depression in freezing point. Hence, the depression in freezing point is maximum for trifluoroacetic acid and minimum for acetic acid.

(ii) Given : Mass of $CH_3CH_2CHClCOOH$ = 20 g

Molar mass of $CH_3CH_2CHClCOOH$

$$= 122.5\ g\ mol^{-1}$$

Number of moles of $CH_3CH_2CHClCOOH$

$$= \frac{20\ g}{122.5\ g\ mol^{-1}} = 0.1632\ mol$$

$$\text{Molality of the solution } (m) = \frac{0.1632}{500} \times 1000$$

$$= 0.3264\ m$$

If α be the degree of dissociation of $CH_3CH_2CHClCOOH$ then

$$CH_3CH_2CHClCOOH \rightleftharpoons CH_3CH_2CHClCOO^- + H^+$$

Initial conc :	C	$C\alpha$	$C\alpha$
At equilibrium :	$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$K_a = \frac{C\alpha \times C\alpha}{C(1-\alpha)} = \frac{C^2\alpha^2}{C(1-\alpha)} \approx C\alpha^2$$

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$\because \quad K_a = C\alpha^2$

$$\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{1.4 \times 10^{-3}}{0.3264}}$$

$= \sqrt{0.004289} = 0.065$

$\therefore \quad \alpha = 0.065$

To calculate van't Hoff factor :

$CH_3CH_2CHClCOOH \rightleftharpoons CH_3CH_2CHClOO^- + H^+$

Initial mol. :	1	0	0
Moles at eq. :	$1 - \alpha$	α	α

Total number of moles $= 1 - \alpha + \alpha + \alpha = 1 + \alpha$

$= 1 + 0.065 = 1.065$

$\because \quad \Delta T_f = iK_f m = (1.065)\ (1.86)(0.3264)$

$= 0.65$ K

Hence, depression in freezing point of water = 0.65 K

OR

(i) Molecular mass of Glauber's salt

$Na_2SO_4 \cdot 10H_2O$

$= [2 \times 23 + 32 + 16 \times 4 + 10 \times (2 \times 1 + 16)]$

$= (46 + 32 + 64 + 180)\ g\ mol^{-1} = 322\ g\ mol^{-1}$

$Na_2SO_4 \cdot 10H_2O$ ionises as :

$Na_2SO_4 \cdot 10H_2O \rightleftharpoons 2Na^+ + SO_4^{2-} + 10H_2O$

$\Rightarrow \quad i = 3$

$$m = \frac{n_B}{W_{A(\text{in kg})}} = \frac{W_B}{M_B \times W_{A(\text{in kg})}}$$

$$= \frac{6.00\ \text{g}}{322\ \text{g mol}^{-1} \times 0.1\ \text{kg}}$$

$= 0.186\ mol\ kg^{-1} = 0.186$ m

Also, $\quad \Delta T_f = i\,K_f \cdot m$

$= 3 \times 1.86\ K\ m^{-1} \times 0.186\ m = 1.04$ K

$\Rightarrow \quad T_f = T_f^\circ - \Delta T_f = (273 - 1.04)$ K $= 271.96$ K

(ii) Here, $n = 2$ because phenol forms dimer on association.

$W_2 = 20$ g, $W_1 = 1$ kg $= 1000$ g, $\Delta T_f = 0.69$ K,

$K_f = 5.1\ K\ m^{-1}$

$$\Delta T_f = \frac{K_f \times W_2 \times 1000}{M_2 \times W_1}$$

$$M_2 = \frac{K_f \times W_2 \times 1000}{\Delta T_f \times W_1} = \frac{5.1 \times 20 \times 1000}{0.69 \times 1000}$$

$= 147.82\ g\ mol^{-1}$

$M_{2(\text{observed})} = 147.82\ g\ mol^{-1}$

$M_{2(\text{calculated})}$:

$C_6H_5OH = 6 \times 12 + 6 \times 1 + 16 = 94\ g\ mol^{-1}$

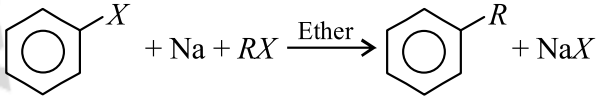
$$i = \frac{M_{2(\text{calculated})}}{M_{2(\text{observed})}} = \frac{94}{147.82} = 0.635$$

$2C_6H_5OH \rightleftharpoons (C_6H_5OH)_2$

$$\alpha = \frac{i-1}{\left(\frac{1}{n} - 1\right)} = \frac{0.635 - 1}{\left(\frac{1}{2} - 1\right)} = \frac{0.365}{0.5} = 0.73 = 73\%$$

37. (i) The first compound is a 2° alkyl halide while all others are 3° alkyl halide. Since 3° alkyl halides are more reactive than 2° alkyl halides in S_N1 reaction, therefore, first compound is the least reactive. Further reactivity increases in the order : chloride < bromide < iodide. Thus, the increasing order of reactivity in S_N1 reactions is (I) < (II) < (III) < (IV).

(ii) A mixture of an alkyl halide and aryl halide gives alkyl arene when treated with sodium in dry ether. This is called Wurtz-Fittig reaction.



(iii) Carbocations are formed in S_N1 reaction which are planar species so an incoming nucleophile can attack from either side and an equimolar mixture of two components are formed, thus racemisation occurs.

(iv) DDT is used as an insecticide and iodoform is used as a mild antiseptic.

(v) Racemic mixture contains equal amount of *d*- and *l*-forms, hence rotation due to one enantiomer is cancelled by another.

OR

(i) (a) $EtOH/H_2O$ being more polar than EtOH facilitates ionization of $(CH_3)_3CBr$ and hence reaction (I) occurs faster than reaction (II).

(b) $(CH_3)_2CHS^-$ being bigger in size than CH_3S^- causes steric hindrance and hence reaction (I) proceeds faster than reaction (II).

(ii) (a) The initially formed less stable 1° carbocation (I) undergoes 1,2-methyl shift to give the more stable 3° carbocation (II) which then attacks the benzene ring to give Friedel–Crafts alkylation product (III).

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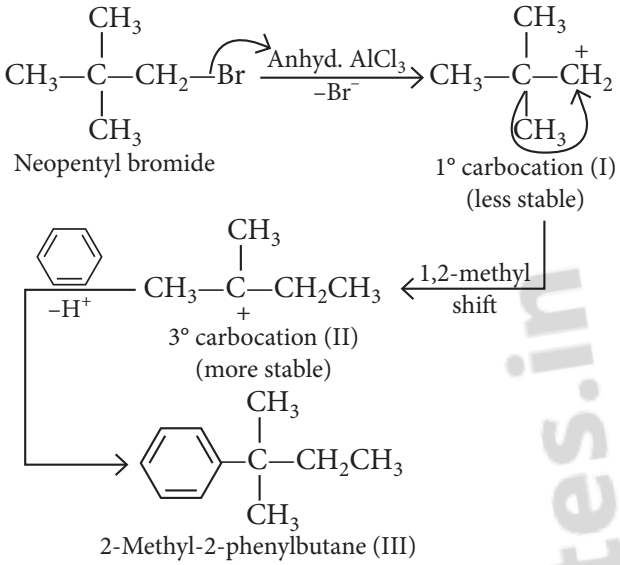
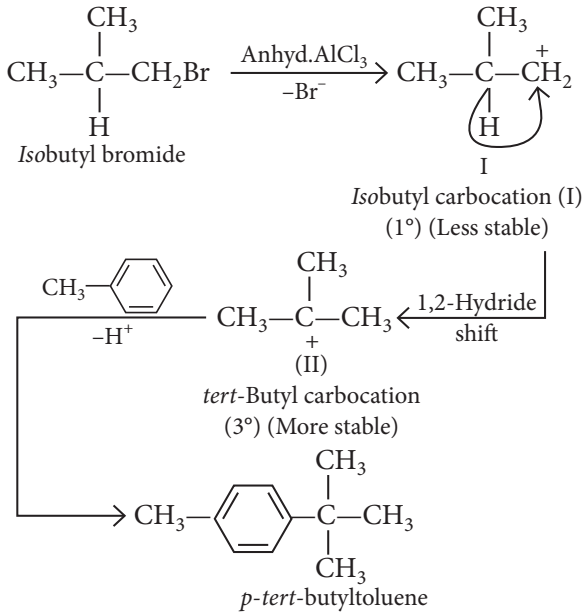


(b) The initially formed less stable 1° carbocation (I) undergoes rearrangement by 1,2-hydride shift to form more stable 3° carbocation (II) which then attacks the toluene ring at the less hindered *p*-position.



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PRACTICE PAPER 2 (Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

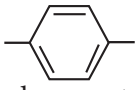
Read the given passage and answer the questions 1 to 5 that follow :

Aldehydes when treated with concentrated alkali undergo self oxidation and reduction (disproportionation) to give one molecule of alcohol and one molecule of carboxylic acid salt. This reaction is called Cannizzaro reaction.

1. Which type of aldehydes undergo Cannizzaro's reaction?

2. Write one chemical reaction to exemplify Cannizzaro reaction.

3. Give an example of intramolecular Cannizzaro reaction.

4.  does not contain any α-hydrogen atom, but still it fails to undergo Cannizzaro reaction. Why?

5. What happens when a mixture of benzaldehyde and formaldehyde is treated with conc. NaOH solution ?

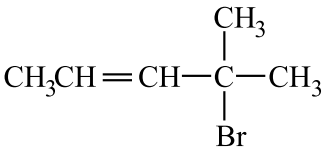
Questions 6 to 10 are one word answers :

6. Which catalyst is used to convert alcohols into gasoline in petroleum industry?

7. The rate constant of a reaction is 3×10^{-4} L mol^{-1} min^{-1}. What is the order of the reaction?

8. Name the actinoid element from which electron starts filling 5*f* sub-shell.

9. Write the IUPAC name of



10. Name the colligative property mostly used for the determination of molecular mass of macromolecules.

Questions 11 to 15 are multiple choice questions :

11. The crystal system of a compound with unit cell dimensions $a = 0.387$ nm, $b = 0.387$ and $c = 0.504$ nm and $\alpha = \beta = 90°$ and $\gamma = 120°$ is

(a) cubic (b) hexagonal

(c) orthorhombic (d) rhombohedral.

12. Why is ferric hydroxide colloid positively charged when prepared by adding ferric chloride to hot water?

(a) Due to precipitation of ferric hydroxide there is an excess of Fe^{3+} ions.

(b) Due to preferential adsorption of Fe^{3+} ions by the sol of $Fe(OH)_3$.

(c) Due to absence of any negatively charged ion.

(d) Due to adsorption of OH^- and Cl^- ions, the remaining sol has only Fe^{3+} ions.

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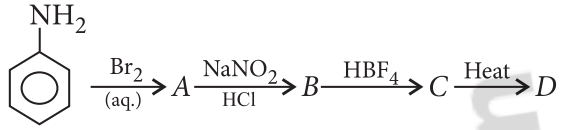
13. _____ being highly reactive is used in the extraction of chromium and manganese.
(a) Al (b) Zn (c) Cu (d) Mg

14. The product '*D*' in the following sequence of reactions is



(a) 2, 4, 6-tribromofluorobenzene
(b) fluorobenzene
(c) *p*-bromofluorobenzene
(d) tribromobenzene.

15. The rate constant for the reaction, $2N_2O_5 \rightarrow 4NO_2 + O_2$ is 2×10^{-5} s^{-1}. If rate of reaction is 1.4×10^{-5} mol L^{-1} s^{-1}, what will be the concentration of N_2O_5 in mol L^{-1}?
(a) 0.8 (b) 0.7 (c) 1.2 (d) 1

Questions 16 to 20 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : Reverse osmosis is used in the desalination of sea water.
Reason (R) : When pressure more than osmotic pressure is applied, pure water is squeezed out of the sea water through the membrane.

17. Assertion (A) : HClO is stronger acid than HBrO.
Reason (R) : Greater is the electronegativity of the halogen, more acidic is the halic acid.

18. Assertion (A) : Acid or enzymatic hydrolysis of sucrose to give an equimolar mixture of glucose and fructose is called inversion.
Reason (R) : Sucrose is the only naturally occurring disaccharide which is reducing sugar.

19. Assertion (A) : H_2–O_2 fuel cell gives a constant voltage throughout its life.
Reason (R) : In this fuel cell, H_2 reacts with OH^- ions yet the overall concentration of OH^- ions does not change.

20. Assertion (A) : In high spin situation, configuration of d^5 ions will be $t_{2g}^3 e_g^2$.
Reason (R) : In high spin situation, pairing energy is less than crystal field energy.

SECTION - B

21. (i) Aromatic 1°-amines cannot be prepared by Gabriel phthalimide reaction. Explain.
(ii) Select the amine which is most reactive towards dil. HCl among the following :
CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$, $C_6H_5NH_2$

OR

How will you convert aniline to (i) *N*–methylaniline (ii) acetanilide?

22. Write the formulae of the following complexes :
(i) Triamminetriaquachromium(III) chloride
(ii) *Tris*(ethane-1, 2-diamine) cobalt (III) chloride
(iii) Diamminesilver(I) dicyanidoargentate(I)
(iv) Mercury(I) tetrathiocyanato-S-cobaltate(III)

23. Draw the structures of
(a) P_4O_{10} (b) $H_4P_2O_7$

24. What are natural and synthetic polymers? Give two examples of each type.

OR

How does the presence of double bonds in natural rubber molecules influence their structure and reactivity?

25. Iron crystallizes in a *bcc* system with a lattice parameter of 2.861 Å. Calculate the density of iron in the *bcc* system.
(Given : atomic weight of Fe = 56, $N_0 = 6.023 \times 10^{23}$ mol^{-1}.)

26. Answer the following :
(a) Why is enthalpy of chemisorption higher than that of physisorption?
(b) How do the free valencies present on the surface of catalyst help in catalysis?

27. Why does paracetamol preferred over aspirin ?

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SECTION - C

28. How can reducing and non-reducing sugars be distinguished? Mention the structural feature characterising reducing sugars.

29. (i) Write one chemical method for the preparation of fluorine.

(ii) Write one use of helium gas and liquid helium each.

(iii) Give one example to explain that sulphuric acid is an oxidising agent.

OR

How will you obtain the following from sulphuric acid?

(i) SO_2 (ii) SO_3 (iii) SO_2Cl_2

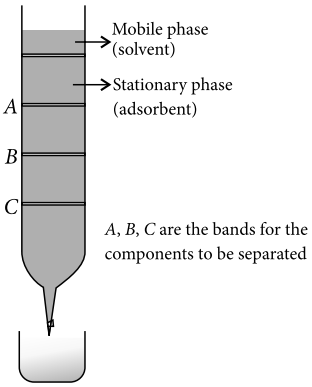
30. Give reasons for the following statements :

(i) Ethyl iodide undergoes S_N2 reaction faster than ethyl bromide.

(ii) (±) -2-Butanol is optically inactive.

(iii) C—*X* bond length in halobenzene is smaller than C—*X* bond length in CH_3—*X*.

31. Column chromatography is used for the refining of metals.



Answer the following :

(i) Which substance can act as stationary phase?

(ii) Which of the three components *A*, *B* and *C* is eluted first of all?

(iii) What is the principle of this technique?

OR

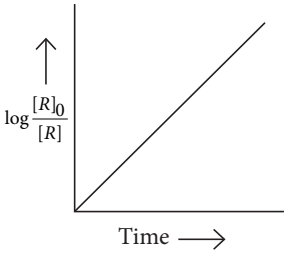
The concept of coupling reaction is useful in making the occurrence of a non-spontaneous reaction possible. Explain giving one suitable example.

32. Answer the following :

(i) There are two —NH_2 groups in semicarbazide. However, only one is involved in the formation of semicarbazone.

(ii) Give the mechanism of esterification of carboxylic acids. Explain.

33. Answer the following questions on the basis of curve for a first order reaction :



(i) What is the relation between slope and rate constant?

(ii) Calculate the rate constant if the slope is $2 \times 10^{-4}\ s^{-1}$.

(iii) Derive the relationship between half life and rate constant.

34. Non-ideal solutions exhibit either positive or negative deviations from Raoult's law. What are these deviations and why are they caused? Explain with one example for each type.

SECTION - D

35. Assign reason for the following :

(i) The enthalpies of atomisation of transition elements are high.

(ii) The transition metals and many of their compounds act as good catalyst.

(iii) Actinoid contraction is greater from element to element than the lanthanoid contraction.

(iv) The E° value for the Mn^{3+}/Mn^{2+} couple is much more positive than that for Cr^{3+}/Cr^{2+}.

(v) Scandium (Z = 21) does not exhibit variable oxidation states and yet it is regarded as transition element.

OR

(i) Write the steps involved in the preparation of

(a) $K_2Cr_2O_7$ from Na_2CrO_4

(b) $KMnO_4$ from K_2MnO_4.

(ii) What is meant by lanthanoid contraction? What effect does it have on the chemistry of the elements which follow lanthanoid contraction?

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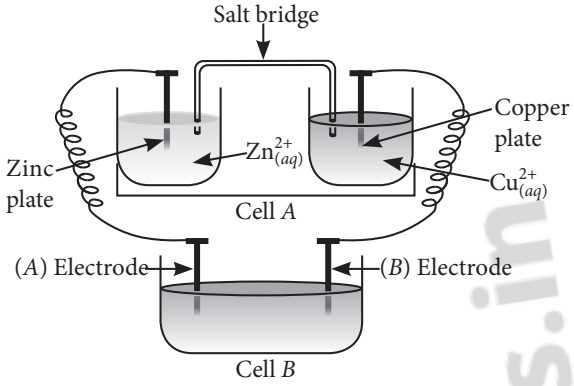
36. (a) Consider the figure and answer the following questions.



(i) Cell '*A*' has E_{cell} = 2 V and cell '*B*' has E_{cell} = 1.1 V. Which of the two cells '*A*' or '*B*' will act as an electrolytic cell? Which electrode reactions will occur in this cell?

(ii) If cell '*A*' has E_{cell} = 0.5 V and cell '*B*' has E_{cell} = 1.1 V then what will be the reactions at anode and cathode?

(b) Conductivity of 0.00241 M acetic acid is 7.896×10^{-5} S cm^{-1}. Calculate its molar conductivity. If Λ°_{m} for acetic acid is 390.5 S cm^2 mol^{-1}, what is its dissociation constant?

OR

(a) An acidic solution of Cu^{2+} salt containing 0.4 g of Cu^{2+} is electrolysed until all the copper is deposited. The electrolysis is continued for seven more minutes with volume of solution kept at 100 mL and the current at 1.2 amp. Calculate the volume of gases evolved at NTP during the entire electrolysis.

(b) Write down the electrode process when aqueous solutions of the given halides are electrolysed:

(i) $NaCl_{(aq)}$ (ii) $NaBr_{(aq)}$

(iii) $NaF_{(aq)}$

37. (a) Write the formula of reagents used in the following reactions :

(i) Bromination of phenol to 2,4,6- tribromophenol

(ii) Hydroboration of propene and then oxidation to propanol.

(b) Arrange the following sets of compounds in the increasing order of their property indicated :

(i) *p*-Nitrophenol, ethanol, phenol (acidic character)

(ii) Propanol, propane, propanal (boiling point)

(c) Write the mechanism (using curved arrow notation) of the following reaction :

$$CH_3-CH_2-OH \xrightarrow[(ii)\ CH_3CH_2OH]{(i)\ H^+}$$

OR

(a) How would you obtain the following :

(i) benzoquinone from phenol

(ii) 2-methylpropan-2-ol from methyl magnesium bromide

(iii) propan-2-ol from propene?

(b) Name the different reagents needed to perform the following reactions :

(i) Reduction of phenol to benzene

(ii) Dehydration of propan-2-ol to propene

(iii) Friedel–Crafts alkylation of anisole to *p*-methoxytoluene

(iv) Dehydrogenation of ethanol to ethanal

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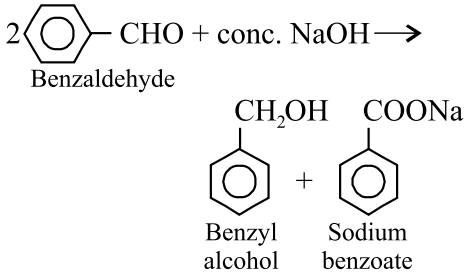
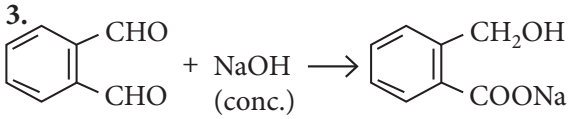
Detailed Solutions

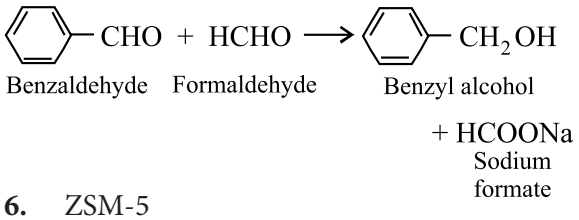
1. Those aldehydes which do not have α-hydrogen undergo Cannizzaro reaction. *e.g.*, HCHO, C_6H_5CHO.

2. 

3. 

4. Because the corresponding hydroxide is strongly stabilized by –*R* effect of –CHO. Consequently the electrophilicity of the carbonyl carbon is reduced and the OH^- does not attack the carbonly group.

5. Sodium formate and benzyl alcohol is produced.



6. ZSM-5

7. Second order

8. Protactinium (Pa)

9. 4-Bromo-4-methylpent-2-ene

10. Osmotic pressure

11. **(b) :** $a = 0.387$ nm; $b = 0.387$ nm; $c = 0.504$ nm

i.e. $a = b \neq c$

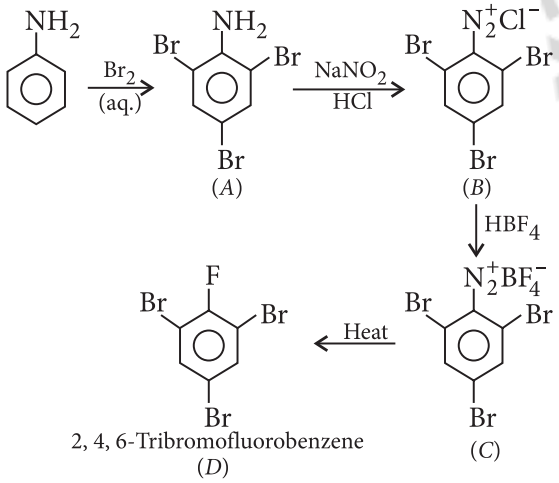
$\alpha = \beta = 90°$ and $\gamma = 120°$

These parameters represent hexagonal system.

12. **(b) :** The adsorption of positively charged Fe^{3+} ions by the sol of hydrated ferric oxide results in positively charged colloid.

13. **(a)**

14. **(a) :**



15. **(b) :** Rate = $k[N_2O_5]$ (first order as unit of rate constant is s^{-1})

$$[N_2O_5] = \frac{\text{rate}}{k} = \frac{1.4 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}}{2 \times 10^{-5} \text{ s}^{-1}} = 0.7 \text{ mol L}^{-1}$$

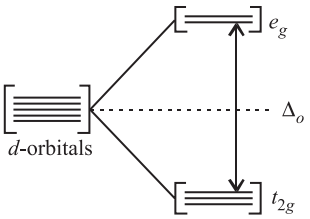
16. **(a) :** If a pressure larger than the osmotic pressure is applied to the solution side, the pure solvent flows out of the solution to the solvent through semi-permeable membrane and this phenomenon is called reverse osmosis.

17. **(a)**

18. **(c) :** Sucrose is a non-reducing sugar.

19. **(b) :** Fuel cells are designed in such a way that the H_2 and O_2 are stored outside the cell and are constantly supplied to electrodes for oxidation and reduction to occur.

20. **(c) :**



d-orbital splitting in an octahedral crystal field

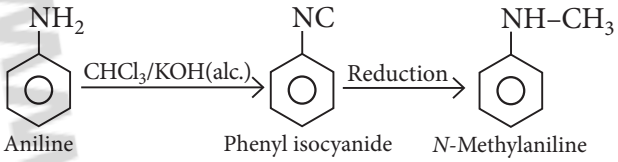
$\therefore$ In high spin situation, $\Delta_o < P$, in d^5 configuration, 4th and 5th electron are added to e_g rather than t_{2g}. So, configuration of d^5 ion will be $t_{2g}^3 e_g^2$.

21. (i) Aromatic 1°-amines cannot be prepared by Gabriel phthalimide reaction because aryl halides do not undergo nucleophilic substitution reactions with potassium phthalimide under mild conditions.

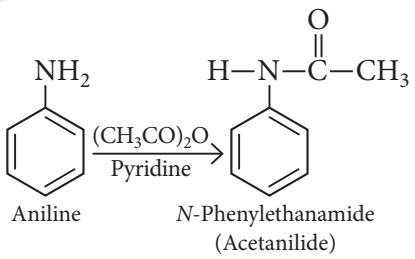
(ii) $(CH_3)_2NH$ is most basic amongst the given amines. Thus, it is most reactive amine towards HCl.

OR

(i)



(ii)



22. (i) $[Cr(NH_3)_3(H_2O)_3]Cl_3$

(ii) $[Co(en)_3]Cl_3$

(iii) $[Ag(NH_3)_2][Ag(CN)_2]$

(iv) $Hg[Co(SCN)_4]$

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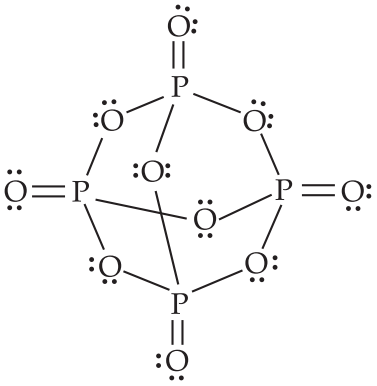
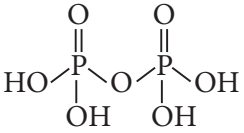
23. (a)



(b)

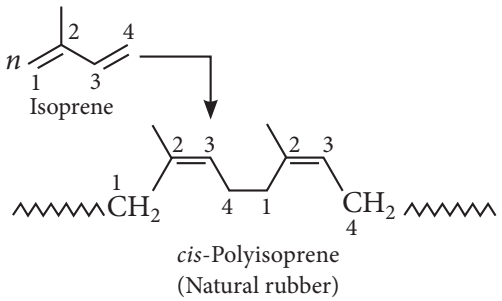


24. Natural polymers : Those polymers which are found in plants and animals are called natural polymers *e.g.*, proteins, starch, etc.

Synthetic polymers : Polymers which are prepared in laboratories are called synthetic polymers *e.g.*, plastics (polythene), synthetic fibres (nylon 6, 6), etc.

OR

Natural rubber is *cis*-polyisoprene and is obtained by 1, 4-polymerisation of isoprene units. In rubber molecule double bonds are located between C_2 and C_3 of each isoprene unit. These *cis*-double bonds do not allow the polymer chains to come closer for effective interactions and hence intermolecular forces are quite weak. Thus, natural rubber *i.e.*, *cis*-polyisoprene has a randomly coiled structure and hence shows elasticity.



25. Density, $\rho = \dfrac{Z \times M}{N_0 \times a^3}$; For *bcc*, $Z = 2$

$$\rho_{Fe} = \frac{2 \times 56.0}{6.023 \times 10^{23} \times (2.861 \times 10^{-8})^3} = 7.94 \text{ g cm}^{-3}$$

26. (a) Since the force of attraction between adsorbate and adsorbent is more in chemisorption as compared to physisorption, so the heat evolved is more, hence, enthalpy is higher.

(b) Free valencies present on the surface of solid catalyst combine temporarily with reacting molecules causing a strain on the system, as a result, reactivity of reactants is increased.

27. Paracetamol is usually preferred over aspirin because aspirin gets hydrolysed to salicylic acid in the stomach. The salicylic acid, thus produced may cause ulcer in the stomach walls, from where bleeding may take place. On the other hand, paracetamol do not cause any harm to the stomach walls.

28. Reducing sugars are those carbohydrates which can reduce reagents like Tollens' reagent (ammoniacal $AgNO_3$), Benedict's solution and Fehling's solution whereas, non-reducing sugars cannot reduce these reagents.

All monosaccharides whether aldoses or ketoses are reducing sugars.

The characteristic structural feature of reducing sugars is that they must have aldehydic group ($-CHO$) or ketonic group ($\diagdown C = O$) in the hemiacetal or hemiketal form.

29. (i) $2K_2MnF_6 + 4SbF_5 \longrightarrow 4KSbF_6 + 2MnF_4$

$2MnF_4 \longrightarrow 2MnF_3 + F_2$

$2K_2MnF_6 + 4SbF_5 \longrightarrow 4KSbF_6 + 2MnF_3 + F_2$

In this reaction, weak Lewis acid MnF_4 is displaced by stronger one, SbF_5 from K_2MnF_6. Being unstable in nature MnF_4 decomposes to MnF_3 and F_2.

(ii) Helium gas is used in gas cooled nuclear reactors and liquid helium is used as cryogenic agent for carrying out various reactions at low temperatures.

(iii) H_2SO_4 oxidises metals, non-metals and other compounds.

$S + 2H_2SO_4 \longrightarrow 3SO_2 + 2H_2O$

$2HI + H_2SO_4 \longrightarrow I_2 + SO_2 + 2H_2O$

OR

(i) SO_2 is obtained by heating copper with conc. H_2SO_4.

$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$

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It can also be obtained by boiling sulphur with conc. H_2SO_4.

$S + 2H_2SO_4 \longrightarrow 3SO_2 + 2H_2O$

(ii) H_2SO_4 when treated with P_2O_5 loses water and forms SO_3.

$H_2SO_4 + P_2O_5 \longrightarrow 2HPO_3 + SO_3$

(iii) SO_2Cl_2 is formed when conc. H_2SO_4 is treated with excess of PCl_5.

$H_2SO_4 + 2PCl_5 \longrightarrow SO_2Cl_2 + 2POCl_3 + 2HCl$

30. (i) The rate of S_N2 reaction depends on the tendency of leaving group to leave and the tendency of attacking nucleophile to attack. Iodide is a better leaving group because of its larger size than bromide therefore, ethyl iodide undergoes S_N2 reaction faster than ethyl bromide.

(ii) (±)-2-Butanol is a racemic mixture. It is a mixture which contains two enantiomers in equal proportions and thus, have zero optical rotation. Therefore, it is optically inactive.

(iii) Due to delocalisation of lone pair of electrons of the *X* atom over the benzene ring, C—*X* bond in halobenzene acquires some double bond character while in CH_3—*X*, C—*X* bond is a pure single bond. Therefore, C—*X* bond in halobenzene is shorter than in CH_3—*X*.

31. (i) Silica gel and alumina can act as stationary phase.

(ii) Component *C* is weakly adsorbed component thus, travels faster than *A* and *B* components with mobile phase and hence, eluted first.

(iii) Different components of the mixture are adsorbed at different rates on the surface of adsorbent. Thus, in mobile phase each component moves on stationary phase to different levels and gets separated.

OR

Coupled reactions : Many of the non-spontaneous reactions (ΔG = positive) can be made spontaneous by coupling these reactions with other reactions which have very large negative value of ΔG.

For example, consider the non-spontaneous reaction : $2Fe_2O_{3(s)} \rightarrow 4Fe_{(s)} + 3O_{2(l)}$,

$\Delta G^\circ = +1487.0 \text{ kJ mol}^{-1}$ ...(i)

Consider another spontaneous reaction :

$2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$, $\Delta G^\circ = -514.4 \text{ kJ mol}^{-1}$ ...(ii)

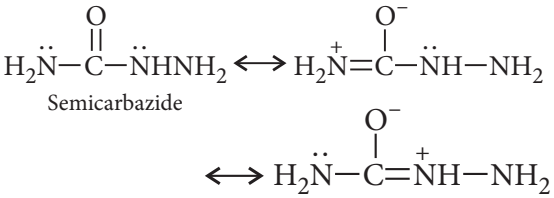
Multiplying equation (ii) by 3 and adding to equation (i),

$2Fe_2O_{3(s)} + 6CO_{(g)} \rightarrow 4Fe_{(s)} + 6CO_{2(g)}$

$\Delta G = 3 \times (-514.4) + 1487.0 = -56.2 \text{ kJ mol}^{-1}$

Thus, Fe_2O_3 can be reduced into Fe by CO spontaneously because ΔG° is negative for the overall reaction.

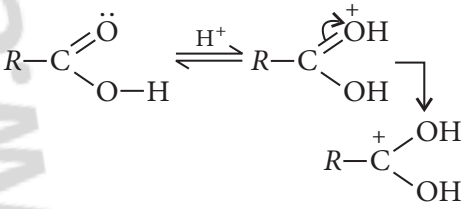
32. (i)



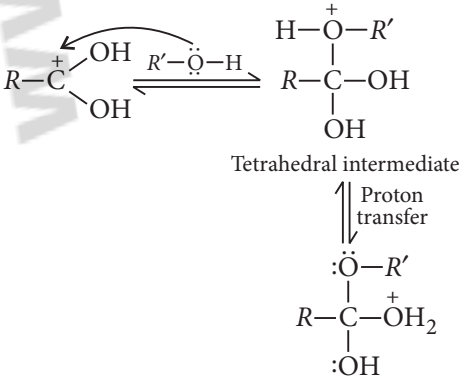
As electron density on one —NH_2 group decreases due to resonance, it does not act as a nucleophile while the lone pair of electrons on the other –NH_2NH_2 group (*i.e.*, attached to >NH) is not involved in resonance and hence is available for nucleophilic attack on the >C═O group of aldehydes and ketones.

(ii) **Mechanism of esterification :** It is a nucleophilic acyl substitution.

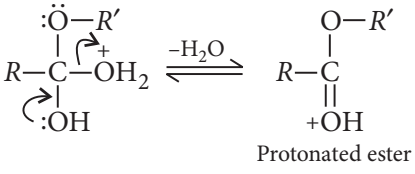
(a) Protonation of carboxyl oxygen :



(b) Nucleophilic addition of alcohol :



(c) Elimination of water molecule :



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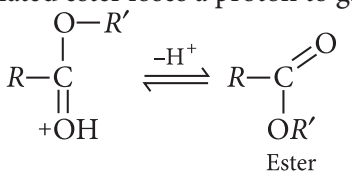
(d) Protonated ester loses a proton to give ester :



Ester

33. (i) $\text{Slope} = \frac{k}{2.303}$

(ii) As slope = $2 \times 10^{-4}\ s^{-1}$

$\therefore \quad k = 2.303 \times 2 \times 10^{-4}\ s^{-1} = 4.606 \times 10^{-4}\ s^{-1}$

(iii) For a first order reaction,

$$t = \frac{2.303}{k}\log\frac{[R]_0}{[R]} \quad \Rightarrow \quad \text{At } t_{1/2}, [R] = \frac{[R]_0}{2}$$

$$t_{1/2} = \frac{2.303}{k}\log 2 \quad \Rightarrow \quad t_{1/2} = \frac{0.693}{k}$$

34. Positive deviation : For non-ideal solution, if the vapour pressure is higher, then it is said to exhibit positive deviation. $A—B$ interactions are weaker than $A—A$ or $B—B$ interactions. Due to this, vapour pressure increases which results in positive deviation. In positive deviation, intermolecular force decreases, volume increases, vapour pressures increases, enthalpy of mixing increases. Therefore, ΔH_{mix} = +ve, ΔV_{mix} = + ve. *e.g.*, ethanol + acetone and carbon disulphide + acetone show positive deviation.

Negative deviation : For non-ideal solution, if the vapour pressure is lower, then it is said to exhibit negative deviation. $A—B$ interactions are stronger than $A—A$ and $B—B$ interactions. Due to this, vapour pressure decreases which results in negative deviation. In negative deviation, intermolecular force increases, volume decreases, vapour pressure decreases and heat is released. Therefore, ΔH_{mix} = –ve, ΔV_{mix} = – ve *e.g.*, phenol + aniline and chloroform + acetone show negative deviation.

35. (i) As transition metals have a large number of unpaired electrons in the *d*-orbitals of their atoms they have strong interatomic attractions or metallic bonds. Hence, they have high enthalpy of atomization.

(ii) Many transition metals and their compounds act as catalysts. The catalytic activity is due to their ability to exhibit multiple oxidation states. For example, V_2O_5 in Contact process and finely divided iron in Haber process.

(iii) The actinoid contraction is more than lanthanoid contraction due to poor shielding by 5*f*-electrons than by 4*f*-electrons.

(iv) Much larger third ionisation energy of Mn (where change is d^5 to d^4) is mainly responsible for this.

(v) Scandium (Z = 21) has incompletely filled 3*d*-orbitals in the ground state ($3d^1$). Hence it is considered as a transition element.

OR

(i) (a) $\underset{\text{Sod. chromate}}{2Na_2CrO_4} + H_2SO_4 \longrightarrow \underset{\text{Sod. dichromate}}{Na_2Cr_2O_7} + Na_2SO_4 + H_2O$

$Na_2Cr_2O_7 + 2KCl \longrightarrow \underset{\text{Pot. dichromate}}{K_2Cr_2O_7} + 2NaCl$

(b) The potassium manganate is oxidised to potassium permanganate by oxidation with chlorine.

$2K_2MnO_{4(aq)} + Cl_{2(g)} \rightarrow 2KMnO_{4(aq)} + 2KCl_{(g)}$

(ii) **Lanthanoid contraction :** The steady decrease in the atomic and ionic radii of lanthanoid elements with increase in atomic number is called lanthanoid contraction.

It is caused due to imperfect shielding of nuclear charge by 4*f*-electrons.

Lanthanum (La) has the largest ionic radius while lutetium (Lu) has the smallest among the 4*f*-series elements.

Consequences of lanthanoid contraction :

(a) The basic strength of oxides and hydroxides of lanthanoids decrease with increasing atomic number.

(b) Atomic and ionic sizes of 4*d* transition series elements and 5*d* series elements are similar. *e.g.*, atomic radii of zirconium(Zr) is same as that of hafnium(Hf).

36. (a) (i) Cell *B* will act as electrolytic cell as it has lower emf.

The electrode reaction will be

$Zn^{2+} + 2e^- \rightarrow Zn$ (Cathode)

$Cu \rightarrow Cu^{2+} + 2e^-$ (Anode)

(ii) With the values of emf, cell *B* will act as galvanic cell due to higher emf and will push electrons into cell *A*.

The electrode reactions will be

$Zn \rightarrow Zn^{2+} + 2e^-$ (Anode)

$Cu^{2+} + 2e^- \rightarrow Cu$ (Cathode)

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(b) $\Lambda_m^c = \frac{\kappa \times 1000}{C} = \frac{(7.896 \times 10^{-5}) \times 1000}{0.00241}$

$= 32.76 \text{ S cm}^2 \text{ mol}^{-1}$

$\alpha = \frac{\Lambda_m^c}{\Lambda_m^o} = \frac{32.76}{390.5} = 8.4 \times 10^{-2}$

$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{0.00241 \times (8.4 \times 10^{-2})^2}{1-(8.4 \times 10^{-2})}$

$= 1.86 \times 10^{-5}$

OR

(a) 0.4 g of $Cu^{2+} = \frac{0.4}{31.75}$

$= 0.0126$ g equivalent

At the same time, the oxygen deposited at anode

$= 0.0126$ g equivalent $= \frac{8}{32} \times 0.0126$

$= 0.00315$ g mol

After the complete deposition of copper, the electrolysis will discharge hydrogen at cathode and oxygen at anode.

The amount of charge passed $= 1.2 \times 7 \times 60$

$= 504$ coulomb

So, oxygen liberated $= \frac{1}{96500} \times 504$

$= 0.00522$ g equivalent

$= \frac{8}{32} \times 0.00523 = 0.001307$ g mol

Hydrogen liberated = 0.00522 g equivalent

$= \frac{1}{2} \times 0.00522 = 0.00261$ g mol

Total gases evolved = (0.00315 + 0.001307 + 0.00261) g mol

$= 0.007067$ g mol

Volume of gases evolved at NTP

$= 22400 \times 0.007067$ mL $= 158.3$ mL

(b) (i) $2H^+ + 2e^- \rightarrow H_2\uparrow$ (Cathode)

$2Cl^- \rightarrow Cl_2 + 2e^-$ (Anode)

(ii) $2H^+ + 2e^- \rightarrow H_2\uparrow$ (Cathode)

$2Br^- \rightarrow Br_2 + 2e^-$ (Anode)

(iii) $2H^+ + 2e^- \rightarrow H_2\uparrow$ (Cathode)

$2OH^- \rightarrow H_2O + \frac{1}{2}O_2 + 2e^-$ (Anode)

37. (a) (i) Br_2/H_2O

(ii) BH_3 in THF/H_2O_2

(b) (i) *p*-Nitrophenol > phenol > ethanol

(ii) Propanol > propanal > propane

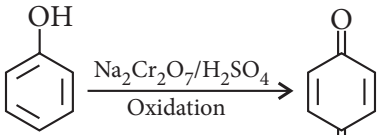
(c) $CH_3-CH_2-\ddot{O}H \xrightarrow{H^+}$

$CH_3-CH_2-\overset{+}{O}H_2 \xrightarrow[-H_2O]{CH_3-CH_2-\ddot{O}H}$ (S_N2 mechanism)

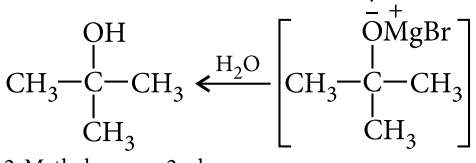
$CH_3-CH_2-\overset{+}{O}(H)-CH_2-CH_3 \xrightarrow{-H^+}$

$CH_3-CH_2-O-CH_2-CH_3$

OR

(a) (i) 

Phenol Benzoquinone

(ii) $CH_3-\overset{\overset{O}{\curvearrowleft}}{C}-CH_3 + CH_3MgBr \rightarrow$ 

2-Methylpropan-2-ol

(iii) $CH_3CH=CH_2 \xrightarrow[H_2O]{H_2SO_4} CH_3-CH(OH)-CH_3$

Propene Propan-2-ol

(b) (i) Zinc dust

(ii) Concentrated H_2SO_4

(iii) CH_3Cl and $AlCl_3$ (anhyd.)

(iv) Cu/573 K

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PRACTICE PAPER 3
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

The graphical representation of Gibbs energy change *vs* absolute temperature for a process is known as Ellingham diagram. Generally, the diagram consists of plots of $\Delta_f G^\circ$ *vs* T for the formation of oxides of elements.

$$2xM_{(s)} + O_{2(g)} \rightarrow 2M_xO_{(s)}$$

Ellingham diagrams help us in predicting the feasibility of thermal reduction of an ore.

1. What is the thermodynamic criterion for the extraction of metal?

2. How is Ellingham diagram helpful in the metallurgy process?

3. Give reason for the following :
Zinc oxide can be reduced to metal by heating with carbon but Cr_2O_3 cannot be reduced by heating with carbon.

4. Although thermodynamically feasible, in practice, magnesium metal is not used for reduction of alumina in the metallurgy of aluminium. Why?

5. What are the Limitations of Ellingham diagram?

Questions 6 to 10 are one word answers :

6. If three elements X, Y and Z crystallized in cubic solid state with X atoms at corners, Y atoms at cube centre and Z atoms at the edge, write the formula of the compound.

7. What is the potential of S.H.E?

8. Name one poisonous gas which can be prepared from chlorine gas.

9. Write down the IUPAC name of the lowest molecular mass tertiary amine.

10. A reaction is 50% completed in 2 hours and 75% completed in 4 hours. What is the order of the reaction?

Questions 11 to 15 are multiple choice questions :

11. If a current of 1.5 ampere flows through a metallic wire for 3 hours, then how many electrons would flow through the wire?
(a) 2.25×10^{22} electrons
(b) 1.13×10^{23} electrons
(c) 1.01×10^{23} electrons
(d) 4.5×10^{23} electrons

12. Which of the following polymers are used as fibre?
(a) Nylon
(b) Polytetrafluoroethane
(c) Terylene (d) Buna-S

13. Arrange the following compounds in decreasing order of their boiling points.
(i) CH_3Br
(ii) CH_3CH_2Br

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(iii) $CH_3CH_2CH_2Br$
(iv) $CH_3CH_2CH_2CH_2Br$
(a) (i) > (ii) > (iii) > (iv)
(b) (iv) > (iii) > (ii) > (i)
(c) (i) > (iii) > (ii) > (iv)
(d) (iii) > (iv) > (i) > (ii)

14. A solution containing 12.5 g of non-electrolyte substance in 185 g of water shows boiling point elevation of 0.80 K. Calculate the molar mass of the substance.
(K_b = 0.52 K kg mol^{-1})
(a) 53.06 g mol^{-1} (b) 25.3 g mol^{-1}
(c) 16.08 g mol^{-1} (d) 43.92 g mol^{-1}

15. The ionic radius of Cl^- ion is 1.81 Å. The interionic distances of NaCl and NaF are 2.79 Å and 2.31 Å respectively. The ionic radius of F^- ion will be
(a) 0.98 Å (b) 0.80 Å
(c) 1.33 Å (d) 2.29 Å

Questions 16 to 20 :
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : Boiling points of alcohols are lower than hydrocarbons.
Reason (R) : Among isomeric alcohols, boiling point decreases in the order : 1° > 2° > 3°.

17. Assertion (A) : According to Freundlich; $\frac{x}{m} = kP^{1/n}$.
Reason (R) : The isotherm shows variation of the amount of gas adsorbed by the adsorbent with temperature.

18. Assertion (A) : Ofloxacin is a bactericidal type of antibiotic.
Reason (R) : Ofloxacin has killing effect on microbes.

19. Assertion (A) : Valine is an essential amino acid.
Reason (R) : The lack of essential amino acids in the diet causes Kwashiorkor.

20. Assertion (A) : Members of 4*d* and 5*d* series of transition elements have nearly same atomic radii.
Reason (R) : Atomic and ionic radii for transition elements are smaller than their corresponding *s*-block elements.

SECTION - B

21. The partial pressure of ethane over a saturated solution containing 6.56×10^{-3} g of ethane is 1 bar. If the solution contains 5.00×10^{-2} g of ethane, then what would be the partial pressure of the gas?

22. Explain the following observations :
(i) Many of the transition elements are known to form interstitial compounds.
(ii) There is a general increase in density from titanium (*Z*=22) to copper (*Z* = 29).

23. (a) What are the constituents of starch?
(b) Write zwitter ion form of aminoacetic acid.

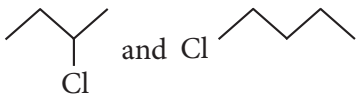
24. Describe the properties of an ore which is to be concentrated by (i) leaching with alkali (ii) floatation (iii) poling (iv) electromagnetic separation.

OR

Name one metal that is refined by each of the following processes : (i) Mond process (ii) electrolysis (iii) van Arkel process (iv) zone refining.

25. Which forces are involved in holding the drugs to the active sites of enzymes?

26. (i) Identify the chiral molecule in the following :



(ii) Out of S_N1 and S_N2, which reaction occurs with :
(a) inversion of configuration
(b) racemisation.

OR

Give one reaction each to convert benzyl bromide into :
(i) benzyl ethyl ether
(ii) benzyl cyanide.

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27. How are the following conversions carried out?
(i) Phenol to toluene
(ii) Ethanol to 1,1-dichloroethane

SECTION - C

28. Write the chemical equations for synthesis of following polymers :
(i) Teflon (ii) PVC
(iii) Glyptal

29. Give reason for the following :
(i) Aniline is a weaker base than cyclohexyl-amine.
(ii) Ammonolysis of alkyl halide does not give a corresponding amine in pure state.

30. Explain the following phenomenon giving reasons :
(i) Rate of physical adsorption decreases with rise of temperature.
(ii) Brownian movement
(iii) Colloidal particles scatter light

31. Give reasons for the following :
(i) In PCl_5, phosphorus is in sp^3d hybridised state but all its five bonds are not equivalent.
(ii) Fluorine show abnormal behaviour.
(iii) OF_2 is called oxygen difluoride and not fluorine oxide.

OR

(i) NH_3 has higher boiling point than PH_3. Give reasons.
(ii) Which poisonous gas is evolved when white phosphorus is heated with conc. NaOH solution? Write the chemical equation.

32. How are the following conversions carried out?
(i) Propene to propan-2-ol
(ii) Benzyl chloride to benzyl alcohol
(iii) Ethyl magnesium chloride to propan-1-ol

33. A zinc rod is dipped in 0.1 M solution of $ZnSO_4$. The salt is 95% dissociated at this dilution at 298 K. Calculate the electrode potential ($E^{\circ}_{Zn^{2+}/Zn}$ = –0.76 V).

OR

(i) What is mercury cell? Give the electrode reaction.
(ii) What would happen if the protective tin coating over an iron bucket is broken in some place?

34. A bottle of commercial sulphuric acid (density = 1.787 g/mL) is labelled as 86 percent by weight.
(i) What is the molarity of the acid?
(ii) What volume of the acid has to be used to make 1 L of 0.2 M H_2SO_4?
(iii) What is the molality of the acid?

SECTION - D

35. (i) The decomposition of Cl_2O_7 at 400 K in the gas phase to Cl_2 and O_2 is a first order reaction.
(a) After 50 seconds at 400 K, the pressure of Cl_2O_7 falls from 0.062 to 0.044 atm. Calculate the rate constant.
(b) Calculate the pressure of Cl_2O_7 after 100 sec of decomposition at this temperature.
(ii) One-fourth of a first order reaction is completed in 32 minutes. What is the half-life period of this reaction?

OR

(a) Explain the following statements :
(i) Average rate of reaction does not give the true picture of the reaction rate.
(ii) A lump of coal burns at a moderate rate in air while coal dust burns explosively.
(iii) In some cases it is found that a large number of colliding molecules have energy more than threshold energy, yet reaction is slow.
(iv) Hydrogen and oxygen do not react at room temperature.
(b) How is rate constant related to concentration of the reactants?

36. (a) (i) Carboxylic acid is a stronger acid than phenol. Why?
(ii) Arrange the following compounds in decreasing order of their acidic strength :
C_6H_5COOH, CH_3COOH, HCOOH
(b) An organic compound (*A*) having molecular formula ($C_8H_{16}O_2$) was hydrolysed with dilute sulphuric acid to give two compounds (*B*) and (*C*). Oxidation of (*C*) with chromic acid produced (*B*). (*C*) on dehydration gives but-1-ene. Write equations for the reactions involved.

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OR

(a) How will you prepare the following compounds starting with benzene:
  (i) Benzaldehyde
  (ii) Acetophenone?

(b) Write the following conversions in not more than two steps :
  (i) Benzaldehdye to α-hydroxyphenylacetic acid
  (ii) Bromobenzene to 1-phenylethanol
  (iii) Benzaldehyde to benzophenone

37. Explain why :

(i) $[Cr(NH_3)_6]^{3+}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic.

(ii) A solution of $[Ni(H_2O)_6]^{2+}$ is green but $[Ni(CN)_4]^{2-}$ is colourless.

(iii) The low spin tetrahedral complexes are rarely observed.

OR

(a) Give one example of each :
  (i) Neutral ligand of carbon
  (ii) Negative ligand of carbon
  (iii) Hexadentate ligand

(b) What are t_2g and eg orbitals?

Hints & Solutions

1. $\Delta G°$ for net reduction reaction must be negative.

2. In Ellingham diagram, elements present below in the diagram reduce the elements present above in the diagram. Thus, it helps in selection of suitable reducing agents as well as optimum temperature.

3. Carbon is suitable reducing agent for reduction of zinc oxide. Reduction of Cr_2O_3 by carbon is not thermodynamically favourable.

4. Below the temperature (1623 K) corresponding to the point of intersection of Al_2O_3 and MgO curves in Ellingham diagram magnesium can reduce alumina. But magnesium is a much costlier metal than aluminium and hence the process will be uneconomical.

5. Limitations of Ellingham diagram :

(i) It does not say about kinetics of reaction.

(ii) It presumes that reactants and products are in equilibrium. But, this is not always true because reactant / product may be solid.

6. XYZ_3

7. Zero

8. Phosgene

9. *N*,*N*-Dimethylmethanamine

10. First order

11. (c) **12. (a)** **13. (b)**

14. (d) :

$$M_B = \frac{K_b \times W_B}{\Delta T_b \times W_A} = \frac{0.52 \times 12.5}{0.80 \times 0.185} = 43.92 \text{ g mol}^{-1}$$

($\because$ 185 g = 0.185 kg)

16. (d) : Alcohols undergo intermolecular H-bonding and hence, their boiling points are higher than those of hydrocarbons.

Among isomeric alcohols, boiling point decreases in the order : 1° > 2° > 3°.

17. (c) : Freundlich adsorption isotherm gives an empirical relationship between the quantity of gas adsorbed by unit mass of solid adsorbent $\left(\frac{x}{m}\right)$ and pressure at a particular temperature.

18. (a)

19. (b) : Valine is an essential amino acid. The amino acids which cannot be synthesised in the body are called essential amino acid.

20. (b) : It is due to lanthanide contraction.

21. According to Henry's law, $m = K_H \times p$

Case I : 6.56×10^{-3} g = $K_H \times 1$ bar

or, $K_H = 6.56 \times 10^{-3}$ g bar^{-1}

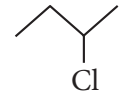
Case II : 5.00×10^{-2} g = $(6.56 \times 10^{-3}$ g bar$^{-1}) \times p$

or $p = \dfrac{5.00\times10^{-2}\,\text{g}}{6.56\times10^{-3}\,\text{g bar}^{-1}} = 7.62\ \text{bar}$

24. **OR**

(i) Ni (ii) Cu (iii) Zr (iv) Ga

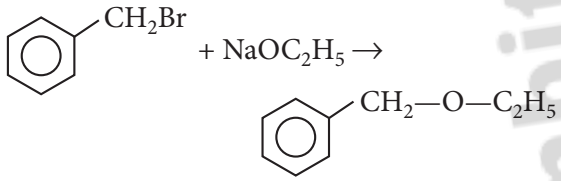
25. Hydrogen bonding, ionic bonding, dipole-dipole interactions and van der Waals' interactions.

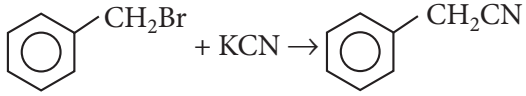
26. (i) 

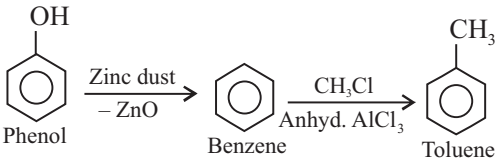
2-Chlorobutane

(ii) (a) S_N2 (b) S_N1

OR

(i) 

(ii) 

27. (i) 

Phenol Benzene Toluene

(ii) $CH_3CH_2OH \xrightarrow{PCC} CH_3C(=O)H \xrightarrow{PCl_5} CH_3CHCl_2$

Ethanol Ethanal 1,1-Dichloroethane

29. (i) In case of aniline, the lone pair of electrons on the nitrogen atom is delocalized over the benzene ring.

(ii) $RNH_2 \xrightarrow{RX} R_2NH \xrightarrow{RX} R_3N \xrightarrow{RX} R_4\overset{+}{N}\overset{-}{X}$

1° 2° 3° Quaternary ammonium salt

This mixture is difficult to separate.

33. The electrode reaction written as reduction reaction is $Zn^{2+} + 2e^- \rightarrow Zn$ ($n = 2$)

Applying Nernst equation, we get

$$E_{Zn^{2+}/Zn} = E^\circ_{Zn^{2+}/Zn} - \frac{0.0591}{2}\log\frac{1}{[Zn^{2+}]}$$

As 0.1 M $ZnSO_4$ solution is 95% dissociated, this means that in the solution,

$$[Zn^{2+}] = \frac{95}{100}\times0.1\ \text{M} = 0.095\ \text{M}$$

$$\therefore\quad E_{Zn^{2+}/Zn} = -0.76 - \frac{0.0591}{2}\log\frac{1}{0.095}$$

$= -0.76 - 0.02955\ (\log 1000 - \log 95)$

$= -0.76 - 0.02955\ (3 - 1.9777)$

$= -0.76 - 0.03021 = -0.79021\ \text{V}$

OR

(i) Mercury cell is a type of dry cell. It is used in small electrical equipments.

Zinc-mercury amalgam is used as anode. A paste of HgO and carbon is used as cathode. The electrolyte is a paste of KOH and ZnO.

Cell reaction :

At anode : $Zn(Hg) + 2OH^- \rightarrow ZnO_{(s)} + H_2O + 2e^-$

At cathode : $HgO_{(s)} + H_2O + 2e^- \rightarrow Hg_{(l)} + 2OH^-$

Net reaction : $Zn(Hg) + HgO \rightarrow ZnO_{(s)} + Hg_{(l)}$

Cell reaction does not involve any of the ions whose concentration changes. Hence, it gives constant potential throughout its use.

(ii) If the protective tin coating over an iron bucket is broken in some places then iron corrodes faster than it does in the absence of tin as oxidation potential of Fe is higher than that of Sn.

34. (i) 86% H_2SO_4 means 100 g of the solution contains 86 g of sulphuric acid.

Volume of the 100 g of acid

$$= \frac{\text{Mass}}{\text{Density}} = \frac{100\ \text{g}}{1.787\ \text{g/mL}} = 55.9\ \text{mL}$$

$$\text{Moles of solute} = \frac{86}{98}$$

$$\text{Molarity} = \frac{\text{Moles of solute}\times1000}{\text{Vol. of solution (mL)}}$$

$$= \frac{86}{98}\times\frac{1000}{55.9} = 15.7\,\text{M}$$

(ii) $M_1V_1 = M_2V_2$

$15.7 \times V_1 = 0.2 \times 1000$

$$\therefore\quad V_1 = \frac{0.2\times1000}{15.7} = 12.74\ \text{mL}$$

(iii) Mass of the solution = 100 g

Mass of the solute = 86 g

Mass of the solvent = 100 – 86 = 14 g

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$$\text{Molality} = \frac{\text{Moles of solute} \times 1000}{\text{Wt. of solvent (g)}}$$

$$= \frac{86}{98} \times \frac{1000}{14} = 62.68 \text{ m}$$

35. (i) $a \propto P_0$ and $(a - x) \propto P_t$

Hence, $k = \frac{2.303}{t} \log \frac{P_0}{P_t}$ (P_t = pressure of Cl_2O_7 at time t)

(a) $t = 50$ s, $P_0 = 0.062$ atm, $P_t = 0.044$ atm

$$k = \frac{2.303}{50 \text{ s}} \log \frac{0.062 \text{ atm}}{0.044 \text{ atm}} = 6.86 \times 10^{-3} \text{ s}^{-1}$$

(b) $t = 100$ s, $P_t = ?$, $k = 6.86 \times 10^{-3}$ s^{-1}

$$6.86 \times 10^{-3} \text{ s}^{-1} = \frac{2.303}{100 \text{ s}} \log \frac{0.062 \text{ atm}}{P_t}$$

$$\log \frac{0.062}{P_t} = 0.2979$$

$$\Rightarrow \quad \frac{0.062}{P_t} = \text{Antilog } 0.2979 = 1.986$$

$$\therefore \quad P_t = 0.0312 \text{ atm}$$

(ii) $k = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{32 \text{ min}} \log \frac{a}{a - a/4}$

$$= \frac{2.303}{32} \log \frac{4}{3} \text{min}^{-1} = 0.009 \text{ min}^{-1}$$

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.009} = 77 \text{ min}$$

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PRACTICE PAPER 4
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

X_2 is greenish yellow gas with an offensive smell used in water purification. It partially dissolves in water to give a solution which turns blue litmus red. When X_2 is passed through NaBr, Br_2 is obtained.

1. Identify X_2.

2. Name the group to which it belongs.

3. Write general electronic configuration of elements of this group.

4. Write chemical equations when X_2 reacts with H_2O.

5. What happens when X_2 reacts with hot and conc. NaOH? Give equation.

Questions 6 to 10 are one word answers :

6. Which concentration term remains unaffected by temperature?

7. Which type of reaction occurs between tertiary butyl alcohol and hydrogen iodide?

8. For the reaction, $Cl_{2(g)} + 2NO_{(g)} \rightarrow 2NOCl_{(g)}$ the rate law is expressed as
Rate = $k[Cl_2]\,[NO]^2$
What is the overall order of this reaction?

9. Name the battery used for low current devices like watches and hearing aids.

10. Name the by-product of manufacture of phenol from cumene.

Questions 11 to 15 are multiple choice questions :

11. During the formation of the N_2O_4 dimer from two molecules of NO_2, the odd electrons, one in each of the nitrogen atoms of the NO_2 molecules, get paired to form a
(a) weak N – N bond, two N – O bonds become equivalent and the other two N – O bonds become nonequivalent
(b) weak N – N bond and all the four N – O bonds become equivalent
(c) strong N – N bond and all the four N – O bonds become non-equivalent
(d) strong N – N bond and all the four N–O bonds become equivalent.

12. The expression to calculate time required for completion of zero order reaction is

(a) $t = \frac{[R]_0}{k}$ (b) $t = [R] - [R]_0$

(c) $t = \frac{k}{[R]_0}$ (d) $t = \frac{[R]_0 - [R]}{[R]_0}$

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13. Glucose $\xrightarrow{HCN}$ $\xrightarrow{Hydrolysis}$ $\xrightarrow{HI/heat}$ *A*

A is

(a) heptanoic acid (b) 2-Iodohexane

(c) heptane (d) heptanol.

14. Which is not the correct statement for ionic solids in which positive and negative ions are held by strong electrostatic attractive forces?

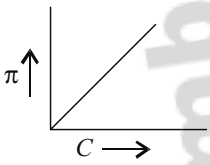
(a) The radius ratio r_+/r_- increases as coordination number increases.

(b) As the difference in size of ions increases, coordination number increases.

(c) When coordination number is eight, r_+/r_- ratio lies between 0.225 to 0.414.

(d) In ionic solid of the type *AX* (ZnS, Wurtzite), the coordination number of Zn^{2+} and S^{2-} respectively are 4 and 4.

15. A graph showing variation of osmotic pressure (π) versus molar concentration '*C*' of an aqueous solution at temperature *T* is given.



The slope of the line represents

(a) solution constant *R*

(b) absolute temperature *T*

(c) *RT*

(d) degree of ionisation.

Questions 16 to 20 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is correct, but Reason (R) is incorrect statement.

(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : 96500 C charge is required for the reduction of one mole of silver ions.
Reason (R) : The amount of electricity (or charge) required for oxidation or reduction depends on the stoichiometry of the electrode reaction.

17. Assertion (A) : Penicillin is an antibiotic.
Reason (R) : The drugs which act on the central nervous system and help in reducing anxiety are called antibiotics.

18. Assertion (A) : If blood cells are placed in pure water, they swell and burst.
Reason (R) : Due to osmosis, the movement of water molecules into the cell, dilutes the salt content.

19. Assertion (A) : Carbonyl compounds react with phenyl hydrazine to form phenyl hydrazones.
Reason (R) : Hydrazones of aldehydes and ketones are prepared in highly acidic medium.

20. Assertion (A) : Bakelite is a thermosetting plastic.
Reason (R) : In thermosetting plastics, the polymeric chains are held together by strong covalent bonds.

SECTION - B

21. Arrange the following carbonyl compounds in increasing order of their reactivity in nucleophilic addition reaction.
Benzaldehyde, *p*-tolualdehyde, *p*-nitro-benzaldehyde, acetophenone.

22. Write the name of the polymer which is used for making non-stick utensils. Write equation for the preparation of this polymer.

23. Predict the products of electrolysis obtained at the electrodes in each case when the electrodes used are of platinum :

(i) An aqueous solution of $AgNO_3$.

(ii) An aqueous solution of H_2SO_4.

24. (i) Give the electronic configuration of the *d*-orbitals of Ti in $[Ti(H_2O)_6]^{3+}$ ion in an octahedral crystal field.

(ii) Why is the given complex coloured? Explain on the basis of distribution of electrons in the *d*-orbitals.

OR

(i) On the basis of crystal field theory, write the electronic configuration of d^4 ion if $\Delta_o < P$.

(ii) Write the hybridization and magnetic behaviour of the complex $[Ni(CO)_4]$. (At. no. of Ni = 28)

25. (a) What happens when protein is denatured?

(b) Name the purines present in DNA.

OR

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Differentiate between nucleotides of DNA and RNA.

26. A solid has *bcc* structure. Calculate the edge length of the cell if the distance of closest approach between two atoms is 1.73 Å.

27. What are fillers and what role these fillers play in soap?

SECTION - C

28. Show how will you synthesize
(i) 1-Phenylethanol from a suitable alkene.
(ii) Cyclohexylmethanol using an alkyl halide by S_N2 reaction.
(iii) Pentan-1-ol using a suitable alkyl halide.

OR

How would you account for the following :
(i) Phenols are much more acidic than alcohols?
(ii) The boiling points of ethers are much lower than those of the alcohols of comparable molecular masses?

29. State Henry's law for solubility of a gas in a liquid and give its important applications.

30. You are provided with the following four reagents $LiAlH_4$, I_2/NaOH, $NaHSO_3$ and Schiff's reagents. Write which two reagents can be used to distinguish between the compounds in each of the following pairs :
(i) CH_3CHO and CH_3COCH_3
(ii) CH_3CHO and C_6H_5CHO
(iii) $C_6H_5COCH_3$ and $C_6H_5COC_6H_5$.

31. Explain why
(i) the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.
(ii) alkyl halides, though polar, are immiscible with water.
(iii) Grignard reagents should be prepared under anhydrous conditions.

OR

(i) Why the boiling points of isomeric haloalkanes decrease with increasing branching?
(ii) Why iodoform behaves as antiseptic?
(iii) Aryl halides cannot be prepared by the reaction of hydrogen halide on phenol. Why?

32. (i) Give the IUPAC name of $[PtCl(NH_2CH_3)(NH_3)_2]Cl$.
(ii) Compare the magnetic behaviour of the complex entities $[Fe(CN)_6]^{4-}$ and $[FeF_6]^{3-}$. [Fe = 26].

33. Complete the following chemical equations :
(i) $MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow$
(ii) $KMnO_4 \xrightarrow{Heat}$
(iii) $Cr_2O_7^{2-} + H_2S + H^+ \longrightarrow$

34. (a) Express the relationship between the rate of production of iodine and the rate of disappearance of hydrogen iodide in the following reaction :

$$2HI \rightarrow H_2 + I_2$$

(b) The rate of formation of a dimer in a second order dimerisation reaction is 9.1×10^{-6} mol $L^{-1}s^{-1}$ at 0.01 mol L^{-1} monomer concentration. Calculate the rate constant.

SECTION - D

35. (i) An organic compound (*A*) with molecular formula $C_9H_{13}N$ dissolves in dil. HCl and releases N_2 with nitrous acid giving an optically active alcohol (*B*). The alcohol (*B*) on oxidation gives dicarboxylic acid which on heating forms an anhydride. Identify the compounds (*A*) and (*B*).
(ii) Give plausible explanations for the following :
(a) Aromatic primary amines cannot be prepared by Gabriel phthalimide synthesis.
(b) Tertiary amines do not undergo acylation reaction.

OR

(i) An aromatic compound, C_7H_8 (*A*) on nitration gives (*B*) as a major product which on reduction with Sn/HCl gives (*C*). The compound (*C*) on treatment

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with $NaNO_2$/HCl at 273 K following by subsequent treatment with CuCl/HCl gives (*D*). When (*D*) is oxidised, it forms *ortho*-substituted monocarboxylic acid with the formula $C_7H_5O_2Cl$. Predict the structures of (*A*), (*B*), (*C*) and (*D*).

(ii) Outline the steps for the conversion of methylcyanide to methylamine.

36. State the principles of the following methods of refining crude metals :

(i) Zone refining

(ii) Liquation method

(iii) Chromatographic method

(iv) Mond process for refining of nickel

(v) Electrolytic refining

OR

(i) Write the principle behind the froth floatation process. What is the role of collectors in this process?

(ii) Giving examples differentiate between calcination and roasting.

(iii) What is the role of silica in the extraction of copper ?

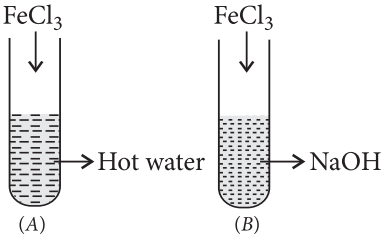
37. (i) What type of colloidal sols are formed in the following?

(a) Sulphur vapours are passed through cold water

(b) White of an egg is mixed with water.

(c) Soap solution

(ii) A colloidal solution of ferric oxide is prepared by two different methods as shown :



(a) What is the charge on colloidal particles in the two test tubes (*A*) and (*B*)?

(b) Give reasons for the origin of charge.

OR

(i) Explain the following observations :

(a) Sun looks red at the time of sunset.

(b) Rate of physical adsorption decreases with rise in temperature.

(c) Physical adsorption is multilayered while chemical adsorption is mono-layered.

(ii) 2.0 g of charcoal is placed in 100 mL of 0.5 M CH_3COOH to form an adsorbed mono-acidic layer of acetic acid molecules and thereby the molarity of CH_3COOH reduces to 0.49 M. The surface area of charcoal is 3×10^2 m^2 g^{-1}. Calculate the surface area of charcoal adsorbed by each molecule of acetic acid.

Hints & Solutions

1. X_2 must be Cl_2 (chlorine) gas as it is greenish yellow gas with offensive smell. Since, it is more reactive than Br_2, so, it can displace Br from NaBr in the form of Br_2.

2. Chlorine belongs to group-17.

3. ns^2np^5 (valence shell)

4. Cl_2 reacts with water giving acidic solution that turns blue litmus red.

$$Cl_2 + H_2O \longrightarrow \underset{\text{(Acidic solution)}}{HClO + HCl}$$

5. When Cl_2 reacts with hot and conc. NaOH, it undergoes disproportionation reaction.

$$6NaOH + 3Cl_2 \rightarrow 5\overset{-1}{NaCl} + \underset{\text{(Sod. chlorate)}}{\overset{+5}{NaClO_3}} + 3H_2O$$
(Hot and conc.)

6. Molality

7. S_N1 reaction

8. Three

9. Mercury cell

10. Acetone

11. **(b)**

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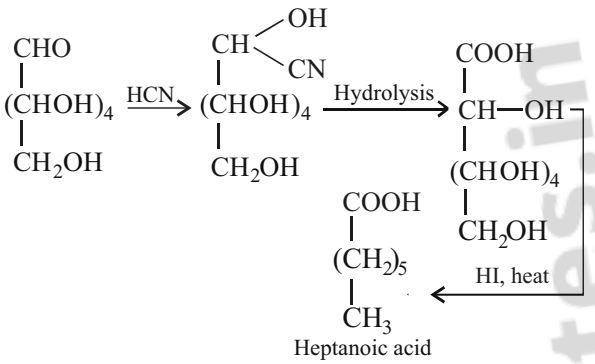
12. (a) : $[R] = [R]_0 - kt$

For completion of reaction, $[R] = 0$

or $t = \dfrac{[R]_0}{k}$

13. (a) :



14. (c) : When coordination number is 8, r_+/r_- ratio lies between 0.732 – 1.

15. (c)

16. (a) : $Ag^+_{(aq)} + e^- \rightarrow Ag_{(s)}$

Charge on one mole of electrons

$= 6.02 \times 10^{23} \times 1.6021 \times 10^{-19}$

$= 96446\ C\ mol^{-1} \approx 96500\ C\ mol^{-1}$

17. (c) : The drugs which act on the central nervous system and help in reducing anxiety are called tranquilizers.

18. (b) : Osmosis is a phenomenon in which the solvent flows from a solution of lower concentration to the solution of higher concentration.

Here, movement of water molecules takes place in highly concentrated salt content of blood cell to such an extent that the cell membrane ruptures.

20. (b) : Thermosetting plastics are cross-linked polymers formed from their monomers by condensation polymerisation. In these polymers, the chains are held together by strong covalent bonds. Bakelite is also a condensation polymer and formed by condensation polymerisation of HCHO and phenol.

23. (i) $AgNO_{3(aq)} \longrightarrow Ag^+_{(aq)} + NO^-_{3(aq)}$

$H_2O_{(l)} \longrightarrow H^+_{(aq)} + OH^-_{(aq)}$

At cathode: $Ag^+_{(aq)} + e^- \longrightarrow Ag_{(s)}$

At anode: $2OH^-_{(aq)} \longrightarrow O_2 + 2H^+_{(aq)} + 4e^-$

or $2H_2O_{(l)} \longrightarrow O_2 + 4H^+_{(aq)} + 4e^-$

(ii) $H_2SO_{4(aq)} \longrightarrow 2H^+_{(aq)} + SO^{2-}_{4(aq)}$

$H_2O_{(l)} \longrightarrow H^+_{(aq)} + OH^-_{(aq)}$

At cathode: $2H^+_{(aq)} + 2e^- \longrightarrow H_{2(g)}$

At anode: $2OH^-_{(aq)} \longrightarrow O_2 + 2H^+_{(aq)} + 4e^-$

or $2H_2O_{(l)} \longrightarrow O_2 + 4H^+_{(aq)} + 4e^-$

25. (a) Protein loses its biological activity due to changes in 2° and 3° structures.

(b) Adenine and guanine

26. Nearest neighbour distance (d) = 1.73 Å

For *bcc*, $d = \dfrac{\sqrt{3}}{2}a$ (a = edge length)

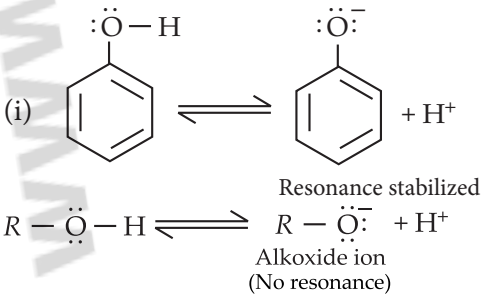
$$a = \frac{1.73 \times 2}{\sqrt{3}} = 2\,\text{Å or}\,200\ \text{pm}$$

27. Substances which are added to soap to change their properties in order to make them more useful for a particular application, are called fillers. For example,

(i) Sodium rosinate, sodium silicate, borax and sodium carbonate are added to laundry soaps to increase their foaming ability.

(ii) Glycerol is added to shaving soaps to prevent them from rapid drying.

28. **OR**



(ii) Ethers have low polarity and as a result do not show any association by intermolecular hydrogen bonding.

30. (i) $LiAlH_4$, Schiff's reagent

(ii) I_2/NaOH, $LiAlH_4$

(iii) I_2/NaOH, $LiAlH_4$

31. **OR**

(i) As the branching increases molecule tends to acquire spherical shape *i.e.*, its surface area decreases, correspondingly the magnitude of van

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der Waals' force also decreases. So, boiling point also decreases.

(ii) Iodoform behaves as antispetic due to liberation of free iodine.

(iii) Due to resonance, in phenol molecule C–O bond has some double bond character due to which it is difficult to break it. Hence, aryl halide cannot be prepared by reaction of hydrogen halide on phenol.

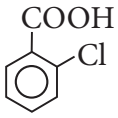
34. (a) Rate of production of iodine

$$= \frac{1}{2} \times \text{Rate of disappearance of HI}$$

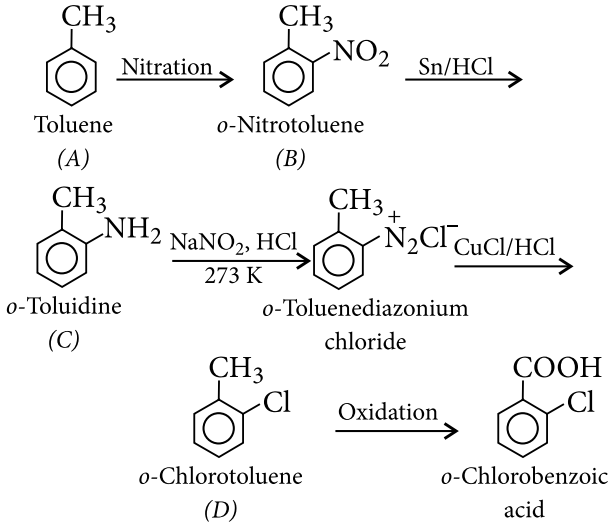
$$\frac{d[I_2]}{dt} = \frac{1}{2} \times \left(\frac{-d[HI]}{dt} \right)$$

35. **OR**

(i) The final product is an *ortho*-substituted monocarboxylic acid. Therefore, it is monocarboxylic aromatic acid with chloro group at the *ortho*-position. Therefore, it is



Thus, the sequence of the reactions can be visualised as follows :



(ii) $CH_3CN + H_2O \xrightarrow{NaOH} CH_3CONH_2$

Methyl cyanide

$CH_3CONH_2 \xrightarrow[Br_2 + aq.\ KOH]{\text{Hoffmann's bromamide reaction}} CH_3NH_2$

Methylamine

36. (i) Zone refining is based on the principle that the impurities are more soluble in the melt than in the solid state of the metal.

(ii) **Refining by liquation :** In this method a low melting metal like tin is heated on a sloping surface of a reverberatory furnace. Metal melts and flows down leaving impurities behind.

(iii) Chromatographic method is based on the principle that different components of a mixture are differently adsorbed on an adsorbent. The adsorbed components are removed (eluted) by using suitable eluent. There are several chromatographic techniques such as paper chromatography, column chromotagraphy, gas chromatography, etc.

(iv) **Mond process :** Impure nickel is treated with carbon monoxide at 60-80°C to form volatile nickel carbonyl which decomposes to give pure nickel at 180°C.

$$\underset{\text{Impure}}{Ni} + 4CO \rightarrow Ni(CO)_4 \rightarrow \underset{\text{Pure}}{Ni} + 4CO$$

(v) **Electrolytic refining :** In this method the impure metal is made the anode, a strip of pure metal is made the cathode, while electrolyte consists of a soluble salt of the metal.

OR

(i) **Froth floatation method :** The method has been in use for removing gangue from sulphide ores. In this process, a suspension of the powdered ore is made with water. To it collectors and froth stabilisers are added.

Collectors (*e.g.*, pine oils, fatty acids, xanthates, etc.) enhance non-wettability of the mineral particles and froth stabilisers (*e.g.*, cresols, aniline) stabilise the froth.

The mineral particles become wet by oils while the gangue particles by water. A rotating paddle agitates the mixture and draws air in it. As a result, froth is formed which carries the mineral particles. The froth is light and skimmed off.

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(ii)

Roasting	Calcination
1. It is the process of heating the ore below its melting point with excess of air. 2. Sulphide ore is converted into oxide form. $2\,ZnS + 3O_2 \xrightarrow{Heat} 2ZnO + 2SO_2\uparrow$ and volatile impurities are removed as oxides SO_2, As_2O_3, etc.	1. It is the process of heating the ore below its melting point in the absence or limited supply of air. 2. Carbonate ores are converted into oxide. $ZnCO_3 \xrightarrow{Heat} ZnO + CO_2\uparrow$ and water and organic impurities are removed.

37. (i) (a) Multimolecular colloid, because sulphur molecules associate together to form colloidal sol.
(b) Macromolecular colloid, because protein molecules present in the white of an egg are macromolecules soluble in water.
(c) Associated colloid, because $RCOO^-$ ions associate together to form micelles.
(ii) (a) Colloidal particles of test tube (*A*) are positively charged whereas colloidal particles of test tube (*B*) are negatively charged.
(b) In test tube (*A*), Fe^{3+} is adsorbed on the precipitate $Fe_2O_3{\cdot}xH_2O$ [or $Fe_2O_3{\cdot}xH_2O/Fe^{3+}$ is formed].
In test tube (*B*), OH^- ion is adsorbed on the precipitate $Fe_2O_3{\cdot}xH_2O$ [or $Fe_2O_3{\cdot}xH_2O/OH^-$ is formed].

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PRACTICE PAPER 5
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

Most of the chemical reactions are accelerated by increase in temperature. The temperature dependence of the rate of a chemical reaction can be accurately explained by Arrhenius equation :

$k = A\ e^{-Ea/RT}$

where A is the Arrhenius factor or the frequency factor. It is also called pre-exponential factor. It is a constant specific to a particular reaction. R is gas constant and E_a is activation energy measured in joules/mole (J mol^{-1}).

1. Draw a graph between log k *vs* $1/T$.

2. What is the effect of temperature on the reaction rate ?

3. For a given reaction,

$$0.02 = 1.61 \exp\left(\frac{-18230.8}{8.314 \times T}\right)$$

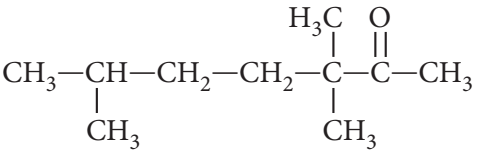
Calculate the temperature at which the reaction occurs.

4. If the activation energy of forward reaction is higher than the backward reaction then the reaction is endothermic or exothermic?

5. Why enzyme-catalyzed reaction is faster than a metal-catalyzed reaction?

Questions 6 to 10 are one word answers :

6. Give the IUPAC name of the following compound :



7. What is responsible for conduction of electricity in molten sodium chloride ?

8. Name the drugs that bind to the receptor site and inhibit its natural function.

9. What is the hybridisation of Fe in $K_3[Fe(CN)_6]$?

10. What is the hybridisation state of the central atom in the conjugate base of NH_4^+ ion?

Questions 11 to 15 are multiple choice questions :

11. Bakelite is an example of

(a) elastomer (b) fibre

(c) thermoplastic (d) thermosetting.

12. Pick up the incorrect statement from the following.

(a) Glucose exists in two different crystalline forms, α-*D*-glucose and β-*D*-glucose.

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(b) Cyclic structure of α-*D*-glucose and β-*D*-glucose is called pyranose structure.
(c) α-*D*-glucose and β-*D*-glucose are enantiomers.
(d) Cellulose is a straight chain polysaccharide made up of only β-glucose units.

13. Which of the following is not a method for coagulation of lyophobic sols?
(a) By electrophoresis
(b) By mixing oppositely charged sols
(c) By adding electrolyte
(d) By adding a protective colloid

14. For an ideal solution with $p_A > p_B$, which of the following is true?
(a) $(x_A)_{liquid} = (x_A)_{vapour}$
(b) $(x_A)_{liquid} > (x_A)_{vapour}$
(c) $(x_A)_{liquid} < (x_A)_{vapour}$
(d) $(x_A)_{liquid}$ and $(x_A)_{vapour}$ do not bear any relationship with each other.

15. Ellingham diagram represents change of
(a) ΔG with temperature
(b) ΔH with temperature
(c) ΔG with pressure
(d) $(\Delta G - T\Delta S)$ with temperature.

Questions 16 to 20 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : In electrolytic refining of metal, impure metal is made cathode while a strip of pure metal is used as anode.
Reason (R) : Impure metal from anode goes into solution and gets deposited on cathode.

17. Assertion (A) : In acidic medium, $K_2Cr_2O_7$ exists as $Cr_2O_7^{2-}$ (orange) while in basic medium it is converted to CrO_4^{2-} (yellow).
Reason (R) : $K_2Cr_2O_7$ is hygroscopic in nature and changes colour on reaction with water.

18. Assertion (A) : Anisole undergoes electrophilic substitution at *ortho* and *para*-positions.
Reason (R) : Anisole is more reactive than phenol towards electrophilic substitution reactions.

19. Assertion (A) : Aliphatic primary amines can be prepared by Gabriel phthalimide synthesis.
Reason (R) : Aryl halides undergo nucleophilic substitution with the anion formed by phthalimide.

20. Assertion (A) : According to crystal field theory, during complex formation, the *d*-orbitals split and form two sets of orbitals t_{2g} and e_g.
Reason (R) : Splitting of *d*-orbitals occurs only in case of strong field ligands.

SECTION - B

21. What is optical isomerism in coordination compounds? Give example.

22. Suggest a way to determine Λ_m° value of water.

OR

Hydrogen is a renewable and non-polluting source of energy. Explain on the basis of electrochemical principle.

23. What happens when an electric field is applied to a colloidal dispersion?

24. What is tincture of iodine? What is its use?

OR

What are artificial sweetening agents? Give two examples.

25. Glucose forms an oxime but glucose pentacetate does not. Explain.

26. Identify *A*, *B*, *C* and *D* in the given sequence of reactions,

$$CH_3CH_2Cl \xrightarrow{[A]} [B] \xrightarrow{\text{Reduction}} CH_3CH_2CH_2NH_2$$

$$[B] \xrightarrow{[C]} [D] \xrightarrow{\text{Reduction}} CH_3CH_2NHCH_3$$

27. Give the synthesis and uses of :
(i) Low density Polyethylene
(ii) Polypropylene

SECTION - C

28. An element *X* crystallises in *fcc* structure. 208 g of it has 4.2832×10^{24} atoms. Calculate the edge of the unit cell, if density of *X* is 7.2 g cm^{-3}.

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29. (a) Name the element that is purified by
(i) zone refining process
(ii) van Arkel method.

(b) The value of $\Delta_f G°$ for formation of Cr_2O_3 is –540 kJ mol^{-1} and that of Al_2O_3 is –827 kJ mol^{-1}. Is the reduction of Cr_2O_3 possible with aluminium?

OR

(a) What is the role of depressant in froth floatation process? Give one example.

(b) Describe the principle involved in Mond's process for refining of nickel.

30. Explain what is observed :
(i) when a beam of light is passed through a colloidal solution.
(ii) an electrolyte is added to hydrated ferric oxide sol.
(iii) electric current is passed through a colloidal solution.

31. Give reasons :
(i) Why silver and gold are known as coinage metals?
(ii) How do the oxides of transition elements in lower oxidation states differ from those in higher oxidation states and why?

32. Write the chemical reactions of the following :
(i) Coupling reaction
(ii) Hoffmann bromamide reaction.

OR

(i) How are 1°, 2° and 3°-amines separated from a mixture of amines?

(ii) Give a chemical test to distinguish between aniline and *N*-methylaniline.

33. (a) Di-*tert*-butyl ether cannot be prepared by Williamson's synthesis. Explain.

(b) Alcohols do not react with NaBr but when H_2SO_4 is added they form alkyl bromides. Explain.

(c) Sodium metal can be used for drying diethylether and benzene but not for ethanol. Explain.

34. Write the structures of compounds *A*, *B* and *C* in each of the following reactions :

(i) $C_6H_5Br \xrightarrow{Mg/\text{dry ether}} A \xrightarrow[(ii)\ H_3O^+]{(i)\ CO_{2(g)}} B \xrightarrow{PCl_5} C$

(ii) $CH_3CN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2/HCl} A \xrightarrow{\text{dil. NaOH}} B \xrightarrow{\Delta} C$

SECTION - D

35. (a) State Raoult's law for solutions of volatile liquids. Explain the meaning of positive and negative deviations from Raoult's law. Also give examples.

(b) Vapour pressure of chloroform ($CHCl_3$) and dichloromethane (CH_2Cl_2) at 298 K are 200 mm Hg and 415 mm Hg respectively. Calculate the vapour pressure of the solution prepared by mixing 25.5 g of $CHCl_3$ and 40 g of CH_2Cl_2 at 298 K.

OR

(a) Define the term osmotic pressure. Describe how the molecular mass of a substance can be determined by a method based on measurement of osmotic pressure.

(b) At 300 K, 36 g of glucose ($C_6H_{12}O_6$) present per litre in its solution has an osmotic pressure of 4.98 bar. If the osmotic pressure of another solution is 1.52 bar at the same temperature, calculate the concentration of the other solution.

36. (i) (a) Does the hydrolysis of XeF_6 lead to a redox reaction?
(b) Draw the structure of $XeOF_4$.
(c) Complete and balance the following equation :
$XeF_4 + H_2O \longrightarrow$

(ii) Account for the following :
(a) H_2S has lower boiling point than H_2O.
(b) Reducing character decreases from SO_2 to TeO_2.

OR

(i) How would you account for the following?
(a) Fluorine atom is more electronegative than iodine atom, yet HF is weaker acid than HI.
(b) Cl has more electron gain enthalpy

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than F.

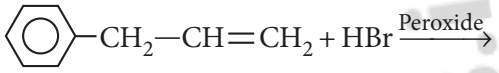
(ii) SF_6 is not easily hydrolysed though thermodynamically it should be. Why?

(iii) Draw the structures of (i) $XeOF_4$ (ii) BrF_3.

37. Complete the following reactions :

(a) $CH_3CH_2COOAg \xrightarrow{Br_2} ? \xrightarrow{\text{alc. KOH}}$

(b) $CH_3CH_2CH_2Cl \xrightarrow{CH_3C\equiv CNa}$

(c)  $+ HBr \xrightarrow{\text{Peroxide}}$

(d) $CH_3\underset{\underset{|}{CH}}{CHI} \xrightarrow[\text{Dry ether}]{Na}$

(e) $CH_3CH{=}CH_2 \xrightarrow{HBr} ? \xrightarrow{\text{aq. KOH}} ?$

OR

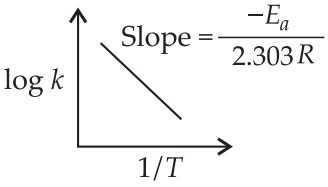
Primary alkyl halide (*A*) C_4H_9Br reacts with alcoholic KOH to give compound (*B*). Compound (*B*) reacts with HBr to give (*C*) which is an isomer of (*A*). When (*A*) reacts with Na metal, it forms a compound (*D*) C_8H_{18} that is different from the compound formed when *n*-butyl bromide reacts with sodium. Give the structural formula of (*A*), (*B*), (*C*), (*D*) and write the equations for all the reactions involved.

Hints & Solutions

1. Arrhenius equation is

$$\log k = \log A - \frac{E_a}{2.303\,RT}$$

Comparing with $y = c + mx$



2. It was found that for a chemical reaction with rise in temperature by 10°C, the rate constant is nearly doubled.

3. $0.02 = 1.61 \exp\left(\frac{-18230.8}{8.314 \times T}\right)$

$$\ln 0.02 = \ln 1.61 + \left(\frac{-18230.8}{8.314 \times T}\right)$$

$$\Rightarrow \log\frac{0.02}{1.61} = \frac{-18230.8}{2.303 \times 8.314\,T}$$

$$T = \frac{-18230.8}{2.303 \times 8.314 \times (-1.90)} \Rightarrow T = 501\text{ K}$$

4. Endothermic

5. Enzyme-catalyzed reactions have lower activation energy.

6. 3,3,6-Trimethylheptan-2-one

7. Free ions

8. Antagonist

9. d^2sp^3

10. sp^3

11. (d)

12. (c) : α-*D*-glucose and β-*D*-glucose are anomers.

13. (d) : Addition of protective colloid is a method of prevention of coagulation.

14. (c) : As *A* is more volatile, vapour phase would be richer in *A*. Hence, $(x_A)_{liquid} < (x_A)_{vapour}$.

15. (a)

16. (d) : In electrolytic refining, impure metal is made anode and a strip of pure metal is made cathode. Impure metal from anode goes into solution and gets deposited on cathode. Impurities get deposited as anode mud.

17. (c) : $K_2Cr_2O_7$ changes colour only with change in pH of aqueous solution.

18. (c) : Methoxy group in anisole is *ortho*, *para*-directing and activates the aromatic ring towards electrophilic substitution. It is less reactive than phenol.

19. (c) : Aliphatic primary amines can be synthesised by Gabriel phthalimide method. Aromatic primary amines cannot be prepared by such method because aryl halides do not undergo nucleophilic substitution with the anion formed by phthalimide.

20. (c) : Splitting of orbitals takes place for both weak and strong field ligands. Pairing of electrons in these orbitals depends on the strength of the field.

21. Coordination compounds having same molecular formula as well as structural formula

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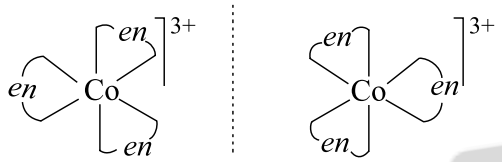
but different direction of rotation of plane polarised light are known as optical isomers.
e.g. $[Co(en)_3]^{3+}$



24. 2-3% solution of iodine in alcohol-water mixture, used as antiseptic.

OR

Saccharine, aspartame, alitame.

27. (i) **Low density Polyethylene :** When ethylene is polymerized at 350-570 K under a pressure of 1000-2000 atm in presence of 0.1% oxygen, polyethylene is obtained.

$$n CH_2 = CH_2 \longrightarrow (-CH_2-CH_2-)_n$$

It is used in making carry bags, milk cartons, toys and electrical insulation.

(ii) **Polypropylene :** When propylene is passed through *n*-hexane containing Ziegler-Natta catalyst (mixture of $TiCl_4$ and triethyl aluminium), polypropylene is formed.

$$n CH_3-CH=CH_2 \xrightarrow[TiCl_4]{Al(C_2H_5)_3} \left[-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-\right]_n$$

It is used in making gramophone records and ropes and in packing of textile materials.

28. For *fcc* structure, $Z = 4$, $d = 7.2$ g/cm^3
Let a be the edge length in pm

We know that density, $d = \dfrac{Z \times M}{a^3 \times 10^{-30}}$

Mass per atom, $M = \dfrac{208}{4.2832 \times 10^{24}}$,

$$a^3 = \frac{Z \times M}{d \times 10^{-30}} = \frac{4 \times 208}{7.2 \times 4.2832 \times 10^{24} \times 10^{-30}}$$

$= 26.97 \times 10^6$ pm$^3 \Rightarrow a = (27 \times 10^6)^{1/3} = 300$ pm

29. (a) (i) Germanium (ii) Zirconium

(b) $\dfrac{4}{3}Al_{(s)} + \dfrac{2}{3}Cr_2O_{3(s)} \longrightarrow \dfrac{2}{3}Al_2O_{3(s)} + \dfrac{4}{3}Cr_{(s)}$

$\Delta_r G^\circ = -287$ kJ mol^{-1}

As $\Delta_r G^\circ$ of the reaction is –ve, reduction of Cr_2O_3 is possible with aluminium.

OR

(a) In froth floatation process, the role of the depressant is to prevent one type of sulphide ore particles from forming the froth with air bubbles.

(b) $Ni + 4CO \xrightarrow{330 - 350\ K} Ni(CO)_4$

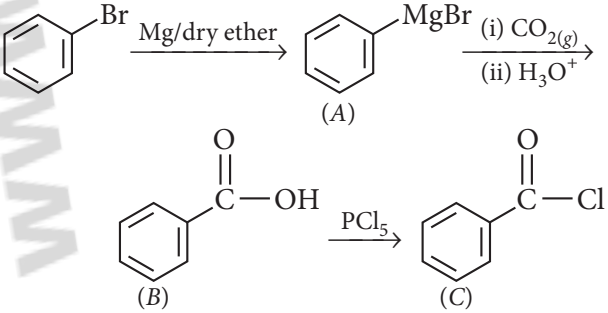
$Ni(CO)_4 \xrightarrow{450 - 470\ K} Ni + 4CO$

31. (i) Since they are noble metals, they are resistant to atmospheric and environmental chemical effects. Hence, they are used in the making of coins.
(ii) In the lower oxidation state, the transition metal oxides are basic and they are acidic if the metal is in higher oxidation state. The oxides are amphoteric when the metal is in intermediate oxidation state.
For example,

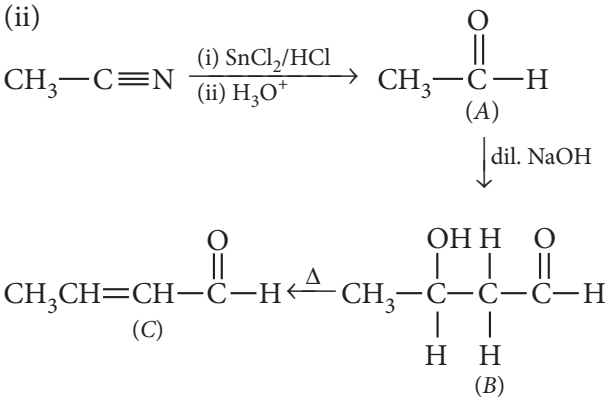
+3	+4	+7
Mn_2O_3	MnO_2	Mn_2O_7
Basic	Amphoteric	Acidic

In case of lower oxide of a transition metal, the metal atom has a low oxidation state. This means some of the valence electrons of the metal atom are not involved in bonding, hence, these can be used for donation. Thus, these are act as bases.

34. (i)



(ii)



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35. (a) Raoult's law states that for a solution of volatile liquids, the partial vapour pressure of each component of the solution is directly proportional to its mole fraction present in solution.

(b) $x_{CH_2Cl_2} = \frac{0.47}{0.683} = 0.688$

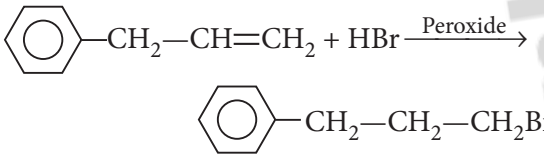
$p_{total} = p_1^\circ + (p_2^\circ - p_1^\circ)\, x_2$

$= 200 + (415 - 200) \times 0.688$

$= 200 + 147.9 = 347.9$ mm Hg

37. (a) $CH_3CH_2COOAg \xrightarrow{Br_2} CH_3CH_2Br \xrightarrow{\text{alc. KOH}} CH_2{=}CH_2$

(b) $CH_3CH_2CH_2Cl \xrightarrow{CH_3C{\equiv}CNa} CH_3C{\equiv}CCH_2CH_2CH_3 + NaCl$

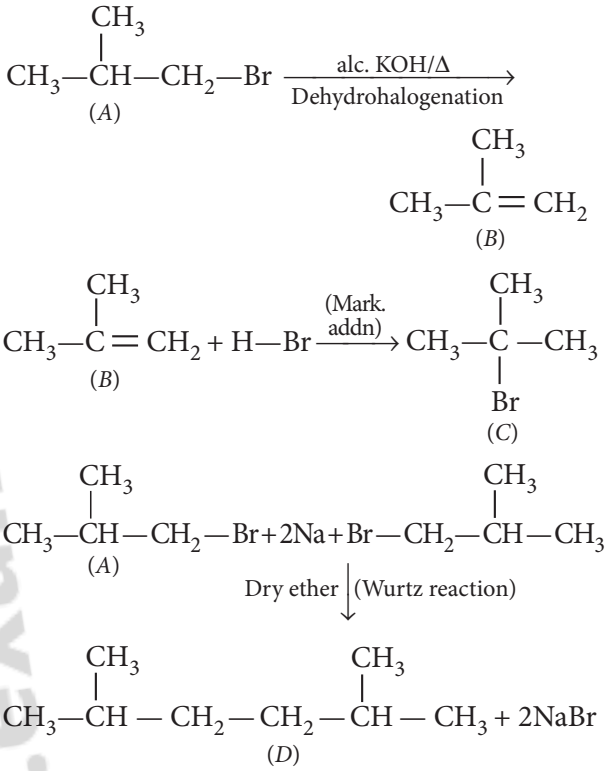
(c) 

(d) $2CH_3CHI(CH_3) \xrightarrow[\text{Dry ether}]{2Na} CH_3{-}CH(CH_3){-}CH(CH_3){-}CH_3 + 2NaI$

(e) $CH_3CH{=}CH_2 \xrightarrow{HBr} CH_3{-}CH(Br){-}CH_3 \xrightarrow{\text{aq. KOH}} CH_3{-}CH(OH){-}CH_3$

OR

Reactions involved are



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PRACTICE PAPER 6
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

Halogens combine amongst themselves to form a number of compounds known as interhalogens of the types XX', XX_3', XX_5' and XX_7' where X is a larger size halogen and X' is smaller size halogen.

1. Which bond is most polar in XX' type compound?

Cl — F, Br — F, I — F, F — F

2. What is the shape of molecule BrF_5?

3. $Cl_2 + 3F_2$ (Excess) $\xrightarrow{573\ K} A$

Identify the product A.

4. How many lone pairs of electrons are present in the molecular shape of BrF_3?

5. What is the hybridisation of central atom in IF_7 molecule?

Questions 6 to 10 are one word answers :

6. On which property colour of colloids depends?

7. Name the product when primary alcohol is oxidised by potassium permanganate.

8. Name the disease caused due to deficiency of vitamin B_{12}.

9. In which disease arsphenamine (salvarsan) is effective?

10. Name the polymer used for making handle of pressure cooker.

Questions 11 to 15 are multiple choice questions :

11. For an endothermic reaction, ΔH represents the enthalpy of the reaction in kJ mol^{-1}. The minimum amount of activation energy will be

(a) less than zero (b) equal to ΔH

(c) less than ΔH (d) more than ΔH.

12. A compound $C_6H_{14}O_2$ has two tertiary alcoholic groups. The IUPAC name of this compound is

(a) 2, 3-dimethyl - 1, 2-butanediol

(b) 3, 3-dimethyl - 1, 2-butanediol

(c) 2, 3-dimethyl - 2, 3-butanediol

(d) 2-methyl - 2, 3-pentanediol.

13. During dissolution when solute is added to the solvent, some solute particles separate out from the solution as a result of crystallisation. At the stage of equilibrium, the concentration of solute in the solution at given temperature and pressure

(a) increases

(b) decreases

(c) remains constant

(d) keeps changing.

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14. What will be the emf of the following concentration cell at 25°C?
$Ag_{(s)} | AgNO_3(0.01\ M) || AgNO_3(0.05\ M) | Ag_{(s)}$
(a) 0.828 V (b) 0.0413 V
(c) –0.0413 V (d) –0.828 V

15. Which of the following pairs of isomers is not correctly matched with its type of isomerism?
(a) $[Cr(NH_3)_6]$ $[Cr(CN)_6]$ and $[Cr(NH_3)_4(CN)_2]$ $[Cr(NH_3)_2(CN)_4]$
– Coordination isomerism
(b) $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$
– Linkage isomerism
(c) $[Co(py)_2(H_2O)_2Cl_2]Cl$ and $[Co(py)_2(H_2O)Cl_3]H_2O$
– Coordination isomerism
(d) $[Pt(NH_3)_4Br_2]Cl_2$ and $[Pt(NH_3)_4Cl_2]Br_2$
– Ionisation isomerism

Questions 16 to 20 :
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : Order of a reaction with respect to any reactant can be zero, positive or fractional.
Reason (R) : Rate of a reaction decreases with increase in concentration of a reactant or a product.

17. Assertion (A) : In close packing of sphere a tetrahedral void is surrounded by four spheres whereas an octahedral void is surrounded by six spheres.
Reason (R) : A tetrahedral void has a tetrahedral shape whereas an octahedral void has an octahedral shape.

18. Assertion (A) : One molar aqueous solution is more concentrated than that of 1 molal aqueous solution.
Reason (R) : Molarity is a function of temperature as volume depends on temperature.

19. Assertion (A) : In the coagulation of a negative sol the flocculating power is in the order :
$Al^{3+} > Ba^{2+} > Na^+$
Reason (R) : Greater the valence of the flocculating ion added, greater is its power to cause precipitation.

20. Assertion (A) : Polysaccharides are non reducing sugars.
Reason (R) : Carbohydrates which yield a large number of monosaccharide units on hydrolysis are called polysaccharides.

SECTION - B

21. (a) Write Nernst equation to calculate the cell potential of the given cell :
$Mg_{(s)} | Mg^{2+}_{(aq)} || Ag^+_{(aq)} | Ag_{(s)}$
(b) Calculate the potential of hydrogen electrode in contact with a solution whose pH is 10.

22. (a) Write the names of any two principal ores of zinc.
(b) What is van Arkel method?
OR
Why is the reduction of a metal oxide easier if the metal formed is in liquid state at the temperature of reduction?

23. Write the structure and IUPAC name of DDT.

24. What are essential and non-essential amino acids? Give two examples of each type.
OR
What are nucleic acids? Mention their two important functions.

25. Write the names and molecular structures of the monomers of the following polymers. Give one use of each specifying the property responsible for this use.
(i) Glyptal (ii) Nylon-6

26. Describe the following with suitable examples :
(i) Tranquilizers
(ii) Antifertility drugs

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27. (i) Why do we not observe Frenkel defect in alkali metal halides?
(ii) Name any one compound which shows both Schottky and Frenkel defect.

SECTION - C

28. In the button cell widely used in watches, the following reaction takes place :

$$Zn_{(s)} + Ag_2O_{(s)} + H_2O_{(l)} \longrightarrow Zn^{2+}_{(aq)} + 2Ag_{(s)} + 2OH^-_{(aq)}$$

Determine $E°$ and $\Delta G°$ for the reaction.
(Given: $E°_{Ag^+/Ag} = 0.80$ V, $E°_{Zn^{2+}/Zn} = -0.76$ V)

29. How will you convert :
(i) Bromomethane to propanone
(ii) 1-Bromopropane to 2-bromopropane?

30. Explain why $[Fe(H_2O)_6]^{3+}$ has magnetic moment value of 5.92 B.M. whereas $[Fe(CN)_6]^{3-}$ has a value of only 1.74 B.M.

OR

$[CoF_6]^{3-}$ is paramagnetic but $[Co(NH_3)_6]^{3+}$ is diamagnetic complex. Why?

31. Explain the following reactions with examples :
(i) Carbylamine reaction
(ii) Coupling reaction

32. Convert :
(i) Diethyl ether into ethylene
(ii) Anisole into toluene

33. Complete the following chemical reactions :
(i) $Pb(NO_3)_2 \xrightarrow{\Delta}$
(ii) $SO_2 + H_2O + Cl_2 \rightarrow$
(iii) $XeF_4 + H_2O \rightarrow$

OR

Draw the structures of the following species :
(i) $XeOF_4$ (ii) H_3PO_4
(iii) $HClO_4$

34. (i) Mention one important application of Henry's law.
(ii) Calculate depression in freezing point when 1.225 g of chlorobutanoic acid is added to 250 g of water.
(Given : $K_a = 1.4 \times 10^{-3}$, $K_f = 1.86$ K kg mol^{-1}.)

SECTION - D

35. (a) Answer the following questions with reasons.
(i) Which element of the first transition series has highest second ionisation enthalpy?
(ii) Which element of the first transition series has highest third ionisation enthalpy?
(iii) Which element of the first transition series has lowest enthalpy of atomisation?
(b) Complete the following reactions :
(i) $SO_2 + Cr_2O_7^{2-} + H^+ \longrightarrow$
(ii) $S_2O_3^{2-} + MnO_4^- + H_2O \longrightarrow$

OR

(a) (i) Which one has higher melting point, Fe or Cu?
(ii) Which one has lower magnetic moment, Co^{2+} or Ni^{2+}?
(b) Give reason for the following:
$E°$ value for $Mn^{3+}|Mn^{2+}$ is higher than those of $Cr^{3+}|Cr^{2+}$ and $Fe^{3+}|Fe^{2+}$.

36. (a) A compound *P* on oxidation gives $Q(C_2H_4O_2)$. *P* reacts with dil. NaOH and on subsequent heating forms *R*. *R* on catalytic hydrogenation gives *S*. Identify *P*, *Q*, *R*, *S* and write down the reactions involved.
(b) Write chemical equations to carry out the following conversions:
(i) Benzene to benzyl alcohol.
(ii) Propanenitrile to 1-phenylpropanone.

OR

(a) Two moles of organic compound '*W*' on treatment with a strong base give two compounds '*X*' and '*Y*'. Compound '*X*' on dehydrogenation with Cu gives '*W*' while acidification of '*Y*' yields carboxylic acid '*Z*' having molecular formula of CH_2O_2. Identify the compounds *W*, *X*, *Y* and *Z*.

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(b) Explain why :

(i) The aldol and ketol readily lose water molecules to give α, β - unsaturated carbonyl compounds.

(ii) Benzaldehyde is less reactive than acetaldehyde towards nucleophilic substitution reaction.

37. Starting from 10 g of a radioactive element, 0.25 g was left after 5 years. Calculate

(i) rate constant for the decay of the radioactive element.

(ii) the amount left after one year.

(iii) the time required for half of the element to decay.

OR

The initial rate of reaction :

$A + 5B + 6C \longrightarrow 3D + 3E$

has been determined by measuring the rate of disappearance of *A* under the following conditions :

Expt. No.	$[A]_0$ (M)	$[B]_0$ (M)	$[C]_0$ (M)	Initial rate (M min^{-1})
1.	0.02	0.02	0.02	2.08×10^{-3}
2.	0.01	0.02	0.02	1.04×10^{-3}
3.	0.02	0.04	0.02	4.16×10^{-3}
4.	0.02	0.02	0.04	8.32×10^{-3}

(i) Determine the order of reaction with respect to each reactant.

(ii) What is the rate constant?

(iii) Calculate the initial rate of the reaction when the concentration of all the reactants is 0.01 M.

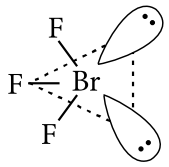
Hints & Solutions

1. Electronegativity difference in I—F is maximum which makes it the most polar bond.

2. Square pyramidal

3. $Cl_2 + \underset{\text{(Excess)}}{3F_2} \xrightarrow{573\ K} 2ClF_3$

4. Two,



BrF_3

Hybridisation-sp^3d

5. sp^3d^3

6. Particles size

7. Carboxylic acid

8. Pernicious anaemia

9. Syphilis

10. Bakelite

11. **(d)** **12.** **(c)** **13.** **(c)**

14. **(b) :** $E = E° - \dfrac{0.0591}{n} \log Q$

$E°$ = 0 for all concentration cells

$$= 0 - \frac{0.0591}{1} \log\left(\frac{0.01}{0.05}\right) = 0.0413\ V$$

15. **(c) :** $[Co(py)_2(H_2O)_2Cl_2]Cl$ and $[Co(py)_2(H_2O)Cl_3]H_2O$ show hydrate isomerism.

16. **(c) :** Rate of a reaction increases with increase in concentration of reactants.

17. **(c) :** Tetrahedral void is so called because it is surrounded by four spheres while octahedral void is so called because it is surrounded by six spheres octahedrally.

18. **(b) :** 1 molar solution has 1 mole of solute per litre of solution or per 1000 cm^3 of solution.

Now, density of water = 1 g/cc

$\therefore$ 1000 cm^3 = 1000 g

Hence, 1 mole is present in less than 1000 g of solvent.

[$\because$ 1000 g of solution = solute and solvent]

Whereas 1 molal solution has 1 mole of solute in l000 g of solvent.

19. **(a) :** This is Hardy–Schulze rule.

20. **(b)**

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21. (a) $E_{cell} = E^{\circ}_{cell} - \frac{2.303\,RT}{nF}\log\frac{[Mg^{2+}]}{[Ag^{+}]^2}$

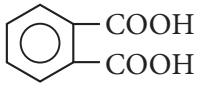
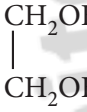
(b) $E_{H^+/H_2} = -0.591$ V

23. 1, 1, 1-Trichloro-2,2-di(4-chlorophenyl)ethane

24. Valine, leucine (essential)
Glycine, alanine (non-essential)

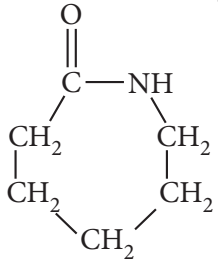
OR

Biomolecules found in nuclei of all living cells in the form of nucleoproteins, DNA and RNA.
Functions : Transfer of hereditary characters and protein synthesis respectively.

25. (i) **Glyptal :** The monomers are phthalic acid,  and ethylene glycol,  .

It is used in adherent paint because it helps in fixing paints.

(ii) **Nylon-6 :** The monomer is caprolactum.



It is used for making tyre cords because it has high tensile strength.

27. (i) We do not observe Frenkel defect in alkali metal halides because the ions are big in size and cannot get into the interstitial sites.

(ii) AgBr

28. $E^{\circ}_{cell} = E^{\circ}_{Zn/Zn^{2+}} + E^{\circ}_{Ag^+/Ag}$
$= 0.76 + 0.80 = 1.56$ V

$\Delta G^{\circ}_{cell} = -nFE^{\circ}_{cell}$
$= -301.08$ kJ mol^{-1}

30. Fe (26) : [Ar] $3d^6\,4s^2$

Fe^{3+} : [Ar] $3d^5$

In $[Fe(H_2O)_6]^{3+}$, magnetic moment of five unpaired electrons

$= \sqrt{n(n+2)}$ B.M. $= \sqrt{35} = 5.92$ B.M.

In $[Fe(CN)_6]^{3-}$, magnetic moment of one unpaired electron

$= \sqrt{n(n+2)} = \sqrt{3} = 1.74$ B.M.

OR

In $[CoF_6]^{3-}$ (sp^3d^2) d-orbital contains 4 unpaired electrons so, it is paramagnetic and in $[Co(NH_3)_6]^{3+}$ (d^2sp^3) d-orbital does not contain any unpaired electron so, it is diamagnetic complex.

33. (i) $2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$

(ii) $SO_2 + 2H_2O + Cl_2 \rightarrow H_2SO_4 + 2HCl$

(iii) $6XeF_4 + 12H_2O \rightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

34. (i) Soft drink and soda water are filled under high pressure.

(ii) Molality of solution, $m = \frac{W_2 \times 1000}{M_2 \times W_1}$

$m = 0.04$ m

Degree of dissociation, $\alpha = \sqrt{\frac{K_a}{C}} = 0.18$

$\alpha = \frac{i-1}{n-1} \Rightarrow 0.18 = \frac{i-1}{2-1}$

or $0.18 = i - 1$ or $i = 1.18$

Using $\Delta T_f = iK_f m = 1.18 \times 1.86 \times 0.04 = 0.0878$ °C.

35. (b) (i) $3SO_2 + Cr_2O_7^{2-} + 2H^+ \longrightarrow 3SO_4^{2-} + 2Cr^{3+} + H_2O$

(ii) $3S_2O_3^{2-} + 8MnO_4^- + H_2O \longrightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

OR

(a) (i) Fe has higher melting point. This is because Fe has four unpaired electrons in 3d-subshell whereas Cu has only one electron in the 4s-subshell. Hence, metallic bonds in Fe are much stronger than those in Cu.

(ii) $Co^{2+} \longrightarrow [Ar]3d^7$ (3 unpaired electrons)

$Ni^{2+} \longrightarrow [Ar]3d^8$ (2 unpaired electrons)

Hence Ni^{2+} has lower magnetic moment than Co^{2+}.

(b) The comparatively high value of E° for Mn^{3+}/Mn^{2+} shows that $Mn^{2+}(d^5)$ is particularly more stable, whereas comparatively low value of E° for Fe^{3+}/Fe^{2+} is due to extra stability of $Fe^{3+}(d^5)$ and that of Cr^{3+} is related to half filled t_{2g} level.

37. (i) $k = \frac{2.303}{t}\log\frac{[A]_0}{[A]}$

$[A]_0 = 10$ g, $[A] = 0.25$ g, $t = 5$ years

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$$k = \frac{2.303}{5}\log\frac{10}{0.25}$$

$$= \frac{2.303}{5}\log 40 = 0.7379 \text{ year}^{-1}$$

OR

(i) Rate law may be written as :

Rate = $k[A]^a[B]^b[C]^c$

Comparing experiments 1 and 2,

$(\text{Rate})_1 = k(0.02)^a(0.02)^b(0.02)^c = 2.08 \times 10^{-3}$ ...(i)

$(\text{Rate})_2 = k(0.01)^a(0.02)^b(0.02)^c = 1.04 \times 10^{-3}$...(ii)

Dividing eq. (i) by eq. (ii) we get,

$$\frac{(\text{Rate})_1}{(\text{Rate})_2} = \frac{(0.02)^a}{(0.01)^a} = \frac{2.08\times10^{-3}}{1.04\times10^{-3}} = 2$$

or $2^a = 2 \quad \therefore \quad a = 1$

Comparing experiments 1 and 3,

$(\text{Rate})_1 = k(0.02)^a(0.02)^b(0.02)^c = 2.08 \times 10^{-3}$...(iii)

$(\text{Rate})_3 = k(0.02)^a(0.04)^b(0.02)^c = 4.16 \times 10^{-3}$...(iv)

Dividing eq. (iv) by eq. (iii) we get,

$$\frac{(\text{Rate})_3}{(\text{Rate})_1} = \frac{(0.04)^b}{(0.02)^b} = \frac{4.16\times10^{-3}}{2.08\times10^{-3}} = 2$$

$2^b = 2 \ \therefore \ b = 1$

Comparing experiments 1 and 4,

$(\text{Rate})_1 = k(0.02)^a(0.02)^b(0.02)^c = 2.08 \times 10^{-3}$...(v)

$(\text{Rate})_4 = k(0.02)^a(0.02)^b(0.04)^c = 8.32 \times 10^{-3}$...(vi)

Dividing eq. (vi) by eq. (v), we get

$$\frac{(\text{Rate})_4}{(\text{Rate})_1} = \frac{(0.04)^c}{(0.02)^c} = \frac{8.32\times10^{-3}}{2.08\times10^{-3}} = 4$$

or $2^c = 2^2$

$\therefore \quad c = 2$

Therefore, the order of the reaction with respect to *A*, *B* and *C* are 1, 1 and 2 respectively.

Rate = $k[A][B][C]^2$

(ii) Substituting the values from expt. 1, we get

$2.08 \times 10^{-3} = k \times 0.02 \times 0.02 \times (0.02)^2$

$$\therefore \quad k = \frac{2.08\times10^{-3}}{(0.02)^4} = \frac{2.08\times10^{-3}}{16\times10^{-8}}$$

$= 1.3 \times 10^4 \text{ M}^{-3} \text{ min}^{-1}$

(iii) Rate of reaction when concentration of all reactants = 0.01 M

Rate = $1.3 \times 10^4 \times (0.01) \times (0.01) \times (0.01)^2$

$= 1.3 \times 10^{-4} \text{ M min}^{-1}$

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PRACTICE PAPER 7
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

Dialkyl ethers react with excess of hydrogen halides to give alkyl halides.

$$R\text{—O—}R + HX \longrightarrow R\text{—}X + R\text{—OH}$$
$$R\text{—OH} + HX \longrightarrow R\text{—}X + H_2O$$

Mixed ether containing primary or secondary alkyl group when heated with hydrogen halide, the lower alkyl group forms halide and higher alkyl group forms an alcohol.

1. What is the order of reactivity of hydrogen halides in this reaction ?

2. Write the main product(s) in the following reaction :

$$CH_3\text{—}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}\text{—O—}CH_3 + HI \longrightarrow$$

3. Phenylmethyl ether reacts with HI to give phenol and methyl iodide and not iodobenzene and methyl alcohol. Why?

4. Write the products in the following reaction :

$$CH_3\text{—}\underset{\displaystyle CH_3}{\overset{|}{CH}}\text{—O—}CH_2\text{—}CH_3 \xrightarrow{HI} ? + ?$$

5. What happens when ethoxy ethane is treated with excess HI?

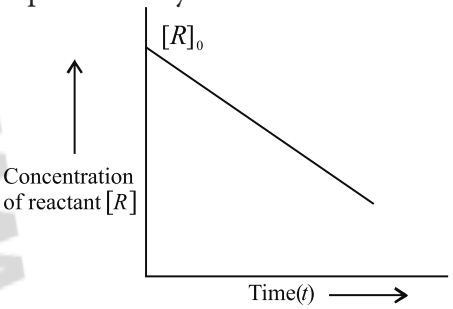
Questions 6 to 10 are one word answers :

6. Name the electrolyte used in nickel -cadmium cell.

7. Name an alloy which contains some of the lanthanoid metals.

8. Name the ore from which Al is extracted.

9. A reaction, Reactant → Product is represented by



Predict the order of the reaction in this case.

10. Name the solid that is separated when salt solution is slowly frozen.

Questions 11 to 15 are multiple choice questions :

11. In nitrogen family, the H – M – H bond angle in the hydrides gradually becomes closer to 90° on going form N to Sb. This shows that gradually

(a) the basic strength of the hydrides increases

(b) almost pure p-orbitals are used for M – H bonds

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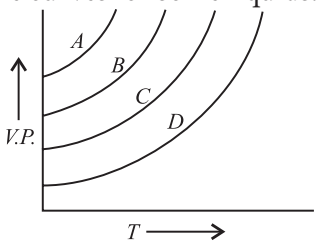
(c) the bond energies of M – H bonds increase
(d) the bond pairs of electrons become nearer to the central atom.

12. Which of the following statements is not correct?
(a) Aldehydes and ketones are functional isomers.
(b) Formaldehyde reacts with ammonia to form hexamethylenetetramine.
(c) $LiAlH_4$ converts ketones into *sec*-alcohols.
(d) Ethanal and propanal give positive iodoform test.

13. Secondary structure of protein refers to
(a) sequence of amino acids in polypeptide chain
(b) bonds between alternate polypeptide chains
(c) folding patterns of polypeptide chain
(d) bonding between NH_3^+ and COO^- of two peptides.

14. Certain crystals produce electric signals on application of pressure. This phenomenon is called
(a) pyroelectricity (b) ferroelectricity
(c) piezoelectricity (d) ferrielectricity.

15. The given graph shows the vapour pressure-temperature curves for some liquids.



Liquids A, B, C and D respectively are
(a) diethyl ether, acetone, ethyl alcohol, water
(b) acetone, ethyl alcohol, diethyl ether, water
(c) water, ethyl alcohol, acetone, diethyl ether
(d) ethyl alcohol, acetone, diethyl ether, water.

Questions 16 to 20 :
(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. **Assertion (A) :** In order to convert R–Cl to pure R–NH_2, Gabriel pthalimide synthesis can be used.
Reason (R) : With proper choice of alkyl halides, pthalimide synthesis can be used to prepare 1°, 2° or 3° amines.

17. **Assertion (A) :** If the activation energy of a reaction is zero, temperature will have no effect on the rate constant.
Reason (R) : Lower the activation energy, faster is the reaction.

18. **Assertion (A) :** Sulphuric acid is more viscous than water.
Reason (R) : Concentrated sulphuric acid has a great affinity for water.

19. **Assertion (A) :** In a Daniel cell, if concentrations of Cu^{2+} and Zn^{2+} ions are doubled, the EMF of the cell will be doubled.
Reason (R) : If the concentration of ions in contact with the metal is doubled, the electrode potential remain unchanged.

20. **Assertion (A) :** The vapour pressure of a liquid decreases if some non-volatile solute is dissolved in it.
Reason (R) : The relative lowering of vapour pressure of a solution containing a non-volatile solute is equal to the mole fraction of the solute in the solution.

SECTION - B

21. A liquid (A) is treated with Na_2CO_3 solution. A mixture of two salts (B) and (C) is produced in the solution. The mixture on acidification with sulphuric acid and distillation produces the liquid A again. Identify (A), (B) and (C) and write the equations involved.

22. (a) What are the products obtained by hydrolysis of lactose?
(b) Why cannot vitamin C be stored in our body?

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23. (a) Dipole moment of phenol is smaller than that of methanol. Why?

(b) Alcohols are comparatively more soluble in water than the hydrocarbons of comparable molecular mass? Explain this fact.

OR

Write the main product(s) in each of the following reactions :

(i) $CH_3-CH=CH_2 \xrightarrow[(ii)\ 3H_2O_2/OH^-]{(i)\ B_2H_6}$

(ii) $C_6H_5-OH \xrightarrow[(ii)\ CO_2,\ H^+]{(i)\ aq.\ NaOH}$

24. Write the net reaction for the formation of aldol and dehydration products obtained from phenyl acetaldehyde.

25. Calculate the mass of ascorbic acid (Vitamin C, $C_6H_8O_6$) to be dissolved in 75 g of acetic acid to lower its melting point by 1.5°C. ($K_f = 3.9$ K kg mol^{-1}).

OR

Henry's law constant for the molality of methane in benzene at 298 K is 4.27×10^5 mm Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg.

26. Answer the following :

(i) Which stoichiometric defect is observed in solids with high co-ordination number?

(ii) What is the effect of increasing temperature on the conductivity of semi conductors?

27. (a) Nb and Ta exhibit similar properties. Give reason.

(b) Among the ionic species, Sc^{3+}, Ce^{4+} and Eu^{2+}, which one is a good oxidising agent?

SECTION - C

28. Explain the following terms with a suitable example in each case :

(i) Shape-selective catalysis

(ii) Dialysis

(iii) Multimolecular colloids

29. Write structures and names of the monomers of the following polymers :

(i) Nylon 6,6 (ii) Nylon 6

(iii) Urea formaldehyde resin

OR

(i) How will you differentiate between low density and high density polythenes?

(ii) Arrange the following polymers in order of increasing intermolecular forces :
Nylon 6,6, Polythene, Neoprene.

30. (i) Give the IUPAC name of $[Co(NH_3)_5(CO_3)]Cl$

(ii) Compare the magnetic behaviour of the complex entities $[Fe(CN)_6]^{4-}$ and $[FeF_6]^{3-}$. [Atomic number of Fe = 26]

31. Mention the action of the following on the human body in bringing relief from a disease.
(i) Brompheniramine (ii) Aspirin
(iii) Equanil

32. (i) Arrange the following in order of increasing boiling point.
1-chloropropane, isopropyl chloride,
1-chlorobutane, isobutyl chloride,

(ii) Dipole moment of chlorobenzene is lower than that of cyclohexyl chloride. Why?

(iii) What do you mean by the term chirality?

OR

(a) Account for the following :

(i) The treatment of alkyl chlorides with aqueous KOH leads to the formation of alcohols but in the presence of alcoholic KOH, alkenes are major products.

(ii) Haloalkanes undergo nucleophilic substitution reactions.

(b) Out of $C_6H_5CH_2Cl$ and $C_6H_5CHClC_6H_5$, which is more easily hydrolysed by aqueous KOH?

33. A pale yellow substance (*A*) when heated with conc. HNO_3, evolves a brown coloured gas (*B*). The substance (*A*) also dissolves in sodium sulphite solution on heating. A clear solution (*C*) is formed which on acidification gives a turbid solution and a pungent smelling gas (*D*) which is formed by heating the substance (*A*) in air. The solution (*C*) decolourises iodide solution. Identify (*A*) to (*D*).

34. The half-life period of a first order reaction is 30 minutes. Calculate the specific reaction rate of the reaction. What fraction of the reactant remains after 70 minutes?

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SECTION - D

35. (i) A cell is prepared by dipping a copper rod in 0.01 M copper sulphate solution, and zinc rod in 0.02 M zinc sulphate solution. The standard reduction potentials of copper and zinc are +0.34 V and –0.76 V respectively.
(a) What will be the cell reaction?
(b) How will the cell be represented?
(c) What will be the emf of the cell?

(ii) The standard cell potential for Daniell cell is +1.10 V. Calculate the standard Gibbs' energy for the reaction.

OR

(i) The electrical resistance of a column of 0.05 mol L^{-1} NaOH solution of diameter 1 cm and length 50 cm is 5.55×10^3 ohm. Calculate its resistivity, conductivity and molar conductivity.

(ii) From the following molar conductivities at infinite dilution calculate Λ_m° for NH_4OH.
Λ_m° for $Ba(OH)_2 = 457.6\ \Omega^{-1}\ cm^2\ mol^{-1}$
Λ_m° for $BaCl_2 = 240.6\ \Omega^{-1}\ cm^2\ mol^{-1}$
Λ_m° for $NH_4Cl = 129.8\ \Omega^{-1}\ cm^2\ mol^{-1}$

36. (a) What is flux? Discuss its types and uses in metallurgical operations.

(b) Write the endothermic reactions taking place in the blast furnance during extraction of iron.

OR

(i) Write the reactions involved during extraction of copper from copper pyrites.

(ii) During the extraction of copper, explain why inner wall of the Bessemer converter is lined with silica?

37. (a) Give plausible explanation for each of the following :
(i) Nitrobenzene will not undergo azo coupling reaction with benzene diazonium chloride.
(ii) Ammonolysis of alkyl halides does not give corresponding amine in pure state.
(iii) Aniline does not undergo Friedel–Crafts reaction.

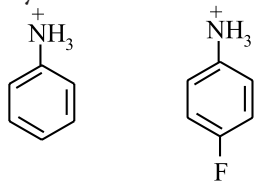
(b) What happens when
(i) aniline is treated with benzaldehyde
(ii) a mixture of alcohol and ammonia is passed over heated aluminium oxide as catalyst?

OR

(a) $\underset{\text{Optically active}}{C_5H_{13}N\ (X)} \xrightarrow{NaNO_2/HCl} \underset{\text{+ Other products}}{Y\ \text{(tertiary alcohol)}}$

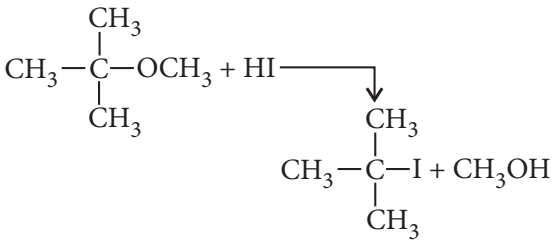
Find *X* and *Y*. Is *Y* optically active? Write the intermediate steps.

(b) Which of the following is more acidic and why?

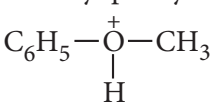


Hints & Solutions

1. HI > HBr > HCl

2.


3. Protonation of anisole (phenyl methyl ether) gives methyl phenyl oxonium ion.



In this ion, the stronger bond is O—C_6H_5. Therefore, attack by I^- ion exclusively breaks the weaker O—CH_3 bond forming methyl iodide and phenol. The phenol formed does not react further to give aryl halides.

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4. $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-O-CH_2CH_3 \xrightarrow{HI}$

$CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3 + CH_3CH_2I$

5. Ethoxyethane reacts with excess HI at high temperature to give ethyl iodide.

$C_2H_5-O-C_2H_5 + HI \xrightarrow{\Delta} C_2H_5I + C_2H_5OH$

$C_2H_5OH + HI \longrightarrow C_2H_5I + H_2O$

6. KOH solution

7. Mischmetal

8. Bauxite

9. Zero order

10. Ice

11. (b) : On moving down the group, electronegativity decreases, consequently bond pair shifts more and more away from the central atom. Hence, H–*M*–H bond angle decreases and becomes closer to 90° in SbH_3 which reveals that almost pure *p*-orbitals are used for *M*–H bonding.

12. (d) : Propanal does not give iodoform test.

13. (c)

14. (c) : Piezoelectric substances produce electric signals when mechanical stress is applied to them.

15. (a) : The vapour pressure increases with decrease in intermolecular forces. When the forces are weak, the liquid has high volatility and maximum vapour pressure. Diethyl ether has highest vapour pressure while water has lowest vapour pressure.

16. (c) : Only primary aliphatic amines can be prepared by Gabriel phthalimide reaction.

17. (b) : According to Arrhenius equation, $k = Ae^{-E_a/RT}$ when $E_a = 0$, $k = A$.

18. (b) : Molecules of sulphuric acid are associated due to large number of intermolecular hydrogen bonding.

19. (d) : On doubling the concentrations, the EMF will remain unchanged because $[Zn^{2+}]/[Cu^{2+}]$ will remain same.

20. (b) : On adding a non-volatile solute in a liquid, the vapour pressure decreases because some molecules of the solvent on the surface are replaced by the molecules of the solute.

21. The liquid (*A*) is bromine which on treatment with sodium carbonate forms a mixture of NaBr (*B*) and $NaBrO_3$(*C*), sodium bromate. The mixture with conc. H_2SO_4 on distillation gives liquid bromine again.

$\underset{(A)}{3Br_2} + 3Na_2CO_3 \longrightarrow \underset{(B)}{5NaBr} + \underset{(C)}{NaBrO_3} + 3CO_2$

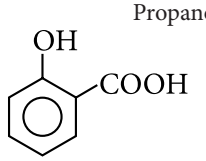
$5NaBr + NaBrO_3 + 3H_2SO_4 \longrightarrow 3Na_2SO_4 + \underset{(A)}{3Br_2} + 3H_2O$

23. (b) Alcohols are relatively more soluble in water than the corresponding hydrocarbons because alcohol molecules can form hydrogen bonds with water.

OR

(i) $CH_3-CH=CH_2 \xrightarrow[(ii)\ 3H_2O_2/OH^-]{(i)\ B_2H_6} CH_3CH_2CH_2OH$

Propanol

(ii) $C_6H_5OH \xrightarrow[(ii)\ CO_2,\ H^+]{(i)\ aq.\ NaOH}$ 

Salicylic acid

25. $W_B = \dfrac{\Delta T_f \times M_B \times W_A}{K_f \times 1000} = \dfrac{1.5 \times 176 \times 75}{3.9 \times 1000} = 5.08\ g$

OR

Applying Henry's law, $p = K_H\, x$

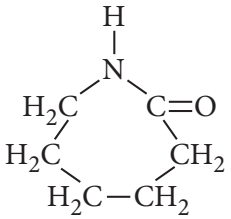
$\therefore\ x = \dfrac{p}{K_H} = \dfrac{760\ mm}{4.27 \times 10^5\ mm} = 1.78 \times 10^{-3}$

26. (i) Schottky defect is observed in solids with high coordination number.

(ii) The increase in temperature increases the conductivity of semiconductors.

29. (i) $HOOC(CH_2)_4COOH$ and $NH_2(CH_2)_6NH_2$

Adipic acid — Hexamethylene diamine

(ii) 

Caprolactam

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(iii) NH_2CONH_2 and HCHO
Urea Formaldehyde

OR

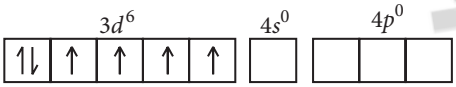
(i)

Low density polythene	High density polythene
1. It is tough but flexible.	It is tougher and harder.
2. Highly branched structure so, it has low density.	It is linearly arranged and has a high density due to close packing.

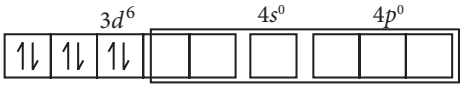
(ii) Neoprene < Polythene < Nylon 6,6

30. (i) Pentaammine carbonato cobalt (III) chloride.

(ii) $[Fe(CN)_6]^{4-}$ ion :

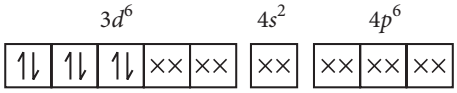
Fe^{2+} ion : 

Fe^{2+} ion hybridised (under the influence of strong field ligand).



Six empty d^2sp^3 hybrid orbitals

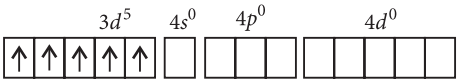
$[Fe(CN)_6]^{4-}$ ion formation :



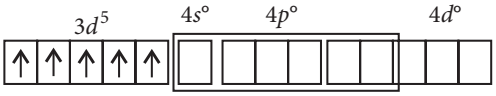
Six pairs of electrons from six CN^- ions (d^2sp^3)

Since the complex ion does not contain any unpaired electron, so it is diamagnetic.

$[FeF_6]^{3-}$ ion :

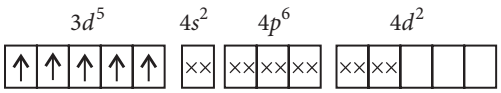
Fe^{3+} ion : 

Fe^{3+} ion hybridised (under the influence of weak field ligand).



Six empty sp^3d^2 hybrid orbitals

$[FeF_6]^{3-}$ ion formation :



Six electron pairs from six F^- ions

As the complex ion contains five unpaired electrons, it is highly paramagnetic in nature.

$$\mu_s = \sqrt{5(5+2)} = \sqrt{35} = 5.9 \text{ BM}$$

31.

S. No.	Drug	Effect
(i)	Brompheniramine	Antihistamine or Antiallergic, it inferes with the natural action of histamine by competing with histamine for binding sites of receptor where histamine exerts its effect.
(ii)	Aspirin	Pain reliever and antipyretic
(iii)	Equanil	For controlling depression and hypertension

33. Substance (*A*) is sulphur.

$$\underset{(A)}{S} + 6HNO_3 \longrightarrow H_2SO_4 + 2H_2O + \underset{(B)}{6NO_2}$$
(Brown colour)

$$\underset{(A)}{S} + Na_2SO_3 \xrightarrow{\Delta} \underset{(C)}{Na_2S_2O_3}$$

$$\underset{(C)}{Na_2S_2O_3} + H_2SO_4 \longrightarrow Na_2SO_4 + SO_2 + S + H_2O$$
(*D*) Turbidity

$$\underset{(A)}{S} + O_2 \longrightarrow \underset{(D)}{SO_2}$$

$$\underset{(C)}{2Na_2S_2O_3} + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$$
Colourless solution

34. For a first order reaction

$$k_1 = \frac{0.693}{t_{1/2}} = \frac{0.693}{30} = 0.0231 \text{ min}^{-1}$$

Suppose the reaction is

	A	$\longrightarrow$	Products
$t = 0$	a		0
$t = 70$ min	$(a - x)$		x

$\therefore$ Fraction of the reactant remains unreacted

$$= \frac{(a-x)}{a}$$

Now, $k_1 = \frac{2.303}{t} \log \frac{a}{a-x}$

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or $0.0231 = \frac{2.303}{70}\log\frac{a}{a-x}$

$\log\frac{a}{a-x} = \frac{0.0231\times 70}{2.303} = 0.7021$

or $\frac{a}{a-x}$ = antilog 0.7021 = 5.036

$\therefore \quad \frac{a-x}{a} = 0.1985 \approx 0.2$

35. (i) (a) $Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

(b) The cell can be represented as

$Zn_{(s)}|Zn^{2+}_{(aq)}\,(0.02\ M)||Cu^{2+}_{(aq)}\,(0.01\ M)|Cu_{(s)}$

(c) $E = E° - \frac{0.059}{n}\log\frac{[Zn^{2+}]}{[Cu^{2+}]}$ = + 1.091 V

(ii) $\Delta_r G° = -nFE° = -$ 212300 J

OR

(i) Resistivity, $\rho = \frac{R\times A}{l} = \frac{5.55\times 10^3 \times \pi}{50\times 4}$

= 87.135 Ω cm

$\text{Conductivity} = \kappa = \frac{1}{\rho} = \left(\frac{1}{87.135}\right) S\ cm^{-1}$

= 0.01148 S cm^{-1}

Molar conductivity,

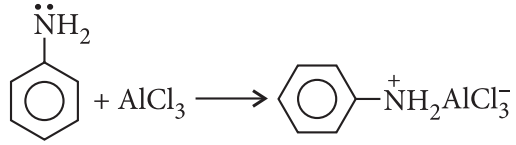
$\Lambda_m = \frac{\kappa\times 1000}{C} = \frac{0.01148\times 1000}{0.05}$

= 229.6 S cm^2 mol^{-1}

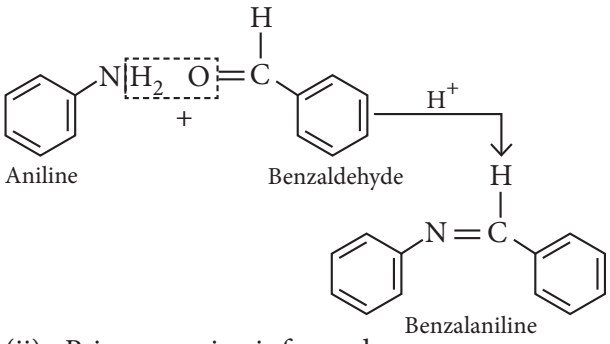
37. (a) (i) Diazonium cation is a weak electrophile and hence reacts with electron rich compounds containing electron donating groups such as – OH, $-NH_2$ and does not react with compounds containing electron withdrawing groups such as $-NO_2$ etc.

(ii) During ammonolysis of alkyl halides, pure amine cannot be prepared because as soon as primary amine is formed it may further react with alkyl halide to give secondary amine which on further reaction with alkyl halide gives *tert*-amine which may finally form quarternary ammonium salt. Thus, a mixture of 1°, 2°, 3° amine and 4° ammonium salt is formed.

(iii) In Friedel — Crafts reaction, $AlCl_3$ is added as a catalyst which is a Lewis acid. It forms a salt with aniline due to which the nitrogen of aniline acquires positive charge. This positively charged nitrogen acts as a strong deactivating group, hence aniline does not undergo Friedel — Crafts reaction.



(b) (i) Benzalaniline is formed.

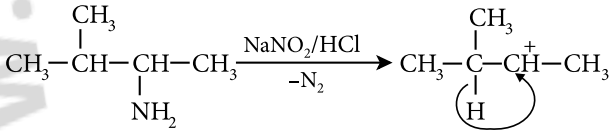


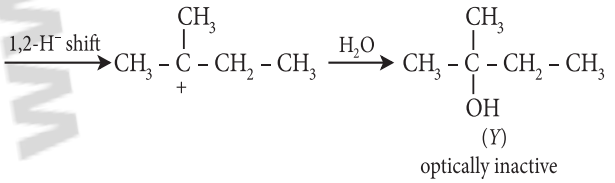
(ii) Primary amine is formed.

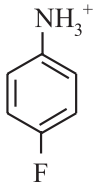
$R—OH + NH_3 \xrightarrow[723\ K]{Al_2O_3} R—NH_2 + H_2O$

1°-amine

OR





(b)  is more acidic due to the strong electron withdrawing power (–*I* effect) of fluorine.

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PRACTICE PAPER 8
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

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SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

Haloalkanes undergo nucleophilic substitution reaction. These reactions proceed by two different mechanisms : S_N1 and S_N2. S_N1 is substitution nucleophilic unimolecular reaction and S_N2 is substitution nucleophilic bimolecular reaction.

1. S_N1 reactions are generally carried out in which type of solvents?

2. S_N2 reactions tend to proceed with which type of nucleophiles?

3. Predict the order of reactivity of the following compounds in S_N1 reaction.
$C_6H_5CH_2Br$, $C_6H_5C(CH_3)(C_6H_5)Br$,
$C_6H_5CH(C_6H_5)Br$, $C_6H_5CH(CH_3)Br$

4. Do vinylic or aryl halides undergo nucleophilic substitution reactions? Give reason for your answer.

5. What happen to the configuration of the products in S_N1 and S_N2 reactions?

Questions 6 to 10 are one word answers :

6. Which oxide of nitrogen combines with Fe (II) ions to form a brown complex?

7. What is the order of the reaction whose k value is 2.3×10^{-5} L^2 mol^{-1} s^{-1}?

8. Which type of drug cimetidine is ?

9. What is the basicity of sulphurous acid ?

10. Give the name of principal ore of copper.

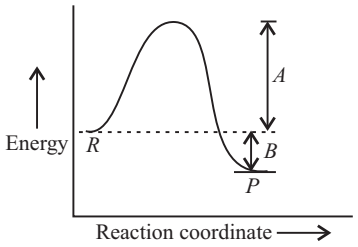
Questions 11 to 15 are multiple choice questions :

11. During the process of electrolytic refining of copper, some metals present as impurity settle as 'anode mud'. These are
(a) Sn and Ag
(b) Pb and Zn
(c) Ag and Au
(d) Fe and Ni.

12. When a lead storage battery is recharged,
(a) lead sulphate is consumed
(b) oxygen gas is evolved
(c) lead sulphate is formed
(d) lead sulphide is formed.

13. H_2S is a toxic gas used in qualitative analysis. If solubility of H_2S in water at STP is 0.195 m, what is the value of K_H?
(a) 0.0263 bar
(b) 69.16 bar
(c) 192 bar
(d) 282 bar

14. The energy diagram of a reaction $P + Q \rightarrow R + S$ is given. What are A and B in the graph?



(a) $A \rightarrow$ activation energy , $B \rightarrow$ heat of reaction
(b) $A \rightarrow$ threshold energy, $B \rightarrow$ heat of reaction
(c) $A \rightarrow$ heat of reaction, $B \rightarrow$ activation energy
(d) $A \rightarrow$ potential energy, $B \rightarrow$ energy of reaction

15. Which of the following statements is not correct for SO_2 gas?
(a) It acts as bleaching agent in moist conditions.
(b) Its dilute solution is used as disinfectant.
(c) Its molecules have linear geometry.
(d) Acidified $KMnO_4$ is decolourised when SO_2 is passed through it.

Questions 16 to 20 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : Maltose and lactose are examples of reducing sugars.

Reason (R) : Maltose and lactose reduce Fehling's solution and Tollens' reagent.

17. Assertion (A) : HNO_3 is stronger acid than HNO_2.

Reason (R) : In HNO_3 there are two nitrogen to oxygen bonds whereas in HNO_2 there is only one.

18. Assertion (A) : Molar conductivity increases with decrease in concentration.

Reason (R) : Conductivity always decreases with decrease in concentration.

19. Assertion (A) : Amalgam of mercury with sodium is an example of solid solutions.
Reason (R) : Mercury is solvent and sodium is solute in the solution.

20. Assertion (A) : A standard hydrogen electrode is also called reversible electrode.

Reason (R) : Standard hydrogen electrode can act both as anode as well as cathode in an electrochemical cell.

SECTION - B

21. Why is freezing point depression of 0.1 M sodium chloride solution nearly twice that of 0.1 M glucose solution?

OR

Non-ideal solutions exhibit either positive or negative deviations from Raoult's law. What are these deviations and why are they caused? Explain with one example for each type.

22. Out of C and CO which is a better reducing agent for FeO ?
(i) In the lower part of blast furnace (higher temperature).
(ii) In the upper part of blast furnace (lower temperature).

23. What do you understand by the terms glycosidic linkage and peptide linkage?

24. In a solid AB having the NaCl structure, A atoms occupy the corners of the cubic unit cells. If all the face-centred atoms along one of the axes are removed then predict the resultant stoichiometry of the solid.

25. Define :
(i) Mole fraction (ii) Molality
How are these two related?

26. (a) Name one narcotic drug used as analgesic.
(b) Give an example of artificial sweetener which can be used in cold food and drinks.

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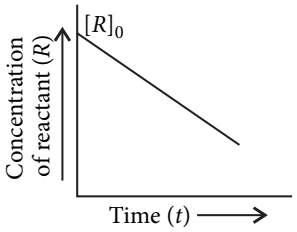
27. The graph for the reaction, $R \longrightarrow P$ is represented by



(a) What will be the unit of rate constant in the reaction?

(b) What does the slope of the graph represent?

SECTION - C

28. The reaction occurs in galvanic cell is

$Zn_{(s)} + 2Ag^{+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + 2Ag_{(s)}$

Give the answers of the following :

(i) Which of the electrode is negatively charged?

(ii) Name the carriers of the current in the cell.

(iii) Write down individual reaction at each electrode.

29. Write chemical equations for the following reactions :

(i) Oxidation of nitrite ion by MnO_4^- in acidic medium.

(ii) Acidification of potassium chromate solution.

(iii) Disproportionation of manganese (VI) in acidic solution.

OR

Account for the following :

(i) Europium (II) is more stable than cerium (II).

(ii) Transition metals have high enthalpies of atomization.

(iii) Actinoid ions are generally coloured.

30. Answer the following questions :

(i) What are biodegradable polymers?

(ii) Identify the aliphatic biodegradable polyester which is used in packaging and orthopaedic devices. Write its full form.

(iii) Write the name and structure of the monomer of nylon 6.

31. Why is aniline acetylated before reacting it with bromine while preparing monobromo-aniline?

OR

Complete the following reactions :

(i) $C_6H_5N_2^+Cl^- + H_3PO_2 + H_2O \rightarrow$

(ii) $C_6H_5NH_2 + (CH_3CO)_2O \rightarrow$

(iii) $C_6H_5N_2^+Cl^- \xrightarrow[\text{(ii) } NaNO_2/Cu,\Delta]{\text{(i) } HBF_4}$

32. How will you bring about the following conversions ?

(a) Toluene to benzaldehyde

(b) Ethanal to but-2-en-1-oic acid

33. Account for the following :

(a) SO_2 is a gas while SeO_2 is a solid.

(b) H_2Te is the strongest reducing agent amongst all the hydrides of group 16 elements.

34. Explain why :

(a) Alkyl halides though polar yet are immiscible in water?

(b) Iodoform is formed by reaction of acetone with hypoiodite ion and not iodide ion?

SECTION - D

35. Answer the following :

(a) Why silica and alumina gels are used for removing moisture and controlling humidity?

(b) On which factors adsorption of a solute from a solution depends?

(c) Define:

(i) Electrophoresis

(ii) Coagulation

OR

(a) Give four examples of heterogeneous catalysis.

(b) What do you mean by activity and selectivity of catalysts?

(c) Explain the following terms :

(i) Dialysis (ii) Tyndall effect

36. Explain why :

(i) $[Cr(NH_3)_6]^{3+}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic.

(ii) A solution of $[Ni(H_2O)_6]^{2+}$ is green but $[Ni(CN)_4]^{2-}$ is colourless.

(iii) The low spin tetrahedral complexes are rarely observed.

OR

(i) $[Ni(Cl)_2(P(CH_3)_3)_2]$ is a paramagnetic complex of Ni(II). Analogous Pd(II) complex is diamagnetic. Explain. How

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many geometrical isomers will be possible for Ni(II) and Pd(II) complexes?

(ii) Write all the geometrical isomers of [Pt (NH_3) (Cl) (Py) (Br)] and how many of these will exhibit optical isomerism?

37. How will you convert :

(i) Phenol to anisole

(ii) Propan-2-ol to 2-methylpropan-2-ol

(iii) Aniline to phenol

(iv) 2-Methyl-1-pentene to 2-methylpentan-2-ol

(v) Phenol to acetophenone?

OR

(i) Ethers are decomposed by H*X*. Arrange reactivity of ethers with HCl, HBr, HI in the increasing order.

(ii) Explain the reaction of $CH_3CH_2OCH_3$ with

(a) limited amount of HI

(b) excess of HI

(iii) Why boiling point of ethers are lower than isomeric alcohols?

Hints & Solutions

1. Polar protic solvents like water, alcohol and acetic acid

2. S_N2 reactions tend to proceed with strong nucleophiles, *e.g.*,CH_3O^-, CN^-, OH^-, etc.

3. $C_6H_5C(CH_3)(C_6H_5)Br > C_6H_5CH(C_6H_5)Br > C_6H_5CH(CH_3)Br > C_6H_5CH_2Br$

4. No, these are unreactive in nucleophilic substitution reactions because of double bond character of C — *X* bond due to resonance.

5. In S_N1 reactions, configuration is retained but in front attack inversion takes place (racemisation and inversion). In S_N2 reactions, inversion of configuration takes place (Walden inversion).

6. NO

7. Third order

8. Antihistamine

9. Two (dibasic)

10. Copper pyrites

11. (c) : In the electrolytic refining of copper the more electropositive impurities like Fe, Zn, Ni, Co, etc. dissolve in the solution and less electropositive impurities such as Ag, Au and Pt collect below the anode in the form of anodic mud.

12. (c) : $2PbSO_{4(s)} + 2H_2O_{(l)} \rightarrow Pb_{(s)} + PbO_{2(s)} + 4H^+_{(aq)} + 2SO^{2-}_{4(aq)}$

13. (d) **14. (a)** **15. (c)**

16. (a)

17. (c) : HNO_3 is stronger acid than HNO_2 but there are three N — O bonds in HNO_3 while two in HNO_2.

18. (b) : $\lambda_m = \kappa V$

With decrease in concentration, total volume *V* of the solution containing one mole of electrolyte also increases, κ decreases with increasing dilution (*V*) of the solution. Though κ decreases, but *V* increases much more, hence molar conductivity (λ_m) increases.

19. (c) : Amalgam of mercury with sodium is an example of liquid in solid type solid solution. Here mercury (liquid metal) acts as solute and sodium as solvent.

20. (a)

21. Sodium chloride being a strong electrolyte completely dissociates in the solution while glucose does not dissociate. The number of particles in sodium chloride solution becomes double than that of glucose solution. Hence freezing point depression of sodium chloride is nearly twice that of glucose solution of same molarity, because it is a colligative property.

OR

When the vapour pressure of a non-ideal solution is either higher or lower than that predicted by Raoult's law, the solution exhibits deviation. These deviations are caused because of unequal intermolecular attractive forces between solute-

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solvent molecules and solute-solute or solvent-solvent molecules.

Examples of positive deviations : Mixture of ethanol and acetone, carbon disulphide and acetone.

Examples of negative deviations : Chloroform and acetone, nitric acid and water.

24. If we remove face-centred atoms of one axis, then two face atoms are removed. Thus, effective no. of *A* atoms becomes 3 and *B* occupies octahedral holes, the effective no. of *B* is 4. Thus, A_3B_4 is the required stoichiometry.

25. Relation between *m* and *x* :

$$x_1 = \frac{n_1}{n_1+n_2}, x_2 = \frac{n_2}{n_1+n_2} \begin{cases} n_1 = \text{No. of moles of solute} \\ n_2 = \text{No. of moles of solvent} \end{cases}$$

$$\frac{x_1}{x_2} = \frac{n_1}{n_2} = \frac{\text{Moles of solute}}{\text{Moles of solvent}} = \frac{w_1}{M_1} \times \frac{M_2}{w_2}$$

or $$\frac{x_1 \times 1000}{x_2 \times M_2} = \frac{w_1 \times 1000}{M_1 \times w_2}$$

$$\therefore \quad \frac{x_1 \times 1000}{(1-x_1)M_2} = \text{Molality}$$

26. (a) Morphine (b) Aspartame

27. (a) The reaction is of zero order. The unit of rate constant of zero order reaction is mol L^{-1} s^{-1}.

(b) Slope of the straight line graph gives rate constant.

$$-k = \frac{d[R]}{dt}$$

29. (i) $5NO_2^- + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 3H_2O + 5NO_3^-$

(ii) $2K_2CrO_4 + 2H^+ \rightarrow K_2Cr_2O_7 + 2K^+ + H_2O$

(iii) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

34. (a) All the alkyl halides are unable to form hydrogen bonding with water or to break the already present hydrogen bonding in water.

(b) Hypoiodite ion can act as an oxidising agent while iodide ion does not.

35. (a) Alumina and silica are good adsorbents. They can adsorb even small amount of moisture present in atmosphere.

(b) Following factors affect adsorption of solute from solution:

(i) The extent of adsorption decreases with an increase in temperature.

(ii) It increases with an increase of surface area of the adsorbent.

(iii) It also depends on the increase in the concentration of the solute in the solution.

(iv) It also depends on nature of adsorbent and adsorbate.

(c) (i) The movement of colloidal particles under an applied electrode potential is called electrophoresis.

(ii) The process of settling of colloidal particles under the effect of gravity is called coagulation.

OR

(a) Examples of heterogenous catalysis :

Oxidation of sulphur dioxide to sulphur trioxide.

$$2SO_{2(g)} + O_{2(g)} \xrightarrow{Pt_{(s)}} 2SO_{3(g)}$$

Formation of ammonia by Haber's process.

$$N_{2(g)} + 3H_{2(g)} \xrightarrow{Fe_{(s)}} 2NH_{3(g)}$$

Oxidation of NH_3 into nitric oxide.

$$4NH_{3(g)} + 5O_{2(g)} \xrightarrow{Pt_{(s)}} 4NO_{(g)} + 6H_2O_{(g)}$$

Hydrogenation of vegetable oils to vegetable ghee.

$$\text{Veg. oil}_{(l)} + H_{2(g)} \xrightarrow{Ni_{(s)}} \text{Veg. ghee}_{(s)}$$

(b) Activity means capacity of a catalyst to increase the rate of the reaction. It depends upon the strength of chemisorption to a large extent.

Selectivity of a catalyst is its ability to direct a reaction to yield a particular product.

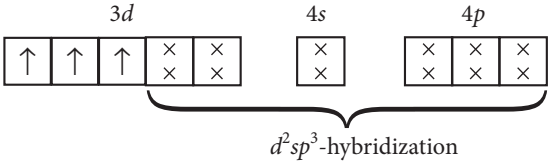
(c) (i) Dialysis is a process of removing a dissolved substance from a colloidal solution by means of diffusion through a semipermeable membrane.

(ii) Scattering of light by the charged particles is called Tyndall effect.

36. (i) The oxidation state of Cr in the compelx is +3.

Cr^{3+} : $Ar[3d]^3$

$[Cr(NH_3)_6]^{3+}$:



d^2sp^3-hybridization

Three unpaired electrons are present, hence it is paramagnetic.

The oxidation state of Ni in the complex is +2.

Ni^{2+} : $[Ar]3d^8$

$[Ni(CN)_4]^{2-}$:

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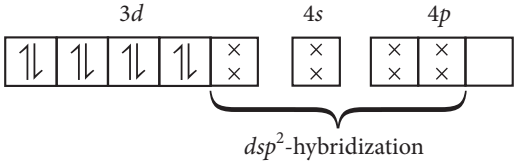
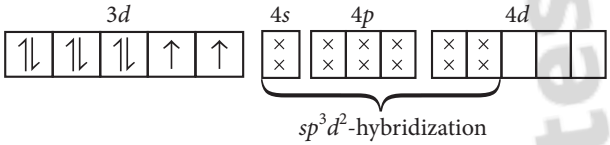


As CN^- is a strong ligand, unpaired electrons are paired up, hence it is diamagnetic.

(ii) H_2O is a weak ligand hence, $[Ni(H_2O)_6]^{2+}$ is an outer-orbital complex. The complex has two unpaired electrons. The *d-d* transition is possible. Hence, the compound is coloured.

$[Ni(H_2O)_6]^{2+}$:



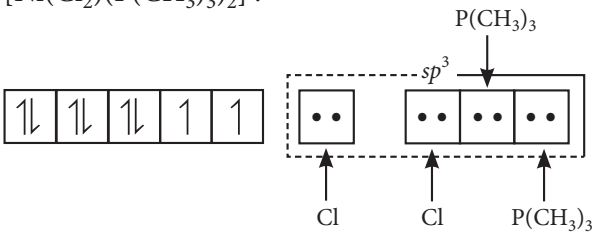
In $[Ni(CN)_4]^{2-}$: CN^- is a strong ligand. The unpaired electrons are paired up. No unpaired electrons are present, *i.e.*, *d-d* transition is not possible, hence the complex is colourless.

(iii) In tetrahedral coordination entity formation, the *d*-orbital splitting is smaller. Consequently, the orbital splitting energies are not sufficiently large to force pairing and therefore, low spin configurations are rarely observed.

OR

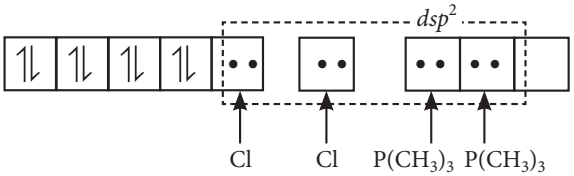
(i) In both Ni(II) and Pd(II), there is d^8 configuration. In Ni(II), value of crystal field splitting energy is less than Pd(II). So in Ni(II) pairing is less favoured while in Pd(II), all electrons are paired.

$[Ni(Cl_2)(P(CH_3)_3)_2]$:

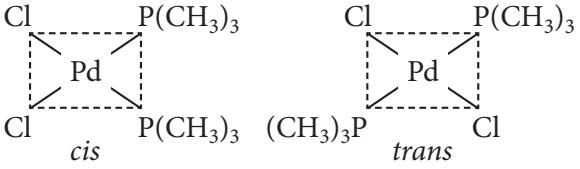


Geometry is tetrahedral, so only one isomer will be possible.

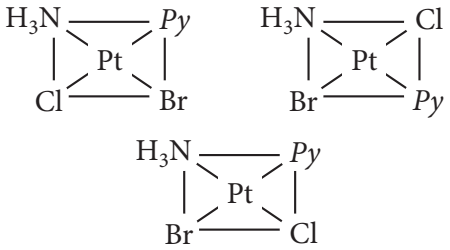
$[Pd(Cl_2)(P(CH_3)_3)_2]$:



So, it has square planar structure *i.e.*, MA_2B_2 type and will exist as *cis* and *trans* isomer.



(ii) Three geometrical isomers are possible.



It does not show optical isomerism.

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PRACTICE PAPER 9
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

Compound *A* is a light green crystalline solid. It gives the following tests :

(i) When it dissolves in dilute sulphuric acid, no gas is produced.

(ii) When a drop of $KMnO_4$ is added to the above solution, the pink colour disappears.

(iii) When a compound *A* is heated strongly, gases *B* and *C* with pungent smell come out and a brown residue *D* is left behind.

1. Identify the compound *A*.

2. Pink colour of potassium permanganate disappears due to formation of which compound?

3. Name one of pungent smelling gases evolved.

4. Give the name of the brown residue '*D*'.

5. The gas *B* decolourises acidified $KMnO_4$ solution. Which property of the gas *B* is indicated by this reaction.

Questions 6 to 10 are one word answers :

6. Name the compound which is formed on bromination of aniline.

7. A solid *XY* has NaCl structure. If radius of X^+ ion is 100 pm, what is the limit of radius of Y^- ion?

8. Name a disaccharide which is non-reducing sugar.

9. Name the alkene which on reductive ozonolysis gives only acetone.

10. Which type of polyesters are biodegradable?

Questions 11 to 15 are multiple choice questions :

11. Which of the following is not correct regarding the adsorption of a gas on the surface of solid?
(a) On increasing pressure, adsorption keeps on decreasing.
(b) Enthalpy and entropy changes are negative.
(c) Chemisorption is more specific than physisorption.
(d) Adsorption increases with increase in surface area.

12. A chelating agent has two or more than two donor atoms to bind to a single metal ion. Which of the following is not a chelating agent?
(a) Thiosulphato
(b) Oxalato
(c) Glycinato
(d) Ethane-1,2-diamine

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13. What would be the equivalent conductivity of a cell in which 0.5 N salt solution offers a resistance of 40 ohm whose electrodes are 2 cm apart and 5 cm^2 in area?

(a) 10 $ohm^{-1}\ cm^2\ eq^{-1}$

(b) 20 $ohm^{-1}\ cm^2\ eq^{-1}$

(c) 30 $ohm^{-1}\ cm^2\ eq^{-1}$

(d) 25 $ohm^{-1}\ cm^2\ eq^{-1}$

14. The term 'broad spectrum antibiotics' means

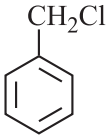
(a) bactericidal antibiotics

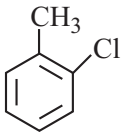
(b) bacteriostatic antibiotics

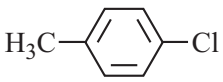
(c) which kill or inhibit a wide range of gram –ve and gram +ve bacteria

(d) which kill or inhibit all types of gram +ve bacteria.

15. The reaction of toluene with chlorine in the presence of iron and in the absence of light yields _____.

(a) 

(b) 

(c) 

(d) mixture of (b) and (c)

Questions 16 to 20 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is correct, but Reason (R) is incorrect statement.

(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : In presence of enzyme, substrate molecule can be attacked by the reagent effectively.

Reason (R) : Active sites of enzymes hold the substrate molecule in a suitable position.

17. Assertion (A) : Anilinium hydrogensulphate, on heating forms a mixture of *ortho*- and *para*-aminobenzene sulphonic acids.

Reason (R) : *p*-Aminobenzene sulphonic acid exists as zwitter ion.

18. Assertion (A) : The boiling points of given compounds follow the order :

$CH_3CH_2CH_2CH_2CH_3 < H_5C_2OC_2H_5 < CH_3CH_2CH_2CHO < CH_3CH_2CH_2CH_2OH$

Reason (R) : Boiling point depends upon molecular mass only.

19. Assertion (A) : When the vapours of a primary or secondary alcohol are passed over heated copper at 573 K, an aldehyde or ketone is formed.

Reason (R) : Reduction of alcohols takes place.

20. Assertion (A) : S_N1 reactions are generally carried out in polar aprotic solvents (like water, alcohol, acetic acid, etc.)

Reason (R) : $C_6H_5CH(C_6H_5)Br$ is more reactive than $C_6H_5CH(CH_3)Br$ in S_N1 reactions.

SECTION - B

21. (a) Of the two bases named below, which one is present in RNA and which one is present in DNA?

(i) Thymine (ii) Uracil

(b) The deficiency of which vitamin causes the following diseases?

(i) Pernicious anaemia

(ii) Xerophthalmia

22. Show that in a first order reaction, time required for completion of 99.9% is 10 times of half-life ($t_{1/2}$) of the reaction.

OR

Show that in a first order reaction, time required for 99% completion is twice the time required for the completion of 90% reaction.

23. Distinguish between hexagonal and monoclinic unit cells.

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24. Account for the following :

Bond angle in NH_4^+ is higher than NH_3.

OR

Complete the following reactions :

(i) $Cu + HNO_3(dilute) \longrightarrow$

(ii) $PH_4I + KOH \longrightarrow$

25. Explain the following observations :

(i) A beam of light passing through a colloidal solution is visible.

(ii) Passing an electric current through a colloidal solution removes colloidal particles from it.

26. Write two advantages of sucralose over the sweetening agents.

27. Write the common and IUPAC names for the isomers of molecular formula C_4H_9Br and classify them as to whether they are 1°, 2° or 3°.

SECTION - C

28. (i) A given ore is a mixed sulphide of lead and zinc (PbS-ZnS). Suggest a method to separate the mixture.

(ii) Write the chemical reactions involved in the extraction of gold by cyanide process.

29. For the complex $[Fe(en)_2Cl_2]Cl$, identify the following :

(i) Type of hybridisation and geometry.

(ii) IUPAC name of the complex.

OR

Draw the structure of optical isomers of :

(i) $[Cr(C_2O_4)_3]^{3-}$ (ii) $[PtCl_2(en)_2]^{2+}$

(iii) $[Cr(NH_3)_2Cl_2(en)]^+$

30. An organic compound (*A*) of molecular formula $C_8H_{16}O_2$ was hydrolysed with dilute sulphuric acid to give a carboxylic acid (*B*) and an alcohol (*C*). Oxidation of (*C*) with chromic acid produced (*B*). (*C*) on dehydration gives but-1-ene. Write equations for the reactions involved.

31. At 380°C, the half-life period for the first order decomposition of H_2O_2 is 360 min. The energy of activation of the reaction is 200 kJ mol^{-1}. Calculate the time required for 75% decomposition at 450°C.

32. (i) At 300 K, 36 g of glucose present per litre in its solution has an osmotic pressure of 4.98 bar. If osmotic pressure of solution is 1.52 bar at the same temperature, what would be its concentration?

(ii) Osmotic pressure is more useful to calculate molecular mass than other colligative properties. Why?

OR

15 g of an unknown molecular mass substance was dissolved in 450 g of water. The resulting solution freezes at – 0.34°C. What is the molar mass of the substance?

(K_f for water = 1.86 K kg mol^{-1})

33. What happens when

(a) Ethylene is treated with dilute alkaline potassium permanganate solution?

(b) Phenol is treated with sodium hydroxide?

(c) Cyclohexanol is treated with conc. H_2SO_4 at 443 K?

34. 1 g of a non-volatile solute was dissolved in 100 g of acetone (molar mass = 58 g mol^{-1}) at 298 K. The vapour pressure of the solution was found to be 192.5 mm of Hg. Calculate the molar mass of solute. (The vapour pressure of pure acetone at 298 K is 195 mm Hg.)

SECTION - D

35. (i) Give one chemical test to distinguish between the following pairs of compounds :

(a) Methylamine and dimethylamine

(b) Secondary and tertiary amines

(ii) Write the chemical reaction stating the reaction conditions required for each of the following conversions :

(a) Methyl bromide to ethylamine

(b) Aniline to phenol

(c) *p*-Toluidine to 2-bromo-4-methylaniline

OR

(i) Write the chemical reaction stating the reaction conditions required for each of the following conversions :

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(a) Aniline to chlorobenzene
(b) Acetaldehyde to ethylamine

(ii) Give one chemical test to distinguish between the following pairs of compounds :
(a) Ethylamine and aniline
(b) Aniline and benzylamine
(c) Aniline and *N*-methylaniline

36. What are primary cells, secondary cells and fuel cells? Explain with example.

OR

(a) Write the Nernst equation.
(b) Calculate the emf of the following cells at 298 K :
(i) $Mg_{(s)} \mid Mg^{2+}(0.001\ M) \parallel Cu^{2+}(0.0001\ M) \mid Cu_{(s)}$
(ii) $Fe_{(s)} \mid Fe^{2+}(0.001\ M) \parallel H^+(1\ M) \mid H_{2(g)}(1\ bar) \mid Pt_{(s)}$

[Given : $E^\circ_{Mg^{2+}/Mg} = -2.37$ V, $E^\circ_{Cu^{2+}/Cu} = 0.34$ V, $E^\circ_{Fe^{2+}/Fe} = -0.44$ V]

37. Answer the following :
(i) Describe the Contact process for the manufacture of sulphuric acid with special reference to the reaction conditions, catalysts used and yield in the process.
(ii) How is XeO_3 obtained? Write the related chemical equations. Draw the structure of XeO_3.

OR

(i) Give the formula and describe the structure of a noble gas species which is isostructural with :
(a) ICl_4^- (b) BrO_3^-
(ii) Does the hydrolysis of XeF_6 lead to a redox reaction?
(iii) What happens when Cl_2 reacts with hot concentrated solution of sodium hydroxide? Is this reaction a disproportionation reaction?

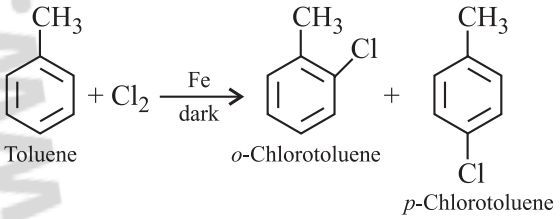
Hints and Solutions

1. Ferrous sulphate ($FeSO_4$)
2. Manganese sulphate
3. Sulphur dioxide or sulphur trioxide
4. Ferric oxide (Fe_2O_3)
5. Reducing agent
6. 2,4,6-Tribromoaniline
7. 136.6 – 241.5 pm
8. Sucrose
9. 2,3-Dimethylbut-2-ene
10. Aliphatic polyesters
11. (a) **12. (a)**
13. (b) : $\kappa = \frac{1}{R} \times \frac{l}{A} = \frac{1}{40} \times \frac{2}{5}$

$\Lambda_{eq} = \kappa \times \frac{1000}{N} = \frac{1}{40} \times \frac{2}{5} \times \frac{1000}{0.5}$

$= 20\ ohm^{-1}\ cm^2\ eq^{-1}$

14. (c)
15. (d) :



16. (a)
17. (b) : Amino group is *ortho* and *para*-directing, hence it directs the SO_3 electrophile to *ortho*- and *para*-positions.
18. (c) : The correct order is :
$CH_3CH_2CH_2CH_2CH_3 < H_5C_2OC_2H_5 < CH_3CH_2CH_2CHO < CH_3CH_2CH_2CH_2OH$
Boiling point depends upon the nature of intermolecular forces, stronger the forces, higher is the boiling point.
$CH_3CH_2CH_2CH_2OH$ molecules are associated due to extensive intermolecular hydrogen bonding, $CH_3CH_2CH_2CHO$ has stronger dipole-dipole

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attractions as compared to $H_5C_2OC_2H_5$ and $CH_3CH_2CH_2CH_2CH_3$ molecules have only van der Waals forces.

19. (c) : When the vapours of a primary or a secondary alcohol are passed over heated copper at 573 K, dehydrogenation takes place and an aldehyde or ketone is formed.

20. (d) : S_N1 reactions are generally carried out in polar protic solvents. Carbocation obtained from $C_6H_5CH(C_6H_5)Br$ is more stable than $C_6H_5CH(CH_3)Br$ as $C_6H_5\overset{+}{C}H(C_6H_5)$ is stabilised by two phenyl groups and hence it is more reactive in S_N1 reaction.

22. $$\frac{t_{99.9\%}}{t_{1/2}} = \frac{6.909}{k} \times \frac{k}{0.693} \simeq 10$$

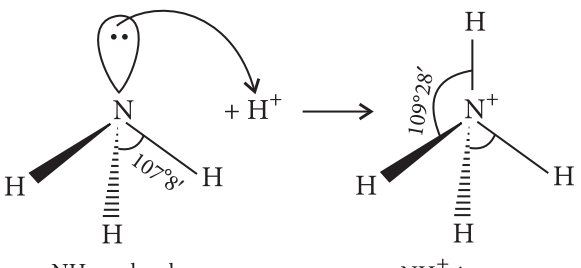
OR

$$\frac{t_{99\%}}{t_{90\%}} = \frac{4.606}{k} \times \frac{k}{2.303} = 2 \Rightarrow t_{99\%} = 2t_{90\%}$$

23.

Hexagonal	**Monoclinic**
$a = b \neq c$	$a \neq b \neq c$
$\alpha = \beta = 90°, \gamma = 120°$	$\alpha = \gamma = 90° \neq \beta$
e.g., Graphite	Monoclinic sulphur

24. N in NH_3 is sp^3–hybridised. It has three bond pairs and one lone pair around N. Due to stronger lone pair-bond pair repulsions than bond pair-bond pair replusions. the tetrahedral angle decreases from 109°28′ to 107°8′. As a result, NH_3 is pyramidal. However, when it reacts with a proton, it forms NH_4^+ ion which has four bond pairs and no lone pair. Due to the absence of lone pair-bond pair repulsions and presence of four identical bond pair-bond pair interaction, NH_4^+ assumes tetrahedral geometry with a bond angle of 109°28′. This explains why the bond angle in NH_4^+ is higher than in NH_3.



NH_3 molecule (Pyramidal) NH_4^+ ion (Tetrahedal)

OR

(i) $3Cu + 8HNO_3(\text{dilute}) \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

(ii) $PH_4I + KOH \longrightarrow KI + H_2O + PH_3$

27. Following are the isomers of molecular formula C_4H_9Br with their common and IUPAC names :

$CH_3CH_2CH_2CH_2Br$; (1-Bromobutane) *n*-Butyl bromide (1° Bromide)

$CH_3CH_2CHBrCH_3$ (2-Bromobutane) *sec*-Butyl bromide (2° Bromide)

$CH_3-CH(CH_3)-CH_2Br$; (1-Bromo-2-methylpropane) *iso*-butyl bromide (1° Bromide)

$(CH_3)_3C-Br$ (2-Bromo-2-methylpropane) *tert*-butyl bromide (3° Bromide)

30. Compound '*A*' ($C_8H_{16}O_2$) on hydrolysis gives an acid '*B*' and an alcohol '*C*'. It shows that '*A*' is an ester. Since the oxidation of alcohol '*C*' also gives the acid '*B*' indicates that '*B*' and '*C*' both contain same number of carbon atoms, *i.e.*, four carbon atoms each and same arrangement of atoms. Formation of but-1-ene on dehydration of '*C*' indicates it to be butan-1-ol, so the possible ester (*A*) could be butyl butanoate.

$CH_3-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_2-CH_2-CH_3$

Butyl butanoate (*A*)

$\xrightarrow{H_2O/H^+}$

$CH_3-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-OH$ + $CH_3-CH_2-CH_2-CH_2-OH$

Butanoic acid (*B*) Butanol (*C*)

$CH_3-CH_2-CH_2-CH_2-OH \xrightarrow{H_2CrO_4}$ (*C*)

$CH_3-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-OH$ (*B*)

$CH_3-CH_2-CH_2-CH_2-OH \xrightarrow[\Delta]{H_2SO_4}$ (*C*)

$CH_3-CH_2-CH=CH_2$ But-1-ene

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31. $k_1 = \dfrac{0.693}{t_{1/2}} = \dfrac{0.693}{360\ \text{min}} = 1.925 \times 10^{-3}\ \text{min}^{-1}$

Applying Arrhenius equation,

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$$

$$\log\frac{k_2}{k_1} = \frac{200\times10^3\ \text{J}}{2.303\times8.314\ \text{J}}\times\left(\frac{723-653}{653\times723}\right)$$

$k_2 = 35.38 \times 1.925 \times 10^{-3}\ \text{min}^{-1} = 6.81 \times 10^{-2}\ \text{min}^{-1}$

For 75% decomposition,

x = 75% of a = 0.75 a

$$t = \frac{2.303}{k}\log\frac{a}{a-x} = \frac{2.303}{6.81\times10^{-2}\ \text{min}^{-1}}\times\log\frac{a}{a-0.75a}$$

$$= \frac{2.303}{6.81\times10^{-2}\ \text{min}^{-1}}\log\frac{1}{0.25}\ \text{min} = 20.36\ \text{min}$$

32. (i) $\pi = CRT$

R and T are same in both cases hence

$$\frac{\pi_1}{\pi_2} = \frac{C_1}{C_2}\ \text{or,}\ C_2 = \frac{\pi_2C_1}{\pi_1}$$

Molarity of first solution

$$C_1 = \frac{36}{180} = 0.2\ \text{mol L}^{-1}$$

$$C_2 = \frac{\pi_2C_1}{\pi_1} = \frac{1.52\times0.2}{4.98} = 0.0610\ \text{M}$$

(ii) Magnitude of osmotic pressure is large even for very dilute solutions and it can be measured at room temperature hence it is more useful for the calculation of molecular mass.

OR

(i) Given : w_2 = 15 g, w_1 = 450 g

K_f = 1.86 K kg mol^{-1}, M_2 = ?

ΔT_f = 0 – (– 0.34) = 0.34°C or 0.34 K

Now, $\Delta T_f = K_f \cdot m$

$$\Rightarrow\ \Delta T_f = K_f \times\frac{w_2}{M_2}\times\frac{1000}{w_1}$$

$$\Rightarrow\ M_2 = \frac{K_f\times w_2\times1000}{\Delta T_f\times w_1}$$

$$\Rightarrow\ M_2 = \frac{1.86\times15\times1000}{0.34\times450}$$

M_2 = 182.35 g mol^{-1}

34. Given, $p_A°$ (acetone) = 195 mm; p_S = 192.5 mm

w_1 = 100 g, M_1 = 58 g mol^{-1} ; w_2 = 1 g, M_2 = ?

$$\frac{p_A° - p_S}{p_A°} = x_2 = \frac{\dfrac{w_2}{M_2}}{\dfrac{w_1}{M_1} + \dfrac{w_2}{M_2}}$$

$$\Rightarrow\ \frac{195 - 192.5}{195} = \frac{\dfrac{1}{M_2}}{\dfrac{1}{M_2} + \dfrac{100}{58}}$$

$$\Rightarrow\ \frac{2.5}{195} = \frac{1}{M_2}\times\frac{58\ M_2}{58 + 100\ M_2}$$

$$\Rightarrow\ 0.0128 = \frac{58}{58 + 100\,M_2}$$

$$\Rightarrow\ 58 + 100\ M_2 = \frac{58}{0.0128}$$

$$58 + 100\ M_2 = 4531.25$$

$$\therefore\quad M_2 = \frac{4473.25}{100} = 44.7325 \simeq 45\ \text{g mol}^{-1}$$

36. **OR**

(a) For the equation,

$M^{n+}_{(aq)} + ne^- \rightarrow M_{(s)}$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log\frac{1}{\left[M^{n+}_{(aq)}\right]}$$

$$= E^\circ_{\text{cell}} - \frac{0.059}{n}\log\frac{1}{\left[M^{n+}_{(aq)}\right]}$$

Concentration of solids is taken as unity.

(b) (i) $\text{Mg}_{(s)}\,|\,\text{Mg}^{2+}(0.001\ \text{M})\,||\,\text{Cu}^{2+}(0.0001\text{M})\,|\,\text{Cu}_{(s)}$

At anode : $\text{Mg}_{(s)} \longrightarrow \text{Mg}^{2+}_{(aq)} + 2e^-$

At cathode : $\text{Cu}^{2+}_{(aq)} + 2e^- \longrightarrow \text{Cu}_{(s)}$

Net cell reaction :

$\text{Mg}_{(s)} + \text{Cu}^{2+}_{(aq)} \longrightarrow \text{Mg}^{2+}{}_{(aq)} + \text{Cu}_{(s)}$

$\therefore\quad n = 2$

Using Nernst equation,

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log\frac{[\text{Mg}^{2+}]}{[\text{Cu}^{2+}]}$$

For the given cell,

$E°_{\text{cell}} = E°_{\text{cathode}} - E°_{\text{anode}} = E°_{\text{Cu}^{2+}/\text{Cu}} - E°_{\text{Mg}^{2+}/\text{Mg}}$

= 0.34 – (–2.37)= 2.71 V

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Given : $[Mg^{2+}] = 0.001$ M, $[Cu^{2+}] = 0.0001$ M
Putting values in Nernst equation at 298 K,

$$E_{cell} = 2.71 - \frac{0.059}{2} \log \frac{0.001}{0.0001}$$

$E_{cell} = 2.71 - 0.0295 \log 10 = 2.71 - 0.0295 = 2.68$ V

(ii) $Fe_{(s)} \mid Fe^{2+}(0.001\ M) \parallel H^+(1\ M) \mid H_{2(g)}(1\ bar) \mid Pt_{(s)}$

At anode : $Fe_{(s)} \longrightarrow Fe^{2+}_{(aq)} + 2e^-$

At cathode : $2H^+_{(aq)} + 2e^- \longrightarrow H_{2(g)}$

Cell reaction : $Fe_{(s)} + 2H^+_{(aq)} \longrightarrow Fe^{2+}_{(aq)} + H_{2(g)}$

$\therefore \quad n = 2$

Using Nernst equation at 298 K,

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2} \log \frac{[Fe^{2+}] \times p_{H_2}}{[H^+]^2}$$

For the given cell,

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{H^+/H_2} - E^\circ_{Fe^{2+}/Fe}$

$= 0 - (-0.44) = +0.44$ V

Given : $[Fe^{2+}] = 0.001$ M; $[H^+] = 1$ M; $p_{H_2} = 1$ bar

Putting values in Nernst equation,

$$E_{cell} = 0.44 - 0.0295 \log \frac{0.001 \times 1}{1^2}$$

$= 0.44 - 0.0295 \log 10^{-3} = 0.44 - [(0.0295) \times (-3)]$

$= 0.44 + 0.0885 = 0.53$ V

37. (i) **Contact process :** It involves three steps :

(a) Burning of sulphur or sulphide ore in air to generate SO_2.

$S + O_2 \rightarrow SO_2$;

$4FeS_2 + 11O_2 \rightarrow 8SO_2 + 2Fe_2O_3$

(b) Conversion of SO_2 to SO_3 by reaction with oxygen in the presence of V_2O_5 catalyst.

$2SO_2 + O_2 \overset{V_2O_5}{\rightleftharpoons} 2SO_3$; $\Delta_r H^\circ = -196$ kJ mol^{-1}

(c) The SO_3 gas from the catalytic converter is absorbed in conc. H_2SO_4 to form oleum ($H_2S_2O_7$). Dilution of oleum with water gives H_2SO_4 of desired concentration.

$SO_3 + H_2SO_4 \rightarrow \underset{\text{(Oleum)}}{H_2S_2O_7}$

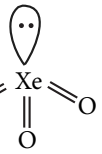
$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$

(ii) XeO_3 can be obtained by hydrolysis of XeF_4 and XeF_6.

$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24\ HF + 3O_2$

$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

Structure :



OR

(ii) Partial hydrolysis of XeF_6 gives $XeOF_4$ and XeO_2F_2

$\overset{+6\ -1}{XeF_6} + \overset{+1-2}{H_2O} \longrightarrow \overset{+6\ -2\ -1}{XeOF_4} + \overset{+1-1}{2HF}$

$\overset{+6\ -1}{XeF_6} + 2\overset{+1-2}{H_2O} \longrightarrow \overset{+6\ -2\ -1}{XeO_2F_2} + \overset{+1-1}{4HF}$

On the other hand, complete hydrolysis XeF_6 gives XeO_3.

$\overset{+6\ -1}{XeF_6} + 3\overset{+1\ -2}{H_2O} \longrightarrow \overset{+6\ -2}{XeO_3} + \overset{+1-1}{6HF}$

As the oxidation state of all elements in the reactants and products of hydrolysis remains the same, therefore, these are not the redox reactions.

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PRACTICE PAPER 10
(Unsolved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

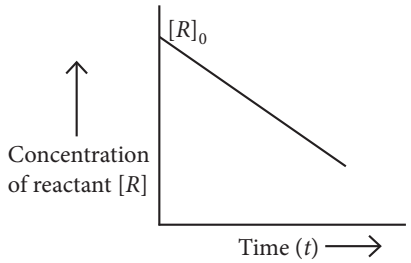
Time allowed : 3 hours **Maximum marks :** 70

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

The integrated rate equations are different for the reaction of different reaction orders.

A reaction, Reactant → Product is represented by



1. Write the expression for $t_{1/2}$ for the reaction.

2. Write the expression for the rate constant for this case.

3. Give an example of the given order reaction.

4. Draw a graph between rate vs. concentration for this type of reaction.

5. What is the intercept of the given order reaction ?

Questions 6 to 10 are one word answers :

6. Give an example of non-narcotic analgesic.

7. Name one vitamin that can be stored in our body.

8. What type of solution obeys Raoult's law at all temperatures and concentrations ?

9. What type of cell nickel-cadmium cell is?

10. Give an example of gel.

Questions 11 to 15 are multiple choice questions :

11. Benzoic acid is treated with $SOCl_2$ and the product (X) formed is reacted with ammonia to give (Y). (Y) on reaction with Br_2 and KOH gives (Z). (Z) in the reaction is

(a) aniline
(b) chlorobenzene
(c) benzamide
(d) benzoyl chloride.

12. Which of the following ores cannot be concentrated by magnetic separation?

(a) Haematite (b) Malachite
(c) Magnetite (d) Siderite

13. A coordination compound $CrCl_3 \cdot 4H_2O$ gives white precipitate of AgCl with $AgNO_3$. The molar conductance of the compound corresponds to two ions. The structural formula of the compound is

(a) $[Cr(H_2O)_4Cl_3]$
(b) $[Cr(H_2O)_3Cl_3]H_2O$
(c) $[Cr(H_2O)_4Cl_2]Cl$
(d) $[Cr(H_2O)_4Cl]Cl_2$

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14. What should be the correct IUPAC name for diethylbromomethane?
(a) 1-Bromo-1,1-diethylmethane
(b) 3-Bromopentane
(c) 1-Bromo-1-ethylpropane
(d) 1-Bromopentane

15. Glyptal polymer is obtained by the following monomers,
(a) malonic acid + ethylene glycol
(b) phthalic acid + ethylene glycol
(c) maleic acid + formaldehyde
(d) acetic acid + phenol.

Questions 16 to 20 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

16. Assertion (A) : Copper dissolves in dilute nitric acid but not in dilute HCl.
Reason (R) : Copper has negative E° value.

17. Assertion (A) : The boiling point of the compounds increases in the order : Isopropylchloride < 1-Chloropropane < 1-Chlorobutane.
Reason (R) : Boiling point depends upon the molecular mass and surface area.

18. Assertion (A) : With Br_2–H_2O, phenol gives 2,4,6-tribromophenol but with Br_2–CS_2, it gives 4-bromophenol as the major product.
Reason (R) : In water, ionisation of phenol is enhanced but in CS_2, it is greatly suppressed.

19. Assertion (A) : Glucose does not form hydrogensulphite addition product with $NaHSO_3$.
Reason (R) : Glucose gives all the reactions of aldehydic group.

20. Assertion (A) : Cross aldol condensation of ethanal and propanal gives a mixture of four products.
Reason (R) : Ethanal and propanal, both contain α-hydrogen atom.

SECTION - B

21. (i) What is the structural difference between a nucleoside and a nucleotide?
(ii) "The two strands of DNA are not identical but are complementary." Explain.

22. Arrange the following in decreasing order of property indicated.
(i) H_2O, H_2S, H_2Se, H_2Te (Acidic character)
(ii) NH_3, PH_3, AsH_3, SbH_3 (Bond angle)

23. What is the formula of a compound in which the element *Y* forms *ccp* lattice and atoms of *X* occupy $2/3^{rd}$ of octahedral voids?

OR

Gold has a close-packed structure which can be viewed as spheres occupying 0.74 of the total volume. What is the radius of gold ion if density of gold is 19.3 g/cc? (Au = 197 amu)

24. Give one chemical test to distinguish between the following pairs of compounds :
(a) Ethanal and propanal
(b) Benzoic acid and ethyl benzoate

25. State the significant of numbers 6 and 6, 6 in polymer nylon-6 and nylon-6, 6.

OR

Define the following with examples :
(i) Semi-synthetic polymers
(ii) Network polymers

26. State reasons for the following :
(i) Rusting of iron is said to be an electrochemical phenomenon.
(ii) For a weak electrolyte, its molar conductance in dilute solution increases

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sharply as its concentration in solution is decreased.

27. Why do we require artificial sweetening agents?

SECTION - C

28. (a) How is leaching carried out in case of low grade copper ores?

(b) Which of the following metals cannot be extracted by the smelting process and why ; Zn, Fe, Pb, Al

29. (a) Arrange the following complex in order of decreasing conductivity :
$[Co(NH_3)_3 Cl_3]$, $[Co(NH_3)_6] Cl_3$, $[Co(NH_3)_4 Cl_2] Cl$, $[Co (NH_3)_5Cl]Cl_2$

(b) Write the name of linkage isomer of $[Co(ONO)(NH_3)_5]^{2+}$.

(c) Though CO is a weak Lewis base yet it forms a number of stable metal carbonyls. Explain.

OR

Using valence bond approach predict the shape and magnetism (*i.e.*, paramagnetic or diamagnetic) of $[Co(C_2O_4)_3]^{3-}$ (atomic number of Co = 27).

30. Explain the following terms :

(i) Micelles (ii) Dialysis

(iii) Hardy-Schulze rule.

31. (a) A solution prepared by dissolving 8.95 mg of a gene fragment in 35.0 mL of water has an osmotic pressure of 0.335 torr at 25°C. Assuming the gene fragment is a non-electrolyte, determine its molar mass.

(b) Osmotic pressure of 1 M KCl solution is higher than 1 M urea solution. Why?

OR

0.5 g KCl was dissolved in 100 g of water and the solution originally at 20°C, froze at –0.24°C. Calculate the percentage ionisation of salt. K_f per 1000 g water = 1.86 K.

32. Account for the following :

(i) Only higher members of group-18 of the periodic table are expected to form compounds.

(ii) Tendency to show – 2 oxidation state diminishes from sulphur to polonium in group 16.

33. Predict the alkenes that would be formed by dehydrohalogenation of the following halides with sodium ethoxide in ethanol and identify the major alkene.

(i) 1-Bromo-1-methylcyclohexane

(ii) 2-Chloro-2-methylpropane

(iii) 3-Bromo-2,2,3-trimethylpentane

34. Answer the following questions :

(i) Why electrophilic substitution in aromatic amines takes place more readily than benzene?

(ii) Why primary amines have higher boiling point than corresponding tertiary amines?

(iii) Write two important uses of *N*, *N*-dimethylamines.

SECTION - D

35. (a) The electronic configurations of two members of the lanthanoid series are as follows : $4f^1 5d^1 6s^2$ and $4f^7 5d^0 6s^2$

What are their atomic numbers? Predict the oxidation states exhibited by these elements in their compounds.

(b) Assign reasons for each of the following statements :

(i) The largest number of oxidation states are exhibited by the elements in the middle of the first row of the transition elements.

(ii) The atomic radii decreases in size with the increasing atomic number in the lanthanoid series.

OR

(a) Explain giving reasons :

(i) Transition elements and their compounds exhibit paramagnetic behaviour.

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(ii) Transition elements and their compounds are found to be good catalysts.

(iii) Enthalpies of atomization of the transition elements are high.

(b) Compare the chemistry of actinoids with that of the lanthanoid in reference to

(i) Atomic and ionic sizes.

(ii) Oxidation states.

36. (i) In a reaction between *A* and *B*, the initial rate of reaction was measured for different initial concentration of *A* and *B* as given below :

A/mol L^{-1}	0.20	0.20	0.40
B/mol L^{-1}	0.30	0.10	0.05
r_0/mol $L^{-1}s^{-1}$	5.07×10^{-5}	5.07×10^{-5}	7.16×10^{-5}

What is the order of reaction with respect to *A* and *B*?

(ii) Can a reaction have zero activation energy.

OR

(i) During nuclear explosion, one of the products is ^{90}Sr with half-life of 28.1 years. If 1 μg of ^{90}Sr was absorbed in the bones of a newly born baby instead of calcium, how much of it will remain after 10 years and 60 years if it is not lost metabolically?

(ii) The decomposition of a hydrocarbon follows the equation,

$k = (4.5 \times 10^{11}s^{-1})e^{-28000K/T}$. Calculate E_a.

37. Answer the following questions :

(a) How is 1-propoxypropane synthesised from propan-1-ol?

(b) Write the chemical equations for the reactions of the hydrogen iodide with

(i) 1-propoxypropane

(ii) Methoxybenzene

(iii) Benzyl ethyl ether.

OR

(a) Write the final product(s) in each of the following reactions :

(i) $CH_3CH_2-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow{Cu/573\ K}$

(ii) $C_6H_5-OH \xrightarrow[(ii)\ H^+]{(i)\ CHCl_3 + aq.\ NaOH}$

(b) Name the reagents which are used in the following conversions :

(i) A primary alcohol to an aldehyde

(ii) Butan-2-one to butan-2-ol

(iii) Phenol to 2, 4, 6-tribromophenol

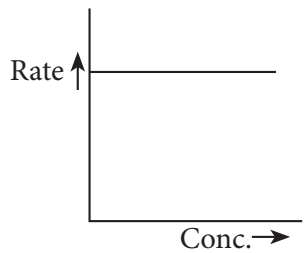
Hints & Solutions

1. It is a zero order reaction, hence $t_{1/2} = \frac{[R]_0}{2k}$

2. $k = \frac{[R]_0 - [R]}{t}$

3. The decomposition of gaseous ammonia on a hot platinum surface at high pressure.

4.



5. $[R]_0$

6. Aspirin

7. Vitamin B_{12}

8. Ideal solution

9. Secondary cell

10. Cheese

11. (a)

12. (b) : Malachite is not a magnetic ore. It is $CuCO_3 \cdot Cu(OH)_2$.

13. (c)

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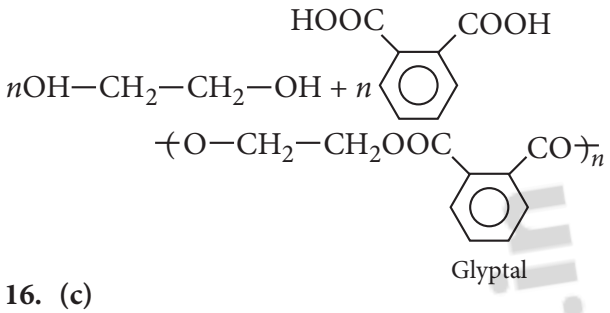
14. (b)

15. (b) :



Glyptal

16. (c)

17. (a)

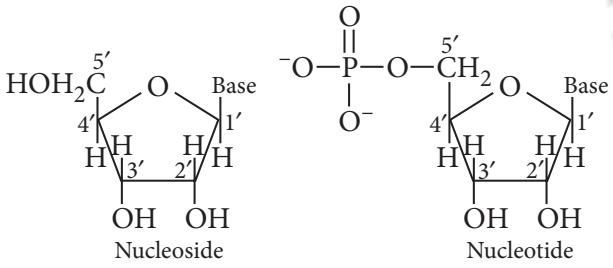
18. (a)

19. (c) : Despite having the aldehyde group glucose does not give 2,4 DNP test, Schiff's test and it does not form the hydrogensulphite addition product with $NaHSO_3$.

20. (a)

21. (i) A unit formed by attachment of a base to C-1′ position of sugar is known as nucleoside.

When nucleoside is linked to phosphoric acid at C-5′ position of sugar moiety, resulting species is called nucleotide.



Nucleoside Nucleotide

Nucleosides contain only two components a pentose sugar and a nitrogenous base.

Nucleotides contain all the three components a pentose sugar, a nitrogenous base and a phosphoric acid group.

(ii) James Watson and Francis Crick gave a double strand helix structure for DNA. Two nucleic acid chains are bonded to each other because the hydrogen bonds are formed between specific pairs of bases. Adenine forms hydrogen bonds with thymine (A═T) whereas cytosine forms hydrogen bonds with guanine (C≡G).

Due to this base pairing principle, the sequence of bases in one strand automatically fixes the sequence of bases in the other strand. Thus, the two strands are complementary and not identical.

22. (i) $H_2Te > H_2Se > H_2S > H_2O$ - acidic character.

(ii) $NH_3 > PH_3 > AsH_3 > SbH_3$ – bond angle

23. No. of *Y* atoms per unit cell in *ccp* lattice

$$= \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$$

[∵ *ccp* is same as face-centred cubic packing. Thus, no. of atoms = 4]

No. of octahedral voids = No. of atoms present in *ccp* lattice

No. of octahedral voids occupied by atoms

$$= \frac{2}{3} \times 4 \ = \frac{8}{3}$$

Therefore, formula of the compound = $X_{\frac{8}{3}}Y_4 = X_2Y_3$

OR

0.74 occupancy of the total volume shows that it has a face-centred cubic cell.

$$\rho = \frac{Z \times M}{N_A \times a^3}$$

$$\Rightarrow a^3 = \frac{4 \times 197}{6.023 \times 10^{23} \times 19.3} = 67.8 \times 10^{-24} \text{ cm}$$

$$\therefore \quad a = 4.07 \times 10^{-8} \text{ cm}$$

For *fcc*, $r = \frac{\sqrt{2}\,a}{4} = 1.439 \times 10^{-8}$ cm

25. Nylon 6 means it is a condensation polymer of one type of monomer molecules containing 6 carbon atoms *i.e.,* caprolactum. Nylon-6, 6 means it is a condensation polymer of two types of monomer molecules each containing 6 carbon atoms *i.e.,* adipic acid and hexamethylene diamirine.

OR

(i) **Semi-synthetic polymers :** These are obtained from natural polymers by chemical modifications *e.g.,* cellulose derivatives as cellulose nitrate and cellulose acetate (rayon).

$$\text{Cellulose} + \underset{\text{Acetic anhydride}}{(CH_3CO)_2O} \xrightarrow{H_2SO_4} \text{Cellulose acetate}$$

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(ii) **Cross-linked or Network polymers :** These are usually formed from bi-functional and tri-functional monomers and contain strong covalent bonds between various linear polymer chains and form a three-dimensional network structure. They are hard and rigid *e.g.*, bakelite, melamine etc.

29. (a) $[Co(NH_3)_6]\,Cl_3 > [Co(NH_3)_5\,Cl]\,Cl_2 > [Co(NH_3)_4\,Cl_2]Cl > [Co(NH_3)_3Cl_3]$

As the number of ions produced increases, conductivity increases.

(b) Pentaamminenitrito-N-cobalt(III) cation $[Co(NO_2)(NH_3)_5]^{2+}$

(c) CO is a weak electron donor and still metal carbonyls are stable. This is because of back bonding resulting from delocalisation of electrons from filled *d*-orbitals of metal into the empty antibonding molecular orbitals of CO ligands.

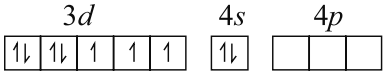
OR

In $[Co(C_2O_4)_3]^{3+}$.

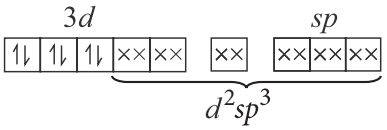
Oxidation state of Co is + 3

Atomic number of Co = 27

Electronic configuration of Co atom :



dsp^3 hybridised orbital of Co(III) :



So, the shape is octahedral and since the complex has no unpaired electrons therefore, it is dimagnetic.

31. (a) $M_B = \dfrac{W_B \times R \times T}{\pi \times V}$

We get,

$$M_B = \frac{8.95\times10^{-3}\,\text{g}\times0.0821\,\text{L atm K}^{-1}\,\text{mol}^{-1}\times298\,\text{K}}{\frac{0.335}{760}\,\text{atm}\times35\times10^{-3}\,\text{L}}$$

M_B = 14193.29 g mol^{-1}

(b) KCl is an electrolyte, it undergoes dissociation in solution, hence number of particles increases. Urea is non-electrolyte.

OR

$$M_B = \frac{K_f \cdot W_B \cdot 1000}{\Delta T_f \cdot W_A}$$

$$= \frac{1.86\times0.5\times1000}{0.24\times100} = 38.75$$

Now, normal molecular mass of KCl

= 39 + 35.5 = 74.5

So, $i = \dfrac{\text{Observed moles of solute}}{\text{Normal moles of solute}} = \dfrac{1+\alpha}{1}$

$\therefore \quad \dfrac{1+\alpha}{1} = 1.92$ or $1+\alpha = 1.92$

or $\alpha = 0.92$

$\therefore$ Percentage ionisation = 92%

36. (i) Let the rate law be $r_0 = k\,[A]^m[B]^n$

$(r_0)_1 = 5.07 \times 10^{-5} = k\,(0.20)^m(0.30)^n$ ...(i)

$(r_0)_2 = 5.07 \times 10^{-5} = k\,(0.20)^m(0.10)^n$ ...(ii)

$(r_0)_3 = 7.16 \times 10^{-5} = k\,(0.40)^m(0.05)^n$ ...(iii)

Dividing equation (i) by equation (ii),

$$\frac{(r_0)_1}{(r_0)_2} = \frac{5.07\times10^{-5}}{5.07\times10^{-5}} = \frac{(0.20)^m(0.30)^n}{(0.20)^m(0.10)^n}$$

$1 = 3^n$ or $3^0 = 3^n \Rightarrow n = 0$

Dividing equation (iii) by equation (ii),

$$\frac{(r_0)_3}{(r_0)_2} = \frac{7.16\times10^{-5}}{5.07\times10^{-5}} = \frac{(0.40)^m(0.05)^n}{(0.20)^m(0.10)^n}$$

$1.412 = 2^m$ or $2^{1/2} = 2^m \Rightarrow m = \dfrac{1}{2}$ or $m = 0.5$

Thus, order of reaction *w.r.t.* A = 0.5, order of reaction *w.r.t.* B = 0

(ii) If E_a = O, then according to Arrhenius equation, $k = Ae^{-Ea/RT} = Ae^{\circ} = A$, *i.e.,* rate constant = collision frequency. This means every collision results into a chemical reaction which cannot be true. Hence, E_a cannot be zero.

37. (a) $CH_3CH_2CH_2OH + HBr \rightarrow CH_3CH_2CH_2Br + H_2O$

$CH_3CH_2CH_2OH + Na \longrightarrow CH_3CH_2CH_2ONa + \dfrac{1}{2}H_2$

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$CH_3CH_2CH_2Br + CH_3CH_2CH_2ONa$

Williamson synthesis ↓

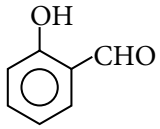
$CH_3CH_2CH_2 — O — CH_2CH_2CH_3 + NaBr$

1-Propoxypropane

OR

(a) (i) $CH_3CH_2CH(OH)CH_3 \xrightarrow[573\ K]{Cu}$ $CH_3CH_2\overset{\overset{O}{\|}}{C}CH_3$

Butan-2-one

(ii) $C_6H_5—OH \xrightarrow[(ii)\ H^+]{(i)\ CHCl_3 + aq.\ NaOH}$ 

Salicylaldehyde

(b) (i) Copper at 573 K

(ii) Sodium borohydride ($NaBH_4$)/C_2H_5OH

(iii) Bromine water

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CBSE SAMPLE PAPER 2019-20
(Solved)

BLUEPRINT

Time Allowed : 3 hours **Maximum Marks :** 70

S. No.	Chapter	VSA-Objective (1 mark)	SA (2 marks)	LA-I (3 marks)	LA-II (5 marks)	Total
1.	Solutions	–	1(2)	1(3)	–	**11(23)**
2.	Electrochemistry	5(5)	–	–	1(5)	
3.	Chemical Kinetics	–	1(2)	1(3)	–	
4.	Surface Chemistry	–	–	1(3)	–	
5.	General Principles and Processes of Isolation of Elements	1(1)	1(2)	–	–	**10(19)**
6.	The *p*-Block Elements	1(1)	1(2)	1(3)	–	
7.	The *d*- and *f*-Block Elements	–	–	–	1(5)	
8.	Coordination Compounds	3(3)	1(2)	–	–	
9.	Haloalkanes and Haloarenes	2(2)	1(2)	–	–	**16(28)**
10.	Alcohols, Phenols and Ethers	1(1)	–	1(3)	–	
11.	Aldehydes, Ketones and Carboxylic Acids	1(1)	1(2)	1(3)	–	
12.	Amines	–	–	–	1(5)	
13.	Biomolecules	4(4)	–	–	–	
14.	Polymers	2(2)	–	–	–	
15.	Chemistry in Everyday Life	–	–	1(3)	–	
	Total	**20(20)**	**7(14)**	**7(21)**	**3(15)**	**37(70)**

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General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) Question paper comprises four sections – A, B, C and D.

(ii) Section A : Q. no. 1 to 20 are very short answer type questions carrying one mark each. Answer these questions in one word or one sentence.

(iii) Section B : Q. no. 21 to 27 are short answer type questions carrying two marks each.

(iv) Section C : Q. no. 28 to 34 are long answer type-I questions carrying three marks each.

(v) Section D : Q. no. 35 to 37 are long answer type-II questions carrying five marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 2 questions of two marks, 2 questions of three marks and all the 3 questions of five marks. You have to attempt only one of the choices in such questions.

(vii) However, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators and log tables is not permitted.

SECTION - A

Read the given passage and answer the questions 1 to 5 that follow :

A Lead storage battery is the most important type of secondary cell having a lead anode and a grid of lead packed with PbO_2 as cathode. A 38% solution of sulphuric acid is used as an electrolyte (Density = 1.294 g mL^{-1}). The battery holds 3.5 L of the acid. During the discharge of the battery, the density of H_2SO_4 falls to 1.139 g mL^{-1} (20% H_2SO_4 by mass).

1. Write the reaction taking place at the cathode when the battery is in use.

2. How much electricity in terms of Faraday is required to carry out the reduction of one mole of PbO_2?

3. What is the molarity of sulphuric acid before discharge?

4. Lead storage battery is considered a secondary cell. Why?

5. Write the products of electrolysis when dilute sulphuric acid is electrolysed using platinum electrodes.

Questions 6 to 10 are one word answers :

6. Name the substance used as depressant in the separation of two sulphide ores by froth floatation method.

7. Name the unit formed by the attachment of a base to 1' position of sugar in a nucleoside.

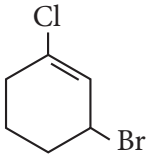
8. Name the species formed when an aqueous solution of amino acid is dissolved in water?

9. What type of reaction occurs in the formation of Nylon 6,6 polymer?

10. Which of the following compounds would undergo cannizzaro reaction :
Benzaldehyde, Cyclohexanone, 2-Methylpentanal

Questions 11 to 15 are multiple choice questions :

11. The IUPAC name of the compound shown below is



(a) 2-bromo-6-chlorocyclohex-1-ene
(b) 6-bromo-2-chlorocyclohexene
(c) 3-bromo-1-chlorocyclohexene
(d) 1-bromo-3-chlorocyclohexene.

12. When one mole of $CoCl_3.5NH_3$ was treated with excess of silver nitrate solution, 2 mol of AgCl was precipitated. The formula of the compound is
(a) $[Co(NH_3)_5Cl_2]Cl$
(b) $[Co(NH_3)_5Cl]Cl_2$
(c) $[Co(NH_3)_4Cl_2](NH_3)Cl$
(d) $[Co(NH_3)_3Cl_3](NH_3)_2$

13. The absorption maxima of several octahedral complex ions are as follows :

S. No.	Compound	λ_{max} nm
1.	$[Co(NH_3)_6]^{3+}$	475
2.	$[Co(CN)_6]^{3-}$	310
3.	$[Co(H_2O)_6]^{3+}$	490

The crystal field splitting is maximum for
(a) $[Co(H_2O)_6]^{3+}$
(b) $[Co(CN)_6]^{3-}$
(c) $[Co(NH_3)_6]^{3+}$
(d) all the complex ions have the same splitting, Δ_o.

14. Predict the number of ions produced per formula unit in an aqueous solution of $[Co(en)_3]Cl_3$.
(a) 4 (b) 3 (c) 6 (d) 2

15. The incorrect statement about LDP is
(a) it is obtained through the free radical addition of ethene
(b) it consists of linear molecules
(c) it is obtained by the H-atom abstraction
(d) peroxide is used as an initiator.

Questions 16 to 20 :

(a) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is correct, but Reason (R) is incorrect statement.
(d) Assertion (A) is incorrect, but Reason (R) is correct statement.

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16. **Assertion :** The two strands in double strand helix structure of DNA are complementary to each other.
Reason : Disulphide bonds are formed between specific pairs of bases.

17. **Assertion :** Glucose reacts with hydroxylamine to form an oxime and also adds a molecule of hydrogen cyanide to give cyanohydrin.
Reason : The carbonyl group is present in the open chain structure of glucose.

18. **Assertion :** The acidic strength of halogen acids varies in the order : HF > HCl > HBr > HI.
Reason : The bond dissociation enthalpy of halogen acids decreases in the order : HF > HCl > HBr > HI.

19. **Assertion :** C_2H_5OH is a weaker base than phenol but is a stronger nucleophile than phenol.
Reason : In phenol the lone pair of electrons on oxygen is withdrawn towards the ring due to resonance.

20. **Assertion :** Aryl halides undergo nucleophilic substitution reactions with ease.
Reason : The carbon halogen bond in aryl halides has partial double bond character.

SECTION - B

21. Calculate the number of lone pairs on central atom in the following molecule and predict the geometry.
XeF_4

22. The rate of a reaction depends upon the temperature and is quantitatively expressed as $k = Ae^{-E_a/RT}$
(i) If a graph is plotted between log k and $1/T$, write the expression for the slope of the reaction.
(ii) If at under different conditions, E_{a_1} and E_{a_2} are the activation energy of two reactions if E_{a_1} = 40 J/mol and E_{a_2} = 80 J/mol and which of the two has a larger value of the rate constant?

23. The experimentally determined molar mass for what type of substances is always lower than the true value when water is used as solvent. Explain. Give one example of such a substance and one example of a substance which does not show a large variation from the true value.

24. Write structure of the products formed :
(a) $CH_3CH_2COOH \xrightarrow{Cl_2,\ Red\ P_4}$
(b) $C_6H_5COCl \xrightarrow[\Delta]{H_2,\ Pd\text{-}BaSO_4}$

25. Draw one of the geometrical isomers of the complex $[Pt(en)_2Cl_2]^{2+}$ which is optically inactive. Also write the name of this entity according to the IUPAC nomenclature.

OR

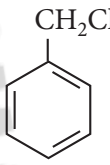
Discuss the bonding in the coordination entity $[Co(NH_3)_6]^{3+}$ on the basis of valence bond theory. Also, comment on the geometry and spin of the given entity. (Atomic no. of Co = 27)

26. What is meant by vapour phase refining? Write any one example of the process which illustrates this technique, giving the chemical equations involved.

OR

Write and explain the reactions involved in the extraction of gold.

27. Which one of the following compounds will undergo hydrolysis at a faster rate by S_N1 mechanism? Justify.



or $CH_3CH_2CH_2Cl$

SECTION - C

28. Calculate the freezing point of a solution containing 0.5 g KCl (Molar mass = 74.5 g/mol) dissolved in 100 g water, assuming KCl to be 92% ionized. (K_f of water = 1.86 K kg/mol).

29. For the reaction $A + B \rightarrow$ products, the following initial rates were obtained at various given initial concentrations.

S. No.	[*A*] mol/L	[*B*] mol/L	Initial rate M/s
1.	0.1	0.1	0.05
2.	0.2	0.1	0.10
3.	0.1	0.2	0.05

Determine the half-life period.

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OR

A first order reaction is 50% complete in 50 minutes at 300 K and the same reaction is again 50% complete in 25 minutes at 350 K. Calculate activation energy of the reaction.

30. Answer the following questions :
(a) Which of the following electrolytes is most effective for the coagulation of AgI/Ag^+ sol?
$MgCl_2$, K_2SO_4, $K_4[Fe(CN)_6]$
(b) What happens when a freshly precipitated $Fe(OH)_3$ is shaken with a little amount of dilute solution of $FeCl_3$.
(c) Out of sulphur sol and proteins, which one forms macromolecular colloids?

31. Account for the following :
(a) Moist SO_2 decolourises $KMnO_4$ solution.
(b) In general, interhalogen compounds are more reactive than halogens (except fluorine).
(c) Ozone acts as a powerful oxidizing agent.

32. Identify the product formed when propan-1-ol is treated with conc. H_2SO_4 at 413 K . Write the mechanism involved for the above reaction.

33. (a) Give chemical tests to distinguish between the following pairs of compounds :
(i) Ethanal and propanone
(ii) Pentan-2-one and pentan-3-one
(b) Arrange the following compounds in increasing order of their acid strength :
Benzoic acid, 4-Nitrobenzoic acid, 3,4-Dinitrobenzoic acid, 4-Methoxybenzoic acid.

OR

Compare the reactivity of benzaldehyde and ethanal towards nucleophilic addition reactions. Write the cross aldol condensation product between benzaldehyde and ethanal.

34. Define and write an example for the following :
(a) Broad spectrum antibiotics
(b) Analgesics

SECTION - D

35. (a) The e.m.f. of the following cell at 298 K is 0.1745 V
$Fe_{(s)} \,|\, Fe^{2+}(0.1\,M) \,||\, H^+(x\,M) \,|\, H_{2\,(g)}(1\,bar) \,|\, Pt_{(s)}$
Given : $E^\circ_{Fe^{2+}/Fe} = -0.44$ V
Calculate the H^+ ions concentration of the solution at the electrode where hydrogen is being produced.
(b) Aqueous solution of copper sulphate and silver nitrate are electrolysed by 1 ampere current for 10 minutes in separate electrolytic cells. Will the mass of copper and silver deposited on the cathode be same or different? Explain your answer.

OR

(a) Calculate the degree of dissociation of 0.0024 M acetic acid if conductivity of this solution is 8.0×10^{-5} S cm^{-1}.
Given : $\lambda^\circ_{H^+} = 349.6$ S cm^2 mol^{-1}, $\lambda^\circ_{CH_3COO^-} = 40.9$ S cm^2 mol^{-1}
(b) Solutions of two electrolytes '*A*' and '*B*' are diluted. The limiting molar conductivity of '*B*' increases to a smaller extent while that of '*A*' increases to a much larger extent comparatively. Which of the two is a strong electrolyte? Justify your answer.

36. An organic compound *A* with molecular formula C_7H_7NO reacts with Br_2/aq. KOH to give compound *B*, which upon reaction with $NaNO_2$ and HCl at 0 °C gives *C*. Compound *C* on heating with CH_3CH_2OH gives a hydrocarbon *D*. Compound *B* on further reaction with Br_2 water gives white precipitate of compound *E*. Identify the compound *A*, *B*, *C*, *D* and *E*, also justify your answer by giving relevant chemical equations.

OR

(a) How will you convert :
(i) Aniline into fluorobenzene
(ii) Benzamide into benzylamine
(iii) Ethanamine to *N*, *N*-diethylethanamine
(b) Write the structures of *A* and *B* in the following :

(i) $CH_3CH_2CN \xrightarrow[\text{Partial hydrolysis}]{OH^-} A \xrightarrow{NaOH + Br_2} B$

(ii) $CH_3CH_2Br \xrightarrow[(ii)\ LiAlH_4]{(i)\ KCN} A \xrightarrow[0°C]{HNO_2} B$

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37. (a) When a chromite ore (*A*) is fused with an aqueous solution of sodium carbonate in free excess of air, a yellow solution of compound (*B*) is obtained. This solution is filtered and acidified with sulphuric acid to form compound (*C*). Compound (*C*) on treatment with solution of KCl gives orange crystals of compound (*D*). Write the chemical formulae of compounds *A* to *D*.

(b) Describe the cause of the following variations with respect to lanthanoids and actinoids :

(i) Greater range of oxidation states of actinoids as compared to lanthanoids.

(ii) Greater actinoid contraction as compared to lanthanoid contraction.

(iii) Lower ionisation enthalpy of early actinoids as compared to the early lanthanoids.

OR

(a) What happens when

(i) Manganate ions (MnO_4^{2-}) undergo disproportionation reaction in acidic medium?

(ii) Lanthanum is heated with sulphur?

(b) Explain the following trends in the properties of the members of the first series of transition elements :

(i) $E°$ (M^{2+}/M) value for copper is positive (+0.34 V) in contrast to the other members of the series.

(ii) Cr^{2+} is reducing while Mn^{3+} is oxidising, though both have d^4 configuration.

(iii) The oxidising power in the series increases in the order :
$VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$.

Detailed Solutions

1. Reaction taking place at cathode when the battery is in use :

$$PbO_{2(s)} + SO_4^{2-}{}_{(aq)} + 4H^+_{(aq)} + 2e^- \rightarrow PbSO_{4(s)} + 2H_2O_{(l)}$$

2. 2 F

3. Mass of solution before discharge

$$= 3500 \text{ mL} \times 1.294 \text{ g mL}^{-1} = 4529 \text{ g}$$

Mass of H_2SO_4 present in this solution

$$= \frac{38}{100} \times 4529 = 1721.02 \text{ g}$$

$$\text{Molarity} = \frac{1721.02}{98 \times 3.5} = 5.02 \text{ M}$$

4. Lead storage battery is rechargeable cell which can be used again and again.

5. At anode : $O_{2(g)}$

At cathode : $H_{2(g)}$

6. Sodium cyanide

7. Nucleoside

Note : Nucleoside contains sugar and base (present at C – 1 of sugar), nucleotide (sugar + base + phosphate group), will only be formed if we place phosphate group at C –5 of sugar of nucleoside.

8. Zwitter ion

9. Condensation Polymerisation

10. Benzaldehyde

11. (c) : 3-Bromo-1-chlorocyclohexene

12. (b) : The formula of the compound is $[Co(NH_3)_5Cl]^{2+}\ 2Cl^-$ or $[Co(NH_3)_5Cl]Cl_2$

13. (b) : Among the given octahedral complexes, $[Co(CN)_6]^{3-}$ ion has maximum crystal field splitting.

14. (a) : In aqueous solution, $[Co(en)_3]Cl_3$ ionises as $[Co(en)_3]^{3+} + 3Cl^-$. Therefore, total number of ions produced per formula unit is 4.

15. (b) : Low density polythene has highly branched structure.

16. (c) : The two strands are complementary to each other because the hydrogen bonds are formed between specific pairs of bases.

17. (a) : Glucose reacts with NH_2OH to form an oxime and adds a molecule of hydrogen cyanide to give cyanohydrin. These reactions confirm the presence of a carbonyl group (>C = O) in glucose.

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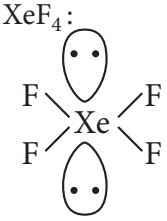
18. (d) : The acidic strength of these acids varies in the order, HF < HCl < HBr < HI. The stability of these halides decreases down the group due to decrease in bond (H—*X*) dissociation enthalpy in the order, HF > HCl > HBr > HI.

19. (d) : Phenol is stronger acid than C_2H_5OH.

20. (d) : Aryl halides are extremely less reactive towards nucleophilic substitution reaction due to resonance effect.

21. Structure of XeF_4 :



Lone Pairs : 2
Geometry : Square planar

22. (i) $\text{Slope} = -\dfrac{E_a}{2.303\,R}$

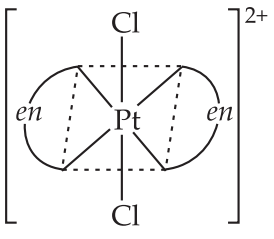
(ii) $\dfrac{k_1}{k_2} = \dfrac{e^{-E_{a_1}/RT}}{e^{-E_{a_2}/RT}} = \dfrac{e^{-40/RT}}{e^{-80/RT}}$ hence, $k_1 > k_2$.

23. When there is dissociation of solute into ions, in dilute solutions (ignoring interionic attractions) the number of particles increases. As the value of colligative properties depends on the number of particles of the solute , the experimentally observed value of colligative property will be higher than the true value, therefore the experimentally determined (observed) molar mass is always lower than the true value.
For KCl(electrolyte), the experimentally determined molar mass is always lower than the true value when water is used as a solvent. Glucose (non-electrolyte) does not show a large variation from the true value.

24. (a) $CH_3-\underset{\displaystyle Cl}{\underset{|}{CH}}-COOH$

(b) C_6H_5CHO

25.



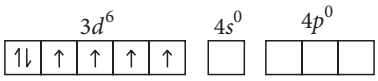
trans-$[PtCl_2(en)_2]^{2+}$

IUPAC Name of the entity :
Dichlorido*bis*(ethane-1,2-diamine)platinum(IV) ion.

OR

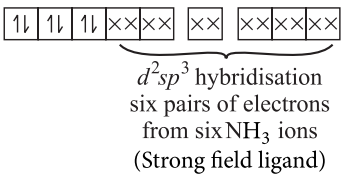
Oxidation of cobalt in $[Co(NH_3)_6]^{3+}$ is +3.

Co^{3+} :

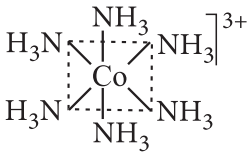


Ground State

$[Co(NH_3)_6]^{3+}$ ion :



Hybridisation – d^2sp^3
Structure – Octahedral
(low spin)
Nature – Diamagnetic



IUPAC name : Hexaamminecobalt(III) ion

26. Vapour phase refining : It is a refining method in which the metal is converted into its volatile compound and collected elsewhere which is then decomposed to give pure metal.
Example : Mond's Process for refining of nickel/ van-Arkel method for refining of zirconium.

Equations involved : $Ni + 4CO \xrightarrow{330\text{-}350\ K} Ni(CO)_4$

$Ni(CO)_4 \xrightarrow{450\text{-}470\ K} \underset{\text{Pure nickel}}{Ni} + 4CO$

OR

Extraction of gold involves leaching the metal with CN^-.
Oxidation reaction :

$4Au_{(s)} + 8CN^-_{(aq)} + 2H_2O_{(aq)} + O_{2(g)} \rightarrow 4[Au(CN)_2]^-_{(aq)} + 4OH^-_{(aq)}$

The metal is recovered by displacement method :

$2[Au(CN)_2]^-_{(aq)} + Zn_{(s)} \rightarrow 2Au_{(s)} + [Zn(CN)_4]^{2-}_{(aq)}$

Zinc acts as a reducing agent.

27. Chlorophenylmethane (benzyl chloride) will undergo hydrolysis at faster rate by S_N1 mechanism than 1-chloropropane.
$CH_3CH_2CH_2Cl$ forms 1° carbocation, which is less stable than benzylic carbocation.

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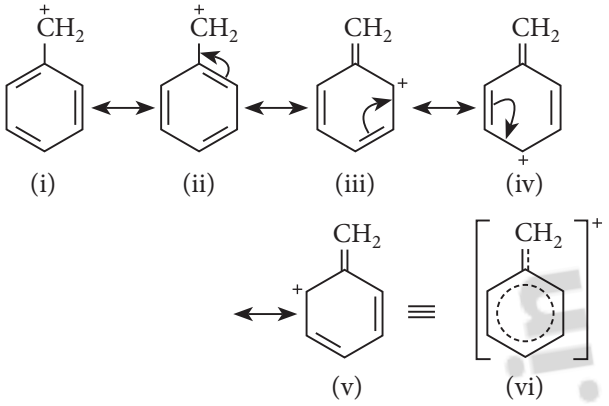

Greater the stability of the carbocation, greater will be its ease of formation from the corresponding halide and faster will be the rate of reaction. The benzylic carbocation formed gets stabilised through resonance.

28. $KCl \rightarrow K^+ + Cl^-$

$n = 2, i = 1 - \alpha + 2\alpha = 1 + \alpha$

$\Delta T_f = iK_f m$

$$= (1+0.92)\times 1.86 \times \frac{0.5\times 1000}{74.5\times 100} = 0.24$$

$\Delta T_f = T_f^\circ - T_f \Rightarrow 0.24 = 0 - T_f, T_f = -0.24°C$

29. Rate $= k\,[A]^x\,[B]^y$

$0.05 = k[0.1]^x\,[0.1]^y$ ...(i)

$0.10 = k[0.2]^x\,[0.1]^y$ ...(ii)

$0.05 = k[0.1]^x\,[0.2]^y$ ...(iii)

Equation (ii) divided by (i)

$$\frac{0.10}{0.05} = (2)^x \Rightarrow x = 1$$

Equation (iii) divided by (i)

$$\frac{0.05}{0.05} = (2)^y \Rightarrow y = 0$$

$\therefore$ rate $= k[A]^1\,[B]^0$

It is a first order reaction.

$$k = \frac{\text{rate}}{[A]} = 0.5\ s^{-1}$$

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.5}$$

$t_{1/2} = 1.386$ s

OR

$$\because\ t_{1/2} = \frac{0.693}{k} \qquad \therefore\ k_2 = \frac{0.693}{25} \text{ at } 350\ K$$

$$k_1 = \frac{0.693}{50} \text{ at } 300\ K \Rightarrow \frac{k_2}{k_1} = 2$$

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log 2 = \frac{E_a}{2.303\times 8.314}\left[\frac{350-300}{350\times 300}\right]$$

$E_a = 12.103$ kJ/mol

30. (a) Coagulation power of an electrolyte depends on the valency of oppositely charged ion, greater the valency of oppositely charged ions, more is the coagulation power. The coagulation power of anions is in the order :

$[Fe(CN)_6]^{4-} > SO_4^{2-} > Cl^-$.

Therefore, $K_4[Fe(CN)_6]$ is most effective for the coagulation of AgI/Ag^+.

(b) $Fe(OH)_3$ is converted into colloidal state by preferential adsorption of Fe^{3+} ions.

$$Fe(OH)_3 + FeCl_3 \rightarrow [Fe(OH)_3.Fe]^{3+} + 3Cl^-$$

Fresh ppt. Electrolyte Colloidal sol

(c) Proteins form macromolecular colloids as these are the polymers of very high molecular mass.

31. (a) Moist sulphur dioxide behaves as a reducing agent, reduces MnO_4^- to Mn^{2+}.

$$2MnO_4^- + 5SO_2 + 2H_2O \rightarrow 5SO_4^{2-} + 2Mn^{2+} + 4H^+$$

(Pink violet) (colourless)

(b) Generally interhalogen compounds are more reactive than halogens (except fluorine) because X –X' bond in interhalogens is weaker than X-X bond in halogens except F-F bond.

(c) Due to the ease with which it liberates atoms of nascent oxygen. ($O_3 \rightarrow O_2 + O$), it acts as a powerful oxidising agent.

32. Propan-1-ol on treatment with conc. H_2SO_4 at 413 K would yield 1-propoxypropane.

$$2CH_3-CH_2-CH_2-OH + H_2SO_4 \xrightarrow{413\ K}$$

Propan-1-ol

$$CH_3-CH_2-CH_2-O-CH_2-CH_2-CH_3$$

1-Propoxypropane

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Mechanism :

$$CH_3-CH_2-CH_2-OH + H^+ \rightleftharpoons$$

Propan-1-ol

$$CH_3-CH_2-CH_2-\overset{+}{O}(H)-H$$

$$CH_3-CH_2-CH_2-\overset{+}{O}(H)-H \rightleftharpoons$$

$$CH_3-CH_2-\overset{+}{C}H_2 + H_2O$$

$$CH_3-CH_2-CH_2-\ddot{O}-H + CH_3-CH_2-\overset{+}{C}H_2$$

Propan-1-ol

$$CH_3-CH_2-CH_2-O-CH_2-CH_2-CH_3 \xleftarrow{-H^+}$$

33. (a) (i)

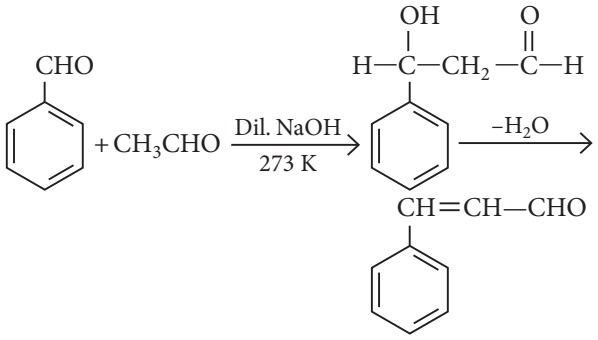
Experiment	Ethanal	Propanone
Tollens' test : Warm the organic compound with freshly prepared ammonical silver nitrate solution (Tollens' reagent).	A bright silver mirror is produced.	No silver mirror is formed.
(ii) **Fehling's solution test :** Heat the organic compound with Fehling's reagent.	A reddish brown precipitate is obtained.	No precipitate is obtained.
Experiment	**Pentan-2-one**	**Pentan 3-one**
Iodoform test : The organic compound is heated with iodine in presence of sodium hydroxide solution.	A yellow precipitate is obtained.	No yellow precipitate is obtained.

(b) The compounds in increasing order of their acidic strength are : 4-Methoxybenzoic acid < Benzoic acid < 4-Nitrobenzoic acid < 3, 4-Dinitrobenzoic acid

OR

The carbon atom of the carbonyl group of benzaldehyde is less electrophilic than carbon atom of the carbonyl group present in ethanal. The polarity of the carbonyl group is reduced in benzaldehyde due to resonance, hence less reactive than ethanal.

The cross aldol condensation reaction between benzaldehyde and ethanol :



34. (a) **Broad spectrum antibiotics :** Antibiotics which kill or inhibit a wide range of Gram-positive and Gram-negative bacteria are said to be broad spectrum antibiotics. *e.g.,* Chloramphenicol, tetracycline, etc.

(b) **Analgesics :** Analgesics are chemicals which reduce pain without causing impairement of consciousness, mental confusion, incoordination or paralysis or some other disturbances of nervous system, *e.g.,* aspirin, paracetamol, etc.

35. (a) $Fe + 2H^+ \rightarrow H_2 + Fe^{2+}$

$$E_{cell} = E^\circ_{cell} - \frac{2.303\,RT}{nF}\log\frac{Fe^{2+} \times P_{H_2}}{[H^+]^2}$$

$$E^\circ_{cell} = E^\circ_{H^+/H_2} - E^\circ_{Fe^{2+}/Fe} = 0 - (-0.44) = 0.44\ V$$

$$0.1745 = 0.44 - \frac{0.0591}{2}\log\frac{[0.1]\times 1}{[x]^2}$$

$\log x = -5$; $\log [H^+] = -5$ $\therefore$ $[H^+] = 10^{-5}$

(b) The mass of copper and silver deposited at the cathode will be different. The amount of different substances deposited by the same quantity of electricity passing through the electrolytic solution are directly proportional to their chemical equivalent weights (Atomic mass of metal/Number of electrons required to reduce the cation).

OR

(a) $\Lambda^\circ(CH_3COOH) = \lambda^\circ_{H^+} + \lambda^\circ_{CH_3COO^-}$

$= 349.6 + 40.9 = 390.5\ S\ cm^2\ mol^{-1}$

$$\Lambda_m = \frac{\kappa \times 1000}{c}$$

$$\Lambda_m = \frac{8.0\times 10^{-5}\,S\,cm^{-1} \times 1000\,cm^3\,L^{-1}}{0.0024\ mol\ L^{-1}}$$

$$= 33.33\,S\,cm^2\,mol^{-1}$$

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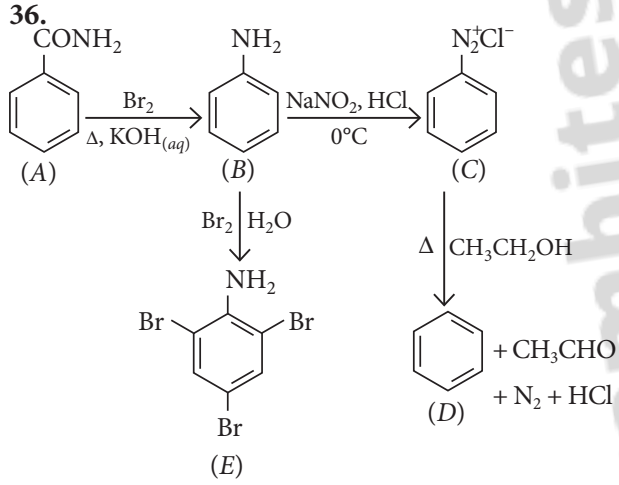
$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} \Rightarrow \alpha = \frac{33.33\ \text{S cm}^2\ \text{mol}^{-1}}{390.5\ \text{S cm}^2\ \text{mol}^{-1}} = 0.085$$

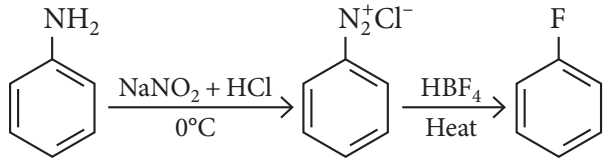
(b) Electrolyte *B* is a strong electrolyte.
Limiting molar conductivity increases only to a smaller extent for a strong electrolyte, as on dilution the interionic interactions are overcome.
Limiting molar conductivity increases to a larger extent for a weak electrolyte, as on dilution the degree of dissociation increases, therefore the number of ions per unit volume of solution increases.

36.

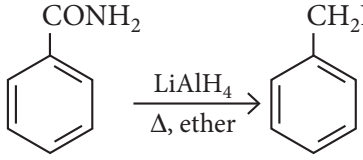


OR

(a) (i)



(ii)



(iii) $CH_3CH_2NH_2 \xrightarrow{C_2H_5Cl} C_2H_5-\underset{\displaystyle H}{\overset{\displaystyle |}{N}}-C_2H_5$

$$\xrightarrow{C_2H_5Cl} C_2H_5-\underset{\displaystyle C_2H_5}{\overset{\displaystyle |}{N}}-C_2H_5$$

(b) (i) *A* : $CH_3CH_2CONH_2$
B : $CH_3CH_2NH_2$

(ii) *A* : $CH_3CH_2CH_2NH_2$
B : $CH_3CH_2CH_2OH$

37. (a) $\underset{(A)}{2FeCr_2O_4} + 4Na_2CO_3 + \frac{7}{2}O_2 \rightarrow \underset{(B)}{Fe_2O_3 + 4Na_2CrO_4} + 4CO_2$

$$\underset{(B)}{2Na_2CrO_4} + H_2SO_4 \rightarrow \underset{(C)}{Na_2Cr_2O_7} + Na_2SO_4 + H_2O$$

$$Na_2Cr_2O_7 + 2KCl \rightarrow \underset{(D)}{K_2Cr_2O_7} + 2NaCl$$

Hence, $A = FeCr_2O_4$; $B = Na_2CrO_4$; $C = Na_2Cr_2O_7$
$D = K_2Cr_2O_7$

(b) (i) 5*f*, 6*d* and 7*s* levels in actinoids are of comparable energies.
(ii) This is due to poorer shielding by 5*f*-electrons in actinoids as compared to shielding by 4*f*-electrons in lanthanoids.
(iii) In actinoids, 5*f*-electrons are more effectively shielded from the nuclear charge than the 4*f*-electrons of the corresponding lanthanoids. Since, the outer electrons are less firmly held, they are available for bonding in the actinoids.

OR

(a) (i) MnO_4^{2-} ions disproportionate in acidic medium to give permanganate ions and manganese(IV) oxide.

$$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

(ii) Lanthanum sulphide is formed when lanthanum is heated with sulphur.

$$2La + 3S \xrightarrow{\text{Heat}} La_2S_3$$

(b) (i) Copper has high enthalpy of atomisation and low enthalpy of hydration. Since, the high energy to transform $Cu_{(s)}$ to $Cu^{2+}_{(aq)}$ is not balanced by hydration enthalpy, therefore $[E^\circ(M^{2+}/M)]$ value for copper is positive (+0.34 V).
(ii) Cr^{2+} is reducing agent as its configuration changes from d^4 to d^3, the latter having more stable half-filled t_{2g} level. On the other hand, the change from Mn^{3+} to Mn^{2+} results in extra stable d^5 configuration.
(iii) This is due to the increasing stability of the species of lower oxidation state to which they are reduced.

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